Surface Production Operations

DESIGN OF OIL HANDLING SYSTEMS AND FACILITIES

Ken Arnold & Maurice Stewart



VOLUME ONE

THIRD EDITION

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Surface Production Operations

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Design of Oil Handling Systems and Facilities

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THIRD EDITION



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Acknowledgments to the Third Edition

A number of people helped to make possible this revised third edition of *Surface Production Operations, Volume 1—Design of Oil and Water Handling Facilities*. A real debt is owed to the 45,000-plus professional men and women of the organizations that I've taught and worked with through my 35-plus years in the oil and gas industry and made a reality the ideas in this book. The companies are too numerous to name, but it's worth emphasizing that a consultant only makes suggestions—it's the engineers, managers, technicians, and operators who are faced with the real challenge. I have been privileged to work with the "best-of-the-best" companies in the world, and this book is dedicated to them for their vision and perseverance.

Although I can't mention everyone who has helped me along the way, I would like to say thank you to my colleagues and friends: Jamin Djuang of PT Loka Datamas Indah; Chang Choon Kiang, Amran Manaf, and Ridzuan Arrifin of Petroleum Training Southeast Asia (PTSEA); Clem Nwogbo of Resourse Plus; Khun Aujchara and Bundit Pattanasak of PTTEP; Al Ducote and Greg Abdelnor of Chevron Nigeria Limited, and David Rodriguez of Chevron Angola (CABGOC).

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importance was the contribution of Heri Wibowo of STC, who was responsible for coordinating the entire typing and drafting effort. Heri was also responsible for editing and pulling it all together at the end. However, we take full responsibility for any errors that still remain in this text.

Lastly, I would like to thank my wife, Dyah who has been my inspiration, providing support and encouragement when needed.

Maurice Stewart

The first editions of this book were based mostly on materials I had developed and gathered over the years based on what was then 20 years worth of experience and interaction with some very talented people at Shell and Paragon Engineering Services (now AMEC Paragon). Maurice provided first drafts of several chapters, additional materials and technical assistance.

The second edition was created by Maurice and I furnishing guidance and technical material to a group of AMEC Paragon engineers who made modifications to the existing chapters. These engineers were: Eric Barron, Jim Cullen, Fernando De La Fuente, Robert Ferguson, Mike Hale, Sandeep Khurana, Kevin Mara, Matt McKinstry, Carl Sikes, Mary Thro, Kirk Trascher and Mike Whitworth. David Arnold pulled it all together.

This edition contains significant amounts of new material which was developed and gathered primarily by Maurice as a result of his years of teaching and consulting using the original editions as a guide. I served mostly as a technical reviewer adding little in the way of new materials. Maurice deserves most of the credit for this edition.

Ken Arnold

About the Book

Surface Production Operations, Volume 1—Design of Oil and Water Handling Facilities, is a complete and up-to-date resource manual for the design, selection, specification, installation, operation, testing, and troubleshooting of oil and water handling facilities. It is the first volume in the Surface Production Operations series and is the most comprehensive book you'll find today dealing with surface production operations in its various stages, from initial entry into the flowline through separation, treating, conditioning, and processing equipment to the exiting pipeline. Featured in this text are such important topics as gas–liquid separation, liquid–liquid separation, oil treating, desalting, water treating, water injection, crude stabilization, and many other related topics.

This complete revision builds upon the classic text to further enhance its use as a facility engineering process design manual of methods and proven fundamentals. This new edition includes important supplemental mechanical and related data, nomographs, illustrations, charts, and tables. Also included are improved techniques and fundamental methodologies to guide the engineer in designing surface production equipment and applying chemical processes to properly detailed equipment.

All volumes of the Surface Production Operations series serve the practicing engineer by providing organized design procedures; details on suitable equipment for application selection; and charts, tables, and nomographs in readily usable form. Facility engineers, designers, and operators will develop a "feel" for the important parameters in designing, selecting,

specifying, operating, and troubleshooting surface production facilities. Readers will understand the uncertainties and assumptions inherent in designing and operating the equipment in these systems and the limitations, advantages, and disadvantages associated with their use.

Preface to the Third Edition

Ken Arnold and I initially wrote the Surface Production Operations two-volume series with the intention of providing facility engineers with a starting point for addressing the design and operation of surface production facilities. This text provides the basic concepts and techniques necessary to design, specify, and manage oil and gas production facilities.

In the early 1980s, Ken and I developed and taught a number of graduate-level production facility design courses. These courses were taught in the petroleum engineering department of the University of Houston, Tulane University, and Louisiana State University. In the mid-1980s, we took our course lecture notes and published the two-volume Surface Production Operations series. These books became the standard for the industry and have been used by thousands in every oil producing region of the world since their first printing.

We developed and taught two 5-day intensive continuing education courses dealing with oil and gas handling facilities; they were based on our production facility design experience, with emphasis on how to design, select, specify, install, operate, test, and troubleshoot. These courses became so well known through presentations in Southeast Asia, Northern and West Africa, the North Sea, Western and Southern Europe, China, Central Asia, the Democratic Republic of Congo, India, Central and South America, Australia, Canada, and throughout the United States, that in the late 1980s, in response to the many requests by international oil and gas companies and design consultants, we developed additional 5-day seminars devoted to all aspects of production facility design. The continuing-education course lecture notes developed for the 20-plus 5-day courses was the starting point for the expansion and extensive revision of this series.

The third edition of *Surface Production Operations, Volume 1—Design of Oil and Water Handling Facilities*, builds upon the classic text to further enhance its use as a production facility engineering design manual. Every chapter has been significantly expanded and thoroughly updated with new material. Every chapter has been carefully reviewed and older material removed and replaced by newer design techniques. It is important to appreciate that not all of the material has been replaced, because much of the so-called older material is still the best available today, and still yields good designs. Additional charts and tables have been included to aid in the design methods or in explaining the design techniques. This book further provides both fundamental theories where applicable and directs application of these theories to applied equations, expressed in both SI and field units, essential in the design effort. A conscious effort has been made to offer guidelines of sound engineering judgment, decisions, and selections with applicable codes, standards, and recommended practices.

Chapter 1
The Production Facility

Introduction

The job of a production facility is to separate the well stream into three components, typically called "phases" (oil, gas, and water), and process these phases into some marketable product(s) or dispose of them in an environmentally acceptable manner. In mechanical devices called "separators," gas is flashed from the liquids and "free water" is separated from the oil. These steps remove enough light hydrocarbons to produce a stable crude oil with the volatility (vapor pressure) to meet sales criteria. Figures 1-1 and 1-2 show typical separators used to separate gas from liquid or water from oil. Separators can be either horizontal or vertical in configuration.The gas that is separated must be compressed and treated for sales. Compression is typically done by engine-driven reciprocating compressors (see Figure 1-3). In large facilities or in booster service, turbine-driven centrifugal compressors, such as that shown in Figure 1-4, are used. Large integral reciprocating compressors are also used (see Figure 1-5).

Usually, the separated gas is saturated with water vapor and must be dehydrated to an acceptable level, normally less than 7 lb/MMscf (110 mg H_2O/Sm^3). This is normally done in a glycol dehydrator, such as that shown in Figure 1-6.

Dry glycol is pumped to the large vertical contact tower, where it strips the gas of its water vapor. The wet glycol then flows through a separator to the large horizontal reboiler, where it is heated and the water boiled off as steam.

In some locations it may be necessary to remove the heavier hydrocarbons to lower the hydrocarbon dew point. Contaminants such as H_2S and CO_2 may be present at levels higher than those acceptable to the gas purchaser. If this is the case, then additional equipment will be necessary to "sweeten" the gas.

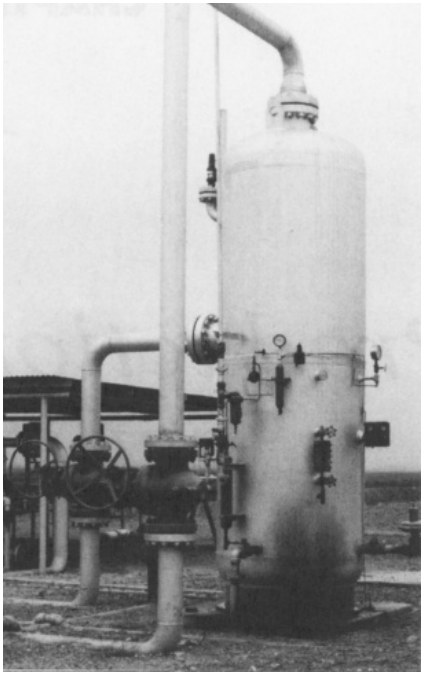


Figure 1-1. A typical vertical two phase separator at a land location. The inlet comes in the left side, gas comes off the top, and liquid leaves the bottom right side of the separator.

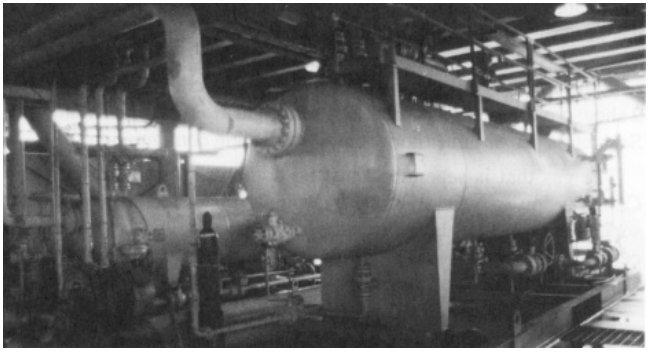


Figure 1-2. A typical horizontal separator on an offshore platform showing the inlet side. Note the drain valves at various points along the bottom and the access platform along the top.

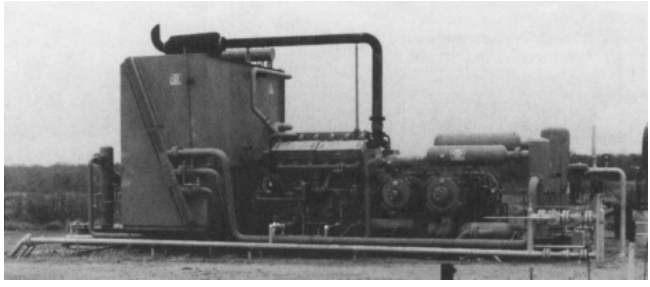


Figure 1-3. Engine-driven reciprocating compressor package. The inlet and inter-stage scrubbers (separators) are at the right. The gas is routed through pulsation bottles to gas cylinders and then to the cooler on the left end of the package. The engine that drives the compressor cylinders is located to the right of the box-like cooler.

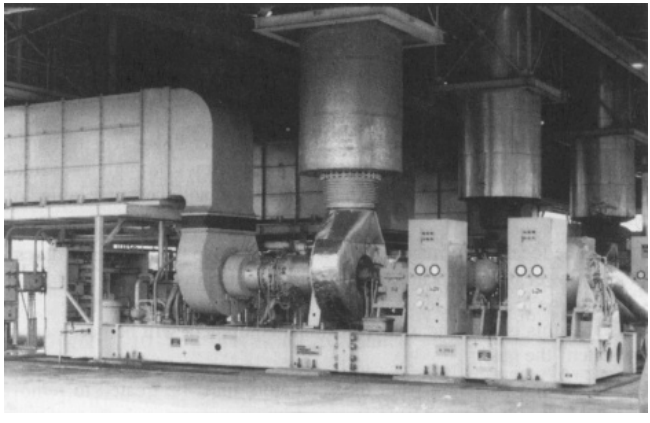


Figure 1-4. Turbine-driven centrifugal compressor package. The turbine draws air in from the large duct on the left. This is mixed with fuel and ignited. The jet of gas thus created causes the turbine blades to turn at high speed before being exhausted vertically upward through the large cylindrical duct. The turbine shaft drives the two centrifugal compressors, which are located behind the control cabinets on the tight end of the skid.

The oil and emulsion from the separators must be treated to remove water. Most oil contracts specify a maximum percent of basic sediment and water (BS&W) that can be in the crude. This will typically vary from 0.5% to 3% depending on location. Some refineries have a limit on salt content in the crude, which may require several stages of dilution with fresh water and subsequent treating to remove the water. Typical salt limits are 10 to 25 pounds of salt per thousand barrels.

Figures 1-7 and 1-8 are typical direct-fired heater-treaters that are used for removing water from the oil and emulsion being treated. These can

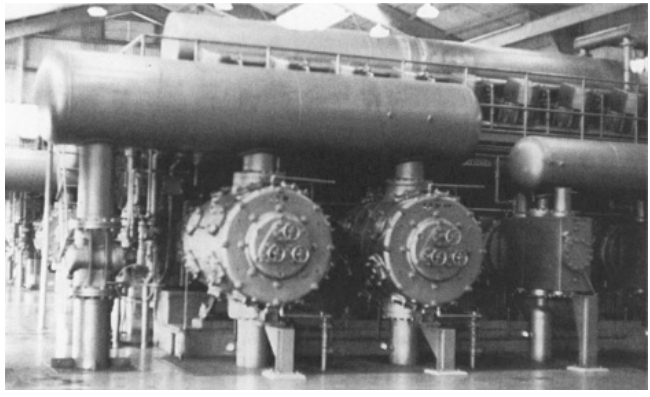


Figure 1-5. A 5500-Bhp integral reciprocating compressor. The sixteen power cylinders located at the top of the unit (eight on each side) drive a crankshaft that is directly coupled to the horizontal compressor cylinders facing the camera. Large cylindrical "bottles" mounted above and below the compressor cylinders filter out acoustical pulsations in the gas being compressed.

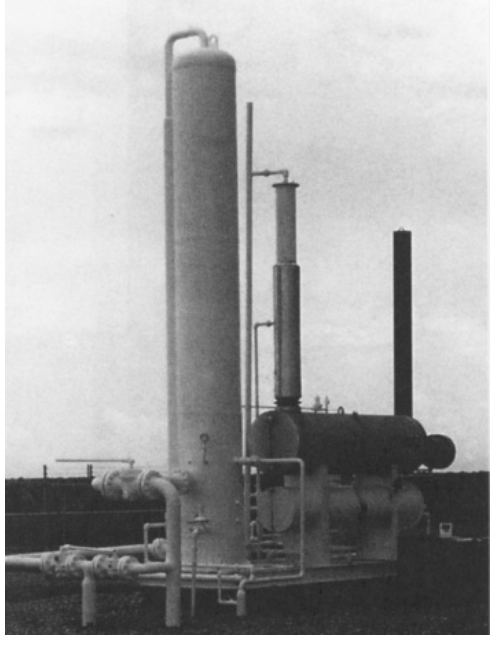


Figure 1-6. A small glycol gas dehydration system. The large vertical vessel on the left is the contact tower where "dry" glycol contacts the gas and absorbs water vapor. The upper horizontal vessel is the "reboiler" or "reconcentrator" where the wet glycol is heated, boiling off the water that exits the vertical pipe coming off the top just behind the contact tower. The lower horizontal vessel serves as a surge tank.

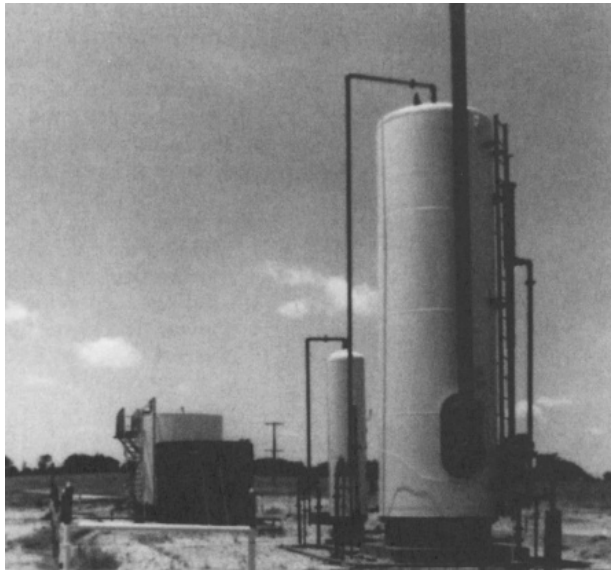


Figure 1-7. A vertical heater-treater. The emulsion to be treated enters on the far side. The fire-tubes (facing the camera) heat the emulsion, and oil exits near the top. Water exits the bottom through the external water leg on the right, which maintains the proper height of the interface between oil and water in the vessel. Gas exits the top. Some of the gas goes to the small "pot" at the lower right where it is scrubbed prior to being used for fuel for the burners.

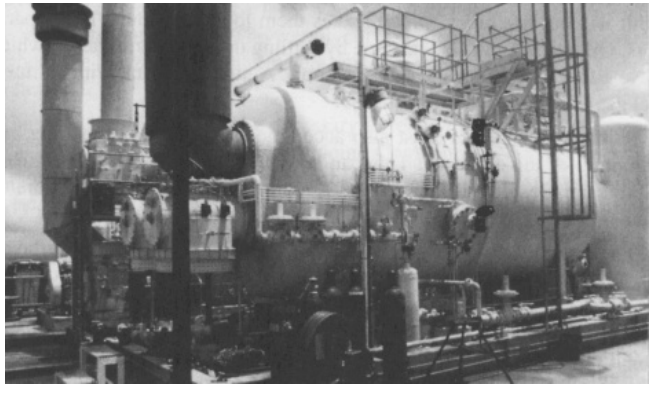


Figure 1-8. A horizontal heater-treater with two burners.

be either horizontal or vertical in configuration and are distinguished by the fire tube, air intakes, and exhausts that are clearly visible. Treaters can be built without fire tubes, which makes them look very much like separators. Oil treating can also be done by settling or in gunbarrel tanks, which have either external or internal gas boots. A gunbarrel tank with an internal gas boot is shown in Figure 1-9.

Production facilities must also accommodate accurate measuring and sampling of the crude oil. This can be done automatically with a Lease Automatic Custody Transfer (LACT) unit or by gauging in a calibrated tank. Figure 1-10 shows a typical LACT unit.

The water that is produced with crude oil can be disposed of overboard in most offshore areas, or evaporated from pits in some locations onshore. Usually, it is injected into disposal wells or used for waterflooding. In any case, water from the separators must be treated to remove small quantities of produced oil. If the water is to be injected into a disposal well, facilities may be required to filter solid particles from it.

Water treating can be done in horizontal or vertical skimmer vessels, which look very much like separators. Water treating can also be done in one of the many proprietary designs discussed in this text such as upflow or downflow CPIs (see Figure 1-11), flotation units (see Figure 1-12), cross-flow coalescers/separators, and hydrocyclones.

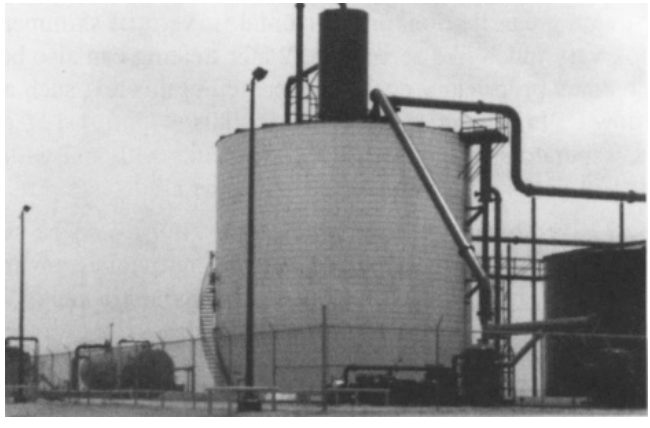


Figure 1-9. A gunbarrel tank for treating oil. The emulsion enters the "gas boot" on top where gas is liberated and then drops into the tank through a specially designed "downcomer" and spreader system. The interface between oil and water is maintained by the external water leg attached to the right side of the tank. Gas from the tank goes through the inclined pipe to a vapor recovery compressor to be salvaged for fuel use.

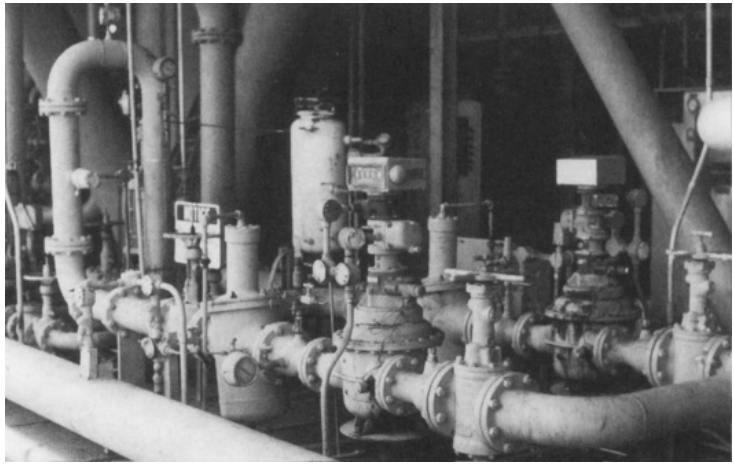


Figure 1-10. A LACT unit for custody transfer of oil. In the vertical loop on left are BS&W probe and a sampler unit. The flow comes through a strainer with a gas eliminator on top before passing through the meter. The meter contains devices for making temperature and gravity corrections, for driving the sampler, and for integrating the meter output with that of a meter prover (not shown).

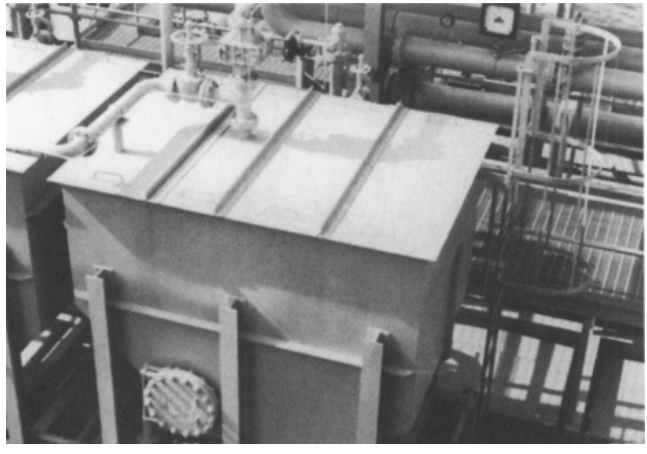


Figure 1-11. A corrugated plate interceptor (CPI) used for treating water. Note that the top plates are removable so that maintenance can be performed on the plates located internally to the unit.

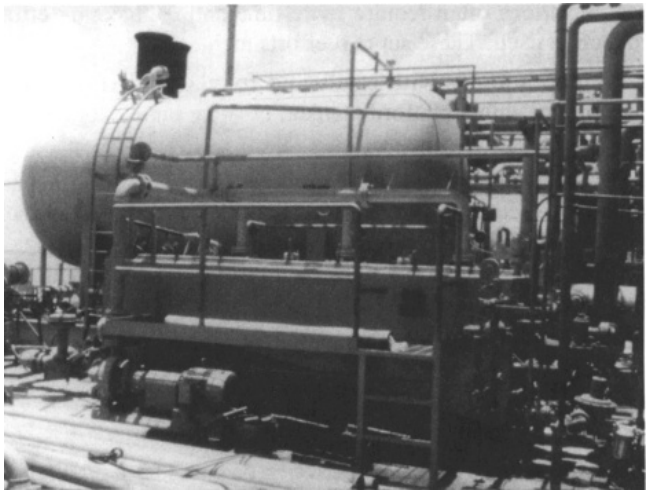


Figure 1-12. A horizontal skimmer vessel for primary separation of oil from water with a gas flotation unit for secondary treatment located in the foreground. Treated water from the flotation effluent is recycled by the pump to each of the three cells. Gas is sucked into the stream from the gas space on top of the water by a venture and dispersed in the water by a nozzle.

Any solids produced with the well stream must also be separated, cleaned, and disposed of in a manner that does not violate environmental criteria. Facilities may include sedimentation basins or tanks, hydrocyclones, filters, etc. Figure 1-13 is a typical hydrocyclone or "desander" installation.

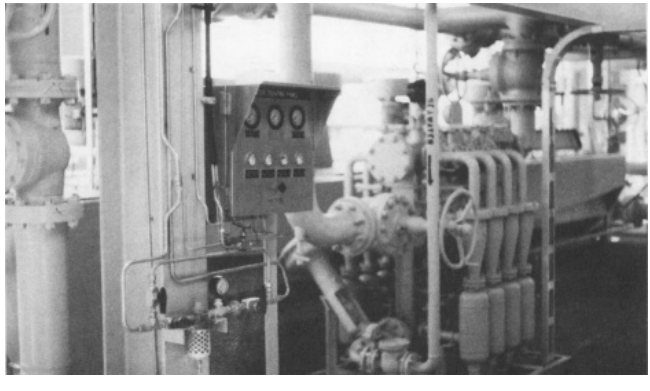


Figure 1-13. Hydrocyclone desanders used to separate sand from produced water prior to treating the water.

The facility must provide for well testing and measurement so that gas, oil, and water production can be properly allocated to each well. This is necessary not only for accounting purposes but also to perform reservoir studies as the field is depleted.

The preceding paragraphs summarize the main functions of a production facility, but it is important to note that the auxiliary systems supporting these functions often require more time and engineering effort than the production itself. These support efforts include

1. Developing a site with roads and foundations if production is onshore, or with a platform, tanker, or some more exotic structure if production is offshore.
2. Providing utilities to enable the process to work: generating and distributing electricity; providing and treating fuel gas or diesel; providing instrument and power air; treating water for desalting or boiler feed, etc. Figure 1-14 shows a typical generator installation, and Figure 1-15 shows an instrument air compressor.
3. Providing facilities for personnel, including quarters (see Figure 1-16), switchgear and control rooms (see Figure 1-17), workshops, cranes, sewage treatment units (see Figure 1-18), drinking water (see Figure 1-19), etc.
4. Providing safety systems for detecting potential hazards (see Figures 1-20 and 1-21), for fighting hazardous situations when they occur (see Figures 1-22 and 1-23), and for personnel protection and escape (see Figure 1-24).

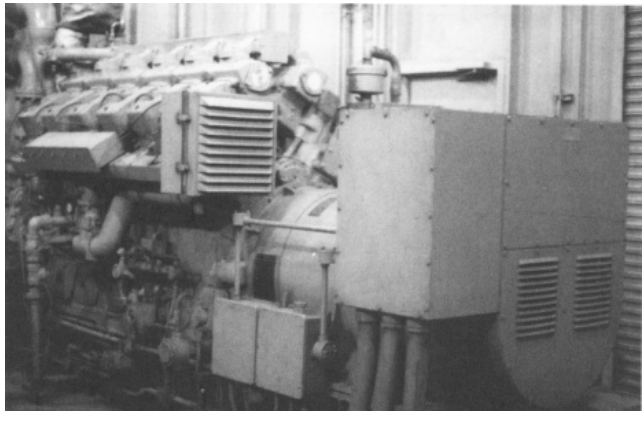


Figure 1-14. A gas-engine-driven generator located in a building on an offshore platform.

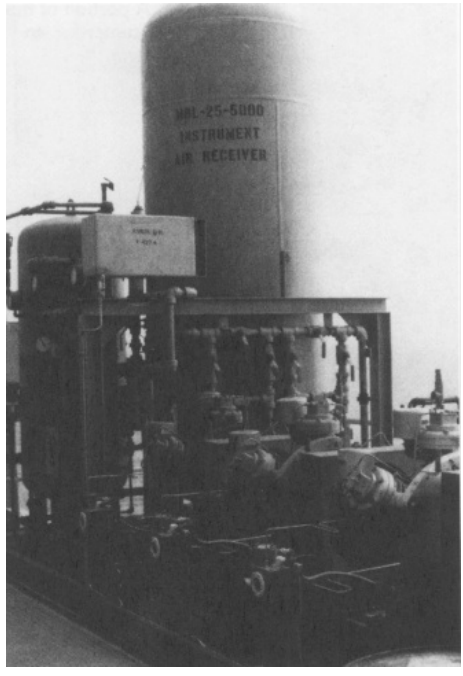


Figure 1-15. A series of three electric-motor-driven instrument air compressors. Note each one has its own cooler. A large air receiver is included to minimize the starting and stopping of the compressors and to assure an adequate supply for surges.

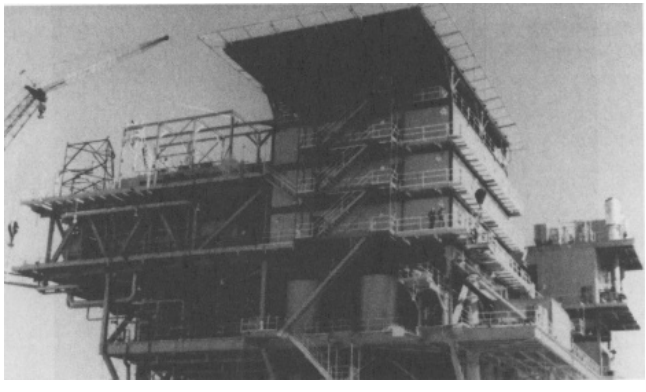


Figure 1-16. A three-story quarters building on a deck just prior to loadout for cross-ocean travel. A helideck is located on top of the quarters.

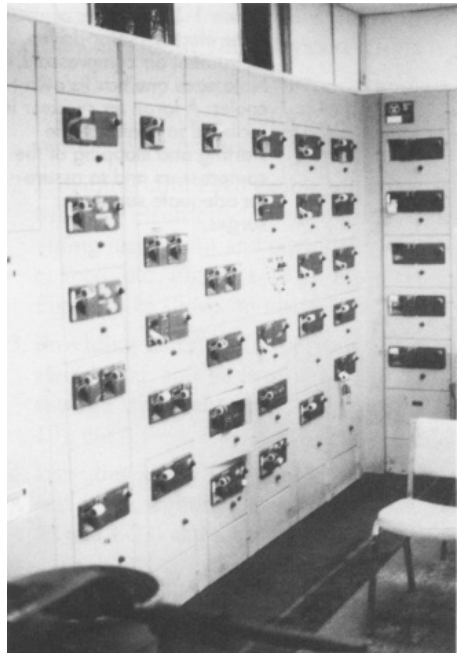


Figure 1-17. A portion of the motor control center for an offshore platform.

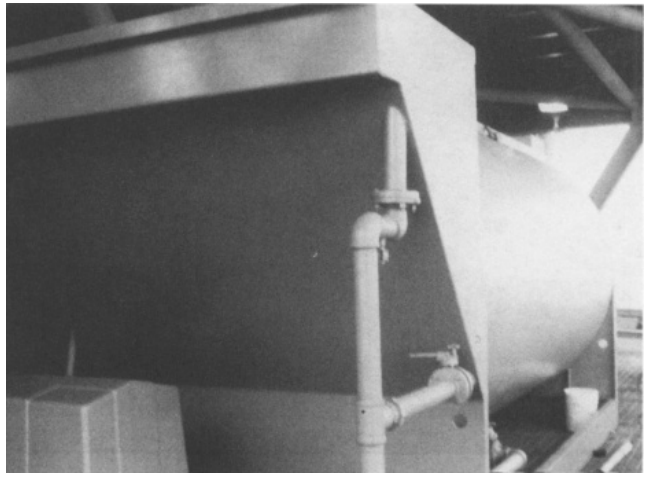


Figure 1-18. An activated sludge sewage treatment unit for an offshore platform.

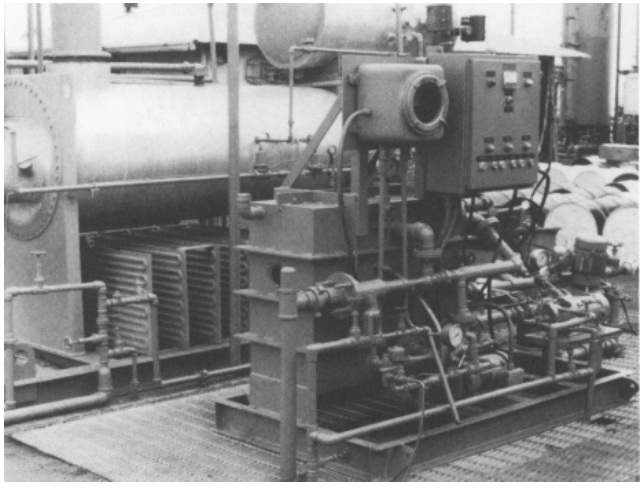


Figure 1-19. A vacuum distillation water-maker system.

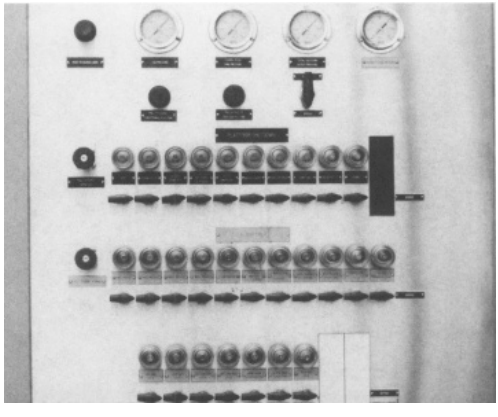


Figure 1-20. A pneumatic shut-in panel with "first-out" indication to inform the operator of which end element caused the shutdown.

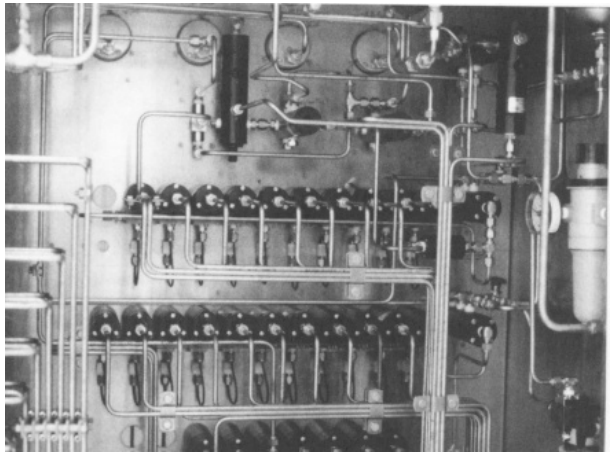


Figure 1-21. The pneumatic logic within the panel shown in Figure 1-20.

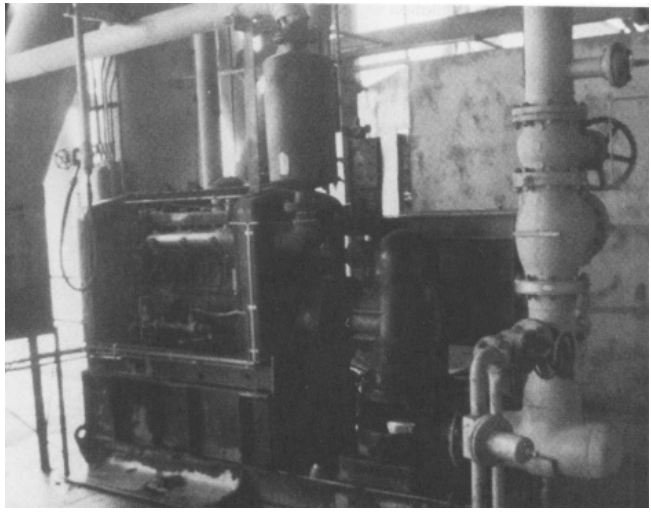


Figure 1-22. Diesel engine driven fire-fighting pump driving a vertical turbine pump through a right angle gear.

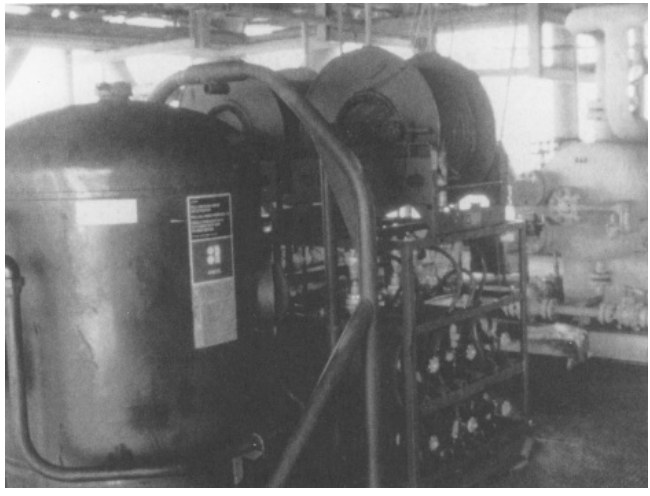


Figure 1-23. A foam fire-fighting station.

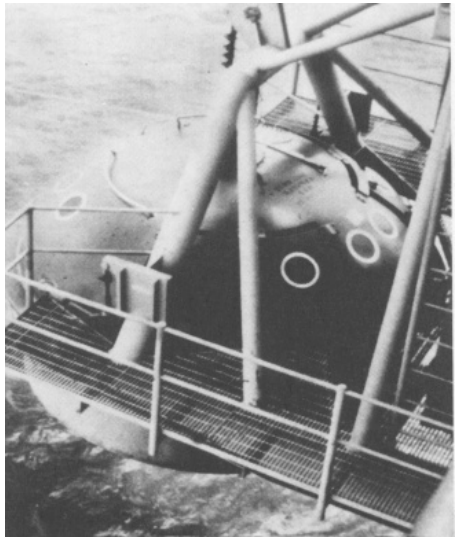


Figure 1-24. An escape capsule mounted on the lower deck of a platform. The unit contains an automatic lowering device and motor for leaving the vicinity of the platform.

Making the Equipment Work

The main items of process equipment have automatic instrumentation that controls the pressure and/or liquid level and sometimes temperature within the equipment. Figure 1-25 shows a typical pressure controller and control valve. In the black box (the controller) is a device that sends a signal to the actuator, which opens and closes the control valve to control pressure. Figure 1-26 shows a self-contained pressure controller, which has an internal mechanism that senses the pressure and opens and closes the valve as required.

Figure 1-27 shows two types of level controllers that use floats to monitor the level. The one on the left is an on/off switch, and the two on the right send an ever-increasing or decreasing signal as the level changes. These floats are mounted in the chambers outside the vessel. It is also possible to mount the float inside. Capacitance and inductance probes and pressure differential measuring devices are also commonly used to measure level.

Figure 1-28 shows a pneumatic-level control valve that accepts the signal from the level controller and opens and closes to allow liquid into or out of the vessel. In older leases it is common to attach the valve to a controller float directly through a mechanical linkage. Some low-pressure installations use a lever-balanced valve such as that shown in Figure 1-29. The weight on the lever is adjusted until the force it exerts



Figure 1-25. A pressure control valve with pneumatic actuator and pressure controller mounted on the actuator. The control mechanism in the box senses pressure and adjusts the supply pressure to the actuator diaphragm causing the valve stem to move up and down as required.

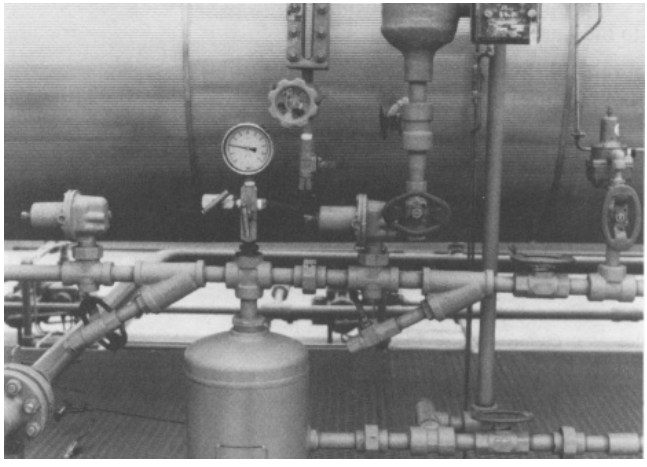


Figure 1-26. Two self-contained pressure regulators in a fuel gas piping system. An internal diaphragm and spring automatically adjust the opening in the valve to maintain pressure.

to keep the valve closed is balanced by the opening force caused by the head of liquid in the vessel.

Temperature controllers send signals to control valves in the same manner as pressure and level controllers.

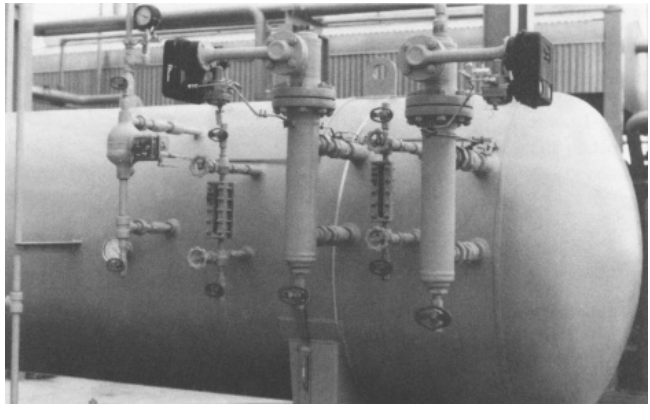


Figure 1-27. Two external level float controllers and an external float switch. The controllers on the right sense the level of fluids in the vessel. The switch on the left provides a high level alarm.

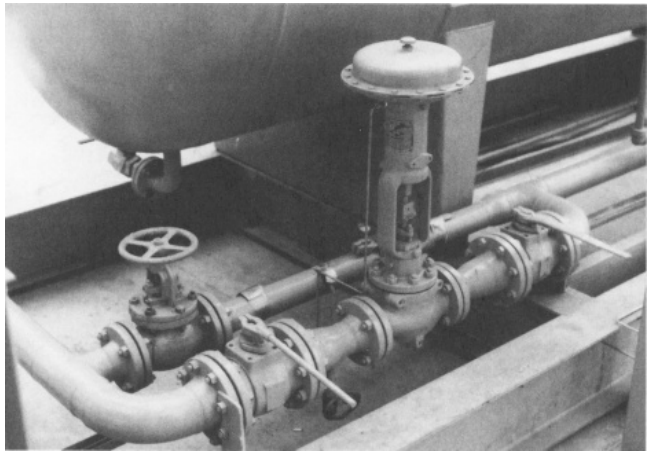


Figure 1-28. A level control valve with bypass. The signal from the controller causes the diaphragm of the actuator and thus the valve stem to move.

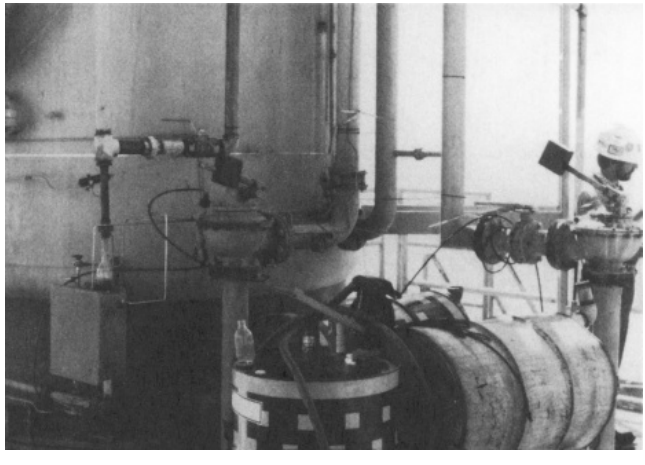


Figure 1-29. Two level-balanced liquid control valves. The position of the weight on the valve lever determines the amount of fluid column upstream of the valve necessary to force the valve to open.

Facility Types

It is very difficult to classify production facilities by type, because they differ due to production rates, fluid properties, sale and disposal requirements, location, and operator preference. Some more or less typical onshore facilities are shown in Figures 1-30, 1-31, and 1-32. In cold weather areas, individual pieces of equipment could be protected as shown in Figure 1-33, or the equipment could be completely enclosed in a building, such as shown in Figure 1-34.

In marsh areas the facilities can be installed on wood, concrete, or steel platforms or on steel or concrete barges, as shown in Figure 1-35.



Figure 1-30. An onshore lease facility showing vertical three-phase separator, a horizontal two-phase separator, a vertical heater-treater, and two storage tanks.



Figure 1-31. An onshore central facility with a large horizontal free water knockout, and a horizontal heater-treater.

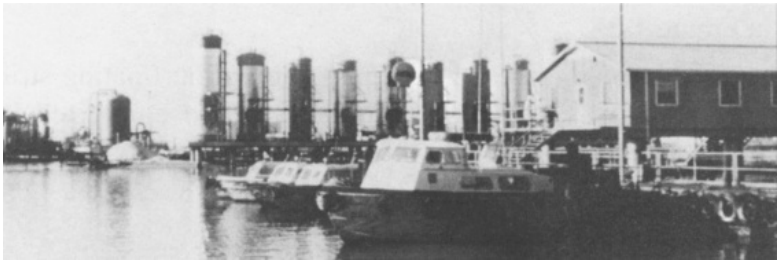


Figure 1-32. A marsh facility where the equipment is elevated on concrete platforms. Note the two large vertical separators in the distance, the row of nine vertical heater-treaters, and the elevated quarters building.

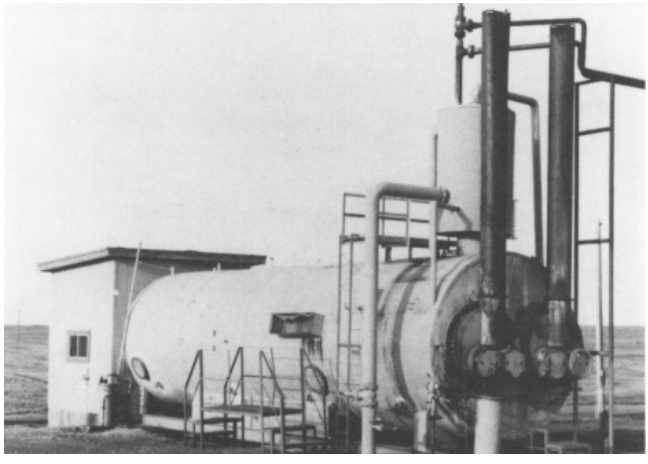


Figure 1-33. In cold weather areas it is sometimes necessary to insulate the vessels and pipe and house all controls in a building attached to the vessel.

In shallow water, facilities can be installed on several different platforms connected by bridges (see Figure 1-36). In deeper water it may be necessary to install all the facilities and the wells on the same platform, as in Figure 1-37. Sometimes, in cold weather areas, the facilities must be enclosed as shown in Figure 1-38.

Facilities have been installed on semi-submersible floating structures, tension leg platforms, tankers (see Figure 1-39) and converted jack-up drilling rigs (see Figure 1-40). Figure 1-41 shows a facility installed on a manmade island.

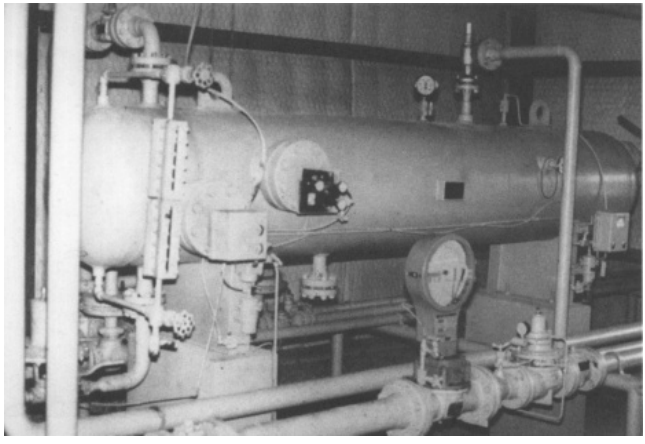


Figure 1-34. An onshore facility in Michigan where the process vessels are enclosed inside an insulated building.

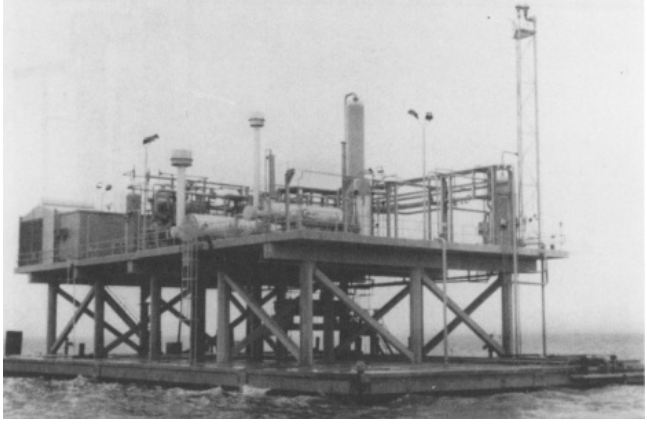


Figure 1-35. In marsh and shallow areas it is sometimes beneficial to build the facilities on a concrete barge onshore and then sink the barge on location.

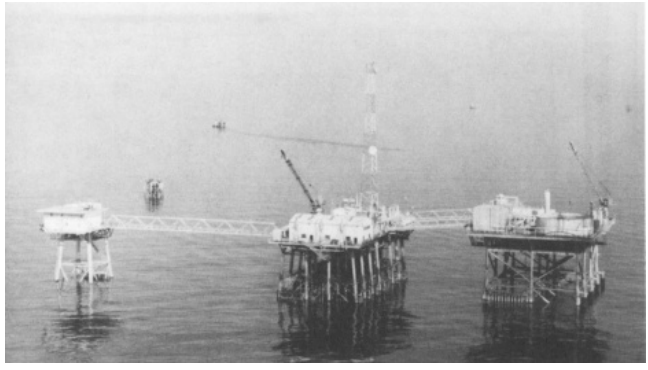


Figure 1-36. In moderate water depths it is possible to separate the quarters (on the left) and oil storage (on the right) from the rest of the equipment for safety reasons.

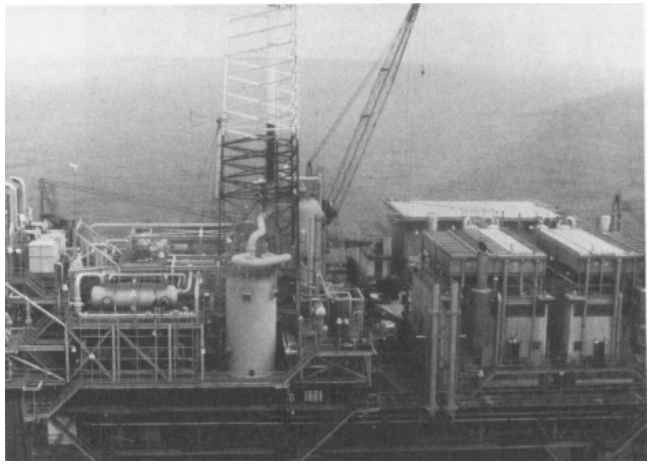


Figure 1-37. In deep waters this is not possible and the facilities can get somewhat crowded.

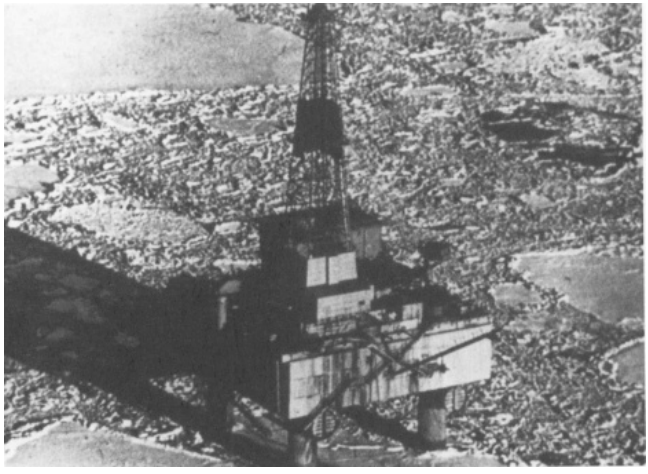


Figure 1-38. In cold weather areas such as this platform in Cook Inlet, Alaska, the facilities may be totally enclosed.



Figure 1-39. A tanker with facilities installed for a location near Thailand.

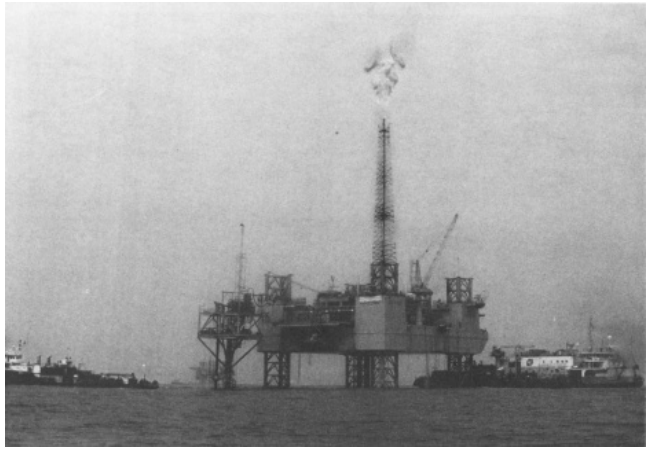


Figure 1-40. This converted jack-up rig was installed off the African coast.

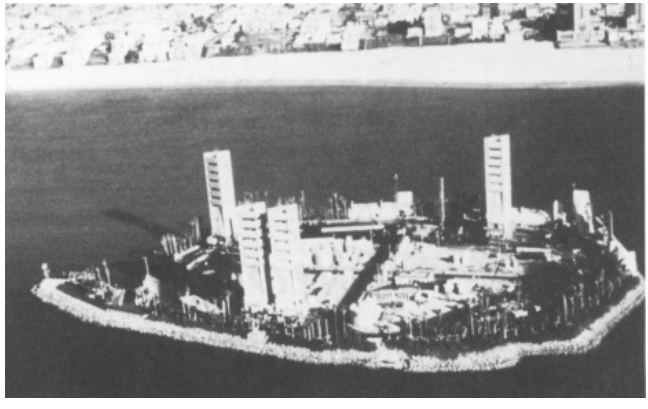


Figure 1-41. Sometimes the facilities must be decorated to meet some group's idea of what is aesthetically pleasing. This facility off California has palm trees, fake waterfalls and drilling derricks disguised as condominiums.

Chapter 2
Process Selection

Introduction

This chapter explains how the various components are combined into a production system. The material is in no way meant to be all-inclusive. Many things must be considered in selecting components for a system, and there is no substitute for experience and good engineering judgment.

A process flowsheet is used to describe the system. Figure 2-1 is a typical flowsheet that will be used as an example for discussion purposes. Another name for a process flowsheet is a process flow diagram (PFD). Regardless what it is called, either a flowsheet or a diagram, the information contained on both is the same. Figure 2-2 defines many of the commonly used symbols in process flowsheets.

Controlling the Process

Before discussing the process itself, it is necessary to understand how the process is controlled.

Operation of a Control Valve

Control valves are used throughout the process to control pressure, level, temperature, or flow. It is beyond the scope of this chapter to discuss the differences between the various types of control valves and the procedures for their sizing.

This section focuses primarily on the functions of this equipment. Figure 2-3 shows the major components of a typical sliding-stem control valve. All control valves have a variable opening or orifice. For a given

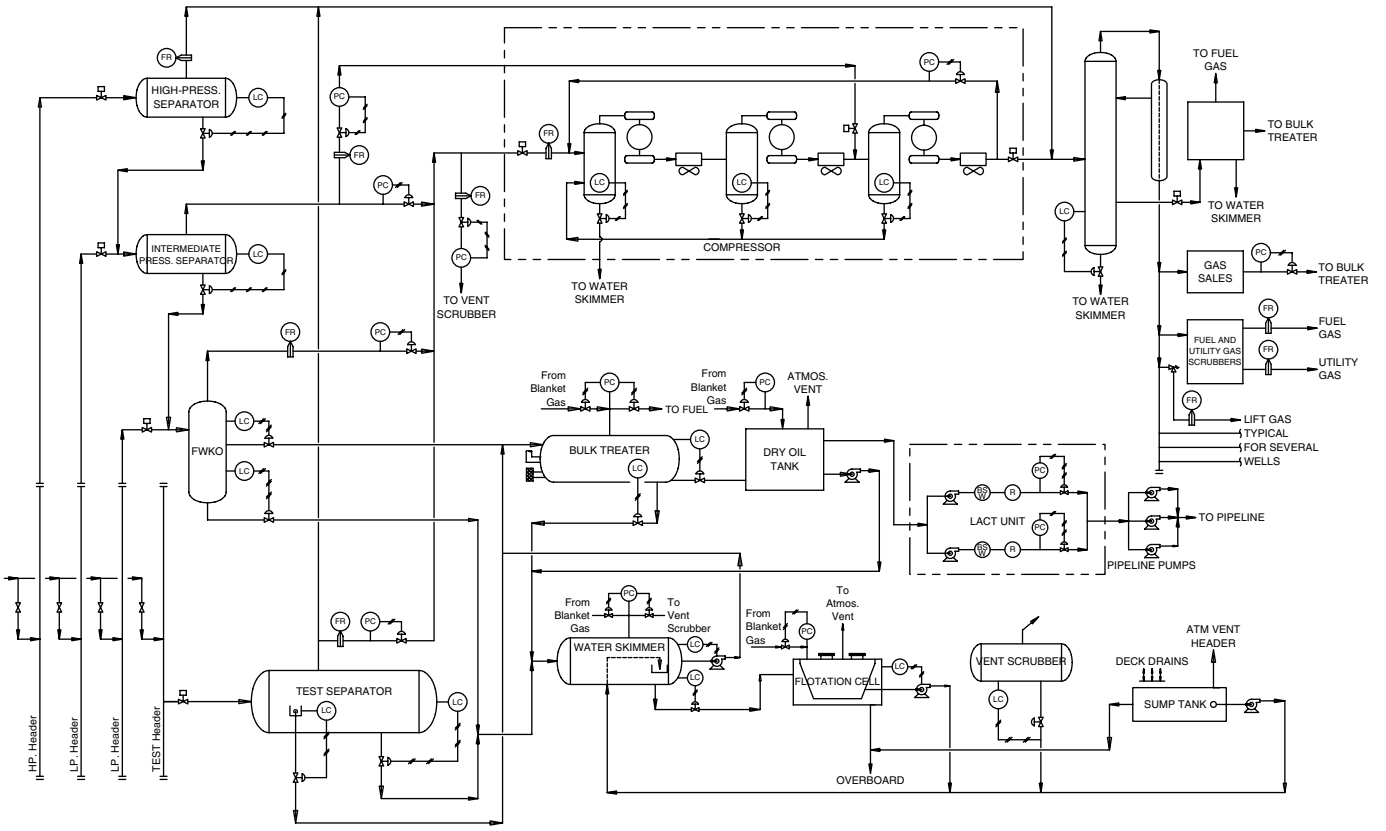


Figure 2-1. Typical flowsheet.

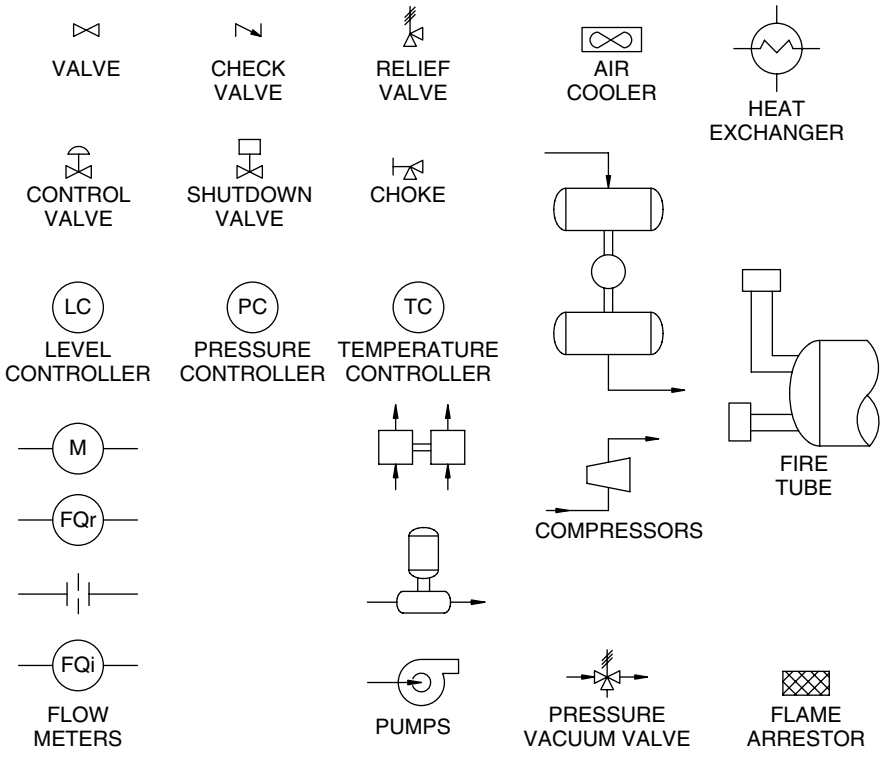


Figure 2-2. Common flowsheet symbols.

pressure drop across the valve, the larger the orifice is, the greater the flow through the valve will be.

Chokes and other flow control devices have either a fixed or a variable orifice. With a fixed pressure drop across the device (i.e., with both the upstream and downstream pressures fixed by the process system), the larger the orifice is, the greater the flow will be. Chokes are used to regulate the flow rate.

In Figure 2-3 the orifice is made larger by moving the valve stem upward. This moves the plug off the seat, creating a larger annulus for flow between the seat and the plug. Similarly, the orifice is made smaller by moving the valve stem downward. The most common way to effect this motion is with a pneumatic actuator, such as that shown in Figure 2-4. Instrument air or gas applied to the actuator diaphragm overcomes a spring resistance and moves the stem either upward or downward.

The action of the actuator must be matched with the construction of the valve body to assure that the required failure mode is met. That is,

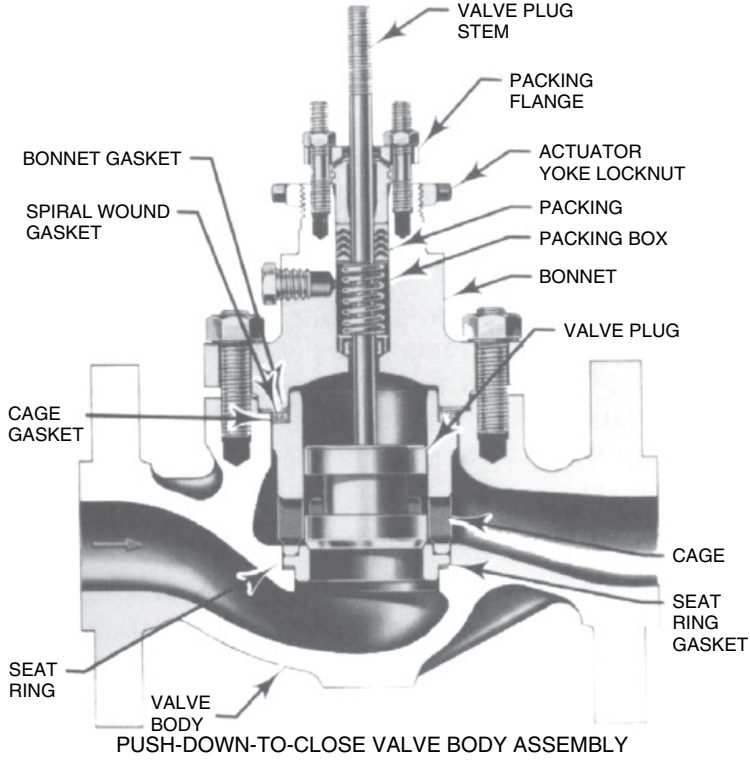


Figure 2-3. Major components of a typical sliding stem control valve. (courtesy of Fisher Controls International, Inc.)

if it is desirable for the valve to fail to close, then the actuator and body must be matched so that on failure of the instrument air or gas, the spring causes the stem to move in the direction that blocks flow (i.e., fully shut). This would normally be the case for most liquid control valves. If it is desirable for the valve to fail to open, as in many pressure control situations, then the spring must cause the stem to move in the fully open direction.

Pressure Control

The hydrocarbon fluid produced from a well is made up of many components ranging from methane, the lightest and most gaseous hydrocarbon, to some very heavy and complex hydrocarbon compounds. Because of this, whenever there is a drop in fluid pressure, gas is liberated. Therefore, pressure control is important.

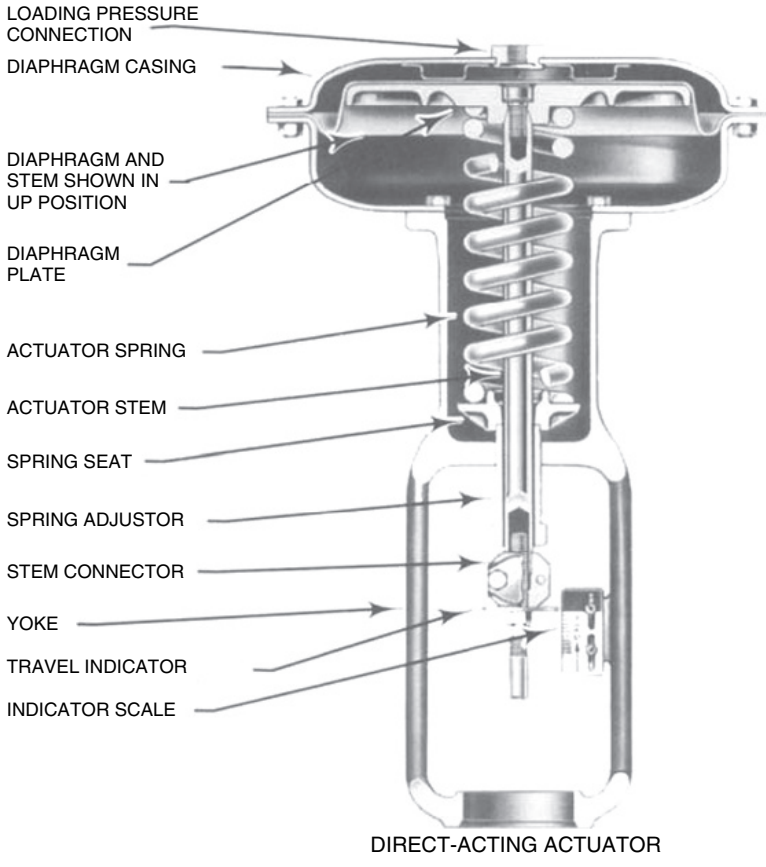


Figure 2-4. Typical pneumatic direct-acting actuator. (courtesy of Fisher Controls International, Inc.)

The most common method of controlling pressure is with a pressure controller and a backpressure control valve. The pressure controller senses the pressure in the vapor space of the pressure vessel or tank. By regulating the amount of gas leaving the vapor space, the backpressure control valve maintains the desired pressure in the vessel. If too much gas is released, the number of molecules of gas in the vapor space decreases, and thus the pressure in the vessel decreases. If insufficient gas is released, the number of molecules of gas in the vapor space increases, and thus the pressure in the vessel increases.

In most instances, there will be enough gas separated or "flashed" from the liquid to allow the pressure controller to compensate for changes in liquid level, temperature, etc., which would cause a change in the number

of molecules of gas required to fill the vapor space at a given pressure. However, under some conditions where there has been only a small pressure drop from the upstream vessel, or where the crude GOR (gas/oil ratio) is low, it may be necessary to add gas to the vessel to maintain pressure control at all times. This is called "make-up" or "blanket" gas. Gas from a pressure source higher than the desired control pressure is routed to the vessel by a pressure controller that senses the vessel pressure automatically, allowing either more or less gas to enter the vessel as required.

Level Control

It is also necessary to control the gas/liquid interface or the oil/water interface in process equipment. This is done with a level controller and liquid dump valve. The most common forms of level controllers are floats and displacers, although electronic sensing devices can also be used. If the level begins to rise, the controller signals the liquid dump valve to open and allow liquid to leave the vessel. If the level in the vessel begins to fall, the controller signals the liquid dump valve to close and decrease the flow of liquid from the vessel. In this manner the liquid dump valve is constantly adjusting its opening to assure that the rate of liquid flowing into the vessel is matched by the rate out of the vessel.

Temperature Control

The way in which the process temperature is controlled varies. In a heater a temperature controller measures the process temperature and signals a fuel valve to let either more or less fuel to the burner. In a heat exchanger the temperature controller could signal a valve to allow more or less of the heating or cooling media to bypass the exchanger.

Flow Control

It is very rare that flow must be controlled in an oil field process. Normally, the control of pressure, level, and temperature is sufficient. Occasionally, it is necessary to assure that flow is split in some controlled manner between two process components in parallel, or perhaps to maintain a certain critical flow through a component. This can become a complicated control problem and must be handled on an individual basis.

Basic System Configuration

Wellhead and Manifold

The production system begins at the wellhead, which should include at least one choke, unless the well is on artificial lift. Most of the pressure drop between the well flowing tubing pressure (FTP) and the initial separator operating pressure occurs across this choke. The size of the opening in the choke determines the flow rate, because the pressure upstream is determined primarily by the well FTP, and the pressure downstream is determined primarily by the pressure control valve on the first separator in the process. For high-pressure wells it is desirable to have a positive choke in series with an adjustable choke. The positive choke takes over and keeps the production rate within limits should the adjustable choke fail.

On offshore facilities and other high-risk situations, an automatic shutdown valve should be installed on the wellhead. (It is required by the authorities having jurisdiction in the United States, Western and Eastern Europe, West Africa, Central Asia, Southeast Asia, and the Middle East.) In all cases, block valves are needed so that maintenance can be performed on the choke if there is a long flowline.

Whenever flows from two or more wells are commingled in a central facility, it is necessary to install a manifold to allow flow from any one well to be produced into any of the bulk or test production systems.

Separation

Initial Separation Pressure

Because of the multicomponent nature of the produced fluid, the higher the pressure at which the initial separation occurs, the more liquid will be obtained in the separator. This liquid contains some light components that vaporize in the stock tank downstream of the separator. If the pressure for initial separation is too high, too many light components will stay in the liquid phase at the separator and be lost to the gas phase at the tank. If the pressure is too low, not as many of these light components will be stabilized into the liquid at the separator and they will be lost to the gas phase.

This phenomenon, which can be calculated using flash equilibrium techniques discussed in Chapter 3, is shown in Figures 2-5 and 2-6. It is important to understand this phenomenon qualitatively. The tendency of any one component in the process stream to flash to the vapor phase depends on its partial pressure. The partial pressure of a component in

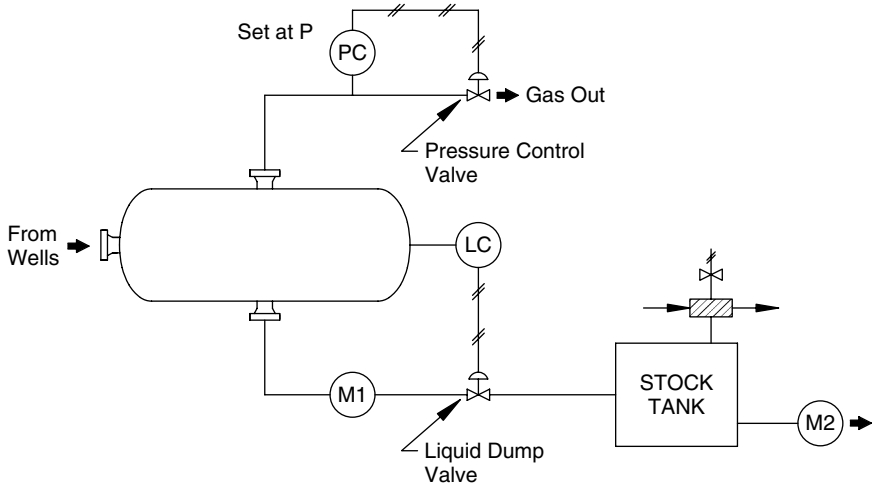


Figure 2-5. Single-stage separation.

a vessel is defined as the number of molecules of that component in the vapor space divided by the total number of molecules of all components in the vapor space times the pressure in the vessel [refer to Eq. (2-1)]:

$$PP_N = \frac{\text{Moles}_N}{\sum \text{Moles}_N}(P), \tag{2-1}$$

where

PP_N = partial pressure of component "N,"
Moles_N = number of moles of component "N,"
$\sum \text{Moles}_N$ = total number of moles of all components,
P = pressure in the vessel, psia (kpa).

Thus, if the pressure in the vessel is high, the partial pressure for the component will be relatively high and the molecules of that component will tend toward the liquid phase. This is seen by the top line in Figure 2-6. As the separator pressure is increased, the liquid flow rate out of the separator increases.

The problem with this is that many of these molecules are the lighter hydrocarbons (methane, ethane, and propane), which have a strong tendency to flash to the gas state at stock-tank conditions (atmospheric pressure). In the stock tank, the presence of these large numbers of molecules creates a low partial pressure for the intermediate-range

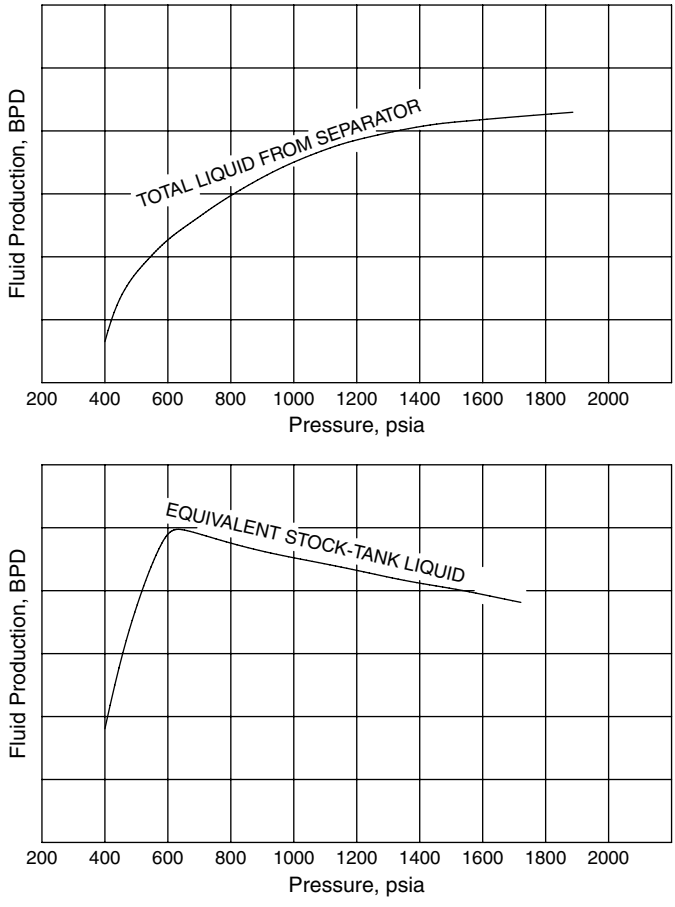


Figure 2-6. Effect of separator pressure on stock-tank liquid recovery.

hydrocarbons (butanes, pentane, and heptane) whose flashing tendency at stock tank conditions is very susceptible to small changes in partial pressure. Thus, by keeping the lighter molecules in the feed to the stock tank, we manage to capture a small amount of them as liquids, but we lose to the gas phase many more of the intermediate-range molecules. That is why beyond some optimum point there is actually a decrease in stock-tank liquids by increasing the separator operating pressure.

Stage Separation

Figure 2-5 deals with a simple single-stage process. That is, the fluids are flashed in an initial separator and then the liquids from that

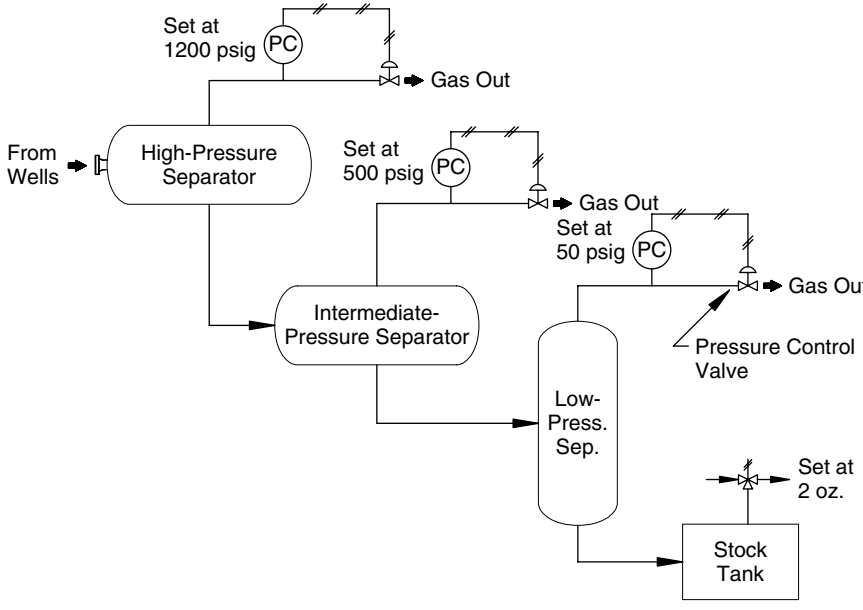


Figure 2-7. Stage separation.

separator are flashed again at the stock tank. Traditionally, the stock tank is not normally considered a separate stage of separation, though it most assuredly is.

Figure 2-7 shows a three-stage separation process. The liquid is first flashed at an initial pressure and then flashed at successively lower pressures two times before entering the stock tank.

Because of the multicomponent nature of the produced fluid, it can be shown by flash calculations that the more stages of separation after the initial separation, the more light components will be stabilized into the liquid phase. This can be understood qualitatively by realizing that in a stage separation process the light hydrocarbon molecules that flash are removed at relatively high pressure, keeping the partial pressure of the intermediate hydrocarbons lower at each stage. As the number of stages approaches infinity, the lighter molecules are removed as soon as they are formed and the partial pressure of the intermediate components is maximized at each stage. The compressor horsepower required is also reduced by stage separation as some of the gas is captured at a higher pressure than would otherwise have occurred. This is demonstrated by the example presented in Table 2-1.

Table 2-1a
Effect of Separation Pressure for a Rich Condensate Stream (Field Units)

Case	Separation Stages (psia)	Liquid Produced (bopd)	Compressor Horsepower Required
I	1215; 65	8,400	861
II	1215; 515; 65	8,496	497
III	1215; 515; 190; 65	8,530	399

Table 2-1b
Effect of Separation Pressure for a Rich Condensate Stream (SI Units)

Case	Separation Stage Pressures (kPa)	Liquid Produced (m^3/hr)	Compressor Power Required (kW)
I	8377; 448	55.6	642
II	8377; 3551; 448	56.3	371
III	8377; 3551; 1310; 448	56.5	298

Selection of Stages

As shown in Figure 2-8, as more stages are added to the process there is less and less incremental liquid recovery. The diminishing income for adding a stage must more than offset the cost of the additional separator, piping, controls, space, and compressor complexities. It is clear that for each facility there is an optimum number of stages. In most cases, the optimum number of stages is very difficult to determine as it may be different from well to well and it may change as the well's flowing pressure declines with time. Table 2-2 is an approximate guide to the number of stages in separation, excluding the stock tank, which field experience indicates is somewhat near optimum. Table 2-2 is meant as a guide and should not replace flash calculations, engineering studies, and engineering judgment.

Fields with Different Flowing Tubing Pressures

The discussion to this point has focused on a situation where all the wells in a field produce at roughly the same flowing tubing pressure, and stage

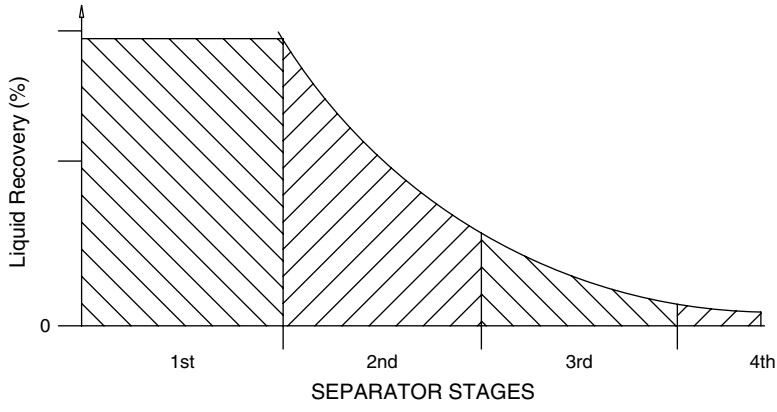


Figure 2-8. Incremental liquid recovery versus number of separator stages.

Table 2-2
Stage Separation Guidelines

Initial Separator Pressure		
(kPa)	**(PSIG)**	**Number of Stages[1]**
170–860	25–125	1
860–2100	125–300	1–2
2100–3400	300–500	2
3400–4800	500–700	2–3[2]

[1] Does not include stock tank.
[2] At flow rates exceeding 650 m^3/hr (100,000 BPD), more stages may be justified.

separation is used to maximize liquid production and minimize compressor horsepower. Often, as in our example flowsheet, stage separation is used because different wells producing to the facility have different flowing tubing pressures. This could be because they are completed in different reservoirs, or are located in the same reservoir but have different water production rates. By using a manifold arrangement and different primary separator operating pressures, there is not only the benefit of stage separation of high-pressure liquids, but also conservation of reservoir energy. High-pressure wells can continue to flow at sales pressure requiring no compression, while those with lower tubing pressures can flow into whichever system minimizes compression.

Determining Separator Operating Pressures

The choice of separator operating pressures in a multistage system is large. For large facilities many options should be investigated before a final choice is made. For facilities handling less than 50,000 bpd, there are practical constraints that help limit the options.

A minimum pressure for the lowest-pressure stage would be in the 25- to 50-psig range. This pressure will probably be needed to allow the oil to be dumped to a treater or tank and the water to be dumped to the water treating system. The higher the operating pressure, the smaller the compressor needed to compress the flash gas to sales. Compressor horsepower requirements are a function of the absolute discharge pressure divided by the absolute suction pressure.

Increasing the low-pressure separator pressure from 50 psig to 200 psig may decrease the compression horsepower required by 33%. However, it may also add backpressure to wells, restricting their flow, and allow more gas to be vented to atmosphere at the tank. Usually, an operating pressure of between 50 and 100 psig is optimum.

As stated before, the operating pressure of the highest-pressure separator will be no higher than the sales gas pressure. A possible exception to this could occur where the gas lift pressure is higher than the sales gas pressure. In choosing the operating pressures of the intermediate stages, it is useful to remember that the gas from these stages must be compressed. Normally, this will be done in a multistage compressor. For practical reasons, the choice of separator operating pressures should match closely and be slightly greater than the compressor inter-stage pressures. The most efficient compressor sizing will be with a constant compressor ratio per stage. Therefore, an approximation of the intermediate separator operating pressures can be derived from

$$R = \left[\frac{P_d}{P_s}\right]^{1/n}, \tag{2-2}$$

where

R = ratio per stage,
P_d = discharge pressure, psia,
P_s = suction pressure, psia,
n = number of stages.

Once a final compressor selection is made, these approximate pressures will be changed slightly to fit the actual compressor configuration.

In order to minimize inter-stage temperatures, the maximum ratio per stage will normally be in the range of 3.6 to 4.0. That means that most

production facilities will have either two- or three-stage compressors. A two-stage compressor only allows for one possible intermediate separator operating pressure. A three-stage allows for either one operating at second- or third-stage suction pressure or two intermediate separators each operating at one of the two compressor intermediate suction pressures. Of course, in very large facilities it would be possible to install a separate compressor for each separator and operate as many intermediate-pressure separators as is deemed economical.

Two-Phase vs. Three-Phase Separators

In our example process the high- and intermediate-stage separators are two-phase, while the low-pressure separator is three-phase. This is called a "free-water knockout" (FWKO) because it is designed to separate the free water from the oil and emulsion, as well as separate gas from liquid. The choice depends on the expected flowing characteristics of the wells. If large amounts of water are expected with the high-pressure wells, it is possible that the size of the other separators could be reduced if the high-pressure separator was three-phase. This would not normally be the case for a facility such as that shown in Figure 2-1 where individual wells are expected to flow at different flowing tubing pressures (FTPs). In some instances, where all wells are expected to have similar FTPs at all times, it may be advantageous to remove the free water early in the separation scheme.

Process Flowsheet

Figure 2-9 is an enlargement of the free-water knockout (FWKO) shown in Figure 2-1. Figure 2-9 illustrates the amount of detail that is expected on a process flowsheet. A flash calculation is needed to determine the amount of gas and liquid that each separator must handle.

In the example process of Figure 2-1, the treater is not considered a separate stage of separation as it operates very close to the FWKO pressure, which is the last stage. Very little gas will flash between the two vessels. In most instances, this gas will be used for fuel or vented and not compressed for sales, although a small compressor could be added to boost this gas to the main compressor suction pressure.

Oil Treating and Storage

Crude requires dehydration before it can go to storage. Water-in-oil emulsions must be broken so as to reduce water cut and reduce salt content.

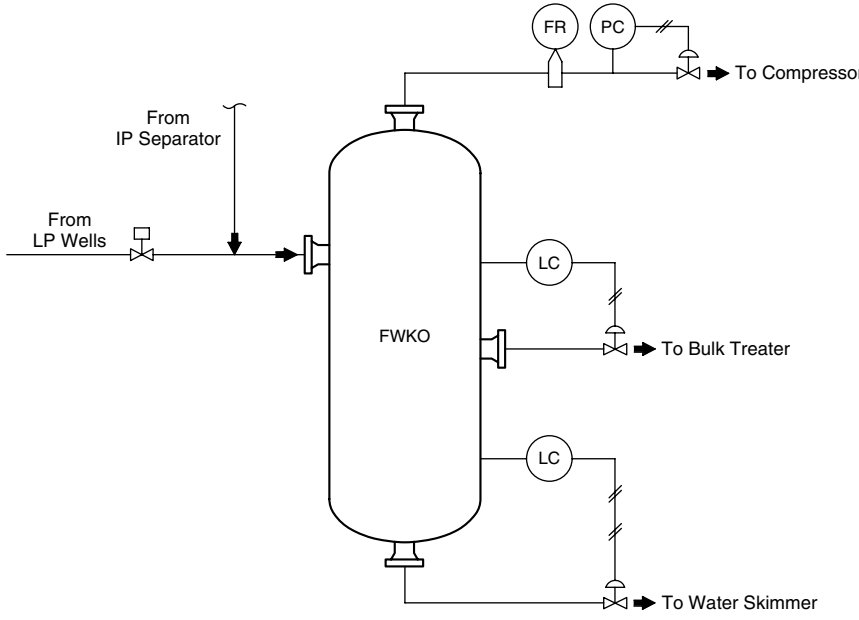


Figure 2-9. Vertical free-water knockout.

Demulsifier chemicals weaken the oil film around the water droplets so that the film will rupture when droplets collide. Droplet collision is accelerated by using heat and electrostatics.

Salt must also be removed from produced crude. This is typically done by mixing 5% fresh water with dehydrated crude and then dehydrating it a second time so as to meet the total suspended solids (TDS) content requirement. Salt content specifications range from 10 to 25 pounds per thousand barrels (PPB). As the last step in production, crude may be run through a stabilizer where its vapor pressure is reduced to allow nonvolatile liquid to be stored in tanks at atmospheric pressure or loaded onto tankers.

Most oil treating on offshore facilities is done in vertical or horizontal treaters, such as those described in Chapter 7. Figure 2-10 is an enlargement of a horizontal oil treater in Figure 2-1. In this case, a gas blanket is provided to assure that there is always sufficient pressure in the treater to allow the water to flow to the water treating system without requiring a pump. In addition, the gas blanket excludes oxygen entry into the system, which could cause scale, corrosion, and bacteria.

At onshore locations the oil may be treated in a "gunbarrel" (or settling/wash tank) with either an internal or external "Gas Boot," as shown in Figure 2-11. The "gunbarrel" with an internal gas boot is used for

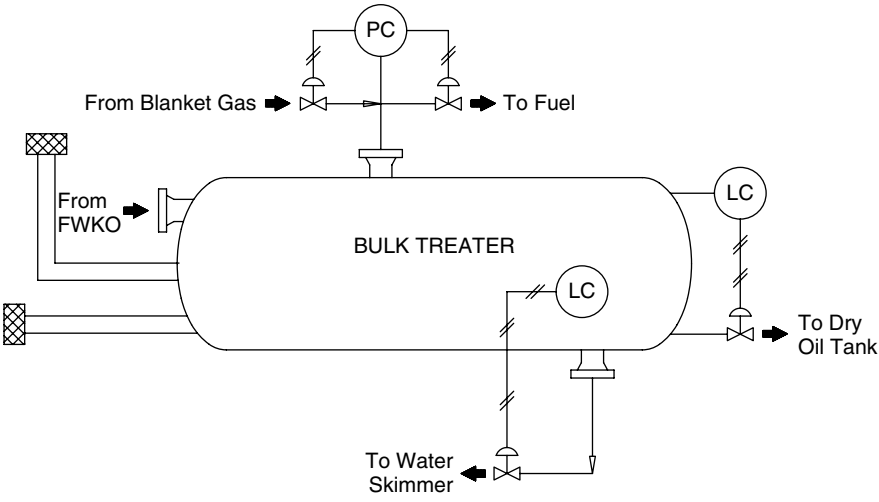


Figure 2-10. Horizontal bulk treater.

low to moderate flow rates while an external gas boot with a wash tank is used in low-pressure, large-flow rate systems. All tanks should have a pressure/vacuum valve with flame arrestor and gas blanket to keep a positive pressure on the system and exclude oxygen. This helps to prevent corrosion, eliminate a potential safety hazard, and conserve some of the hydrocarbon vapors.

Figure 2-12 shows a typical pressure/vacuum valve. Pressure in the tank lifts a weighted disk or pallet, which allows the gas to escape. If there is a vacuum in the tank because the gas blanket failed to maintain a slight positive pressure, the greater ambient pressure lifts another disk, which allows air to enter. Although we wish to exclude air, it is preferable to allow a small controlled volume into the tank rather than allow the tank to collapse. The savings associated with keeping a positive pressure on the tank is demonstrated in Table 2-3.

Figure 2-13 shows a typical flame arrestor. The tubes in the device keep a vent flame from traveling back into the tank. Flame arrestors have a tendency to plug with paraffin and thus must be installed where they can be inspected and maintained. Since they can plug, a separate relieving device (most often a gauge hatch set to open a few ounces above the normal relieving device) must always be installed.

The oil is skimmed off the surface of the gunbarrel and the water exits from the bottom through either a water leg or an interface controller and dump valve. It must be pointed out that since the volume of the liquid is fixed by the oil outlet, gunbarrels cannot be used as surge tanks.

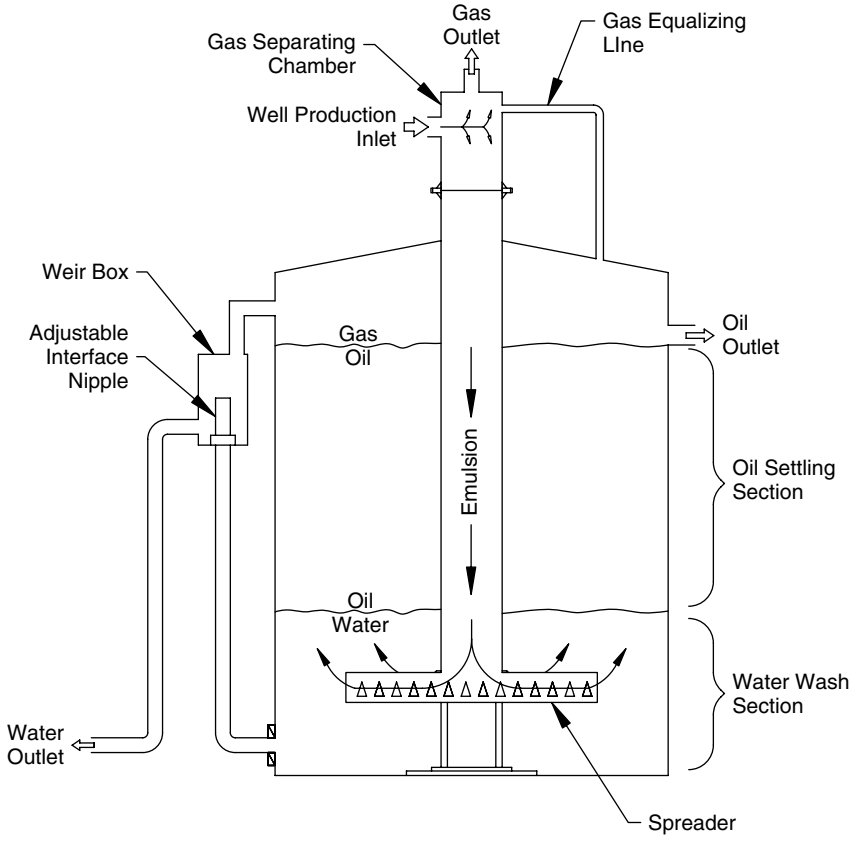


Figure 2-11. "Gunbarrel" with an internal "Gas Boot."

Flow from the treater or gunbarrel goes to a settling tank from which it either flows into a barge or truck or is pumped into a pipeline.

Lease Automatic Custody Transfer (LACT)

In large facilities oil is typically sold through a LACT unit, which is designed to meet API Standards and whatever additional measuring and sampling standards are required by the crude purchaser. The value received for the crude will typically depend on its gravity, basic settlement and water (BS&W) content, and volume.

Therefore, the LACT unit must not only measure the volume accurately, but must continuously monitor the BS&W content and take a

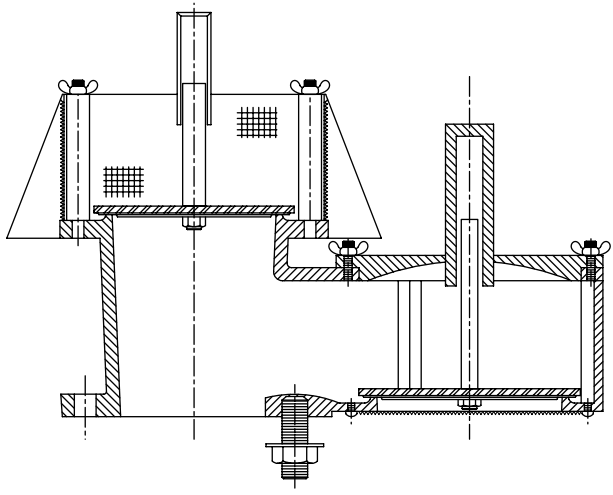


Figure 2-12. Typical pressure/vacuum valve. (courtesy of Groth Equipment Corp.)

Table 2-3
Tank Breathing Loss

	Breathing Loss		
Nominal Capacity (bbl)	**Open Vent (bbl/yr)**	**Pressure Valve (bbl/yr)**	**Barrels Saved**
5,000	235	154	81
10,000	441	297	144
20,000	625	570	255
55,000	2,000	1,382	618

sufficiently representative sample so that the gravity and BS&W can be measured.

Figure 2-14 shows schematically the elements of a typical LACT unit. The crude first flows through a strainer/gas eliminator to protect the meter and to assure that there is no gas in the liquid. An automatic BS&W probe is mounted in a vertical run. When BS&W exceeds the sales contract quality, this probe automatically actuates the diverter valve, which blocks the liquid from going further in the LACT unit and sends it back to the process for further treating. Some sales contracts allow for the BS&W probe to merely sound a warning so that the operators can manually take corrective action. The BS&W probe must be mounted in a vertical run if it is to get a true reading of the average quality of the stream.

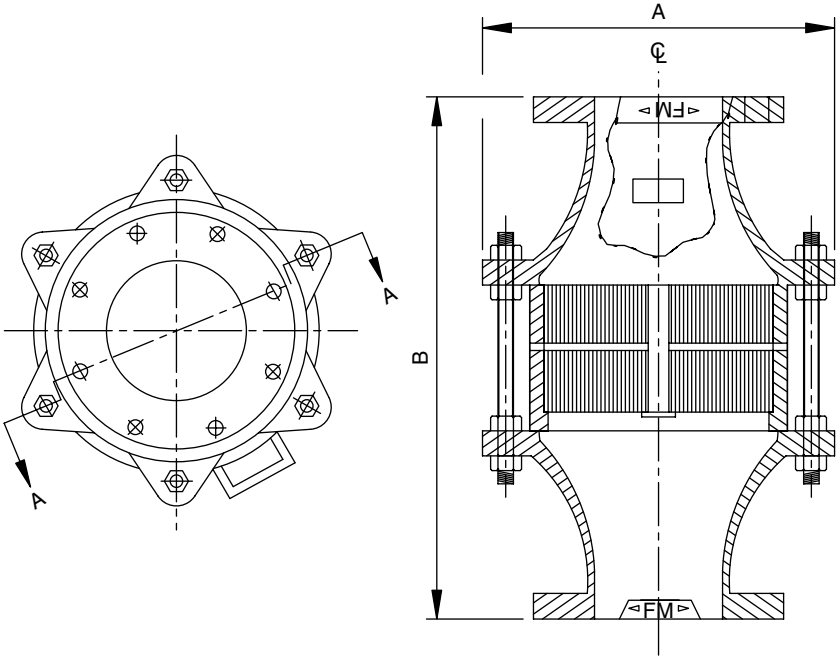


Figure 2-13. Typical frame arrestor. (courtesy of Groth Equipment Corp.)

Downstream of the diverter a sampler in a vertical run takes a calibrated sample that is proportional to the flow and delivers it to a sample container. The sampler receives a signal from the meter to assure that the sample size is always proportional to flow even if the flow varies. The sample container has a mixing pump so that the liquid in the container can be mixed and made homogeneous prior to taking a sample of this fluid. It is this small sample that will be used to convert the meter reading for BS&W and gravity.

The liquid then flows through a positive displacement meter. Most sales contracts require the meter to be proven at least once a month and a new meter factor calculated. On large installations a meter prover such as that shown in Figure 2-14 is included as a permanent part of the LACT skid or is brought to the location when a meter must be proven. The meter prover contains a known volume between two detector switches. This known volume has been measured in the factory to $\pm 0.02\%$ when measured against a calibrated "prover tank" that has been calibrated by the National Bureau of Standards (USA) or other authority having jurisdiction. A spheroid pig moves back and forth between the detectors as the four-way valve is automatically switched. The volume recorded

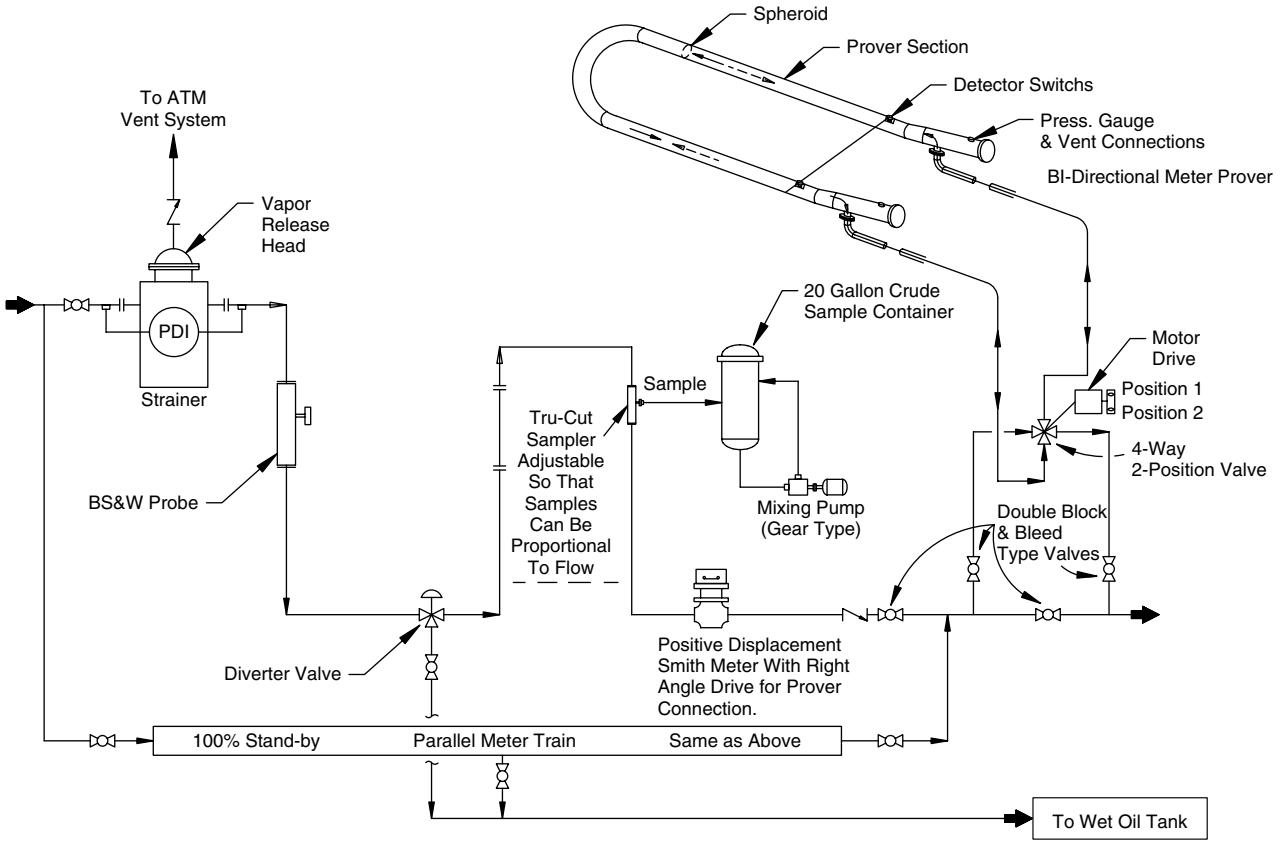


Figure 2-14. Typical LACT unit schematic.

by the meter during the time the pig moves between detectors for a set number of traverses of the prover is recorded electrically and compared to the known volume of the meter prover.

On smaller installations, a master meter that has been calibrated using a calibrated prover may be brought to the location to run in series with the meter to be proven. In many onshore locations, a truck-mounted meter prover is used. The sales meter must have a proven repeatability of ±0.02% when calibrated against a master meter or ±0.05% when calibrated against a tank or meter prover.

Pumps

Pumps are normally needed to move oil through the LACT unit and deliver it at pressure to a pipeline downstream of the unit. Pumps are sometimes used in water treating and disposal processes. In addition, many small pumps may be required for pumping skimmed oil to higher-pressure vessels for treating, glycol heat medium and cooling water service, firefighting, etc.

Water Treating

Chapter 9 describes choosing a process for this subsystem, including a vessel and open drains. Figure 2-15 shows an enlargement of the water treating system for the example.

Compressors

Figure 2-16 shows the configuration of the typical three-stage reciprocating compressor in our example flowsheet. Gas from the FWKO enters the first-stage suction scrubber. Any liquids that may have come through the line are separated at this point and the gas flows to the first stage. Compression heats the gas, so there is a cooler after each compression stage. At the higher pressure more liquids may separate, so the gas enters another scrubber before being compressed and cooled again.

In the example, gas from the intermediate-pressure separator can be routed to either the second-stage or third-stage suction pressure, as conditions in the field change.

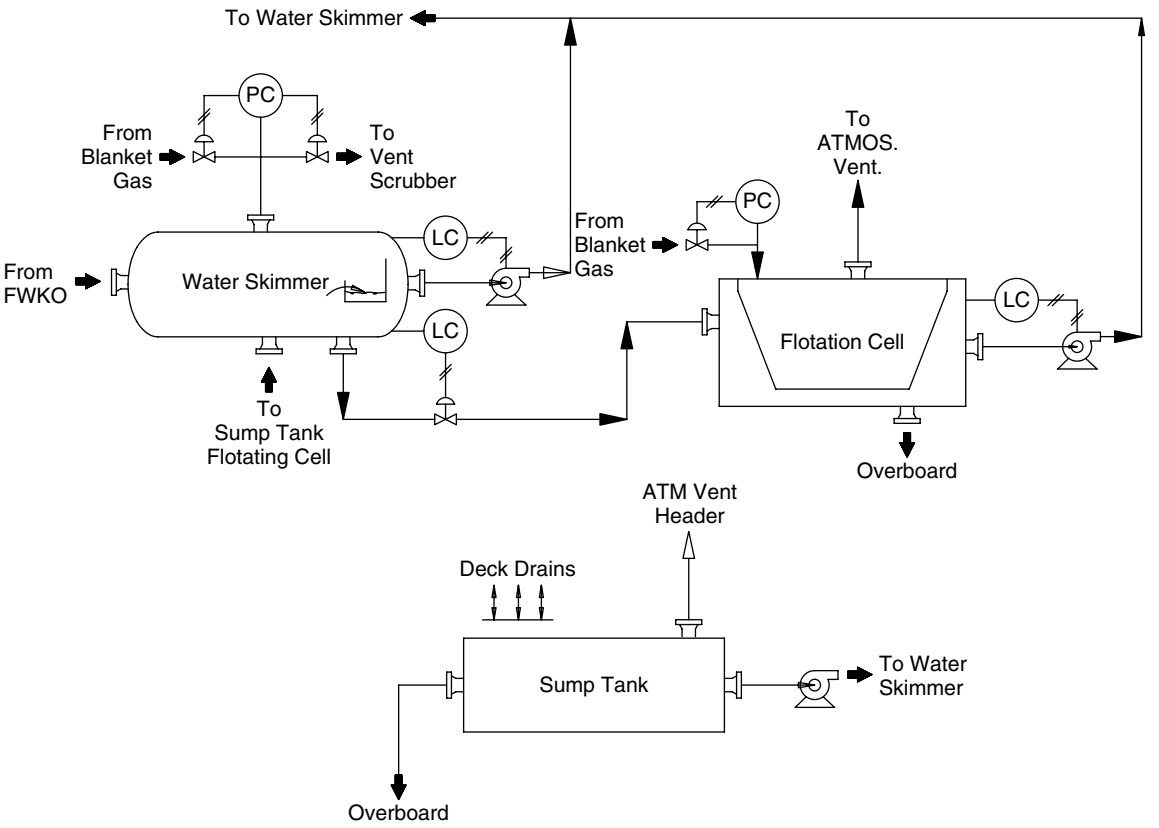


Figure 2-15. Water treating system.

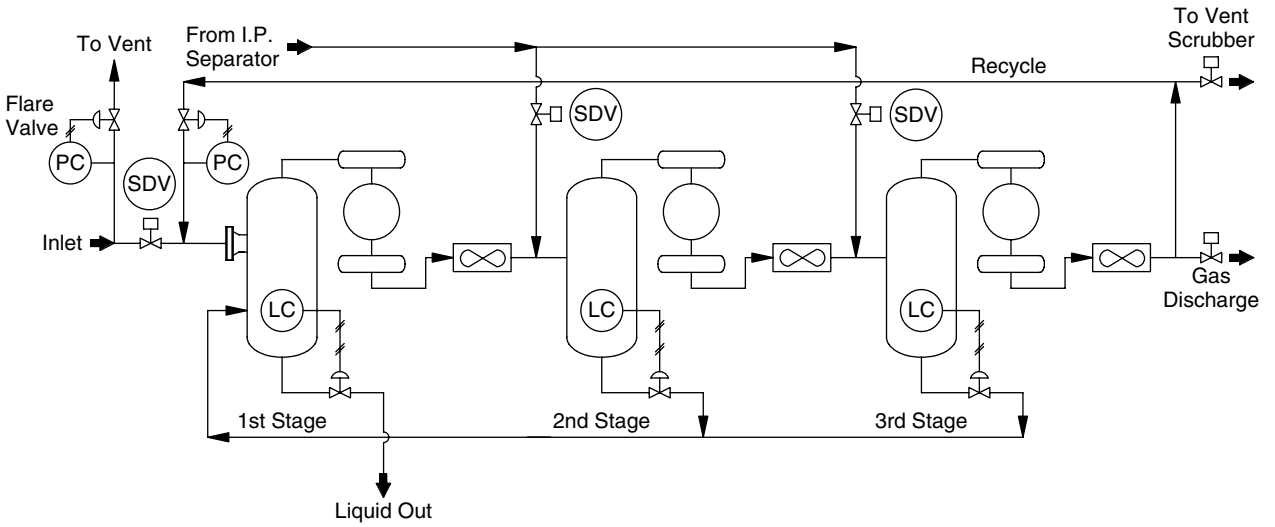


Figure 2-16. Three-stage compressor.

Worldwide accident records indicate that compressors are the single most hazardous piece of equipment in the process. The compressor is equipped with an automatic suction shut-in valve on each inlet and a discharge shut-in valve so that when the unit shuts down, or when an abnormal condition is detected, the shut-in valves actuate to isolate the unit from any new sources of gas. Many operators prefer, and in some cases regulations require, that an automatic blowdown valve also be installed so that as well as isolating the unit, all the gas contained within the unit is vented safely at a remote location.

Compressors in oil field service should be equipped with a recycle valve and a vent valve, such as shown in Figure 2-16. Compressor operating conditions are typically not well known when the compressor is installed, and even if they were, they are liable to change greatly as wells come on and off production. The recycle valve allows the compressor to be run at low throughput rates by keeping the compressor loaded with its minimum required throughput. In a reciprocating compressor, this is done by maintaining a minimum pressure on the suction. In a centrifugal compressor, this is done by a more complex surge control system.

The vent valve allows production to continue when the compressor shuts down. Many times a compressor will only be down for a short time, and it is better to vent the gas rather than automatically shut-in production. The vent valve also allows the compressor to operate when there is too much gas to the inlet. Under such conditions the pressure will rise to a point that could overload the rods on a reciprocating compressor.

The two basic types of compressors used in production facilities are reciprocating and centrifugal. Reciprocating compressors compress the gas with a piston moving linearly in a cylinder. Because of this, the flow is not steady, and care must be taken to control vibrations. Centrifugal compressors use high-speed rotating wheels to create a gas velocity that is converted into pressure by stators.

Reciprocating compressors are particularly attractive for low-horsepower ($<2{,}000$ hp), high-ratio applications, although they are available in sizes up to approximately 10,000 hp. They have higher fuel efficiencies than centrifugals, and much higher turndown capabilities.

Centrifugal compressors are particularly well suited for high-horsepower ($>4{,}000$ hp) or for low-ratio (<2.5) in the 1,000-hp and greater sizes. They are less expensive, take up less space, weigh less, and tend to have higher availability and lower maintenance costs than reciprocating compressors. Their overall fuel efficiency can be increased if use is made of the high-temperature exhaust heat in the process.

Gas Dehydration

Removing most of the water vapor from the gas is required by most gas sales contracts, because it prevents hydrates from forming when the gas is cooled in the transmission and distribution systems and prevents water vapor from condensing and creating a corrosion problem. Dehydration also increases line capacity marginally.

Most sales contracts in the southern United States call for reducing the water content in the gas to less than 7 lb/MMscf. In colder climates, sales requirements of 3 to 5 lb/MMscf are common. The following methods can be used for drying the gas:

1. Cool to the hydrate formation level and separate the water that forms. This can only be done where high water contents (±30 lb/MMscfd) are acceptable.
2. Use a Low-temperature Exchange (LTX) unit designed to melt the hydrates as they are formed. Figure 2-17 shows the process. LTX

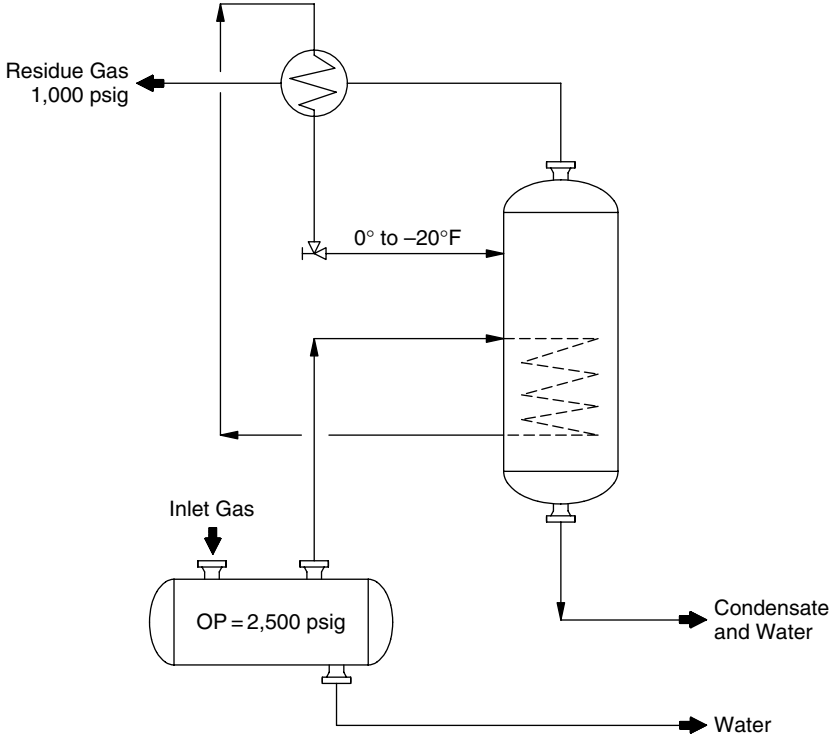


Figure 2-17. Low-temperature exchange unit.

units require inlet pressures greater than 2,500 psi to work effectively. Although they were common in the past, they are not normally used because of their tendency to freeze and their inability to operate at lower inlet pressures as the well FTP declines.

3. Contact the gas with a solid bed of $CaCl_2$. The $CaCl_2$ will reduce the moisture to low levels, but it cannot be regenerated and is very corrosive.
4. Use a solid desiccant, such as activated alumina, silica gel, or molecular sieve, which can be regenerated. These are relatively expensive units, but they can get the moisture content to very low levels. Therefore, they tend to be used on the inlets to low-temperature gas processing plants, but are not common in production facilities.
5. Use a liquid desiccant, such as methanol or ethylene glycol, which cannot be regenerated. These are relatively inexpensive. Extensive use is made of methanol to lower the hydrate temperature of gas well flowlines to keep hydrates from freezing the choke.
6. Use a glycol liquid desiccant, which can be regenerated. This is the most common type of gas dehydration system and is the one shown on the example process flowsheet.

Figure 2-18 shows how a typical bubble-cap glycol contact tower works. Wet gas enters the base of the tower and flows upward through the bubble caps. Dry glycol enters the top of the tower and, because of the down-comer weir on the edge of each tray, flows across the tray and down to the next. There are typically six to eight trays in most applications. The bubble caps assure that the upward-flowing gas is dispersed into small bubbles to maximize its contact area with the glycol.

Before entering the contactor the dry glycol is cooled by the outlet gas to condense water vapor and hydrocarbon liquids as much as possible before it enters the tower. The wet glycol leaves from the base of the tower and flows to the reconcentrator (reboiler) by way of heat exchangers, a gas separator, and filters, as shown in Figure 2-19. In the reboiler the glycol is heated to a sufficiently high temperature to drive off the water as steam. The dry glycol is then pumped back to the contact tower.

Most glycol dehydrators use triethylene glycol, which can be heated to 340° F to 400° F in the reconcentrator and work with gas temperatures up to 120° F. Tetraethylene glycol is more expensive, but it can handle hotter gas without high losses and can be heated in the reconcentrator to 400° F to 430° F.

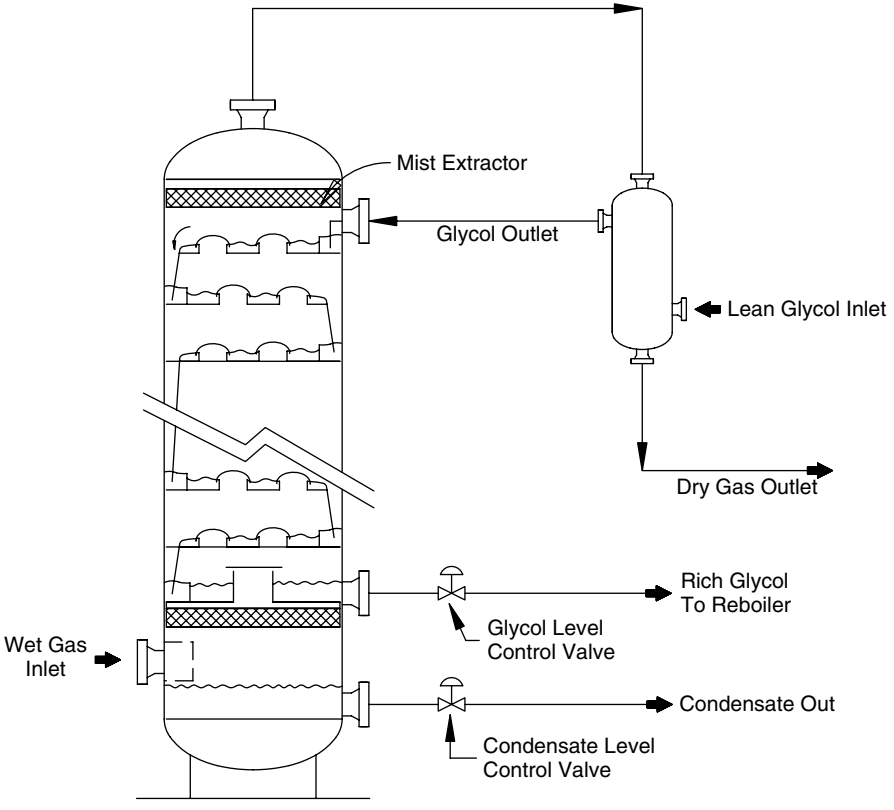


Figure 2-18. Typical glycol contact tower.

Well Testing

It is necessary to keep track of the gas, oil, and water production from each well to be able to manage the reserves properly, evaluate where further reserve potential may be found, and diagnose well problems as quickly as possible. Proper allocation of income also requires knowledge of daily production rates as the royalty or working interest ownership may be different for each well.

In simple facilities that contain only a few wells, it is attractive to route each well to its own separator and/or treater and measure its gas, oil, and water production on a continuous basis. In facilities that handle production from many wells, it is sometimes more convenient to enable each well to flow through the manifold to one or more test subsystems

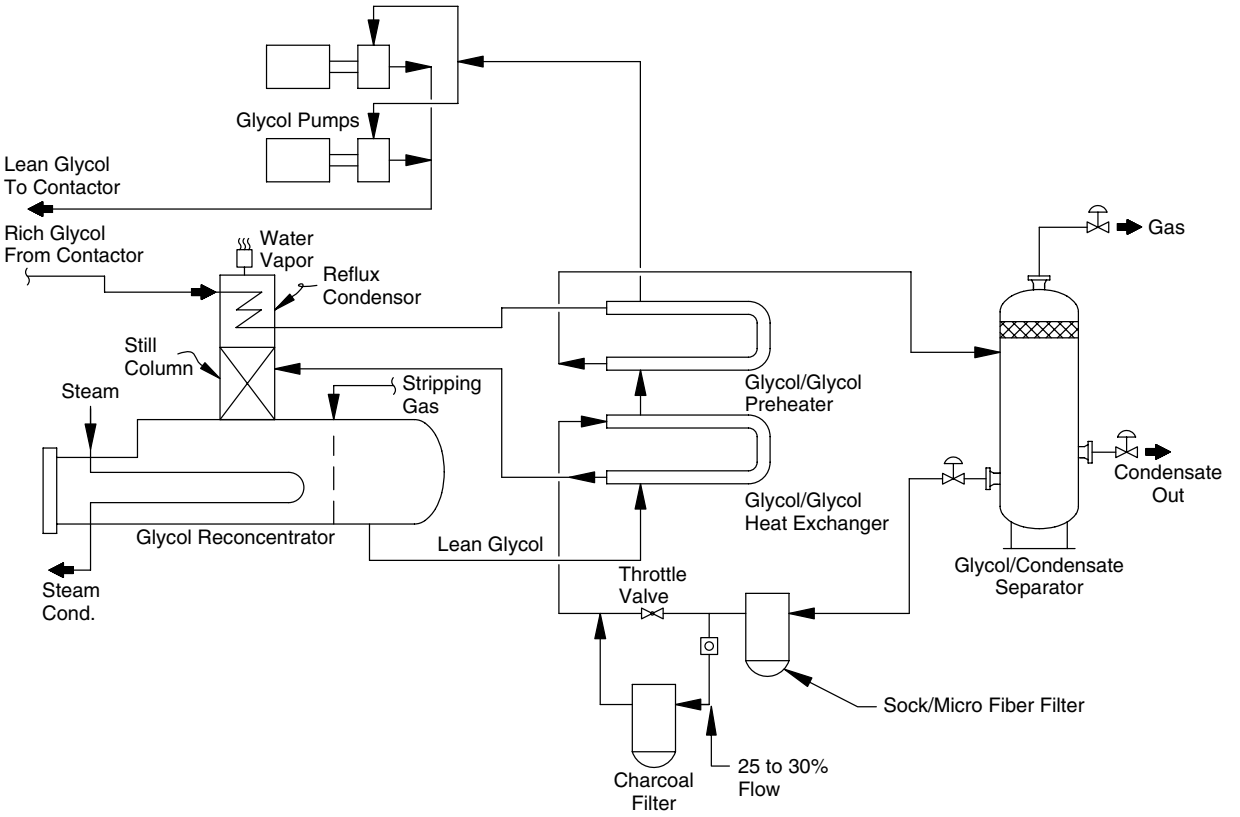


Figure 2-19. Typical glycol reconcentrator.

on a periodic basis. Total production from the facility is then allocated back to the individual wells on the basis of these well tests.

The frequency with which wells must be tested and the length of the test depend upon well properties, legal requirements, requirements for special studies, etc. Most oil wells should be tested at least twice a month for 4 to 12 hours. Gas wells should be tested at least once a month. Due to the need to put troublesome wells on long-term tests, the need to repeat tests whose results might be suspect, and the need to test several wells whenever there is an unexpected change in total production, one test system can handle approximately 20 oil wells.

In order to obtain a valid test, the test system should operate at the same pressure as the system to which the well normally flows. That is, if a well normally flows to a high-pressure separator, the first vessel in the test system should operate at that pressure. If other wells normally flow to an intermediate- or low-pressure separator, the first vessel in the test system must be able to operate at that pressure as well. Thus, in our example facility, Figure 2-1, either we must install separate high-, intermediate-, and low-pressure test systems, or we must arrange the gas backpressure valves on the first vessel in the test system so that the vessel can operate at any of the three pressures by just switching a valve.

A test system can be made up of any of the components we have discussed (e.g., separators, FWKOs, treaters) arranged in any combination that makes sense to obtain the required data. A three-phase separator could be used where oil/water emulsions are not considered severe. The amount of oil in the water outlet is insignificant and can be neglected. The water in the oil outlet can be determined from a net oil computer, which automatically corrects for the water, or by taking a sample and measuring its oil content. This would be particularly well suited for gas wells.

A vertical treater could be used where it was considered necessary to heat the emulsion in order to measure its water content. Standard treaters are low-pressure vessels with limited gas and free-water capacity. For this reason they would tend to be used on low-pressure oil wells. If it is desirable to use the treater on a higher-pressure oil well, this could be done by including a separator upstream of and in series with the treater. If a great deal of free water is expected, the treater could be designed with a large FWKO section, or a three-phase separator could be installed upstream.

Some facilities use a high-pressure three-phase separator for the high- and intermediate-pressure wells that do not make much water and a treater for the low-pressure wells. Figure 2-20 shows an enlargement of the well test separator.

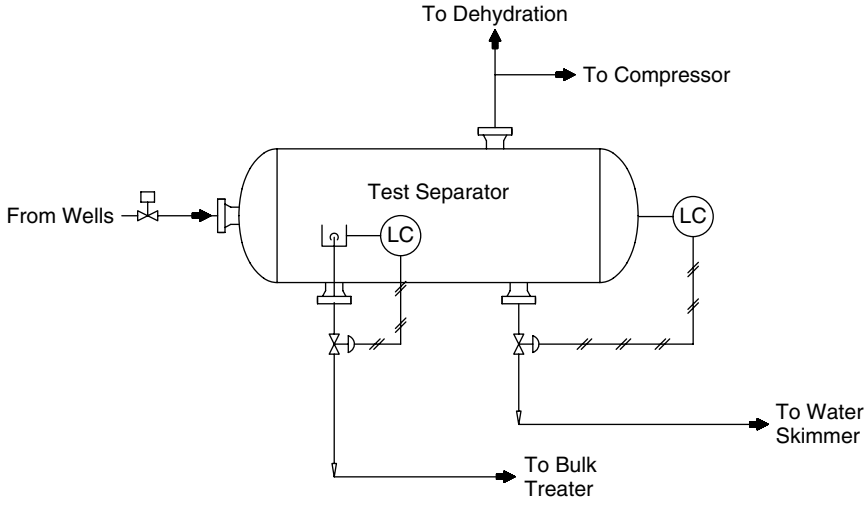


Figure 2-20. Well test system.

Gas Lift

We must comment a bit about gas lift systems because they are in widespread use and have a significant impact on the facility process. Figure 2-21 is a diagram of a gas lift system from the facility engineer's perspective. High-pressure gas is injected into the well to lighten the column of fluid and allow the reservoir pressure to force the fluid to the surface. The gas that is injected is produced with the reservoir fluid into the low-pressure system. Therefore, the low-pressure separator must have sufficient gas separation capacity to handle gas lift as well as formation gas.

If gas lift is to be used, it is even more important from a production standpoint that the low-pressure separator be operated at the lowest practical pressure. Figure 2-22 shows the effects of wellhead backpressure for a specific set of wells. It can be seen that a 1-psi change in well backpressure will cause between a 2- and 6-BFPD change in well deliverability.

The higher the injected gas pressure into the casing is, the deeper the last gas lift valve can be set. As shown in Figure 2-23, for a typical well the higher the design injection is, the higher the flow rate. Most gas sales contracts are in the 1,000- to 1,200-psi range. Therefore, the process must be designed to deliver the sales gas at this pressure. As seen from Figure 2-23, at about this range a rather large change in gas injection pressure is necessary for a small change in well deliverability. In the range of pressures under consideration (approximately 65-psia suction, 1,215-psia

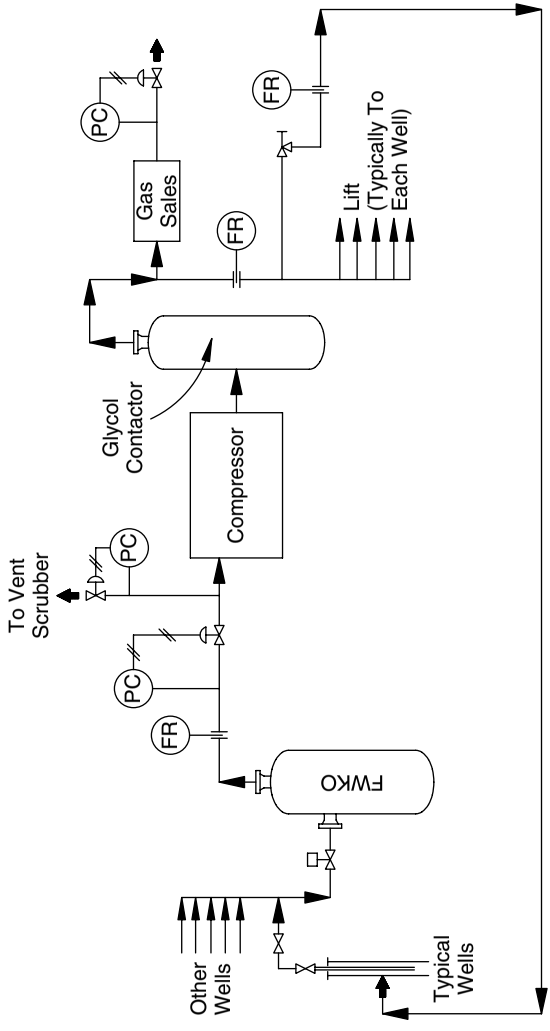
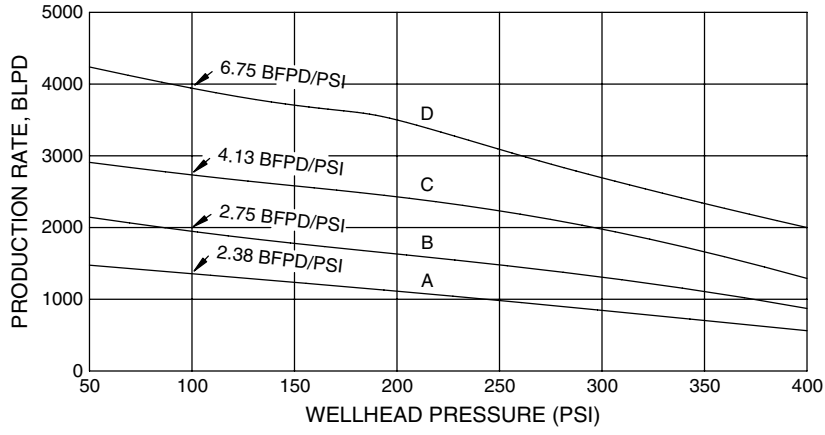
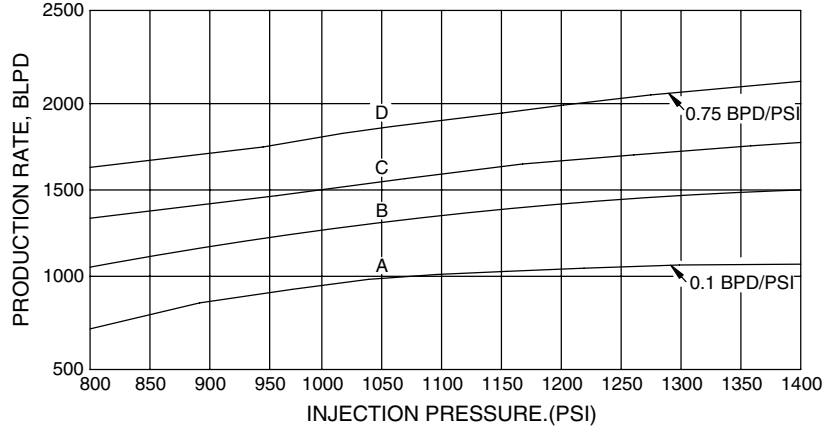


Figure 2-21. Gas lift system.



Note: These curves are for a specific set of tubing size, casing pressure, and fluid out.

Figure 2-22. Effect of wellhead backpressure on total fluid production rate for a specific set of wells.



Note: These curves are for a specific set of tubing size, casing pressure, and fluid out.

Figure 2-23. Effect of gas lift injection pressure on total fluid production rate for a specific set of wells.

discharge), a 1-psi change in suction pressure (i.e., low-pressure separation operating pressure) is equivalent to a 19-psi change in discharge pressure (i.e., gas lift injection pressure) as it affects compressor ratio and thus compressor horsepower requirements. A comparison of Figures 2-22 and 2-23 shows that a 1-psi lowering of suction pressure in this typical case is more beneficial than a 19-psi increase in discharge pressure for the wells with a low productivity index (PI) but not as beneficial for

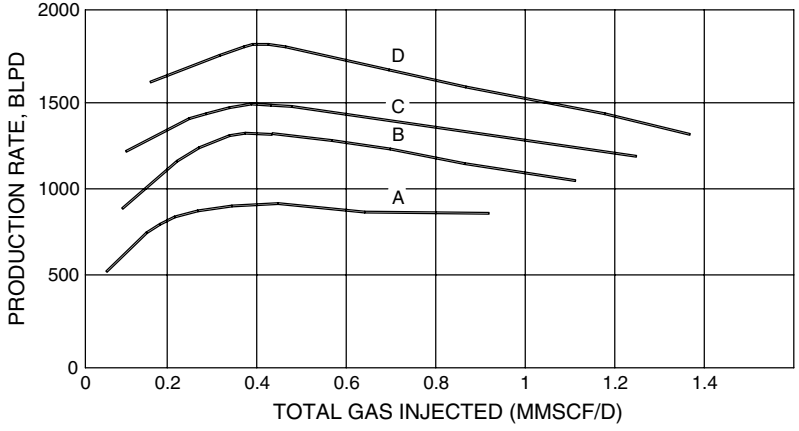


Figure 2-24. Effect of gas lift injection rate on total fluid production rate for a specific set of wells.

the high-PI wells. The productivity index is the increase in fluid flow into the bottom of the well (in barrels per day) for a 1-psi drawdown in bottom-hole pressure.

Figure 2-24 shows the effect of gas injection rate. As more gas is injected, the weight of fluid in the tubing decreases and the bottom-hole flowing pressure decreases. This is balanced by the friction drop in the tubing. As more gas lift gas is injected, the friction drop of the mixture returning to the surface increases exponentially. At some point the friction drop effect is greater than the effect of lowering fluid column weight. At this point, injecting greater volumes of gas lift gas causes the bottom-hole pressure to increase and thus the production rate to decrease.

Each gas lift system must be evaluated for its best combination of injection rate, separator pressure, and injection pressure, taking into account process restraints (e.g., need to move the liquid through the process) and the sales gas pressure. In the vast majority of cases, a low-pressure separator pressure of about 50 psig and a gas lift injection pressure of 1,000 to 1,400 psig will prove to be near optimum.

Offshore Platform Considerations

Overview

An increasing amount of the world's oil and gas comes from offshore fields. This trend will accelerate as onshore fields are depleted. A growing amount of engineering effort is being spent on offshore facility designs.

Thus, it is appropriate that this section describe platforms that accommodate simultaneous drilling and production operations.

Modular Construction

Modules are large boxes of equipment installed in place and weighing from 300 to 2,000 tons each. Modules are constructed, piped, wired, and tested in shipyards or in fabrication yards and then transported on barges and set on the platform, where the interconnections are made (Figure 2-25). Modular construction is used to reduce the amount of work and the number of people required for installation and start-up.

Equipment Arrangement

The equipment arrangement plan shows the layout of all major equipment. Each platform has a unique layout requirement based on drilling and well-completion needs that differ from installation to installation. Layouts can be on one level or multiple levels. An example layout is shown in Figure 2-26; the right-hand module contains the flare drums, water skimmer tank, and some storage vessels. In addition, it provides support for the flare boom.

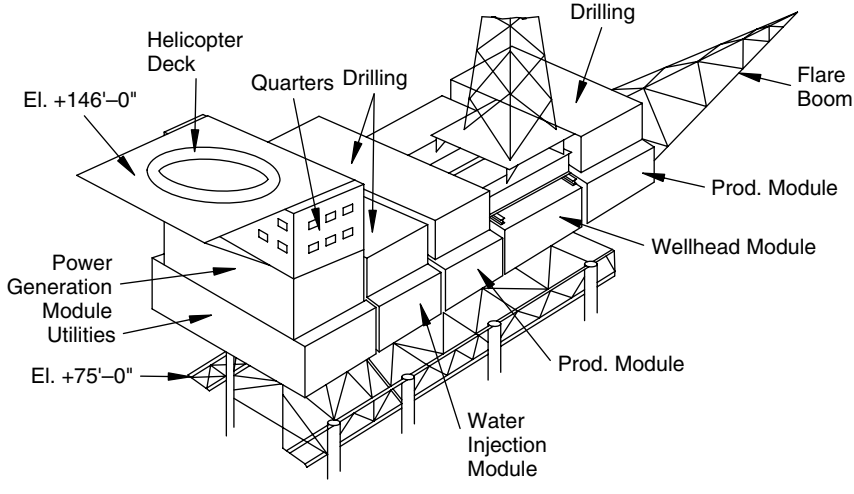


Figure 2-25. Schematic of a large offshore platform, illustrating the concept of modularization.

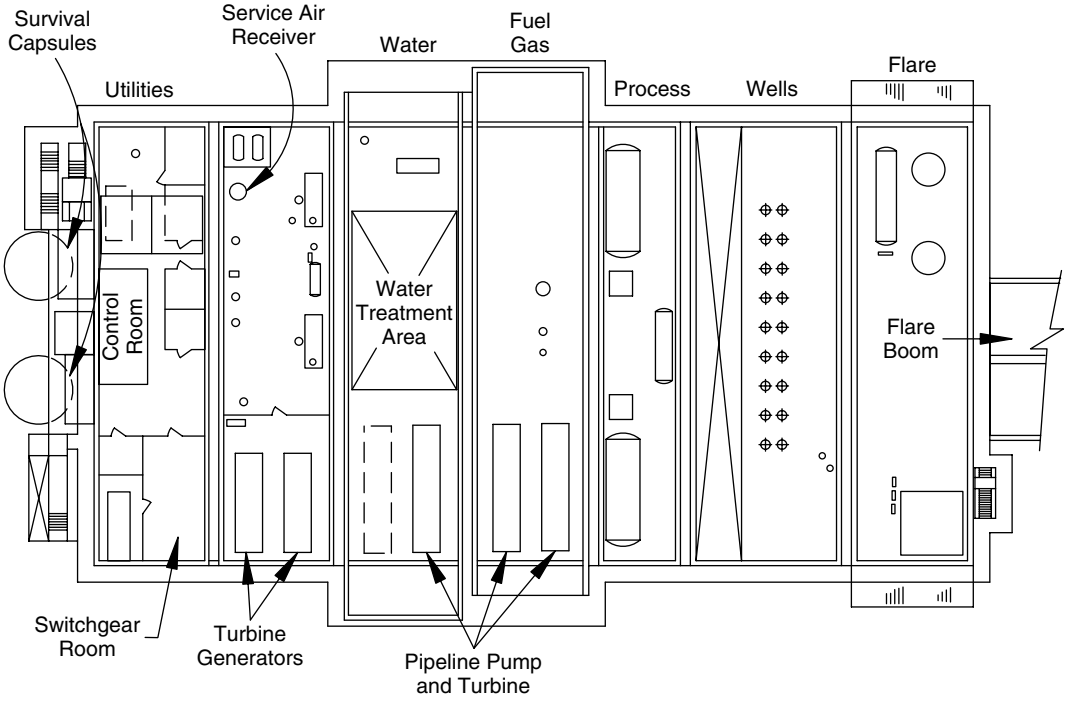


Figure 2-26. Equipment arrangement plan of a typical offshore platform illustrating the layout of the lower deck.

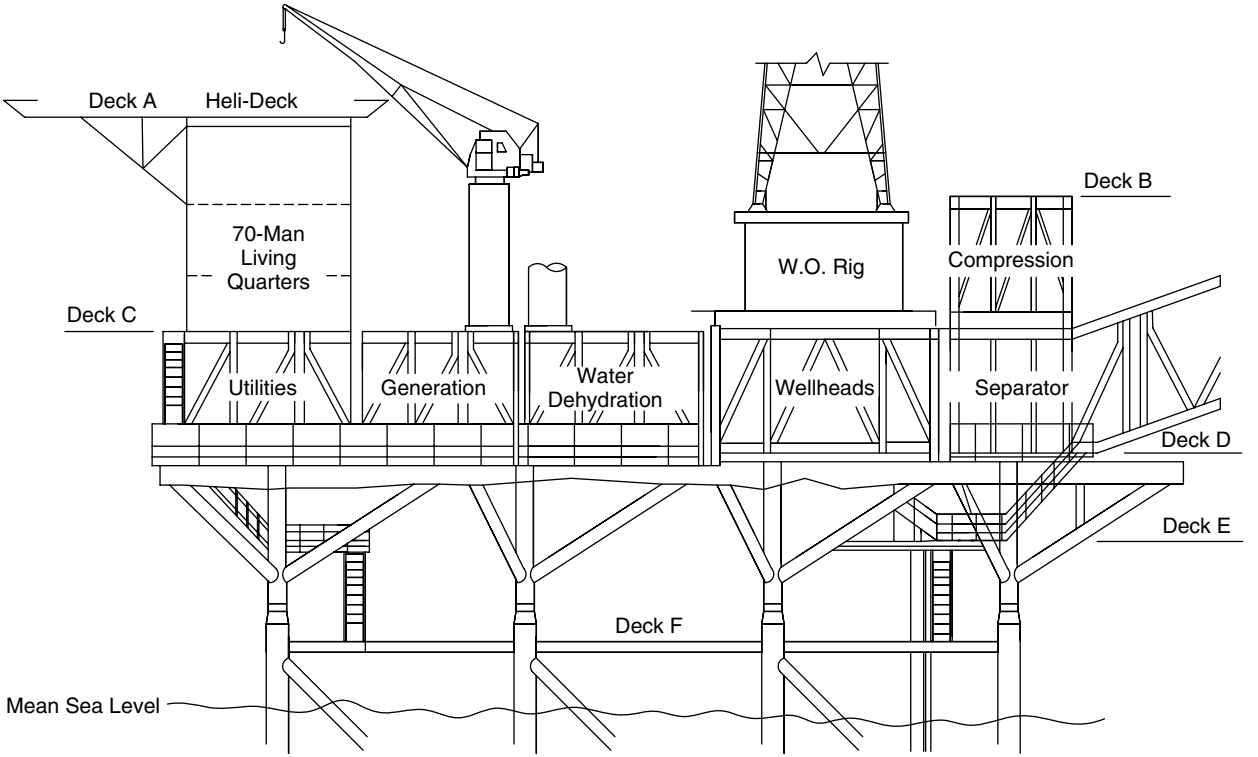


Figure 2-27. Typical elevation view of an offshore platform showing the relationship among the major equipment modules.

The adjacent wellhead module consists of a drilling template with conductors through which the wells will be drilled. The third unit from the right contains the process module, which houses the separators and other processing equipment. The fourth and fifth modules from the right contain turbine-driven pumps, fuel gas scrubbers, and the produced-water treating area. The last two modules house utilities such as power generators, air compressors, potable water makers, a control room, and switchgear and battery rooms. The living quarters are located over the last module. Figure 2-27 shows an elevation of a platform in which the equipment arrangement is essentially the same.

Chapter 3
Basic Principles

Introduction

Before describing the equipment used in the process and design techniques for sizing and specifying that equipment, it is necessary to review some basic principles and fluid properties. We will also discuss some of the common calculation procedures, conversions, and operations used to describe the fluids encountered in the process.

Basic Oil-Field Chemistry

Elements, Compounds, and Mixtures

Matter is anything that possesses mass and occupies space. Matter is made up of elements, compounds, and mixtures. An element is the simplest form of matter. There are over 100 known elements or combinations of elements. Table 3-1 illustrates the five most abundant elements on earth.

Elements in the free or uncombined state make up only a fraction of matter. Most matter exists as compounds or mixtures of compounds. A compound is a more complex form of matter made up of chemically combined elements. Molecules of compounds are identical to each other in composition and properties. Typical compounds are methane (CH_4), carbon dioxide (CO_2), sodium chloride (NaCl), and water (H_2O). A mixture consists of two or more elements or compounds that are mixed physically but still maintain their chemical identity. Mixtures can always be separated into their component parts. Typical mixtures are natural gas, air, oil, coal, or any alloys.

Table 3-1
Five Most Abundant Elements

Element	Percent
Oxygen	49.2
Silicon	25.7
Aluminum	7.5
Iron	4.7
Calcium	3.3
All others	9.6
Total	100.0

Atomic and Molecular Weights

An atom is the smallest part of an element that can be divided and still retain all the properties of that element. An atom is the smallest unit of matter that can enter into combination with itself or atoms of other elements. The three basic particles of atoms are

- Protons: particle consisting of a positive electrical charge located in the nucleus of the atom.
- Neutrons: particle with no electrical charge located in the nucleus of the atom.
- Electrons: particle consisting of a negative electrical charge located in various orbits and rotating around the nucleus of the atom.

Protons and neutrons weigh essentially the same. Electrons are smaller, weigh less, and exert electrical charges equal but opposite to that of protons. Characteristics of atoms are

- All atoms contain the same number of electrons as they do protons; thus, all atoms are electrically neutral.
- Each element has an atomic number, which indicates the number of protons as well as the number of electrons in an atom.
- Each atom has an atomic weight that approximates the number of protons plus the number of neutrons in the atom.
- Since nearly all the weight of the atom is concentrated in the nucleus, the electron can be considered weightless.

Atomic weights are relative weights of one atom to another. For example, hydrogen has an atomic weight equal to 1, oxygen 16, carbon 12, and sulfur 32.

A molecule is the smallest unit of matter formed by the combination of atoms. A molecule can be a combination of two atoms of the same

element as in most gases or different atoms to form complex molecules as in hydrocarbons. The molecular weight is the relative weight of one molecule to another.

Elements have different atomic structures because of the varying numbers of protons and neutrons in the nucleus and the electrons orbiting the nucleus. For example, carbon has six protons and six neutrons in the nucleus and six electrons orbiting the nucleus (two electrons in the first orbit and four electrons in the second orbit). The number of electrons of an element involved in forming a compound is known as the valence number. Elements have a tendency to lose or gain electrons in the outer orbit of their atom. Those atoms that lose one, two, or three electrons become positively charged or form positive ions. Those atoms that gain one, two, or three electrons become negatively charged or form negative ions. When an atom gains or loses electrons in an outer orbit and becomes an ion, it changes the properties it possessed as an atom.

Compounds are formed by the electrovalent union when a positively charged ion (sodium) and a negatively charged ion (chlorine) electrostatically attract each other to form a compound (sodium chloride). The molecular weight of a compound is the sum of the atomic weight of the various atoms making up that compound. Covalent union occurs when atoms share electrons. For example, carbon, which has four electrons in its outer orbit, shares each of its electrons with four hydrogen atoms, each of which is seeking one electron to complete its outer orbit. This union forms the compound methane (CH_4). Most atoms, such as hydrogen, oxygen, nitrogen, chlorine, etc., share their electrons with a similar atom to form elemental gases of hydrogen (H_2), oxygen (O_2), nitrogen (N_2), and chlorine (Cl_2). These gases exist in nature as a covalent union.

Hydrocarbon Nomenclature

Hydrocarbons are covalent compounds composed of carbon and hydrogen atoms. Atoms combine in a number of ways to satisfy the valence requirements. Hydrocarbons are separated into "families" of homologous series. Carbon atoms link together to form chains and/or ring structures. Natural gas and crude oil consist primarily of "straight chain" hydrocarbon molecules. Most of the hydrocarbons of concern fall in the paraffin series.

Natural gas consists of methane (CH_4), which is the main component, and other components, including ethane, propane, *n*-butane, *i*-butane, *n*-pentane, *i*-pentane, hexanes, heptanes, octanes, and heavier hydrocarbons. Hexanes and heavier are referred to as hexane plus. Natural gas also includes diluents of carbon dioxides, nitrogen, and smaller quantities of hydrogen sulfide, water vapor, and oxygen.

Table 3-2
Seven Most Common Paraffin Molecules and Their MW

Name	Molecular Weight
Methane	16
Ethane	30
Propane	44
Butane	58
Pentane	72
Hexane	86
Heptane	100

Paraffin Series: (C_nH_{2n+2})

Hydrocarbons in this series are saturated, that is, all four carbon bonds are connected either to another carbon atom or to a hydrogen atom, with one such atom for each bond. All names end in –ane. The number of hydrogen atoms is two times the number of carbon atoms plus two more for the ends of the chain. Paraffins are the most stable hydrocarbons because all valence bonds are fully satisfied, as indicated by the single line linkage. Each successive molecule is created by adding one carbon and two hydrogens to the previous molecule (refer to Table 3-2).

The abbreviations C_3 for propane, C_4 for butane, etc. are often used. Statements like "propane plus fraction" (C_{3+}) refer to a mixture composed of propane and larger atoms.

When a molecule contains four or more carbon atoms, there are different ways these can be connected without affecting the formula. Compounds that have the same chemical formula but different atomic structure are called isomers. Isomers possess different physical and chemical properties.

Paraffin Compounds

In upstream oil and gas operations, we are primarily concerned with

- Paraffin series with 15 or fewer carbon atoms,
- Nitrogen,
- Water,
- Contaminants such as H_2S and CO_2,
- Mercury.

Paraffin hydrocarbons are less reactive with other materials than many hydrocarbons. They are conditioned by use of alcohols, glycols, and amines in which they are soluble and react to some degree.

Acids and Bases

All inorganic compounds of hydrogen, except water and hydroxides, are acids (refer to Table 3-3). They consist of hydrogen combined with an acid radical (anion).

The acidity or alkalinity of a material is measured using a scale known as pH. The scale runs between 0 and 14. A pH of 7 is neutral. Acids have a pH less than 7; bases (alkaline solutions) have a pH greater than 7.

Fluid Analysis

An example fluid analysis of a gas well is shown in Table 3-4.

Note that only paraffin hydrocarbons are shown. This is not correct, even though they may be the predominant series present. Also note that all molecules heptane and larger are lumped together as a heptanes plus fraction. The hydrocarbon portion of an analysis is usually obtained from a chromatograph, as shown in Figure 3-1.

Physical Properties

An accurate estimate of physical properties is essential if one is to obtain reliable calculations. Physical and chemical properties depend upon

- Pressure,
- Temperature,
- Composition.

Table 3-3
Common Acids

Acid Radical	Symbol	Acid	Formula
Chloride	Cl^{-1}	Hydrochloric	HCl
Carbonate	CO_3^{-2}	Carbonic	H_2CO_3
Sulfate	SO_4^{2-}	Sulfuric	H_2SO_4
Nitrate	NO_3^{-1}	Nitric	HNO_3
Phosphate	PO_4^{-3}	Phosphoric	H_3PO_4

Table 3-4
Example Fluid Analysis of Gas Well

Component	Mol %
Methane (C_1)	35.78
Ethane (C_2)	21.46
Propane (C_3)	11.40
i-Butane (i-C_4)	5.35
n-Butane (n-C_4)	10.71
i-Pentane (i-C_5)	3.81
n-Pentane (n-C_5)	3.07
Hexanes (C_6)	3.32
Heptanes plus (C_{7+})	3.24
Nitrogen	0.20
Carbon dioxide	1.66
Total	100.00

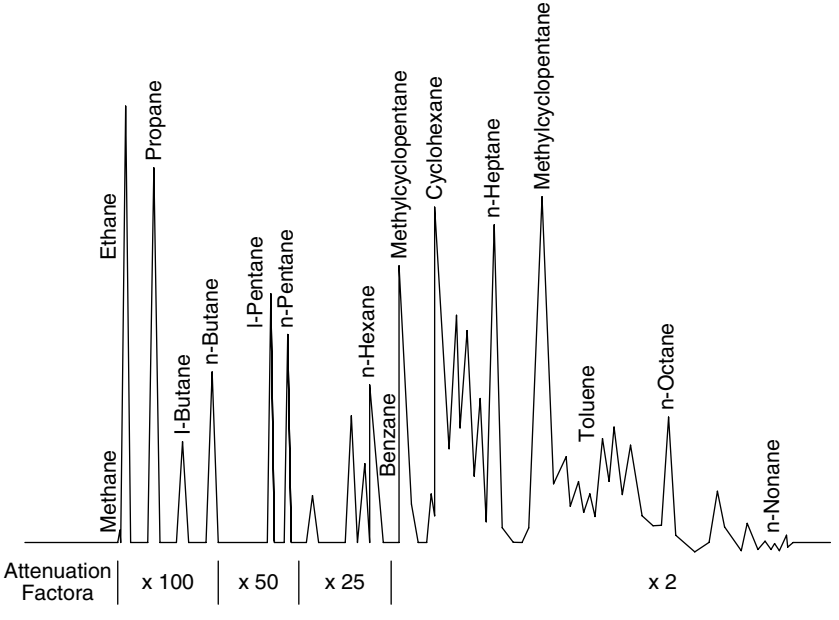


Figure 3-1. Chromatograph of condensed liquid.

Hydrocarbon streams are mixtures of hydrocarbons that contain various amounts of impurities such as hydrogen sulfide, carbon dioxide, and water. A single-component system composed entirely of a simple molecule, such as methane or propane, behaves in a very predictable,

correctable manner. The performance of a single-component system can be accurately correlated in tabular form. For all others, one must use pressure, volume, and temperature (PVT) equations of state or a weighted average (assumes that the contribution of individual molecules is in proportion to their relative quantity in the mixture). The more dissimilar the molecules, the less accurate the prediction becomes. Table 3-5 lists properties of some hydrocarbon molecules which are important for process calculations.

Water in liquid or vapor form is present to some degree in all systems. Liquid water is basically immiscible in hydrocarbons. Since phase behavior calculations are not applicable to water, special procedures must be used.

Equations of state use the values of P, V, and T at the critical point. A specific critical point exists for each hydrocarbon component. For a pure component the critical values represent the maximum pressure and temperature at which a two-phase, vapor-liquid system can exist. Above P_c and T_c, only a single phase is possible. For mixtures, pseudocritical values are calculated. These values are not a point on the phase diagram but a correlation value that allows one to perform routine calculations.

Any equation correlating P, V, and T is called an "equation of state." Equation (3-1) is called the "ideal gas law" or "general gas law."

Table 3-5
Properties of Some Hydrocarbon Molecules

		Critical Temperature		Critical Pressure	
Compound	**Molecular Weight**	**°R**	**K**	**psia**	**MPa**
C_1	16.043	343	191	666	4.60
C_2	30.070	550	305	707	4.88
C_3	44.097	666	370	617	4.25
iC_4	58.124	734	408	528	3.65
nC_4	58.124	765	425	551	3.80
iC_5	72.151	829	460	491	3.39
nC_5	72.151	845	470	489	3.37
nC_6	86.178	913	507	437	3.01
nC_7	100.205	972	540	397	2.74
nC_8	114.232	1024	569	361	2.49
nC_9	128.259	1070	595	332	2.29
nC_{10}	142.286	1112	618	305	2.10

Table 3-6
Universal Gas Constant

P	*V*	*T*	*R*
Kpa	M^3	°K	8.314 (KPa)(M^3)/(Kmol)(K)
MPa	M^3	°K	0.00831 (MPa)(m^3)/(Kmol)(K)
bar	M^3	°K	0.08314 (bar)(M^3)/(Kmol)(K)
psi	ft^3	°R	10.73(psia)(ft^3)/(lb-mol)(°R)
lb/ft^2	ft^3	°R	1545(psfa)(ft^3l/(lb-mol)(°R)

$$PV = nRT, \tag{3-1}$$

where

P = absolute pressure,
V = volume,
n = number of moles of gas of volume V at P and T,
R = universal gas constant (see Table 3-6),
T = absolute temperature.

Equation (3-1) is valid up to pressures of about 60 psia (500 KPa) (4 bara). As pressure increases above this level, its accuracy becomes less and the system should be considered a nonideal gas equation of state. Table 3-6 lists the values of the universal gas constant for different unit systems.

Molecular Weight and Apparent Molecular Weight

The number of moles is defined as follows:

$$\text{mole} = \frac{\text{weight}}{\text{molecular weight}} \tag{3-2}$$

expressed as

$$n = \frac{m}{M}, \tag{3-3}$$

or, in units, as

$$\text{lb-mole} = \frac{\text{lb}}{\text{lb/lb-mole}}. \tag{3-4}$$

Example 3-1: Molecular weight calculation

Given: Determine the molecular weight of methane, CH_4.

Solution:

Element	No. of Atoms		Atomic Weight		Product
C	1	×	12	=	12
H	4	×	1	=	4
		Molecular weight		=	16 lb/lb-mole

Up to now we have addressed only pure substances. We now have to consider hydrocarbon mixtures. However, first we must discuss apparent molecular weight and specific gravity.

It is not correct to say that a hydrocarbon mixture has a molecular weight; rather, it has an apparent molecular weight. Apparent molecular weight is defined as the sum of the products of the mole fractions of each component times the molecular weight of that component. This is shown in Eq. (3-5):

$$\mathrm{MW} = \sum y_i \, (\mathrm{MW})_i, \tag{3-5}$$

where

y_i = molecular fraction of ith component,
MW_i = molecular weight of ith component,
$\sum y_i = 1$.

Now let's look at an example of the application of apparent molecular weight.

Example 3-2: Determine the apparent molecular weight of dry air, which is a gas mixture consisting of nitrogen, oxygen, and small amounts of Argon.

Given: Determine the apparent molecular weight of air given its approximate composition.

Gas Composition	
Component	**Mole Fraction**
Nitrogen	0.79
Oxygen	0.20
Argon	0.01
Total =	1.00

Solution:

(1) Look up the molecular weight of each component from the physical constant table:

$$(\mathrm{MW})_N = 28;\ (\mathrm{MW})_O = 32;\ (\mathrm{MW})_A = 40.$$

(2) Multiply the mole fraction of each component by its molecular weight:

$$\begin{aligned}(\mathrm{MW})_{\mathrm{AIR}} &= \sum y_i(\mathrm{MW})_i = y_N(\mathrm{MW})_N + y_0(\mathrm{MW})_0 + y_A(\mathrm{MW})_A. \\ &= (0.79 \times 28) + (0.20 \times 32) + (0.01 \times 40) \\ &= 29 \text{ lb/lb-mole.}\end{aligned}$$

Gas Specific Gravity and Density

The specific gravity of a gas is the ratio of the density of the gas to the density of air at standard conditions of temperature and pressure.

$$S = \frac{\rho_g}{\rho_{\mathrm{air}}} \tag{3-6}$$

where

ρ_g = density of gas,
ρ_{air} = density of air.

Both densities must be computed at the same pressure and temperature, usually at standard conditions.

It may be related to the molecular weight by Equation 3.7. The derivation of Equation (3-7) follows; its derivation begins with the following equations:

$$\rho_g = \frac{(\mathrm{MW})_g P}{RT} \quad \text{and} \quad \rho_{\text{air}} = \frac{(\mathrm{MW})_{\text{air}} P}{RT}.$$

Substituting into the above equation

$$S = \frac{\dfrac{(\mathrm{MW})_g P}{RT}}{\dfrac{(\mathrm{MW})_{\text{air}} P}{RT}} = \frac{(\mathrm{MW})_g}{(\mathrm{MW})_{\text{air}}},$$

we see from the previous example that $(\mathrm{MW})_{\text{air}} = 29$; thus,

$$S = \frac{(\mathrm{MW})_g}{29}. \tag{3-7}$$

Example 3-3: Calculate the specific gravity of a natural gas with the following composition.

Given:

Component	Mole Fraction (Y_i)
Methane (C_1)	0.85
Ethane (C_2)	0.09
Propane (C_3)	0.04
n-Butane (*n*-C_4)	0.02
Total	1.00

Solution:

(1)

Component	Mole Fraction		MW, $(\mathrm{MW})_i$		$Y_i\ (\mathrm{MW})_i$
C_1	0.85	×	16.0	=	13.60
C_2	0.09	×	30.0	=	2.71
C_3	0.04	×	44.0	=	1.76
n-C_4	0.02	×	58.0	=	1.16
	1.00		$(\mathrm{MW})_g$	=	19.23

(2) $$S = \frac{(\mathrm{MW})_g}{29} = \frac{19.23}{29} = 0.66.$$

In most calculations the specific gravity of the gas is always referred to in terms of standard conditions of temperature and pressure and therefore is always given, Eq. (3-8), once the molecular weight of the gas is known. The density of a gas at any condition of temperature and pressure can be determined by remembering that the density of air at standard conditions of temperature and pressure (60°F and 14.7 psia) is 0.00764 lb/ft^3. The density of gas is thus given as

$$\rho_g = 2.70\frac{SP}{TZ}, \tag{3-8}$$

$$\rho_g = 0.093\frac{(\mathrm{MW})P}{TZ}, \tag{3-9}$$

where

ρ_g = density of gas, lb/ft^3,
S = specific gravity of gas (air = 1),
P = pressure, psia,
T = temperature, °R,
Z = gas compressibility factor,
MW = gas molecular weight.

The density of gas at standard conditions of temperature and pressure is, by definition,

$$\rho_{std} = 0.0762\,\mathrm{S};$$

the volume of a pound of gas is given by the specific volume as

$$\overline{V} = \frac{1}{\rho};$$

the equation of state for a gas is given for engineering calculations as

$$PV = nZRT.$$

Thus, for a given number of moles of gas:

$$\frac{PV}{ZT} = nR = \text{constant}.$$

For any gas, comparing its equation of state at standard conditions to that at actual conditions:

$$\frac{P_{\text{std}}\overline{V}_{\text{std}}}{T_{\text{std}}Z_{\text{std}}} = \frac{P\overline{V}}{TZ},$$

$$\frac{(14.7)}{(520)(1.0)(0.0762)S} = \frac{P}{\rho TZ},$$

$$\rho = 2.70\frac{SP}{TZ},$$

$$S = \frac{\text{MW}}{29},$$

$$\rho = 0.093\frac{(\text{MW})P}{TZ}.$$

Nonideal Gas Equations of State

The ideal gas equations of state describes most real gases at low pressure but does not yield reasonable results at higher pressures. Many PVT equations have been developed to describe nonideal, real gas behavior. Each is empirical in that it correlates a specific set of data using one, or more, empirical constants. Unfortunately, there is no correlation that is equally good for all gas mixtures. Some of the more common equations of state that attempt to define the relationship between V, T, and P for the real gases follow:

Van der Waals

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT, \tag{3-10}$$

where

a, b = correlation factors,
V = molar volume.

Redlich–Kwong (RK)

$$P = \frac{RT}{V-b} - \frac{a}{T^{0.5}V\,(V-b)}, \tag{3-11}$$

where

$a,\ b$ = correlation factors,
V = molar volume.

Peng–Robinson (PR)

$$P = \frac{RT}{V-b} - \frac{a\,(T)}{V\,(V+b)+b\,(V-b)}. \tag{3-12}$$

Benedict–Webb–Rubin (BWR)

$$\begin{aligned} P = {} & RT\rho + \left(B_0RT - A_0 - C_0/T^2\right)\rho^2 \\ & + (bRT - a)\,\rho^3 + a\alpha\rho^6 \\ & + \left(c\rho^3/T^2\right)\left(1+\gamma\rho^2\right)e^{\gamma\rho^2} \end{aligned} \tag{3-13}$$

where

$A_0, B_0, C_0, a, b, c, \alpha$, and γ are correlation constants,
P = absolute pressure,
T = absolute temperature,
ρ = molar density.

A number of modifications to the above equations have been published in an attempt to improve the validity of the equation. The above equations are the basis of most computer programs. However, the real accuracy may be no better than some simpler methods when the designer considers the quality of the compositional data usually obtained from a drill stem test in the early stages of a project. Fortunately, all of the ideal equations of state can be approximated to the compressibility equation of state by multiplying the "RT" part of the equation by Z:

$$PV = ZnRT, \tag{3-14}$$

where

$$Z = \frac{\text{actual gas volume}}{\text{ideal gas volume}}. \tag{3-15}$$

If the gas acted as if it were an ideal gas, then the Z factor would be 1. The typical range of $Z = 0.8$ to 1.2. Figure 3-2 shows the Z factor as a function of pressure at constant temperature, various temperatures, and pseudo-critical properties. Figures 3-3 through 3-6 are some correlations for the compressibility factor, Z, which have proven useful for natural gas calculations.

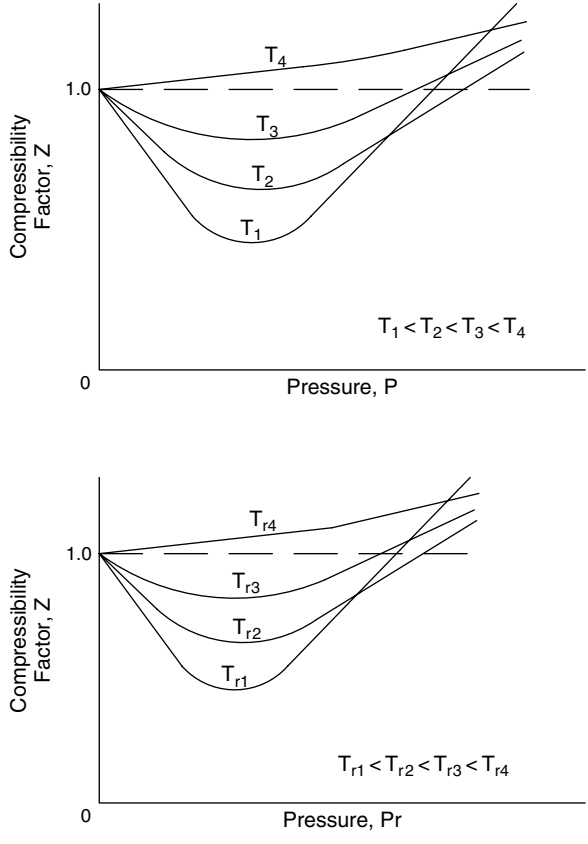


Figure 3-2. Top: Z factor as a function of pressure for a single hydrocarbon component at various temperatures; bottom: relationship for hydrocarbon mixtures is similar but dependent on reduced properties.

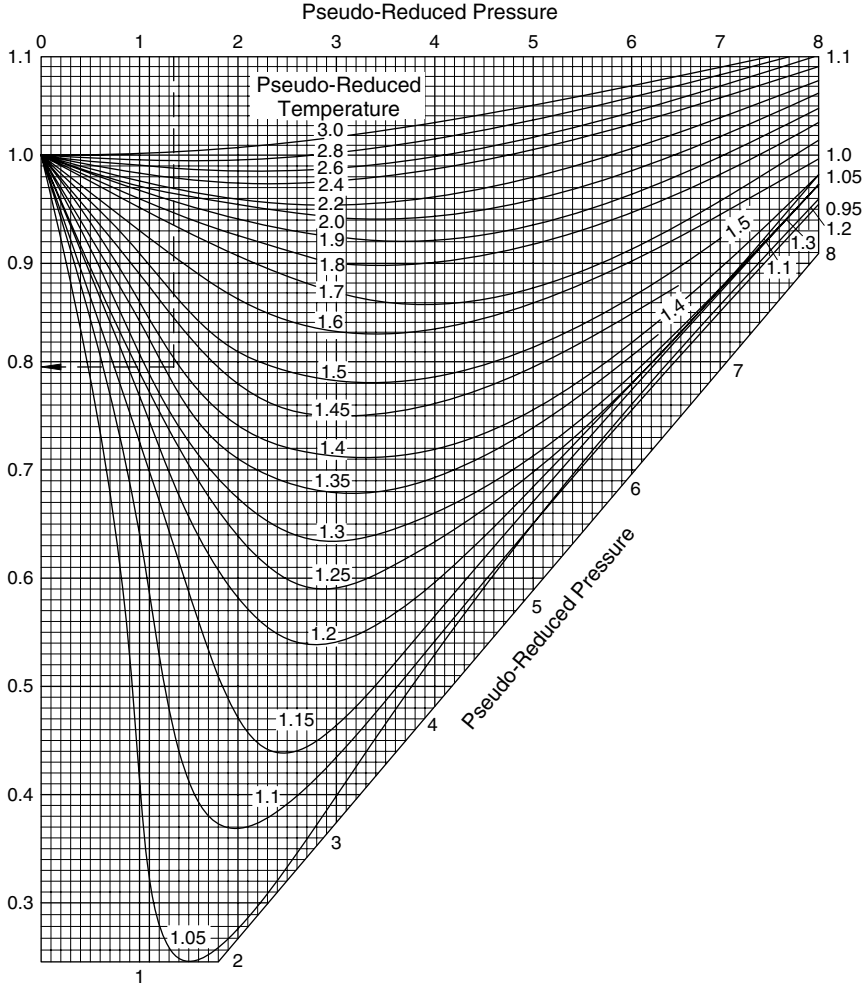


Figure 3-3. Compressibility factor for lean, sweet natural gas. (After Katz et al.)

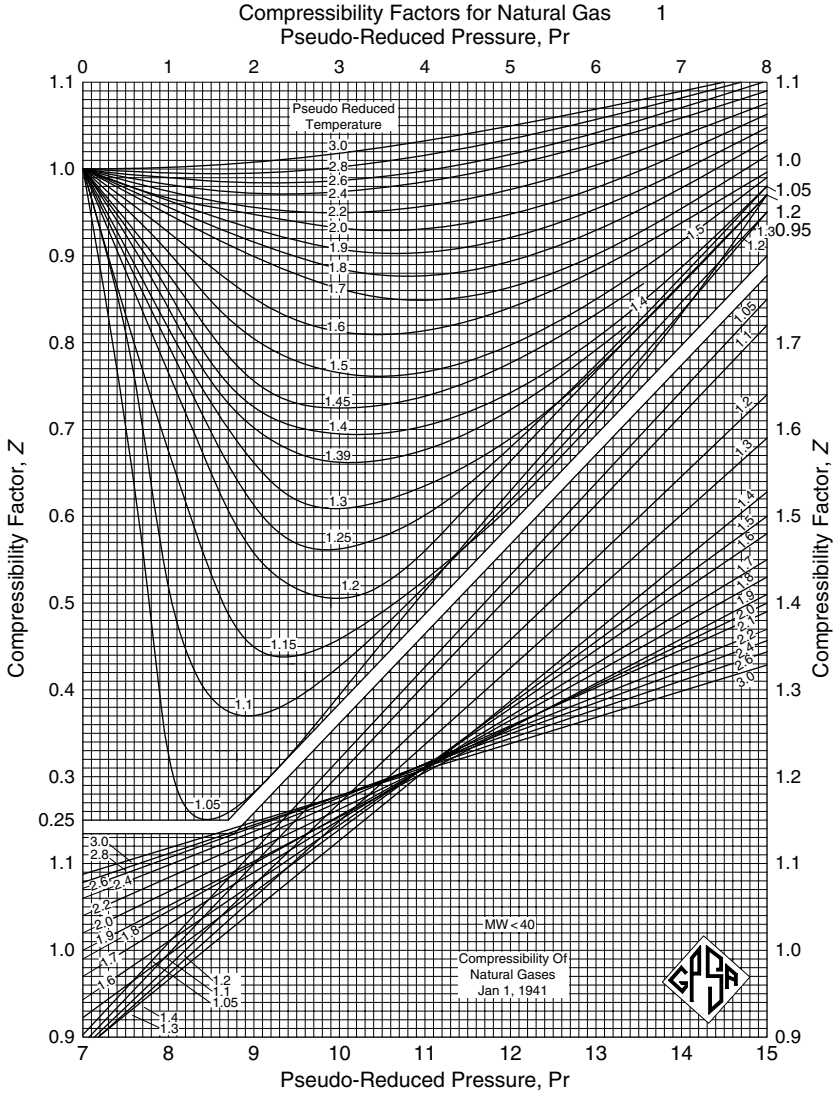


Figure 3-4. Compressibility factors for natural gas. (Courtesy of GPSA *Engineering Data Book*.)

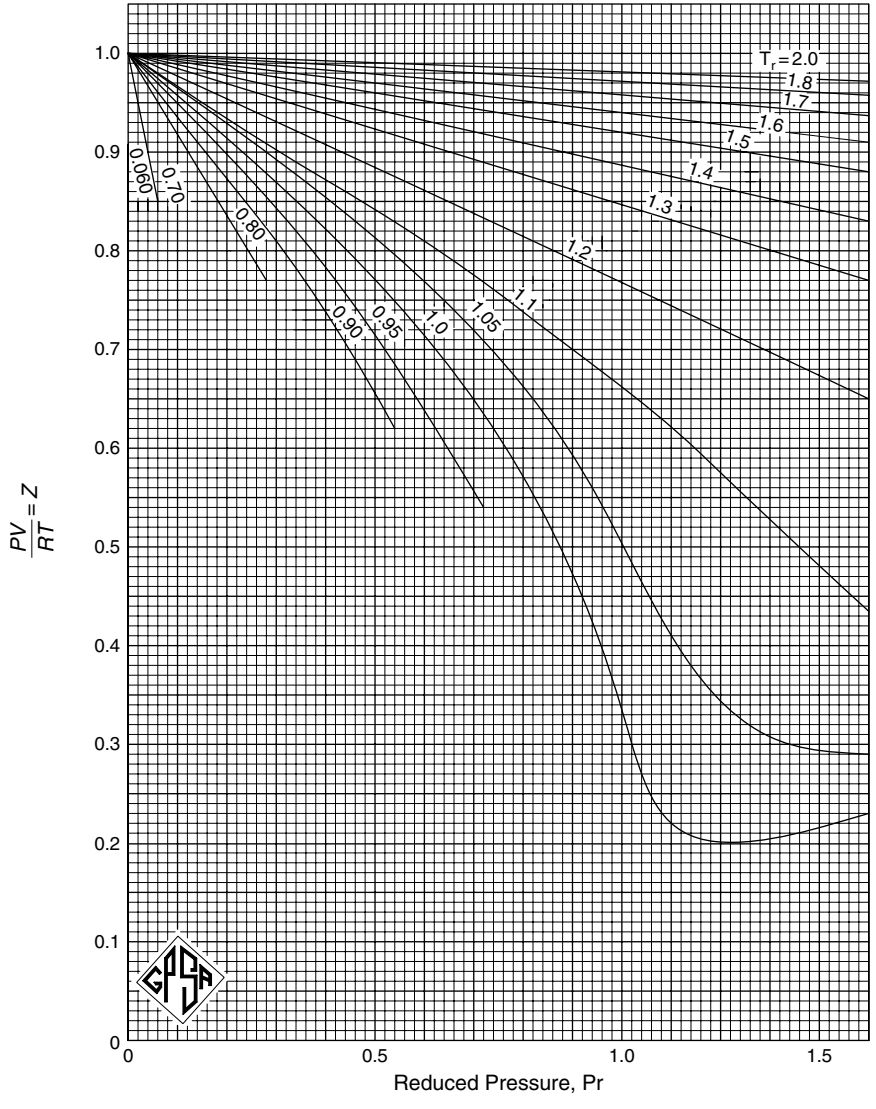


Figure 3-5. Compressibility factors for natural gas at low reduced pressures. (Courtesy of GPSA *Engineering Data Book*.)

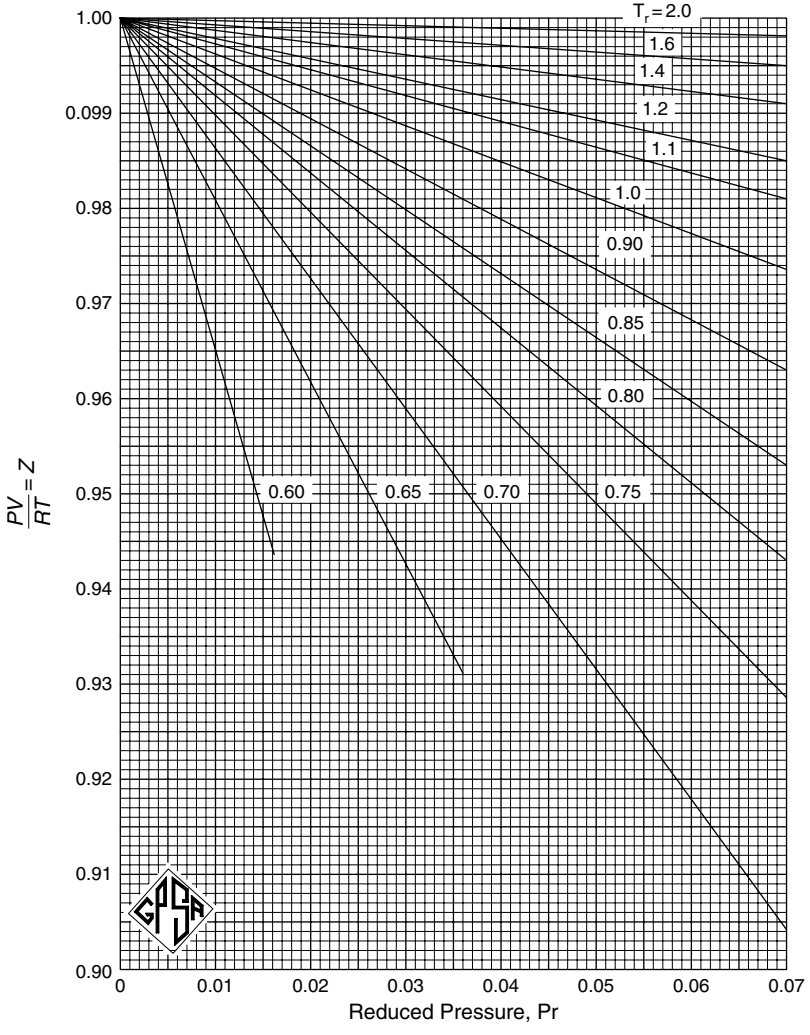


Figure 3-6. Compressibility factors for natural gas at near atmospheric pressure. (Courtesy of GPSA *Engineering Data Book*.)

They depend on the reduced pressures and temperatures which are described below.

Reduced Properties

Reduced properties are used to correlate experimental data. Equations (3-39) and (3-40) can be used to calculate the reduced properties of a flow stream.

$$P_r = \frac{P}{P_c} \tag{3-16}$$

and

$$T_r = \frac{T}{T_c}, \tag{3-17}$$

where

P_r = reduced pressure,
P_c = critical pressure,
T_r = reduced temperature,
T_c = critical temperature.

The reduced pressure and temperature values are not truly a pressure and temperature but a ratio. Thus, it is a nondimensional term and does not take the unit "degrees."

Pseudo-critical properties allow one to evaluate gas mixtures. Equations (3-18) and (3-19) can be used to calculate the pseudo-critical properties:

$$P'_c = \sum y_i P_{ci} \tag{3-18}$$

and

$$T'_c = \sum y_i T_{ci}, \tag{3-19}$$

where

P'_c = pseudo-critical pressure,
T'_c = pseudo-critical temperature,
P_{ci} = critical pressure at component I,
T_{ci} = critical temperature at component i,
y_i = mole fraction of each component in the mixture,
$\sum y_i = 1$.

Pseudo-reduced properties are the same equations as for the reduced properties, except that pseudo-critical properties have been substituted for actual critical properties. Equations (3-20) and (3-21) are used to calculate the pseudo-reduced properties:

$$P_r^l = \frac{P}{P_c^l} \tag{3-20}$$

and

$$T_r^l = \frac{T}{T_c^l}. \tag{3-21}$$

Example 3-4: Calculate the pseudo-critical temperature and pressure for the following natural gas stream composition:

Component	Mole Fraction, y_i
Methane (C_1)	0.85
Ethane (C_2)	0.09
Propane (C_3)	0.04
n-Butane (nC_4)	0.06
	1.00

Solution:

Mole Fraction T_c, °R						
Component	**(y_i)**		**(T_{ci})**			**(y_i)(T_{ci})**
C_1	0.85	×	343.2		=	291.7
C_2	0.09	×	551.8		=	49.7
C_3	0.04	×	660.0		=	26.4
nC_4	0.02	×	765.0		=	15.2
				T_c	=	383.0 °R

Mole Fraction P_c/psia						
Component	**(Y_i)**		**(P_{ci})**			**(y_i)(P_{ci})**
C_1	0.85	×	667.9		=	567.7
C_2	0.09	×	707.0		=	63.6
C_3	0.04	×	617.0		=	24.7
nC_4	0.02	×	549.8		=	11.0
				P_c	=	667.0 psia

Example 3-5: Calculate the volume of 1 lb mole of the natural gas stream given in the previous example at 120 °F and 1500 psia.

Solution:

(1) Calculate the pseudo-reduced temperature and pressure:

$$T_r^l = \frac{T}{T_c^l} = \frac{(460+120)}{383} = 1.51,$$

$$P_c^l = \frac{P}{P_c^l} = \frac{1500}{667} = 2.25.$$

(2) From the Z factor correlation with $P_r' = 2.25$ and $T_r' = 1.51$, read $Z = 0.811$ from Figure 3-3.

(3) Using the most useful form of the compressibility equation of state

$$V = \frac{ZnRT}{P}$$

$$= \frac{(0.811)\,(1)\,(10.73)\,(120+460)}{1500}$$

$$= 3.37\,\text{ft}^3.$$

Figure 3-7 allows one to determine the pseudo-critical properties when gas composition is not known. The correlation is based on gas gravity. Use the condensate well gas curve for wet gas systems and retrograde gas systems. Use the miscellaneous gases curve for dry gas systems.

The compressibility factor for a natural gas can be approximated from Figures 3-8 through 3-13, which are from the *Engineering Data Book* of the Gas Processor Suppliers Association.

The Wichert and Aziz equations can be used to account for the effect of acid gases. These equations utilize an adjustment of the P_c and T_c and a correction parameter, ε, found from Figure 3-14. The correction parameter is used to adjust the pseudo-criticals. The adjusted values are then used to find the reduced pressure and temperature. The adjustment equations are

$$P_c'' = \frac{P_c' T_c''}{T_c' + B\,(1-B)\,\varepsilon} \tag{3-22}$$

and

$$T_c'' = T_c' - \varepsilon, \tag{3-23}$$

where

T_c'' and P_c'' = adjusted critical values,
T_c' and P_c' = pseudo-criticals,
B = mol fraction of H_2S in gas,
ε = correction factor from Figure 3-13, or
$= 120(A^{0.9} - A^{1.6}) + 15(B^{0.5} - B^4)$,
A = mol fraction of H_2S plus CO_2 in gas.

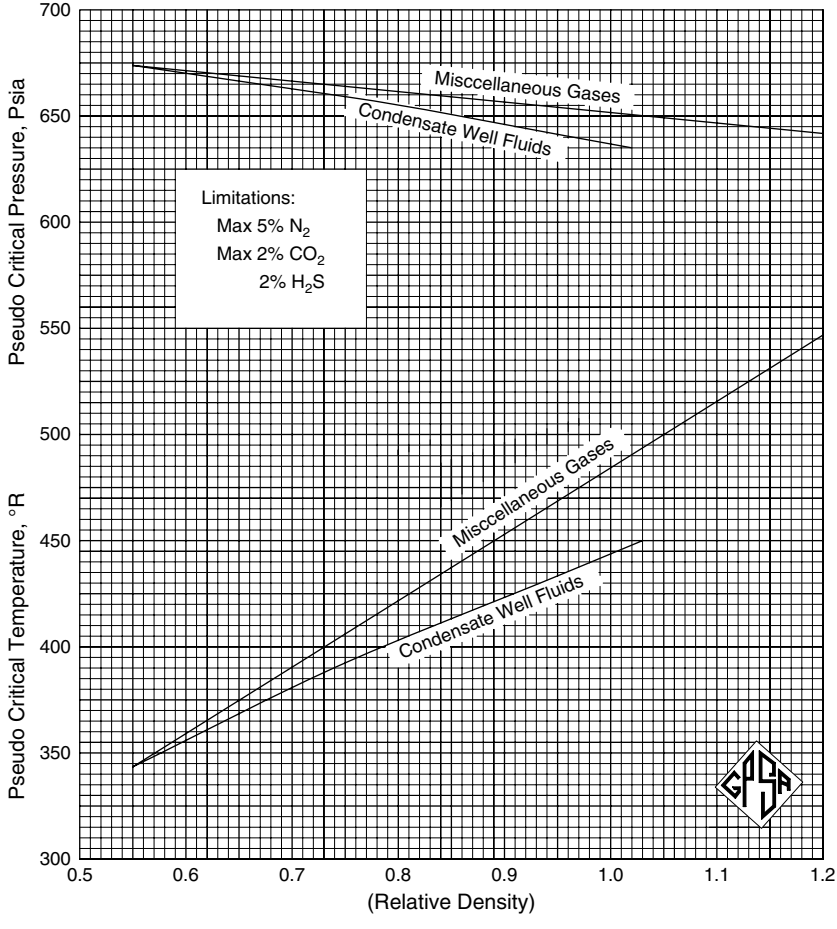


Figure 3-7. Pseudo-critical properties of gases given their gas gravities.

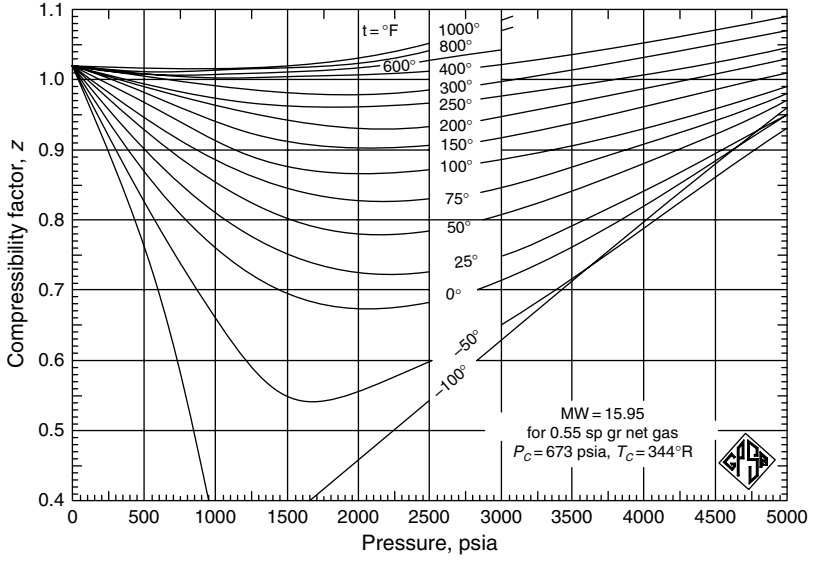


Figure 3-8. Compressibility factor for specific gravity = 0.55 gases. (Courtesy of GPSA *Engineering Data Book*.)

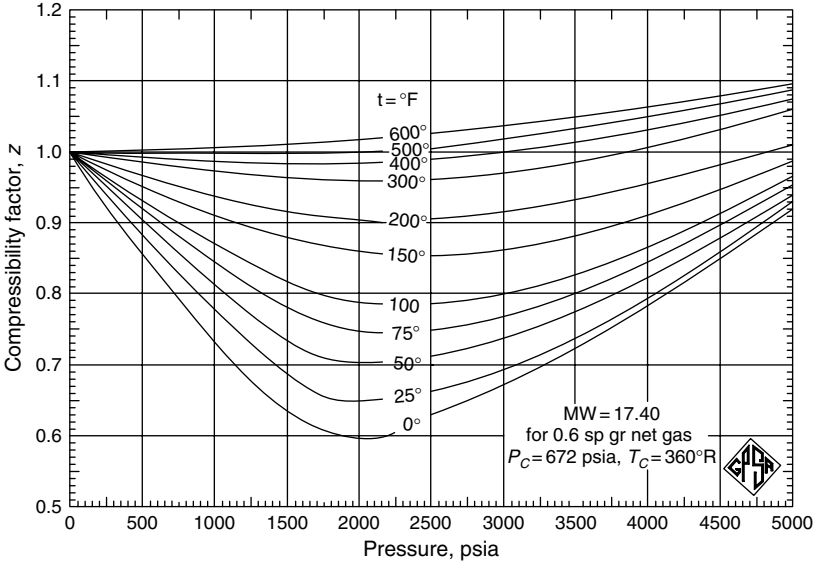


Figure 3-9. Compressibility factor for specific gravity = 0.6 gases. (Courtesy of GPSA *Engineering Data Book*.)

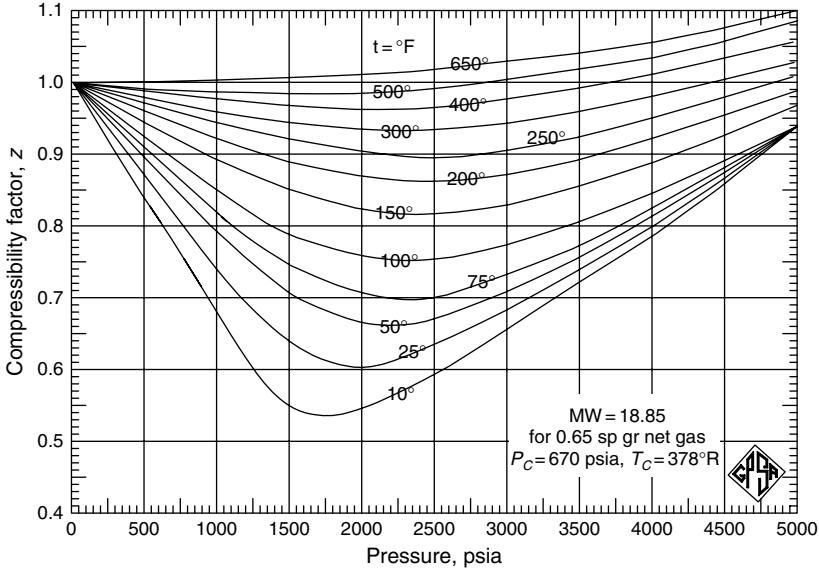


Figure 3-10. Compressibility factor for specific gravity = 0.65 gases. (Courtesy of GPSA *Engineering Data Book*.)

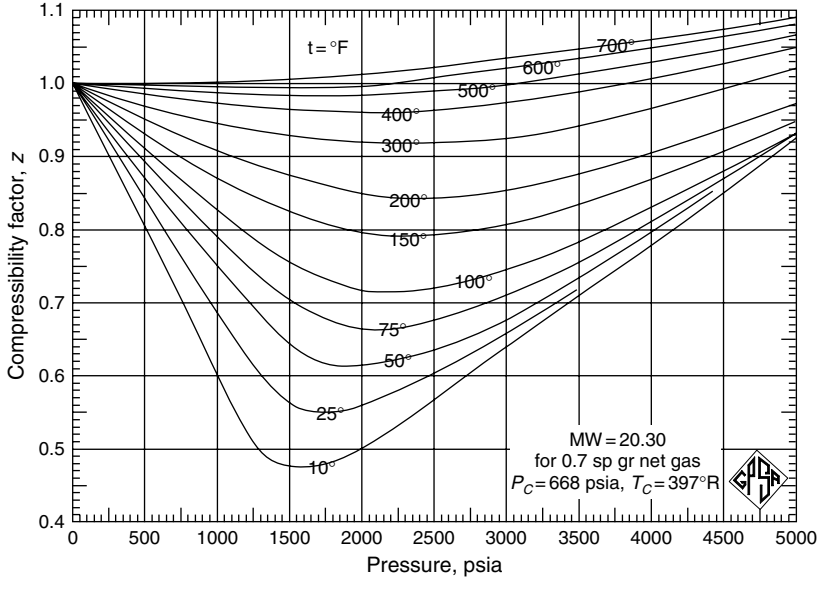


Figure 3-11. Compressibility factor for specific gravity = 0.7 gases. (Courtesy of GPSA *Engineering Data Book*.)

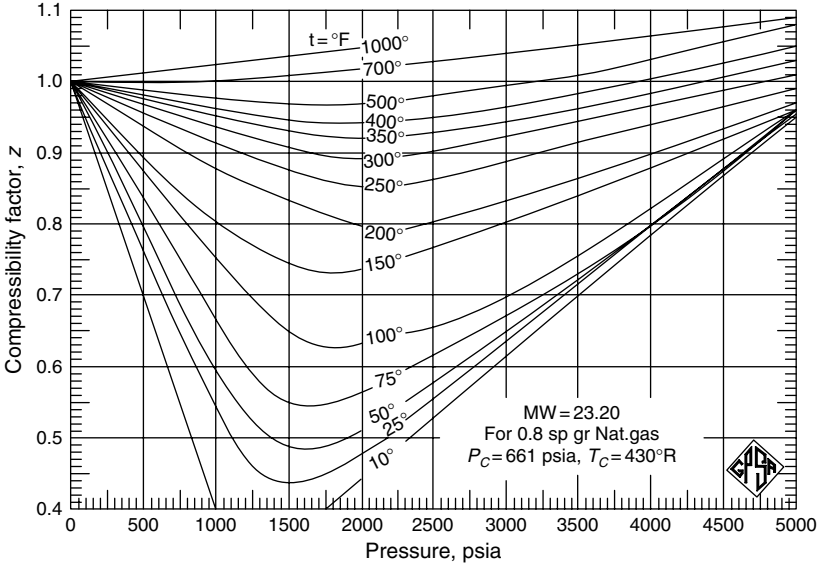


Figure 3-12. Compressibility factor for specific gravity = 0.8 gases. (Courtesy of GPSA *Engineering Data Book*.)

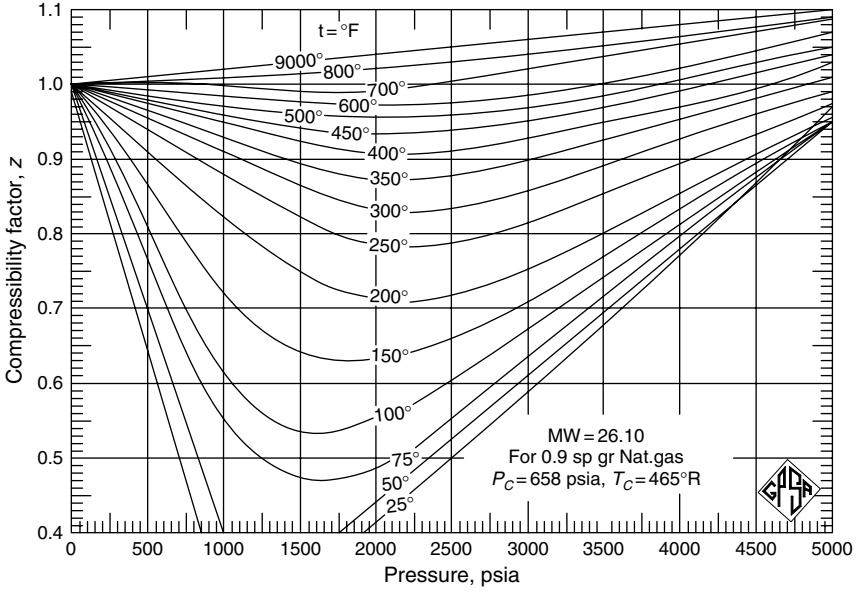


Figure 3-13. Compressibility factor for specific gravity = 0.9 gases. (Courtesy of GPSA *Engineering Data Book*.)

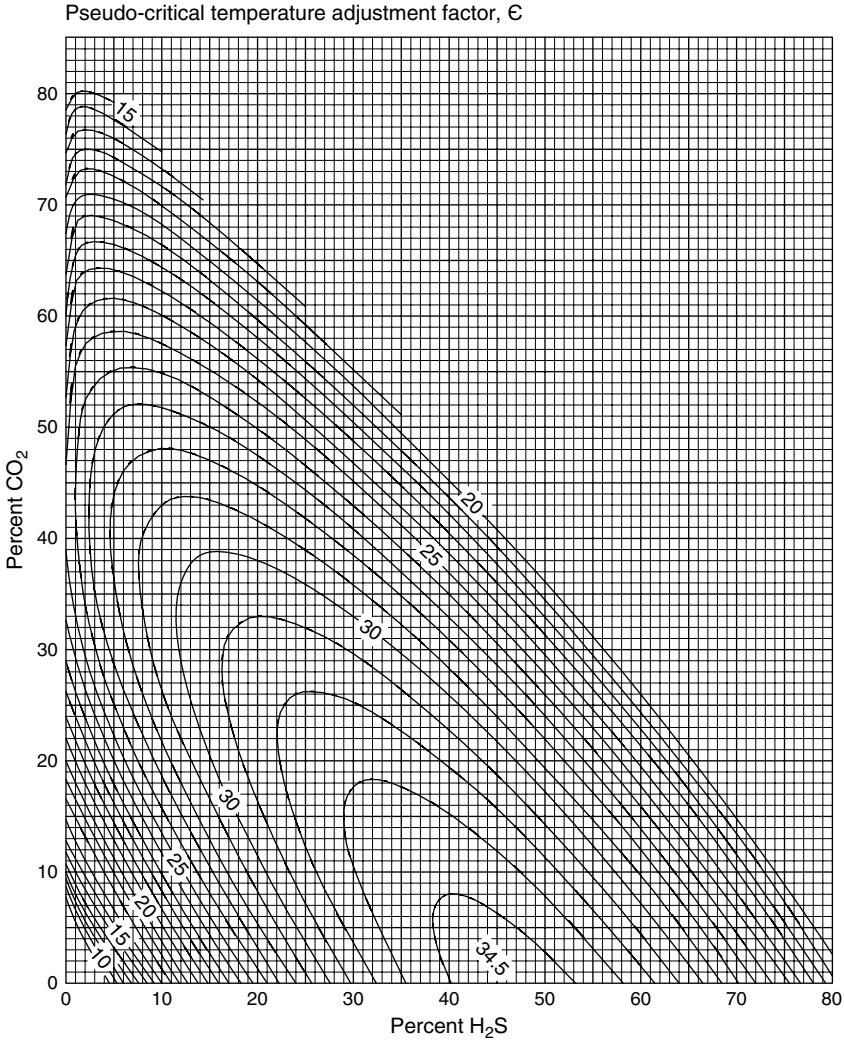


Figure 3-14. Correction factor chart for sour gases.

Example 3-6: A sour natural gas has the following composition. Determine the compressibility factor for the gas at 100°F and 1000 psia:

Component	Mole Fraction (y_i)		P_c, psia (P_{ci})			(y_i)(P_{ci})
CO_2	0.10	×	1071		=	107.1
H_2S	0.20	×	1300		=	250.0
N_2	0.05	×	493.1		=	24.7
CH_4	0.60	×	666.4		=	399.8
C_2H_6	0.05	×	706.5		=	35.3
				P'_c	=	826.9

Component	Mole Fraction (y_i)		T_c,° R, (T_{ci})			(y_i)(T_{ci})
CO_2	0.10	×	547.5		=	54.8
H_2S	0.20	×	672.1		=	134.4
N_2	0.05	×	227.2		=	11.4
CH_4	0.60	×	343.0		=	205.8
C_2H_6	0.05	×	549.6		=	27.5
				T'_c	=	433.9

Solution:

(1) Read correction parameter, $\varepsilon = 29.8$ from Figure 3-14 at $H_2S = 20\%$ and $CO_2 = 10\%$.

(2) Adjusted pseudo-critical temperature is

$$T_c^{ll} = T_c^{l} - \varepsilon = 433.9 - 29.8 = 404.1^{\circ}\text{R}.$$

(3) Adjusted pseudo-critical pressure is

$$P_c^{ll} = \frac{P_c^{ll} T_c^{ll}}{T_c^{l} + B(1-B)\varepsilon}$$

$$= \frac{(826.9)(404.1)}{433.9 + (0.2)(1-0.2)(29.8)}$$

$$= 761.7 \text{ psia}.$$

(4) Pseudo-reduced temperatures and pressure are

$$T_r^l = \frac{560}{404.1} = 1.385,$$

$$P_r^l = \frac{1000}{761.7} = 1.313.$$

(5) The Z factor = 0.831 from correlation chart.

Liquid Density and Specific Gravity

The specific gravity of a liquid is the ratio of the density of the liquid at 60 °F to the density of pure water. API gravity is related to the specific gravity by the following equation:

$$^\circ\text{API} = (141.5/\text{SG}) - 131.5, \tag{3-24}$$

where SG = specific gravity of a liquid (water = 1).

Liquids are relatively incompressible when compared to gases. The calculation of the properties of liquid mixtures is complicated by two factors: the presence of high vapor pressure components like methane and ethane, and the shrinkage in volume upon mixing two hydrocarbon liquids. Both effects can be understood by remembering that liquid are made up of molecules of different sizes and shapes, possessing different amounts of energy.

The complexity of these calculations depends on the presence of methane, ethane, CO_2, sulfur compounds, and nonparaffins. Specific gravity of a liquid is the ratio of the density of the liquid to the density of water:

$$\text{SG} = \frac{\rho_0}{\rho_w}, \tag{3-25}$$

where SG = specific gravity of liquid.

The density of crude oil is sometimes shown in °API. This term is defined by the equation

$$\text{SG} = \frac{141.5}{131.5 + {}^\circ\text{API}} \tag{3-26}$$

or

$$^\circ\text{API} = \frac{141.5}{\text{SG}} - 131.5, \tag{3-27}$$

where SG = specific gravity of a liquid (water = 1).

By definition, 1 API barrel = 42 U.S. gallons at 60 °F:

$$1 \text{ API bbl} = 42 \text{ U.S. gallons} = 35 \text{ U.K. (Imperial) gallons}$$
$$= 5.61 \text{ ft}^3 = 0.159 \text{ m}^3 = 159 \text{ l}.$$

In most calculations the specific gravity of liquids is normally referenced to actual temperature and pressure conditions. Figure 3-15 can be used to approximate how the specific gravity of a liquid decreases with increasing temperature, assuming no phase changes. In most practical pressure drop calculations associated with production facilities, the

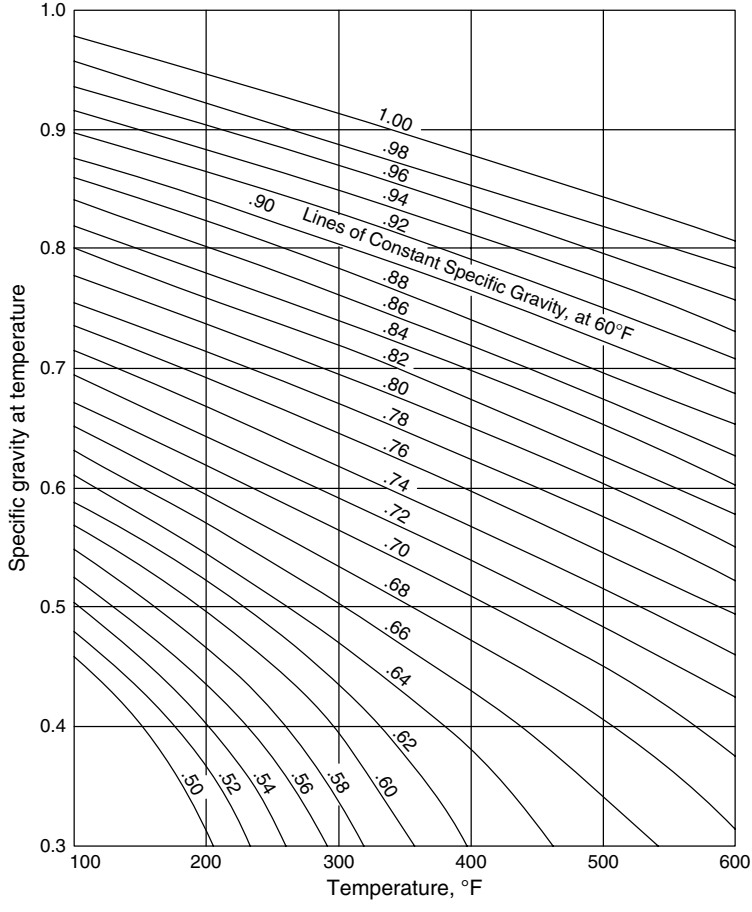


Figure 3-15. Approximate specific gravity of petroleum fractions. (Courtesy of GPSA *Engineering Data Book*.)

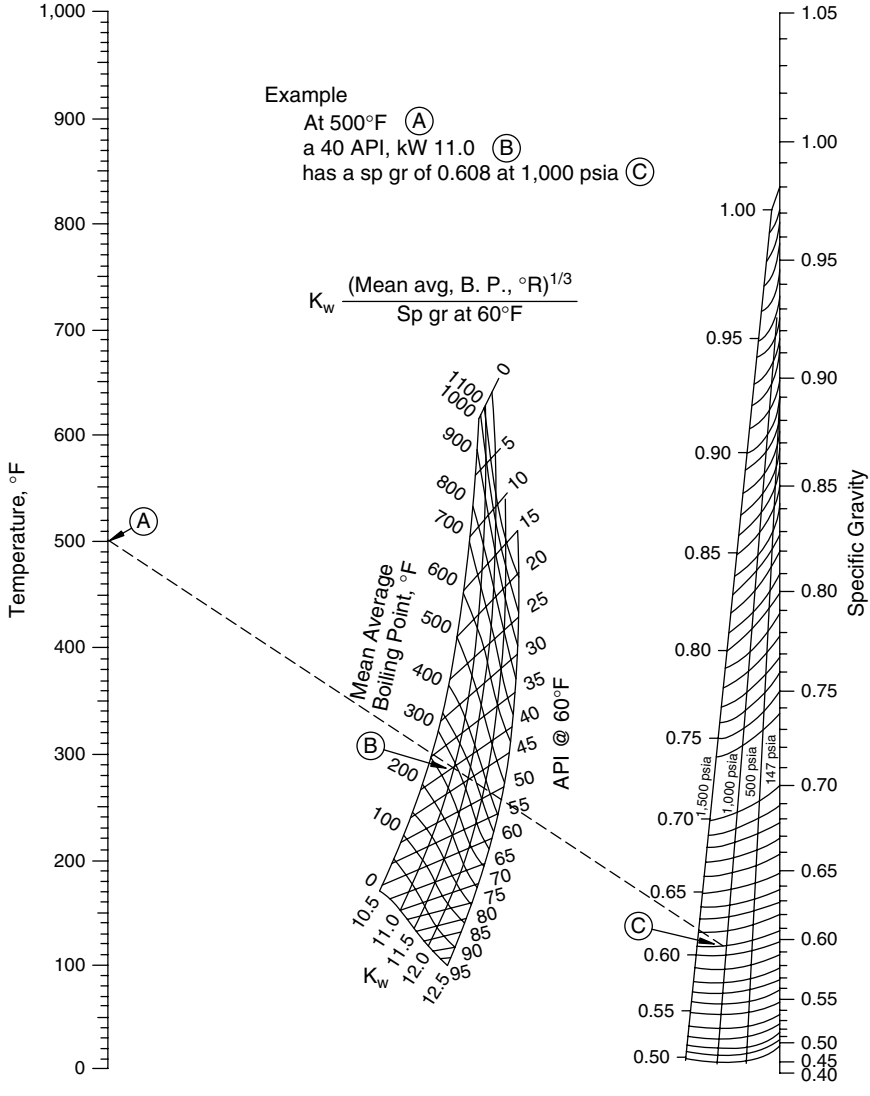


Figure 3-16. Specific gravity of petroleum fractions. (Courtesy of Petroleum Refinery: Ritter, Lenory, and Schweppe, 1958.)

difference in specific gravity caused by pressure changes will not be severe enough to be considered if there are no phase changes.

For hydrocarbons, which undergo significant phase changes, Figure 3-16 can be used as an approximation of the specific gravity at a given pressure and temperature, once the API gravity of the liquid is known.

It should be pointed out that both Figures 3-15 and 3-16 are approximations only for the liquid component. Where precise calculation is required for a hydrocarbon, it is necessary to consider the gas that is liberated with decreasing pressure and increasing temperature. Thus, if a hydrocarbon is heated at constant pressure, its specific gravity will increase as the lighter hydrocarbons are liberated. The change in the molecular makeup of the fluid is calculated by a "flash calculation," which is described in more detail later in this chapter.

Viscosity

This property of a fluid indicates its resistance to flow. It is an important property used in flow equations and sizing of process equipment. It is a dynamic property in that it can be measured only when the fluid is in motion. Viscosity is a number that represents the drag forces caused by the attractive forces in adjacent fluid layers. It might be considered as the internal friction between molecules, separate from that between the fluid and the pipe wall.

The correlations that follow have proven useful and reliable for most calculations. The unit of viscosity in the petroleum industry is the centipoise, where

$$\begin{aligned} 1 \text{ centipoise (cp)} &= 0.01 \text{ dynes/cm}^2 \\ &= 0.000672 \text{ lbm/ft-sec.} \end{aligned}$$

There are two expressions of viscosity, absolute (or dynamic) viscosity, μ, and kinematic viscosity. These expressions are related by the following equation:

$$\gamma = \frac{\mu}{\rho}, \tag{3-28}$$

where

μ = absolute viscosity, centipoise,
γ = kinematic viscosity, centistokes,
ρ = density, g/cm^3,
1 centistoke = $0.01 \text{ cm}^2/\text{s}$
$= 1.0 \times 10^{-6} \text{ m}^2/\text{s}$.

Fluid viscosity changes with temperature. Liquid viscosity decreases with increasing temperature, whereas gas viscosity decreases initially with increasing temperature and then increases with further increasing temperature.

Gas Viscosity

Figure 3-17 can be used to estimate the viscosity of a hydrocarbon gas at various conditions of temperature and pressure if the specific gravity of the gas at standard conditions is known. It is useful when the gas

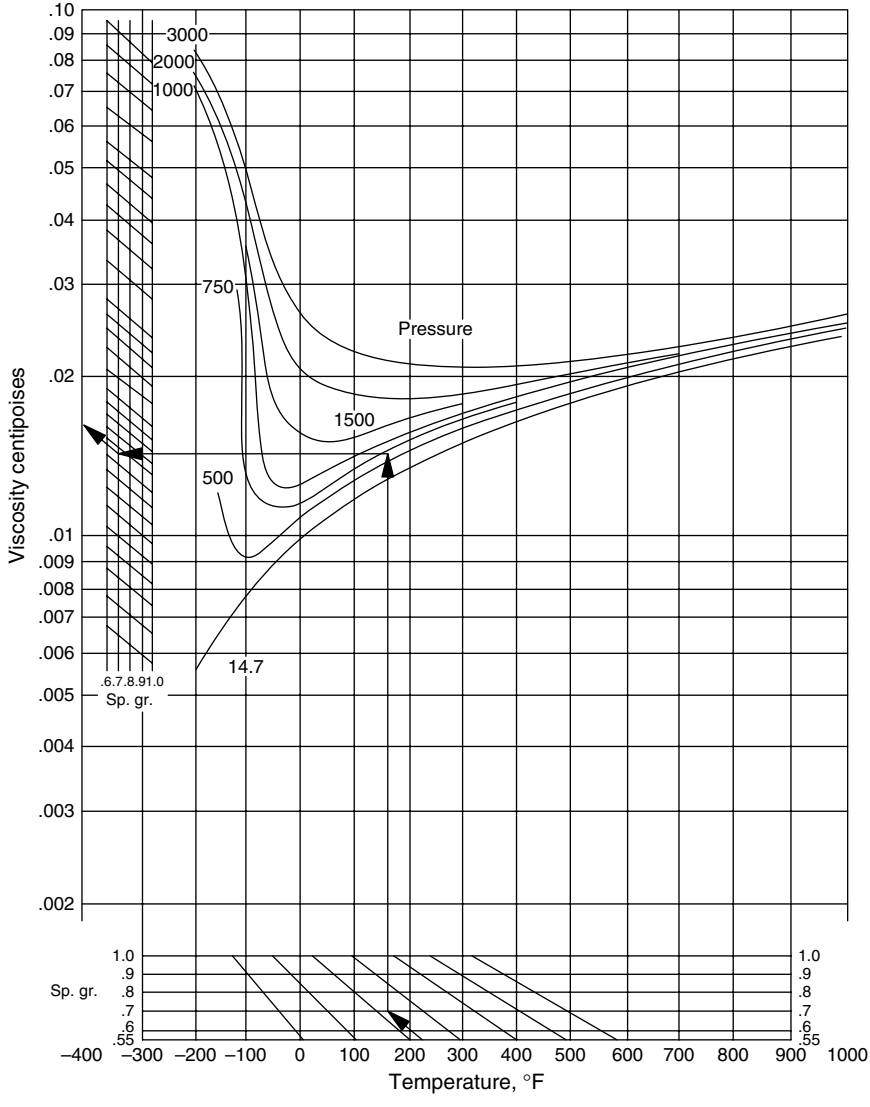


Figure 3-17. Hydrocarbon gas viscosity. (Courtesy of GPSA *Engineering Data Book*.)

composition is not known. It does not make corrections for H_2S, CO_2, and N_2. It is useful for determining viscosities at high pressure. Unfortunately, it is an approximate correlation and thus yields less accurate results than other correlations, but for most engineering calculations yields results within acceptable limits. When compared to liquid viscosity, gas viscosity is very low, which indicates the relatively large distances between molecules.

Liquid Viscosity

The best way to determine the viscosity of a crude oil at any temperature is by measurement. If the viscosity is known at only one temperature, Figure 3-18 can be used to determine the viscosity at another temperature by striking a line parallel to that for crudes "A," "C," and "D." Care must be taken to assure that the crude does not have its pour point within the temperature range of interest. If it does, its temperature-viscosity relationship may be as shown for crude "B."

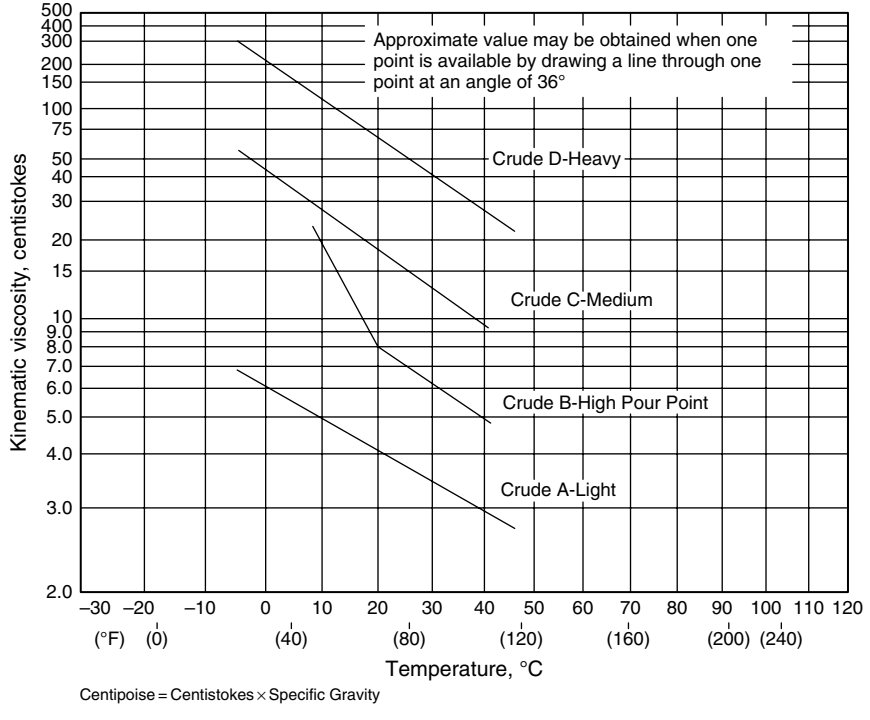


Figure 3-18. Typical viscosity-temperature curves for crude oils. (Courtesy of ASTM D-341.)

Solid phase high-molecular-weight hydrocarbons, otherwise known as paraffins, can dramatically affect the viscosity of the crude sample. The cloud point is the temperature at which paraffins first become visible in a crude sample. The effect of the cloud point on the temperature-viscosity curve is shown for crude "B" in Figure 3-18. This change in the temperature-viscosity relationship can lead to significant errors in estimation. Therefore, care should be taken when one estimates viscosities near the cloud point.

The pour point is the temperature at which the crude oil becomes a solid and ceases to flow, as measured by a specific ASTM procedure (D97). Estimations of viscosity near the pour point are highly unreliable and should be considered accordingly.

In the absence of any laboratory data, correlations exist that relate viscosity and temperature, given the oil gravity. The following equation relating viscosity, gravity, and temperature was developed by Beggs and Robinson after observing 460 oil systems:

$$\mu = 10^{x} - 1, \tag{3-29}$$

where

μ = oil viscosity, cp,
T = oil temperature, °F,
$x = y(T)^{-1.163}$,
$y = 10^{z}$,
$z = 3.0324 - 0.02023\text{G}$,
G = oil gravity, °F.

The data set from which this relationship was obtained included a range of between 16 and 58 °API and 70 to 295 °F. It has been the author's experience that the correlation tends to overstate the viscosity of the crude oil when dealing in temperature ranges below 100 to 150 °F. Figure 3-19 is a graphical representation of another correlation.

Oil-Water Mixture Viscosity

The viscosity of produced water depends on the amount of dissolved solids in the water as well as the temperature, but for most practical situations it varies from 1.5 to 2 centipoise at 50 °F, 0.7 to 1 centipoise at 100 °F, and 0.4 to 0.6 centipoise at 150 °F.

When an emulsion of oil and water is formed, the viscosity of the mixture may be substantially higher than either the viscosity of the oil or that of the water taken by themselves. Figure 3-20 shows some experimental

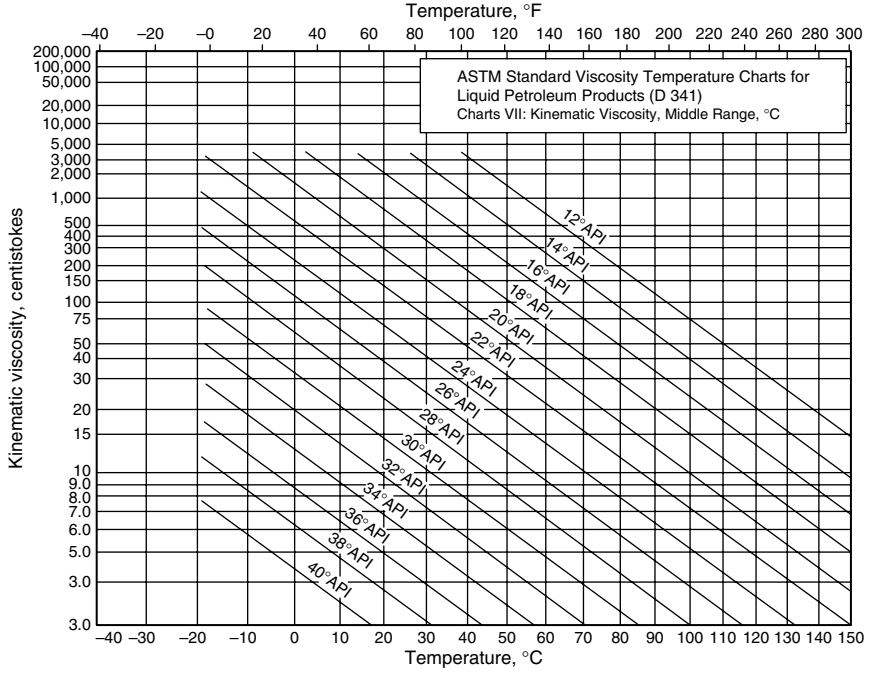


Figure 3-19. Oil viscosity vs. gravity and temperature. (Courtesy of Paragon Engineering Services, Inc.)

data for a mixture of produced oil and water taken from a south Louisiana field. Produced oil and water were mixed vigorously by hand, and viscosity was measured for various percentages of water. For 70% water cut, the emulsion began to break before viscosity readings could be made, and for water cuts greater than this, the oil and water began to separate as soon as the mixing was stopped. Thus, at approximately 70% water cut, it appears as if oil ceases to be the continuous phase and water becomes continuous.

The laboratory data plotted in Figure 3-20 agree closely with the modified Vand's equation assuming a 70% breakover point. This equation allows one to determine the effective viscosity of an oil-water mixture and is written in the form

$$\mu_{\text{eff}} = \left(1 + 2.5\varnothing + 10\varnothing^2\right)\mu_c, \tag{3-30}$$

where

μ_{eff} = effective viscosity,
μ_c = viscosity of the continuous phase,
$\varnothing$ = volume fraction of the discontinuous phase.

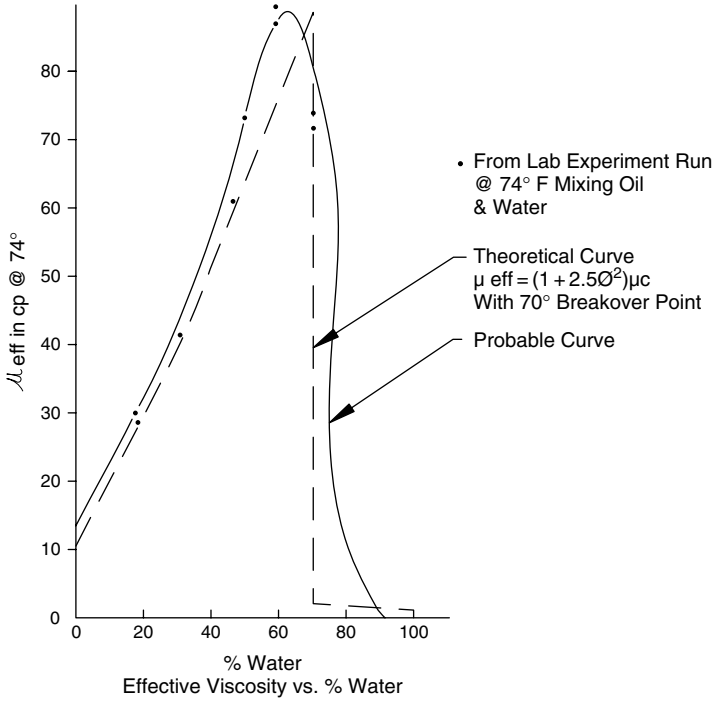


Figure 3-20. Effective viscosity of an oil–water mixture.

Phase Behavior

Before studying the properties of gases and liquids, we need to understand the relationship between the phases. Phase defines any homogeneous and physically distinct part of a system that is separated from other parts of the system by definite bounding surfaces:

- Solid (ice),
- Liquid (liquid water),
- Vapor (water vapor).

The energy possessed by any substance depends on its phase. Solids have a definite shape and are hard to the touch. They are composed of molecules with very low energy that stay in one place even though they vibrate. Liquids have a definite volume but no definite shape. Liquids assume the shape of the container but will not necessarily fill that container. Liquid molecules possess more energy than a solid (allows movement from place to another). By virtue of the energy there is more space between molecules, and liquids are less dense than solids. Vapors

do not have a definite volume or shape and will fill a container in which they are placed. Vapor molecules possess more energy than liquids (very active) and are less dense than liquids.

Factors important to the physical behavior of molecules are

- Pressure:
  - Reflection of the number of times the molecules of a gas strike the walls of its container,
  - Reflection of the number of molecules present.
- Temperature:
  - Reflection of the average kinetic energy of the molecules of the material,
  - As heat is added, the kinetic energy of the molecules is increased and thus increases the temperature.
- Intermolecular forces:
  - Forces of attraction and repulsion between molecules,
  - Forces change as the distance between molecules changes,
  - Attractive force increases as the distance between the molecules decreases until the molecules get so close together that their electronic fields overlap.

Our primary concern is the difference in energy level between phases. Energy is added to melt a solid to form a liquid. Additional energy will cause the liquid to vaporize. One needs to know the phase or phases that exist at given conditions of pressure, volume, and temperature so as to determine the corresponding energy level. To do this, we separate substances into three classifications:

- Pure substance (single-component systems),
- Two substances,
- Multicomponent.

Phase diagrams illustrate the phase that a particular substance will take under specified conditions of pressure, temperature, and volume.

System Components

Natural gas systems are composed primarily of the lighter alkane series of hydrocarbons, with methane (CH_4) and ethane (C_2H_6) comprising 80% to 90% of the volume of a typical mixture. Methane and ethane exist as gases at atmospheric conditions.

Propane (C_3H_8), butane (n-C_4H_{10} and i-C_4H_{10}), and heavier hydrocarbons may be extracted from the gas system and liquefied for transportation and storage. These are the primary components of liquefied petroleum gas, or *LPG*.

The intermediate-weight hydrocarbons (pentane through decane) exist as volatile liquids at atmospheric conditions. These components are commonly referred to as pentanes-plus, condensate, natural gasoline, and natural gas liquids (NGL).

Natural gas systems can also contain non-hydrocarbon constituents, including hydrogen sulfide (H_2S), carbon dioxide (CO_2), nitrogen (N_2), and water vapor. These constituents may occur naturally in gas reservoirs, or they may enter the system as contaminants during production, processing, and transportation. In addition, operators may intentionally add odorants, tracers (such as helium), or other components.

Dry, or lean, natural gas systems have high concentrations of the lighter hydrocarbons (methane and ethane), while wet, or rich, gas systems have higher concentrations of the intermediate-weight hydrocarbons. Lean gases burn with a low air-to-gas ratio and display a colorless to blue or yellow flame, whereas rich gases require comparatively higher amounts of air for combustion and burn with an orange flame. Intermediate-weight hydrocarbons may condense from rich gases upon cooling.

Table 3-7 shows typical compositions for a lean gas and a rich gas, illustrating the wide diversity in composition that can exist between different types of gas systems.

Single-Component Systems

A pure component of a natural gas system exhibits a characteristic *phase behavior*, as shown in Figure 3-21. Depending on the component's pressure and temperature, it may exist as a vapor, a liquid, or some equilibrium combination of vapor and liquid.

Table 3-7
Typical Compositions of Natural Gas Systems

	Lean Gas, Mole Percent	Rich Gas Component, Mole Percent
Methane	86.00	68.51
Ethane	5.81	9.05
Propane	3.58	5.34
Butanes	1.72	4.48
Pentanes and heavier	0.72	12.50
CO_2	0.10	0.01
N_2	2.00	0.11
H_2S	0.07	0.00
Total	100.00	100.00

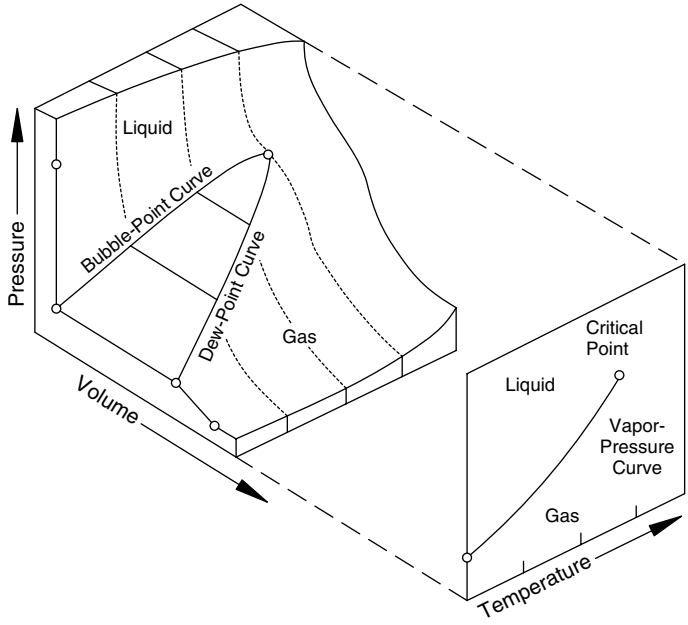


Figure 3-21. Characteristic pressure-volume-temperature phase behavior of a pure component.

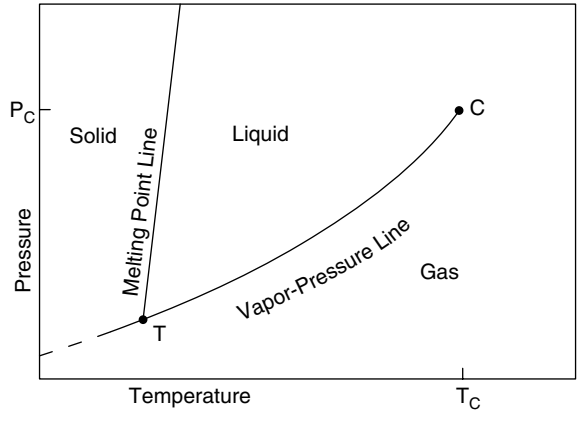


Figure 3-22. Pressure/temperature behavior of a pure component.

Figure 3-22 shows the pressure/temperature behavior of a pure, single-component system. (Note that, in addition to the vapor and liquid phases, this particular component may also exist as a solid; the most common example of this type of three-phase system, of course, is water.)

The *vapor pressure line* in Figure 3-22 divides the liquid region from the vapor region. This line represents the locus of temperatures and pressures at which vapor and liquid exist in equilibrium.

The critical point (C) is the point at which the intensive, or mass-independent, properties of the liquid and vapor phases become identical. This point marks the end of the vapor pressure line and identifies the critical pressure and critical temperature of the pure component. The triple point (T) is the point where liquid, gas, and solid phases exist at equilibrium.

Table 3-8 lists critical pressures and critical temperatures, along with molecular weights, of some pure components present in many natural gas systems.

Multicomponent Systems

In reality, natural gas systems are not pure substances. Rather, they are mixtures of various components, with phase behavior characteristics that

Table 3-8
Critical Properties and Molecular Weights of Typical Natural Gas Components

Component	P_c, psia	T_c, °R	Molecular Weight
Methane (CH_4)	673.1	343.2	16.04
Ethane (C_2H_6)	708.3	549.9	30.07
Propane (C_3H_8)	617.4	666.0	44.09
i-Butane (*i*-C_4H_{10})	529.1	734.6	58.12
n-Butane (*n*-C_4H_{10})	550.1	765.7	58.12
i-Pentane (*i*-C_5H_{12})	483.5	829.6	72.15
n-Pentane (*n*-C_5H_{12})	489.8	846.2	72.15
n-Hexane (*n*-C_6H_{14})	440.1	914.2	86.17
n-Heptane (*n*-C_7H_{16})	395.9	972.4	100.2
n-Octane (*n*-C_8H_{18})	362.2	1024.9	114.2
n-Nonane (*n*-C_9H_{20})	334.0	1073.0	128.3
n-Decane (*n*-$C_{10}H_{22}$)	312.0	1115.0	142.3
Air	547.0	239.0	28.97
Carbon dioxide (CO_2)	1070.2	547.5	44.01
Helium (He)	33.2	9.5	4.003
Hydrogen (H)	189.0	59.8	2.016
Hydrogen sulfide (H_2S)	1306.5	672.4	34.08
Nitrogen (N_2)	492.2	227.0	28.02
Oxygen (O_2)	736.9	278.6	32.00
Water (H_2O)	3209.5	1165.2	18.02

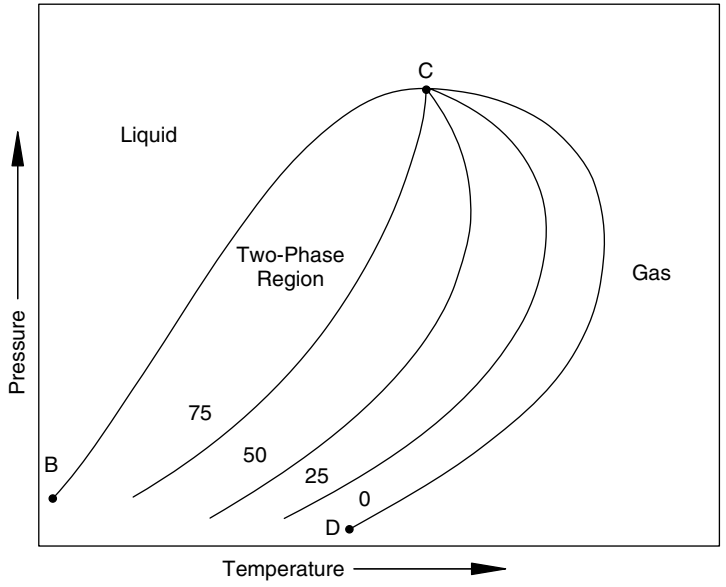


Figure 3-23. Phase envelope of a multicomponent mixture.

differ from those of a single-component system. Instead of having a vapor pressure curve, a mixture exhibits a *phase envelope*, as shown in Figure 3-23.

The phase envelope (curve BCD in Figure 3-23) separates the liquid and gas phases. The area within this envelope is called the *two-phase region* and represents the pressure and temperature ranges at which liquid and gas exist in equilibrium.

The upper line of the two-phase region (curve BC) is the *bubble-point line*. This line indicates where the first bubble of vapor appears when the pressure of the liquid phase mixture is lowered at constant temperature, or when the temperature increases at constant pressure.

The lower section of the phase envelope (curve CD) is the *dewpoint line*. When the pressure of a mixture in the gaseous phase is changed at constant temperature, or when the temperature is lowered at constant pressure, the first drop of liquid forms on this line. (Note that for certain temperatures on this particular envelope, there are two dew points.) The bubble-point line and the dewpoint line meet at the critical point (*C*).

The highest pressure in the two-phase region is called the *cricondenbar*, while the highest temperature in the two-phase region is called the

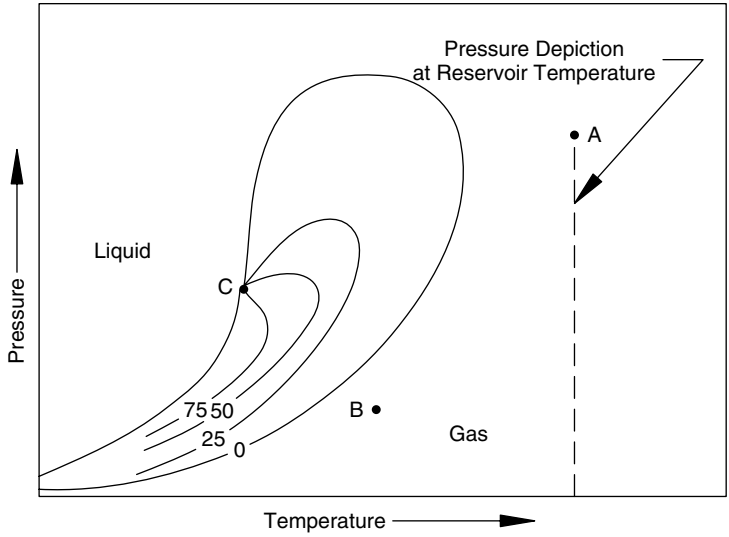


Figure 3-24. Lean or dry natural gas where separator operating pressure and temperature fall outside the phase envelope.

cricondentherm. The cricondentherm for a rich gas is greater than the cricondentherm for a lean gas.

Lean Gas Systems

For most natural gas systems, the reservoir temperature is higher than the cricondentherm. Figure 3-24 illustrates this normal situation. As the reservoir is produced, its pressure declines at constant temperature. There are no phase changes in the reservoir.

Figure 3-24 also shows the pressure and temperature at surface conditions for the produced gas (point B). For a lean or dry gas, the usual transition from reservoir conditions to surface conditions does not intersect the phase envelope. The gas is produced without a phase change.

Rich Gas Systems

Figure 3-25 illustrates another common situation. The gas is sufficiently rich in intermediates and heavier components so that surface conditions fall within the phase envelope. Both liquid and vapor are produced. Such

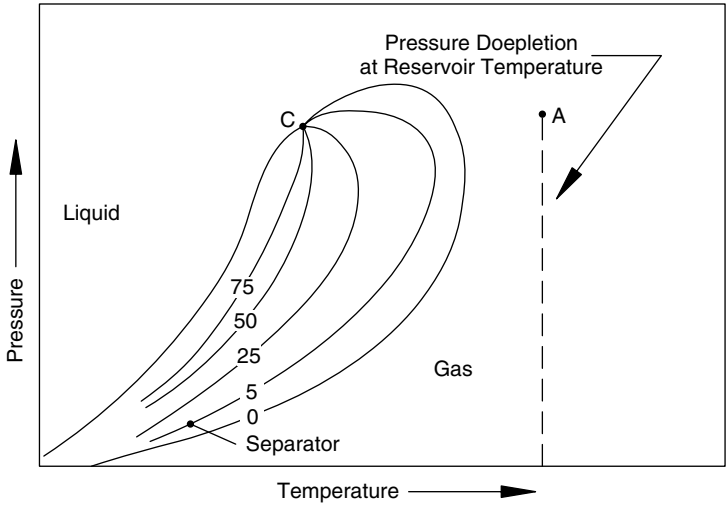


Figure 3-25. Rich natural gas where the separator operating pressure and temperature fall inside the phase envelope.

a gas is called a rich, or wet, gas. (Note that the term "wet," in this context, does not refer to the water content of the gas.) It would be normal in this system for liquid to begin forming near the surface or, even more commonly, in the surface equipment. This liquid, called *condensate*, may be responsible for a significant portion of the well's economic benefits. Liquid content, or *yield*, is normally expressed in terms of condensate volume per volume of gas (e.g., STB/MSCF or m_3/m_3).

Retrograde Systems

In some cases, the gas composition and the reservoir pressure and temperature may result in *retrograde* behavior. If the reservoir temperature is less than the cricondentherm but greater than the critical temperature, and the reservoir pressure is initially above the phase envelope, the reservoir fluid enters the two-phase region by forming liquid as the pressure decreases. As pressure decreases further, the liquid revaporizes and the mixture again enters the gas phase region. Retrograde behavior is shown in Figure 3-26 by the line drawn from point A.

Although most reservoirs that produce condensate are simple wet gas reservoirs, many of them are erroneously called retrograde reservoirs. True retrograde reservoirs have the characteristic of forming liquid as

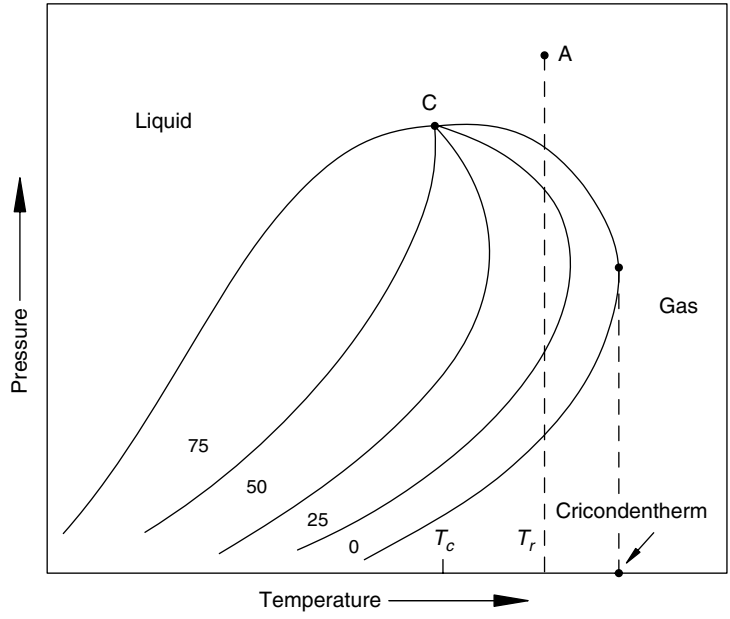


Figure 3-26. Retrograde gas where the reservoir temperature is less than the cricondentherm and greater than the critical temperature.

pressure decreases at constant temperature. Retrograde reservoirs can build up a liquid saturation in the reservoir, whereas for a wet gas reservoir, properly designed tubing strings do not allow this saturation build-up to occur.

To determine whether a natural gas system is wet or retrograde, we can use a windowed *PVT* (pressure/volume/temperature) cell to observe its phase behavior over a range of pressures and temperatures.

Application of Phase Envelopes

Proper analysis of many petroleum problems requires knowledge about the phase envelope. Behavior of a reservoir fluid during production is determined by the shape of its phase diagram and the position of its critical point as shown in Figure 3.27.

Reservoirs can be characterized by the shape of the reservoir fluid phase envelope and the initial conditions of pressure and temperature as described in the following sections.

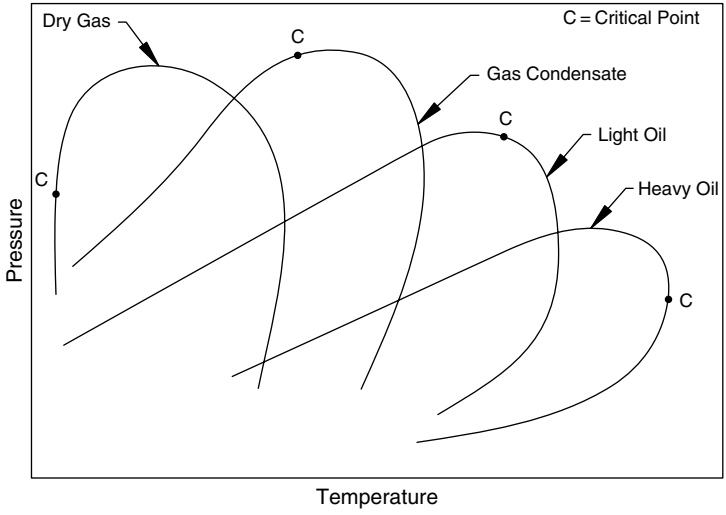


Figure 3-27. Characteristic phase envelope for four reservoirs.

Black Oil Reservoir

Phase Diagram Characteristics

Black oil reservoirs are also known as low-shrinkage crude oil. Its phase diagram characteristics are shown in Figure 3-28. Its temperature is less than the critical temperature. The reservoir is initially undersaturated and could dissolve more gas if gas were present. No gas will form in the reservoir until the pressure reaches the bubble point, at which point it becomes saturated and contains as much dissolved gas as it can hold. Any reduction in pressure will release gas to form a free gas phase in the reservoir. Additional gas evolves from the oil as it moves from the reservoir, through the wellbore, to the surface, thus causing some shrinkage of the oil. Separator conditions lie well within the phase envelope, indicating that a relatively large amount of liquid arrives at the surface.

Field Characteristics

Black oil reservoirs exhibit an initial producing gas-oil ratio of 2000 SCF/STB or less. Producing gas-oil ratios will increase during production when reservoir pressure falls below bubble-point pressure. These reservoir's exhibit a stock-tank oil gravity of 35 °API or heavier. Stock-tank gravity will slightly decrease with time until late in the life of the

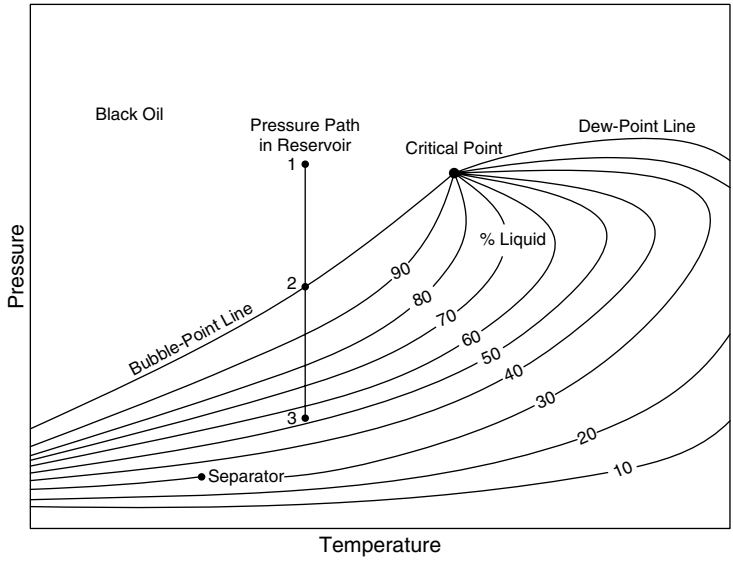


Figure 3-28. Phase diagram of a typical black oil reservoir with a line of isothermal reduction of reservoir pressure and surface separation conditions.

reservoir when it will increase. A stock-tank oil color that is very dark indicates the presence of heavy hydrocarbons, black with a greenish cast or brown.

Laboratory Analysis

Black oil reservoirs exhibit an initial oil formation volume factor of 2.0 reservoir BBL/STB or less, where the oil formation volume factor is the quantity of reservoir liquid in barrels required to produce one stock-tank barrel. The volume of oil at the bubble point (point 2 in the figure) shrinks by one-half or less on its trip to the stock tank. The laboratory-determined composition of heptanes plus will be higher than 30 mole percent, thus indicating a large quantity of heavy hydrocarbons in black oils.

Volatile Oil Reservoir

Phase Diagram Characteristics

Volatile oil reservoirs are also known as high-shrinkage crude oils. Their phase diagram characteristics are shown in Figure 3-29. They contain

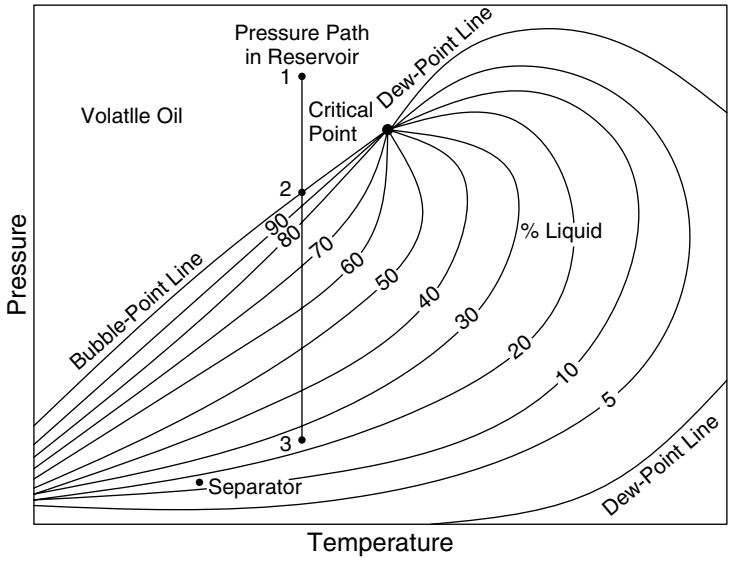


Figure 3-29. Phase diagram of a typical volatile oil reservoir with a line of isothermal reduction of reservoir pressure and surface conditions.

relatively fewer heavy molecules and more intermediates (defined as ethane through hexanes) than black oil reservoirs. Their temperature is less than the critical temperature. The temperature range covered by the phase envelope is somewhat smaller, but of more interest is the position of the critical point. Critical temperature is much lower than for a black oil and is close to reservoir temperature. Iso-Vols are not evenly spaced but are shifted upwards toward the bubble-point line. A small reduction in pressure below the bubble point causes the release of a large amount of gas in the reservoir. An Iso-Vol with a much lower percent liquid crosses the separator conditions—hence the name "volatile oil."

Field Characteristics

Volatile oil reservoirs exhibit an initial producing gas-oil ratio between 2000 and 3300 SCF/STB. Producing gas-oil ratio increases as production proceeds and reservoir pressure falls below the bubble-point pressure. The volatile oil reservoir exhibits a stock-tank oil gravity of 40°API or higher (contains fewer heavier hydrocarbon molecules). Stock-tank gravity will increase during production as reservoir pressure falls below the bubble point. A stock-tank oil is deeply colored (usually brown, orange, or sometimes green).

Laboratory Analysis

Volatile oil reservoirs exhibit an initial oil formation volume factor greater than 2.0 reservoir BBL/STB. The volume of oil at the bubble point (point 2 in the figure) shrinks by more than one-half, often three-quarters, on its trip to the stock tank. The laboratory-determined composition of heptanes plus will be between 12.5 to 30 mole percent. When the heptanes plus concentration is greater than 12.5 mole percent, the reservoir fluid is almost always liquid and exhibits a bubble point. When the heptanes plus concentration is less than 12.5 mole percent, the reservoir fluid is almost always gas and exhibits a dew point. Any exceptions to this rule normally do not meet the rule of thumb with respect to gravity or color.

Retrograde Gas Reservoir

Phase Diagram Characteristics

The retrograde gas reservoir is also known as a retrograde gas condensate reservoir. Its phase diagram characteristics are shown in Figure 3-30. Its temperature is between that of the critical and the cricondentherm.

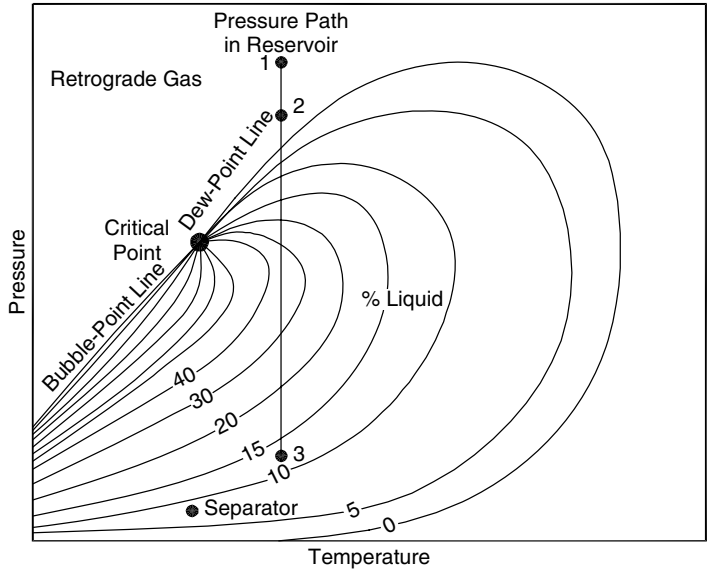


Figure 3-30. Phase diagram of a typical retrograde reservoir with a line of isothermal reduction of reservoir pressure and surface conditions.

Initially, the retrograde gas is totally gas in the reservoir (refer to point 1 in Figure 3-30). As reservoir pressure decreases, the retrograde gas exhibits a dew point (refer to point 2). As pressure is reduced, liquid condenses from the gas to form a free liquid in the reservoir. This liquid will normally not flow and cannot be produced. The liquid will come from the heaviest ends in the dense phase fluid. As the pressure declines below the dew point, liquid formation increases as long as the pressure is in the retrograde region. Below the retrograde region some vaporization occurs.

Field Characteristics

Retrograde gas reservoirs exhibit an initial producing gas-oil ratio (GOR) between 3300 and 150,000 SCF/STB. GORs as high as 150,000 SCF/STB have been observed. Gases with high gas-oil ratios (greater than 50,000 SCF/STB) have cricondentherms close to reservoir temperature and drop very little retrograde liquid in the reservoir. The reservoir fluid can be treated as if it were a wet gas. Producing gas-oil ratios for a retrograde gas will increase after production begins when reservoir pressure falls below the dewpoint pressure of the gas. Stock-tank liquid gravities are between 40 and 60 °API and increase as reservoir pressure falls below the dewpoint pressure. The liquid can be lightly colored, brown, orange, greenish, or water-white.

Laboratory Analysis

Retrograde gases exhibit a dew point when pressure is reduced at reservoir temperature. The heptanes plus fraction is less than 12.5 mole percent. An initial producing gas-oil ratio of 3300 to 5000 SCF/STB indicates a very rich retrograde gas, one that will condense sufficient liquid to fill 35% or more of the reservoir volume. Even this quantity of liquid seldom will flow and normally cannot be produced. The surface gas is very rich in intermediates and often is processed to remove liquid propane, butanes, pentanes, and heavier hydrocarbons. These liquids are called plant liquids. The gas-oil ratios in the rules of thumb discussed above do not include any of these plant liquids.

Wet Gas Reservoir

Phase Diagram Characteristics

In a wet gas reservoir, the surface liquid is normally called condensate and the reservoir gas is sometimes called condensate-gas, which leads to

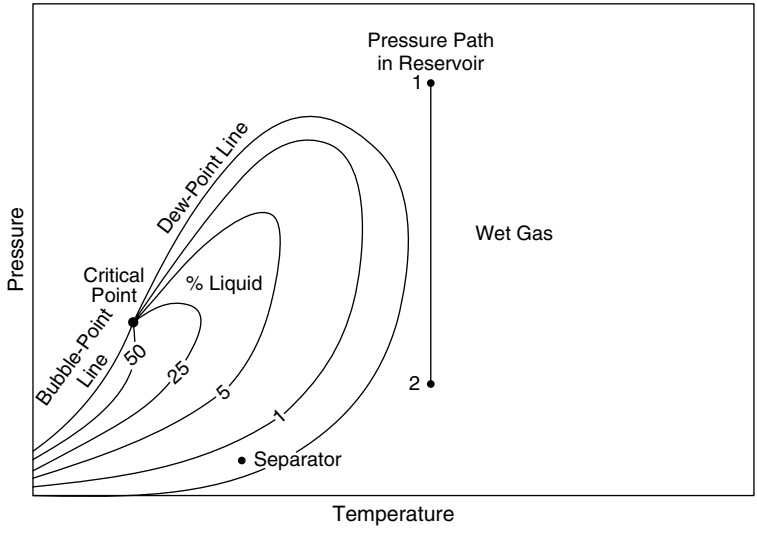


Figure 3-31. Phase diagram of a typical wet gas reservoir with a line of isothermal reduction of reservoir pressure and surface conditions.

confusion between wet gases and retrograde gases. The word "wet" does not mean that the gas is wet with water but refers to the hydrocarbon liquid, which condenses at surface conditions. In fact, reservoir gas is normally saturated with water. Its phase diagram characteristics are shown in Figure 3-31. Notice the entire phase diagram lies below the reservoir temperature. The fluid in the reservoir exists solely as a gas in the reservoir throughout the reduction in reservoir pressure. The pressure path does not enter the phase envelope, and thus no liquid is formed in the reservoir. Separator conditions lie within the phase envelope, causing some hydrocarbon liquid to be formed at the surface.

Field Characteristics

Wet gas reservoirs exhibit usually producing gas-oil ratio above 50,000 SCF/STB. Producing gas-oil ratios will remain constant throughout the life of the reservoir. They produce stock-tank liquids with the same range of gravities (40 to 60°API) as the liquids from retrograde gas reservoirs. The gravity of the stock-tank liquid does not change during the life of the reservoir. Stock-tank liquid is usually water-white.

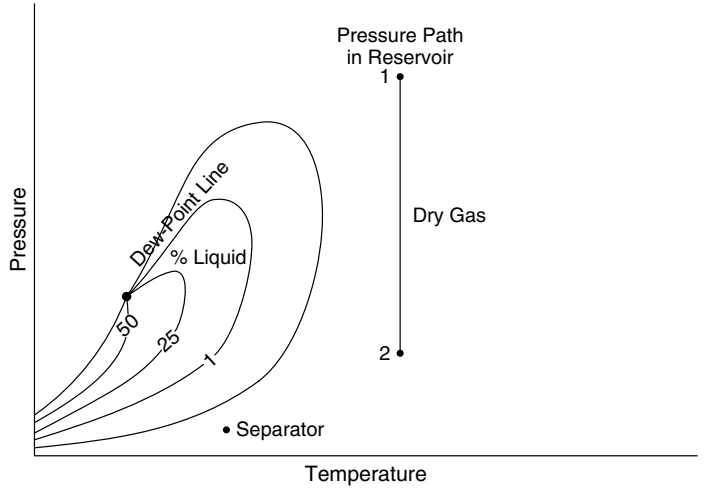


Figure 3-32. Phase diagram of a typical dry gas reservoir with a line of isothermal reduction of reservoir pressure and surface conditions.

Dry Gas Reservoir

Phase Diagram Characteristics

The word "dry" indicates that the gas does not contain enough of the heavier molecules to form hydrocarbon liquid at the surface. Usually some liquid water is condensed at the surface. This is often called simply a "gas reservoir." This leads to confusion because wet gas reservoirs sometimes are called gas reservoirs. In addition, a retrograde gas usually exists as gas in the reservoir. Dry gas is primarily methane with some intermediates. Its phase diagram characteristics are shown in Figure 3-32. The hydrocarbon mixture is solely gas in the reservoir, and normal surface separator conditions fall outside the phase envelope. Thus, theoretically, no hydrocarbon liquid is formed at the surface.

Information Required for Design

The two most common mistakes in the front end of a project are

- Failure to obtain good reservoir fluid samples,
- Failure to determine the phase behavior characteristics of a sample.

These mistakes often cause one to make a series of unnecessary "assumptions" that usually prove to be very expensive. The minimum phase diagram information required for a black oil reservoir is a section of

the bubble-point curve. The minimum information required for a volatile oil, condensate gas, and wet gas and dry gas reservoirs are the upper section of the phase diagram and the critical point, cricondenbar, and cricondentherm.

Flash Calculations

The amount of hydrocarbon fluid that exists in the gaseous phase or the liquid phase at any points at the process is determined by a flash calculation. As explained in Chapter 2, for a given pressure and temperature, each component in the gas phase will depend not only on pressure and temperature, but also on the partial pressure of the component. Therefore, the amount of gas depends upon the total composition of the fluids as the mole fraction of any one component in the gas phase is the function of the mole fraction of every other component in this phase.

This is best understood by assigning an equilibrium "*K*" value to each component. The *K* value is a strong function of temperature and pressure and of the composition of the vapor and liquid phase. It is defined as

$$K_N = \frac{V_N/V}{L_N/L}, \tag{3-31}$$

where

K_N = constant for component N at a given temperature and pressure,
V_N = moles of component N in the vapor phase,
V = total moles in the vapor phase,
L_N = moles of component N in the liquid phase,
L = total mole in the liquid phase.

The Gas Processors Suppliers Association (GPSA) present graphs of *K* values for the important components in a hydrocarbon mixture such as those in Figures 3-33 to 3-45. The *K* values are for specific "convergence" pressure. A procedure in the GPSA's *Engineering Data Book* for calculating convergence pressure based on simulating a binary fluid system with the lightest hydrocarbon component, which makes up at least 0.1 mole percent in the liquids and a pseudo-heavy component having the same average weight and critical temperature as the remaining heavier hydrocarbons. The convergence pressure is then read from a graph of convergence pressure versus operating temperature for common pseudo-binaries.

In most oil-field applications the convergence pressure will be between 2000 and 3000 psia, except at very low pressures, where a psia between 500 and 1500 is possible. If the operating pressure is much less than the convergence pressure, the equilibrium constant is not greatly affected

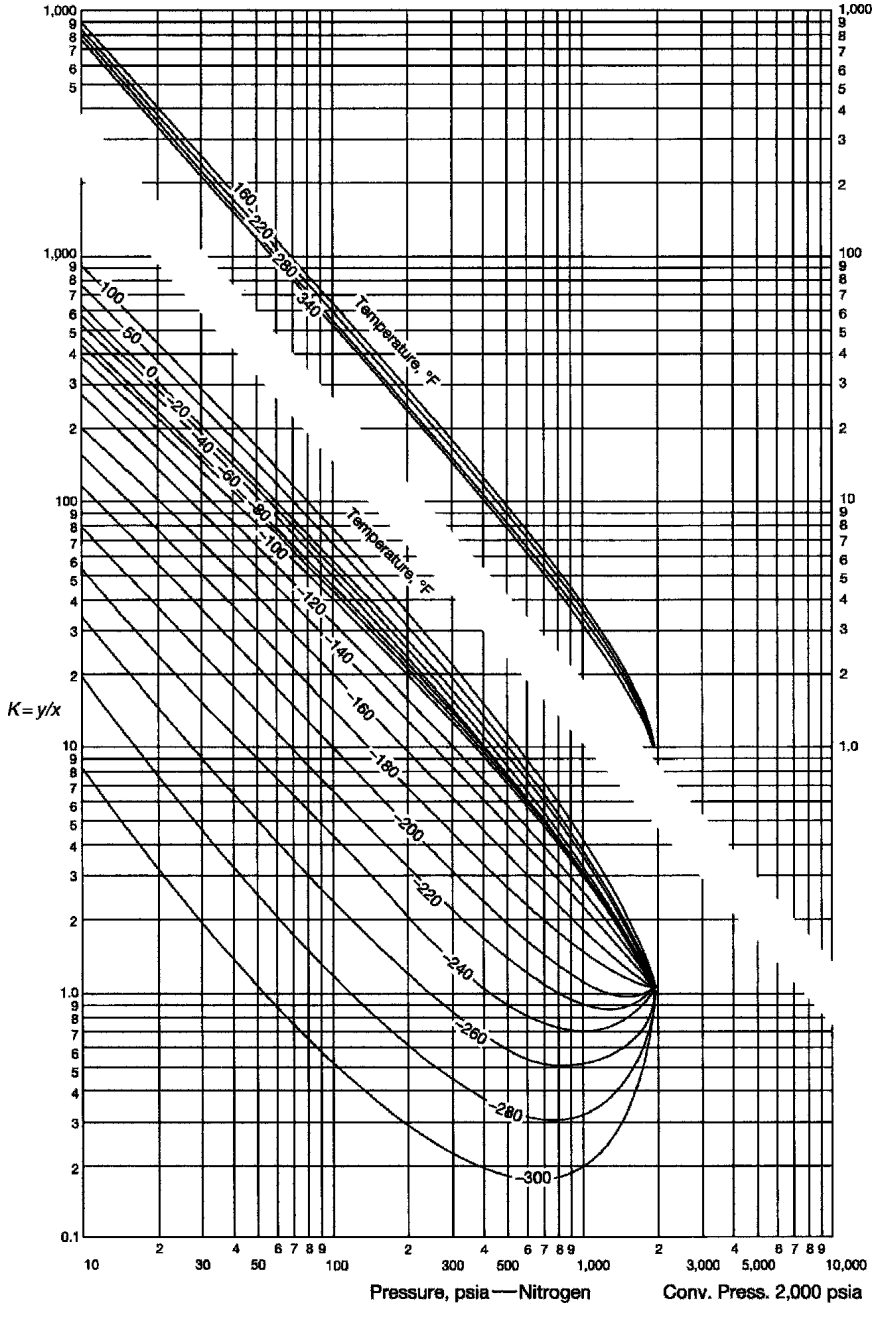


Figure 3-33. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book*.)

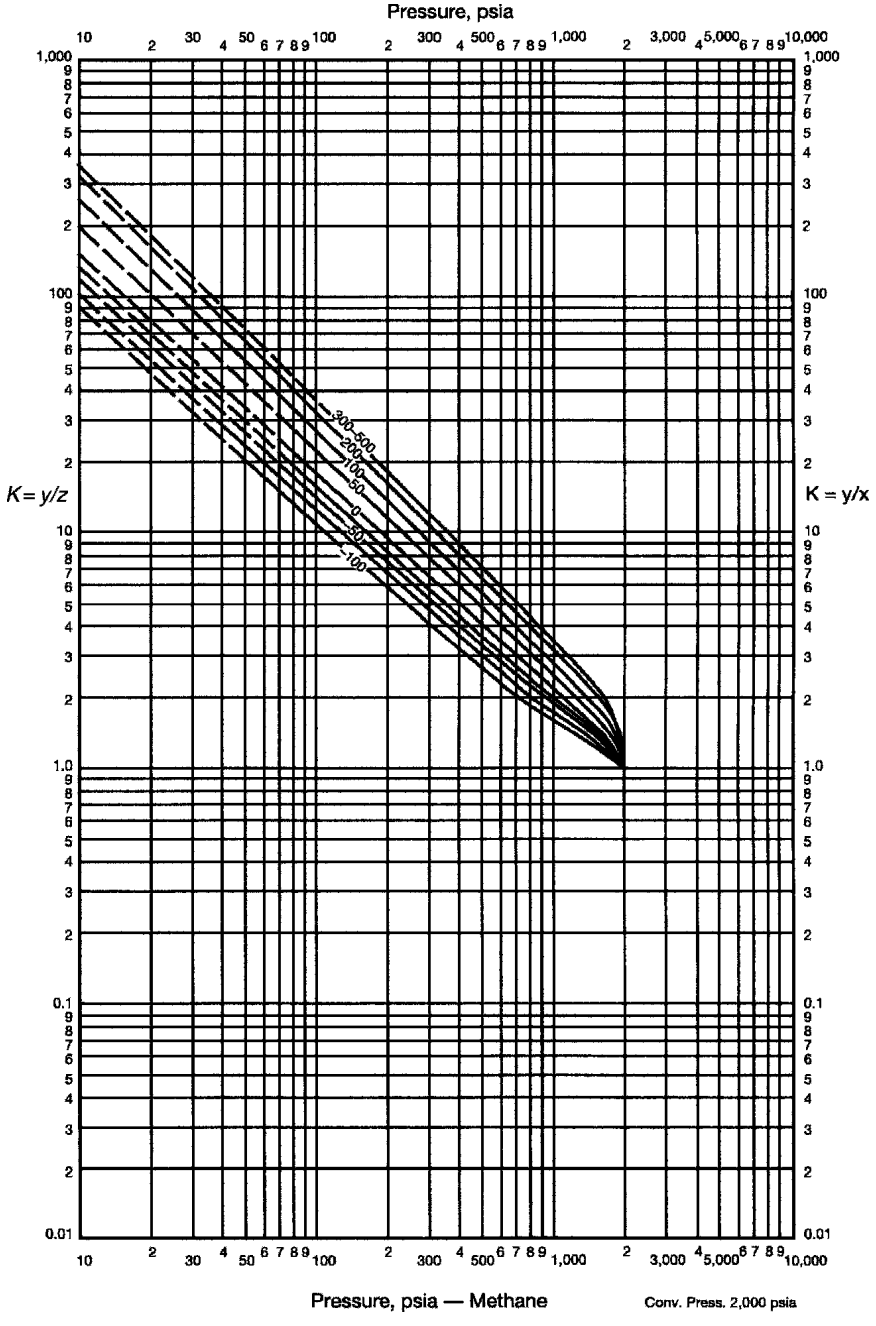


Figure 3-34. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book*.)

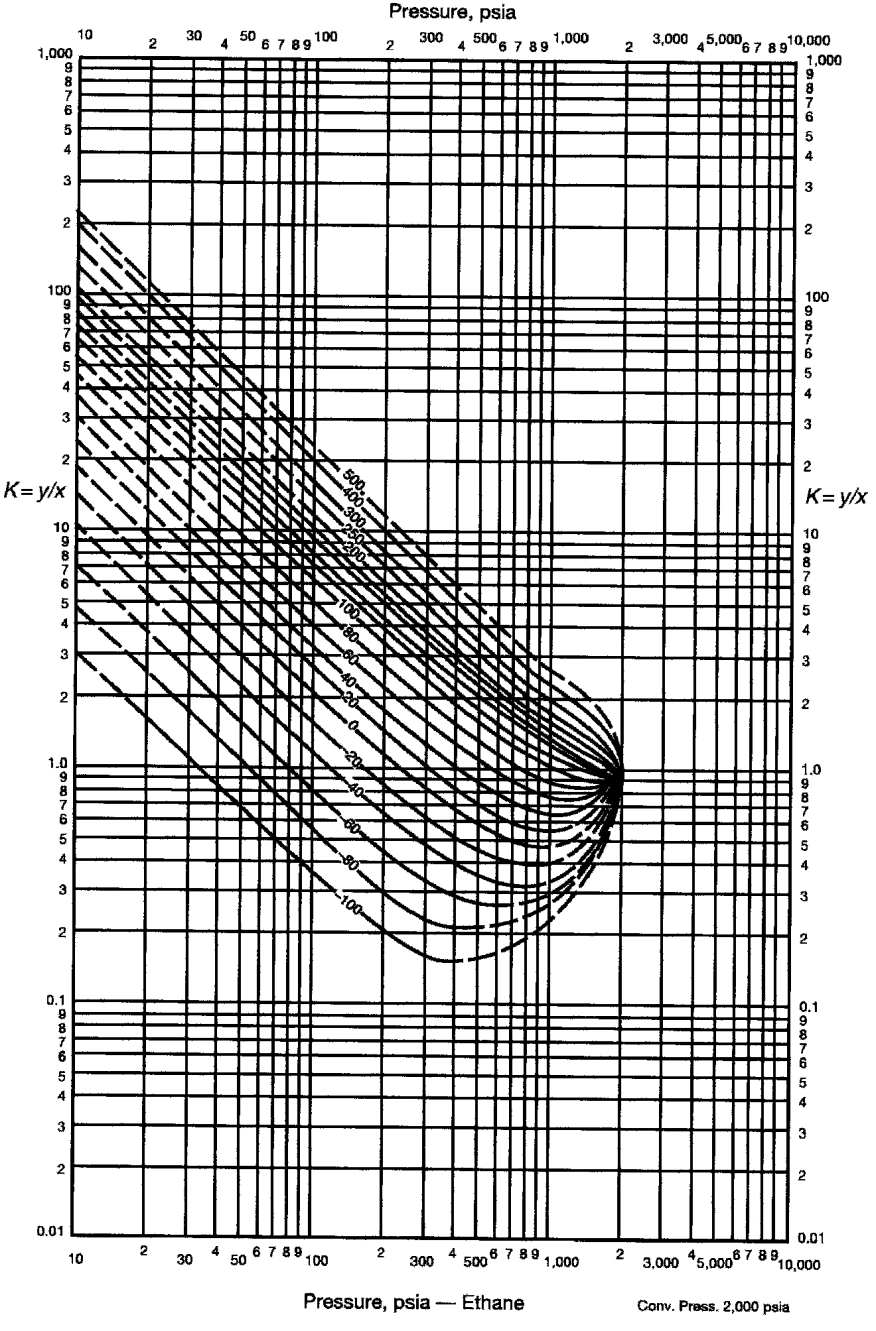


Figure 3-35. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book*.)

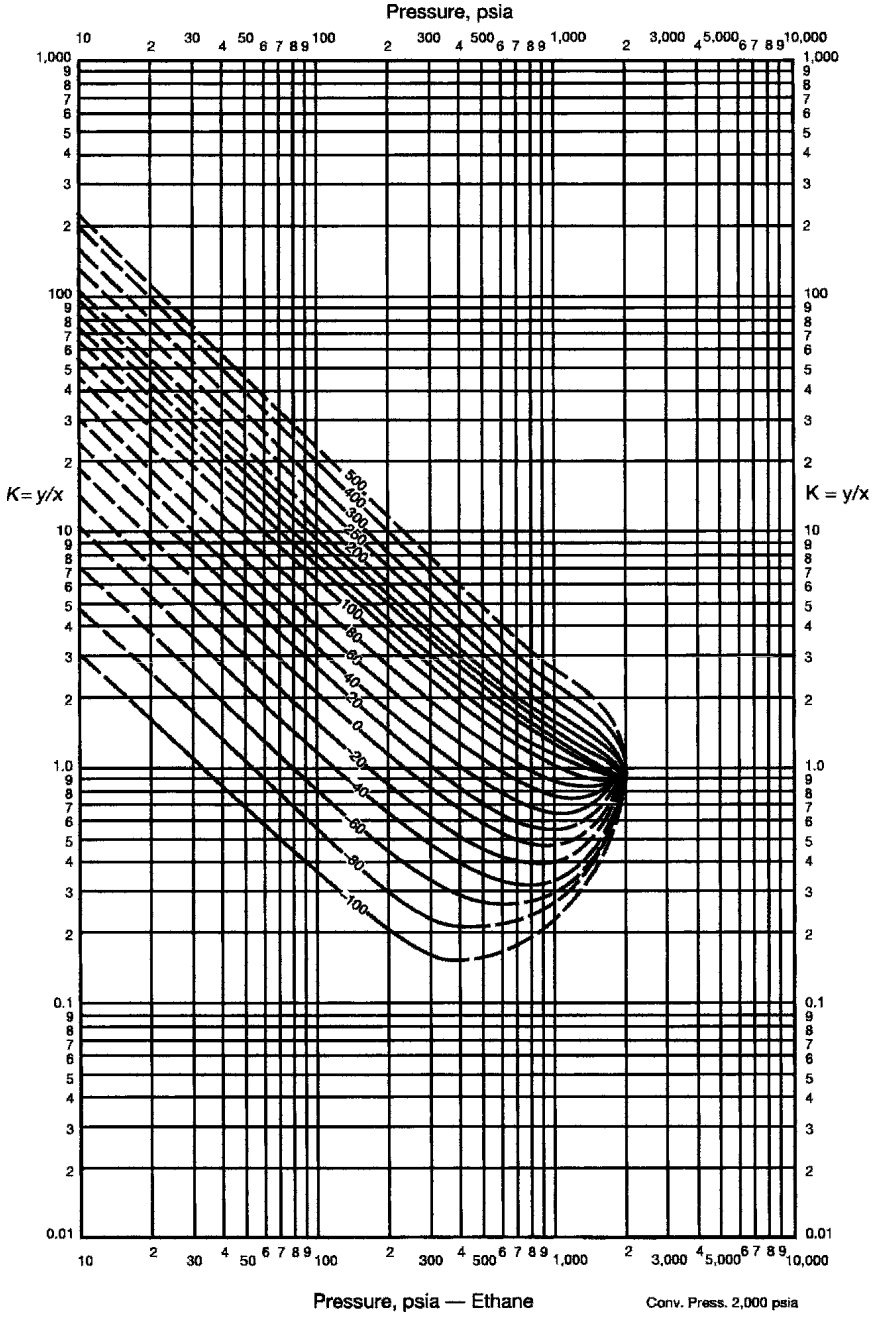


Figure 3-36. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book*.)

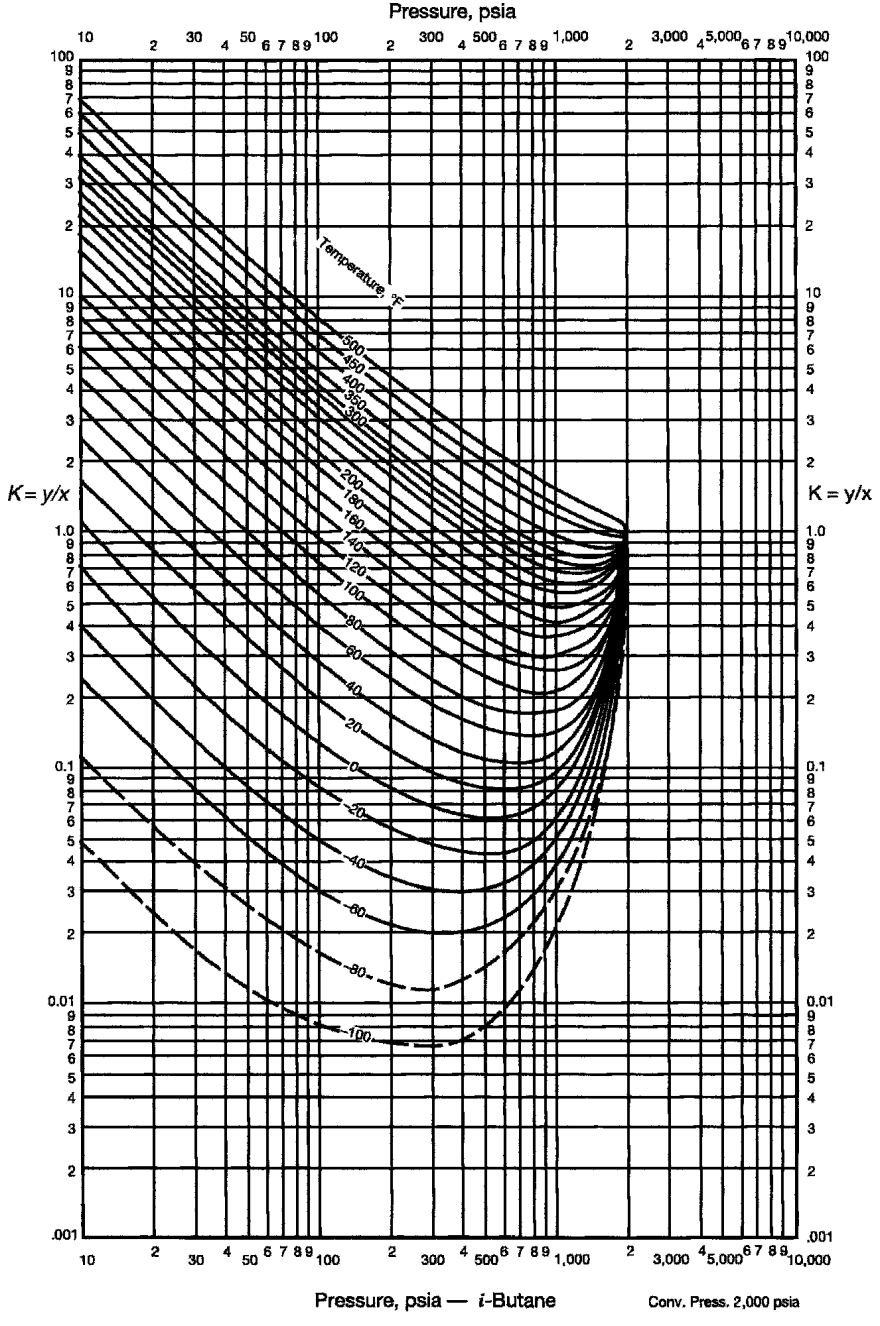


Figure 3-37. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book*.)

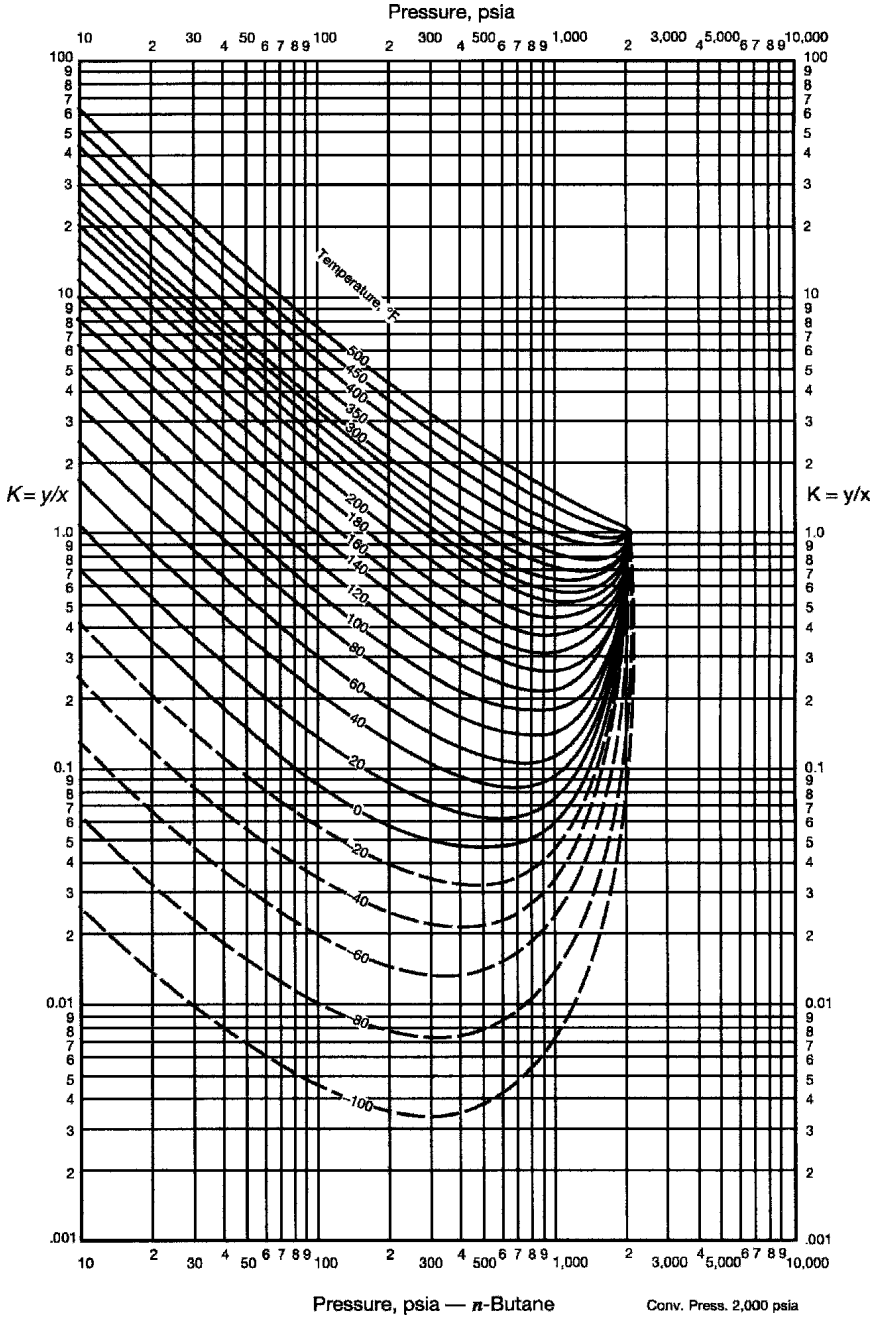


Figure 3-38. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book.*)

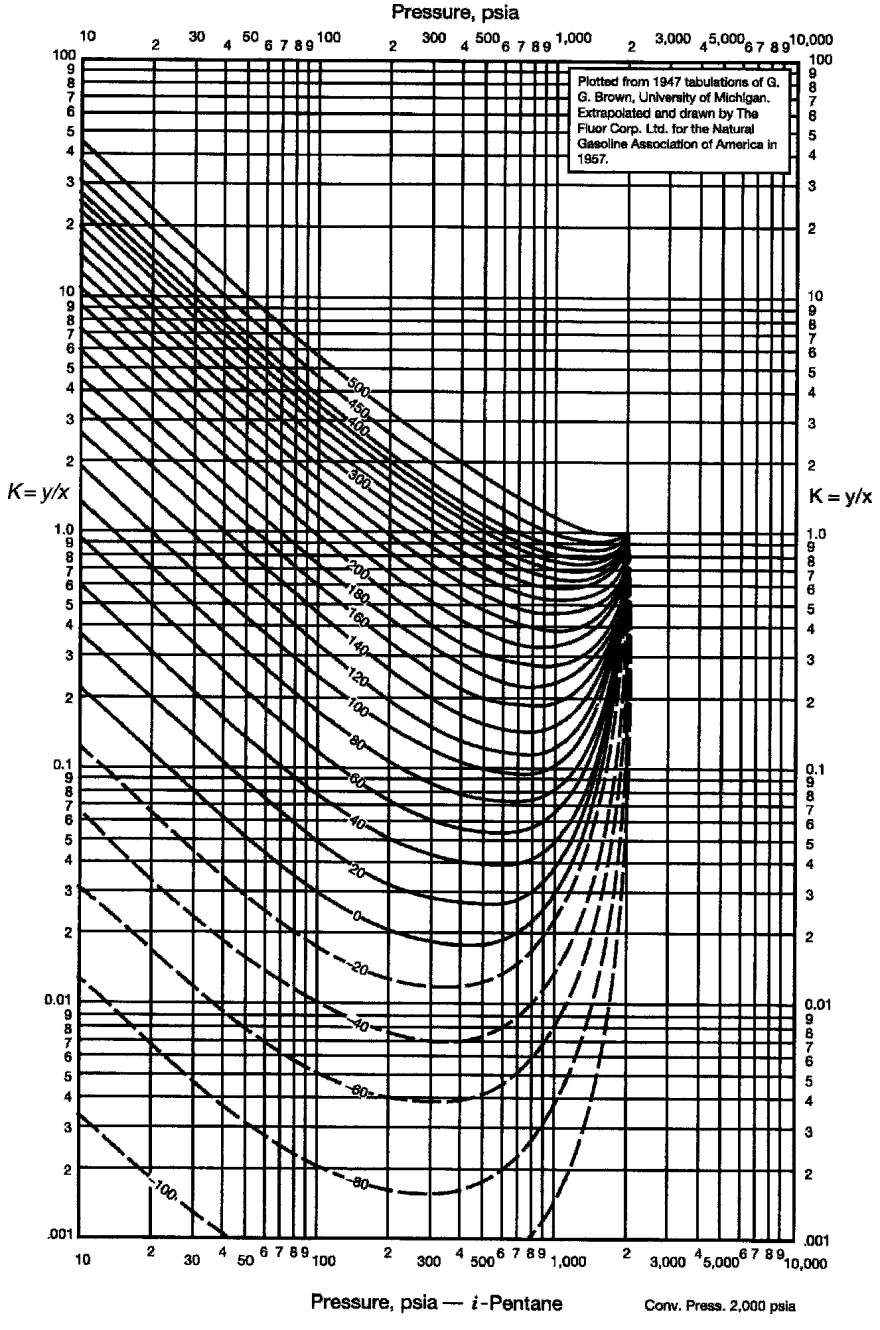


Figure 3-39. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book*.)

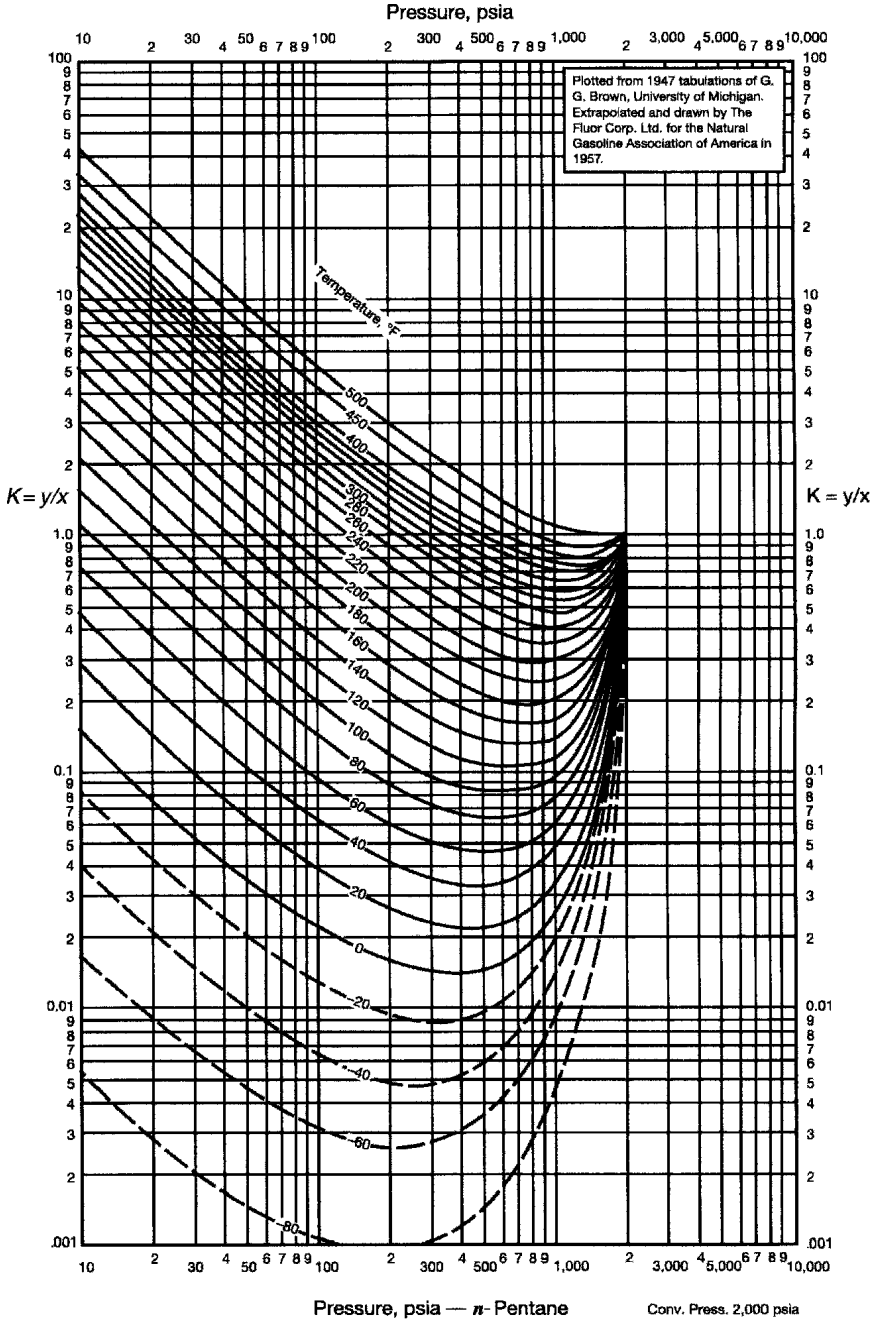


Figure 3-40. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book*.)

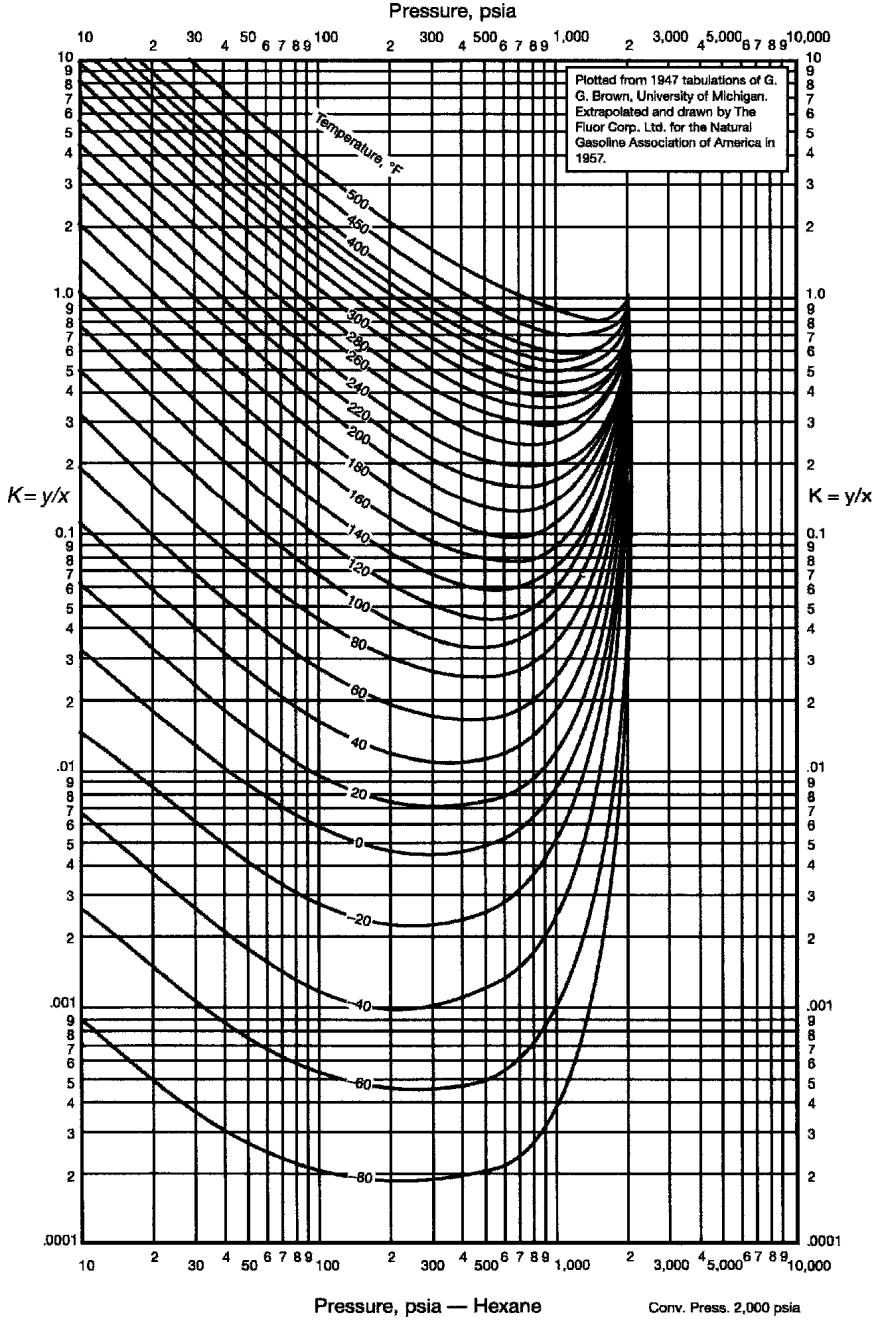


Figure 3-41. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book*.)

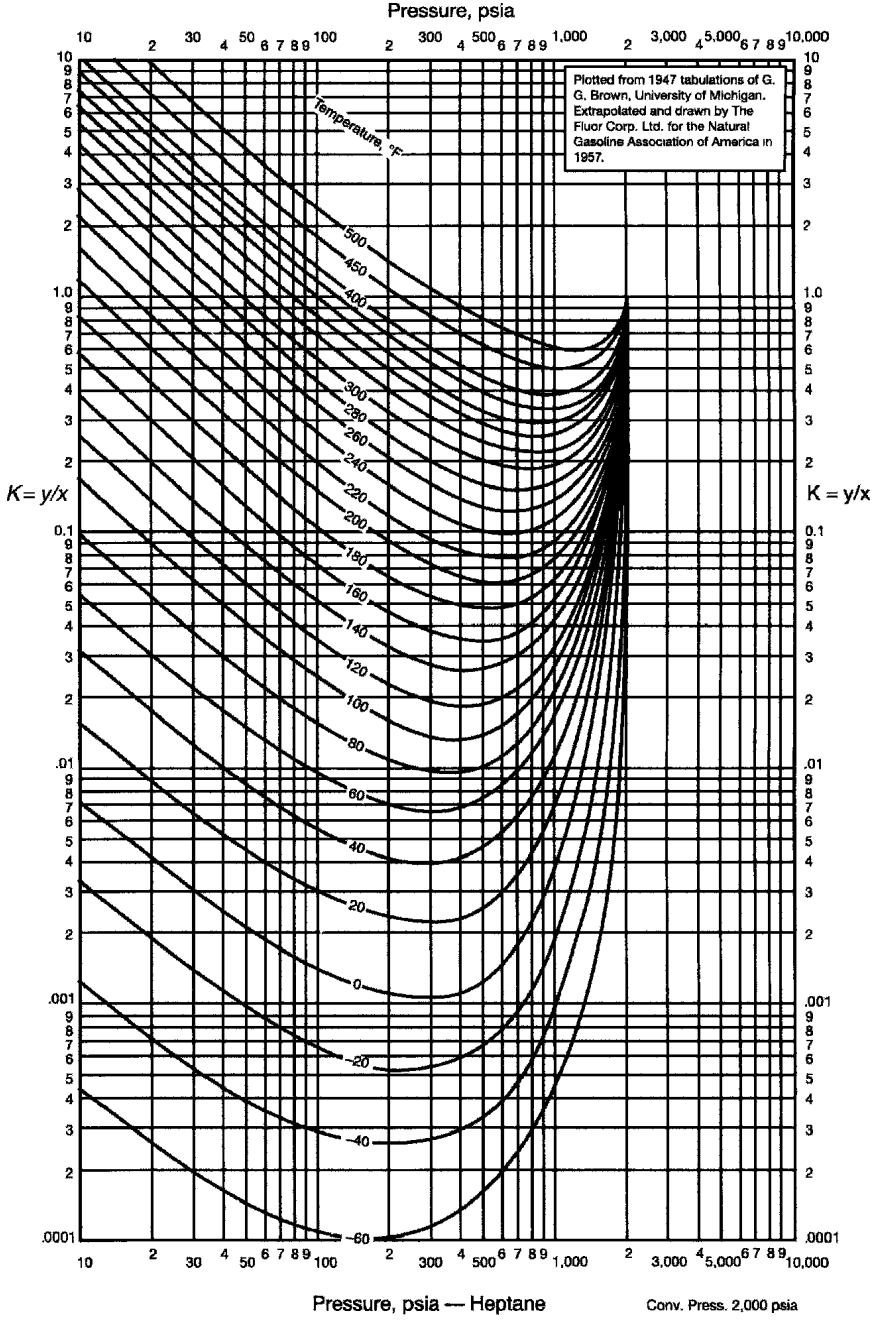


Figure 3-42. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book.*)

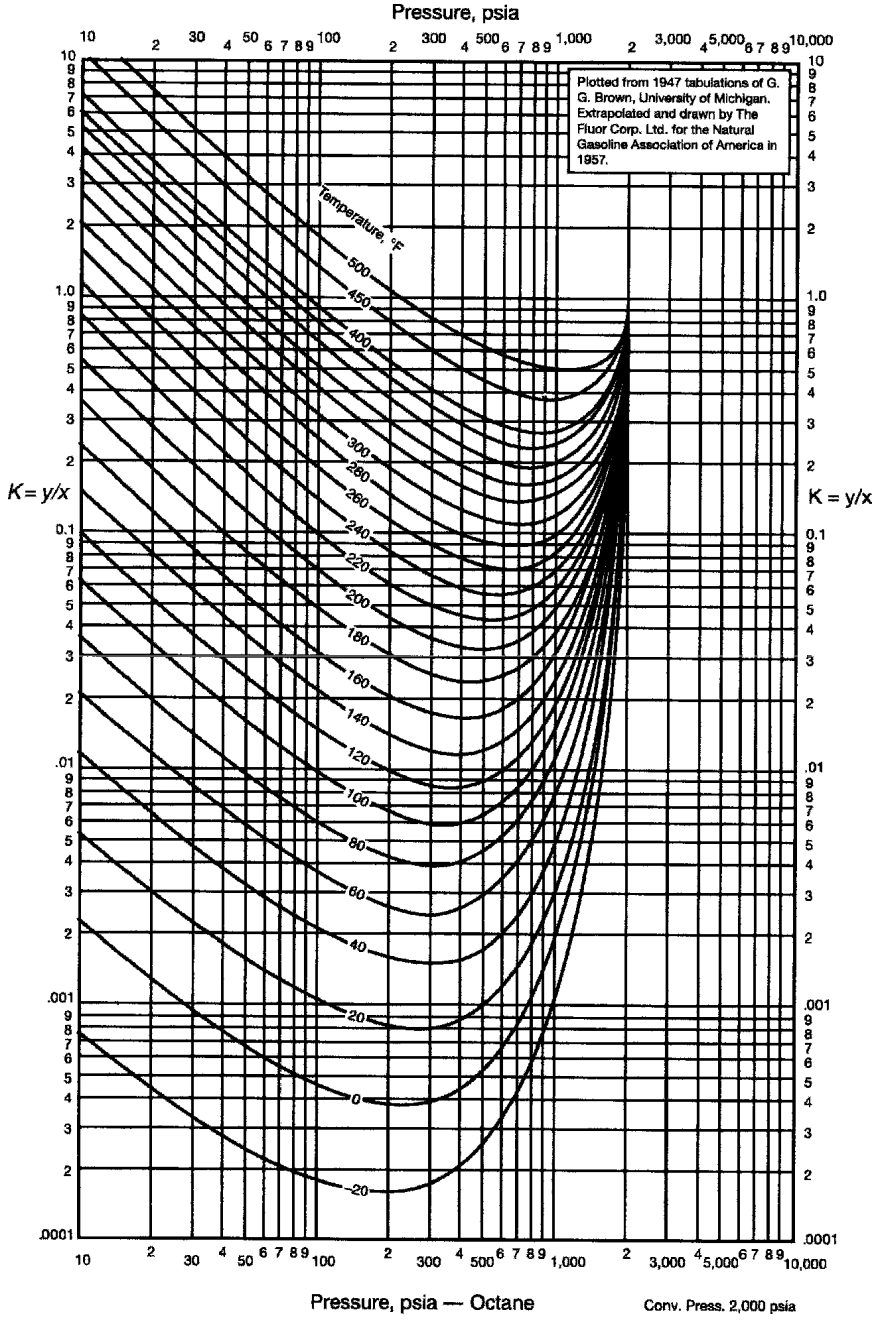


Figure 3-43. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book*.)

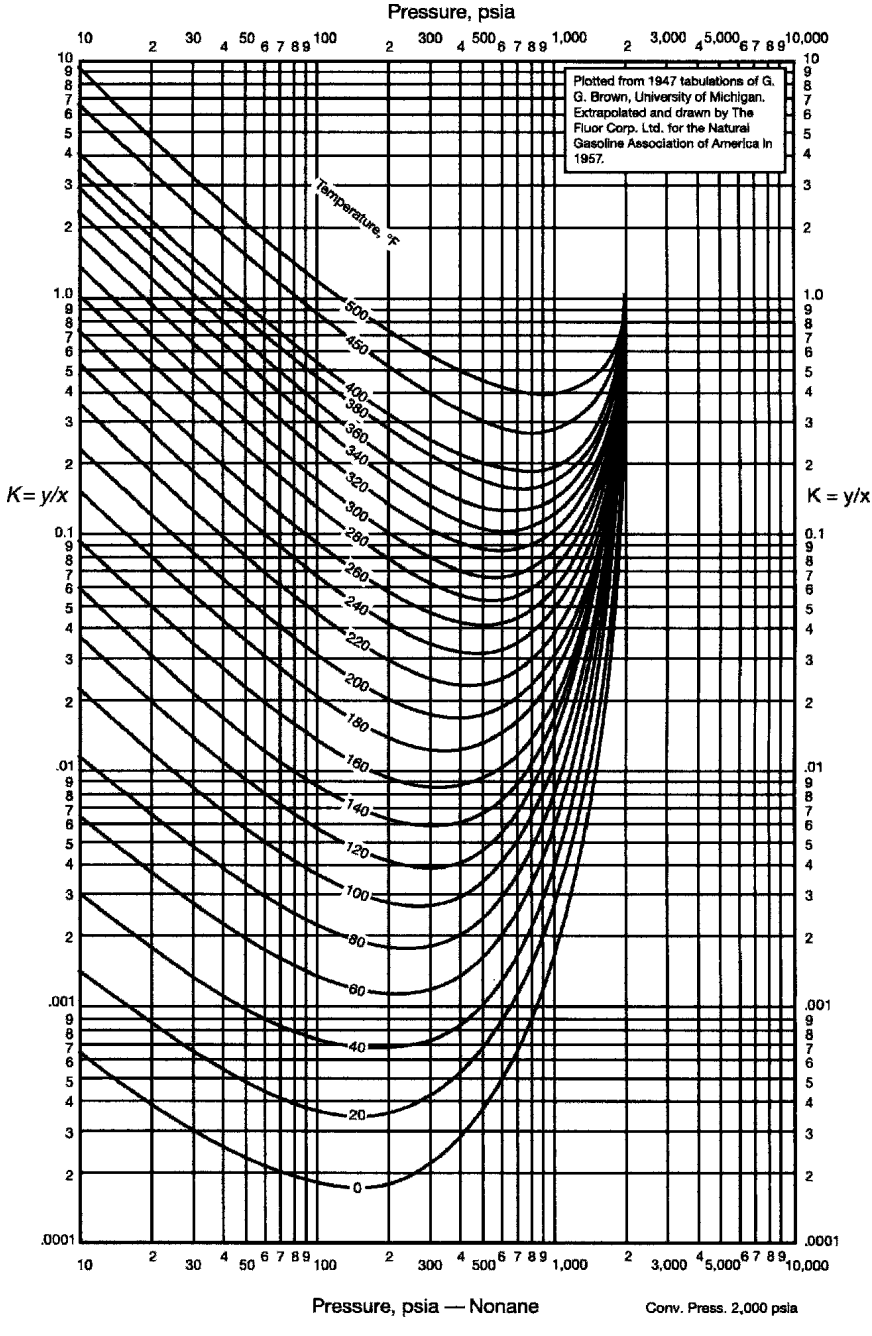


Figure 3-44. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book.*)

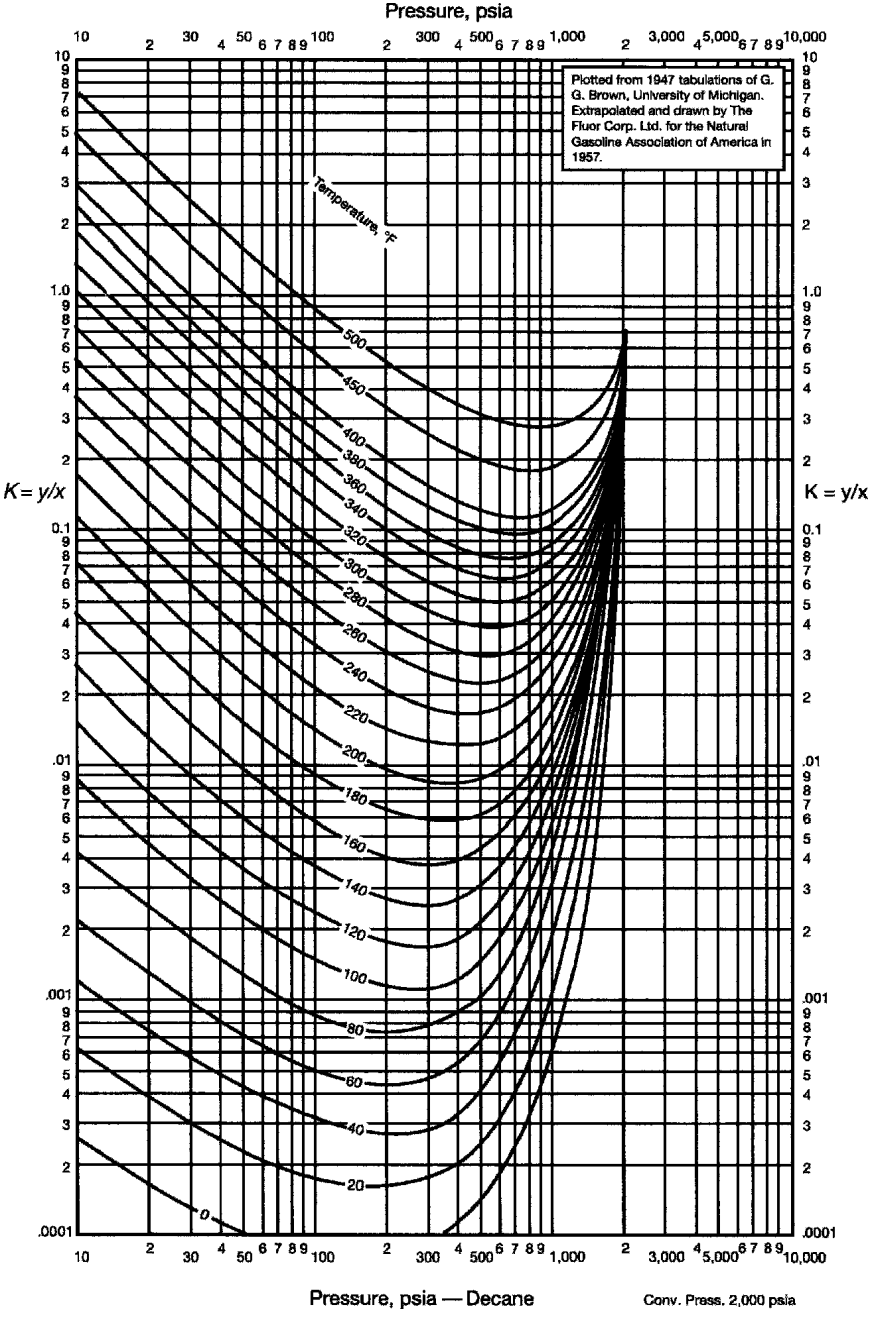


Figure 3-45. *K* values for hydrocarbon mixtures. (Courtesy of GPSA *Engineering Data Book*.)

by the choice of convergence pressure. Therefore, using a convergence pressure of 2000 psia is a good first approximation for most flash calculations. Where greater precision is required, the convergence pressure should be calculated.

If K_N for each component and the ratio of total moles of vapor to total moles of liquid (V/L) are known, than the moles of the component N in vapor phase (V_N) and the moles in the liquid phase (L_N) can be calculated from

$$V_N = \frac{K_N F_N}{\frac{1}{(V/L)} + K_N}, \tag{3-32}$$

$$L_N = \frac{F_N}{K_N(V/L) + 1}, \tag{3-33}$$

where F_N = total moles of component N in the fluid.

Equations (3-32) and (3-33) are derived as follows:

$$K_N = \left(\frac{V_N}{L_N}\right)\frac{1}{(V/L)},$$

$$L_N = \left(\frac{V_N}{K_N}\right)\frac{1}{(V/L)},$$

$$F_N = L_N + V_N,$$

$$F_N = \left(\frac{V_N}{K_N}\right)\frac{1}{(V/L)} + V_N,$$

$$F_N = \left[\frac{1}{K_N(V/L)} + 1\right]V_N,$$

$$V_N = \frac{F_N}{\frac{1}{K_N(V/L)} + 1}$$

$$V_N = \frac{K_N F_N}{\frac{1}{(V/L)} + K_N},$$

$$V_N = L_N K_N (V/L),$$

from the definitions of K_N

$$F_N = L_N + V_N$$

$$F_N = L_N + L_N K_N (V/L),$$

$$L_N = \frac{F_N}{K_N(V/L)+1}.$$

To solve either Eq. (3-32) or (3-33), it is necessary to first know the quantity (V/L), but since both V and L are determined by summing V_N and L_N, it is necessary to use an iterative solution. This is done by estimating (V/L), calculating V_N and L_N for each component, summing up to obtain the total moles of gas (V) and liquid (L), and then comparing the calculated (V/L) to assumed value. In doing the procedure, it is helpful to use the relationship

$$L = \frac{F}{1+(V/L)}. \tag{3-34}$$

Equation (3-34) is derived as follows:

$$F = V + L,$$
$$V = (V/L)L,$$
$$F = (V/L)L + L = [(V/L)+1],$$
$$L = \frac{F}{1+(V/L)}.$$

Once an assumed value of V/L is made, it is easy to calculate the corresponding assumed value of L. This is best illustrated by the example in Table 3-9. The mole fraction (column 2) for each component is given from test data. Column 3 is determined from the graphs for K_N, assuming a convergence pressure of 2000 psia. Column 4 is derived from Eq. (3-33), assuming $F = 100$ moles and $V/L = 1.5$. That is, $L = 40$ moles. With this assumption it is calculated that $L_N = 44.5$ moles, which is plotted in Figure 3-46 as point "1."

Another assumption is then made that $V/L = 0.5$ (i.e., $L = 66.7$ moles) and in column 5, L is calculated to be 60.87. This is plotted as point "2." From Figure 3-46, point "3," which represents the intersection of assumed and calculated values, indicated $L \cong 50$, which corresponds to a $V/L \cong 1.0$. This is tabulated in column 6. It can be seen that L_N is calculated as 49.9. Column 7, which characterizes the composition of the gas stream, is obtained by the difference between column 6 and column 2.

Table 3-9
Flash Calculation at 1000 psia and 100°F

(1) Component	(2) Mole Fraction Percent	(3) K_N	(4) $V/L = 1.5$ $L = 40L_N$	(5) $V/L = 0.5$ $L = 66.7L_N$	(6) $V/L = 1$ $L = 50L_N$	(7) V_N
CO_2	0.22	1.88*	0.06	0.11	0.08	0.14
N_2	0.09	4.00	0.01	0.03	0.02	0.07
Methane	63.35	2.80	12.18	26.40	16.67	46.68
Ethane	4.21	0.98	1.73	2.84	2.15	2.06
Propane	2.09	0.38	1.33	1.76	1.51	0.58
i-Butane	0.68	0.22	0.51	0.61	0.56	0.12
n-Butane	1.08	0.18	0.85	0.99	0.92	0.16
i-Pentane	0.47	0.10	0.41	0.45	0.43	0.04
n-Pentane	0.38	0.09	0.33	0.36	0.35	0.03
Hexane	1.36	0.05	1.20	1.33	1.30	0.06
Heptane+	26.07	0.006**	25.84	25.99	25.91	0.16
	100.00		44.45	60.87	49.90	50.10
	100.00					

*Calculated as $K_{CO_2} = (K_{C_1} + K_{C_2})^{1/2}$.
**Simulated as decane.

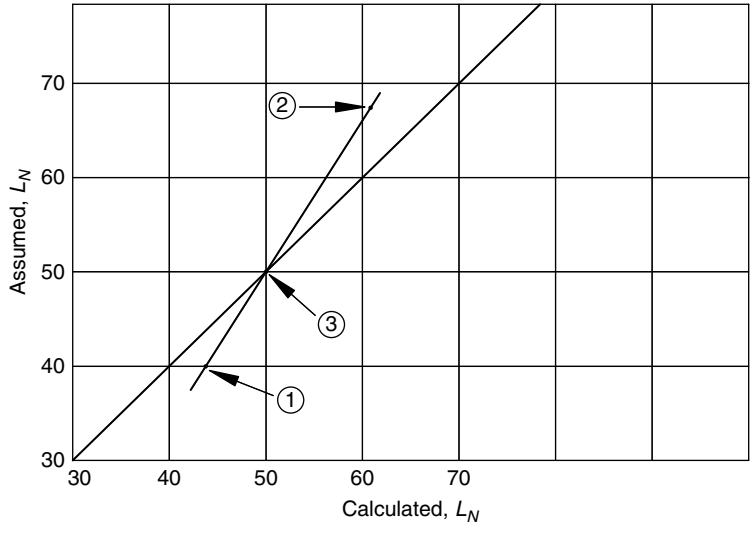


Figure 3-46. Interpolation of flash calculation results.

Characterizing the Flow Stream

Once a flash calculation is made and the molecular composition of the liquid and gas components have been determined, it is possible to determine the properties and flow rates of both the gas and the liquid streams.

Molecular Weight of Gas

The molecular weight of a stream is calculated from the weighted average gas molecular weight given by

$$\mathrm{MW} = \sum [V_N \times (\mathrm{MW})_N]. \tag{3-35}$$

The molecular weight of the gas stream of Table 3-9 is calculated in Table 3-10. Column 2 lists the molecular weight of the components from standard reference sources. Column 3 lists the number of moles of each component for 100 moles of feed. This is the same as column 7 in Table 3-9. Column 4 is derived from column 2 times column 3. The molecular weight of the gas is

$$\mathrm{MW} = \frac{911.5}{50.1} = 18.19.$$

The gas's specific gravity can be determined from the molecular weight from Eq. (3-7), as shown in Table 3-10:

$$S = \frac{18.19}{29} = 0.63.$$

Gas Flow Rate

If the flow of the inlet stream is known in moles per day, then the number of moles per day of gas flow can be determined from

$$V = \frac{F}{1 + \frac{1}{(V/L)}}, \tag{3-36}$$

where

V = gas flow rate, moles/day,
F = total stream flow rate, moles/day,
L = liquid flow rate, moles/day.

Table 3-10
Gas Flow Characterization

(1) Component	(2) MW_N	(3) V_N Moles	(4) $V_N \times MW_N$
CO_2	44.01	0.14	6.2
N_2	28.01	0.07	2.0
Methane	16.04	46.68	748.7
Ethane	30.07	2.06	61.9
Propane	44.10	0.58	25.6
i-Butane	58.12	0.12	7.0
n-Butane	58.12	0.16	9.3
i-Pentane	72.15	0.04	2.9
n-Pentane	72.15	0.03	2.2
Hexane	86.18	0.06	5.2
Heptane+	253.00*	0.16	40.5
		50.10	911.5

*From PVT analysis of feed stream.

Equation (3-36) is derived from the following:

$$F = V + L,$$
$$V = (V/L)L,$$
$$F = V + \frac{V}{(V/L)} = V\left[1 + \frac{1}{(V/L)}\right],$$
$$V = \frac{F}{1 + \frac{1}{(V/L)}}.$$

Once the mole flow rate of gas is known, then the flow rate in standard cubic feet can be determined by recalling that one mole of gas occupies 380 cubic feet at standard conditions. Therefore:

$$Q_g = \frac{380V}{1{,}000{,}000}, \tag{3-37}$$

where Q_g = gas flow rate, MMscfd.

Assuming a feed flow rate of 10,000 moles per day for the stream being flashed in Table 3-9:

$$V = \frac{10{,}000}{1 + \frac{1}{50.1/49.9}} = 5010 \text{ moles/day},$$

$$Q_g = \frac{380V}{1{,}000{,}000}(5010) = 1.90 \text{ MMscfd.}$$

Liquid Molecular Weight

The molecular weight of the liquid stream is calculated from the weighted average liquid component molecular weight given by

$$\text{MW} = \frac{\sum [L_N \times (\text{MW})_N]}{L}. \tag{3-38}$$

This is calculated in Table 3-11. Column 2 is as in Table 3-10, and column 3 is the liquid stream composition for 100 moles of feed as calculated in Table 3-9, column 7. Column 4 is column 2 times column 3 and represents the weight of each component in the liquid stream. The molecular weight of the liquid is

$$\text{MW} = \frac{7212}{49.90} = 145.$$

Table 3-11
Liquid Flow Characterization

(1) Component	(2) MW_N	(3) L_N	(4) $L_N \times (MW_N)$	(5) SG_N	(6) $\frac{L_N \times (MW)_N}{SG_N}$
CO_2	44.01	0.08	3	0.83**	4
N_2	28.01	0.02	1	0.81**	1
Methane	16.04	16.67	2.6	0.30**	891
Ethane	30.07	2.15	65	0.36**	179
Propane	44.10	1.51	67	0.51**	1.5
i-Butane	58.12	0.56	33	0.56**	58
n-Butane	58.12	0.92	53	0.58**	92
i-Pentane	72.15	0.43	3.1	0.62	50
n-Pentane	72.15	0.35	25	0.63	40
Hexane	86.18	1.30	112	0.66	170
Heptane$^+$	253.00*	25.91	6.5	0.86*	7.3
			721.2		923.8

* From PVT analysis of feed stream.
** Pseudo-value at saturation pressure.

Specific Gravity of Liquid

Remembering that the weight of each component is the number of moles of that component times its molecular weight, the specific gravity of the liquid is given by

$$\mathrm{SG} = \frac{\sum [L_N \times (\mathrm{MW})_N]}{\sum \dfrac{[L_N \times (\mathrm{MW})_N]}{(\mathrm{SG})_N}}. \tag{3-39}$$

Equation (3-39) is derived as follows:

ρ in lb/ft^3, volume in ft^3,

$$\mathrm{SG} = \frac{\rho}{62.4},$$

$$\rho = \frac{\sum \mathrm{pounds}_N}{\sum Volume_N},$$

$$\mathrm{SG} = \frac{1}{62.4} \frac{\sum \mathrm{pounds}_N}{\sum Volume_N},$$

$$Volume_N = \frac{\mathrm{pounds}_N}{\rho_N},$$

$$Volume_N = \frac{\mathrm{pounds}_N}{62.4(\mathrm{SG})_N},$$

$$\sum Volume_N = \frac{1}{62.4} \sum \frac{\mathrm{pounds}_N}{(\mathrm{SG})_N},$$

$$\mathrm{SG} = \frac{\sum \mathrm{pounds}_N}{\sum \dfrac{\mathrm{pounds}_N}{(\mathrm{S.G})_N}},$$

$$\mathrm{pounds}_N = L_N \times (\mathrm{MW})_N,$$

$$\mathrm{SG} = \frac{\sum [L_N \times (\mathrm{MW})_N]}{\sum \dfrac{[L_N \times (\mathrm{MW})_N]}{(\mathrm{SG})_N}}.$$

Column 5 lists a specific gravity for each component in the liquid phase at standard conditions except as noted. It would be more accurate

to adjust these gravities for the actual pressure and temperature of the fluid being flashed. This will have a marginal effect on the results as the characterization of the heptanes is of overriding importance to the calculation. If this is not known for the pressure and temperature of the flash, it can be approximated from known conditions using Figure 3-16.

Column 6 is derived by dividing column 4 by column 5. The liquid's specific gravity is determined by dividing the sum of column 4 by column 6.

$$\text{SG} = \frac{721.2}{923.8} = 0.78.$$

The specific gravity in °API can be calculated from Eq. (3-24) as

$$^{\circ}\text{API} = \frac{141.5}{0.78} - 131.5 = 49.9.$$

Liquid Flow Rate

The liquid flow rate in moles per day for the given inlet stream can be determined from Eq. (3-40). In our example, for an inlet stream rate of 10,000 moles/day, the liquid flow rate is

$$L = \frac{10{,}000}{1 + \frac{50.1}{49.9}} = 4990 \text{ moles/day}.$$

The liquid flow rate in barrels per day can be derived from

$$Q_1 = \frac{L \times (\text{MW})}{350(\text{SG})}, \tag{3-40}$$

where

Q_1 = liquid flow rate, bpd,
SG = specific gravity of liquid (water = 1).

Equation (3-40) is derived as follows: there are 350 pounds per barrel of water and 350 (SG) pounds per barrel of liquid.

Pounds of liquid $= L \times (\text{MW})$,

$$Q_1 = \frac{\text{Pounds}}{\text{Pounds per Barrel}}$$

$$Q_1 = \frac{L \times (\text{MW})}{350\,(\text{SG})}.$$

For our example:

$$Q_1 = \frac{(4990)(145)}{(350)(0.78)} = 2650 \text{ bpd}.$$

The Flow Stream

Many times the designer is given the mole fraction of each component in the feed stream but is not given the mole's flow rate for the stream. It may be necessary to estimate the total number of moles in the feed stream (F) from an expected stock-tank oil flow rate. As a first approximation, it can be assumed that all the oil in the stock tank can be characterized by the C_7^+ component of the stream. Thus, the feed rate in moles per day can be approximated as

$$L \cong \frac{350(\text{SG})_7 Q_1}{(\text{MW})_7}, \tag{3-41}$$

where

L = liquid flow rate, moles per day,
$(\text{SG})_7$ = specific gravity of C_7^+,
$(\text{MW})_7$ = molecular weight of C_7^+,
Q_1 = flow rate of liquid, BPD.

The mole flow rate of the feed stream is then calculated as

$$F = \frac{L}{(\text{mole fraction})_7} \tag{3-42}$$

where

F = flow rate feed stream, moles per day,
$(\text{mole fraction})_7$ = mole fraction of the C_7^+ component in the feed stream.

For our example, if the mole feed rate of 10,000 moles per day was not given but was required to design for 2500 bpd of stock-tank liquid, the mole feed rate flash calculations would be approximated as

$$L = \frac{350(0.86)(2500)}{253} = 2974,$$

$$F = \frac{2974}{0.2607} = 11{,}400 \text{ moles/day}.$$

The flash calculation could then proceed. The calculated flow rates for each stream in the process could then be used in a ratio to reflect the error between assumed stock-tank flow rate and desired stock-tank flow rate.

Approximate Flash Calculations

Most often, flash calculations are too involved and are subject to many arithmetic mistakes to be done by hand as previously detailed. They are usually done using a standard a computer program. Sometimes it is necessary to get a quick estimate of the volume of gas that is expected to be flashed from an oil stream at various pressures.

Figure 3-47 was developed by flashing several crude oils of different gravities at different pressure ranges. The curves are approximate. The

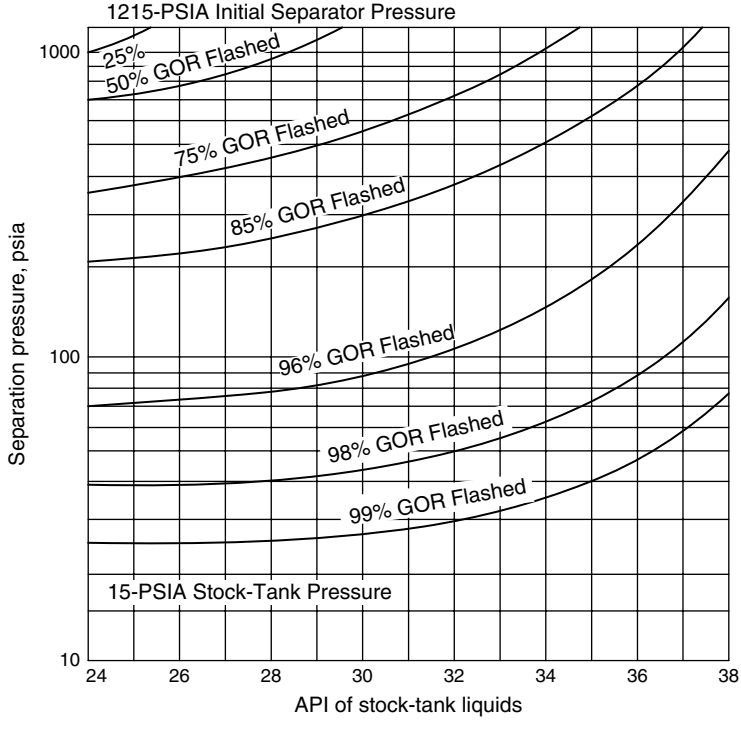


Figure 3-47. Preliminary estimation of % GOR flashed for given API of stock-tank liquids and separation pressures—Gulf Coast crudes.

actual shape would depend on the initial separation pressure, the number and pressure of intermediate flashes, and the temperature.

Use of the curve is best explained by an example. Suppose a 30°API crude with a GOR of 500 is flashed at 1000 psia, 500 psia, and 50 psia before going to a stock tank. Roughly 50% of the gas that will eventually be flashed from the crude, or 250 ft^3/B, will be liberated as gas in the 1000-psia separator. Another 25% (75% − 50%), or 125 ft^3/B, will be separated at 500 psia, and 23% (98% − 75%), or 115 ft^3/B, will be separated at 50 psia. The remaining 10 ft^3/B (100% − 98%) will be vented from the stock tank.

It must be stressed that Figure 3-47 is only to be used where a quick approximation, which could be subject to error, is acceptable. It cannot be used for estimating gas flashed from condensate produced in gas wells.

Other Properties

The iterative manual flash calculation detailed in the previous sections shows one of many methods for calculating equilibrium conditions. Flash calculations are inherently rigorous and best performed by sophisticated simulation software, such as HYSIM or other similar programs. For preliminary considerations, however, correlations and simplified methods for flash calculation such as that detailed earlier may used.

Once the equilibrium conditions (and, therefore, the gas and the liquid compositions) are known, several very useful physical properties are obtainable, such as the dew point, the bubble point, the heating value (net and gross), and k, the ratio of gas-specific heats. These properties are described next:

Dew point: the point at which liquid first appears within a gas sample.
Bubble point: the point at which gas first appears within a liquid sample.
Net heating value: heat released by combustion of gas sample with water vapor as a combustion product; also known as the lower heating value (LHV).
Gross heating value: heat released by combusting of gas sample with liquid water as a combustion product; also known as the higher heating value (HHV).
***k*:** ratio of heat capacity at constant pressure (C_P) to heat capacity at constant volume (C_V). Often used in compressor calculation of horsepower requirement and volumetric efficiencies. This ratio is relatively constant for natural gas molecular weight and ranges between 1.2 and 1.3 (see Figure 3-48).

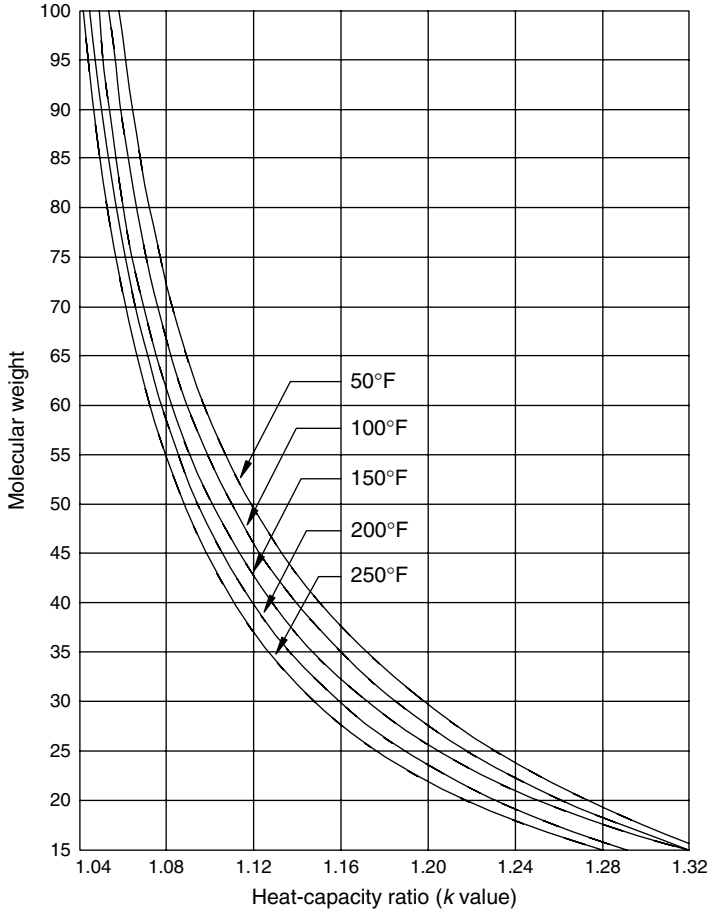


Figure 3-48. Approximate heat-capacity ratios of hydrocarbon gases. (Courtesy of GPSA *Engineering Data Book*.)

A more precise definition of the dew point makes a distinction between the hydrocarbon dew point, which represents the condensation of a hydrocarbon liquid, and the water dew point, which represents the condensation of liquid water. Often, sales gas contracts specify control of the water dew point for hydrate and corrosion control and not the hydrocarbon dew point. In such cases, hydrocarbons will often condense in the pipeline as the gas cools (assuming that separation has occurred at a higher temperature than ambient), and provisions to separate this "condensate" must be provided.

The bubble point can also be referred to as the "true vapor pressure." A critical distinction lies here between the true vapor pressure and the

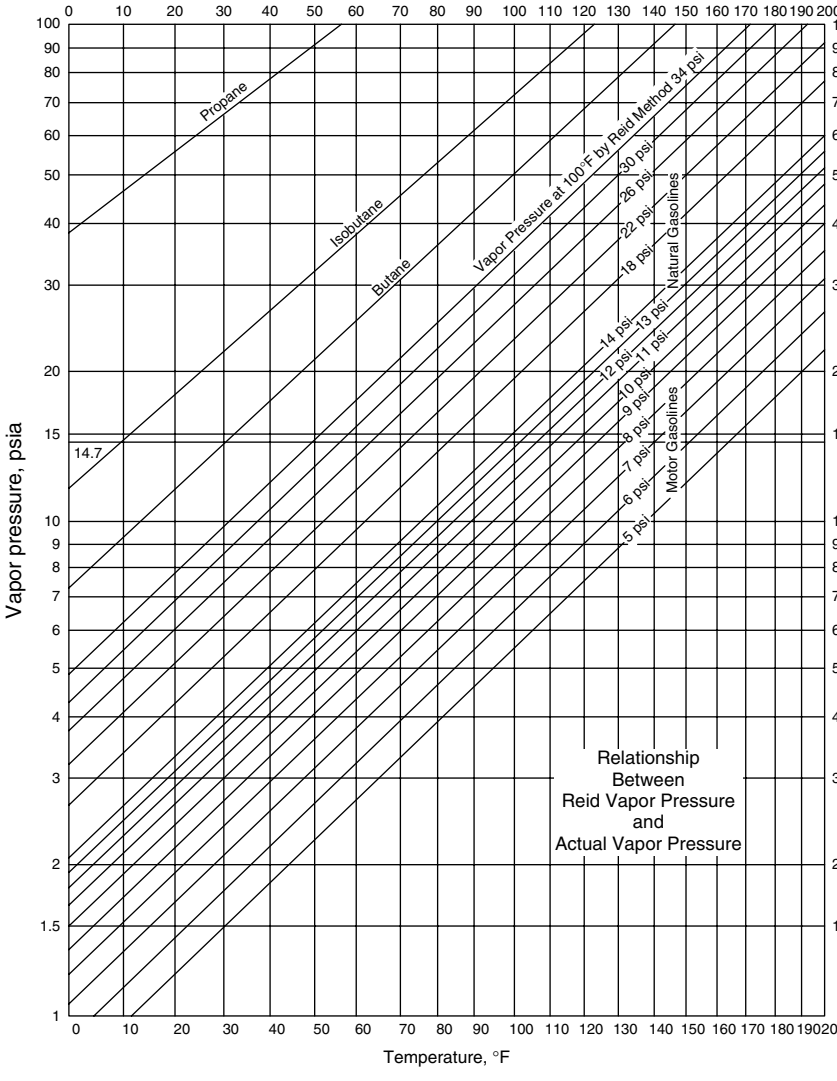


Figure 3-49. Relationship between Reid vapor pressure and actual vapor pressure. (Courtesy of GPSA *Engineering Data Book*.)

Reid vapor pressure (RVP). The Reid vapor pressure is measured according to a specific ASTM standard (D323) and lies below the true vapor pressure.

The approximate relationship between the two pressures is shown in Figure 3-49. (Note that an RVP below atmospheric pressure does not indicate that vapors will be absent from a sample at atmospheric pressure.)

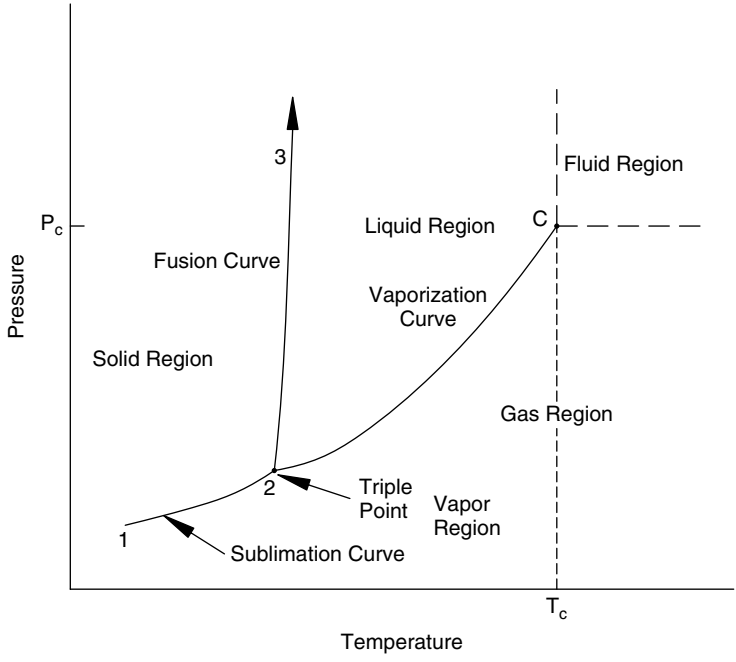


Figure 3-50. Pressure-temperature diagram for a pure substance. (From the introduction to *Chemical Engineering Thermodynamics*, 4th edition, by J. M. Smith and H. C. Van Ness, New York: McGraw Hill, Inc., 1987.)

The bubble point and the dew point are equivalent for a single-component mixture only. A graphic representation of the bubble/dew point for a pure substance is shown as the vaporization curve in Figure 3-50.

A basic representation of the equilibrium information for a specific fluid composition can be found in a P-H (pressure-enthalpy) diagram, which is highly dependent on the sample composition. This diagram can be used to investigate thermodynamic fluid properties as well as thermodynamic phenomena such as retrograde condensation and the Joule–Thomson effect. Please note, however, that a P-H diagram is unlikely to be available for anything but a single component of the mixture, unless the diagram is created by simulation software packages such as those mentioned above. A P-H diagram for propane is shown in Figure 3-51; a P-H diagram for a 0.6 specific gravity natural gas is shown in Figure 3-52.

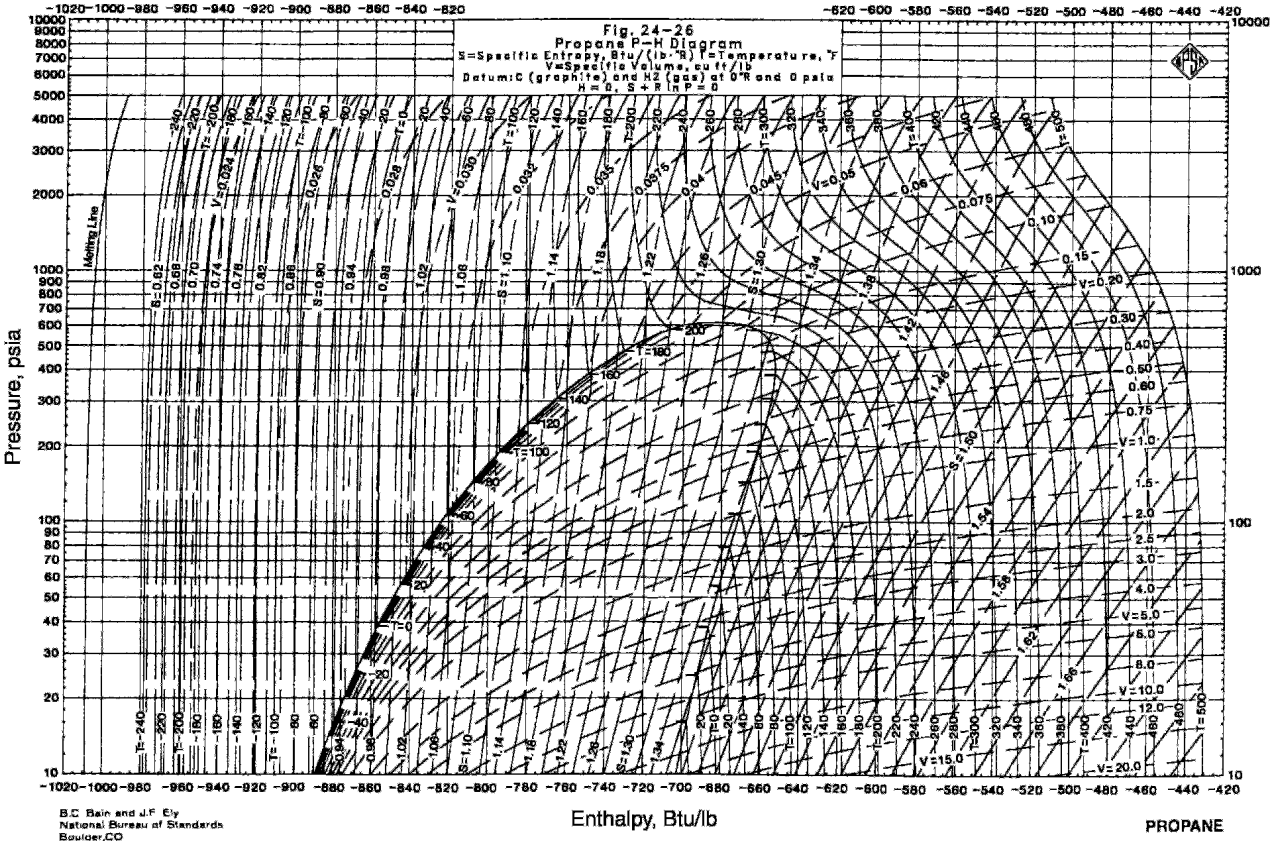


Figure 3-51. A P-H diagram for propane. (Courtesy of GPSA *Engineering Data Book*.)

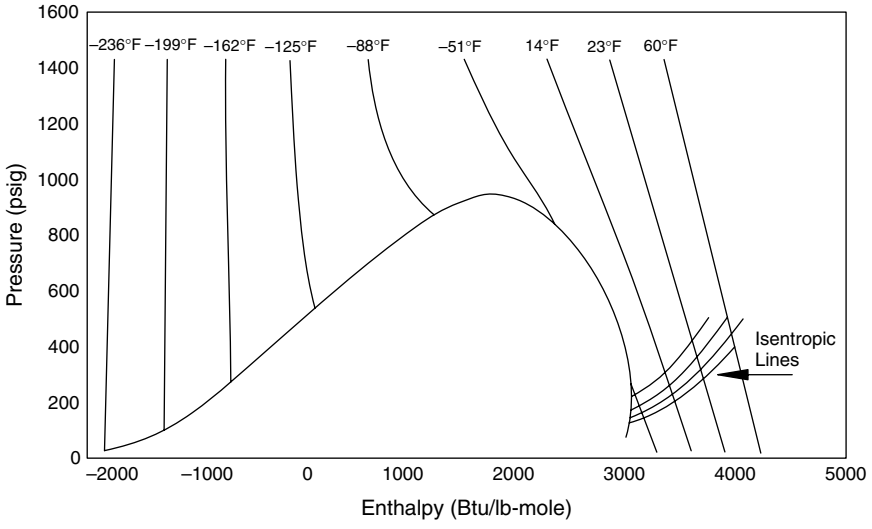


Figure 3-52. A P-H diagram for 0.6 specific gravity natural gas.

Exercises

Problem 1.

Given:
A well stream with the following data and with the data shown in Table 3-12:

$P_o = 315$ psia,
$T_o = 90\,^\circ$F.

Determine:

a. Molecular weight of gas
b. Specific gravity of gas (SG)
c. Gas flow rate (Q_g)
d. Specific gravity of liquid (SG_L)
e. API gravity
f. Molecular weight of liquid
g. Liquid flow rate (Q_L)
h. Liquid flow rate for 15-MMscfd separator gas
i. Gas flow rate for 1000-BPD separator liquid

Table 3-12
Well Stream Composition

Component	Moles/Day	K_N
CO_2	10	4.8
N_2	4	29.5
C_1	2893	9.1
C_2	192	1.8
C_3	95	0.6
iC_4	31	0.28
nC_4	49	0.21
iC_5	21	0.092
nC_5	17	0.071
C_6	64	0.026
C_{7+}	1190	0.0006

The molecular weight of C_7 is 253. Its specific gravity is 0.8604. The V/L is between 1.5 and 2.0.

Problem 2.

Given:

a. A loosely consolidated water-drive reservoir
b. Well stream properties:

$(SG)_o = 30\,°\text{API}\ (0.875)$,
$SG = 0.6$,
$SITP = 4000\,\text{psig}$,
$FTP_i = 1200\,\text{psig}$,
$FTP_f = 100\,\text{psig}$,
$T_o = 90\,°\text{F}$.

c. Production forecast (per well):

of wells $= 10$,
$Q_{oi} = 500\,\text{BOPD}$,
$Q_{of} = 50\,\text{BOPD}$,
$Q_{wi} = 0\,\text{BWPD}$,
$Q_{wf} = 1000\,\text{BWPD}$,
Gas-oil ratio $= 1000$.

d. A field reservoir engineer provided a production schedule that is typical of the area (Figure 3-53). To simplify the calculation, the percent peak production is summarized in Table 3-13.

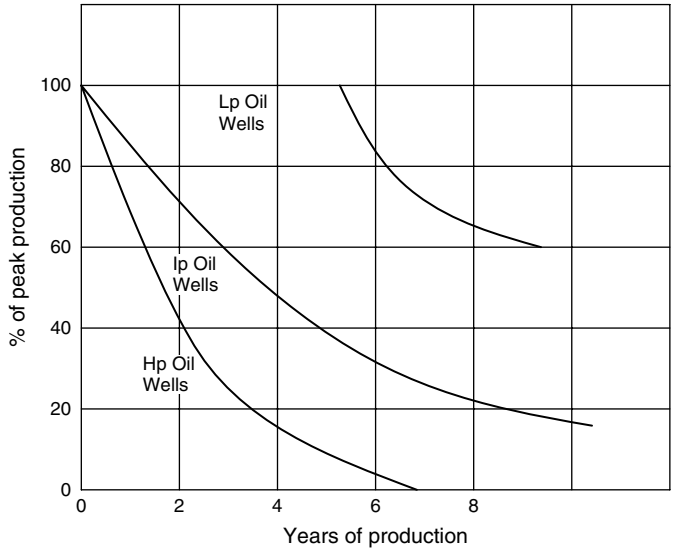


Figure 3-53. Exercise 2—production schedule.

Table 3-13
Percent of Peak Production

Year	HP Wells	IP Wells	LP Wells
0	100.0	100.0	100.0
1	52.5	78.0	100.0
2	35.0	62.5	100.0
3	24.0	51.5	100.0
4	8.0	42.0	100.0
5	2.0	34.0	100.0
6	0	28.0	89.0
7	0	24.0	76.0
8	0	20.0	69.0
9	0	18.0	65.0
10	0	18.0	62.0

The reservoir engineer also provided a "quick look" (Figure 3-54) graph that can be used to estimate the percent of GOR flashed for a given API or stock-tank liquid and separation pressures.

e. Gas lift requirements
When wells drop to low pressure and are gas lifted, they will require 350 MCFD of gas lift gas per well. Based on data received from the

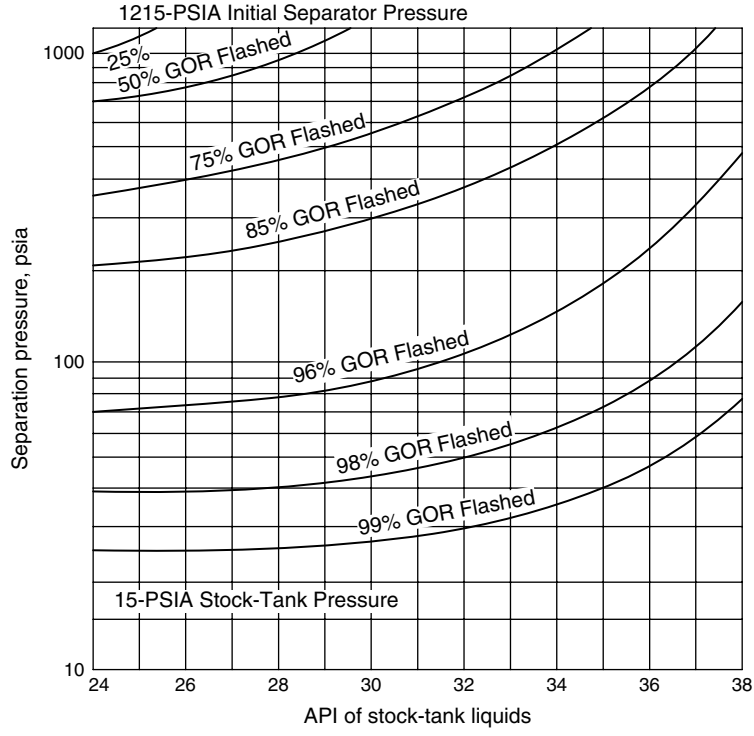


Figure 3-54. Exercise 2—"quick look" estimate of percent GOR flashed for a given API stock-tank liquid and separation pressure.

field reservoir engineer (Figure 3-55), wells will eventually be gas lifted but not initially.

f. Gas sales pressure is 900 psig.
   Assume the maximum pressure drop through the system is 50 psig.
g. Oil pipeline quality is less than 1% BS&W.
   There will be no custody transfer at the facility, but oil will have to be pumped from an atmospheric storage tank (14.7 psig) into an oil pipeline.
h. The authorities having jurisdiction limit the produced water effluent to 15 mg/l of oil in water.
i. There are no sand, foam, or salt problems. The reservoir is sweet; thus, the amounts of H_2S and CO_2 are negligible. Determine:

   1) Number of stages of separation.
   2) Operating pressures for each stage.
   3) Percent of GOR flashed at each stage.

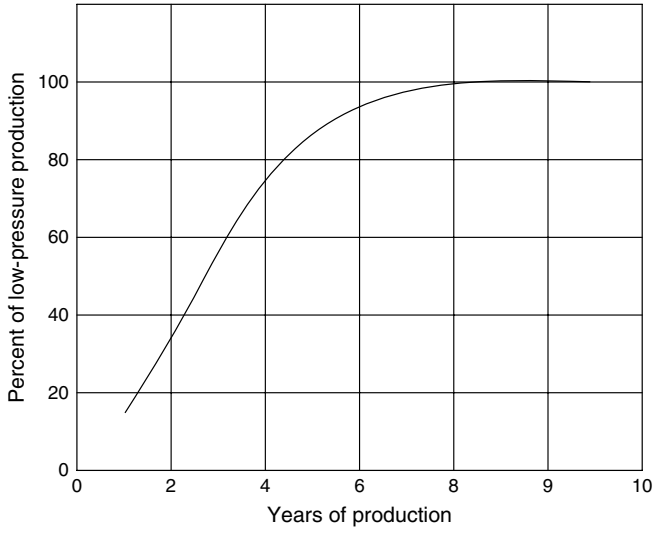


Figure 3-55. Exercise 2—gas lift requirements.

4) Separator type (vertical/horizontal) and phase (two phase/ three phase) for each separator.
5) Gas and liquid capacity, gas viscosity, surge and retention time for each separator.
6) Size of each separator based on your selection criteria.
7) Size of the oil treating system.
8) Size of the water treating system.
9) Draw a process flow diagram of your system illustrating sizes of equipment and pressure, level, temperature, and flow controls.

Problem 3.

Calculate the density at 86 °F and 1015 psia of a gas with the following composition:

Component	Mol %
C_1	80
C_2	15
C_3	5
	100

Problem 4.

Calculate the specific gravity of the gas in problem 3.

Problem 5.

Calculate the density of a sour gas at 100 °F and 2000 psia with the following composition (MW of mixture = 23.12):

Component	Mol %
N_2	9.6
H_2S	3.2
CO_2	12.6
C_1	66.5
C_2	5.1
C_3	1.7
iC_4	0.4
nC_4	0.5
iC_5	0
nC_5	0.2
C_6	0.2
C_{7+}	0
	100

Problem 6.

Calculate the viscosity of the gas stream in problem 5 at 2000 psia and 100 °F.

Problem 7.

Estimate the surface tension of a liquid with an MW = 120 in N/m if the liquid is composed primarily of paraffin hydrocarbons.

Problem 8.

The following data of a producing stream are available:

Q_L = 50 BPD,
Q_g = 20 MMscfd,

$$
\begin{aligned}
T_{\text{Wellhead}} &= 140\,^{\circ}\text{F},\\
P_{\text{Wellhead}} &= 1100\,\text{psia},\\
T_{\text{Reservoir}} &= 196\,^{\circ}\text{F},\\
P_{\text{Reservoir}} &= 6000\,\text{psia},\\
P_{(\text{Cricondenbar})} &= 4500\,\text{psia},\\
T_{(\text{Cridondentherm})} &= 187\,^{\circ}\text{F}.
\end{aligned}
$$

a. Calculate GOR.
b. Draw the above conditions on a graph (reservoir pressure vs. reservoir temperature).

Problem 9.

Identify the following components of this pressure/temperature phase diagram (Figure 3-56):

a) Two-phase region
b) Critical point
c) Dewpoint curve
d) Bubble-point curve
e) Cricondentherm
f) Cricondenbar
g) Liquid phase
h) Vapor phase

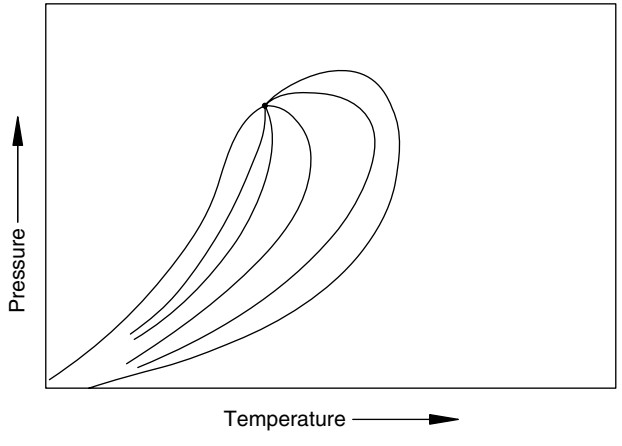


Figure 3-56. Exercise 9—pressure-temperature phase diagram.

Problem 10.

The phase behavior shown on this pressure-temperature diagram (Figure 3-57) is indicative of a (choose one)

a) Lean gas system
b) Rich gas system
c) Retrograde system

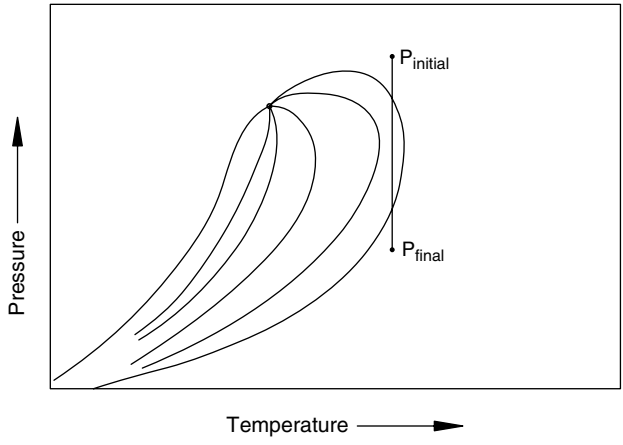


Figure 3-57. Exercise 10—pressure-temperature phase diagram.

References

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Chapter 4
Two-Phase Oil and Gas Separation

Introduction

Produced wellhead fluids are complex mixtures of different compounds of hydrogen and carbon, all with different densities, vapor pressures, and other physical characteristics. As a well stream flows from the hot, high-pressure petroleum reservoir, it experiences pressure and temperature reductions. Gases evolve from the liquids and the well stream changes in character. The velocity of the gas carries liquid droplets, and the liquid carries gas bubbles. The physical separation of these phases is one of the basic operations in the production, processing, and treatment of oil and gas.

In oil and gas separator design, we mechanically separate from a hydrocarbon stream the liquid and gas components that exist at a specific temperature and pressure. Proper separator design is important because a separation vessel is normally the initial processing vessel in any facility, and improper design of this process component can "bottleneck" and reduce the capacity of the entire facility.

Downstream equipment cannot handle gas-liquid mixtures. For example, pumps require gas-free liquid, to avoid cavitation, while compressors and dehydration equipment require liquid-free gas. Product specifications set limits on impurities, such as oil, generally cannot contain more than 1% basic sediment and water (BS&W), while gas sales contracts generally require that gas contain no free liquids. In addition, measurement devices for gases or liquids are highly inaccurate when another phase is present.

Separators are classified as "two-phase" if they separate gas from the total liquid stream and "three-phase" if they also separate the liquid stream into its crude oil and water components. This chapter deals with two-phase separators. In addition, it discusses the requirements of good separation design and how various mechanical devices take advantage of the physical forces in the produced stream to achieve good separation.

Separators are sometimes called "gas scrubbers" when the ratio of gas rate to liquid rate is very high. A "slug catcher," commonly used in gas gathering pipelines, is a special case of a two-phase gas-liquid separator that is designed to handle large gas capacities and liquid slugs. Some operators use the term "traps" to designate separators that handle flow directly from wells. In any case, they all have the same configuration and are sized in accordance with the same procedure.

Phase Equilibrium

Chapter 3 described the phase relationships of a production system through the use of phase equilibrium diagrams. The phase equilibrium diagram is a useful tool to visualize phase behavior. Equilibrium is a theoretical condition that describes an operating system that has reached a "steady-state" condition whereby the vapor is condensing to a liquid at exactly the same rate at which liquid is boiling to vapor. Simply stated, phase equilibrium is a condition where the liquids and vapors have reached certain pressure and temperature conditions at which they can separate. In most production systems, true equilibrium is never actually reached; however, vapors and liquids move through the system slow enough that a "pseudo" or "quasi" equilibrium is assumed. This assumption simplifies process calculations.

Figure 4-1 illustrates several operating points on a generic phase equilibrium diagram. Point A represents the operating pressure and temperature in the petroleum reservoir. Point B represents the flowing conditions at the bottom of the production tubing of a well. Point C represents the flowing conditions at the wellhead. Typically, these conditions are called flowing tubing pressure (FTP) and flowing tubing temperature (FTT). Point D represents the surface conditions at the inlet of the first separator. In Figure 4-1, the reservoir fluid is shown as a liquid; however, reservoir fluids can be either a liquid, a vapor or a mixture of the two depending on the reservoir pressure, temperature, and fluid composition.

As discussed in Chapter 3, flash calculations is a useful tool, if the reservoir composition is known, to create a phase equilibrium diagram that would include determination of the "pseudo" critical pressure and temperature, bubble point, and dew point. Flash calculations are also used to determine the vapor-liquid ratio, which allows one to determine the gas and liquid loads, which in turn are used to size a separator. When the reservoir composition is unknown, precise details about the phase equilibrium diagram cannot be determined and other tools, similar to those discussed in Chapter 3, must be employed to predict the separator loads and size.

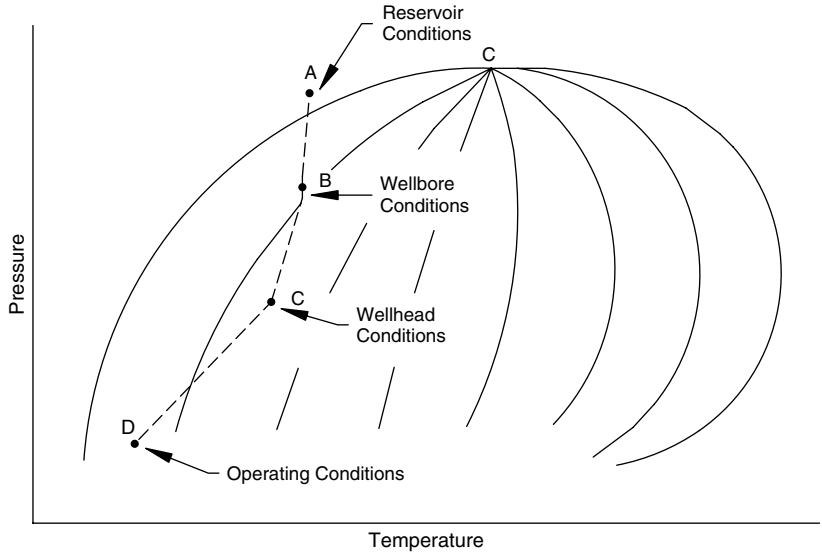


Figure 4-1. Phase equilibrium phase diagram for a typical production system.

Factors Affecting Separation

Characteristics of the flow stream will greatly affect the design and operation of a separator. The following factors must be determined before separator design:

- Gas and liquid flow rates (minimum, average, and peak),
- Operating and design pressures and temperatures,
- Surging or slugging tendencies of the feed streams,
- Physical properties of the fluids such as density and compressibility factor,
- Designed degree of separation (e.g., removing 100% of particles greater than 10 microns),
- Presence of impurities (paraffin, sand, scale, etc.),
- Foaming tendencies of the crude oil,
- Corrosive tendencies of the liquids or gas.

Functional Sections of a Gas-Liquid Separator

Regardless of the size or shape of a separator, each gas-liquid separator contains four major sections. Figures 4-2 and 4-3 illustrate the four major sections of a horizontal and vertical two-phase separator, respectively.

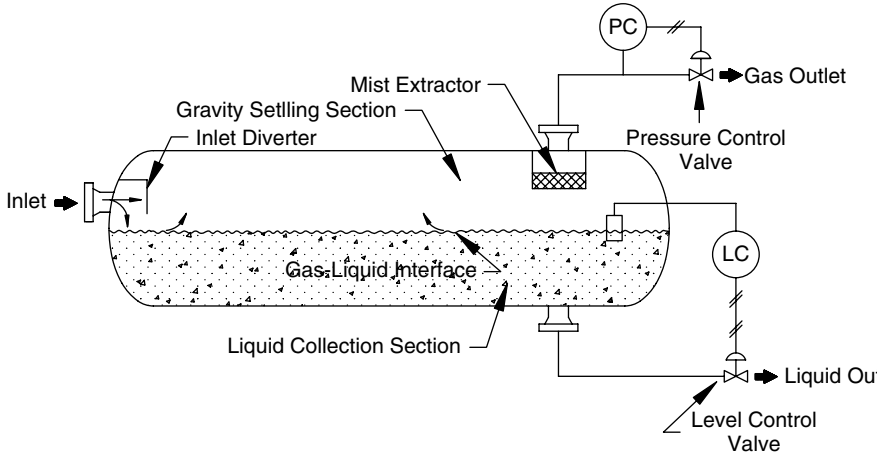


Figure 4-2. Horizontal separator schematic.

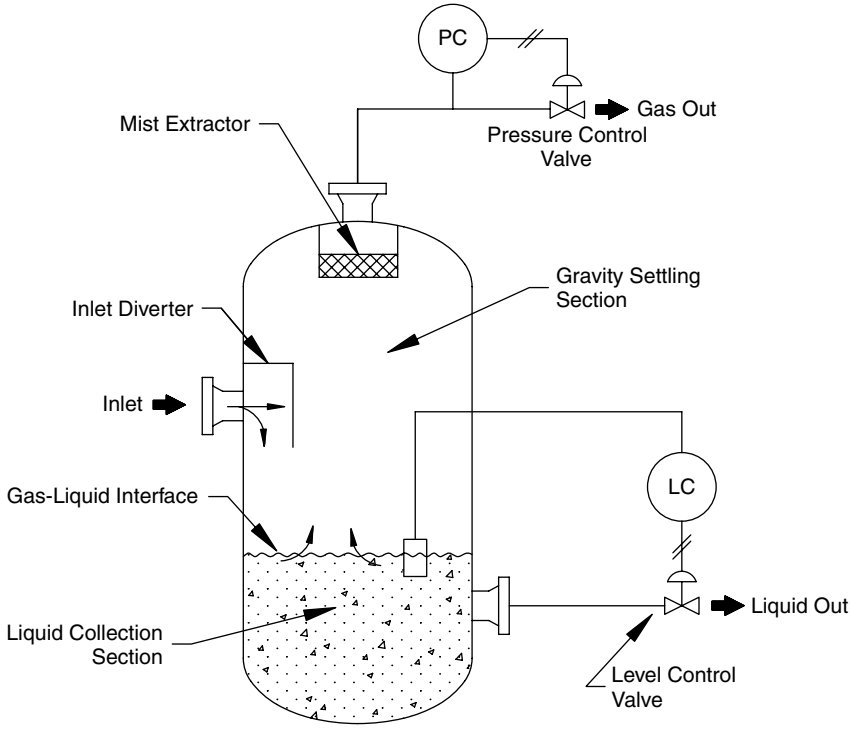


Figure 4-3. Vertical separator schematic.

Inlet Diverter Section

The inlet stream to the separator is typically a high-velocity turbulent mixture of gas and liquid. Due to the high velocity, the fluids enter the separator with a high momentum. The inlet diverter, sometimes referred to as the primary separation section, abruptly changes the direction of flow by absorbing the momentum of the liquid and allowing the liquid and gas to separate. This results in the initial "gross" separation of liquid and gas.

Liquid Collection Section

The liquid collection section, located at the bottom of the vessel, provides the required retention time necessary for any entrained gas in the liquid to escape to the gravity settling section. In addition, it provides a surge volume to handle intermittent slugs. The degree of separation is dependent on the retention time provided. Retention time is affected by the amount of liquid the separator can hold, the rate at which the fluids enter the vessel, and the differential density of the fluids. Liquid-liquid separation requires longer retention times than gas-liquid separation.

Gravity Settling Section

As the gas stream enters the gravity settling section, its velocity drops and small liquid droplets that were entrained in the gas and not separated by the inlet diverter are separated out by gravity and fall to the gas-liquid interface. The gravity settling section is sized so that liquid droplets greater than 100 to 140 microns fall to the gas-liquid interface while smaller liquid droplets remain with the gas. Liquid droplets greater than 100 to 140 microns are undesirable as they can overload the mist extractor at the separator outlet.

Mist Extractor Section

Gas leaving the gravity settling section contains small liquid droplets, generally less than 100 to 140 microns. Before the gas leaves the vessel, it passes through a coalescing section or mist extractor. This section uses coalescing elements that provide a large amount of surface area used to coalesce and remove the small droplets of liquid. As the gas flows through the coalescing elements, it must make numerous directional changes. Due to their greater mass, the liquid droplets cannot follow the rapid changes

in direction of flow. These droplets impinge and collect on the coalescing elements, where they fall to the liquid collection section.

Equipment Description

Separators are designed and manufactured in horizontal, vertical, spherical, and a variety of other configurations. Each configuration has specific advantages and limitations. Selection is based on obtaining the desired results at the lowest "life-cycle" cost.

Horizontal Separators

Figure 4-4 is a cutaway of a horizontal two-phase separator. The fluid enters the separator and hits an inlet diverter, causing a sudden change in momentum. The initial gross separation of liquid and vapor occurs at the inlet diverter. The force of gravity causes the liquid droplets to fall out of the gas stream to the bottom of the vessel, where it is collected. The liquid collection section provides the retention time required to let entrained gas evolve out of the oil and rise to the vapor space and reach a state of "equilibrium." It also provides a surge volume, if necessary, to handle intermittent slugs of liquid. The liquid leaves the vessel through the liquid dump valve. The liquid dump valve is regulated by a level controller. The level controller senses changes in liquid level and controls the dump valve accordingly.

Gas and oil mist flow over the inlet diverter and then horizontally through the gravity settling section above the liquid. As the gas flows through this section, small droplets of liquid that were entrained in the

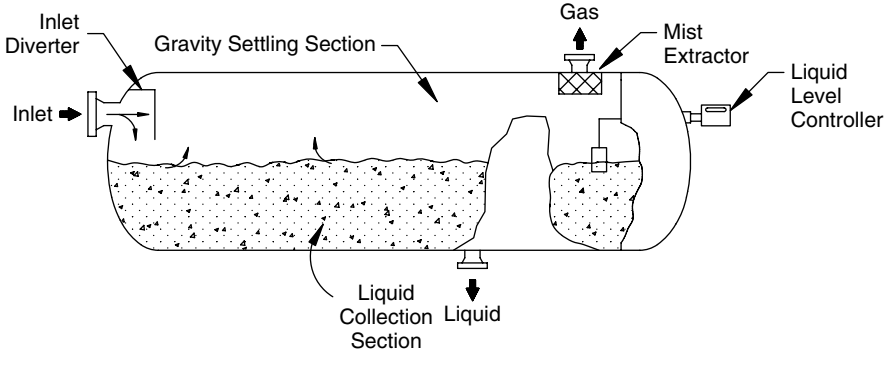


Figure 4-4. Cutaway view of a horizontal two-phase separator.

gas and not separated by the inlet diverter are separated out by gravity and fall to the gas-liquid interface.

Some of the drops are of such a small diameter that they are not easily separated in the gravity settling section. Before the gas leaves the vessel, it passes through a coalescing section or mist extractor. This section uses elements of vanes, wire mesh, or plates to provide a large amount of surface area used to coalesce and remove the very small droplets of liquid in one final separation before the gas leaves the vessel.

The pressure in the separator is maintained by a pressure controller mounted on the gas outlet. The pressure controller senses changes in the pressure in the separator and sends a signal to either open or close the pressure control valve accordingly. By controlling the rate at which gas leaves the vapor space of the vessel, the pressure in the vessel is maintained. Normally, horizontal separators are operated half full of liquid to maximize the surface area of the gas-liquid interface.

Horizontal separators are smaller and thus less expensive than a vertical separator for a given gas and liquid flow rate. Horizontal separators are commonly used in flow streams with high gas-liquid ratios and foaming crude.

Vertical Separators

Figure 4-5 is a cutaway of a vertical two-phase separator. In this configuration the inlet flow enters the vessel through the side. As in the horizontal separator, the inlet diverter does the initial gross separation. The liquid flows down to the liquid collection section of the vessel. There are seldom any internals in the liquid collection section except possibly a still well for the level control float or displacer. The still well usually consists of walled box or tube, open on the top and bottom. Its function is to stop wave action in the separator from interfering with the level controller's operation. Liquid continues to flow downward through this section to the liquid outlet. As the liquid reaches equilibrium, gas bubbles flow counter to the direction of the liquid flow and eventually migrate to the vapor space. The level controller and liquid dump valve operate the same as in a horizontal separator.

The gas flows over the inlet diverter and then vertically upward toward the gas outlet. Secondary separation occurs in the upper gravity settling section. In the gravity settling section the liquid droplets fall vertically downward counter-current to the upward gas flow. The settling velocity of a liquid droplet is directly proportional to its diameter. If the size of a liquid droplet is too small, it will be carried up and out with the vapor. Thus, a mist extractor section is added to capture small liquid droplets.

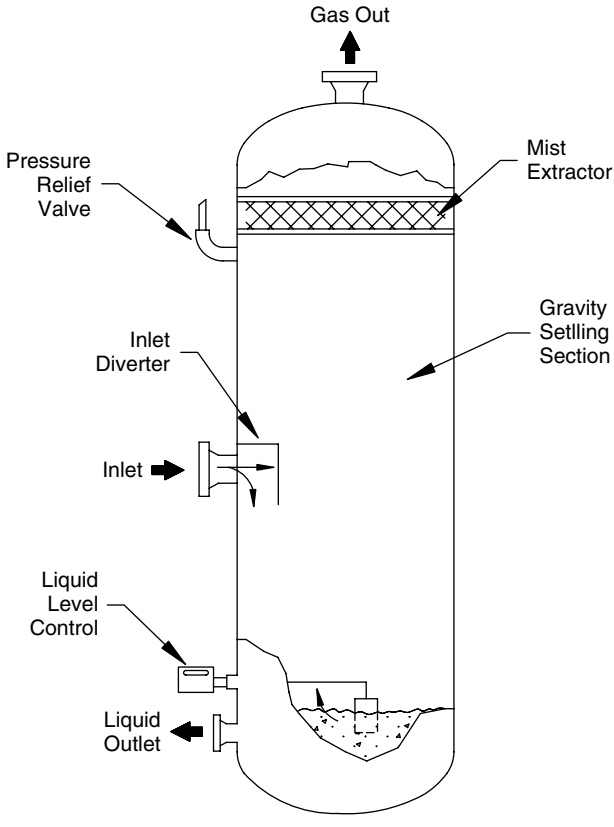


Figure 4-5. Cutaway view of a vertical two-phase separator.

Gas goes through the mist extractor section before it leaves the vessel. Pressure and level are maintained as in a horizontal separator.

Vertical separators are commonly used in flow streams with low to intermediate gas-liquid ratios. They are well suited for production containing sand and other sediment and thus are often fitted with a false cone bottom to handle sand production.

Spherical Separators

A typical spherical separator is shown in Figure 4-6. The same four sections can be found in this vessel. Spherical separators are a special case of a vertical separator where there is no cylindrical shell between the two heads. Fluid enters the vessel through the inlet diverter where the

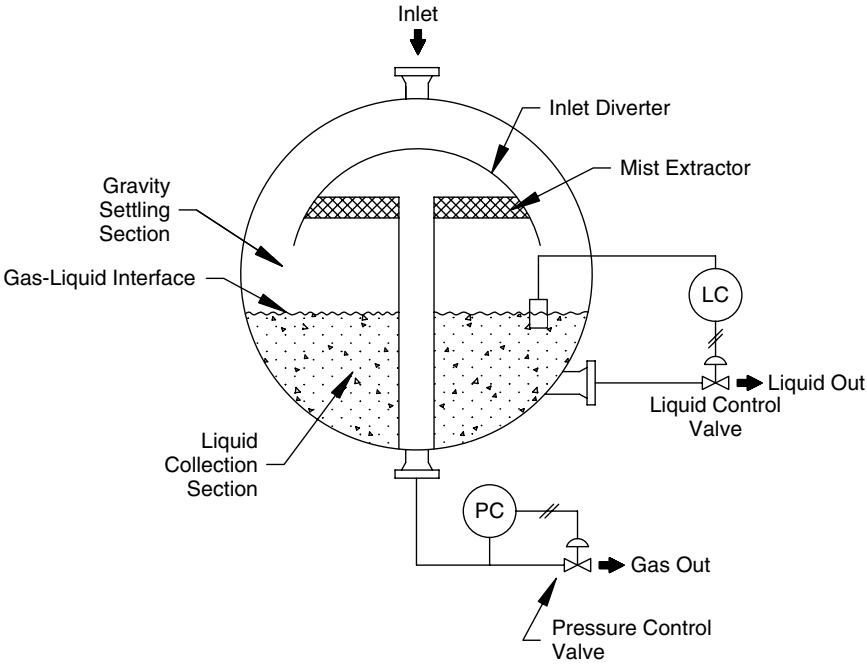


Figure 4-6. Spherical separator schematic.

flow stream is split into two streams. Liquid falls to the liquid collection section, through openings in a horizontal plate located slightly below the gas-liquid interface. The thin liquid layer across the plate makes it easier for any entrained gases to separate and rise to the gravity settling section. Gases rising out of the liquids pass through the mist extractor and out of the separator through the gas outlet. Liquid level is maintained by a float connected to a dump valve. Pressure is maintained by a back pressure control valve while the liquid level is maintained by a liquid dump valve.

Spherical separators were originally designed to take advantage, theoretically, of the best characteristics of both horizontal and vertical separators. In practice, however, these separators actually experienced the worst characteristics and are very difficult to size and operate. They may be very efficient from a pressure containment standpoint, but because (1) they have limited liquid surge capability and (2) they exhibit fabrication difficulties, they are seldom used in oil field facilities. For this reason we will not be discussing spherical separators any further.

Centrifugal Separators

Centrifugal separators, sometimes referred to as cylindrical cyclone separators (CCS), work on the principle that droplet separation can be enhanced by the imposition of a radial or centrifugal force. This centrifugal force may range from 5 times the gravitational force in large-diameter units, to 2,500 times the gravitational force in small, high-pressure units.

As shown in Figure 4-7, the centrifugal separator consists of three major sections: inclined tangential inlet, tangential liquid outlet, and axial gas outlet. The basic flow pattern involves a double vortex, with the gas spiraling downward along the wall, and then upward in the center. The spiral velocity in the separator may reach several times the inlet velocity. The flow patterns are such that the radial velocities are directed toward the walls, thus causing droplets to impinge on the vessel walls, and run down to the bottom of the unit.

The units are designed to handle liquid flow rates between 100 to 50,000 bpd in sizes ranging from 2 to 12 inches in diameter. Centrifugal separators are designed to provide bulk gas-liquid separation. They are

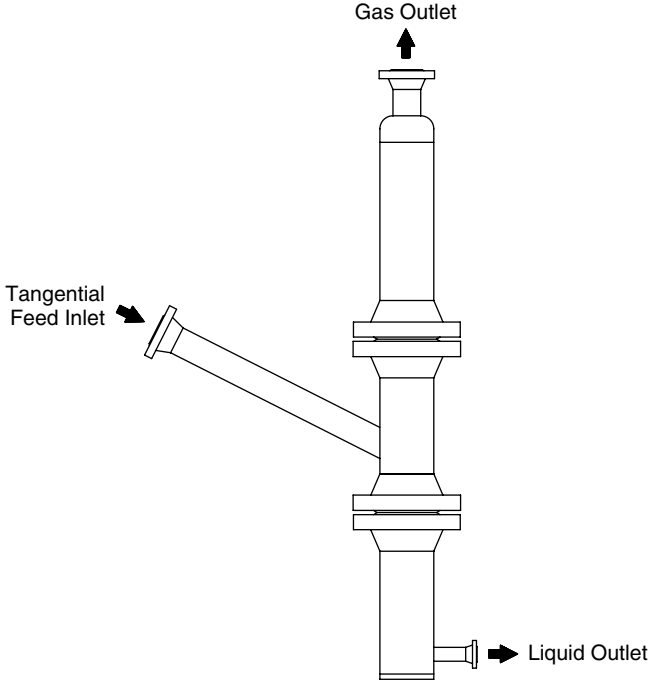


Figure 4-7. Cylindrical cyclone separator.

best suited for fairly clean gas streams. Fluids are introduced tangentially into the separator via an inclined feed pipe. The high-velocity swirling flow creates a radial acceleration field that causes the gas to flow to the axial core region due to differences in gas and liquid density. The gas exits through an axial outlet located at the top of the separator, and the liquid leaves through a tangential outlet at the bottom. The feed pipe is inclined at an optimal angle to stratify the inlet flow phases and preferentially direct the liquid flow toward the liquid outlet. To obtain optimal separation performance, the separator requires the liquid level to be maintained within a particular range, which is usually just below the inlet level. The method of level control is dependent on the application, that is, phase composition and location within the process. Control can be achieved by a control valve on either the liquid or the gas outlet lines, or in some applications a level control valve on the liquid outlet line and a pressure control valve on the gas outlet line.

The major benefits of centrifugal separators are: no moving parts; low maintenance; compact, in terms of space and weight; insensitive to motion; and low cost when compared to conventional separator technology. Although such designs can result in significantly smaller sizes, they are not commonly used in production operations because (1) their design is rather sensitive to flow rate and (2) they require greater pressure drop than the standard configurations previously described. Since separation efficiency decreases as velocity decreases, the centrifugal separator is not suitable for widely varying flow rates. These units are commonly used to recover glycol carryover downstream of a glycol contact tower. In recent years, demand for using centrifugal separators on floating production facilities has increased because space and weight considerations are overriding on such facilities. The design of these separators is proprietary and, therefore, will not be covered.

Venturi Separators

Like the centrifugal, the venturi separator increases droplet coalescence by introducing additional forces into the system. Instead of centrifugal forces, the venture acts on the principle of accelerating the gas linearly through a restricted flow path with a motive fluid to promote the coalescence of droplets.

Venturi separators are normally best suited for applications that contain a mixture of solids and liquids. They are not normally cost-effective for removing liquid entrainment alone, because of the high-pressure drop and need for a motive fluid. Even with solids present, the baffle-type

units are more suitable for entrained particulates down to 15 microns in diameter.

Double-Barrel Horizontal Separators

Figure 4-8 illustrates a double-barrel horizontal separator, which is a variation of the horizontal separator. Double-barrel horizontal separators are commonly used in applications where there are high gas flow rates and where there is a possibility of large liquid slugs, e.g., slug catchers. Single-barrel horizontal separators can handle large gas flow rates but offer poor liquid surge capabilities. The double-barrel horizontal separator partially alleviates this shortcoming. In these designs the gas and liquid chambers are separated as shown in Figure 4-8. The flow stream enters the vessel in the upper barrel and strikes the inlet diverter. The gas flows through the gravity settling section, where it encounters the baffle-type mist extractors en route to the gas outlet. Figure 4-9 is a cutaway view of a double-barrel separator fitted with a baffle-type mist extractor.

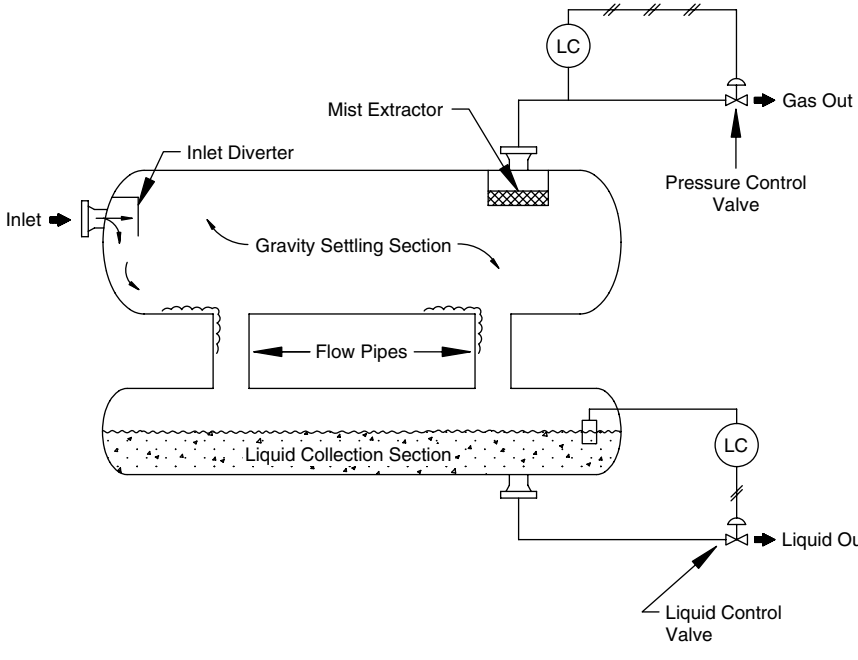


Figure 4-8. Double-barrel horizontal separator.

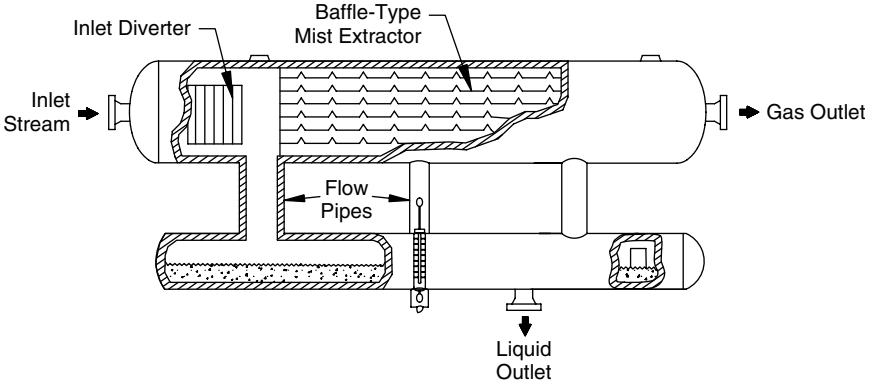


Figure 4-9. Cutaway view of a horizontal double-barrel separator fitted with a baffle-type mist extractor in the gravity settling section.

The baffles help the free liquids to fall to the lower barrel through flow pipes. The liquids drain through a flow pipe or equalizing tube into the lower barrel. Small amounts of gas entrained in the liquid are liberated in the liquid collection barrel and flow up through the flow pipes or equalizing tubes. In this manner the liquid accumulation is separated from the gas stream so that there is no chance of high gas velocities re-entraining liquid as it flows over the interface. Because of their additional cost, and the absence of problems with single-vessel separators, they are not widely used in oil field systems. However, in gas handling, conditioning, and processing systems, two-barrel separators are typically used as gas scrubbers on the inlet of compressors, glycol contact towers, and gas treating systems where the liquid flow rate is extremely low relative to the gas flow rate.

Horizontal Separator with a "Boot" or "Water Pot"

Figure 4-10 shows a special case of a two-barrel separator. It is a single-barrel separator with a liquid "boot" or "water pot" at the outlet end. The main body of the separator operates essentially dry as in a two-barrel separator. The small amounts of liquid in the bottom flow to the boot end, which provides the liquid collection section. These vessels are less expensive than two-barrel separators, but they also contain less liquid handling capability. It is used when there are very low liquid flow rates, especially where the flow rates are low enough that the "boot" can serve as a liquid-liquid separator as well.

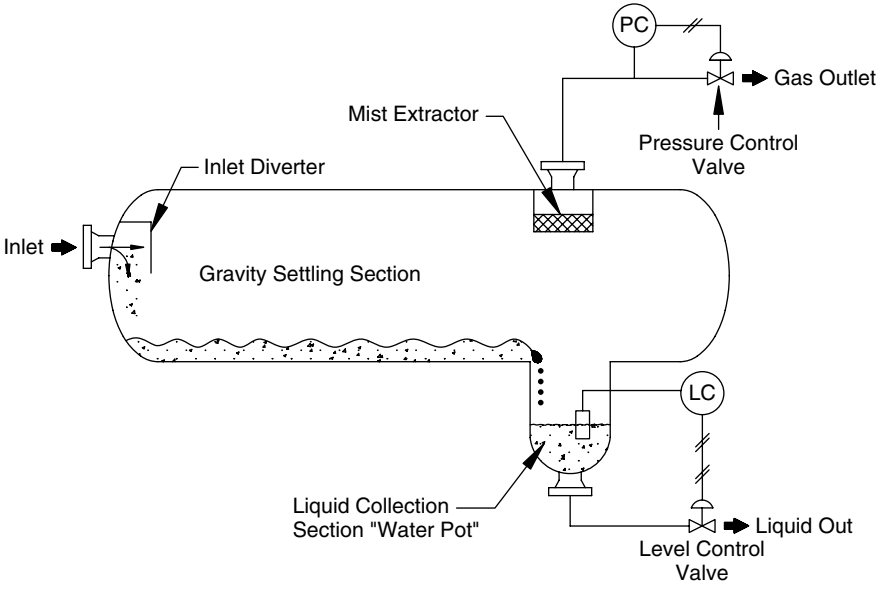


Figure 4-10. Single-barrel horizontal separator with a liquid "boot."

Filter Separators

Another type of separator that is frequently used in some high-gas/low-liquid flow applications is a filter separator. They can be either horizontal or vertical in configuration. Filter separators are designed to remove small liquid and solid particles from the gas stream. These units are used in applications where conventional separators employing gravitational or centrifugal force are ineffective. Figure 4-11 shows a horizontal two-barrel filter separator design. Filter tubes in the initial separation section cause coalescence of any liquid mist into larger droplets as the gas passes through the tubes. A secondary section of vanes or other mist extractor elements removes these coalesced droplets. Filter separators are commonly used on compressor inlets in field compressor stations, final scrubbers upstream of glycol contact towers, and instrument/fuel gas applications. The design of filter separators is proprietary and dependent upon the type of filter element employed. Some filter elements can remove 100% of 1-micron particles and 99% of 1/2-micron particles when they are operated at rated capacity and recommended filter-change intervals.

Figure 4-12 shows a typical filter element, which consists of a perforated metal cylinder with gasketed ends for compression sealing. A fiberglass cylinder, typical 1/2-inch (1.25-cm) thick, surrounds the perforated metal

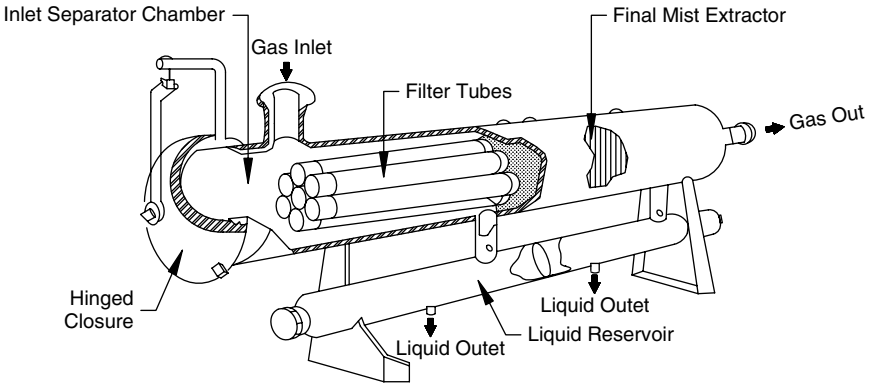


Figure 4-11. Typical horizontal two-barrel filter separator.

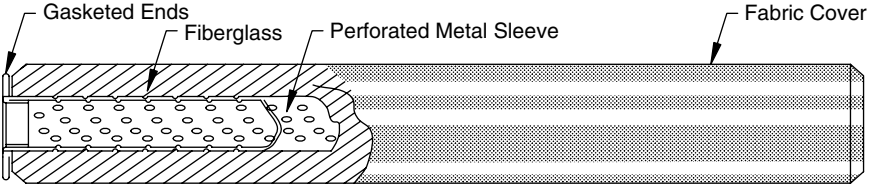


Figure 4-12. Typical filter element.

cylinder. Gas flow is from outside the fiberglass cylinder to the center of the perforated metal tube. A micron fiber fabric layer is located on both sides of the fiberglass cylinder. This prevents migration of fiberglass fibers into the gas stream. The filter elements are securely held over openings in the vessel tube sheet by a center rod. This rod centers each element over its tube-sheet opening and provides the compression for sealing the element between the tube sheet and plate, which closes the opposite end.

In applications where there is very little liquid flow, often a horizontal separator will be designed with a liquid sump on the outlet end to provide the required liquid retention time. This results in an overall smaller diameter for the vessel.

Scrubbers

A scrubber is a two-phase separator that is designed to recover liquids carried over from the gas outlets of production separators or to catch liquids condensed due to cooling or pressure drops. Liquid loading in

a scrubber is much lower than that in a separator. Typical applications include: upstream of mechanical equipment such as compressors that could be damaged, destroyed or rendered ineffective by free liquid; downstream of equipment that can cause liquids to condense from a gas stream (such as coolers); upstream of gas dehydration equipment that would lose efficiency, be damaged, or be destroyed if contaminated with liquid hydrocarbons; and upstream of a vent or flare outlet.

Vertical scrubbers are most commonly used. Horizontal scrubbers can be used, but space limitations usually dictate the use of a vertical configuration.

Slug Catchers

A "slug catcher," commonly used in gas gathering pipelines, is a special case of a two-phase gas-liquid separator that is designed to handle large gas capacities and liquid slugs on a regular basis. Since the gathering systems are designed to handle primarily gas, the presence of liquid restricts flow and causes excessive pressure drop in the piping. Pigging is periodically used to sweep the lines of liquids. When the pigs sweep the liquids out of the gathering lines, large volumes of liquids must be handled by the downstream separation equipment. The separators used in this service are called slug catchers. There are numerous slug catcher designs. Figure 4-13 is a schematic of a two-phase horizontal slug catcher with liquid "fingers." Gas and liquid slug from the gathering system enters the horizontal portion of the two-phase vessel, where primary gas-liquid separation is accomplished. Gas exits the top of the separator through the mist extractor while the liquid exits the bottom of the vessel through a series of large-diameter tubes or "fingers." The tubes provide a large liquid holding volume and routes the liquid to a three-phase free-water knockout (FWKO) for further liquid-liquid separation. The design of an FWKO is discussed in Chapter 5.

Selection Considerations

The geometry of and physical and operating characteristics give each separator type advantages and disadvantages. Horizontal separators are smaller, more efficient at handling large volumes of gas, and less expensive than vertical separators for a given gas capacity. In the gravity settling section of a horizontal vessel, the liquid droplets fall perpendicular to the gas flow and thus are more easily settled out of the gas continuous phase. Also, since the interface area is larger in a horizontal separator than a

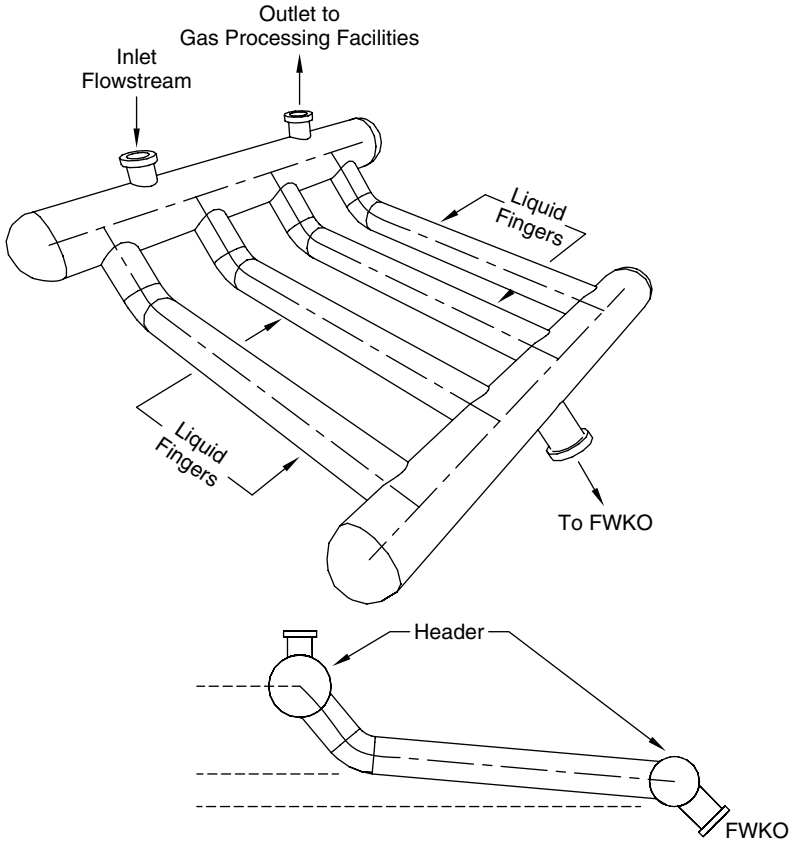


Figure 4-13. Schematic of a two-phase horizontal slug catcher with liquid "fingers."

vertical separator, it is easier for the gas bubbles, which come out of solution as the liquid approaches equilibrium, to reach the vapor space. Horizontal separators offer greater liquid capacity and are best suited for liquid-liquid separation and foaming crude.

Thus, from a pure gas/liquid separation process, horizontal separators would be preferred. However, they do have the following drawbacks, which could lead to a preference for a vertical separator in certain situations:

1. Horizontal separators are not as good as vertical separators in handling solids. The liquid dump line of a vertical separator can be placed at the center of the bottom head so that solids will not build up in the separator but continue to the next vessel in the process. As

an alternative, a drain could be placed at this location so that solids could be disposed of periodically while liquid leaves the vessel at a slightly higher elevation.

2. In a horizontal vessel, it is necessary to place several drains along the length of the vessel. Since the solids will have an angle of repose of 45° to 60°, the drains must be spaced at very close intervals. Attempts to lengthen the distance between drains, by providing sand jets in the vicinity of each drain to fluidize the solids while the drains are in operation, are expensive and have been only marginally successful in field operations.
3. Horizontal vessels require more plan area to perform the same separation as vertical vessels. While this may not be of importance at a land location, it could be very important offshore. If several separators are used, however, this disadvantage may be overcome by stacking one horizontal separator on top of each other.
4. The ability of a separator to absorb a slug of liquid is called the surge capacity of a separator. Smaller, horizontal vessels can have less liquid surge capacity than vertical vessels sized for the same steady-state flow rate. For a given change in liquid surface elevation, there is typically a larger increase in liquid volume for a horizontal separator than for a vertical separator sized for the same flow rate. However, the geometry of a horizontal vessel causes any high-level shut-down device to be located close to the normal operating level. In very large diameter [greater than 6 ft (1.8 m)] horizontal vessels and in vertical vessels, the shut-down device could be placed much higher, allowing the level controller and dump valve more time to react to the surge. In addition, surges in horizontal vessels could create internal waves, which could activate a high-level sensor prematurely.

It should be pointed out that vertical vessels also have some drawbacks that are not process related and must be considered in making a selection. These are as follows:

1. The relief valve and some of the controls may be difficult to service without special ladders and platforms.
2. The vessel may have to be removed from a skid for trucking due to height restrictions.

Generally, horizontal separators are less expensive than equally sized vertical separators. Since vertical separators are supported only by the

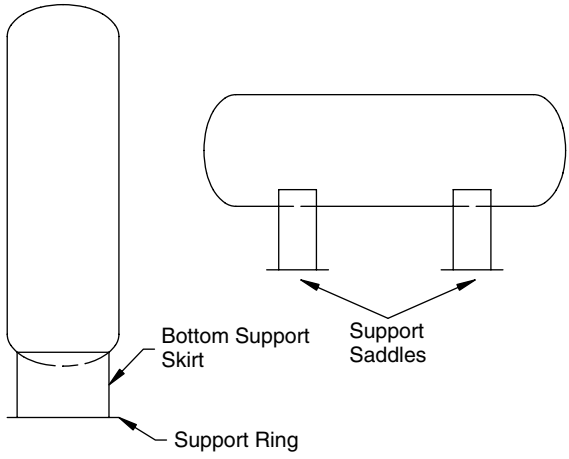


Figure 4-14. Comparison of vertical and horizontal support structures.

bottom skirt (refer to Figure 4-14), the walls of vertical separators must be somewhat thicker than a similarly sized and rated horizontal separator, which may be supported by saddles. Furthermore, large vertical separators, when exposed to high winds, can be subjected to large lateral (wind) loads. When this is the case, the vertical separator's wall thickness must be increased, which in turn increases the cost of the overall vessel.

The same discussion regarding gas capacity applies equally to the double-barrel horizontal separator. The addition of the second barrel increases the vessel's surge capacity.

Spherical separators have more gas capacity than similarly sized vertical separators but less than similarly sized horizontal separators. They have less surge capacity than similarly sized horizontal separators. Installation and operation of level controls on spherical separators are difficult. Few spherical separators are still in existence today.

Overall, horizontal vessels are the most economical for normal oil-gas separation, particularly where there may be problems with emulsions, foam, or high gas-oil ratios (GOR). Vertical vessels work most effectively in low-GOR applications. They are also used in some very high GOR applications, such as scrubbers where only fluid mists are being removed from the gas and where extra surge capacity is needed to allow shutdown to activate before liquid is carried out the gas outlet (e.g., compressor suction scrubber).

Vessel Internals

Inlet Diverters

Inlet diverters serve to impart flow direction of the entering vapor/liquid stream and provide primary separator between the liquid and vapor. There are many types of inlet diverters. Three main types are baffle plates (shown in Figure 4-15), centrifugal diverters (shown in Figure 4-16), and elbows (shown in Figure 4-17).

A baffle plate can be a spherical dish, flat plate, angle iron, cone, elbow, or just about anything that will accomplish a rapid change in direction and velocity of the fluids and thus disengage the gas and liquid. At the same velocity the higher-density liquid possesses more energy and, thus, does not change direction or velocity as easily as the gas. Thus, the gas tends to flow around the diverter while the liquid strikes the diverter and then falls to the bottom of the vessel. The design of the baffles is governed principally by the structural supports required to resist the impact-momentum load. The advantage of using devices such as a half-sphere elbow or cone is that they create less disturbance than plates or angle iron, cutting down on re-entrainment or emulsifying problems.

Centrifugal inlet diverters use centrifugal force, rather than mechanical agitation, to disengage the oil and gas. These devices can have a cyclonic chimney or may use a tangential fluid race around the walls (refer to Figure 4-18). Centrifugal inlet diverters are proprietary but generally use an inlet nozzle sufficient to create a fluid velocity of about 20 f/s (6 m/s) around a chimney whose diameter is no longer than two-thirds that of the vessel diameter. Centrifugal diverters can be designed to efficiently separate the liquid while minimizing the possibility of foaming or emulsifying

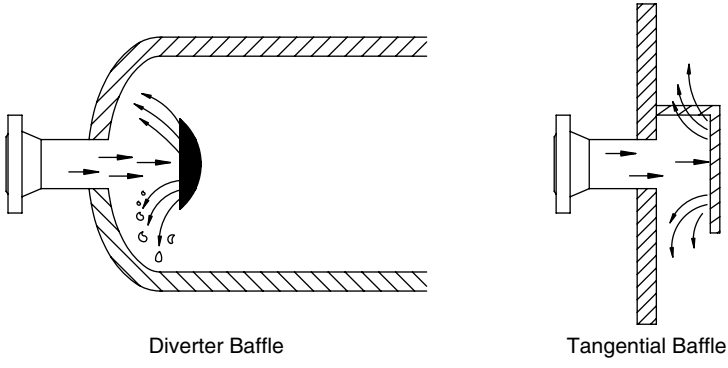


Figure 4-15. Baffle plates.

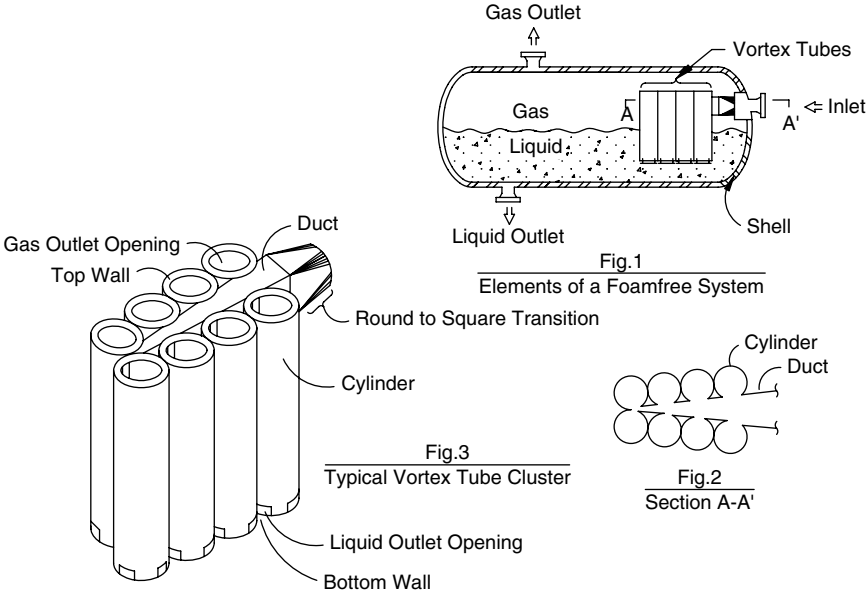


Figure 4-16. Three views of an example centrifugal inlet diverter. (Courtesy of Porta-Test Systems, Inc.)

problems. The disadvantage is that their design is rate sensitive. At low velocities they will not work properly. Thus, they are not normally recommended for producing operations where rates are not expected to be steady.

Wave Breakers

In long horizontal vessels, usually located on floating structures, it may be necessary to install wave breakers. The waves may result from surges of liquids entering the vessel. Wave breakers are nothing more than perforated baffles or plates that are placed perpendicular to the flow located in the liquid collection section of the separator. These baffles dampen any wave action that may be caused by incoming fluids. The wave action in the vessel must be maintained so that liquid level controllers, level safety switches, and weirs perform properly. On floating or compliant structures where internal waves may be set up by the motion of the foundation, wave breakers may also be required perpendicular to the flow direction. The wave actions in the vessel must be eliminated so level controls, level switches, and weirs may perform properly. Figure 4-19

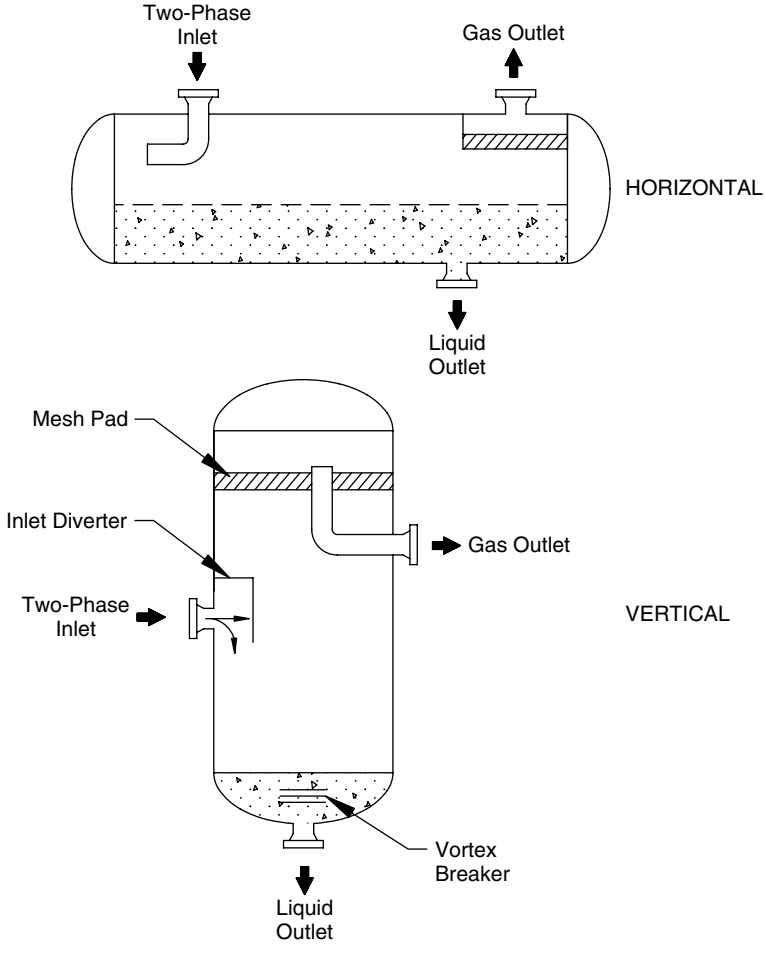


Figure 4-17. Elbow inlet diverter.

is a three-dimensional view of a horizontal separator fitted with an inlet diverter, de-foaming element, mist extractor, and wave breakers.

Defoaming Plates

Foam at the interface may occur when gas bubbles are liberated from the liquid. Foam can severely degrade the performance of a separator. This foam can be stabilized with the addition of chemicals at the inlet. Many times a more effective solution is to force the foam to pass through

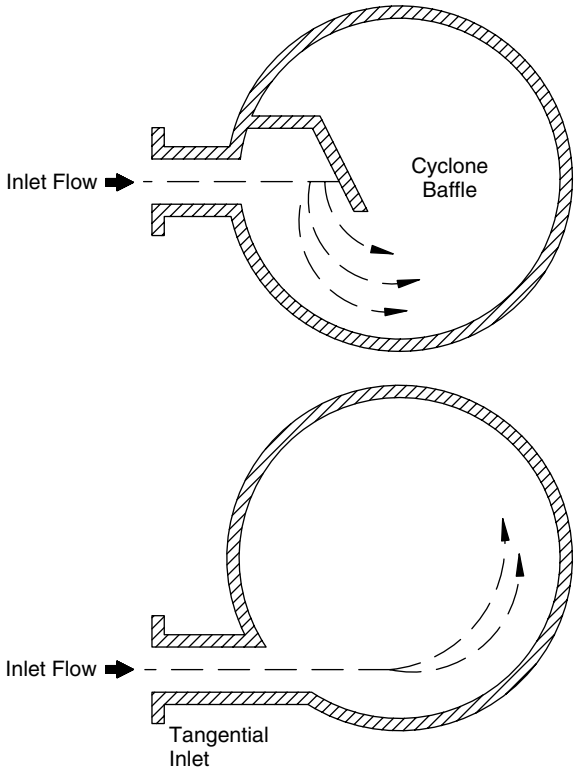


Figure 4-18. Centrifugal inlet diverters. (Top) Cyclone baffle. (Bottom) Tangential raceway.

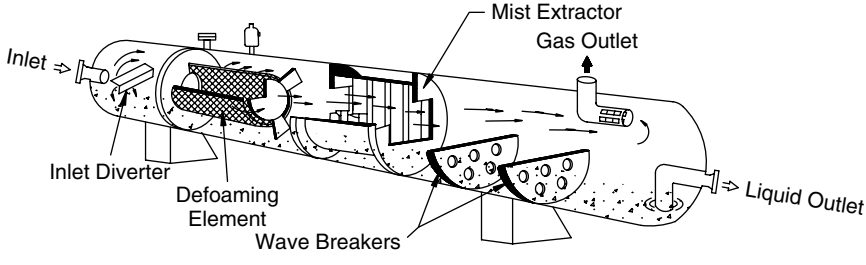


Figure 4-19. Three-dimensional view of a horizontal separator fitted with an inlet diverter, defoaming element, mist extractor, and wave breaker.

a series of inclined parallel plates or tubes as shown in Figure 4-20. These closely spaced, parallel plates or tubes provide additional surface area, which breaks up the foam and allows the foam to collapse into the liquid layer.

Vortex Breaker

Liquid leaving a separator may form vortices or whirlpools, which can pull gas down into the liquid outlet. Therefore, horizontal separators are often equipped with vortex breakers, which prevent a vortex from developing when the liquid control valve is open. A vortex could suck some gas out of the vapor space and re-entrain it in the liquid outlet. One type of vortex breaker is shown in Figure 4-21. It is a covered cylinder with radially directed flat plates. As liquid enters the bottom of the vortex breaker, any circular motion is prevented by the flat plates. Any tendency to form vortices is removed. Figure 4-22 illustrates other commonly used vortex breakers.

Stilling Well

A stilling well, which is simply a slotted pipe fitting surrounding an internal level control displacer, protects the displacer from currents, waves,

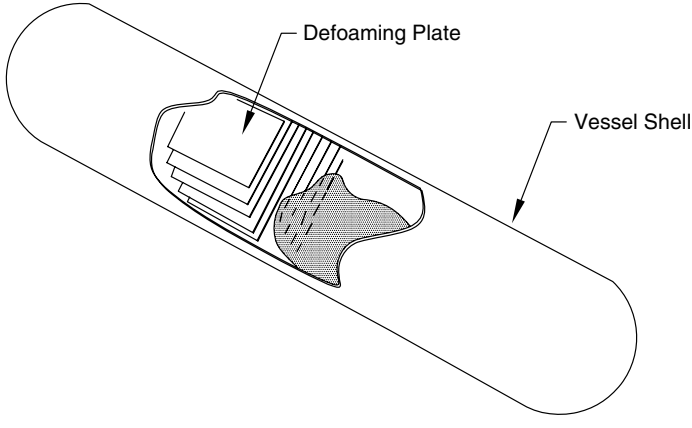


Figure 4-20. Defoaming plates.

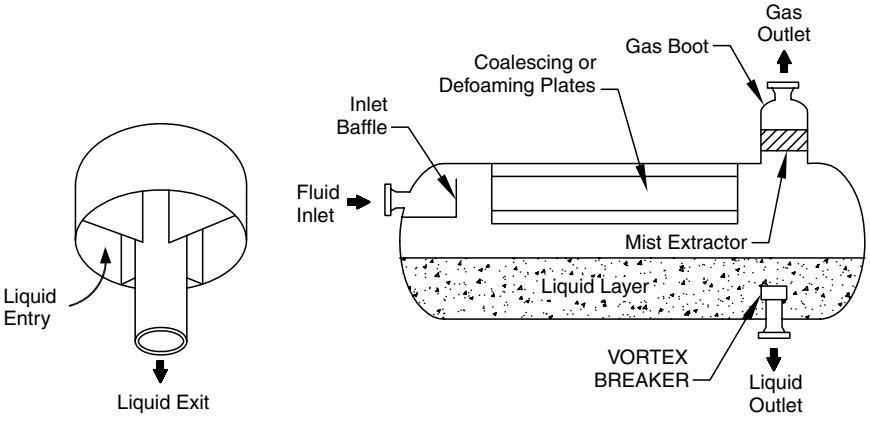


Figure 4-21. Vortex breaker.

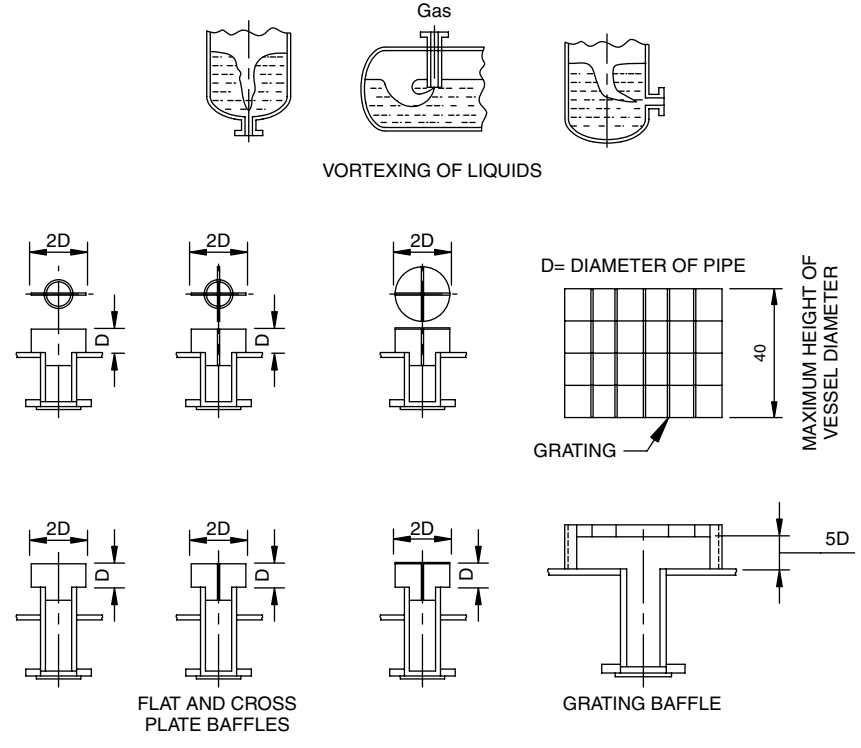


Figure 4-22. Typical vortex breakers.

and other disturbances that could cause the displacer to sense an incorrect level measurement.

Sand Jets and Drains

In horizontal separators, one worry is the accumulation of sand and solids at the bottom of the vessel. If allowed to build up, these solids will upset the separator operations by taking up vessel volume. Generally, the solids settle to the bottom and become well packed.

To remove the solids, sand drains are opened in a controlled manner, and then high-pressure fluid, usually produced water, is pumped through the jets to agitate the solids and flush them down the drains. The sand jets are normally designed with a 20-ft/s (6-m/s) jet tip velocity and aimed in such a manner to give good coverage of the vessel bottom. To prevent the settled sand from clogging the sand drains, sand pans or sand troughs are used to cover the outlets. These are inverted troughs with slotted side openings (refer to Figure 4-23). To assure proper solids removal without upsetting the separation process, an integrated system, consisting of a drain and its associated jets, should be installed at intervals not exceeding 5 ft (1.5 m). Field experience indicates it is not possible to mix and fluff the bottom of a long horizontal vessel with a single sand jet header.

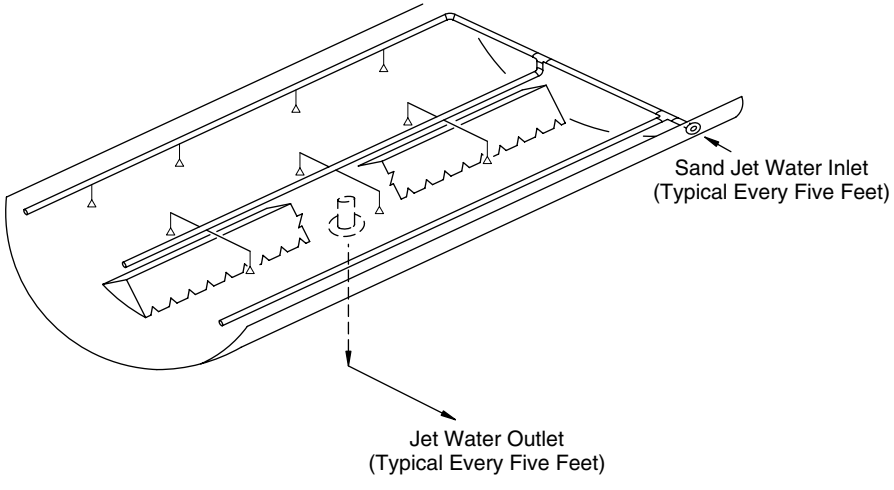


Figure 4-23. Schematic of a horizontal separator fitted with sand jets and inverted trough.

Mist Extractors

Introduction

There are many types of equipment, known as mist extractors or mist eliminators, designed to remove the liquid droplets and solid particles from the gas stream. Before a selection can be made, one must evaluate the following factors:

- Size of droplets the separator must remove
- Pressure drop that can be tolerated in achieving the required level of removal
- Susceptibility of the separator to plugging by solids, if solids are present
- Liquid handling capability of the separator
- Whether the mist extractor/eliminator can be installed inside existing equipment, or if it requires a standalone vessel instead
- Availability of the materials of construction that are comparable with the process
- Cost of the mist extractor/eliminator itself and required vessels, piping, instrumentation, and utilities

Gravitational and Drag Forces Acting on a Droplet

All mist extractor types are based on the some kind of intervention in the natural balance between gravitational and drag forces. This is accomplished in one or more of the following ways:

- Overcoming drag force by reducing the gas velocity (gravity separators or settling chambers)
- Introducing additional forces (venturi scrubbers, cyclones, electrostatic precipitators)
- Increasing gravitational force by boosting the droplet size (impingement-type)

The relevant laws of fluid mechanics and the principle forces acting on a liquid droplet falling through the continuous gas phase are discussed below. As the gas in a vessel flows upward, there are two opposing forces acting on a liquid droplet: a gravitational force (or negative buoyant force) acting downward to accelerate the droplet, and an opposing drag force acting to slow the droplet's rate of fall. An increase in the upward gas velocity increases the drag force on the droplet. The drag force continues to reduce the rate of fall until a point is reached when the downward velocity reaches zero, and the droplet becomes stationary. When the gravitational or negative buoyant force equals the drag force,

the acceleration of the liquid droplet becomes zero and the droplet will settle at a constant "terminal" or "settling" velocity. Additional increases in gas velocity result in an initial reduction in settling velocity of the droplet. Further increase causes the droplet to move upward at increasing velocities until a point is reached where the droplet velocity approaches the gas velocity. The same theory is applicable to horizontal gas flow as well. The primary difference is that the gravitational and drag forces are operating at 90 degrees to each other. Thus, there is always a net force acting in the downward direction.

Impingement-Type

The most widely used type of mist extractor is the impingement-type because it offers good balance between efficiency, operating range, pressure drop requirement, and installed cost. These types consist of baffles, wire meshes, and micro-fiber pads. Impingement-type mist extractors may involve just a single baffle or disc installed in a vessel. As illustrated in Figure 4-24, as the gas approaches the surface of the baffle or disc (commonly referred to as a target), fluid streamlines spread around the baffle or disc. Ignoring the eddy streams formed around the target, one can assume that the higher the stream velocity, the closer to the target these streamlines start to form. A droplet can be captured by the target in an impingement-type mist extractor/eliminator via any of the following three mechanisms: inertial impaction, direct interception, and diffusion (refer to Figure 4-24).

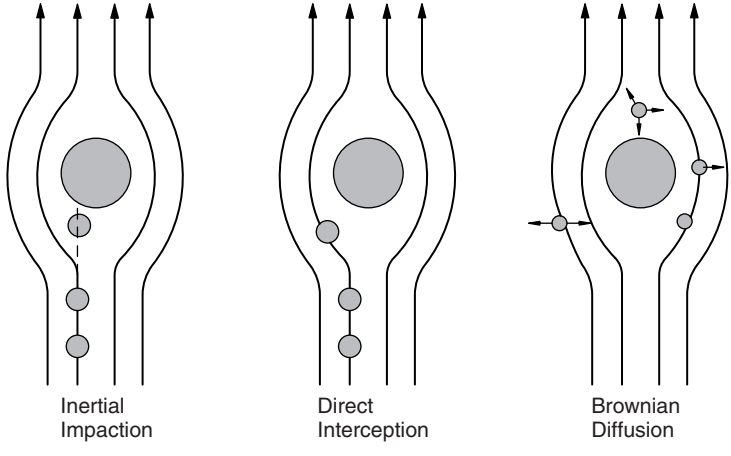


Figure 4-24. The three primary mechanisms of mist capture via impingement are inertial impaction (left), direct interception (center), and Brownian diffusion (right).

- **Inertial impaction.** Because of their mass, particles 1 to 10 microns in diameter in the gas stream have sufficient momentum to break through the gas streamlines and continue to move in a straight line until they impinge on the target. Impaction is generally the most important mechanism in wire mesh pads and impingement plates.
- **Direct interception.** There are also particles in the gas stream that are smaller, between 0.3 to 1 microns in diameter, than those above. These do not have sufficient momentum to break through the gas streamlines. Instead, they are carried around the target by the gas stream. However, if the streamline in which the particle is traveling happens to lie close enough to the target so that the distance from the particle centerline to the target is less than one-half the particle's diameter, the particle can touch the target and be collected. Interception effectiveness is a function of pore structure. The smaller the pores, the greater the media to intercept particles.
- **Diffusion.** Even smaller particles, usually smaller than 0.3 microns in diameter, exhibit random Brownian motion caused by collisions with the gas molecules. This random motion will cause these small particles to strike the target and be collected, even if the gas velocity is zero. Particles diffuse from the streamlines to the surface of the target where the concentration is zero. Diffusion is favored by low-velocity and high-concentration gradients.

Baffles

This type of impingement mist extractor consists of a series of baffles, vanes, or plates between which the gas must flow. The most common is the vane or chevron-shape, as shown in Figures 4-25 and 4-26. The vanes force the gas flow to be laminar between parallel plates that contain directional changes. The surface of the plates serves as a target for droplet impingement and collection. The space between the baffles ranges from 5 to 75 mm, with a total depth in the flow direction of 150 to 300 mm. Figures 4-27 and 4-28 illustrate a vane mist extractor installed in a vertical and horizontal separator, respectively. Figure 4-29 shows a vane mist extractor made from an angle iron. Figure 4-30 illustrates an "arch" plate mist extractor. As gas flows through the plates, droplets impinge on the plate surface. The droplets coalesce, fall, and are routed to the liquid collection section of the vessel. Vane-type eliminators are sized by their manufacturers to assure both laminar flow and a certain minimum pressure drop. Vane or chevron-shaped mist extractors remove liquid

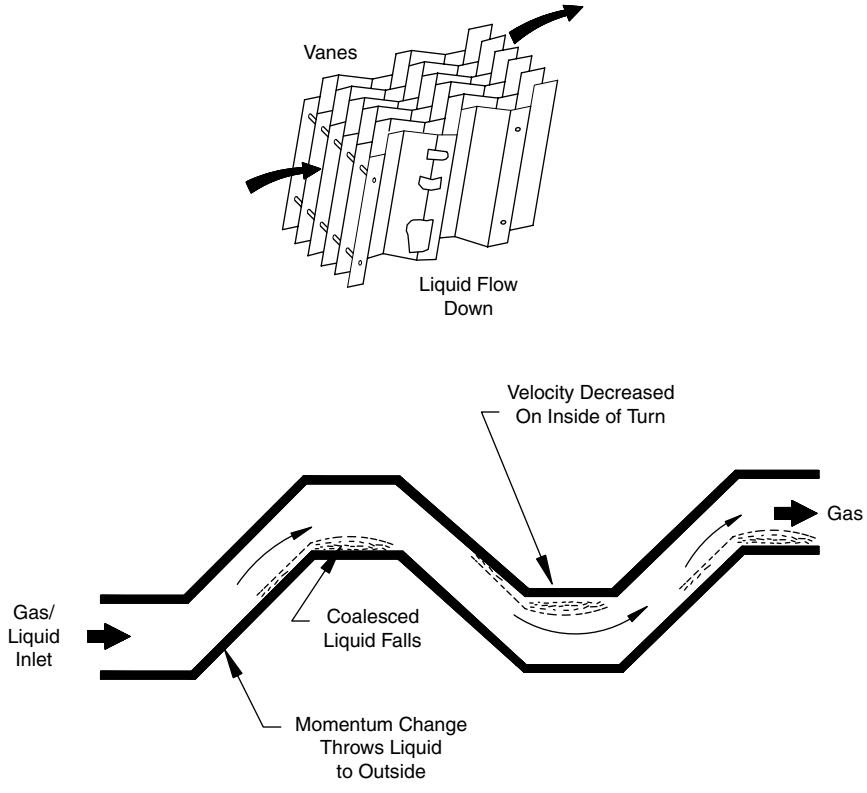


Figure 4-25. Typical vane-type mist extractor/eliminator.

droplets 10 to 40 microns and larger. Their operation is usually dictated by a design velocity expressed as follows:

$$V = K\sqrt{\left[\frac{(\rho_l - \rho_g)}{\rho_l}\right]}, \tag{4-1}$$

where

V = gas velocity,
K = Souders–Brown coefficient,
ρ_l = liquid or droplet density,
ρ_g = gas density.

The "K" factor or Souders–Brown coefficient, is determined experimentally for each plate geometry. Its value ranges from 0.3 to 1.0 ft/s (0.09

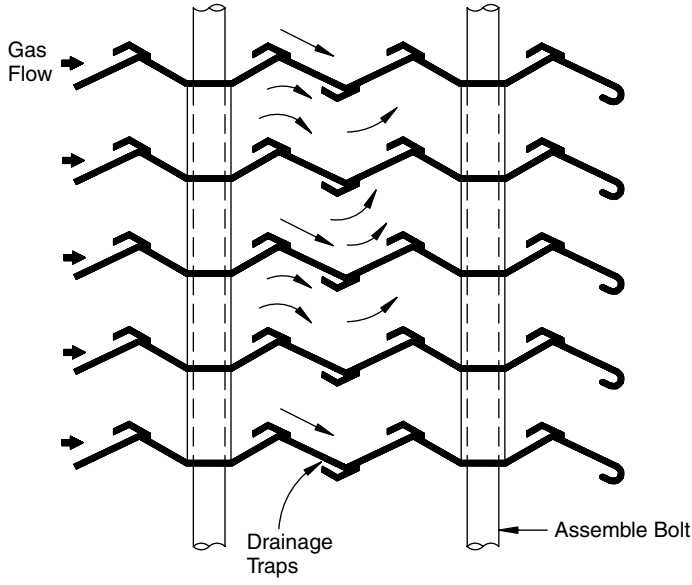


Figure 4-26. Vane-type element with corrugated plates and liquid drainage trays.

to 0.3 m/s) in typical designs. Since impaction is the primary collection mechanism, at too low a value of "*K*" the droplets can remain in the gas streamlines and pass through the device uncollected. The upper limit is set to minimize re-entrainment, which is caused either by excessive breakup of the droplets as they impinge onto the plates or by shearing of the liquid film on the plates.

Higher gas velocities can be handled if the vanes are installed in a horizontal gas flow, instead of vertical up-flow. In the horizontal configuration the liquid can easily drain downward due to gravity and thus out of the path of the incoming gas, which minimizes re-entrainment of the liquid.

The vane type appears most often in process systems where the liquid entrainment is contaminated with solids, or where high liquid loading exists. Vane-type mist extractors are less efficient in removing very small droplets than other impaction-types such as wire mesh or micro-fiber. Standard designs are generally limited to droplets larger than 40 microns. However, high-efficiency designs provide droplet removal down to less than 15 microns in diameter. The pressure drop is low, often less than 10–15 mm H_2O.

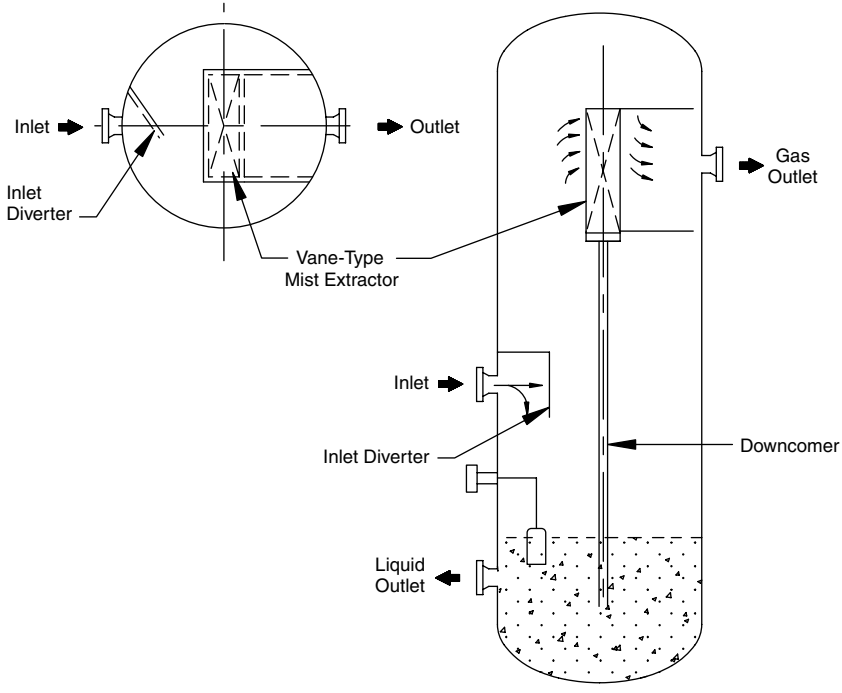


Figure 4-27. Cutaway view of a vertical separator fitted with a vane-type mist extractor.

Wire-Mesh

The most common type of mist extractor found in production operations is the knitted-wire-mesh type (refer to Figure 4-31). These units outnumber all other types of mist extractors. They are knitted (rather than woven) wire, and these devices have high surface area and void volume. Whereas woven wire has one set of wires running perpendicular to a second set of wires, knitted wire instead has a series of interlocking loops just like cloth fiber. This makes the knitted product sufficiently flexible and yet structurally stable.

The wire-mesh mist extractor is often specified by calling for a certain thickness (usually 3 to 7 inches) and mesh density (usually 10 to 12 pounds per cubic foot). They are usually constructed from wires of diameter ranging from 0.10 to 0.28 mm, with a typical void volume fraction of 0.95 to 0.99. The wire pad is placed between top and bottom support grids to complete the assembly. The grids must be strong enough to span between the supports and have sufficient free area for flow. Wire-mesh pads are mounted near the outlet of a separator, generally on a

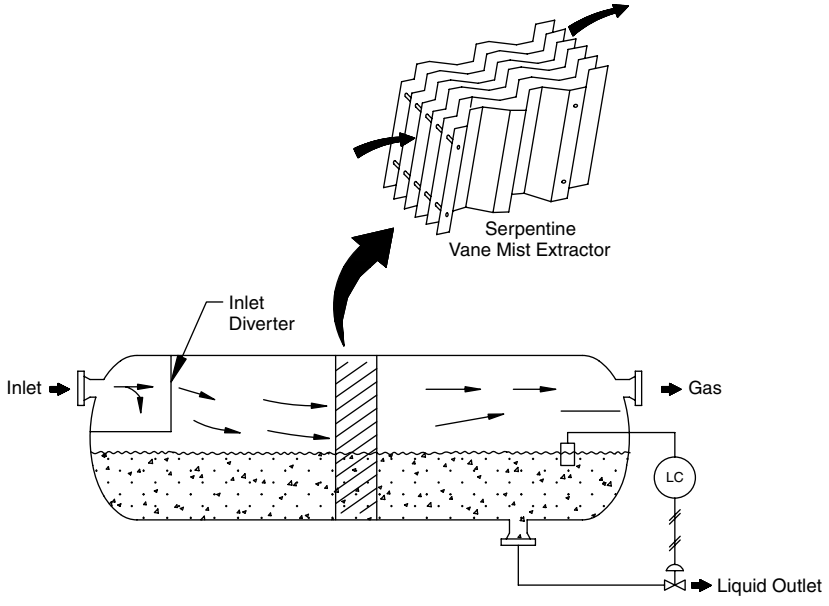


Figure 4-28. Cutaway view of a horizontal separator fitted with a vane-type mist extractor.

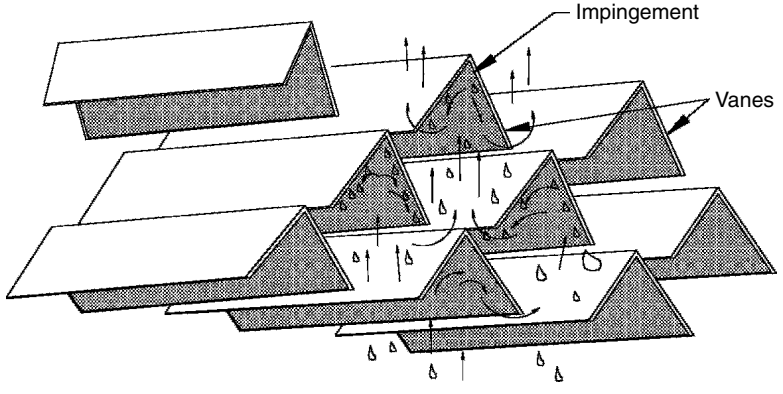


Figure 4-29. A vane-type mist extractor made from angle iron.

support ring (vertical separator) or frame (horizontal separator). (Refer to Figures 4-32 and 4-33, respectively.)

Wire-mesh mist extractors are normally installed in vertical upward gas flow, although horizontal flows are employed in some specialized applications. In a horizontal flow the designer must be careful because liquid droplets captured in the higher elevation of the vertical mesh may

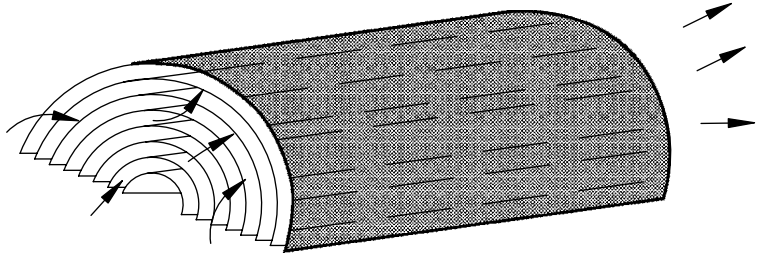


Figure 4-30. An "arch" plate-type mist extractor.



Figure 4-31. Example wire-mesh mist extractor. (Photo courtesy of ACS Industries, LP, Houston, Texas.)

drain downward at an angle as they are pushed through the mesh, resulting in re-entrainment.

The effectiveness of wire-mesh depends largely on the gas being in the proper velocity range [refer to Eq. (4-1)]. If the velocities are too high, the liquids knocked out will be re-entrained. If the velocities are low, the vapor just drifts through the mesh element without the droplets impinging and coalescing. The lower limit of the velocity is normally set at 30% of design velocity, which maintains a reasonable efficiency. The upper limit is governed by the need to prevent re-entrainment of liquid droplets from the downstream face of the wire-mesh device.

The pressure drop through a wire-mesh unit is a combination of "dry" pressure drop due to gas flow only, plus the "wet" pressure drop due to liquid holdup. The "dry" pressure drop may be calculated from the following equation:

$$\Delta P_{dry} = \frac{fHa\rho_g V^2}{(981{*}10^{30})}, \tag{4-2}$$

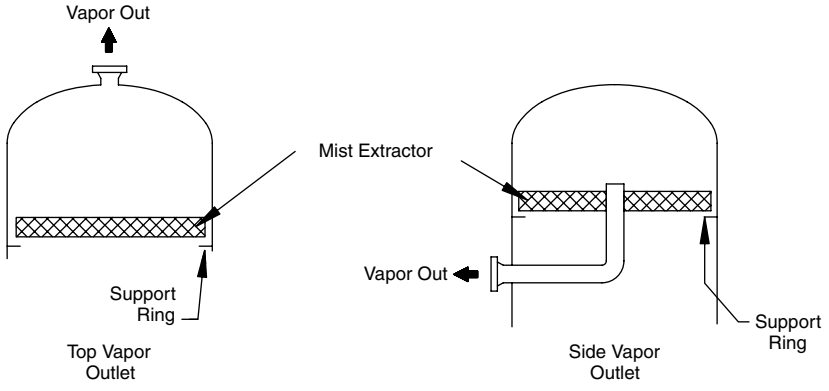


Figure 4-32. Vertical separators fitted with wire-mesh pads supported by support rings.

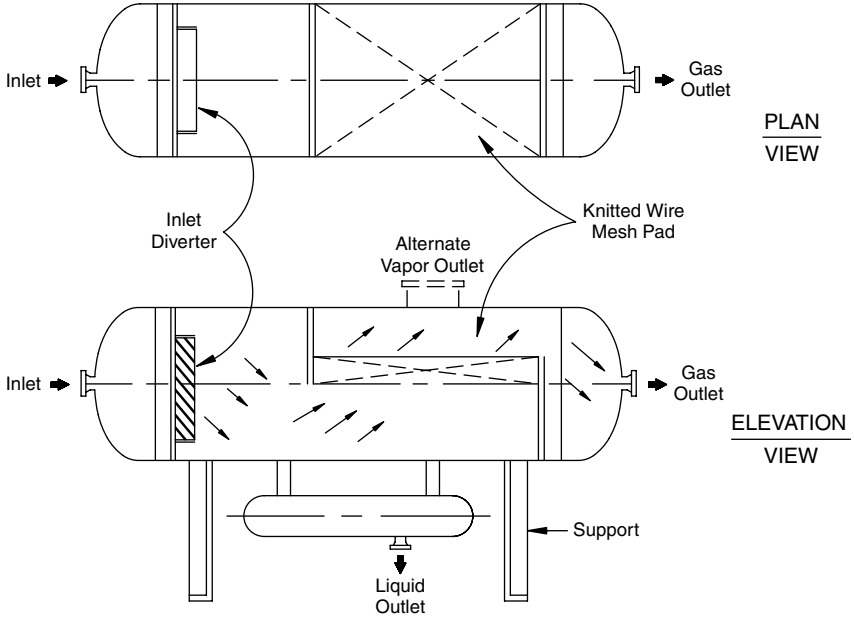


Figure 4-33. Horizontal separator fitted with wire-mesh pads supported by a frame.

where

f = friction factor from Figure 4-34,
H = thickness of mesh pad, inches,

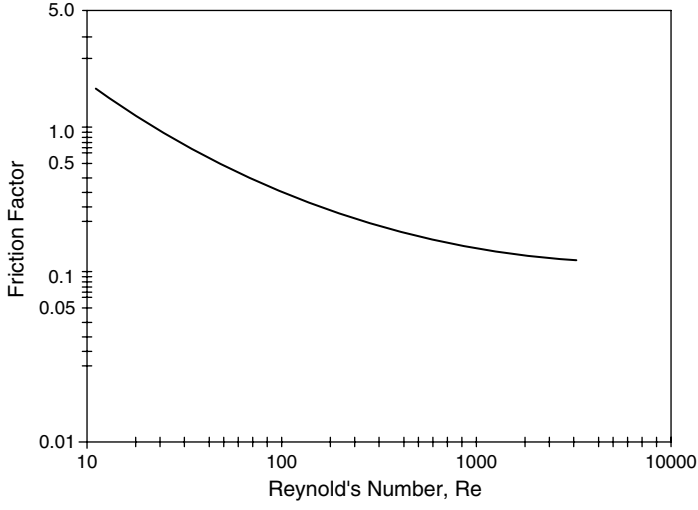


Figure 4-34. Friction factor versus Reynolds number for a dry knitted wire-mesh extractor.

a = surface area, in.^{2},
ρ_g = gas density, lb/ft^3,
V = gas velocity, ft/s.
ΔP_{dry} = pressure drop, psi

The "wet" pressure drop, a function of liquid loading as well as wire-mesh pad geometry, may be obtained experimentally over a range of gas velocities and liquid loadings. There are also correlations available for the various wire-mesh geometries.

Whether installed inside a piece of process equipment or placed inside a separate vessel of its own, a wire-mesh or baffle-type mist extractor offers low-pressure drop. To ensure a unit's operation at design capacity and high mist elimination efficiency, the flow pattern of the gas phase must be uniform throughout the element.

When there are size limitations inside a process vessel, an integral baffle plate can be used on the downstream side face of the wire-mesh element as a vapor distributor. Even here the layout of the drum must be such that the flow stream enters the mesh pad with flow-pattern streamlines that are nearly uniform.

When knockout drums are equipped with vanes or wire-mesh pads, one can use any one of the four following design configurations: horizontal or vertical vessels, with horizontal or vertical vane or mesh elements. The classic configuration is the vertical vessel with horizontal element. In order to achieve uniform flow, one has to follow a few design criteria (refer to Figure 4-35).

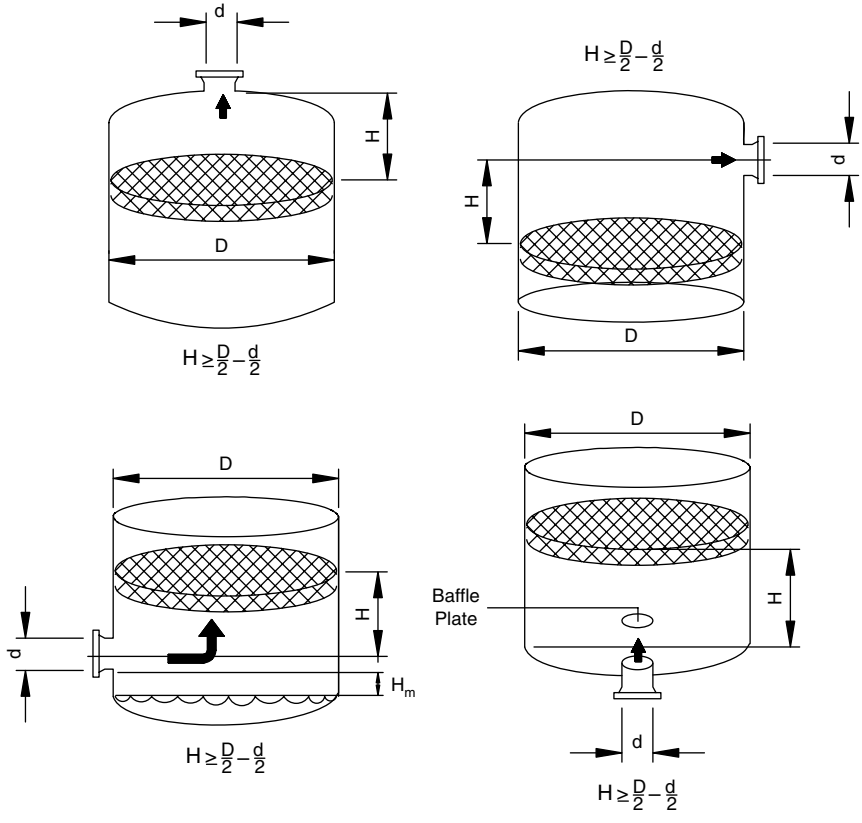


Figure 4-35. Dimensions for the placement of a wire-mesh mist extractor. [*H* represents minimum height, and H_m must be at least 1 foot (305 mm).]

A properly sized wire-mesh unit can remove 100% of liquid droplets larger than 3 to 10 microns in diameter. Although wire-mesh eliminators are inexpensive, they are more easily plugged than the other types. Wire-mesh pads are not the best choice if solids can accumulate and plug the pad.

Micro-Fiber

Micro-fiber mist extractors use very small diameter fibers, usually less than 0.02 mm, to capture very small droplets. Gas and liquid flow is horizontal and co-current. Because the micro-fiber unit is manufactured from densely packed fiber, drainage by gravity inside the unit is limited. Much of the liquid is eventually pushed through the micro-fiber and

drains on the downstream face. The surface area of a micro-fiber mist extractor can be 3 to 150 times that of a wire-mesh unit of equal volume.

There are two categories of these units, depending on whether droplet capture is via inertial impaction, interception, or Brownian diffusion. Only the diffusion type can remove droplets less than 2 microns. As with wire-mesh pads, micro-fiber units that operate in the inertial impaction mode have a minimum velocity below which efficiency drops off significantly. Micro-fiber units that operate in the diffusion mode have no such lower velocity limit. In fact, efficiency continues to improve as the gas velocity is reduced to zero.

For impaction-type micro-fiber units, the maximum velocity is usually set by the onset of re-entrainment, just as in the case of wire-mesh and vane devices. For micro-fiber units operating in the diffusion mode, the upper velocity can be set by re-entrainment, by loss of efficiency, or by pressure drop. Typical velocity ranges from 20 to 60 ft/min (60 to 180 m/min) for impaction-type units, compared to 1 to 4 ft/min (3 to 12 m/min) for units in the diffusion mode.

As with other mist extractors, each micro-fiber supplier has developed data on the capacity, pressure drop, and efficiency correlations for its products.

Table 4-1 summarizes the major parameters that should be considered when selecting a mist extractor. For more detailed information, see Fabian (1993).

Other Configurations

Some separators use centrifugal mist extractors, discussed earlier in this chapter, that cause liquid droplets to be separated by centrifugal force (refer to Figures 4-36 and 4-37). These units can be more efficient than either wire-mesh or vanes and are the least susceptible to plugging. However, they are not in common use in production operations because their removal efficiencies are sensitive to small changes in flow. In addition, they require relatively large pressure drops to create the centrifugal force. To a lesser extent, random packing is sometimes used for mist extraction, as shown in Figure 4-38. The packing acts as a coalescer.

Final Selection

The selection of a type of mist extractor involves a typical cost-benefit analysis. Wire-mesh pads are the cheapest, but mesh pads are the most susceptible to plugging with paraffins, gas hydrates, etc. With age, mesh pads also tend to deteriorate and release wires and/or chunks of the pad into the gas stream. This can be extremely damaging to downstream

Table 4-1
Features of Impingement-Type Mist Extractors

Consideration	Wire-Mesh	Vane	Micro-fiber
Cost	Lowest	2–3 times wire-mesh unit	Highest
Efficiency	100% (for droplets larger than 3–10 μ	100% (for mists > 20–40 μ)	Up to 99.9% (for mists < 3 μ)
Pressure drop	< 25 mm H_2O	< 15 mm H_2O	100–300 mm
Gas capacity	Very good	Up to twice that of a wire-mesh unit	Lowest
Liquid capacity	Good	Best	Lowest
Solids	Good	Best	Soluble particles with sprays only

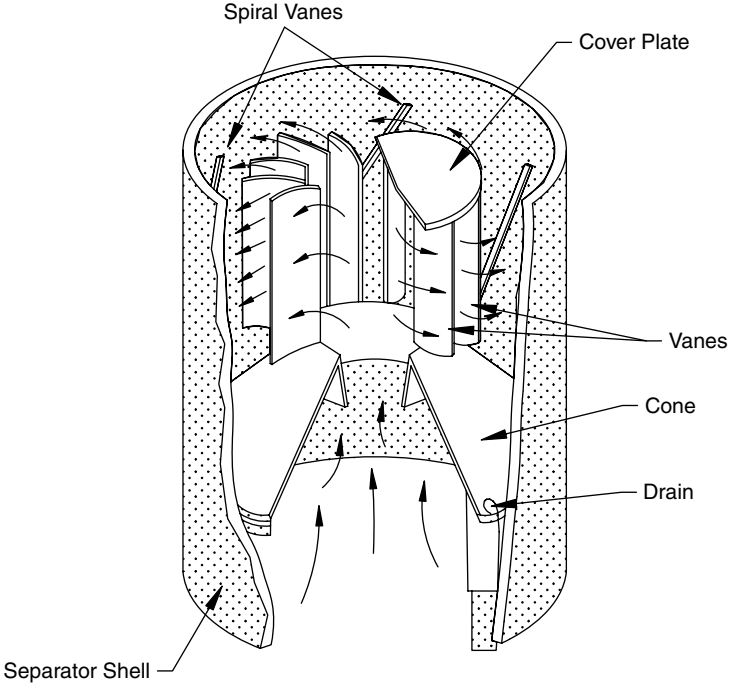


Figure 4-36. Centrifugal mist extractor.

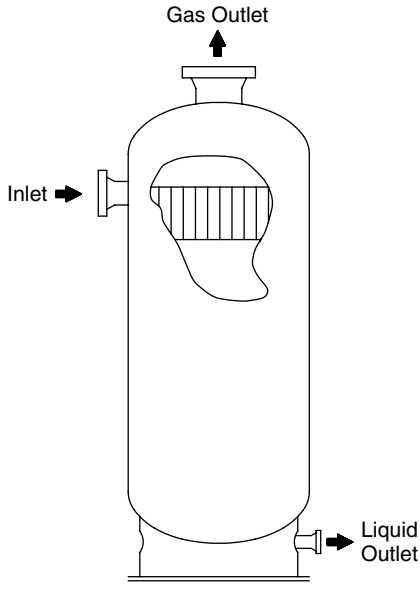


Figure 4-37. Vertical separator fitted with a centrifugal mist element. (Courtesy of Peerless Manufacturing Co.)

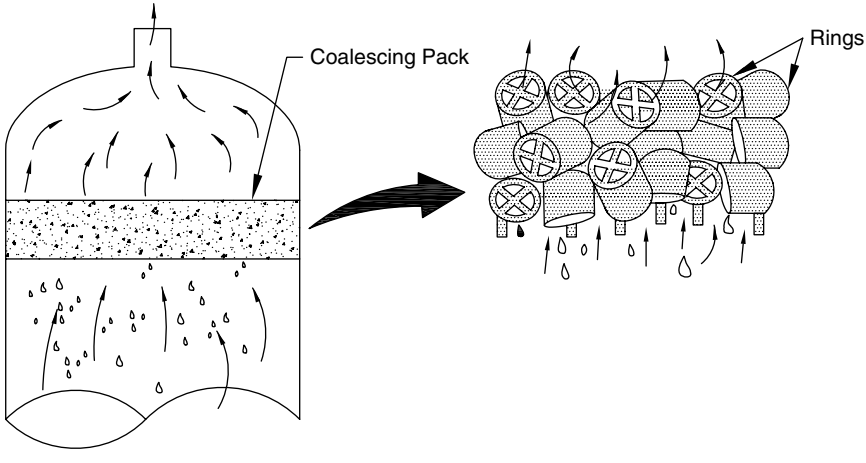


Figure 4-38. A coalescing pack mist extractor.

equipment, such as compressors. Vane units, on the other hand, are more expensive. Typically, vane units are less susceptible to plugging and deterioration than mesh pads. Micro-fiber units are the most expensive and are capable of capturing very small droplets but, like wire mesh pads, are susceptible to plugging. The selection of a type of mist extractor is affected by the fluid characteristics, the system requirements, and the cost.

It is recommended that the sizing of mist extractors should be left to the manufacturer. Experience indicates that if the gravity settling section is designed to remove liquid droplets of 500 microns or smaller diameter, there will be sufficient space to install a mist extractor.

Potential Operating Problems

Foamy Crude

The major cause of foam in crude oil is the presence of impurities, other than water, which are impractical to remove before the stream reaches the separator. One impurity that almost always causes foam is CO_2. Sometimes completion and workover fluids, that are incompatible with the wellbore fluids, may also cause foam. Foam presents no problem within a separator if the internal design assures adequate time or sufficient coalescing surface for the foam to "break."

Foaming in a separating vessel is a threefold problem:

1. Mechanical control of liquid level is aggravated because any control device must deal with essentially three liquid phases instead of two.
2. Foam has a large volume-to-weight ratio. Therefore, it can occupy much of the vessel space that would otherwise be available in the liquid collecting or gravity settling sections.
3. In an uncontrolled foam bank, it becomes impossible to remove separated gas or degassed oil from the vessel without entraining some of the foamy material in either the liquid or gas outlets.

The foaming tendencies of any oil can be determined with laboratory tests. Only laboratory tests, run by qualified service companies, can qualitatively determine an oil's foaming tendency. One such test is ASTM D 892, which involves bubbling air through the oil. Alternatively, the oil may be saturated with its associated gas and then expanded in a gas container.

This alternative test more closely models the actual separation process. Both of these tests are qualitative. There is no standard method of measuring the amount of foam produced or the difficulty in breaking the foam. Foaming is not possible to predict ahead of time without laboratory tests. However, foaming can be expected where CO_2 is present in small quantities (1–2%). It should be noted that the amount of foam is dependent on the pressure drop to which the inlet liquid is subjected, as well as the characteristics of the liquid at separator conditions.

Comparison of foaming tendencies of a known oil to a new one, about which no operational information is known, provides an understanding of the relative foam problem that may be expected with the new oil as weighed against the known oil. A related amount of adjustment can then be made in the design parameters, as compared to those found satisfactory for the known case.

The effects of temperature on a foamy oil are interesting. Changing the temperature at which a foamy oil is separated has two effects on the foam. The first effect is to change the oil viscosity. That is, an increase in temperature will decrease the oil viscosity, making it easier for the gas to escape from the oil. The second effect is to change the gas-oil equilibrium. A temperature increase will increase the amount of gas, which evolves from the oil.

It's very difficult to predict the effects of temperature on the foaming tendencies of an oil. However, some general observations have been made. For low API gravity crude (heavy oils) with low GORs, increasing the operating temperature decreases the oils' foaming tendencies. Similarly, for high API crude (light oils) with high GORs, increasing the operating temperature decreases the oils' foaming tendencies. However, increasing the operating temperature for a high API gravity crude (light oil) with low GORs may increase the foaming tendencies. Oils in the last category are typically rich in intermediates, which have a tendency to evolve to the gas phase as the temperature increases. Accordingly, increasing the operating temperature significantly increases gas evolution, which in turn increases the foaming tendencies.

Foam depressant chemicals often will do a good job in increasing the capacity of a given separator. However, in sizing a separator to handle a specific crude, the use of an effective depressant should not be assumed because characteristics of the crude and of the foam may change during the life of the field. Also, the cost of foam depressants for high-rate production may be prohibitive. Sufficient capacity should be provided in the separator to handle the anticipated production without use of a foam depressant or inhibitor. Once placed in operation, a foam depressant may allow more throughput than the design capacity.

Paraffin

Separator operation can be adversely affected by an accumulation of paraffin. Coalescing plates in the liquid section and mesh pad mist extractors in the gas section are particularly prone to plugging by accumulations of paraffin. Where it is determined that paraffin is an actual or potential problem, the use of plate-type or centrifugal mist extractors should be considered. Manways, handholes, and nozzles should be provided to allow steam, solvent, or other types of cleaning of the separator internals. The bulk temperature of the liquid should always be kept above the cloud point of the crude oil.

Sand

Sand can be very troublesome in separators by causing cutout of valve trim, plugging of separator internals, and accumulation in the bottom of the separator. Special hard trim can minimize the effects of sand on the valves. Accumulations of sand can be removed by periodically injecting water or steam in the bottom of the vessel so as to suspend the sand during draining. Figure 4-23 is a cutaway of a sand wash and drain system fitted into a horizontal separator fitted with sand jets and an inverted trough.

Sometimes a vertical separator is fitted with a cone bottom. This design would be used if sand production was anticipated to be a major problem. The cone is normally at an angle of between 45° and 60° to the horizontal. Produced sand may have a tendency to stick to steel at 45°. If a cone is installed, it could be part of the pressure-containing walls of the vessel (refer to Figure 4-39), or for structural reasons, it could be installed internal to the vessel cylinder (refer to Figure 4-40). In such a case, a gas equalizing line must be installed to assure that the vapor behind the cone is always in pressure equilibrium with the vapor space.

Plugging of the separator internals is a problem that must be considered in the design of the separator. A design that will promote good separation and have a minimum of traps for sand accumulation may be difficult to attain, since the design that provides the best mechanism for separating the gas, oil, and water phases probably will also provide areas for sand accumulation. A practical balance for these factors is the best solution.

Liquid Carryover

Liquid carryover occurs when free liquid escapes with the gas phase and can indicate high liquid level, damage to vessel internals, foam, improper

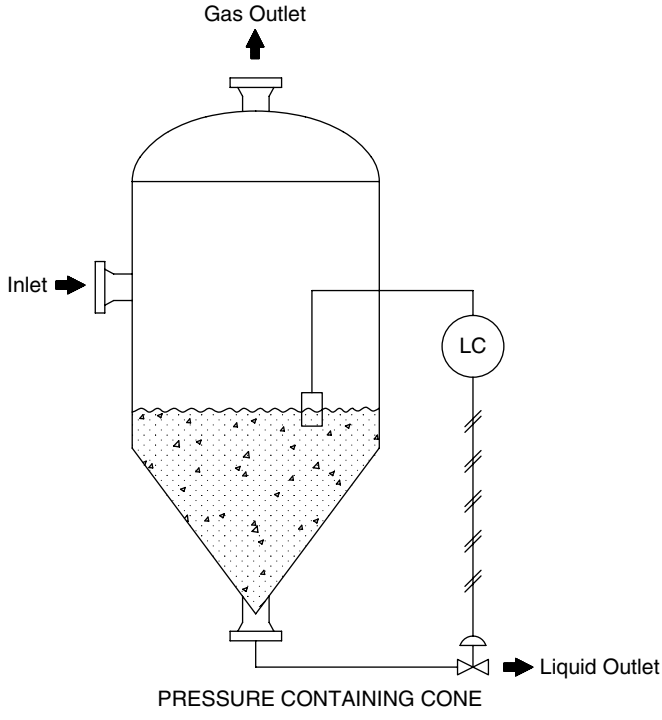


Figure 4-39. Vertical separator with a pressure containing cone bottom used to collect solids.

design, plugged liquid outlets, or a flow rate that exceeds the vessel's design rate. Liquid carryover can usually be prevented by installing a level safety high (LSH) sensor that shuts in the inlet flow to the separator when the liquid level exceeds the normal maximum liquid level by some percentage, usually 10–15%.

Gas Blowby

Gas blowby occurs when free gas escapes with the liquid phase and can be an indication of low liquid level, vortexing, or level control failure. This could lead to a very dangerous situation. If there is a level control failure and the liquid dump valve is open, the gas entering the vessel will exit the liquid outlet line and would have to be handled by the next downstream vessel in the process. Unless the downstream vessel is designed for the gas blowby condition, it can be over-pressured. Gas blowby can usually be prevented by installing a level safety low sensor (LSL) that shuts

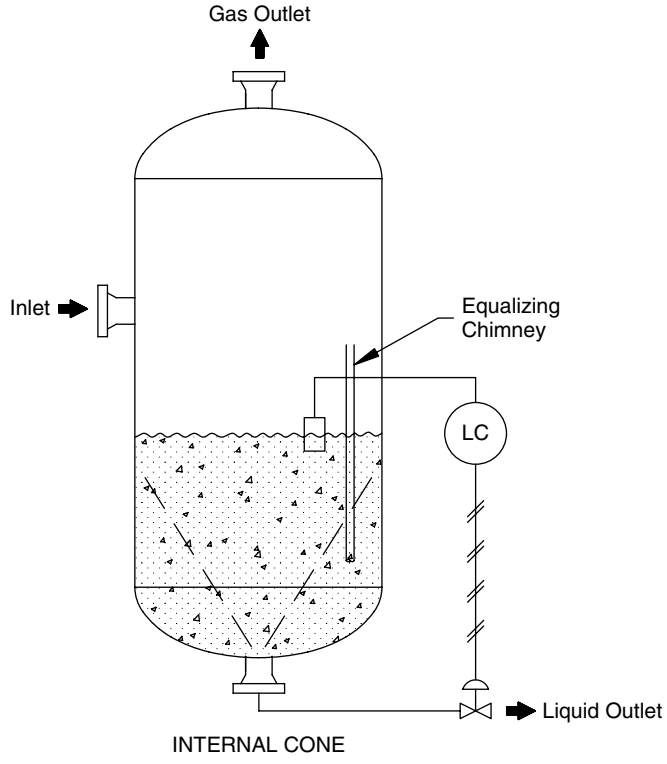


Figure 4-40. Vertical separator fitted with an internal cone bottom and an equalizing line.

in the inflow and/or outflow to the vessel when the liquid level drops to 10–15% below the lowest operating level. In addition, downstream process components should be equipped with a pressure safety high (PSH) sensor and a pressure safety valve (PSV) sized for gas blowby.

Liquid Slugs

Two-phase flow lines and pipelines tend to accumulate liquids in low spots in the lines. When the level of liquid in these low spots rises high enough to block the gas flow, then the gas will push the liquid along the line as a slug. Depending on the flow rates, flow properties, length and diameter of the flow line, and the elevation change involved, these liquid slugs may contain large liquid volumes.

Situations in which liquid slugs may occur should be identified prior to the design of a separator. The normal operating level and the high-level shutdown on the vessel must be spaced far enough apart to accommodate the anticipated slug volume. If sufficient vessel volume is not provided, then the liquid slugs will trip the high-level shutdown.

When liquid slugs are anticipated, slug volume for design purposes must be established. Then the separator may be sized for liquid flow-rate capacity using the normal operating level. The location of the high-level set point may be established to provide the slug volume between the normal level and the high level. The separator size must then be checked to ensure that sufficient gas capacity is provided even when the liquid is at the high-level set point. This check of gas capacity is particularly important for horizontal separators because, as the liquid level rises, the gas capacity is decreased. For vertical separators, sizing is easier as sufficient height for the slug volume may be added to the vessel's seam-to-seam length.

Often the potential size of the slug is so great that it is beneficial to install a large pipe volume upstream of the separator. The geometry of these pipes is such that they operate normally empty of liquid, but fill with liquid when the slug enters the system. This is the most common type of "slug catcher" used when two-phase pipelines are routinely pigged. Figure 4-13 is a schematic of a liquid finger slug catcher.

Design Theory

Settling

In the gravity settling section of a separator, liquid droplets are removed using the force of gravity. Liquid droplets, contained in the gas, settle at a terminal or "settling" velocity. At this velocity, the force of gravity on the droplet or "negative buoyant force" equals the drag force exerted on the droplet due to its movement through the continuous gas phase. The drag force on a droplet may be determined from the following equation:

$$F_D = C_D A d \rho \left[\frac{V_t^2}{2g}\right], \tag{4-3}$$

where

F_D = drag force, lb_f(N),
C_D = drag coefficient,
A_d = cross-sectional area of the droplet, ft^2 (m^2),
ρ = density of the continuous phase, lb/ft^3 (kg/m^3),
V_t = terminal (settling velocity) of the droplet, ft/s (m/s),
g = gravitational constant, $32.2\, lb_m ft/lb_f s^2$ (m/s^2).

If the flow around the droplet were laminar, then Stokes' law would govern and

$$C_D = \frac{24}{\mathrm{Re}}, \tag{4-4}$$

where Re = Reynolds number, which is dimensionless.

It can be shown that in such a gas the droplet settling velocity would be given by:

Field Units

$$V_t = \frac{1.78 \times 10^{-6}\,(\Delta \mathrm{SG})\, d_m^2}{\mu}. \tag{4-5a}$$

SI Units

$$V_t = \frac{5.56 \times 10^{-7}\,(\Delta \mathrm{SG})\, d_m^2}{\mu}, \tag{4-5b}$$

where

ΔSG = difference in specific gravity relative to water of the droplet and the gas,
d_m = droplet diameter, microns,
μ = viscosity of the gas, cp.

Equations (4-5a) and (4-5b) are derived as follows: for low Reynolds number flows, i.e., Re < 1,

$$C_D = \frac{24}{\mathrm{Re}}.$$

The drag force is then

$$\begin{aligned} F_D &= C_D A_d \rho_g \frac{V^2}{2g} \\ &= \frac{24}{\mathrm{Re}} \left(\pi \frac{{D_m}^2}{4} \right) \rho_g \frac{V^2}{2g} \\ &= \frac{24}{\frac{\rho_g}{g} \frac{VD_m}{\mu'}} \left(\pi \frac{{D_m}^2}{4} \right) \rho_g \frac{V^2}{2g}, \end{aligned}$$

where

D_m = droplet diameter, ft (m),
μ' = viscosity, lb-sec/ft^2(kg-s/m^2),
$F_D = 3\pi\mu' V D_m$ (Stokes' law).

The buoyant force on a sphere from Archimedes' principles is

$$F_B = \left(\rho_l - \rho_g\right) \frac{\pi {D_m}^3}{6}.$$

When the drag force is equal to the buoyancy force, the droplet's acceleration is zero so that it moves at a constant velocity. This is the terminal velocity.

Field Units

$$\begin{aligned} F_D &= F_B, \\ 3\pi\mu' V D_m &= \left(\rho_1 - \rho_g\right) \frac{\pi {D_m}^3}{6}, \\ V_t &= \frac{\left(\rho_1 - \rho_g\right) {D_m}^2}{18\mu'}, \\ \mu' &= \mu(2.088 \times 10^{-5}), \end{aligned}$$

where

μ = viscosity, cp,
$D_m = (d_m)(3.281 \times 10^{-6})$,

where

d_m = diameter, micron,
$\rho_1 = 62.4 \times \text{SG}$,
$\rho_g = 62.4 \times \text{SG}$,

where

SG = specific gravity relative to water,

$$V_t = \frac{62.4\,(\Delta\text{SG})\left(3.281 \times 10^{-6} \times d_m\right)^2}{18\,(\mu)\,(2.088 \times 10^{-5})},$$

$$V_t = \frac{1.78 \times 10^{-6}\,(\Delta\text{SG})\,d_m^2}{\mu}.$$

SI Units

$$F_D = F_B,$$

$$3\pi\mu' V D_m = \left(\rho_1 - \rho_g\right)\frac{\pi {D_m}^3}{6},$$

$$V_t = \frac{\left(\rho_l - \rho_g\right){D_m}^2}{18\mu},$$

$$\mu' = \mu(0.0001),$$

where

μ = viscosity, cp,
$D_m = (d_m)(1 \times 10^{-6})$,

where

d_m = diameter, micron,
$\rho_l = 1000 \times \text{SG}$,
$\rho_g = 1000 \times \text{SG}$,

where

SG = specific gravity relative to water,

$$V_t = \frac{1000\,(\Delta\text{SG})\left(1 \times 10^{-6} \times d_m\right)^2}{18\,(\mu)\,(0.0001)},$$

$$V_t = \frac{5.56 \times 10^{-7}\,(\Delta\text{SG})\,d_m^2}{\mu}.$$

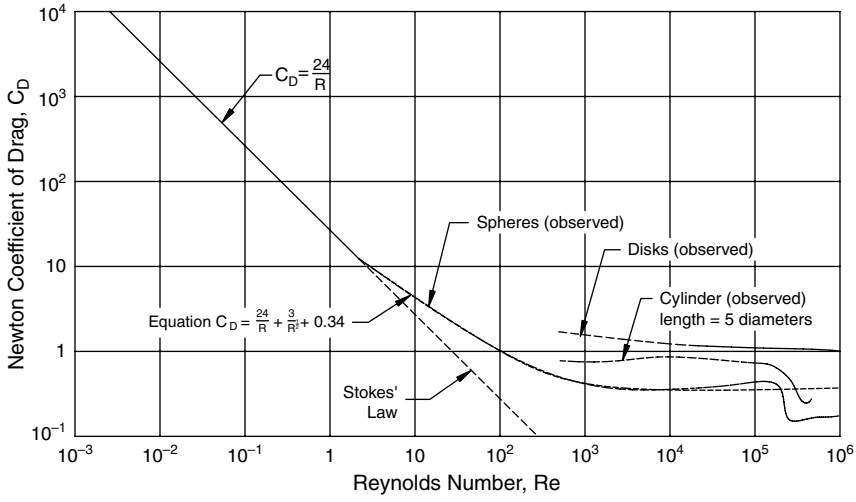


Figure 4-41. Coefficient of drag for varying magnitudes of the Reynolds number.

Unfortunately, for production facility designs it can be shown that Stokes' law does not govern, and the following more complete formula for drag coefficient must be used (refer to Figure 4-41):

$$C_D = \frac{24}{\mathrm{Re}} + \frac{3}{\mathrm{Re}^{1/2}} + 0.34. \tag{4-6}$$

Equating drag and buoyant forces, the terminal settling velocity is given by

Field Units

$$V_t = 0.0119\left[\left(\frac{\rho_l - \rho_g}{\rho_g}\right)\frac{d_m}{C_D}\right]^{1/2} \tag{4-7a}$$

SI Units

$$V_t = 0.0036\left[\left(\frac{\rho_l - \rho_g}{\rho_g}\right)\frac{d_m}{C_D}\right]^{1/2}, \tag{4-7b}$$

where

ρ_l = density of liquid, lb/ft^3(kg/m^3),
ρ_g = density of the gas at the temperature and pressure in the separator, lb/ft^3(kg/m^3).

Equations (4-7a) and (4-7b) are derived as follows:

C_D = constant.

The drag force is then:

Field Units

$$F_D = C_D A_d \rho_g \frac{V^2}{2g}$$
$$= C_D \left(\frac{\pi {D_m}^2}{4}\right) \rho_g \frac{V^2}{2g}.$$

When $F_B = F_D$,

$$F_D = \left(\frac{\pi {D_m}^2}{4}\right) \rho_g \frac{V^2}{2g},$$

$$F_B = (\rho_l - \rho_g) \frac{\pi {D_m}^3}{6},$$

$$V_t = 6.55 \left[\left(\frac{\rho_l - \rho_g}{\rho_g}\right) \frac{D_m}{C_D}\right]^{1/2},$$

$$D_m = d_m (3.281 \times 10^{-6}),$$

$$V_t = 0.0119 \left[\left(\frac{\rho_l - \rho_g}{\rho_g}\right) \frac{d_m}{C_D}\right]^{1/2}.$$

For $C_D = 0.34$,

$$V_t = 0.0204 \left[\left(\frac{\rho_l - \rho_g}{\rho_g}\right) d_m\right]^{1/2}.$$

SI Units

$$F_D = C_D A_d \rho_g \frac{V^2}{2g}$$

$$= C_D \left(\frac{\pi {D_m}^2}{4} \right) \rho_g \frac{V^2}{2g}.$$

When $F_B = F_D$,

$$F_D = \left(\frac{\pi {D_m}^2}{4} \right) \rho g \frac{V^2}{2g},$$

$$F_B = \left(\rho_l - \rho_g \right) \frac{\pi {D_m}^3}{6},$$

$$V_t = 3.61 \left[\left(\frac{\rho_l - \rho_g}{\rho_g} \right) \frac{D_m}{C_D} \right]^{1/2},$$

$$D_m = d_m (1 \times 10^{-6}),$$

$$V_t = 0.0036 \left[\left(\frac{\rho_l - \rho_g}{\rho_g} \right) \frac{d_m}{C_D} \right]^{1/2}.$$

For $C_D = 0.34$,

$$V_t = 0.0062 \left[\left(\frac{\rho_l - \rho_g}{\rho_g} \right) d_m \right]^{1/2}.$$

Equations (4-6) and (4-7) can be solved by an iterative process. Start by assuming a value of C_D, such as 0.34, and solve Eq. (4-7) for V_t. Then, using V_t, solve for R_e. Then, Eq. (4-6) may be solved for C_D. If the calculated value of C_D equals the assumed value, the solution has been reached. If not, then the procedure should be repeated using the calculated C_D as a new assumption. The original assumption of 0.34 for C_D was used because this is the limiting value for large Reynolds numbers. The iterative steps are shown below:

Field Units

1. Start with

$$V_1 = 0.0204 \left[\frac{\left(\rho_l - \rho_g \right) d_m}{\rho_g} \right]^{1/2}.$$

2. Calculate

$$\mathrm{Re} = 0.0049 \frac{\rho_g d_m V}{\mu}.$$

3. From Re, calculate C_D using

$$C_D = \frac{24}{\text{Re}} + \frac{3}{\text{Re}^{1/2}} + 0.34.$$

4. Recalculate V_t using

$$V_t = 0.0119 \left[\left(\frac{\rho_l - \rho_g}{\rho_g} \right) \frac{d_m}{C_D} \right]^{1/2}.$$

5. Go to step 2 and iterate.

SI Units

1. Start with

$$V_1 = 0.0062 \left[\frac{(\rho_l - \rho_g)\, d_m}{\rho_g} \right]^{1/2}.$$

2. Calculate

$$\text{Re} = 0.001 \, \frac{\rho_g d_m V}{\mu}.$$

3. From Re, calculate C_D using

$$C_D = \frac{24}{\text{Re}} + \frac{3}{\text{Re}^{1/2}} + 0.34.$$

4. Recalculate V_t using

$$V_t = 0.0036 \left[\left(\frac{\rho_l - \rho_g}{\rho_g} \right) \frac{d_m}{C_D} \right]^{1/2}.$$

5. Go to step 2 and iterate.

Droplet Size

The purpose of the gravity settling section of the vessel is to condition the gas for final polishing by the mist extractor. To apply the settling equations to separator sizing, a liquid droplet size to be removed must be selected. From field experience, it appears that if 140-micron droplets are removed in this section, the mist extractor will not become flooded and will be able to perform its job of removing those droplets between 10- and 140-micron diameters. The gas capacity design equations in this section are all based on 140-micron removal. In some cases, this will give an overly conservative solution. The techniques used here can be easily modified for any droplet size.

In this book we are addressing separators used in oil field facilities. These vessels usually require a gravity settling section. There are special cases where the separator is designed to remove only very small quantities of liquid that could condense due to temperature or pressure changes in a stream of gas that has already passed through a separator and a mist extractor. These separators, commonly called "gas scrubbers," could be designed for removal of droplets on the order of 500 microns without fear of flooding their mist extractors. Fuel gas scrubbers, compressor suction scrubbers, and contact tower inlet scrubbers are examples of vessels to which this might apply.

Flare or vent scrubbers are designed to keep large slugs of liquid from entering the atmosphere through the vent or relief systems. In vent systems the gas is discharged directly to the atmosphere, and it is common to design the scrubbers for removal of 300- to 500-micron droplets in the gravity settling section. A mist extractor is not included because of the possibility that it might get plugged, thus creating a safety hazard. In flare systems, where the gas is discharged through a flame, there is the possibility that burning liquid droplets could fall to the ground before being consumed. It is still common to size the gravity settling section for 300- to 500-micron removal, which the API guideline for refinery flares indicates is adequate to ensure against a falling flame. In critical locations, such as offshore platforms, many operators include a mist extractor as an extra precaution against a falling flame. If a mist extractor is used, it is necessary to provide safety relief protection around the mist extractor in the event that it becomes plugged.

Retention Time

To assure that the liquid and gas reach equilibrium at separator pressure, a certain liquid storage is required. This is defined as "retention time" or

Table 4-2
Retention Time for Two-Phase Separators

°API Gravity	Retention Time (Minutes)
35+	0.5 to 1
30	2
25	3
20+	4+

1. If foam exists, increase above retention times by a factor of 2 to 4.
2. If high CO_2 exists, use a minimum of 5-minute retention time.

the average time a molecule of liquid is retained in the vessel, assuming plug flow. The retention time is thus the volume of the liquid storage in the vessel divided by the liquid flow rate.

For most applications retention times between 30 s and 3 min have been found to be sufficient. Where foaming crude is present, retention times up to four times this amount may be needed. In the absence of liquid or laboratory data, the guidelines presented in Table 4-2 can be used.

Liquid Re-entrainment

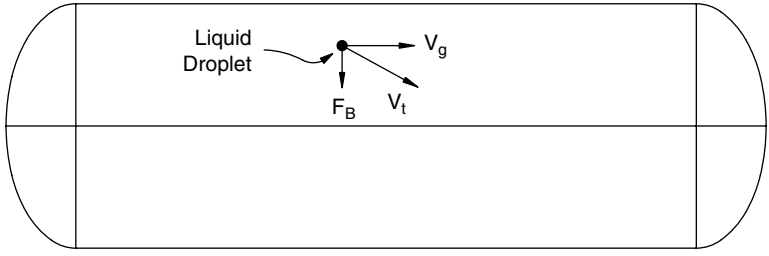
Liquid re-entrainment is a phenomenon caused by high gas velocity at the gas-liquid interface of a separator. Momentum transfer from the gas to the liquid causes waves and ripples in the liquid, and then droplets are broken away from the liquid phase.

The general rule of thumb that calls for limiting the slenderness ratio to a maximum of 4 or 5 is applicable for half-full horizontal separators. Liquid re-entrainment should be particularly considered for high-pressure separators sized on gas-capacity constraints. It is more likely at higher operating pressures (>1,000 psig or >7,000 kPa) and higher oil viscosities (<30° API). For more specific limits, see Viles (1993).

Separator Design

Horizontal Separators Sizing—Half Full

The guidelines presented in this section can be used for the initial sizing of a horizontal separator 50% full of liquid. They are meant to complement,



Legend:

F_B = Buoyant Force
V_g = Gas Velocity
V_t = Terminal or Settling Velocity Relative to Gas

Figure 4-42. Model of a horizontal separator.

and not replace, operating experience. Determination of the type and size of separator must be on an individual basis. All the functions and requirements should be considered, including the uncertainties in design flow rates and fluid properties. For this reason, there is no substitute for good engineering evaluations of each separator by the design engineer. The "trade-off" between design size and details and uncertainties in design parameters should not be left to manufacturer recommendations or rule of thumb.

When sizing a horizontal separator, it is necessary to choose a seam-to-seam vessel length and a diameter. This choice must satisfy the conditions for gas capacity that allow the liquid droplets to fall from the gas to the liquid volume as the gas traverses the effective length of the vessel. It must also provide sufficient retention time to allow the liquid to reach equilibrium. Figure 4-42 shows a vessel 50% full of liquid, which is the model used to develop sizing equations for a horizontal separator.

Gas Capacity Constraint

The principles of liquid droplets settling through a gas can be used to develop an equation to size a separator for a gas flow rate. The gas capacity constraint equations are based on setting the gas retention time equal to the time required for a droplet to settle to the liquid interface. For

a vessel 50% full of liquid, and separation of 100-micron liquid droplets from the gas, the following equation may be derived:

Field Units

$$dL_{\text{eff}} = 420\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2} \tag{4-8a}$$

SI Units

$$dL_{\text{eff}} = 34.5\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}, \tag{4-8b}$$

where

d = vessel internal diameter, in. (mm),
L_{eff} = effective length of the vessel where separation occurs, ft (m),
T = operating temperature, °R(°K),
Q_g = gas flow rate, MMscfd (scmh),
P = operating pressure, psia (kPa),
Z = gas compressibility,
C_D = drag coefficient,
d_m = liquid droplet to be separated, micron,
ρ_g = density of gas, lb/ft^3(kg/m^3),
ρ_l = density of liquid, lb/ft^3(kg/m^3).

Equations (4-8a) and (4-8b) are derived as follows: assume horizontal vessel is half full of liquid. Determine gas velocity, V_g. A is in ft^2 (m^2), D in ft (m), d in inches (mm), Q in ft^3/s (m^3/s).

Field Units

$$V_g = \frac{Q}{A_g},$$

$$A_g = \frac{1}{2}\left(\frac{\pi}{4}D^2\right)$$

$$= \frac{1}{2}\left(\frac{\pi}{4}\frac{d^2}{144}\right)$$

$$= \frac{d^2}{367},$$

Q_g is in MMscfd,

$$
\begin{aligned}
Q &= Q_g \times 10^6 \frac{\text{scf}}{\text{MMscf}} \times \frac{\text{day}}{24\,\text{hr}} \times \frac{\text{hr}}{3{,}600\,\text{s}} \times \frac{14.7}{\text{P}} \times \frac{\text{TZ}}{520} \\
&= 0.327 \frac{\text{TZ}}{P} Q_g, \\
V_g &= \frac{\left(0.327 \frac{TZ}{P} Q_g\right)(367)}{d^2}, \\
V_g &= 120 \frac{TZQ_g}{Pd^2}.
\end{aligned}
$$

SI Units

$$
\begin{aligned}
V_g &= \frac{Q}{A_g}, \\
A_g &= \frac{1}{2}\left(\frac{\pi}{4} D^2\right) \\
&= \frac{1}{2}\frac{\pi}{4}\left(\frac{d}{1000}\right)^2 \\
&= 3.927 \times 10^{-7} \times d^2,
\end{aligned}
$$

Q_g is in scm/hr,

$$
\begin{aligned}
Q &= Q_g \times \frac{101.3}{P} \times \frac{TZ}{288.6} \times \frac{1\,\text{hr}}{3600\,\text{s}} \\
&= 9.75 \times 10^{-5} (TZQ_g)/P, \\
V_g &= \left[\frac{9.75 \times 10^{-5} \frac{TZ}{P} Q_g}{3.927 \times 10^{-7} d^2}\right], \\
V_g &= 248.3 \frac{TZQ_g}{Pd^2}.
\end{aligned}
$$

Set the residence time of the gas equal to the time required for the droplet to fall to the gas-liquid interface:

Field Units

$$t_g = \frac{L_{\text{eff}}}{V_g}, \quad t_d = \frac{D}{2V_t} = \frac{d}{24V_t},$$

$$t_g = \frac{L_{\text{eff}}}{120\left(\frac{TZQ_g}{Pd^2}\right)}.$$

Recalling that

$$V_t = 0.0119\left[\left(\frac{\rho_1 - \rho_g}{\rho_g}\right)\frac{d_m}{C_D}\right]^{1/2},$$

we have

$$t_d = \frac{d}{(24)\,(0.0119)}\left[\left(\frac{\rho_g}{\rho_1 - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}.$$

Setting $t_g = t_d$,

$$\frac{L_{\text{eff}}}{120\left(\frac{TZQ_g}{Pd^2}\right)} = \frac{d\left[\left(\frac{\rho_g}{\rho_1 - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}}{(24)\,(0.0119)},$$

$$L_{\text{eff}}d = 420\frac{TZQ_g}{P}\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}.$$

SI Units

$$t_g = \frac{L_{\text{eff}}}{V_g}, \quad t_d = \frac{D}{2V_t} = \frac{d}{2000V_t},$$

$$t_g = \frac{L_{\text{eff}}}{248.30\left(\frac{TZQ_g}{Pd^2}\right)},$$

$$V_t = 0.0036\left[\left(\frac{\rho_l - \rho_g}{\rho_g}\right)\frac{d_m}{C_D}\right]^{1/2},$$

$$t_d = \frac{d}{2000 \times 0.0036}\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}.$$

Setting $t_g = t_d$,

$$\frac{L_{\text{eff}}}{248.3\left(\frac{TZQ_g}{Pd^2}\right)} = \frac{d\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}}{2000 \times 0.0036},$$

$$L_{\text{eff}}d = 34.5\frac{TZQ_g}{P}\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}.$$

Liquid Capacity Constraint

Two-phase separators must be sized to provide some liquid retention time so the liquid can reach phase equilibrium with the gas. For a vessel 50% full of liquid, with a specified liquid flow rate and retention time, the following may be used to determine vessel size.

Field Units

$$d^2 L_{\text{eff}} = \frac{t_r Q_l}{0.7} \tag{4-9a}$$

SI Units

$$d^2 L_{\text{eff}} = 42{,}441 t_r Q_{\text{l}}, \tag{4-9b}$$

where

t_r = desired retention time for the liquid, min,
Q_l = liquid flow rate, bpd (m^3/hr).

Equations (4-9a) and (4-9b) are derived as follows [where the t is in s, V is in ft^3(m^3), and Q is in ft^3/s(m^3/ min)].

Field Units

$$t = \frac{V}{Q},$$

$$V = \frac{1}{2}\left(\frac{\pi D^2 L_{\text{eff}}}{4}\right)$$

$$= \frac{\pi d^2 L_{\text{eff}}}{(2)\,(4)\,(144)}$$

$$= 2.73 \times 10^{-3} d^2 L_{\text{eff}},$$

Q_1 is in BPD,

$$Q = Q_1 \times 5.62\left(\frac{\text{ft}^3}{\text{barrel}}\right)\left(\frac{\text{day}}{24\,\text{hr}}\right)\left(\frac{\text{hr}}{3{,}600\,\text{s}}\right)$$

$$= 6.50 \times 10^{-5} Q_l,$$

$$t = 42.0\frac{d^2 L_{\text{eff}}}{Q_l} \qquad (t = 60 t_r),$$

$$d^2 L_{\text{eff}} = \frac{t_r Q_l}{0.7}.$$

SI Units

$$V = \frac{1}{2}\left(\frac{\pi D^2 L_{\text{eff}}}{4}\right)$$

$$= \frac{\pi}{8} L_{\text{eff}}\left(\frac{d}{1000}\right)^2$$

$$= 3.927 \times 10^{-7} d^2 L_{\text{eff}},$$

Q_l is in m^3/min,

$$Q = Q_l \times \frac{1\,\text{hr}}{60\,\text{min}} = \frac{Q_l}{60},$$

$$t = \frac{\text{Vol}}{Q} = \frac{3.927 \times 10^{-7} d^2 L_{\text{eff}}}{\frac{Q_l}{60}},$$

$$t = 2.3562 \times 10^{-5}\frac{d_2 L_{\text{eff}}}{Q_l},$$

$$d^2 L_{\text{eff}} = 42{,}441 t_r Q_{l.}$$

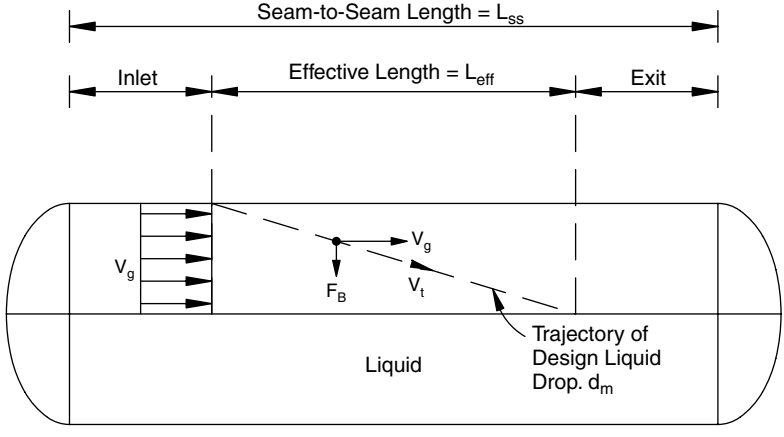
Seam-to-Seam Length

The effective length may be calculated from Eqs. (4-8a and 4-8b) and (4-9a and 4-9b). From this, a vessel seam-to-seam length may be determined. The actual required seam-to-seam length is dependent on the physical design of the internals of the vessel.

As shown in Figure 4-43, for vessels sized on a gas capacity basis, some portion of the vessel length is required to distribute the flow evenly near the inlet diverter. Another portion of the vessel length is required for the mist extractor. The length of the vessel between the inlet diverter and the mist extractor with evenly distributed flow is the L_{eff} calculated from Eqs. (4-8a) and (4-8b). As a vessel's diameter increases, more length is required to evenly distribute the gas flow. However, no matter how small the diameter may be, a portion of the length is still required for the mist extractor and flow distribution. Based on these concepts coupled with field experience, the seam-to-seam length of a vessel may be estimated as the larger of the following.

Field Units

$$L_{\text{ss}} = L_{\text{eff}} + \frac{d}{12} \qquad \text{for gas capacity} \tag{4-10a}$$



Legend:

V_g = Average Gas Velocity = $\frac{Q}{A}$

V_t = Terminal or Setting Velocity Relative to Gas

F_B = Buoyant Force

Figure 4-43. Approximate seam-to-seam length of a horizontal separator one-half full.

SI Units

$$L_{ss} = L_{eff} + \frac{d}{1000} \qquad \text{for gas capacity} \tag{4-10b}$$

For vessels sized on a liquid capacity basis, some portion of the vessel length is required for inlet diverter flow distribution and liquid outlet. The seam-to-seam length should not exceed the following:

$$L_{ss} = (4/3)L_{eff}. \tag{4-11}$$

Slenderness Ratio

Equations (4-8a and 4-8b) and (4-9a and 4-9b) allow for various choices of diameter and length. For each vessel design, a combination of L_{eff} and d exists that will minimize the cost of the vessel. It can be shown that the smaller the diameter, the less the vessel will weigh and thus the lower its cost. There is a point, however, where decreasing the diameter increases the possibility that high velocity in the gas flow will create waves and re-entrain liquids at the gas-liquid interface. Experience has shown that if the gas capacity governs and the length divided by the diameter, referred to as the "slenderness ratio," is greater than 4 or 5, re-entrainment could become a problem. Equation (4-11) indicates that slenderness ratios must be at least 1 or more. Most two-phase separators are designed for slenderness ratios between 3 and 4. Slenderness ratios outside the 3 to 4 range may be used, but the design should be checked to assure that re-entrainment will not occur.

Procedure for Sizing Horizontal Separators—Half Full

1. The first step in sizing a horizontal separator is to establish the design basis. This includes specifying the maximum and minimum flow rates, operating pressure and temperature, droplet size to be removed, etc.
2. Prepare a table with calculated values of L_{eff} for selected values of d that satisfy Eqs. (4-8a) and (4-8b), and the gas capacity constraint. Calculate L_{ss} using Eqs. (4-10a) and (4-10b).

   Field Units

$$L_{eff}d = 420\frac{TZQ_g}{P}\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}$$

SI Units

$$L_{\text{eff}}d = 34.5\frac{TZQ_g}{P}\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}$$

3. For the same values of d, calculate values of L_{eff} using Eqs. (4-9a) and (4-9b) for liquid capacity and list these values in the same table. Calculate L_{ss} using Eq. (4-11).

Field Units

$$d^2 L_{\text{eff}} = \frac{t_r Q_l}{0.7}$$

SI Units

$$d^2 L_{\text{eff}} = 42,441 t_r Q_l$$

4. For each d, the larger L_{eff} should be used.
5. Calculate the slenderness ratio, $12L_{\text{eff}}/d_o (1{,}000L_{\text{eff}}/d_o)$, and list for each d. Select a combination of d and L_{ss} that has a slenderness ratio between 3 and 4. Lower ratios can be chosen if dictated by available space, but they will probably be more expensive. Higher ratios can be chosen if the vessel is checked for re-entrainment.
6. When making a final selection, it is always more economical to select a standard vessel size. Vessels with outside diameters up through 24 inches (600 mm) have nominal pipe dimensions. Vessels with outside diameters larger than 24 inches (600 mm) are typically rolled from plate with diameter increments of 6 inches (150 mm). The shell seam-to-seam length is expanded in 2.5-ft (750-mm) segments and is usually from 5 ft to 10 ft (1500 mm to 3000 mm). Standard separator vessel sizes may be obtained from API 12J.

Horizontal Separators Sizing Other Than Half Full

The majority of oil field two-phase separators are designed with the liquid level at the vessel centerline, that is, 50% full of liquid. For a vessel other than 50% full of liquid, Eqs. (4-12a and 4-12b) and (4-13a and 4-13b) apply. These equations were derived using the actual gas and liquid areas to calculate gas velocity and liquid volume (refer to Figure 4-44).

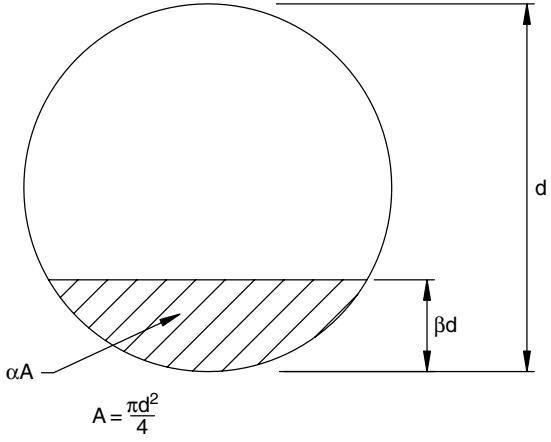


Figure 4-44. Definition of parallel areas.

Gas Capacity Constraint

Field Units

$$dL_{\text{eff}} = 420\left(\frac{1-\beta}{1-\alpha}\right)\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l-\rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}, \qquad (4\text{-}12a)$$

where

$$\left(\frac{1-\beta}{1-\alpha}\right) = \text{design constant}$$

$$= \text{Figure 4-45.}$$

SI Units

$$dL_{\text{eff}} = 34.5\left(\frac{1-\beta}{1-\alpha}\right)\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l-\rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}, \qquad (4\text{-}12b)$$

where

$$\left(\frac{1-\beta}{1-\alpha}\right) = \text{design constant}$$

$$= \text{Figure 4-46.}$$

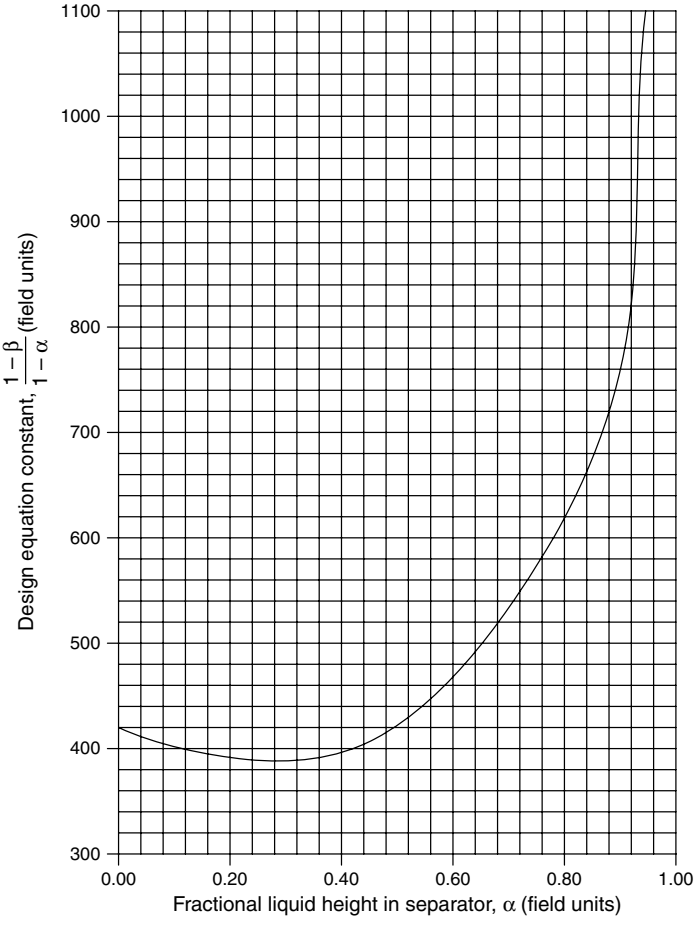


Figure 4-45. Gas capacity constraint design constant $[(1-\beta)/(1-\alpha)]$ vs. liquid height of a cylinder for a horizontal separator other than 50% full of liquid (field units).

Liquid Capacity Constraint

Field Units

$$d^2 L_{\text{eff}} = \frac{t_r Q_l}{1.4\alpha}, \tag{4-13a}$$

where

α = design constant

If β is known, α can be determined from Figure 4-47.

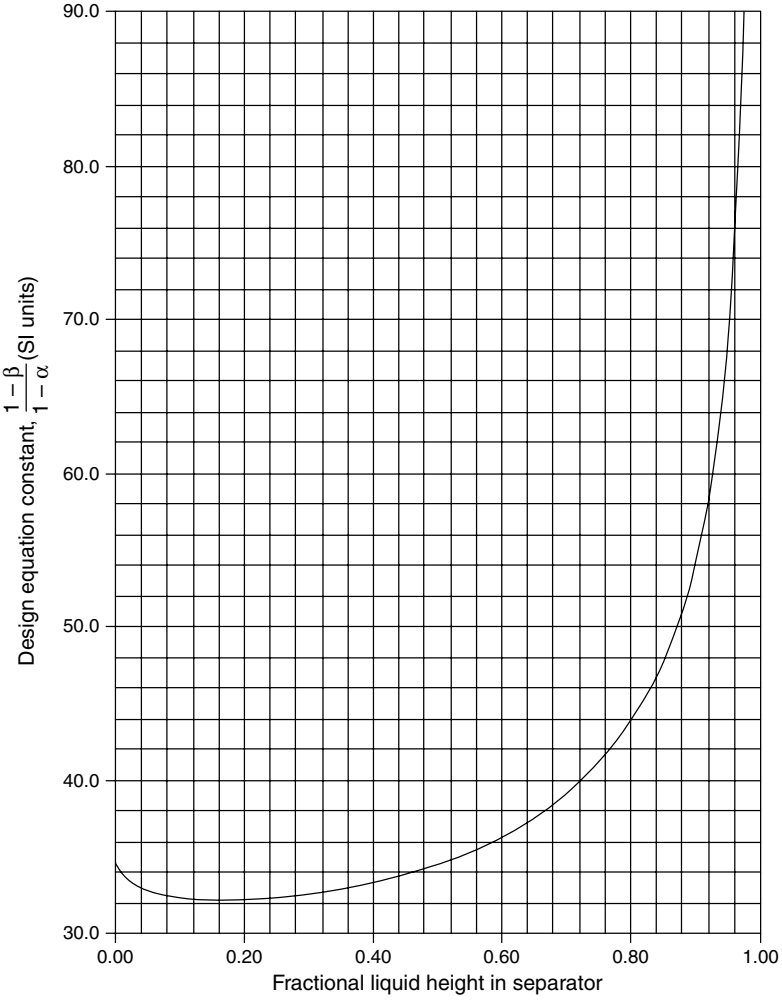


Figure 4-46. Gas capacity constraint design constant $[(1-\beta)/(1-\alpha)]$ vs. liquid height of a cylinder for a horizontal separator other than 50% full of liquid (SI units).

SI Units

$$d^2 L_{\text{eff}} = \frac{21{,}221 t_r Q_l}{\alpha}, \tag{4-13b}$$

where

α = design constant

If β is known, α can be determined from Figure 4-48.

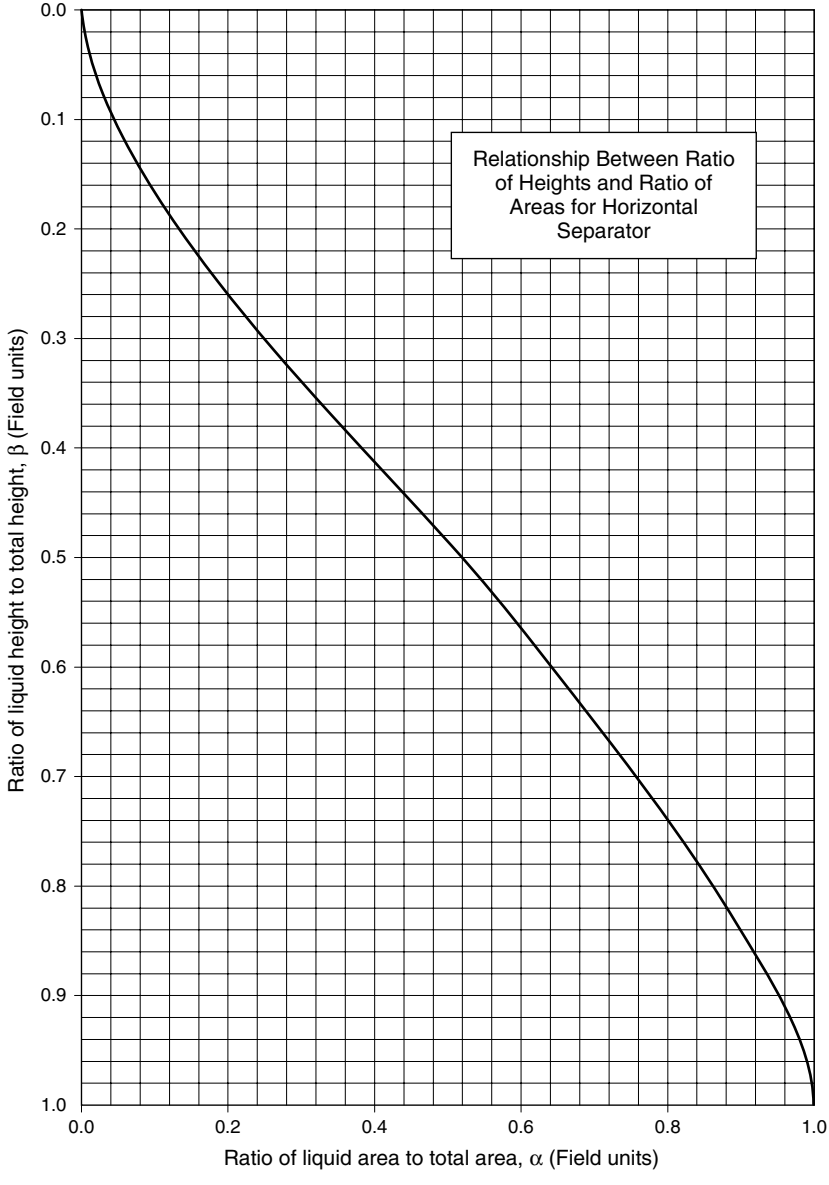


Figure 4-47. Liquid capacity constraint design constant—ratio of areas (α) vs. ratio of heights (β) for a horizontal separator other than 50% full of liquid (field units).

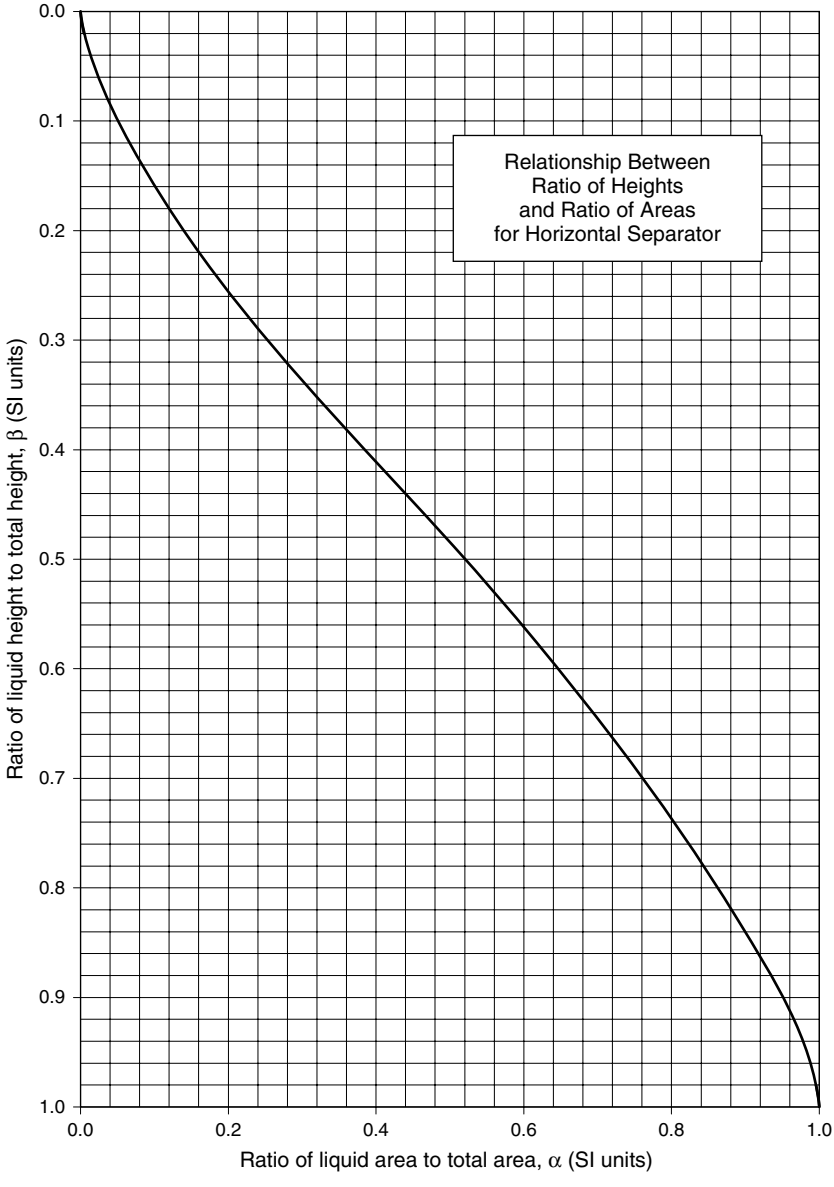


Figure 4-48. Liquid capacity constraint design constant—ratio of areas (α) vs. ratio of heights (β) for a horizontal separator other than 50% full of liquid (SI units).

Vertical Separators' Sizing

The guidelines presented in this section can be used for initial sizing of a vertical two-phase separator. They are meant to complement, and not replace, operating experience. Determination of the type and size of separator must be on an individual basis. All the functions and requirements should be considered, including the uncertainties in design flow rates and properties. For this reason, there is no substitute for good engineering evaluations of each separator by the design engineer. The "trade-off" between design size and details and uncertainties should not be left to manufacturer recommendations or rules of thumb.

In vertical separators, a minimum diameter must be maintained to allow liquid droplets to separate from the vertically moving gas. The liquid retention time requirement specifies a combination of diameter and liquid volume height. Any diameter greater than the minimum required for gas capacity can be chosen. Figure 4-49 shows the model used for a vertical separator.

Gas Capacity Constraint

The principles of liquid droplets settling through a gas can be used to develop an equation to size a separator for a gas flow rate. By setting the gas retention time equal to the time required for a droplet to settle to the liquid interface, the following equation may be derived.

Field Units

$$d^2 = 5{,}040\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2} \tag{4-14a}$$

SI Units

$$d^2 = 34{,}444\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2} \tag{4-14b}$$

Equations (4-14a) and (4-14b) may be derived as follows: for the droplets to fall, the gas velocity must be less than the terminal velocity of the droplet. Recall that

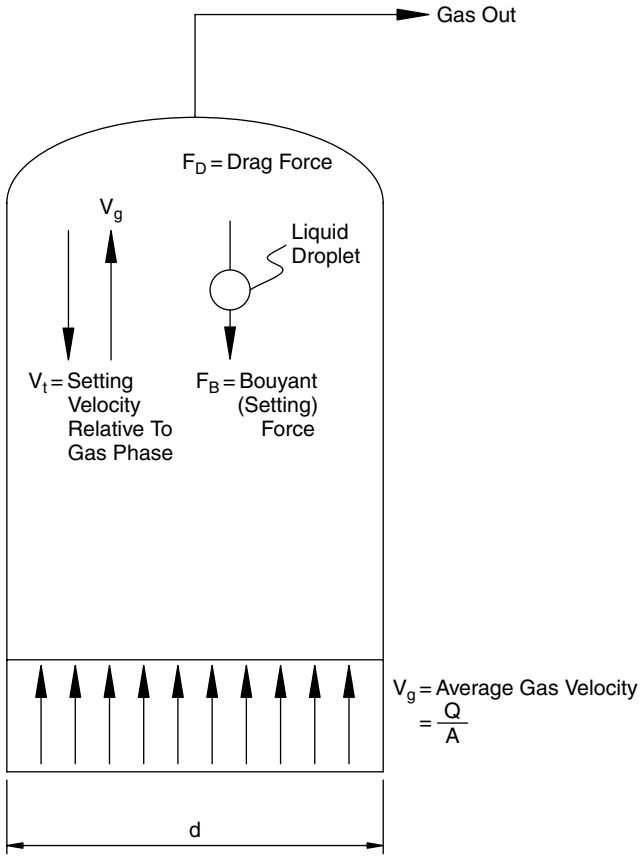


Figure 4-49. Model of a vertical separator.

Field Units

$$V_t = 0.0119\left[\left(\frac{\rho_l - \rho_g}{\rho_g}\right)\frac{d_m}{C_D}\right]^{1/2}$$

SI Units

$$V_t = 0.0036\left[\left(\frac{\rho_l - \rho_g}{\rho_g}\right)\frac{d_m}{C_D}\right]^{1/2}$$

Determine gas velocity, V_g. A is in ft^2 (m^2), D in ft (m), d in inches (mm), Q in ft^3/s (m^3/s).

Field Units

$$V_g = \frac{Q}{A_g},$$

$$A_g = \left(\frac{\pi}{4}D^2\right)$$

$$= \left(\frac{\pi}{4}\frac{d^2}{144}\right)$$

$$= \frac{d^2}{183},$$

Q_g is in MMscfd,

$$Q = Q_g \times 10^6 \left(\frac{\text{scf}}{\text{MMscf}}\right)\left(\frac{\text{day}}{24\,\text{hr}}\right)\left(\frac{\text{hr}}{3{,}600\,\text{s}}\right) \times \frac{14.7}{P} \times \frac{TZ}{520}$$

$$= 0.327\frac{TZ}{P}Q_g,$$

$$V_g = \frac{\left(0.327\frac{TZ}{P}Q_g\right)(183)}{d^2},$$

$$V_g = 60\frac{TZQ_g}{Pd^2},$$

$$V_t = V_g,$$

$$0.0119\left[\left(\frac{\rho_l - \rho_g}{\rho_g}\right)\frac{d_m}{C_D}\right]^{1/2} = \frac{60TZQ_g}{Pd^2},$$

$$d^2 = 5{,}040\frac{TZQ_g}{P}\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}.$$

SI Units

$$V_g = \frac{Q}{A_g},$$

$$A_g = \left(\frac{\pi}{4}D^2\right)$$

$$= \frac{\pi}{4}\left(\frac{d^2}{1000^2}\right)$$

$$= 7.855 \times 10^{-7}d^2,$$

$$Q_g = \text{scm/s},$$

$$Q = Q_g \times \frac{101.3}{P} \times \frac{TZ}{288.6} \times \frac{1\,\text{hr}}{3600\,\text{s}}$$

$$= 9.75 \times 10^{-5} \frac{TZ}{P} Q_g,$$

$$V_g = \frac{9.75 \times 10^{-5} \frac{TZQ_g}{P}}{7.855 \times 10^{-7} d^2},$$

$$V_g = 124 \frac{TZQ_g}{Pd^2},$$

$$V_t = V_{g,}$$

$$0.0036 \left[\left(\frac{\rho_l - \rho_g}{\rho_g} \right) \frac{d_m}{C_D} \right]^{1/2} = 124 \frac{TZQ_g}{Pd^2},$$

$$d^2 = 34{,}444 \frac{TZQ_g}{P} \left[\left(\frac{\rho_g}{\rho_l - \rho_g} \right) \frac{C_D}{d_m} \right]^{1/2}.$$

Liquid Capacity Constraint

Two-phase separators must be sized to provide some liquid retention time so the liquid can reach phase equilibrium with the gas. For a specified liquid flow rate and retention time, the following may be used to determine a vessel size.

Field Units

$$d^2 h = \frac{t_r Q_l}{0.12} \tag{4-15a}$$

SI Units

$$d^2 h = \frac{t_r Q_l}{4.713 \times 10^{-8}}, \tag{4-15b}$$

where h = height of the liquid volume, in. (mm).

Equations (4-15a) and (4-15b) are derived as follows: where t is in s, V is in ft^3(m^3), Q is in ft^3/s (m^3/s), and h is in inches (mm).

Field Units

$$t = \frac{V}{Q},$$

$$V = \frac{\pi D^2 h}{(4)\,(12)}$$

$$= \frac{\pi \times d^2 \times h}{4 \times 144 \times 12}$$

$$= 4.55 \times 10^{-4} d^2 h,$$

Q_1 is in BPD,

$$Q = Q_l \times 5.61 \left(\frac{\text{ft}^3}{\text{barrel}}\right)\left(\frac{\text{day}}{24\,\text{hr}}\right)\left(\frac{\text{hr}}{3{,}600\,\text{s}}\right)$$

$$= 6.49 \times 10^{-5} Q_l,$$

$$t = \frac{V}{Q} = \frac{4.55 \times 10^{-4} d^2 h}{6.49 \times 10^{-5} Q_l},$$

$$t = 7.00 \frac{d^2 h}{Q_l},$$

t_r is in min $(t = 60 t_r)$,

$$d^2 h = \frac{t_r Q_l}{0.12}.$$

SI Units

$$t = \frac{V}{Q},$$

$$V = \frac{\pi D^2}{4} \times \frac{h}{1000}$$

$$= \frac{\pi d^2 h}{4 \times 1000^2 \times 1000}$$

$$= 7.854 \times 10^{-10} d^2 h,$$

$$Q_l = \text{m}^3/\text{hr},$$

$$Q = Q_l \times \frac{1\,\text{hr}}{3600\,\text{s}}$$

$$= \frac{Q_l}{3600},$$

$$t = \frac{V}{Q} = \frac{7.853 \times 10^{-10} d^2 h}{\frac{Q_l}{3600}},$$

$$t = 2.828 \times 10^{-6} \frac{d^2 h}{Q_l},$$

t_r is in min,

$$d^2 h = \frac{t_r Q_l}{4.713 \times 10^{-8}}.$$

Seam-to-Seam Length

As with horizontal separators, the specific design of the vessel internals will affect the seam-to-seam length. The seam-to-seam length of vertical vessels may be estimated based on the diameter and liquid height. As shown in Figure 4-50, allowance must be made for the gas separation section and mist extractor and for any space below the water outlet. For screening purposes, the following may be used to estimate L_{ss}.

Field Units

$$L_{ss} = \frac{h + 76}{12} \qquad \text{(for diameters} \leq 36\,\text{in.)} \tag{4-16a}$$

SI Units

$$L_{ss} = \frac{h + 1930}{1000} \qquad \text{(for diameters} \geq 194\,\text{mm)} \tag{4-16b}$$

Field Units

$$L_{SS} = \frac{h + d + 40}{12} \qquad \text{(for diameters} > 36\,\text{in.)} \tag{4-17a}$$

SI Units

$$\frac{h + d + 1016}{1000} \qquad \text{(for diameters} > 194\,\text{mm)}, \tag{4-17b}$$

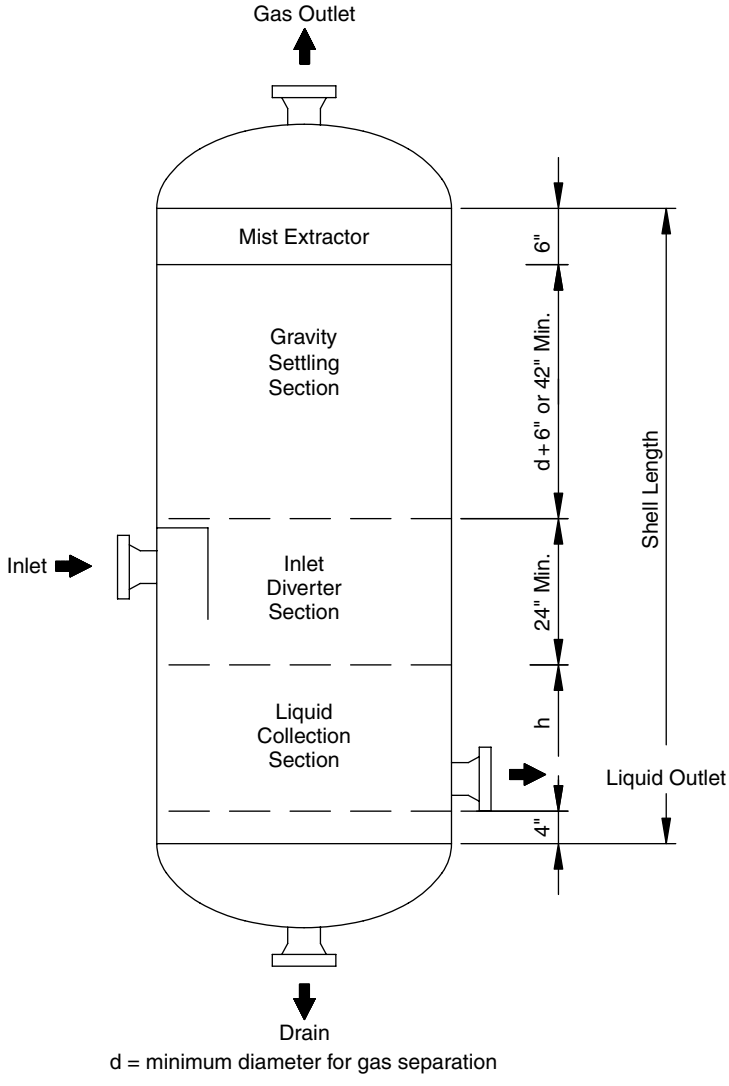


Figure 4-50. Approximate seam-to-seam shell length for a vertical separator.

where

h = height of liquid level, in. (mm),
d = vessel ID, in. (mm).

The larger of the L_{ss} values from Eqs. (4-16a and 4-16b) and (4-17a and 4-17b) should be used.

Slenderness Ratio

As with horizontal separators, the larger the slenderness ratio, the less expensive the vessel will be. In vertical separators whose sizing is liquid dominated, it is common to choose slenderness ratios no greater than 4 to keep the height of the liquid collection section to a reasonable level. Choices of between 3 and 4 are common, although height restrictions may force the choice of a lower slenderness ratio.

Procedure for Sizing Vertical Separators

1. The first step in sizing a vertical separator is to establish the design basis. This includes specifying the maximum and minimum flow rates, operating pressure and temperature, droplet size to be removed, etc.
2. Equations (4-14a) and (4-14b) may be used to determine the minimum required d. Any diameter larger than this value may be used.
3. For a selected d, Eqs. (4-15a) and (4-15b) may be used to determine h.
4. From d and h, the seam-to-seam length may be estimated using Eqs. (4-16a and 4-16b) and (4-17a and 4-17b). The larger value of L_{ss} should be used.
5. Check the slenderness ratio to determine if it is less than 4.
6. When making a final selection, it is always more economical to select a standard vessel size. Vessels with outside diameters up through 24 inches (600 mm) have nominal pipe dimensions. Vessels with outside diameters larger than 24 inches (600 mm) are rolled from plate with diameter increments of 6 inches (150 mm). The shell seam-to-seam length is expanded in 2.5-ft (750-mm) segments and is usually from 5 ft to 10 ft (1500 mm to 3000 mm). Standard separator vessel sizes may be obtained from API 12J.

Examples

Example 4-1: Sizing a Vertical Separator (Field Units)

Given:

Gas flow rate:	10 MMSCFD at 0.6 specific gravity
Oil flow rate:	2,000 BOPD at 40°API
Operating pressure:	1,000 psia
Operating temperature:	60°F
Droplet size removal:	140 microns
Retention time:	3 min

Solution:

1. Calculate C_D.

$$\rho_l = 62.4\left[\left(\frac{141.5}{(131.5+40)}\right)\right]$$

$$= 51.5\frac{\text{lb}}{\text{ft}^3},$$

$$\rho_g = 2.70\frac{SP}{TZ},$$

$$Z = 0.84 \text{ (from Chapter 3)},$$

$$\rho_g = 2.70\frac{(0.6)\,(1{,}000)}{(520)\,(0.84)} = 3.71\,\text{lb/ft}^3,$$

$$d_m = 140 \text{ micron},$$

$$\mu = 0.013\,\text{cp (from Chapter 3)}.$$

Assume $C_D = 0.34$,

$$V_t = 0.0119\left[\left(\frac{51.5-3.71}{3.71}\right)\frac{140}{0.34}\right]^{1/2},$$

$$V_t = 0.867\,\text{ft/s},$$

$$\text{Re} = 0.0049\left[\frac{(3.71)\,(140)\,(0.866)}{0.013}\right] = 169.54,$$

$$C_D = \frac{24}{169.54} + \frac{3}{(169.54)^{1/2}} + 0.34,$$

$$C_D = 0.712.$$

Repeat using $C_D = 0.712$.

$$V_t = 0.599\,\text{ft/s},$$

$$\text{Re} = 117,$$

$$C_D = 0.822.$$

Repeat:

$$V_t = 0.556,$$

$$\text{Re} = 110,$$

$$C_D = 0.844.$$

Repeat:

$$V_t = 0.548,$$
$$\text{Re} = 108,$$
$$C_D = 0.851.$$

Repeat:

$$V_t = 0.545,$$
$$\text{Re} = 108,$$
$$C_D = 0.854\text{—OK}.$$

2. Gas capacity constraint

$$d^2 = 5{,}040\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2},$$

$Z = 0.84$ (from Chapter 3),

$$d^2 = 5{,}040\left[\frac{(520)\,(0.84)\,(10)}{1{,}000}\right]\left[\left(\frac{3.71}{51.5 - 3.71}\right)\frac{0.851}{140}\right]^{1/2},$$

$d = 21.9$ in.

3. Liquid capacity constraint

$$d^2 h = \frac{t_r Q_l}{0.12}$$

4. Compute combinations of d and h for various t_r (Table 4-3).
5. Compute seam-to-seam length (Table 4-3).

$$L_{ss} = \frac{h + 76}{12} \quad \text{or} \quad L_{ss} = \frac{h + d + 40}{12},$$

where d is the minimum diameter for gas capacity

6. Compute slenderness ratio: $12L_{ss}/d$. Choices in the range of 3 to 4 are most common (Table 4-3).
7. Choose a reasonable size with a diameter greater than that determined by the gas capacity. A 36-in. diameter by 10-ft. seam-to-seam separator provides slightly more than 3 minutes' retention time with a diameter greater than 21.8 in. and a slenderness ratio of 3.2.

Table 4-3
Vertical Separator Example Diameter vs. Length for Liquid Capacity Constraint

t_r (min)	d (in.)	h (in.)	L_{ss} (ft.)	$S_R\left(\frac{12L_{ss}}{d}\right)$
3	24	86.8	13.6	6.8
	30	55.6	11.0	4.4
	36	38.6	9.6	3.2
	42	28.3	8.7	2.5
	48	21.7	8.1	2.0
2	24	57.9	11.2	5.6
	30	37.0	9.4	3.8
	36	25.7	8.5	2.8
	42	18.9	7.9	2.3
1	24	28.9	8.7	4.4
	30	18.5	7.9	3.2
	36	12.9	7.4	2.5

Example 4-2: Sizing a Vertical Separator (SI Units)

Given:

Gas flow rate:	11,803 scm/hr at 0.6 specific gravity
Oil flow rate:	317.6 m^3/hr at 40°API
Operating pressure:	6900 kPa
Operating temperature:	15.6°C
Droplet size removal:	140 microns
Retention time:	3 minutes

Solution:

1. Calculate C_D:

$$\rho_l = 1000\left[\frac{141.5}{131.5+40}\right] = 825\frac{\text{kg}}{\text{m}^3},$$

$$\rho_g = 3.492\frac{SP}{TZ},$$

$$Z = 0.84 \text{ (from Chapter 3)},$$

$$\rho_g = 3.492\frac{(0.6)\,(6900)}{(288.6)\,(0.84)} = 59.6\frac{\text{kg}}{\text{m}^3},$$

$d_m = 140$ micron,
$\mu = 0.013$ cp (from Chapter 3).

Assume $C_D = 0.34$.

$$V_t = 0.0036\left[\left(\frac{825-59.6}{59.6}\right)\frac{140}{0.34}\right]^{1/2},$$

$$V_t = 0.2618\,\text{m/s},$$

$$\text{Re} = 0.001\left[\frac{(59.6)\,(140)\,(0.2618)}{0.013}\right] = 168,$$

$$C_D = \frac{24}{168} + \frac{3}{(168)^{1/2}} + 0.34,$$

$$C_D = 0.714.$$

Repeat using $C_D = 0.714$.

$$V_t = 0.1812\,\text{m/s},$$
$$\text{Re} = 116,$$
$$C_D = 0.86.$$

Repeat:

$$V_t = 0.1686,$$
$$\text{Re} = 108,$$
$$C_D = 0.851\text{—OK}.$$

2. Gas capacity constraint

$$d^2 = 34{,}444\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2},$$

$Z = 0.84$ (from Chapter 3),

$$d^2 = 34{,}444\left[\frac{(288.8)\,(0.84)\,(11803)}{6900}\right]\left[\left(\frac{59.6}{825-59.6}\right)\frac{0.851}{140}\right]^{1/2},$$

$$d = 557.5\,\text{mm}.$$

3. Liquid capacity constraint

$$d^2h = \frac{t_r Q_L}{4{,}713 \times 10^{-8}}$$

4. Compute combinations of d and h for various t_r (Table 4-4).
5. Compute seam-to-seam length (Table 4-4).

$$L_{ss} = \frac{h + 1930}{1000} \quad \text{or} \quad = \frac{h + d + 1016}{1000},$$

where d is the minimum diameter for gas capacity.
6. Compute slenderness ratio,

$$\frac{L_{ss}}{[(1000)\,(d)]}.$$

Choices in the range of 3 to 4 are most common (Table 4-4).
7. Choose a reasonable size with a diameter greater than that determined by the gas capacity. A 914 mm diameter by 3 m seam-to-seam separator provides slightly more than 3 minutes' retention time with a diameter greater than 557.5 mm and a slenderness ratio of 3.2.

Table 4-4
Vertical Separator Example Diameter vs. Length for Liquid Capacity Constraint

t_r (min)	d (mm)	h (mm)	L_{ss} (m)	$S_R \frac{L_{ss}}{d(1000)}$
3	609.6	2268	4.2	6.8
	762	1453	3.4	4.4
	914.4	1009	2.9	3.2
	1066.8	741	2.7	2.5
	1219.2	667	2.6	2.0
2	609.6	1513	3.4	5.6
	762	968	2.9	3.8
	914.4	672	2.6	2.8
	1066.8	494	2.4	2.3
1	609.6	767	2.7	4.4
	762	484	2.4	3.2
	914.4	336	2.3	2.5

Example 4-3: Sizing a Horizontal Separator (Field Units)

Given:

Gas flow rate:	10 MMscfd at 0.6 specific gravity
Oil flow rate:	2,000 BOPD at 40°API
Operating pressure:	1,000 psia
Operating temperature:	60°F
Droplet size removal:	140 microns
Retention time:	3 minutes

Solution:

1. Calculate C_D (same as Examples 4-1 and 4-2).

$$C_D = 0.851$$

2. Gas capacity constraint

$$dL_{\text{eff}} = 420\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_{l-}\rho_g}\right)\frac{C_D}{d_m}\right]^{1/2},$$

$$Z = 0.84 \text{ (from Chapter 3)},$$

$$dL_{\text{eff}} = 420\left[\frac{(520)\,(0.84)\,(10)}{1{,}000}\right]\left[\left(\frac{3.71}{51.5-3.71}\right)\frac{0.851}{140}\right]^{1/2}$$

$$= 55.04.$$

3. Liquid capacity constraint

$$d^2L_{\text{eff}} = \frac{t_r Q_l}{0.7}$$

4. Compute combinations of d and L_{ss} for gas and liquid capacity.
5. Compute seam-to-seam length for various d (Table 4-5).

$$L_{\text{ss}} = L_{\text{eff}} + \frac{d}{12}$$

6. Compute slenderness ratios, $12L_{\text{ss}}/d$. Choices in the range of 3 to 4 are common.
7. Choose a reasonable size with a diameter and length combination above both the gas capacity and the liquid capacity constraint lines. A 36-in. × 10-ft separator provides about 3 minutes' retention time.

Table 4-5
Horizontal Separator Example Diameter vs. Length

d (ft)	Gas L_{eff} (ft)	Liquid L_{eff} (ft)	L_{ss} (ft)	$12L_{ss}/d$
16	2.5	33.5	44.7	33.5
20	2.0	21.4	28.5	17.1
24	1.7	14.9	19.9	9.9
30	1.3	9.5	12.7	5.1
36	1.1	6.6	9.1*	3.0
42	0.9	4.9	7.4*	2.1
48	0.8	3.7	6.2*	1.6

*$L_{ss} = L_{eff} + 2.5$ governs.

Example 4-4: Sizing a Horizontal Separator (SI Units)

Given:

Gas flow rate:	11,803 scf/hr at 0.6 specific gravity
Oil flow rate:	13.25 m^3/hr at 40°API
Operating pressure:	6900 kPa
Operating temperature:	15.6°C
Droplet size removal:	140 microns
Retention time:	3 minutes

Solution:

1. Calculate C_D (same as Examples 4-1 and 4-2).

$$C_D = 0.85$$

2. Gas capacity constraint

$$dL_{\text{eff}} = 34.5\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2},$$

$Z = 0.84$ (from Chapter 3),

$$dL_{\text{eff}} = 34.5\left[\frac{(288.6)\,(0.84)\,(11803)}{6900}\right]\left[\left(\frac{59.6}{825 - 59.6}\right)\frac{0.851}{140}\right]^{1/2},$$

$$dL_{\text{eff}} = 311.3.$$

Table 4-6
Horizontal Separator Example Diameter vs. Length

d (mm)	Gas L_{eff} (m)	Liquid L_{eff} (m)	L_{ss} (m)	$S_R \frac{L_{ss}}{[(1000)(d)]}$
406.4	0.77	10.21	13.62	33.5
508	0.61	6.54	8.72	17.1
609.6	0.51	4.54	6.05	9.9
762	0.41	2.91	3.87	5.1
914.4	0.39	2.02	*2.78	3.0
1066.8	0.29	1.48	*2.24	2.1
1219.2	0.2	1.13	*1.90	1.6

*$L_{ss} = L_{eff} + 2.5$ governs.

3. Liquid capacity constraint

$$d^2 L_{\text{eff}} = 42.441 t_r Q_l$$

4. Compute combinations of d and L_{ss} for gas and liquid capacity.
5. Compute seam-to-seam length for various d (Table 4-6).

$$L_{ss} = L_{\text{eff}} + \frac{d}{1000}$$

6. Compute slenderness ratios:

$$\frac{L_{ss}}{[(1000)\,(d)]}.$$

Choices in the range of 3 to 4 are common.

7. Choose a reasonable size with a diameter and length combination above both the gas capacity and the liquid capacity constraint lines. A 914 mm- by 3-m separator provides about 3 minutes' retention time.

Nomenclature

A_d = cross-sectional area of the droplet, ft^2 (m^2)
A_g = cross-sectional area of vessel available for gas settling, ft^2 (m^2)
A_l = cross-sectional area of vessel available for liquid retention, ft^2 (m^2)
A_T = total cross-sectional area of vessel, ft^2 (m^2)

API = API gravity of oil, °API
CA = corrosion allowance, in (mm)
C_D = drag coefficient, dimensionless
D_m = droplet diameter, ft (m)
D = vessel's internal diameter, ft (m)
D_h = hydraulic diameter, ft (m)
d = vessel's internal diameter, in. (mm)
d_m = droplet's diameter, micron (μ)
$d_{\min}$ = min allowable vessel internal diameter to avoid re-entrainment, in. (mm)
d_o = vessel's external diameter, in. (mm)
E = joint efficiency, dimensionless
F_B = buoyant force, lb (N)
F_D = drag force, lb (N)
g = gravitational constant, $32.2 \text{lb}_\text{m}\text{ft}/\text{lb}_\text{f}\text{s}^2$ ($9.81\,\text{m}/\text{s}^2$)
H = height of liquid volume, ft (m)
h = height of liquid volume, in. (mm)
H_l = height of liquid in horizontal vessel, ft (m)
h_l = height of liquid in horizontal vessel, in. (mm)
L_{eff} = effective length of the vessel, ft (m)
L_{ss} = vessel length seam-to-seam, ft (m)
N_μ = viscosity number, dimensionless
P = operating pressure, psia (kPa)
P_b = pressure base, 14.7 psia (100 kPa)
P_c = gas pseudo-critical pressure, psia (kPa)
P_{cc} = corrected pseudo-critical pressure, psia (kPa)
P_d = design pressure, psia (kPa)
P_r = gas reduced pressure, dimensionless
Q = flow rate, ft^3/s (m^3/s)
Q_g = gas flow rate, MMscfd (std m^3/hr)
Q_l = liquid flow rate, BPD (std m^3/hr)
r = vessel external radius, in. (mm)
Re = Reynolds number, dimensionless
S = allowable stress, psia (kPa)
T = operating temperature, °R (K)
t = shell thickness, in. (mm)
T_b = temperature base, 520°R (288.15 K)
T_c = gas pseudo-critical temperature, °R (K)
T_{cc} = corrected pseudo-critical temperature, °R (K)
t_d = droplet settling time, s
t_g = gas retention time, s
T_r = gas reduced temperature, dimensionless
t_r = liquid retention time, min

V_g = gas velocity, ft/s (m/s)
V_l = average liquid velocity, ft/s (m/s)
V_t = terminal settling velocity of the droplet, ft/s (m/s)
W = vessel weight, lb (kg)
Y_{CO_2} = gas mole fraction CO_2
Y_{H_2S} = gas mole fraction H_2S
Z = gas compressibility factor, dimensionless
α = fractional cross-sectional area of liquid, dimensionless
β = fractional height of liquid within the vessel = h_l/d
ΔSG = difference in specific gravity relative to water of the droplet and the gas
Δ_ρ = density difference, liquid and gas lbm/ft^3 (kg/m^3)
ΔT = Wichert–Aziz correction, °R (K)
θ = angle used in determining α, radians degrees
μ = gas viscosity, cp
μ_l = dynamic viscosity of the liquid, lbm/ft-s (kg/m-s)
μ' = gas viscosity, cp (lb-s/ft^2)
ρ = density of the continuous phase, lb/ft^3 (kg/m^3)
ρ_g = density of the gas at the temperature and pressure in the separator, lb/ft^3 (kg/m^3)
ρ_l = density of liquid, lb/ft^3 (kg/m^3)
ρ_m = gas density, g/cm^3
ρ_r = reduced density
ρ_{r+1} = value of reduced density for iteration "$r+1$"
σ = surface tension lbm/s^2 (kg/s^2)

Review Questions

1. The advantage(s) of a vertical separator is (are)
   a) requires less plan area than a horizontal separator of equal size
   b) less expensive than equally sized horizontal separator
   c) have less liquid surge capacity than horizontal vessels sized for the same steady-state flow rate
   d) more efficient from a pure gas-liquid separation process
2. Scrubbers
   a) are two-phase separators
   b) are usually installed downstream of production separators
   c) protect compression equipment from liquid carryover

d) all of the above
e) B and C only

3. A separation vessel that removes entrained mist, rust, and/or scale with filter elements is a

a) cyclone mist extractor
b) filter separator
c) slug catcher
d) horizontal double-barrel separator
e) wire-mesh mist extractor

4. A propriety scrubber that separates liquid droplets and dust from a gas stream by a swirling action is called a(n)

a) filter scrubber
b) impingement-type separator
c) cyclone mist extractor
d) centrifugal cyclone separator
e) spherical separator

5. List the four functional sections of a gas-liquid separator.

6. The inlet diverter

a) abruptly changes the direction of flow by absorbing the momentum of the liquid
b) uses the inertia of the incoming fluid to achieve an initial free liquid separation
c) lowers the temperature of the incoming fluid
d) lowers both the specific gravity and viscosity of the oil
e) is sized so that liquid droplets greater than 100 to 140 microns fall to the gas-liquid interface

7. When selecting a mist extractor, which of the following factors should be evaluated:

a) size of droplets the separator must remove
b) maximum pressure drop that can be tolerated to achieve the required level of removal
c) liquid handling capability of the separator
d) susceptibility of the separator to plugging of solids, if solids are present
e) whether the mist extractor can be installed inside existing equipment, or if it requires a standalone vessel

8. In the gravity settling section of a separator, the velocity where the drag forces acting on the liquid droplet are equal to the buoyant forces is called

   a) coalescing velocity
   b) settling velocity
   c) interface velocity
   d) stall velocity
   e) gas-liquid velocity

9. Gas blowby can be an indication of

   a) low liquid level
   b) level control valve failure
   c) vortexing
   d) all of the above
   e) A and C only

10. For most two-phase separator applications, retention times

    a) range between 30 seconds and 3 minutes
    b) are dependent upon API gravity
    c) are lower for horizontal separators
    d) determine the volume of the liquid collection section
    e) A, B, and D

11. Micro-fiber mist extractors

    a) use very small diameter fibers to capture very small droplets
    b) surface area can be 3 to 150 times that of a wire-mesh unit equal volume
    c) are prone to plugging by the accumulation of paraffins
    d) are the most expensive type of mist extractor
    e) gas and liquid flow is horizontal and co-current

12. Which of the following can cause crude oil to foam in a separator?

    a) CO_2
    b) completion and workover fluids that are incompatible with the wellbore fluids
    c) paraffin hydrocarbons
    d) A and B
    e) all of the above

13. Which of the following factors affect gas-liquid separation?

    a) Gas and liquid flow rates (minimum, average, and peak)
    b) physical properties of the fluids, such as specific gravity and compressibility
    c) operating and design pressures and temperatures
    d) foaming tendencies of the crude oil
    e) all of the above

14. Slug catchers

    a) are a special case of a two-phase gas-liquid separator
    b) are designed to handle large gas capacities and liquid slugs on a regular basis
    c) can be designed in either vertical or horizontal configurations
    d) sometimes include liquid "fingers"
    e) all of the above

Exercises

Problem 1.

Determine the size of a vertical two-phase separator given the following data:

Q_g	= 1.6 MMscfd,
Q_o	= 3,900 BOPD,
Q_w	= 3,000 BWPD,
P_o	= 455 psia,
T_o	= 90°F,
S_g	= 0.6,
$(\text{SG})_o$	= 30°API,
$(\text{SG})_w$	= 1.07,
droplet size removal	= 100 microns,
retention time	= 2 min.

Problem 2.

Determine the size of a horizontal two-phase separator given the following data:

Q_g	= 1.6 MMscfd,
Q_o	= 3,900 BOPD,

Q_w	= 3,000 BWPD,
P_o	= 455 psia,
T_o	= 90°F,
S_g	= 0.6,
$(SG)_o$	= 30°API,
$(SG)_w$	= 1.07,
droplet size removal	= 100 microns,
retention time	= 2 min.

Problem 3.

Determine the size of a vertical two-phase separator given the following data:

Q_g	= 2.75 MMscfd,
Q_o	= 5,000 BOPD,
Q_w	= 1,000 BWPD,
P_o	= 1,015 psia,
T_o	= 90°F,
S_g	= 0.6,
$(SG)_o$	= 30°API,
$(SG)_w$	= 1.07,
droplet size removal	= 100 microns,
retention time	= 2 min.

Problem 4.

Determine the size of a horizontal two-phase separator given the following data:

Q_g	= 2.75 MMscfd,
Q_o	= 5,000 BOPD,
Q_w	= 1,000 BWPD,
P_o	= 1,015 psia,
T_o	= 90°F,
S_g	= 0.6,
$(SG)_o$	= 30°API,
$(SG)_w$	= 1.07,
droplet size removal	= 100 microns,
retention time	= 2 min.

Problem 5.

Determine the size of a double-barrel horizontal two-phase separator given the following data:

Q_g	= 100 MMscfd,
Q_o	= 4,000 BOPD,
Q_w	= 2,000 BWPD,
P_o	= 1,000 psia,
T_o	= 90°F,
S_g	= 0.6,
$(SG)_o$	= 45°API,
$(SG)_w$	= 1.07,
droplet size removal	= 100 microns,
retention time	= 2 min.

Problem 6.

Determine the size of a vertical two-phase separator given the following data:

Q_g	= 1,960 scm/hr,
Q_o	= 620 m^3/hr,
Q_w	= 475 m^3/hr,
P_o	= 3,140 kPa,
T_o	= 35°C,
S_g	= 0.6,
$(SG)_o$	= 30°API,
$(SG)_w$	= 1.07,
droplet size removal	= 100 microns,
retention time	= 2 min.

Problem 7.

Determine the size of a horizontal two-phase separator given the following data:

Q_g	= 1,960 scm/hr,
Q_o	= 620 m^3/hr,

Q_w	$= 475\,m^3/hr$,
P_o	$= 3{,}140\,kPa$,
T_o	$= 35°C$,
S_g	$= 0.6$,
$(SG)_o$	$= 30°API$,
$(SG)_w$	$= 1.07$,
droplet size removal	$= 100$ microns,
retention time	$= 2$ min.

Problem 8.

Determine the size of a vertical two-phase separator given the following data:

Q_g	= 3,365 scm/hr,
Q_o	$= 795\,m^3/hr$,
Q_w	$= 160\,m^3/hr$,
P_o	$= 150\,kPa$
T_o	$= 35°C$,
S_g	$= 0.6$,
$(SG)_o$	$= 30°API$,
$(SG)_w$	$= 1.07$,
droplet size removal	$= 100$ microns,
retention time	$= 2$ min.

Problem 9.

Determine the size of a horizontal two-phase separator given the following data:

Q_g	$= 3{,}365$ scm/hr,
Q_o	$= 795\,m^3/hr$,
Q_w	$= 160\,m^3/hr$,
P_o	$= 150\,kPa$,
T_o	$= 35°C$,
S_g	$= 0.6$,
$(SG)_o$	$= 30°API$,
$(SG)_w$	$= 1.07$,
droplet size removal	$= 100$ microns,
retention time	$= 2$ min.

Problem 10.

Determine the size of a horizontal double-barrel two-phase separator given the following data:

Q_g	= 120,000 scm/hr,
Q_o	= 635 m^3/hr,
Q_w	= 317.6 m^3/hr,
P_o	= 6,900 kPa,
T_o	= 35°C,
S_g	= 0.6,
$(SG)_o$	= 45° API,
$(SG)_w$	= 1.07,
droplet size removal	= 100 microns,
retention time	= 2 min.

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Chapter 5
Three-Phase Oil and Water Separation

Introduction

This chapter discusses the concepts, theory, and sizing equations for the separation of two immiscible liquid phases (in this case, those liquids are normally crude oil and produced water). The separator design concepts presented in Chapter 4 relate to the two-phase separation of liquid and gas and are applicable to the separation of gas that takes place in three-phase separators, gas scrubbers, and any other device in which gas is separated from a liquid phase.

When oil and water are mixed with some intensity and then allowed to settle, a layer of relatively clean free water will appear at the bottom. The growth of this water layer with time will follow a curve as shown in Figure 5-1. After a period of time, ranging anywhere from 3 minutes to 30 minutes, the change in the water height will be negligible. The water fraction, obtained from gravity settling, is called "free water." It is normally beneficial to separate the free water before attempting to treat the remaining oil and emulsion layers.

"Three-phase separator" and "free-water knockout" are terms used to describe pressure vessels that are designed to separate and remove the free water from a mixture of crude oil and water. Because flow normally enters these vessels directly from either (1) a producing well or (2) a separator operating at a higher pressure, the vessel must be designed to separate the gas that flashes from the liquid as well as separate the oil and water.

The term "three-phase separator" is normally used when there is a large amount of gas to be separated from the liquid, and the dimensions of the vessel are determined by the gas capacity equations discussed in Chapter 4. "Free-water knockout" is generally used when the amount of

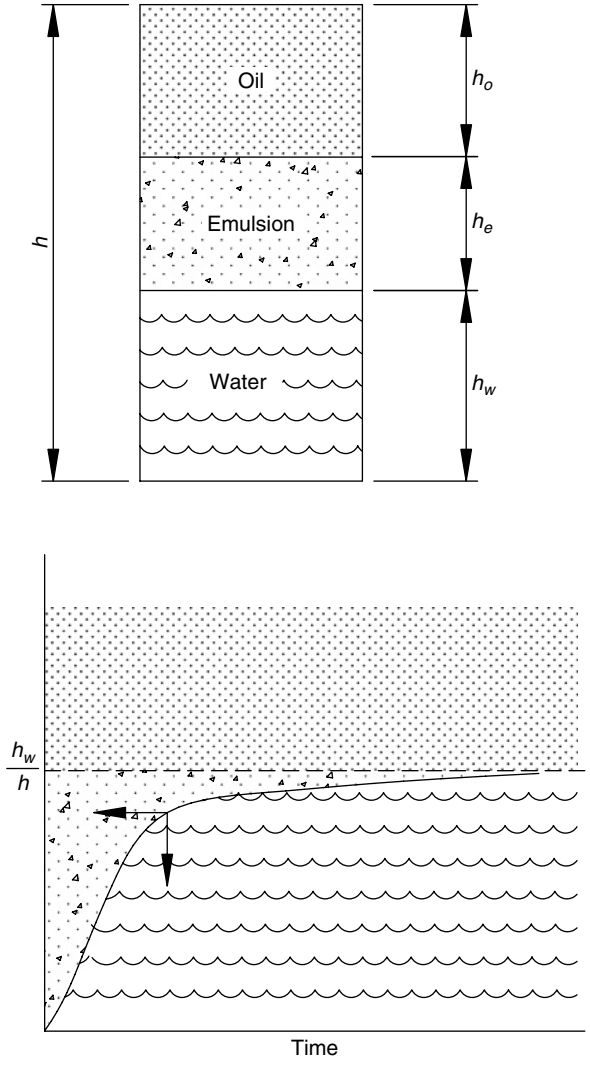


Figure 5-1. Growth of water layer with time.

gas is small relative to the amount of oil and water, and the dimensions of the vessel are determined by the oil–water separation equations discussed in this chapter. No matter what name is given to the vessel, any vessel that is designed to separate two immiscible liquid phases will employ the concepts described in this chapter. For purposes of this chapter, we will call such a vessel a "three-phase separator."

The basic design aspects of three-phase separation are identical to those discussed for two-phase separation in Chapter 4. The only additions are that more concern is placed on liquid-liquid settling rates and that some means of removing the free water must be added. Liquid-liquid settling rates will be discussed later in this chapter. Water removal is a function of the control methods used to maintain separation and removal from the oil. Several control methods are applicable to three-phase separators. The shape and diameter of the vessel will, to a degree, determine the types of control used.

Equipment Description

Horizontal Separators

Three-phase separators are designed as either horizontal or vertical pressure vessels. Figure 5-2 is a schematic of a typical horizontal three-phase separator. The fluid enters the separator and hits an inlet diverter. This sudden change in momentum does the initial gross separation of liquid and vapor as discussed in Chapter 4. In most designs the inlet diverter contains a down-comer that directs the liquid flow below the oil–water interface.

This forces the inlet mixture of oil and water to mix with the water continuous phase in the bottom of the vessel and rise through the oil–water

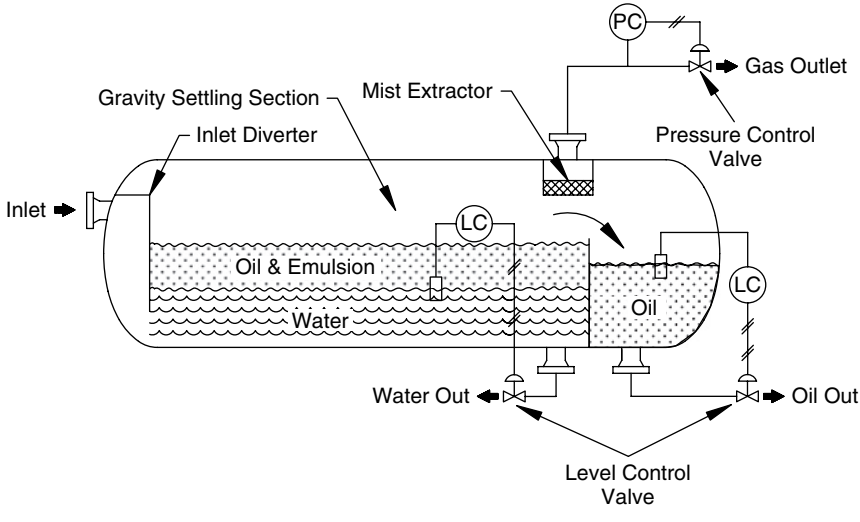


Figure 5-2. Schematic of a horizontal three-phase separator with interface level control and weir.

interface. This process is called "water washing," and it promotes the coalescence of water droplets, which are entrained in the oil continuous phase. Figure 5-3 illustrates the principles of "water washing." The inlet diverter assures that little gas is carried with the liquid, and the water wash assures that the liquid does not fall on top of the gas–oil or oil–water interface, mixing the liquid retained in the vessel and making control of the oil–water interface difficult.

The liquid collecting section of the vessel provides sufficient time so that the oil and emulsion form a layer or "oil pad" on top of the free water. The free water settles to the bottom. Figure 5-4 is a cutaway view of a typical horizontal three-phase separator with an interface level controller and weir. The weir maintains the oil level, and the level controller maintains the water level. The oil is skimmed over the weir. The level of the oil downstream of the weir is controlled by a level controller that operates the oil dump valve.

The produced water flows from a nozzle in the vessel located upstream of the oil weir. An interface level controller senses the height of the oil–water interface. The controller sends a signal to the water dump valve,

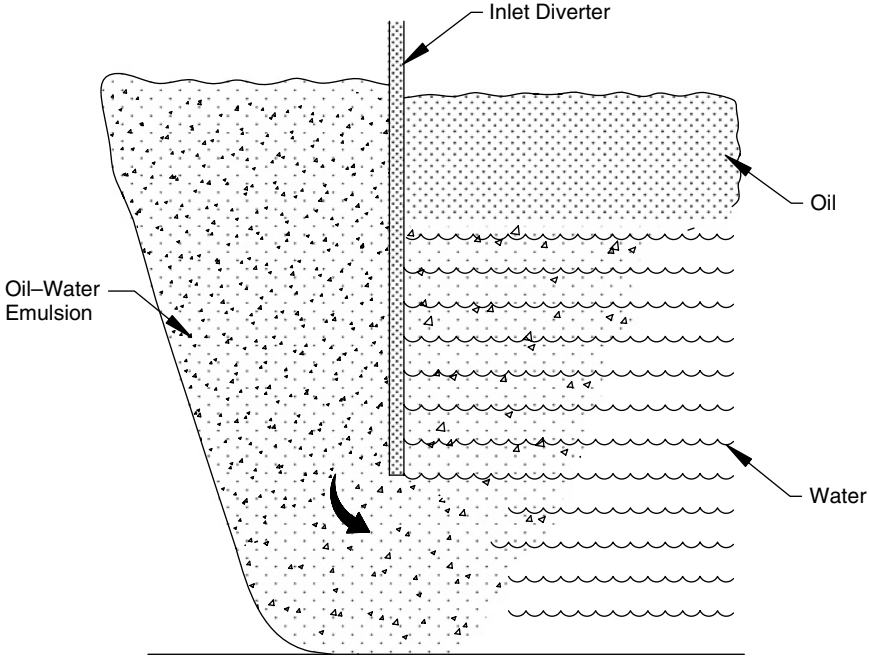


Figure 5-3. Inlet diverter illustrating the principles of "water washing."

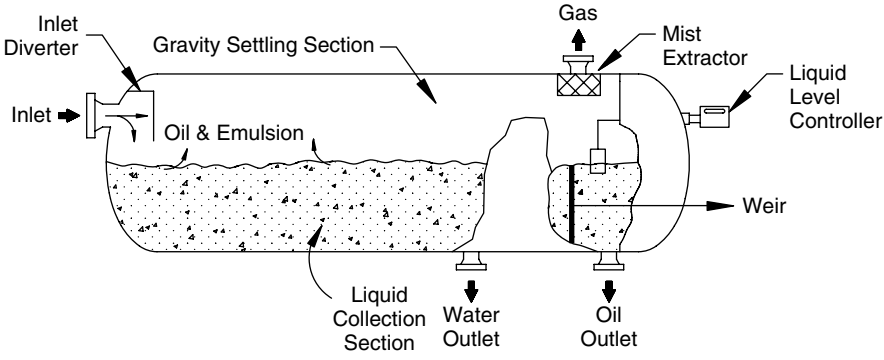


Figure 5-4. Cutaway view of a horizontal three-phase separator with interface level control and weir.

thus allowing the correct amount of water to leave the vessel so that the oil–water interface is maintained at the design height.

The gas flows horizontally and out through a mist extractor to a pressure control valve that maintains constant vessel pressure. The level of the gas–oil interface can vary from 50% to 75% of the diameter depending on the relative importance of liquid–gas separation. The most common configuration is half-full, and this is used for the design equations in this section. Similar equations can be developed for other interface levels.

Figure 5-5 shows an alternate configuration known as a "bucket and weir" design. Figure 5-6 is a cutaway view of a horizontal three-phase separator with a bucket and weir. This design eliminates the need for a liquid interface controller. Both the oil and water flow over weirs where level control is accomplished by a simple displacer float. The oil overflows the oil weir into an oil bucket where its level is controlled by a level controller that operates the oil dump valve. The water flows under the oil bucket and then over a water weir. The level downstream of this weir is controlled by a level controller that operates the water dump valve. As shown in Figures 5-5 and 5-6, the back of the oil bucket is higher than the front of the bucket. This differential height configuration assures oil will not flow over the back of the bucket and out with the water should the bucket become flooded.

The height of the oil weir controls the liquid level in the vessel. The difference in height of the oil and water weirs controls the thickness of the oil pad due to specific gravity differences. It is critical to the operation of the vessel that the water weir height is sufficiently below the oil weir height so that the oil pad thickness provides sufficient oil retention time. If the water weir is too low and the difference in specific

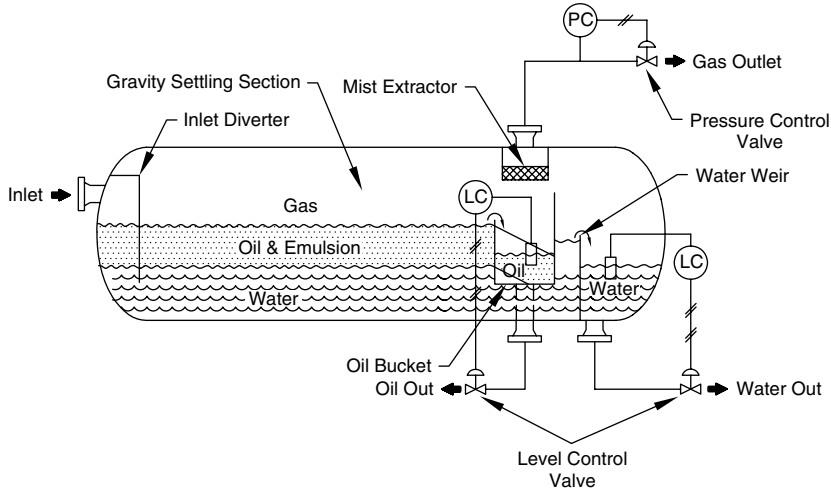


Figure 5-5. Schematic of a horizontal three-phase separator with a "bucket and weir."

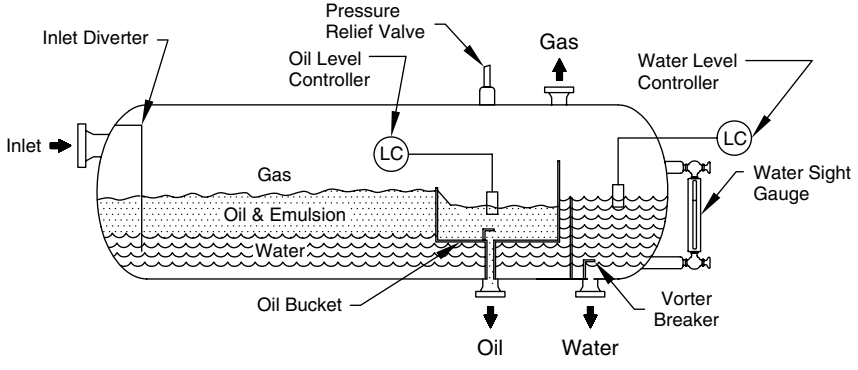


Figure 5-6. Cutaway view of a horizontal three-phase separator with a "bucket and weir."

gravity is not as great as anticipated, then the oil pad could grow in thickness to a point where oil will be swept under the oil box and out the water outlet. Normally, either the oil or the water weir is made adjustable so that changes in oil or water specific gravities or flow rates can be accommodated.

To obtain a desired oil pad height, the water weir should be set a distance below the oil weir. This distance is calculated by using Eq. (5-1),

which is developed by equating the static heads at point "A."

$$\Delta h = h_o \left[1 - \left(\frac{\rho_o}{\rho_w} \right) \right], \tag{5-1}$$

where

Δh = distance below the oil weir, in (mm),
h_o = desired oil pad height, in (mm),
ρ_o = oil density, lb/ft^3(kg/m^3),
ρ_w = water density, lb/ft^3(kg/m^3).

This equation neglects the height of the oil and water flowing over the weir and presents a view of the levels when there is no inflow. A large inflow of oil will cause the top of the oil pad to rise; the oil pad will thus get thicker, and the oil bucket must be deep enough so that oil does not flow under it. Similarly, a large inflow of water will cause the level of water flowing over the water weir to rise, and there will be a large flow of oil from the oil pad over the oil weir until a new h_w is established. These dynamic effects can be minimized by making the weirs as long as possible.

Derivation of Equation (5-1)

ρ is in lb/ft^3(kg/m^3), h is in in. (mm). Setting the pressures at point "A" in Figure 5-7 results in

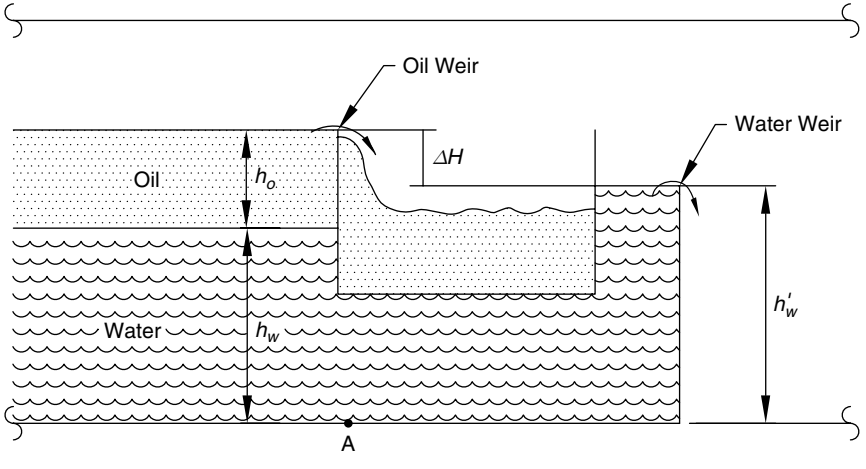


Figure 5-7. Determination of oil pad height.

$$\rho_o h_o + \rho_w h_w = \rho_w h'_w,$$

$$h_w = \frac{\rho_w h'_w - p_o h_o}{\rho_w} = hw' - \frac{\rho_o}{\rho_w} h_o,$$

$$\Delta h = h_o + h_w - h'_w,$$

$$\Delta h = h_o - \frac{\rho_o}{\rho w} h_o = h_o \left[1 - \frac{\rho_o}{\rho_w} \right].$$

Three-phase separators with a bucket and weir design are most effective with high water-to-oil flow rates and/or small density differences. Interface control design has the advantage of being easily adjustable to handle unexpected changes in oil or water specific gravity or flow rates. Interface control should be considered for applications with high oil flow rates and/or large density differences. However, in heavy oil applications or where large amounts of emulsion or paraffin are anticipated, it may be difficult to sense interface level. In such a case bucket and weir control is recommended.

Free-Water Knockout

The term "free-water knockout" (FWKO) is reserved for a vessel that processes an inlet liquid stream with little entrained gas and makes no attempt to separate the gas from the oil. Figure 5-8 illustrates a horizontal FWKO. Figure 5-9 illustrates a vertical FWKO. The major difference between a conventional three-phase separator and an FWKO is that in the latter there are only two fluid outlets; one for oil and very small amounts of gas and the second for the water. FWKOs are usually operated as packed vessels. Water outflow is usually controlled with an interface level control. It should be clear that the principles of operation of such a vessel are the same as those described above. The design of an FWKO is

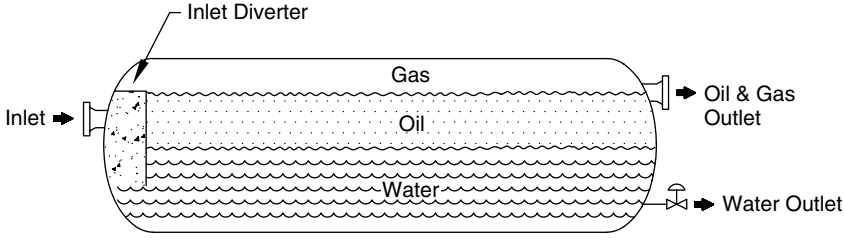


Figure 5-8. Schematic of a horizontal FWKO.

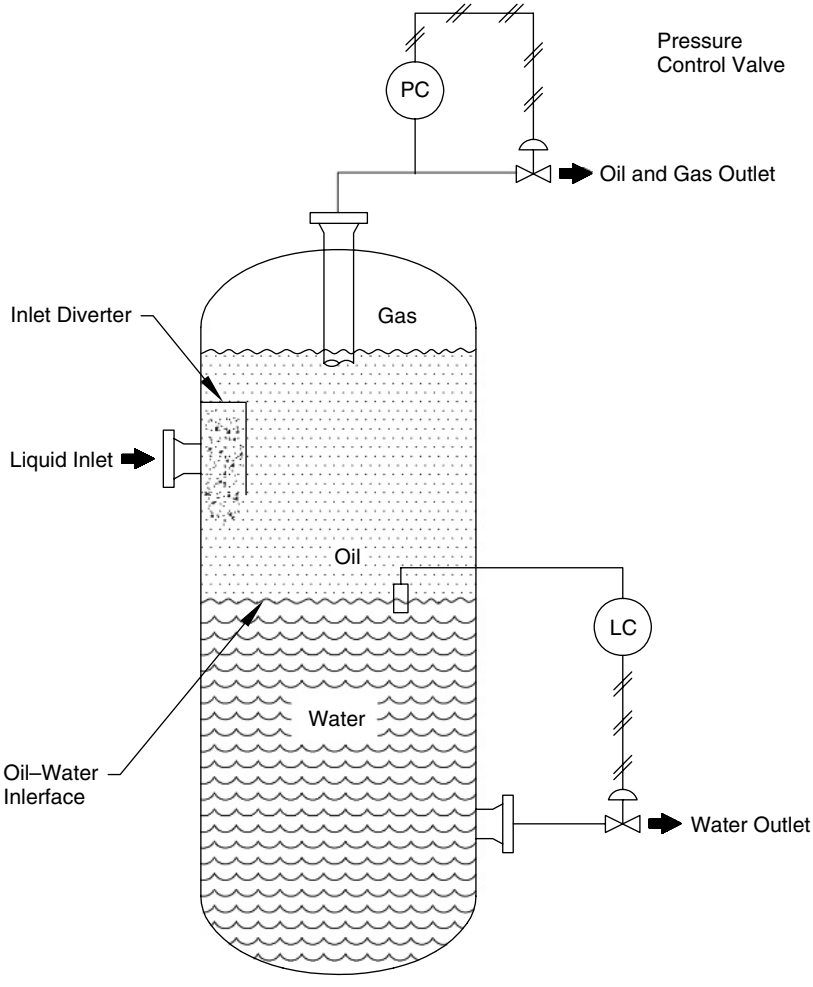


Figure 5-9. Schematic of a vertical FWKO.

the same as that of a three-phase separator. Since there is very little gas, the liquid capacity constraint always dictates the size.

Flow Splitter

Figure 5-10 illustrates a typical flow splitter. A "flow splitter" is a special version of a free-water knockout. Basically, it is an FWKO where the oil outlet is split among two or more outlet lines that are directed to several downstream process components. This vessel contains several

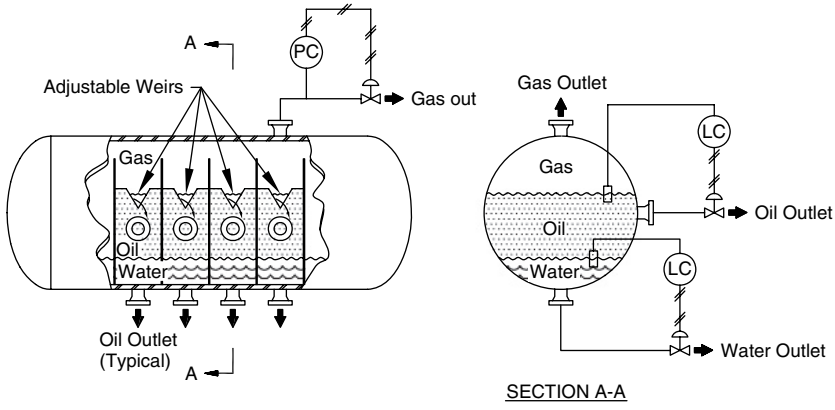


Figure 5-10. Schematic of a flow splitter with four compartments.

compartments, which are sealed from each other. Each compartment has its own level control and outlet oil valve. Unlike the FWKO, which may be operated as a packed vessel, the flow splitter must be operated with a gas blanket. Adjustable weirs separate the compartments from water and oil outside the compartments. Oil flows over the weirs into the individual compartments. The water level control is used to maintain the top of the oil layer above the highest weir. Individual level controls in each compartment assure oil leaves the compartments at the same rate at which it enters.

The flow of liquid across the notched weir is directly proportional to the difference in height between the liquid upstream of the weir and the bottom of the notch. When the weirs of different compartments are set at different heights, the flow into each compartment is different. The water level control holds the water level constant, which assures all oil that enters the separator leaves through the compartments in proportions related to the weir heights.

Horizontal Three-Phase Separator with a Liquid "Boot"

Figure 5-11 shows a horizontal three-phase separator with a water "boot" on the bottom of the vessel barrel. The "boot" collects small amounts of water that settle out in the liquid collection section and travel to the outlet end of the vessel. These vessels are a special case of three-phase separators. In this case, the flow rate of both oil and water can provide enough retention time for separation of oil and water, and there is no need to use the main body of the separator to provide oil retention time.

Figure 5-12 shows a horizontal two-phase separator with a liquid boot. Because the water flow rate is so low relative to the oil flow rate, the

small amount of water retention time provided by the boot is sufficient. Thus the diameter of the main body of the vessel can be smaller. The liquid boot collects small amounts of liquid in the liquid collection section. These vessels are a special case of two-barrel two-phase separators, which are typically used in dry gas applications and should only be used where separation of the two liquid phases is relatively easy.

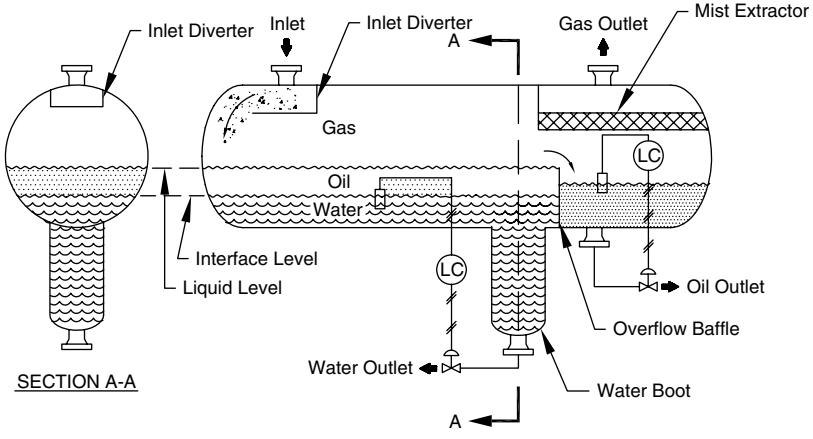


Figure 5-11. Schematic of a horizontal three-phase separator with a "water boot."

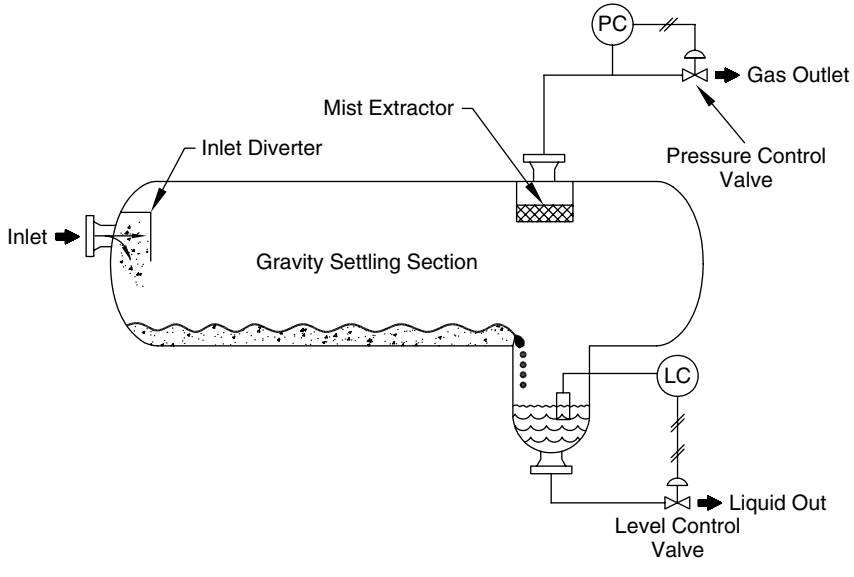


Figure 5-12. Schematic of a horizontal two-phase separator with a "liquid boot."

Vertical Separators

Figure 5-13 shows a typical configuration for a vertical three-phase separator. Flow enters the vessel through the side as in the horizontal separator. The inlet diverter separates the bulk of the gas. A down-comer is required to route the liquid through the oil–gas interface so as not to disturb the oil skimming action taking place. A chimney is needed to equalize gas pressure between the lower section and the gas section.

The spreader, or down-comer, outlet is located just below the oil–water interface, thus "water washing" the incoming stream. From this point as the oil rises, any free water trapped within the oil phase separates out. The water droplets flow countercurrent to the oil. Similarly, the water flows downward and oil droplets trapped in the water phase tend to rise

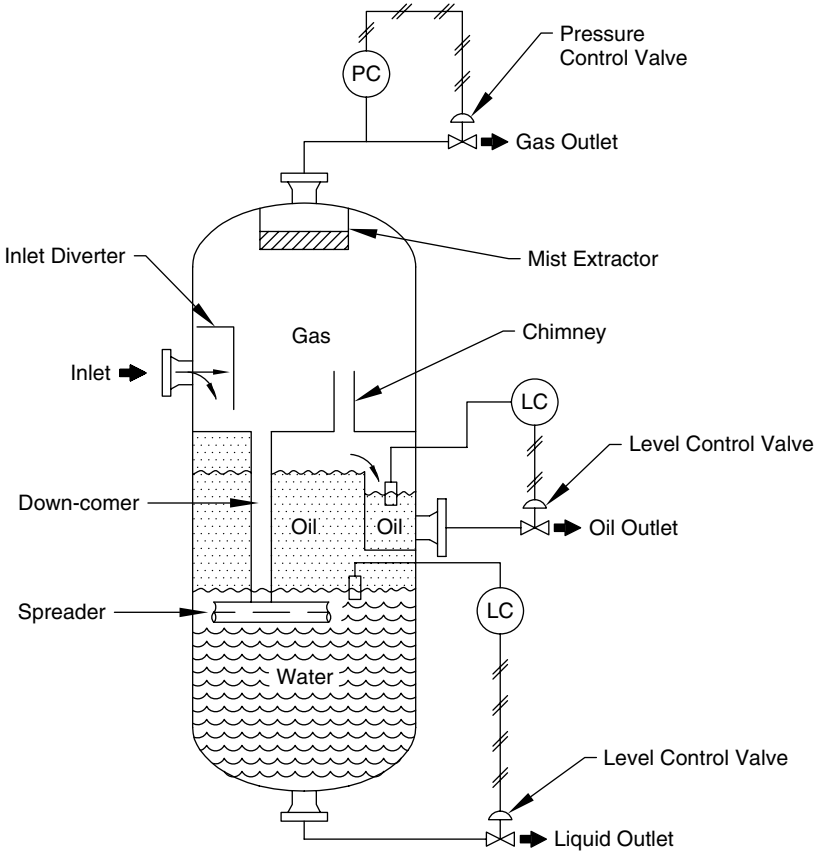


Figure 5-13. Schematic of a vertical three-phase separator with interface level control.

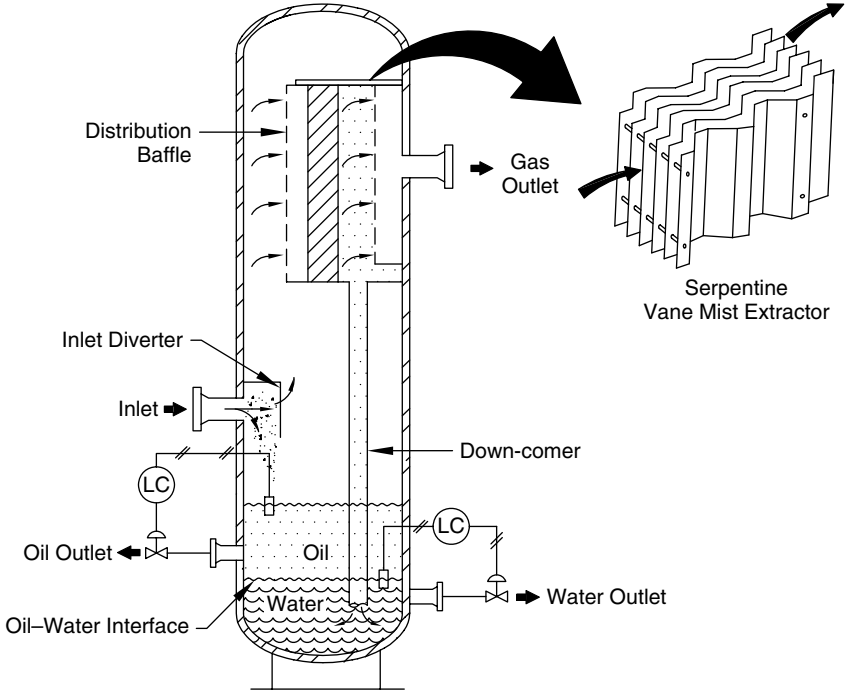


Figure 5-14. Cutaway view of a vertical three-phase separator without water washing and with vane mist extractor.

countercurrent to the water flow. Figures 5-14 and 5-15 are views of vertical three-phase separators without water washing and with interface control.

Figure 5-16 shows the three different methods of control that are often used on vertical separators.

The first is strictly level control. A regular displacer float is used to control the gas–oil interface and regulate a control valve dumping oil from the oil section. An interface float is used to control the oil–water interface and regulate a water outlet control valve. Because no internal baffling or weirs are used, this system is the easiest to fabricate and handles sand and solids production best.

The second method shown uses a weir to control the gas–oil interface level at a constant position. This results in a better separation of water from the oil as all the oil must rise to the height of the oil weir before exiting the vessel. Its disadvantages are that the oil box takes up vessel volume and costs money to fabricate. In addition, sediment and solids could collect in the oil box and be difficult to drain, and a separate

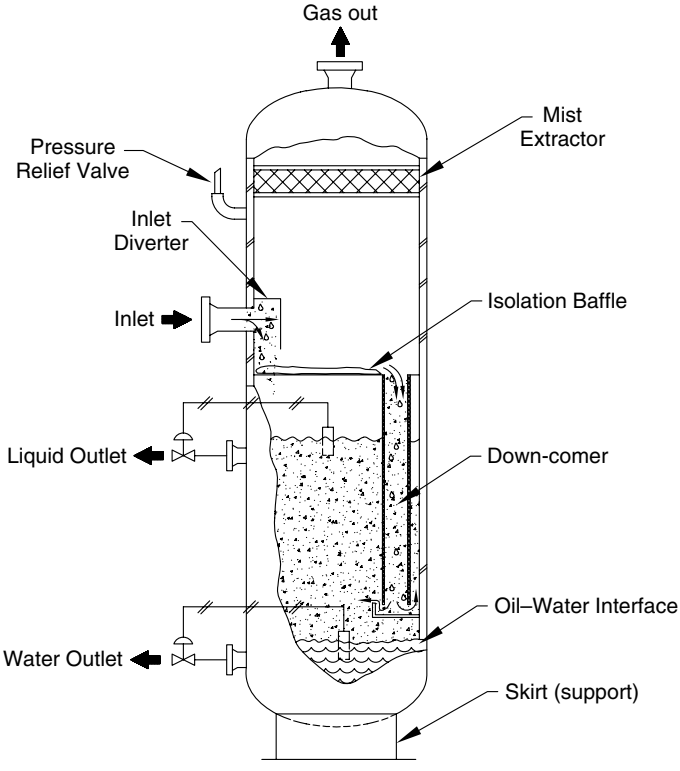


Figure 5-15. Cutaway view of a vertical three-phase separator without water washing and with wire-mesh mist extractor.

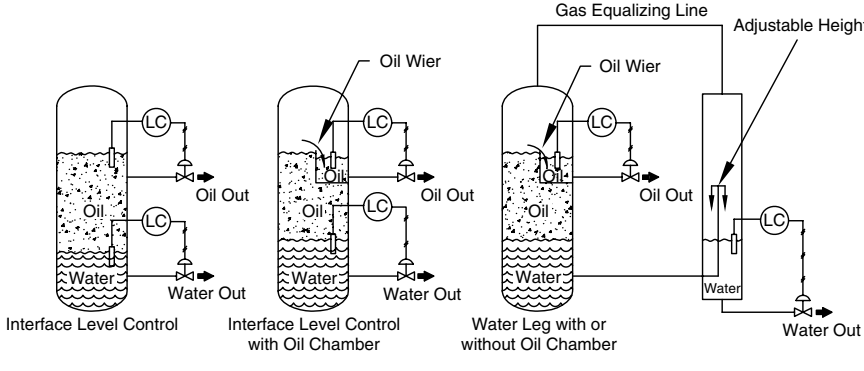


Figure 5-16. Liquid level control schemes.

low-level shut-down may be required to guard against the oil dump valve's failing to open.

The third method uses two weirs, which eliminates the need for an interface float. Interface level is controlled by the height of the external water weir relative to the oil weir or outlet height. This is similar to the bucket and weir design of horizontal separators. The advantage of this system is that it eliminates the interface level control. The disadvantage is that it requires additional external piping and space. In cold climates the water leg is sometimes installed internal to the vessel so that the vessel insulation will prevent it from freezing.

Selection Considerations

The geometry and physical and operating characteristics give each separator type advantages and disadvantages. Gravity separation is more efficient in horizontal vessels than in vertical vessels. In the gravity settling section of a horizontal vessel, the settling velocity and flow velocity are perpendicular rather than countercurrent in a vertical vessel. Horizontal separators have greater interface areas, which enhances phase equilibrium. This is especially true if foam or emulsion collect at the gas–oil interface. Thus, from a process perspective, horizontal vessels are preferred. However, they do have several drawbacks, which could lead to a preference for a vertical vessel in certain situations:

1. Horizontal separators are not as good as vertical separators in handling solids. The liquid dump valve of a vertical separator can be placed at the center of the bottom head so that solids will not build up in the separator, but continue to the next vessel in the process. As an alternative, a drain could be placed at this location so that solids could be disposed of periodically while liquid leaves the vessel at a slightly higher elevation. In a horizontal vessel, it is necessary to place several drains along the length of the vessel. Since the solids will have an angle of repose of 45° to 60°, the drains must be spaced at very close intervals [usually no farther than 5 ft (1.5 m) apart]. Attempts to lengthen the distance between drains, by providing sand jets in the vicinity of each drain to fluidize the solids while the drains are in the operation, are expensive and have been only marginally successful in field operations.
2. Horizontal vessels require more plan area to perform the same separation as vertical vessels. While this may not be of importance at

a land location, it could be very important offshore. If several separators are used, however, this disadvantage may be overcome by stacking horizontal separators on top of each other.

3. Small-diameter horizontal vessels [3-ft (1.5-m) diameter and smaller] have less liquid surge capacity than vertical vessels sized for the same steady-state flow rate. For a given change in liquid surface elevation, there is typically a larger increase in liquid volume for a horizontal separator than for a vertical separator sized for the same flow rate. However, the geometry of a small horizontal vessel causes any high-level shut-down device to be located close to the normal operating level. In very large diameter [greater than 6 ft (1.8 m)] horizontal vessels and in vertical vessels, the shut-down could be placed much higher, allowing the level controller and dump valve more time to react to the surge. In addition, surges in horizontal vessels could create internal waves, which could activate a high-level sensor prematurely.
4. Care should be exercised when selecting small-diameter [5 ft (1.5 m)] horizontal separators. The level controller and level switch elevations must be considered. The vessel must have a sufficiently large diameter so that the level switches may be spaced far enough apart, vertically, so as to avoid operating problems. This is important if surges in the flow of slugs of liquids are expected to enter the separator.

It should be pointed out that vertical vessels have some drawbacks that are not process related and that must be considered when making a selection. For example, the relief valve and some of the controls may be difficult to service without special ladders and platforms. The vessel may have to be removed from the skid for trucking due to height restrictions.

In summary, horizontal vessels are most economical for normal oil–water separation, particularly where there may be problems with emulsions, foam, or high gas–liquid ratios. Vertical vessels work most effectively in low gas–oil ratio (GOR) applications and where solids production is anticipated.

Vessel Internals

Vessel internals common to both two-phase and three-phase separators, such as inlet diverters, wave breakers, de-foaming plates, vortex breakers, stilling wells, sand jets and drains, and mist extractors, are covered in Chapter 4: Two-Phase Oil and Gas Separation and will not be repeated here. Additional internals that aid in the separation of oil and water are presented in this section.

Coalescing Plates

It is possible to use various plate or pipe coalescer designs to aid in the coalescing of oil droplets in the water and water droplets in the oil. The installation of coalescing plates in the liquid section will cause the size of the water droplets entrained in the oil phase to increase, making gravity settling of these drops to the oil–water interface easier. Thus, the use of coalescing plates (Figure 5-17), which are described in Chapter 7, will often lead to the ability to handle a given flow rate in a smaller vessel. However, because of the potential for plugging with sand, paraffin, or corrosion products, the use of coalescing plates should be discouraged, except for instances where the savings in vessel size and weight are large enough to justify the potential increase in operating costs and decrease in availability.

Turbulent Flow Coalescers

Turbulent flow coalescers, which were marketed under the name SP Packs and are described further in chapter 8, utilized the turbulence created by flow in a serpentine pipe path to promote coalescence. As shown in Figure 5-18, SP Packs took up more space in the vessel than plate coalescers, but, since they did not have small clearances, they were not susceptible to plugging. Despite the design advantages, the units were not well received and, as such, are no longer being manufactured.

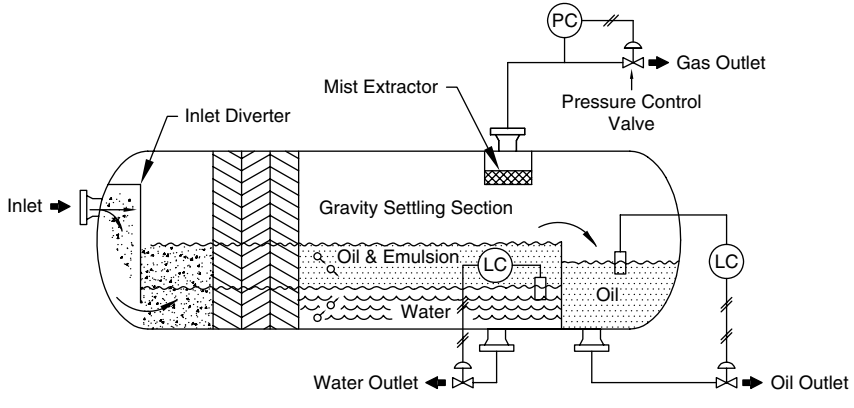


Figure 5-17. Schematic of a horizontal three-phase separator fitted with coalescing plates.

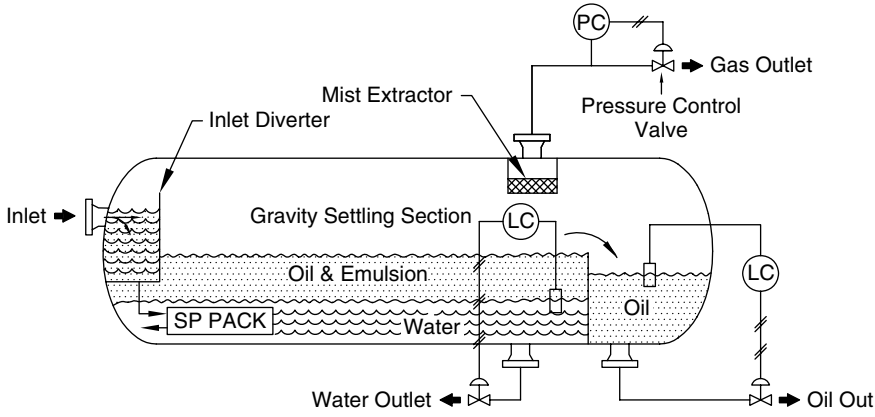


Figure 5-18. Schematic of a horizontal three-phase separator fitted with free-flow turbulent coalescers (SP Packs).

Potential Operating Problems

Emulsions

Three-phase separators may experience the same operating problems as two-phase separators. In addition, three-phase separators may develop problems with emulsions which can be particularly troublesome in the operation of three-phase separators. Over a period of time an accumulation of emulsified materials and/or other impurities may form at the interface of the water and oil phases. In addition to adverse effects on the liquid level control, this accumulation will also decrease the effective oil or water retention time in the separator, with a resultant decrease in water–oil separation efficiency. Addition of chemicals and/or heat often minimizes this difficulty.

Frequently, it is possible to appreciably lower the settling time necessary for water–oil separation by either the application of heat in the liquid section of the separator or the addition of de-emulsifying chemicals. The treating of emulsions is discussed in more detail in Chapter 7.

Design Theory

Gas Separation

The concepts and equations pertaining to two-phase separation described in Chapter 4 are equally valid for three-phase separation.

Oil–Water Settling

It can be shown that flow around settling oil drops in water or water drops in oil is laminar and thus Stokes' law governs. The terminal drop velocity is

Field Units

$$V_t = \frac{1.78 \times 10^{-6} (\Delta SG)\, d_m^2}{\mu} \tag{5-2a}$$

SI Units

$$V_t = \frac{5.56 \times 10^{-7} (\Delta SG)\, d_m^2}{\mu}, \tag{5-2b}$$

where

V_t = terminal settling velocity, ft/s (m/s),
ΔSG = difference in specific gravity relative to water between the oil and the water phases,
d_m = drop size, microns,
μ = viscosity of continuous phase, cp.

Water Droplet Size in Oil

It is difficult to predict the water droplet size that must be settled out of the oil phase to coincide with the rather loose definition of "free oil." Unless laboratory or nearby field data are available, good results have been obtained by sizing the oil pad such that water droplets 500 microns and larger settle out. As shown in Figure 5-19, if this criterion is met, the emulsion to be treated by downstream equipment should contain less than 5% to 10% water. In heavy crude oil systems, it is sometimes necessary to design for 1,000-micron water droplets to settle. In such cases the emulsion may contain as much as 20% to 30% water.

Oil Droplet Size in Water

From Eqs. (5-2a) and (5-2b) it can be seen that the separation of oil droplets from the water is easier than the separation of water droplets from the oil. The oil's viscosity is on the order of 5 to 20 times that of water.

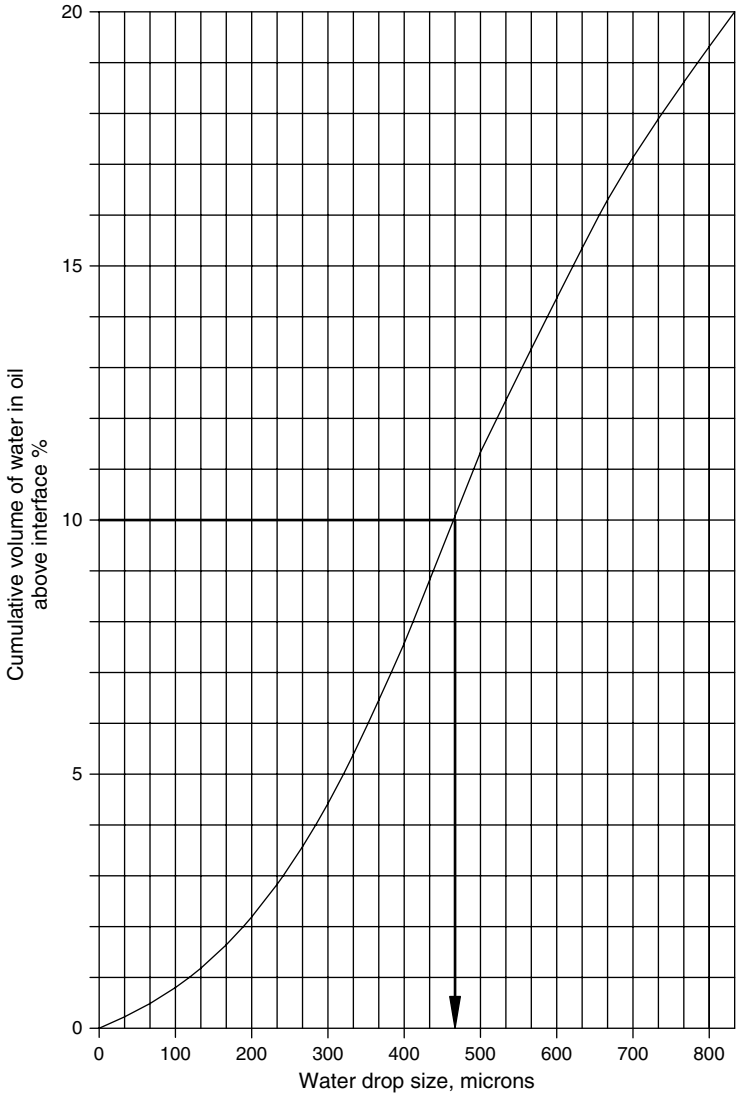


Figure 5-19. Example water droplet size distribution. Size distribution varies widely for different process conditions and crude and water properties.

Thus, the terminal settling velocity of an oil droplet in water is much larger than that of a water droplet in oil. The primary purpose of three-phase separation is to prepare the oil for further treating. Field experience indicates that oil content in the produced water from a three-phase separator, sized for water removal from oil, can be expected to be between a

few hundred and 2,000 mg/l. This water will require further treating prior to disposal and is discussed Chapter 8. Sizing for oil droplet removal from the water phase does not appear to be a meaningful criterion.

Occasionally, the viscosity of the water phase may be as high as, or higher than, the liquid hydrocarbon phase viscosity. For example, large glycol dehydration systems usually have a three-phase flash separator. The viscosity of the glycol/water phase may be rather high. In cases like this, the settling equation should be applied to removing oil droplets of approximately 200 microns from the water phase.

If the retention time of the water phase is significantly less than the oil phase, then the vessel size should be checked for oil removal from the water. For these reasons, the equations are provided so the water phase may be checked. However, the separation of oil from the water phase rarely governs the vessel size and may be ignored for most cases.

Retention Time

A certain amount of oil storage is required to assure that the oil reaches equilibrium and that flashed gas is liberated. An additional amount of storage is required to assure that the free water has time to coalesce into droplet sizes sufficient to fall in accordance with Eqs. (5-2a) and (5-2b). It is common to use retention times ranging from 3 minutes to 30 minutes depending upon laboratory or field data. If this information is not available, the guidelines presented in Table 5-1 can be used. Generally, the retention time must be increased as the oil gravity or viscosity increases.

Similarly, a certain amount of water storage is required to assure that most of the large droplets of oil entrained in the water have sufficient time to coalesce and rise to the oil–water interface. It is common to use retention times for the water phase ranging from 3 minutes to 30 minutes

Table 5-1
Oil Retention Time

°API Gravity	Minutes
Condensate	2–5
Light crude oil (30°–40°)	5–7.5
Intermediate crude oil (20°–30°)	7.5–10
Heavy crude oil (less than 20°)	10+

Note: If an emulsion exists in inlet stream, increase above retention times by a factor of 2 to 4.

depending upon laboratory or field data. If this information is not available, a water retention time of 10 minutes is recommended for design.

The retention time for both the maximum oil rate and the maximum water rate should be calculated, unless laboratory data indicate that it is unnecessary to take this conservative design approach.

Separator Design

The guidelines presented here can be used for initial sizing of a horizontal three-phase separator 50% full of liquid. They are meant to complement, and not replace, operating experiences. Determination of the type and size of the separator must be made on an individual basis. All the functions and requirements should be considered including the likely uncertainties in design flow rates and properties. For this reason, there is no substitute for good engineering evaluations of each separator by the design engineer. The "trade-off" between design size and details and uncertainties in design parameters should not be left to manufacturer recommendations or rules of thumb.

Horizontal Separator Sizing—Half-Full

For sizing a horizontal three-phase separator it is necessary to specify a vessel diameter and a seam-to-seam vessel length. The gas capacity and retention time considerations establish certain acceptable combinations of diameter and length. The need to settle 500-micron water droplets from the oil and 200-micron oil droplets from the water establishes a maximum diameter corresponding to the given liquid retention time.

Gas Capacity Constraint

The principles of liquid droplets settling through a gas, which were derived in Chapter 4, can be used to develop an equation to size a separator for a gas flow rate. By setting the gas retention time equal to the time required for a drop to settle to the liquid interface, the following equations may be derived:

Field Units

$$dL_{\text{eff}} = 420\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_1 - \rho_g}\right)\frac{C_d}{d_m}\right]^{1/2} \tag{5-3a}$$

SI Units

$$dL_{\text{eff}} = 34.5\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_d}{d_m}\right]^{1/2}, \tag{5-3b}$$

where

d = vessel inside diameter, in. (mm),
L_{eff} = vessel effective length, ft (m),
T = operating temperature, °R(°K),
Z = gas compressibility,
Q_g = gas flow rate, MMscfd (scm/hr),
P = operating pressure, psia (kPa),
ρ_g = density of gas, lb/ft^3(kg/m^3),
ρ_l = density of liquid, lb/ft^3(kg/m^3),
C_D = drag coefficient,
d_m = liquid drop to be separated, microns.

Retention Time Constraint

Liquid retention time constraints can be used to develop the following equation, which may be used to determine acceptable combinations of d and L_{eff}.

Field Units

$$d^2 L_{\text{eff}} = 1.42\left[(Q_w)(t_r)_w + (Q_o)(t_r)_o\right] \tag{5-4a}$$

SI Units

$$d^2 L_{\text{eff}} = 4.2 \times 10^4\left[(Q_w)(t_r)_w + (Q_o)(t_r)_o\right], \tag{5-4b}$$

where

Q_w = water flow rate, BPD (m^3/hr),
$(t_r)_w$ = water retention time, min,
Q_o = oil flow rate, BPD (m^3/hr),
$(t_r)_o$ = oil retention time, min.

Derivation of Equations (5-4a) and (5-4b)

t is in s, V in $ft^3(m^3)$, Q in $ft^3/s\,(m^3/s)$, D in ft (m), d in in. (mm), L_{eff} in ft (m).

Field Units

$$t = \frac{Vol}{Q},$$

$$Vol = \frac{1}{2}\left(\frac{\pi D^2 L_{\text{eff}}}{4}\right)$$

$$= \frac{\pi d^2 L_{\text{eff}}}{(2)\,(4)\,(144)}$$

$$= 2.73 \times 10^{-3} d^2 L_{\text{eff}},$$

$$(Vol)_o = 2.73 \times 10^{-3} d^2 L_{\text{eff}} \left(\frac{A_o}{A_l}\right),$$

$$(Vol)_w = 2.73 \times 10^{-3} d^2 L_{\text{eff}} \left(\frac{A_w}{A_l}\right),$$

Q_o and Q_w are in BPD,

$$Q = Q_o \times \frac{561\,\text{ft}^3}{\text{barrel}} \times \frac{\text{day}}{24\,\text{hr}} \times \frac{\text{hr}}{3{,}600\,\text{s}},$$

For oil:

$$Q = 6.49 \times 10^{-5} Q_o$$

$$t_o = \frac{(Vol)_o}{Q_o} = 42\frac{d^2 L_{\text{eff}}}{Q_o}\left(\frac{A_o}{A_l}\right)$$

For water:

$$Q = 6.49 \times 10^{-5} Q_w$$

$$t_w = \frac{(Vol)_w}{Q_w} = 42\frac{d^2 L_{\text{eff}}}{Q_w}\left(\frac{A_w}{A_l}\right)$$

SI Units

$$t = \frac{Vol}{Q},$$

$$\begin{aligned} Vol &= \frac{1}{2}\left(\frac{\pi D^2 L_{\text{eff}}}{4}\right) \\ &= \frac{\pi d^2 L_{\text{eff}}}{(2)\,(4)\,(1000^2)} \\ &= 3.927 \times 10^{-7} d^2 L_{\text{eff}}, \end{aligned}$$

$$(Vol)_o = 3.927 \times 10^{-7} d^2 L_{\text{eff}} \left(\frac{A_o}{A_l}\right),$$

$$(Vol)_w = 3.927 \times 10^{-7} d^2 L_{\text{eff}} \left(\frac{A_w}{A_l}\right),$$

Q_o and Q_w are in m^3/hr,

For oil:

$$Q = Q_0 \times \frac{1\,\text{hr}}{3600\,\text{s}} = \frac{Q_o}{3600},$$

$$t_o = \frac{(Vol)_o}{Q_o} = 0.0014 \frac{d^2 L_{\text{eff}}}{Q_o}\left(\frac{A_w}{A_l}\right)$$

For water:

$$Q = \frac{Q_w}{3600},$$

$$t_w = \frac{(Vol)_w}{Q_w} = 0.0014 \frac{d^2 L_{\text{eff}}}{Q_w}\left(\frac{A_w}{A_l}\right)$$

A_o, A_w, and A_l are cross-sectional areas of oil, water, and liquid, respectively.

Field Units

Rearranging the equation for t_o and t_w

$$42\left(\frac{A_o}{A_l}\right)=\frac{t_o Q_o}{d^2 L_{\text{eff}}},\qquad 42\left(\frac{A_w}{A_l}\right)=\frac{t_w Q_w}{d^2 L_{\text{eff}}},$$

$(t_r)_o$ and $(t_r)_w$ are in min,

$$0.7\left(\frac{A_o}{A_l}\right)=\frac{(t_r)_o Q_o}{d^2 L_{\text{eff}}},\qquad 0.7\left(\frac{A_w}{A_l}\right)=\frac{(t_r)_w Q_w}{d^2 L_{\text{eff}}},$$

Adding the two equation:

$$0.7\left(\frac{A_o+A_w}{A_l}\right)=\frac{(t_r)_o Q_o+(t_r)_w Q_w}{d^2 L_{\text{eff}}},$$

$$A_o+A_w=A_l$$

$$d^2 L_{\text{eff}}=1.42\left[(t_r)_o Q_o+(t_r)_w Q_w\right].$$

SI Units

Rearranging the equations for t_o and t_w:

$$0.0014\left(\frac{A_o}{A_l}\right)=\frac{t_o Q_o}{d^2 L_{\text{eff}}},\qquad 0.0014\left(\frac{A_w}{A_l}\right)=\frac{t_w Q_w}{d^2 L_{\text{eff}}},$$

$(t_r)_o$ and $(t_r)_w$ are in min,

$$2.356\times 10^{-5}\left(\frac{A_o}{A_l}\right)=\frac{(t_r)_o Q_o}{d^2 L_{\text{eff}}},\qquad 2.356\times 10^{-5}\left(\frac{A_w}{A_l}\right)=\frac{(t_r)_w Q_w}{d^2 L_{\text{eff}}},$$

Adding two equations:

$$2.356\times 10^{-5}\left(\frac{A_o+A_w}{A_l}\right)=\frac{(t_r)_o Q_o+(t_r)_w Q_w}{d^2 L_{\text{eff}}},$$

$$A_o+A_w=A_l$$

$$d^2 L_{\text{eff}}=4.2\times 10^4\left[(t_r)_o Q_o+(t_r)_w Q_w\right].$$

Settling Water Droplets from Oil Phase

The velocity of water droplets settling through oil can be calculated using Stokes' law. From this velocity and the specified oil phase retention time, the distance that a water droplet can settle may be determined. This settling distance establishes a maximum oil pad thickness given by the following formula:

Field Units

$$h_o = \frac{0.00128\,(t_r)_o\,(\Delta\text{SG})\,d_m^2}{\mu} \tag{5-5a}$$

SI Units

$$h_o = \frac{0.033\,(t_r)_o\,(\Delta\text{SG})\,d_m^2}{\mu} \tag{5-5b}$$

Derivation of Equations (5-5a) and (5-5b)

t_w, t_o are in s, V in ft/s (m^3/s), h_o in in. (mm), d_m in microns, μ in cp, and

$$t_w = t_o.$$

Field Units

$$t_w = \frac{h_o/12}{V_t}, \qquad V_t = \frac{1.78 \times 10^{-6}\,(\Delta\text{SG})\,d_m^2}{\mu},$$

$$t_w = 46{,}800\frac{\mu h_o}{(\Delta\text{SG})\,d_m^2},$$

t_r is in min,

$$t_o = 60(t_r)_o$$

$$46{,}800\frac{\mu h_o}{(\Delta\text{SG})\,d_m^2} = 60(t_r)_{o,}$$

$$h_o = \frac{0.00128\,(t_r)_o\,(\Delta\text{SG})\,d_m^2}{\mu}.$$

SI Units

$$t_w = \frac{\frac{h_o}{1000}}{V_t}, \qquad V_t = \frac{5.556 \times 10^{-7}\,(\Delta SG)\,d_m^2}{\mu},$$

$$t_w = 1800\frac{\mu h_o}{(\Delta SG)\,d_m^2},$$

t_r is in min,

$$t_o = 60(t_r)_o,$$

$$1800\frac{\mu h_o}{(\Delta SG)\,d_m^2} = 60(t_r)_o,$$

$$h_o = \frac{0.033\,(t_r)_o\,(\Delta SG)\,d_m^2}{\mu}.$$

This is the maximum thickness the oil pad can be and still allow the water droplets to settle out in time $(t_r)_o$. For $d_m = 500$ microns, the following equation may be used.

Field Units

$$(h_o)_{\max} = 320\frac{(t_r)_o\,(\Delta SG)}{\mu} \tag{5-6a}$$

SI Units

$$(h_o)_{\max} = 8250\frac{(t_r)_o\,(\Delta SG)}{\mu} \tag{5-6b}$$

For a given oil retention time $[(t_r)_o]$ and a given water retention time $[(t_r)_w]$, the maximum oil pad thickness constraint establishes a maximum diameter in accordance with the following procedure:

1. Compute $(h_o)_{\max}$. Use 500-micron droplet if no other information is available.
2. Calculate the fraction of the vessel cross-sectional area occupied by the water phase. This is given by

$$\frac{A_w}{A} = 0.5\frac{Q_w\,(t_r)_w}{(t_r)_o\,Q_o + (t_r)_w\,Q_w}. \tag{5-7}$$

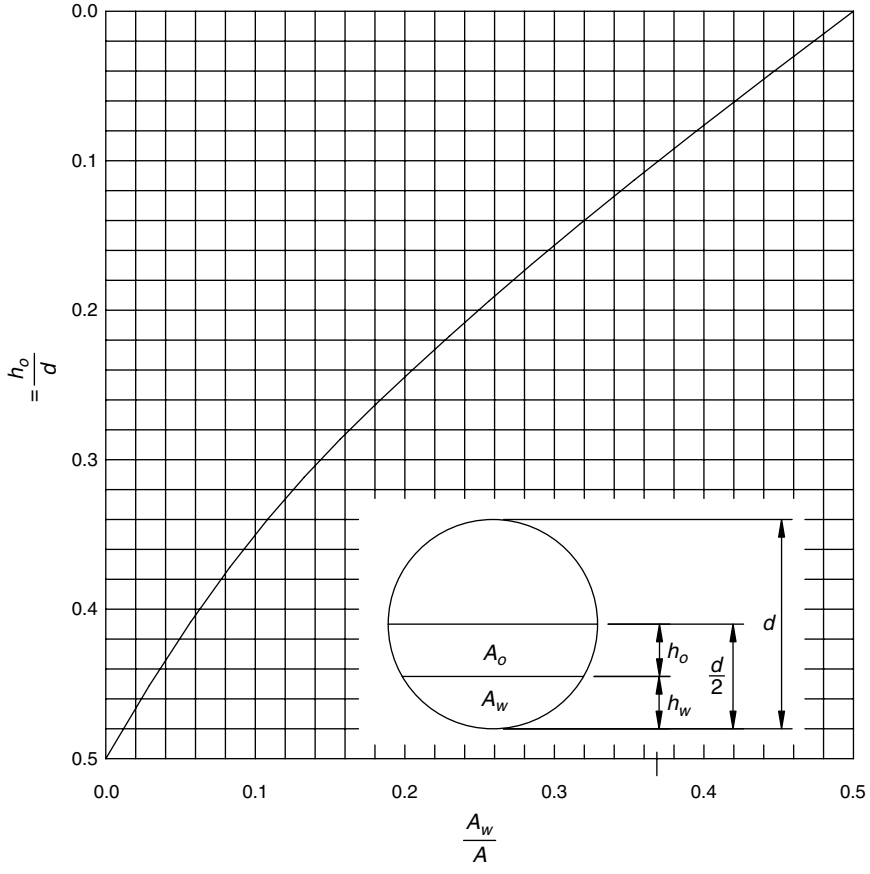


Figure 5-20. Coefficient "β" for a cylinder half filled with liquid.

3. From Figure 5-20, determine the coefficient β.
4. Calculate $d_{\max}$ from

$$d_{\max} = \frac{(h_o)_{\max}}{\beta}, \tag{5-8}$$

where

$$\beta = \frac{(h_o)}{d}.$$

Any combination of d and L_{eff} that satisfies all three of Eqs. (5-3), (5-4), and (5-5) will meet the necessary criteria.

Derivation of Equation (5-7)

A_o and A_w are in $ft^2(m^2)$, Q in ft^3/s (m^3/s), t in s, L_{eff} in ft (m).

Field Units

$$A = \frac{Qt}{L_{\text{eff}}},$$

$$Q = 6.49 \times 10^{-5} Q_o,$$

$$t_o = 60(t_r)_o,$$

$$A_o = 3.89 \times 10^{-3} \frac{Q_o (t_r)_o}{L_{\text{eff}}},$$

$$Q = 6.49 \times 10^{-5} Q_w$$

$$t_w = 60(t_r)_w$$

$$A_w = 3.89 \times 10^{-3} \frac{Q_w (t_r)_w}{L_{\text{eff}}}$$

For A vessel 1/2 full of liquid:

$$A = 2(A_o + A_w),$$

$$\frac{A_w}{A} = 0.5 \frac{Q_w (t_r)_w}{(t_r)_o Q_o + (t_r)_w Q_w}.$$

SI Units

$$A = \frac{Qt}{L_{\text{eff}}},$$

$$Q = \frac{Q_o}{3600}, \quad Q = \frac{Q_w}{3600}$$

$$t_o = 60(t_r)_o, \quad t_w = 60(t_r)_w$$

$$A_o = 0.0167 \frac{Q_o (t_r)_o}{L_{\text{eff}}}, \quad A_w = 0.0167 \frac{Q_w (t_r)_w}{L_{\text{eff}}}$$

$$A = 2(A_o + A_w),$$

$$\frac{A_w}{A} = 0.5 \frac{Q_w (t_r)_w}{(t_r)_o Q_o + (t_r)_w Q_w}.$$

Separating Oil Droplets from Water Phase

Oil droplets in the water phase rise at a terminal velocity defined by Stokes' law. As with water droplets in oil, the velocity and retention time may be used to determine a maximum vessel diameter from Eqs. (5-4a) and (5-4b). It is rare that the maximum diameter determined from a 200-micron oil droplet rising through the water phase is larger than a 500-micron water droplet falling through the oil phase. Therefore, the maximum diameter determined from a 500-micron water droplet settling through the oil phase normally governs the vessel design. For $d_m = 200$ microns, the following equations may be used:

Field Units

$$(h_w)_{\max} = \frac{(51.2(t_r)_w(\Delta SG))}{\mu_w} \tag{5-9a}$$

SI Units

$$(h_w)_{\max} = \frac{(1{,}520(t_r)_w(\Delta SG))}{\mu_w} \tag{5-9b}$$

The maximum diameter may be found from the following equation:

$$d_{\max} = \frac{(h_w)_{\max}}{\beta}. \tag{5-10}$$

Seam-to-Seam Length

The effective length may be calculated from Eqs. (5-4a) and (5-4b). From this, a vessel seam-to-seam length may be estimated. The actual required seam-to-seam length is dependent on the physical design of the vessel.

For vessels sized based on gas capacity, some portion of the vessel length is required to distribute the flow evenly near the inlet diverter. Another portion of the vessel length is required for the mist extractor. The length of the vessel between the inlet and the mist extractor with evenly distributed flow is the L_{eff} calculated from Eqs. (5-3a) and (5-3b). As a vessel's diameter increases, more length is required to evenly distribute the gas flow. However, no matter how small the diameter may be, a portion of the length is still required for the mist extractor and flow distribution. Based on these concepts coupled with field experience, the

seam-to-seam length of a vessel may be estimated as the larger of the following:

$$L_{ss} = 4/3L_{eff}. \tag{5-11}$$

Field Units

$$L_{ss} = L_{eff} + d/12 \tag{5-12a}$$

SI Units

$$L_{ss} = L_{eff} + d/1000 \tag{5-12b}$$

For vessels sized on a liquid capacity basis, some portion of the vessel length is required for inlet diverter flow distribution and liquid outlet. The seam-to-seam length should not exceed the following:

$$L_{ss} = 4/3L_{eff}. \tag{5-13}$$

Slenderness Ratio

For each vessel design, a combination of L_{eff} and d exists that will minimize the cost of the vessel. In general, the smaller the diameter of a vessel, the less it will cost. However, decreasing the diameter increases the fluid velocities and turbulence. As a vessel diameter decreases, the likelihood of the gas re-entraining liquids or destruction of the oil/water interface increases. Experience indicates that the ratio of the seam-to-seam length divided by the outside diameter should be between 3 and 5. This ratio is referred to as the "slenderness ratio" (SR) of the vessel. Slenderness ratios outside the 3 to 5 range may be used but are not as common. Slenderness ratios outside the 3 to 5 range may be used, but the design should be checked to assure that re-entrainment will not occur.

Procedure for Sizing Three-Phase Horizontal Separators—Half-Full

1. The first step in sizing a horizontal separator is to establish the design basis. This includes specifying the maximum and minimum flow rates, operating pressure and temperature, droplet size to be removed, etc.

2. Select a $(t_r)_o$ and a $(t_r)_w$.
3. Calculate $(h_o)_{\max}$. Use a 500-micron droplet if no other information is available.

Field Units

$$(h_o)_{\max} = 1.28 \times 10^{-3} \frac{(t_r)_o\,(\Delta \mathrm{SG})\,d_m^2}{\mu}$$

For 500 microns,

$$(h_o)_{\max} = 320 \frac{(t_r)_o\,(\Delta \mathrm{SG})}{\mu}.$$

SI Units

$$(h_o)_{\max} = 0.033 \frac{(t_r)_o\,(\Delta \mathrm{SG})\,d_m^2}{\mu}$$

For 500 microns,

$$(h_o)_{\max} = 8250 \frac{(t_r)_o\,(\Delta \mathrm{SG})}{\mu}.$$

4. Calculate A_w/A:

$$\frac{A_w}{A} = (0.5)\,\frac{Q_w\,(t_r)_w}{(t_r)_o\,Q_o + (t_r)_w\,Q_w}.$$

5. Determine β from curve.
6. Calculate $d_{\max}$:

$$d_{\max} = \frac{(h_o)_{\max}}{\beta}.$$

Note: $d_{\max}$ depends on Q_o, Q_w, $(t_r)_o$, and $(t_r)_w$.

7. Calculate combinations of d, L_{eff} for d less than $d_{\max}$ that satisfy the gas capacity constraint. Use 100-micron droplet if no other information is available.

Field Units

$$dL_{\text{eff}} = 420\left(\frac{TZQ_g}{P}\right)\left[\left(\frac{\rho_g}{\rho_1 - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}$$

SI Units

$$dL_{\text{eff}} = 34.5\left(\frac{TZQ_g}{P}\right)\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}$$

8. Calculate combinations of d, L_{eff} for d less than $d_{\max}$ that satisfy the oil and water retention time constraints.

Field Units

$$d^2 L_{\text{eff}} = 1.42[(t_r)_o Q_o + (t_r)_w Q_w]$$

SI Units

$$d^2 L_{\text{eff}} = 4.2 \times 10^4[(t_r)_o Q_o + (t_r)_w Q_w]$$

9. Estimate seam-to-seam length.

Field Units

$$L_{\text{ss}} = L_{\text{eff}} + \frac{d}{12} \qquad \text{(gas capacity)}$$

$$L_{\text{ss}} = \frac{4}{3}L_{\text{eff}} \qquad \text{(liquid capacity)}$$

SI Units

$$L_{\text{ss}} = L_{\text{eff}} + \frac{d}{1000} \qquad \text{(gas capacity)}$$

$$L_{\text{ss}} = \frac{4}{3}L_{\text{eff}} \qquad \text{(liquid capacity)}$$

10. Select a reasonable diameter and length. Slenderness ratios ($12\,L_{ss}/d$) on the order of 3 to 5 are common.
11. When making a final selection, it is always more economical to select a standard vessel size. API sizes for small separators can be found in API Spec. 12J. In larger sizes in most locations, heads come in outside diameters, which are multiples of 6 in. (150 mm). The width of steel sheets for the shells is usually 10 ft (3000 mm), thus it's common practice to specify L_{ss} in multiples of five.

Horizontal Separators Sizing Other Than Half-Full

For three-phase separators other than 50% full of liquid, equations can be derived similarly, using the actual oil and water areas. The equations are derived using the same principles as discussed in Chapter 4 and this chapter.

Gas Capacity Constraint

Field Units

$$d\,L_{\text{eff}} = 420\left(\frac{1-\beta}{1-\alpha}\right)\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_1-\rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}, \tag{5-14a}$$

where

$\dfrac{1-\beta}{1-\alpha}$ = design constant found from Figures 5-21 and 5-23.

SI Units

$$d\,L_{\text{eff}} = 34.5\left(\frac{1-\beta}{1-\alpha}\right)\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l-\rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}, \tag{5-14b}$$

where

$\dfrac{1-\beta}{1-\alpha}$ = design constant found from Figures 5-22 and 5-24.

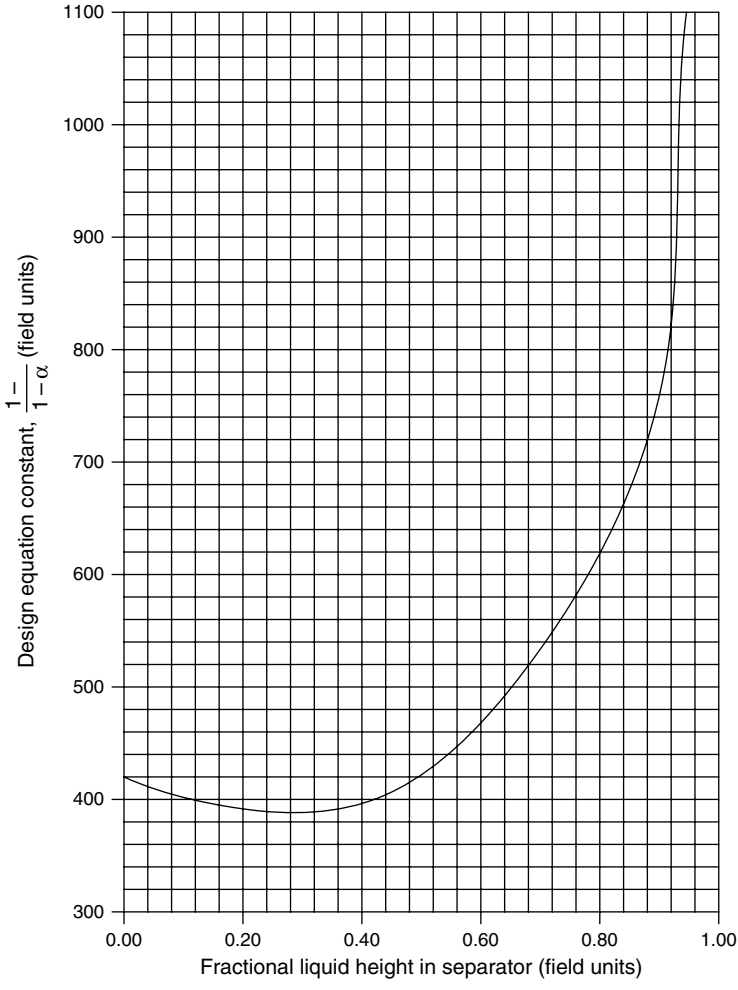


Figure 5-21. Gas capacity constraint design constant $[(1-\beta)/(1-\alpha)]$ vs. liquid height of a cylinder for a horizontal separator other than 50% full of liquid (field units).

Retention Time Constraint

Field Units

$$d^2 L_{\text{eff}} = \frac{(t_r)_o\, Q_o + (t_r)_w\, Q_w}{1.4\,\alpha}, \tag{5-15a}$$

where $\alpha =$ design constant found in Figure 5-23.

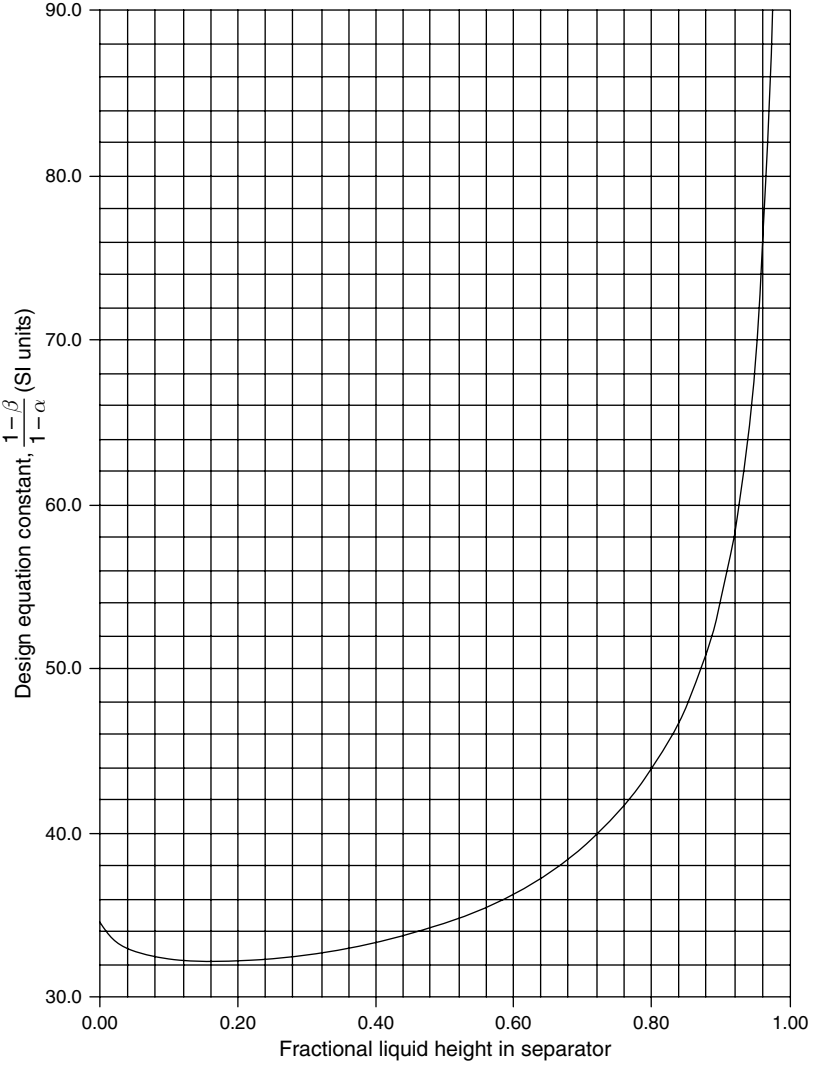


Figure 5-22. Gas capacity constraint design constant $[(1-\beta)/(1-\alpha)]$ vs. liquid height of a cylinder for a horizontal separator other than 50% full of liquid (SI units).

SI Units

$$d^2 L_{\text{eff}} = 21.000 \frac{(t_r)_o Q_o + (t_r)_w Q_w}{\alpha} \tag{5-15b}$$

where $\alpha =$ design constant found in Figure 5-24.

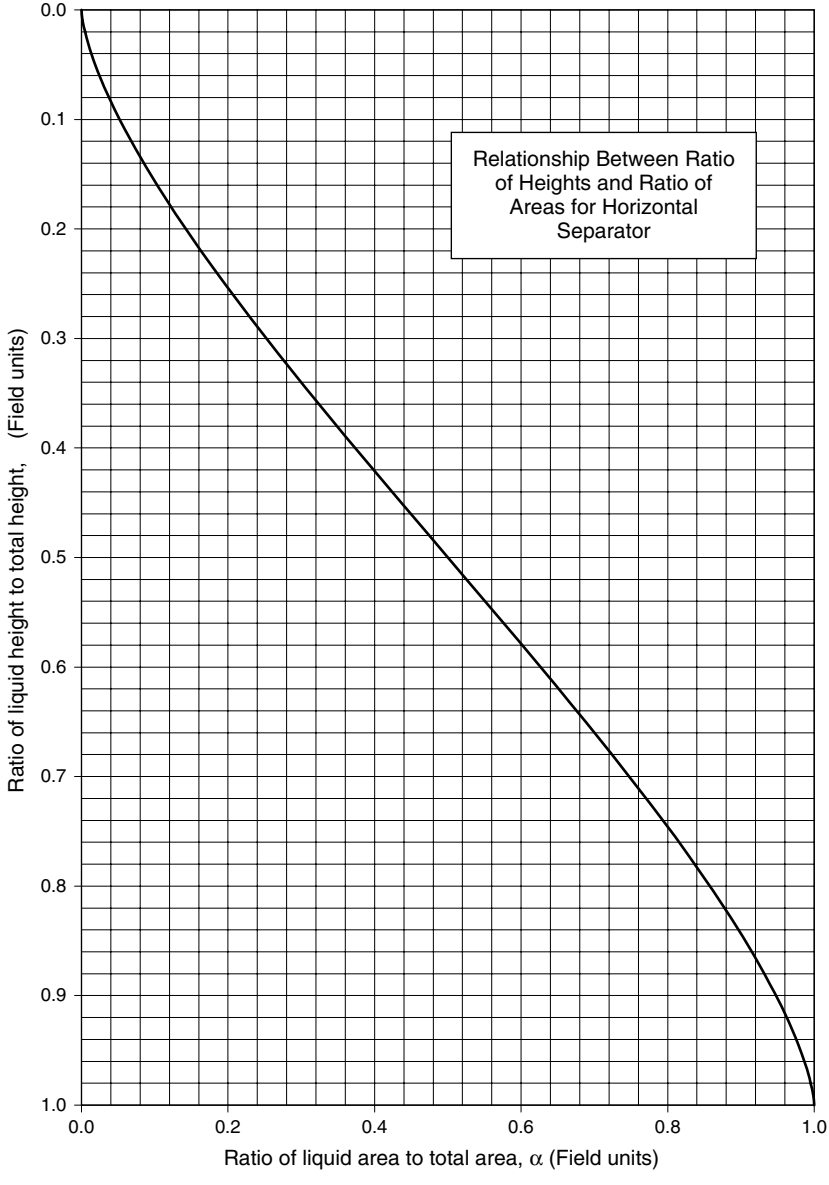


Figure 5-23. Retention time constraint design constant—ratio of areas (α) vs. ratio of heights (β) for a horizontal separator other than 50% full of liquid (field units).

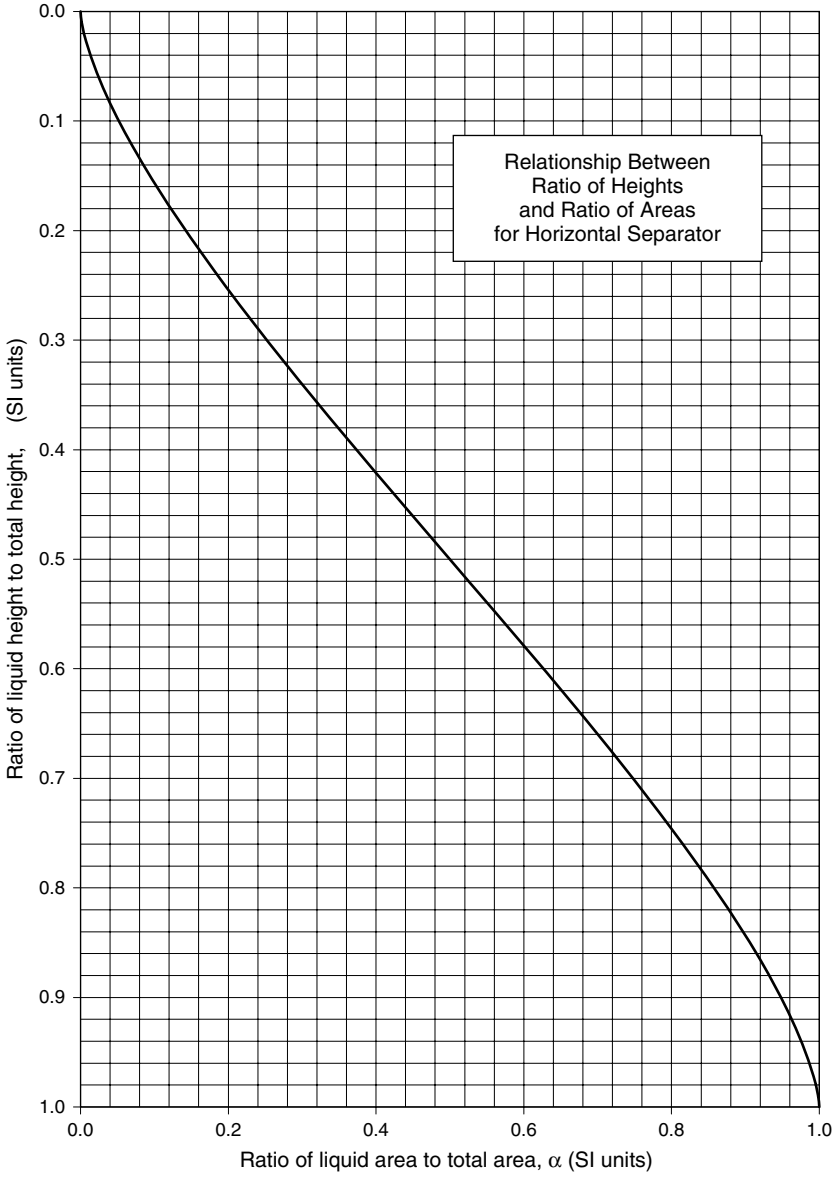


Figure 5-24. Retention time constraint design constant—ratio of areas (α) vs. ratio of heights (β) for a horizontal separator other than 50% full of liquid (SI units).

Settling Equation Constraint

From the maximum oil pad thickness, liquid flow rates, and retention times, a maximum vessel diameter may be calculated. The fractional cross-sectional area of the vessel required for water retention may be determined as follows:

$$\alpha_w = \frac{\alpha_1 Q_w (t_r)_w}{Q_o (t_r)_o + Q_w (t_r)_w}, \tag{5-16}$$

where

α_l = fractional area of liquids,
α_w = fractional area of water.

The fractional height of the vessel required for the water can be determined by solving the following equation by trial and error:

$$\alpha_w = \frac{1}{80} \cos^{-1} [1 - 2\beta_w] - \left(\frac{1}{\pi}\right) [1 - 2\beta_w], \tag{5-17}$$

where β_w represents the fractional height of water.

A maximum vessel diameter may be determined from the fractional heights of the total liquids and water as follows:

$$d_{\max} = \frac{(h_o)_{\max}}{\beta_1 - \beta_w}, \tag{5-18}$$

where $d_{\max}$ is the maximum vessel internal diameter in inches (mm).

Any vessel diameter less than this maximum may be used to separate specified water droplet size in the specified oil retention time.

Vertical Separators' Sizing

As with vertical two-phase separators, a minimum diameter must be maintained to allow liquid droplets to separate from the vertically moving gas. The vessel must also have a large enough diameter to allow water droplets to settle in the upward-flowing oil phase and to allow oil droplets to rise in the downward-moving water phase. The liquid retention time requirement specifies a combination of diameter and liquid volume height. Any diameter greater than the minimum required for gas capacity and for liquid separation can be chosen.

Gas Capacity Constraint

By setting the gas velocity equal to the terminal settling velocity of a droplet, the following may be derived:

Field Units

$$d^2 = 5{,}040\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2} \tag{5-19a}$$

SI Units

$$d^2 = 34{,}500\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2} \tag{5-19b}$$

For 100-micron droplet removal, Eqs. (5-19a) and (5-19b) are reduced to the following:

Field Units

$$d^2 = 504\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_1 - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2} \tag{5-20a}$$

SI Units

$$d^2 = 3{,}450\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_1 - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2} \tag{5-20b}$$

Settling Water Droplets from Oil Phase

The requirement for settling water droplets from the oil requires that the following equation must be satisfied:

Field Units

$$d^2 = 6{,}690\frac{Q_o\mu}{(\Delta\text{SG})\,d_m^2}. \tag{5-21a}$$

SI Units

$$d^2 = 6.37 \times 10^8 \frac{Q_o \mu}{(\Delta SG)\, d_m^2}. \tag{5-21b}$$

Derivation of Equations (5-21a) and (5-21b)

Field Units

V_t is in ft/s (m/s), V_o is in ft/s (m/s), d_m is in micron, μ is in cp.

$$V_t = V_o,$$

$$V_t = \frac{1.78 \times 10^{-6} (\Delta SG)\, d_m^2}{\mu},$$

Q is in ft^3/s, A is in ft^2,

$$V_o = \frac{Q}{A},$$

Q_o is in BPD,

$$Q = Q_o \times 5.61 \frac{ft^3}{barrel} \times \frac{day}{24\,hr} \times \frac{hr}{3{,}600\,s}$$

$$= 6.49 \times 10^{-5} Q_o,$$

D is in ft, d is in in.,

$$A = \frac{\pi D^2}{4} = \frac{\pi d^2}{(4)\,(144)},$$

$$V_o = 0.0119 \frac{Q_o}{d^2},$$

$$\frac{1.78 \times 10^{-6} (\Delta SG)\, d_m^2}{\mu} = 0.0119 \frac{Q_o}{d^2},$$

$$d^2 = 6{,}690 \frac{Q_o \mu}{(\Delta SG)\, d_m^2}.$$

SI Units

$$V_t = V_o,$$

$$V_t = \frac{5.556 \times 10^{-7}\,(\Delta \mathrm{SG})\,d_m^2}{\mu},$$

Q is in m^3/s, A is in. (m^2),

$$V_o = \frac{Q}{A},$$

Q_o is in. (m^3/s),

$$Q = Q_o \times \frac{1\,\mathrm{hr}}{3600\,\mathrm{s}}$$

$$= \frac{Q_o}{3600},$$

D is in. (m), d is in. (mm),

$$A = \frac{\pi D^2}{4} = \frac{\pi d^2}{4\,(1000)^2},$$

$$V_o = 354\frac{Q_o}{d^2},$$

$$\frac{5.556 \times 10^{-7}\,(\Delta \mathrm{SG})\,d_m^2}{\mu} = 353.6\frac{Q_o}{d^2},$$

$$d^2 = 6.37 \times 10^8 \frac{Q_o \mu}{(\Delta \mathrm{SG})\,d_m^2}.$$

For 500-micron droplets, Eqs. (5-21a) and (5-21b) become

Field Units

$$d^2 = 0.0267\left(\frac{Q_o \mu}{\Delta \mathrm{SG}}\right). \tag{5-22a}$$

SI Units

$$d^2 = 2550\left(\frac{Q_o \mu}{\Delta \mathrm{SG}}\right). \tag{5-22b}$$

Settling Oil from Water Phase

The requirement for separating oil from water requires that the following equation must be satisfied:

Field Units

$$d^2 = 6{,}690 \frac{Q_o \mu}{(\Delta \mathrm{SG})\, d_m^2}. \tag{5-21a}$$

SI Units

$$d^2 = 6.37 \times 10^8 \frac{Q_o \mu}{(\Delta \mathrm{SG})\, d_m^2}. \tag{5-21b}$$

For 200-micron droplets, Eqs. (5-21a) and (5-21b) become

Field Units

$$d^2 = 0.167 \left(\frac{Q_o \mu}{(\Delta \mathrm{SG})} \right). \tag{5-23a}$$

SI Units

$$d^2 = 1.59 \times 10^4 \left(\frac{Q_o \mu}{(\Delta \mathrm{SG})} \right). \tag{5-23b}$$

Retention Time Constraint

Field Units

$$h_o + h_w = \frac{[(t_r)_o Q_o + (t_r)_w Q_w]}{0.12 d^2} \tag{5-24a}$$

SI Units

$$h_o + h_w = \frac{[(t_r)_o Q_o + (t_r)_w Q_w]}{4.713 \times 10^{-8} d^2}, \tag{5-24b}$$

where

h_o = height of oil pad, in. (mm),
h_w = height from water outlet to interface, in. (mm).

(Note: this height must be adjusted for cone bottom vessels.)

Derivation of Equations (5-24a) and (5-24b)

From two-phase separator design:

Field Units

$$d^2 h = \frac{t_r Q_1}{0.12}.$$

Thus,

$$d^2 h_o = \frac{(t_r)_o Q_o}{0.12},$$
$$d^2 h_w = \frac{(t_r)_w Q_w}{0.12},$$
$$h_o + h_w = \frac{(t_r)_o Q_o + (t_r)_w Q_w}{0.12 d^2}.$$

SI Units

$$d^2 h = \frac{t_r Q_l}{4.713 \times 10^{-8}}.$$

Thus,

$$d^2 h_o = \frac{(t_r)_o Q_o}{4.713 \times 10^{-8}},$$
$$d^2 h_w = \frac{(t_r)_w Q_w}{4.713 \times 10^{-8}},$$
$$h_o + h_w = \frac{(t_r)_o Q_o + (t_r)_w Q_w}{4.713 \times 10^{-8} d^2}.$$

Seam-to-Seam Length

As with horizontal three-phase separators, the specific design of the vessel internals will affect the seam-to-seam length. The seam-to-seam length (L_{ss}) of vertical vessels may be estimated based on the diameter and liquid height. As shown in Figure 5-25, allowance must be made for

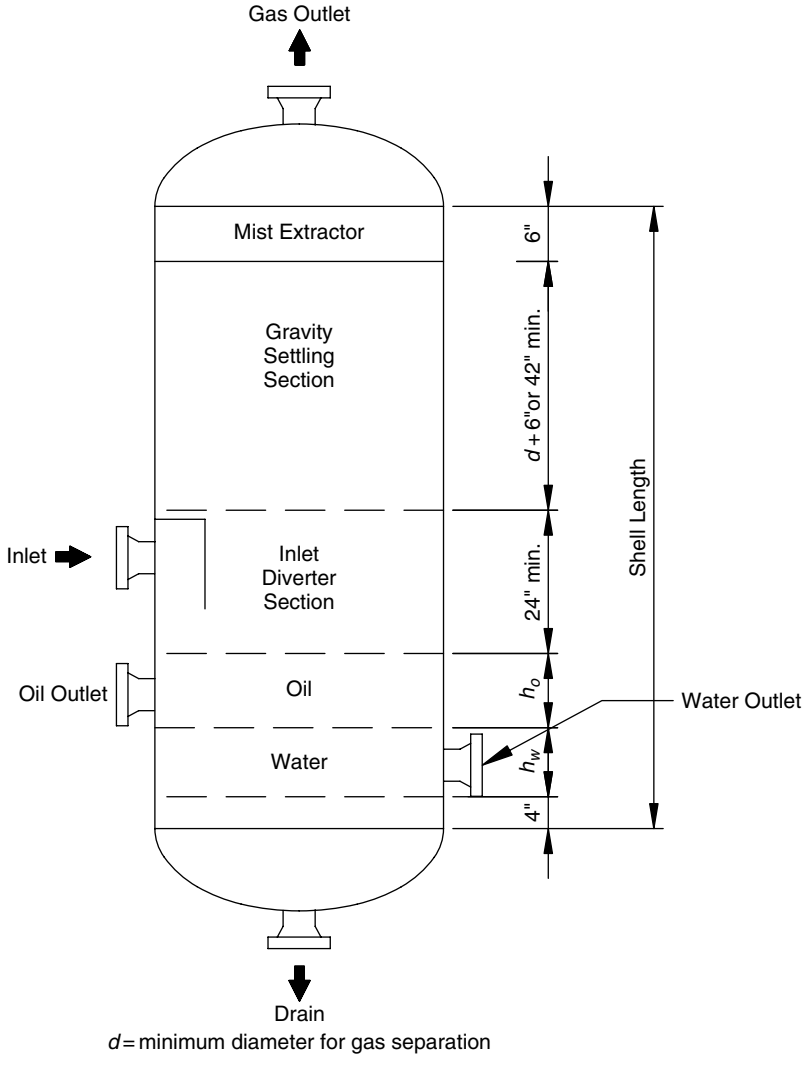


Figure 5-25. Approximate seam-to-seam shell length for a vertical three-phase separator.

the gravity settling (gas separation) section, inlet diverter, mist extractor, and any space below the water outlet. For screening purposes, the larger L_{ss} values from Eqs. (5-25a and 5-25b) and (5-26a and 5-26b) should be used.

Field Units

$$L_{ss} = \frac{h_o + h_w + 76}{12} \quad \text{(for diameters} \leq 36\,\text{in.)}, \tag{5-25a}$$

$$L_{ss} = \frac{h_o + h_w + d + 40}{12} \quad \text{(for diameters} > 36\,\text{in.)}. \tag{5-26a}$$

SI Units

$$L_{ss} = \frac{h_o + h_w + 1930}{1000} \quad \text{(for diameters} \leq 914\,\text{mm)}, \tag{5-25b}$$

$$L_{ss} = \frac{h_o + h_w + d + 1016}{1000} \quad \text{(for diameters} > 914\,\text{mm)}, \tag{5-26b}$$

where

h_o = height of oil pad, in. (mm),
h_w = height from water outlet to interface, in. (mm),
d = vessel's internal diameter, in. (mm).

The larger of the L_{ss} values from Eqs. (2-25a and 2-25b) and (5-26a and 5-26b) should be used.

Slenderness Ratio

As with horizontal three-phase separators, the larger the slenderness ratio, the less expensive the vessel. In vertical separators whose sizing is liquid dominated, it is common to choose slenderness ratios no greater than 4 to keep the height of the liquid collection section to a reasonable level. Choices between 1.5 to 3 are common, although height restrictions may force the choice of a lower slenderness ratio.

Procedure for Sizing Three-Phase Vertical Separators

1. The first step in sizing a vertical separator is to establish the design basis. This includes specifying the maximum and minimum flow rates, operating pressure and temperature, droplet size to be removed, etc.
2. Equations (5-19a) and (5-19b) may be used to calculate the minimum diameter for a liquid droplet to fall through the gas phase. Use Eqs. (5-20a) and (5-20b) for 100-micron droplets if no other information is available.

   Field Units

$$d^2 = 5{,}040\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2} \tag{5-19a}$$

   SI Units

$$d^2 = 34{,}500\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2} \tag{5-19b}$$

   For 100 microns:

   Field Units

$$d^2 = 504\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_1 - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2} \tag{5-20a}$$

   SI Units

$$d^2 = 3{,}500\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_1 - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2} \tag{5-20b}$$

3. Equations (5-21a) and (5-21b) may be used to calculate the minimum diameter for water droplets to fall through the oil phase. Use Eqs. (5-22a) and (5-22b) for 500-micron droplets if no other information is available.

   Field Units

$$d^2 = 6{,}690\frac{Q_o\mu}{(\Delta\text{SG})\,d_m^2} \tag{5-21a}$$

SI Units

$$d^2 = 6.37 \times 10^8 \frac{Q_o \mu}{(\Delta SG)\, d_m^2} \tag{5-21b}$$

For 500-micron droplets:

Field Units

$$d^2 = (0.0267)\left(\frac{Q_o \mu}{\Delta SG}\right). \tag{5-22a}$$

SI Units

$$d^2 = 2550\left(\frac{Q_o \mu}{\Delta SG}\right). \tag{5-22b}$$

4. Equations (5-21a) and (5-21b) may be used to calculate the minimum diameter for oil droplets to rise through the water phase. Use Eqs. (5-23a) and (5-23b) for 200-micron droplets if no other information is available.

Field Units

$$d^2 = 6{,}690 \frac{Q_o \mu}{(\Delta SG)\, d_m^2} \tag{5-21a}$$

SI Units

$$d^2 = 6.37 \times 10^8 \frac{Q_o \mu}{(\Delta SG)\, d_m^2} \tag{5-21b}$$

For 200-micron droplets:

Field Units

$$d^2 = 0.167\left(\frac{Q_o \mu}{(\Delta SG)}\right). \tag{5-23a}$$

SI Units

$$d^2 = 1.59 \times 10^4 \left(\frac{Q_o \mu}{(\Delta \text{SG})} \right). \tag{5-23b}$$

5. Select the largest of the three diameters calculated in steps 2–4 as the minimum diameter. Any value larger than this minimum may be used for the vessel diameter.
6. For the selected diameter, and assumed values of $(t_r)_o$ and $(t_r)_w$, Eqs. (5-24a) and (5-24b) may be used to determine $h_o + h_w$.

Field Units

$$h_o + h_w = \frac{[(t_r)_o \, Q_o + (t_r)_w \, Q_w]}{0.12 \, d^2} \tag{5-24a}$$

SI Units

$$h_o + h_w = \frac{[(t_r)_o \, Q_o + (t_r)_w \, Q_w]}{4.713 \times 10^{-8} \, d^2} \tag{5-24b}$$

7. From d and $h_o + h_w$ the seam-to-seam length may be estimated using Eqs. (5-25a and 5-25b) and (5-26a and 5-26b). The larger value of L_{ss} should be used.

Field Units

$$L_{ss} = \frac{h_o + h_w + 76}{12} \quad (\text{for diameters} \leq 36 \, \text{in.}) \tag{5-25a}$$

SI Units

$$L_{ss} = \frac{h_o + h_w + 1930}{1000} \quad (\text{for diameters} \leq 914 \, \text{mm}) \tag{5-25b}$$

Field Units

$$L_{ss} = \frac{h_o + h_w + d + 40}{12} \quad (\text{for diameters} > 36 \, \text{in.}) \tag{5-26a}$$

SI Units

$$L_{ss} = \frac{h_o + h_w + d + 1016}{1000} \quad \text{(for diameters} > 914\,\text{mm)} \qquad \text{(5-26b)}$$

8. Check the slenderness ratios. Slenderness ratios between 1.5 and 3 are common. The following equations may be used:

Field Units

$$\text{SR} = \frac{12\,L_{ss}}{d}. \qquad \text{(5-27a)}$$

SI Units

$$\text{SR} = \left(\frac{L_{ss}}{(1000)\,d}\right). \qquad \text{(5-27b)}$$

9. If possible, select a standard-size diameter and seam-to-seam length.

Examples

Example 5-1: Sizing a vertical three-phase separator (field units)

Given:

Q_o = 5,000 BOPD,
Q_w = 3,000 BWPD,
Q_g = 5 MMscfd,
P_o = 100 psia,
T_o = 90 °F,
Oil = 30 °API,
$(\text{SG})_w$ = 1.07,
S_g = 0.6,
$(t_r)_o = (t_r)_w$ = 10 min,
μ_o = 10 cp,
μ_w = 1 cp,
ρ_g = 0.3 lb/ft^3,
ρ_l = 54.7 lb/ft^3,
C_D = 2.01

Droplet removal = 100 microns liquids, 500 microns water, 200 microns oil.

Solution:

1. Calculate difference in specific gravities.

$$^{\circ}\text{API} = \frac{141.5}{(\text{SG})_o} - 131.5$$
$$= 0.876,$$
$$\Delta\text{SG} = 1.07 - 0.876 = 0.194.$$

2. Calculate the minimum diameter required to settle a liquid droplet through the gas phase [Eq. (5-19a)].

$$d^2 = 5040\left[\frac{(550)\,(0.99)\,(5)}{(100)}\right]\left[\left(\frac{0.3}{(54.7)-(0.3)}\right)\frac{2.01}{100}\right]^{1/2},$$
$$d = 34.9\,\text{in.}$$

3. Calculate the minimum diameter required for water droplets to settle through the oil phase [Eq. (5-21a)].

$$d^2 = 6{,}690\left[\frac{Q_o\mu}{(\Delta\text{SG})\,d_m^2}\right]$$
$$= 6{,}690\left[\frac{(5{,}000)\,(10)}{(0.194)\,(500)^2}\right],$$
$$d = 83.0\,\text{in.}$$

4. Calculate the minimum diameter required for oil droplets to rise through the water phase [Eq. (5-23a)].

$$d^2 = 6{,}690\left[\frac{Q_o\mu}{(\Delta\text{SG})\,d_m^2}\right]$$
$$= 6{,}690\left[\frac{(3{,}000)\,(1)}{(0.194)\,(200)^2}\right],$$
$$d = 50.8\,\text{in.}$$

5. Select the largest diameter from steps 2–4 as the minimum inside diameter required.

$$d_{\min} = 83.0\,\text{in.}$$

6. Calculate $h_o + h_w$.

$$h_o + h_w = \frac{(t_r)_o\,(Q_o) + (t_r)_w\,Q_w}{0.12\,d^2},$$

$$h_o + h_w = \frac{(10)\,(5{,}000 + 3{,}000)}{0.12\,d^2}$$

$$= \frac{667{,}000}{d^2}.$$

Refer to Table 5-2 for results.

Table 5-2
Vertical Three-Phase Separator Capacity Diameter vs. Length for Retention Time Constraint $(t_r)_o = (t_r)_w = 10$ min

d_o (in.)	$h_o + h_w$ (in.)	L_{ss} (ft)	SR $\left(\frac{12L_{ss}}{d_o}\right)$
84	94.5	18.2	2.6
90	82.3	17.7	2.4
96	72.3	17.4	2.2
102	64.1	17.2	2.0

7. Compute seam-to-seam length (L_{ss}). Select the larger value from Eq. (5-25a) or (5-26a).

$$L_{ss} = \frac{h_o + h_w + 76}{12} \qquad \text{(for diameters} \leq 36\,\text{in.)},$$

$$L_{ss} = \frac{h_o + h_w + d + 40}{12} \qquad \text{(for diameters} > 36\,\text{in.)}.$$

Refer to Table 5-2 for results.

8. Compute the slenderness ratio.

Slenderness ratio $= \frac{12L_{ss}}{d}$

Choices in the range of 1.5 to 3 are common.

Refer to Table 5-2 for results.

9. Make final selection: compute combinations of d and $h_o + h_w$ for diameters greater than the minimum diameter. See Table 5-2 for results. Select 90 in outside diameter (OD) $\times$ 20 ft seam-to-seam length (s/s).

Example 5-2: Sizing a vertical three-phase separator (SI units)

Given:

$Q_o = 33\,\text{m}^3/\text{hr}$,
$Q_w = 19.8\,\text{m}^3/\text{hr}$,
$Q_g = 5902\,\text{sm}^3/\text{hr}$,
$P_o = 690\,\text{kPa}$,
$T_o = 32.20\,^\circ\text{C}$,
Oil $= 30\,^\circ\text{API}$,
$(\text{SG})_w = 1.07$,
$S_g = 0.6$,
$(t_r)_o = (t_r)_w = 10$ min,
$\mu_o = 10\,\text{cp}$,
$\mu_w = 1\,\text{cp}$,
$\rho_g = 49\,\text{kg/m}^3$,
$\rho_l = 866\,\text{kg/m}^3$,
$C_D = 2.01$

Droplet removal = 100 microns liquids, 500 microns water, 200 microns oil. Vessel is half-full of liquid.

Solution:

1. Calculate difference in specific gravities.

$$^\circ\text{API} = \frac{141.5}{(\text{SG})_o} - 131.5$$
$$= 0.876,$$
$$\Delta\text{SG} = 1.07 - 0.876 = 0.194.$$

2. Calculate minimum diameter required to settle a liquid droplet through the gas phase [Eq. (5-19b)].

$$d^2 = 34{,}500\left[\frac{TZQ_g}{P}\right]\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}$$
$$= 34{,}500\left[\frac{(306)\,(0.99)\,(5902)}{(690)}\right]\left[\left(\frac{49}{866-49}\right)\frac{2.01}{100}\right]^{1/2},$$
$$d = 886\,\text{mm}.$$

3. Calculate the minimum diameter required for water droplets to settle through the oil phase [Eq. (5-21b)].

$$d^2 = 6.37 \times 10^8 \left(\frac{Q_o \mu}{\Delta \text{SG} d_m^2} \right)$$

$$= 6.37 \times 10^8 \left(\frac{(33)\,(10)}{0.194\,(500)^2} \right),$$

$$d = 2081\,\text{mm}.$$

4. Calculate the minimum diameter required for oil droplets to rise through the water phase [Eq. (5-23b)].

$$d^2 = 6.37 \times 10^8 \left[\frac{Q_o \mu}{(\Delta \text{SG})\, d_m^2} \right]$$

$$= 6.37 \times 10^8 \left[\frac{(19.8)\,(1)}{(0.194)\,(200)^2} \right],$$

$$d = 1274\,\text{mm}.$$

5. Select the largest diameter from steps 2–4 as the minimum diameter inside required.

$$d_{\min} = 2081\,\text{mm}$$

6. Calculate $h_o + h_w$.

$$h_o + h_w = \frac{(10)\,(33 + 19.8)}{4.713 \times 10^{-8} d^2}$$

$$= \frac{1.12 \times 10^{10}}{d^2}$$

Refer to Table 5-3 for result.

7. Compute seam-to-seam length (L_{ss}). Select the larger value from Eq. (5-25b) or (5-26b).

$$L_{\text{ss}} = \frac{h_o + h_w + 1930}{1000} \qquad \text{(for diameters} \leq 914\,\text{mm)},$$

$$L_{\text{ss}} = \frac{h_o + h_w + d + 1016}{1000} \qquad \text{(for diameters} > 914\,\text{mm)}.$$

Refer to Table 5-3 for results.

Table 5-3
Vertical Three-Phase Separator Capacity Diameter vs. Length for Retention Time Constraint $(t_r)_o = (t_r)_w = 10$ min

d_o (mm)	$h_o + h_w$ (mm)	L_{ss} (m)	$SR\left(\frac{L_{ss}}{(1000)d_o}\right)$
2286	2144	5.4	2.4
2438	1884	5.3	2.2
2591	1669	5.3	2.0

8. Compute the slenderness ratio.

$$\text{Slenderness ratio} = \left(\frac{L_{ss}}{(1000)\,d_o}\right)$$

Choices in the range of 1.5 to 3 are common.

Refer to Table 5-3 for results.

9. Make final selection: compute combinations of d and $h_o + h_w$ for diameters greater than the minimum diameter. See Table 5-3 for results. Select 2286 outside diameter (OD) × 5.4 m (s/s).

Example 5-3: Sizing a horizontal three-phase separator (field units)

Given:

Q_o = 5,000 BOPD,
Q_w = 3,000 BWPD,
Q_g = 5 MMscfd,
P = 100 psia,
T = 90°F,
Oil = 30°API,
$(SG)_w$ = 1.07,
S_g = 0.6,
$(t_r)_o = (t_r)_w$ = 10 min,
μ_o = 10 cp,
μ_w = 1 cp,

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil. Vessel is half-full of liquids.

Solution:

1. Calculate difference in specific gravities.

$$^{\circ}\text{API} = \frac{141.5}{(\text{SG})_o} - 131.5,$$

$$(\text{SG})_o = \frac{141.5}{30 + 131.5} = 0.876,$$

$$\Delta\text{SG} = 1.07 - 0.876 = 0.194.$$

2. Calculate maximum oil pad thickness $(h_o)_{\max}$. Use 500-micron droplet size if no other information is available.

$$(h_o)_{\max} = \left(1.28 \times 10^{-3}\right) \frac{(t_r)_o \, (\Delta\text{SG}) \, d_m^2}{\mu}$$

$$= 0.00128 \frac{(10)\,(0.194)\,(500)^2}{10}$$

$$= 62.1$$

3. Calculate $\dfrac{A_w}{A}$:

$$\frac{A_w}{A} = 0.5 \frac{Q_w \, (t_r)_w}{(t_r)\, Q_o + (t_r)_w \, Q_w}$$

$$= 0.5 \frac{(19.8)\,(10)}{(33)\,(10) + (19.8)\,(10)}$$

$$= 0.1875.$$

4. Determine β from Figure 5-20. With $Aw/A = 0.1875$, read $\beta = 0.257$.

5. Calculate $d_{\max}$.

$$d_{\max} = \frac{(h_o)_{\max}}{\beta}$$

$$= \frac{62.1}{0.257},$$

$$d_{\max} = 241.6 \text{ in.}$$

6. Calculate combinations of d, L_{eff} for d less than $d_{\max}$ that satisfy the gas capacity constraint. Use 100-micron droplet size if no other information is available.

$$dL_{\text{eff}} = 420\left(\frac{TZQ_g}{P}\right)\left[\left(\frac{\rho_g}{\rho_l - \rho_g}\right)\frac{C_D}{d}\right]^{1/2}$$

$$= 420\left(\left(\frac{(550)\,(0.99)\,(5)}{(100)}\right)\left(\frac{0.3}{(54.7)-(0.3)}\right)\frac{2.01}{100}\right)^{1/2}$$

$$= 120$$

Refer to Table 5-4 for results.

Table 5-4
Horizontal Three-Phase Separator Diameter vs. Length for Gas Capacity Constraint

d (in.)	L_{eff} (ft)
60	1.7
72	1.4
84	1.2
96	1.1

Since the values of L_{eff} are low, the gas capacity does not govern.

7. Calculate combinations of d, L_{eff} for d less than $d_{\max}$ that satisfy the oil and water retention time constraints.

$$d^2L_{\text{eff}} = 1.42\left[Q_w(t_r)_w + Q_o(t_r)_o\right]$$

$$= (1.42)(10)(8{,}000)$$

$$= 113{,}600$$

Refer to Table 5-5 for results.

8. Estimate seam-to-seam length.

$$L_{\text{ss}} = L_{\text{eff}} + \frac{d}{12} \qquad \text{(for gas capacity)},$$

$$L_{\text{ss}} = \frac{4}{3}L_{\text{eff}} \qquad \text{(for liquid capacity)}.$$

Table 5-5
Horizontal Three-Phase Separator Capacity Diameter vs. Length for Liquid Retention Time Constraint $(t_r)_o = (t_r)_w = 10$ min

d (in.)	L_{eff} (ft)	L_{ss} (ft)	$SR\left(\frac{12L_{ss}}{d}\right)$
60	31.6	42.1	8.4
72	21.9	29.2	4.9
84	16.1	21.5	3.1
96	12.3	16.4	2.1
108	9.7	13.0	1.4

Refer to Table 5-5 for results.

9. Select slenderness ratio ($12\,L_{ss}/d$). Choices in the range of 3 to 5 are common.
10. Choose a reasonable size that does not violate gas capacity restraint or oil pad thickness restraint. Possible choices are 72 in. diameter by 30 ft seam-by-seam and 84 in. diameter by 25 ft seam-by-seam.

Example 5-4: Sizing a horizontal three-phase separator (SI units)

Given:

$Q_o = 33\,m^3/hr$,
$Q_w = 19.8\,m^3/hr$,
$Q_g = 5902\,sm^3/hr$,
$P = 690\,kPa$,
$T = 32.20\,°C$,
Oil $= 30\,°API$,
$(SG)_w = 1.07$,
$S_g = 0.6$,
$(t_r)_o = (t_r)_w = 10$ min,
$\mu_o = 10\,cp$,
$\mu_w = 1\,cp$,
$\rho_g = 49\,kg/m^3$,
$\rho_l = 866\,kg/m^3$,
$C_D = 2.01$

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil.

Solution:

1. Calculate difference in specific gravities.

$$^{\circ}\text{API} = \frac{141.5}{(\text{SG})_o} - 131.5,$$

$$(\text{SG})_o = \frac{141.5}{30 + 131.5} = 0.876,$$

$$\Delta\text{SG} = 1.07 - 0.876 = 0.194.$$

2. Calculate maximum oil pad thickness $(h_o)_{\max}$. Use 500-micron droplet size if no other information is available.

$$(h_o)_{\max} = 0.033 \frac{(t_r)_o \, (\Delta\text{SG}) \, d_m^2}{\mu}$$

$$= 0.033 \frac{(10)\,(0.194)\,(500)^2}{10}$$

$$= 1600$$

3. Calculate $\dfrac{A_w}{A}$:

$$\frac{A_w}{A} = 0.5 \frac{Q_w \, (t_r)_w}{(t_r)\, Q_o + (t_r)_w \, Q_w}$$

$$= 0.5 \frac{(19.8)\,(10)}{(33)\,(10) + (19.8)\,(10)}$$

$$= 0.1875.$$

4. Determine β from Figure 5-19. With $A_w/A = 0.1875$, read $\beta = 0.257$.
5. Calculate $d_{\max}$.

$$d_{\max} = \frac{(h_o)_{\max}}{\beta}$$

$$= \frac{1600}{0.257},$$

$$d_{\max} = 6226\,\text{mm}.$$

6. Calculate combinations of d, L_{eff} for d less than $d_{\max}$ that satisfy the gas capacity constraint. Use 100-micron droplet size if no other information is available.

$$dL_{\text{eff}} = 34.5\left(\frac{TZQ_g}{P}\right)\left[\left(\frac{\rho_g}{\rho_l-\rho_g}\right)\frac{C_D}{d_m}\right]^{1/2}$$

$$= 34.5\left(\left(\frac{(306)\,(0.99)\,(5902)}{(690)}\right)\left(\frac{49}{866-49}\right)\frac{2.01}{100}\right)^{1/2}$$

$$= 3095$$

Refer to Table 5-6 for results.

Table 5-6
Horizontal Three-Phase Separator Diameter vs. Length for Gas Capacity Constraint

d (mm)	L_{eff} (m)
1524	0.52
1829	0.43
2134	0.37
2438	0.34

Since the values of L_{eff} are low, gas capacity does not govern.

7. Calculate combinations of d, L_{eff} for d less than $d_{\max}$ that satisfy the oil and water retention constraints.

$$d^2L_{\text{eff}} = 42{,}000[Q_w(t_r)_w + Q_o(t_r)_o]$$

$$= 42{,}000[10][52.8]$$

$$= 22{,}410{,}432$$

Refer to Table 5-7 for results.

8. Estimate seam-to-seam length.

$$L_{\text{ss}} = L_{\text{eff}} + \frac{d}{1000} \quad \text{(for gas capacity)},$$

$$L_{\text{ss}} = \frac{4}{3}L_{\text{eff}} \quad \text{(for liquid capacity)}.$$

Refer to Table 5-7 for results.

Table 5-7
Horizontal Three-Phase Separator Capacity Diameter vs. Length for Liquid Retention Time Constraint $(t_r)_o = (t_r)_w = 10$ min

d (mm)	L_{eff} (m)	L_{ss} (m)	SR $\left(\frac{L_{ss}}{(d)(1000)}\right)$
1524	9.65	12.87	8.4
1828.8	6.70	8.93	4.9
2133.6	4.62	6.56	3.1
2438.4	3.77	5.03	2.1
2743.2	2.98	3.97	1.4

9. Select slenderness ratio, $\frac{L_{ss}}{(d)\,(1000)}$. Choices in the range of 3 to 5 are common.
10. Choose a reasonable size that does not violate gas capacity restraint or oil pad thickness restraint. Possible choices are 1828 mm × 8.93 m and 2133.6 mm × 6.56 m.

Nomenclature

A = cross-sectional area of the droplet, ft^2 (m^2)
A_g = cross-sectional area of vessel available for gas settling, ft^2 (m^2)
A_l = cross-sectional area of vessel available for liquid retention, ft^2 (m^2)
A_T = total cross-sectional area of vessel, ft^2 (m^2)
A_w = cross-sectional area of vessel available for water retention, ft^2 (m^2)
B = weir width, ft (m)
C = coefficient of discharge, dimensionless
API = API gravity of oil, °API
CA = corrosion allowance, in. (mm)
C_D = drag coefficient, dimensionless
D = drop diameter, ft (m)
D = vessel internal diameter, ft (m)
D_h = hydraulic diameter, ft (m)
d = vessel internal diameter, in. (mm)
d_l = water leg standpipe internal diameter, in. (mm)
d_m = drop diameter, micron (μm)
d_{max} = maximum vessel internal diameter, in. (mm)
D_m = droplet diameter micron (μm)

d_{min} = minimum allowable vessel internal diameter to avoid re-entrainment, in. (mm)
d_o = vessel external diameter, in. (mm)
E = joint efficiency, dimensionless
F = height of liquid over weir, ft (m)
f = moody friction factor for pipe
F_B = buoyant force, lb (N)
F_D = drag force, lb (N)
G = oil gravity, degrees API
g = acceleration of gravity, 32.2 ft/s^2, (9.8 m/s^2)
g = gravitational constant, 32.2lb$_m$ft/lb$_f$s^2 (9.81 m/s^2)
H = height of liquid volume, ft (m)
h = height of liquid volume, in. (mm)
H_l = height of liquid in horizontal vessel, ft (m)
h_l = height of liquid in horizontal vessel, in. (mm)
H_o = height of oil pad, ft (m)
h_o = height of oil pad, in (mm)
$(h_o)_{max}$ = maximum oil pad thickness, in. (mm)
H_w = height from water outlet to interface, ft (m)
H'_w = height of water in standpipe, ft (m)
h_w = height from water outlet to interface, in. (mm)
$(h_w)_{max}$ = maximum water height, in. (mm)
h'_w = height of water weir, in. (mm)
L_{eff} = effective length of the vessel, ft (m)
L_{ell} = equivalent length of ell, ft (m)
L_{ent} = equivalent length of inward projecting pipe entrance, ft (m)
L_{equiv} = equivalent length of pipe entrance, ell, and pipe exit, ft (m)
L_{exit} = equivalent length of pipe exit, ft (m)
L_{ss} = vessel length seam-to-seam, ft (m)
M = slope of straight line
N_μ = viscosity number, dimensionless
P = operating pressure, psia (kPa)
P_b = pressure base, 14.7 psia (100 kPa)
P_c = gas pseudo-critical pressure, psia (kPa)
P_{cc} = corrected pseudo-critical pressure, psia (kPa)
P_d = design pressure, psia (kPa)
P_r = gas reduced pressure, dimensionless
Q = flow rate, ft^3/s (m^3/s)
Q_g = gas flow rate, MMscfd (std m^3/hr)
Q_l = liquid flow rate, BPD (m^3/hr)
Q_o = oil flow rate, BPD (m^3/hr)
Q_w = water flow rate, BPD (m^3/hr)

r = vessel external radius, in. (mm)
Re_f = film Reynolds number, dimensionless
Re = Reynold's number, dimensionless
SR = Slenderness ratio, dimensionless
S = allowable stress, psia (kPa)
SG = oil specific gravity
T = operating temperature, °R (K)
T = temperature, °F (°C)
T_b = temperature base, 520°R (288.15 K)
T_c = gas pseudo-critical temperature, °R (K)
T_{cc} = corrected pseudo-critical temperature, °R (K)
t_d = droplet settling time, s
t_g = gas retention time, s
T_n = temperature corresponding to μn, °R (K)
T_r = gas reduced temperature, dimensionless
t = shell thickness, in. (mm)
t_o = oil retention time or settling time, s
t_r = liquid retention time, min
$(t_r)_o$ = oil retention time, min
$(t_r)_w$ = water retention time, min
t_w = water retention time or settling time, s
V = volume, ft^3 (m^3)
V_g = gas velocity, ft/s (m/s)
$(V_g)_{max}$ = maximum gas velocity, no re-entrainment, ft/s (m/s)
V_l = average liquid velocity, ft/s (m/s)
V_o = oil volume, ft^3 (m^3)
V_t = terminal settling velocity of the droplet, ft/s (m/s)
V_w = water volume, ft^3 (m^3)
W = vessel weight, lb (kg)
Y_{CO_2} = gas mole fraction CO_2
Y_{H_2S} = gas mole fraction H_2S
Z = gas compressibility factor, dimensionless
Z_L = depth of liquid for weir calculations, ft (m)
α = fractional cross-sectional area of liquid
α_l = fractional area of liquids
α_o = fractional area of oil
α_w = fractional area of water
ß = fractional height of liquid within the vessel = h_l/d_l
$ß_l$ = fractional height of liquid
$ß_w$ = fractional height of water
Δ_h = height difference between oil weir and water weir, in. (mm)
ΔL = total equivalent length for water leg, ft (m)
ΔP = pressure loss in standpipe, psi (kPa)

ΔSG = difference in specific gravity relative to water of the drop and the gas
Δ_ρ = density difference, liquid and gas, $lb_m/ft^3 (kg/m^3)$
ΔT = Wichert–Aziz correction, °R (K)
θ = angle used in determining α, radians or degrees
μ = viscosity of continuous phase, cp (Pa s)
μ_l = dynamic viscosity of the liquid, lb_m/ft-s (kg/m-s)
μ_n = oil viscosity at T_n, cp (Pa s)
μ_o = viscosity of oil phase, cp (Pa s)
μ_w = viscosity of water phase, cp (Pa s)
μ' = gas viscosity, cp $(lb - s/ft^2)$
δ = kinematic viscosity, cs
ρ = density of the continuous phase, $lb/ft^3 (kg/m^3)$
ρ_g = density of the gas at the temperature and pressure in the separator, $lb/ft^3 (kg/m^3)$
ρ_l = density of liquid, $lb/ft^3 (kg/m^3)$
ρ_o = oil density, $lb/ft^3 (kg/m^3)$
ρ_w = water density, $lb/ft^3 (kg/m^3)$
ρ_m = gas density, g/cm^3
ρ_r = reduced density
ρ_{r+1} = value of reduced density for iteration "$r + 1$"
σ = surface tension $lb_m/s^2 (kg/s^2)$

Review Questions

1. Three-phase separators
   a) experience the same operating problems as two-phase separators
   b) may develop problems with emulsions
   c) have two liquid outlets
   d) B and C
   e) all of the above
2. The inlet diverter in a three-phase separator
   a) extends below the oil–water interface
   b) promotes the coalescence of water droplets that are entrained in the oil continuous phase
   c) assures liquid does not fall on top of the gas–oil or oil–water interface
   d) B and C
   e) all of the above

3. The oil–water interface is maintained by

   a) liquid interface level controller
   b) bucket and weir
   c) bubble tube
   d) A and B
   e) all of the above

4. The bucket and weir design

   a) eliminates the need for a liquid interface controller
   b) prevents gas blowby
   c) requires two interface level control valves
   d) requires one level control valve and one interface level control valve
   e) none of the above

5. The inlet diverter

   a) contains a down-comer that directs the liquid flow below the oil–water interface
   b) forces the inlet mixture of oil and water to mix with the water continuous phase in the bottom of the vessel
   c) promotes the coalescence of water droplets that are entrained in the oil continuous phase
   d) none of the above
   e) all of the above

6. A flow splitter

   a) is a special version of a free-water knockout
   b) is an FWKO where the oil outlet is split among two or more outlet lines that are directed to several downstream process components
   c) contains several compartments that are sealed from each other
   d) must be operated with a gas blanket
   e) all of the above

7. Characteristics of horizontal vessels include that

   a) they are not as good as vertical vessels in handling solids
   b) they require more plan area to perform the same separation as vertical vessels
   c) they have greater interface areas, which enhance phase equilibrium
   d) gravity separation is more efficient than a vertical vessel
   e) all of the above

8. Which of the following criteria are important when sizing a three-phase separator?

   a) gas capacity
   b) oil and water retention times
   c) settling water droplets from the oil phase
   d) rising oil droplets from the water phase
   e) all of the above

9. In the absence of laboratory data, what maximum water droplet size is suggested to settle through the oil phase?

   a) 500 microns
   b) 200 microns
   c) 100 microns
   d) it doesn't matter

10. Common vessel internals include

    a) coalescing plates
    b) sand jets and drains
    c) vortex breakers
    d) inlet diver
    e) level controllers
    f) all of the above
    g) A, B, and E

Exercises

Problem 1.

Determine the size of a vertical three-phase separator given the following data:

Q_g = 6.6 MMscfd,
Q_o = 5,000 BOPD,
Q_w = 6,000 BWPD,
P_o = 65 psia,
T_o = 90°F,
S_g = 0.6,
$(SG)_o$ = 30 °API,
$(SG)_w$ = 1.07,
μ_o = 10 cp,
μ_w = 1 cp,

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil,

$(t_r)_o = 10$ min,
$(t_r)_w = 10$ min.

Problem 2.

Determine the size of a horizontal three-phase separator given the following data:

$Q_g = 6.6$ MMscfd,
$Q_o = 5{,}000$ BOPD,
$Q_w = 6{,}000$ BWPD,
$P_o = 65$ psia,
$T_o = 90°$F,
$S_g = 0.6$,
$(\text{SG})_o = 30°$API,
$(\text{SG})_w = 1.07$,
$\mu_o = 10$ cp,
$\mu_w = 1$ cp,

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil,

$(t_r)_o = 10$ min,
$(t_r)_w = 10$ min.

Problem 3.

Determine the size of a vertical three-phase separator given the following data:

$Q_g = 1.65$ MMscfd,
$Q_o = 3{,}900$ BOPD,
$Q_w = 3{,}000$ BWPD,
$P_o = 455$ psia,
$T_o = 90°$F,
$S_g = 0.6$,
$(\text{SG})_o = 30°$API,
$(\text{SG})_w = 1.07$,
$\mu_o = 10$ cp,
$\mu_w = 1$ cp,

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil,

$(t_r)_o = 10$ min,
$(t_r)_w = 10$ min.

Problem 4.

Determine the size of a horizontal three-phase separator given the following data:

$Q_g = 1.6$ MMscfd,
$Q_o = 3{,}900$ BOPD,
$Q_w = 3{,}000$ BWPD,
$P_o = 455$ psia,
$T_o = 90°$F,
$S_g = 0.6$,
$(\text{SG})_o = 30°$API,
$(\text{SG})_w = 1.07$,
$\mu_o = 10$ cp,
$\mu_w = 1$ cp,

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil,

$(t_r)_o = 10$ min,
$(t_r)_w = 10$ min.

Problem 5.

Determine the size of a vertical FWKO separator given the following data:

$Q_g = 1.5$ MMscfd,
$Q_o = 2{,}000$ BOPD,
$Q_w = 5{,}000$ BWPD,
$P_o = 65$ psia,
$T_o = 165°$F,
$S_g = 0.6$,
$(\text{SG})_o = 30°$API,
$(\text{SG})_w = 1.07$,
$\mu_o = 10$ cp,
$\mu_w = 1$ cp,

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil,

$(t_r)_o = 10\,\text{min},$
$(t_r)_w = 10\,\text{min}.$

Problem 6.

Determine the size of a vertical three-phase separator given the following data:

$Q_g = 7790\ \text{sm}^3/\text{hr},$
$Q_o = 33\ \text{m}^3/\text{hr},$
$Q_w = 40\ \text{m}^3/\text{hr},$
$P_o = 450\ \text{kPa},$
$T_o = 32°\text{C},$
$S_g = 0.6,$
$(\text{SG})_o = 30°\text{API},$
$(\text{SG})_w = 1.07,$
$\mu_o = 10\,\text{cp},$
$\mu_w = 1\,\text{cp},$

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil,

$(t_r)_o = 10\,\text{min},$
$(t_r)_w = 10\,\text{min}.$

Problem 7.

Determine the size of a horizontal three-phase separator given the following data:

$Q_g = 7790\ \text{sm}^3/\text{hr},$
$Q_o = 33\ \text{m}^3/\text{hr},$
$Q_w = 40\ \text{m}^3/\text{hr},$
$P_o = 480\ \text{kPa},$
$T_o = 32°\text{C},$
$S_g = 0.6,$
$(\text{SG})_o = 30°\text{API},$
$(\text{SG})_w = 1.07,$
$\mu_o = 10\,\text{cp},$
$\mu_w = 1\,\text{cp},$

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil,

$(t_r)_o$ = 10 min,
$(t_r)_w$ = 10 min.

Problem 8.

Determine the size of a vertical three-phase separator given the following data:

Q_g = 7790 sm^3/hr,
Q_o = 33 m^3/hr,
Q_w = 40 m^3/hr,
P_o = 480 kPa,
T_o = 32°C,
S_g = 0.6,
$(SG)_o$ = 30°API,
$(SG)_w$ = 1.07,
μ_o = 10 cp,
μ_w = 1 cp,

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil,

$(t_r)_o$ = 10 min,
$(t_r)_w$ = 10 min.

Problem 9.

Determine the size of a horizontal three-phase separator given the following data:

Q_g = 1950 sm^3/hr,
Q_o = 26 m^3/hr,
Q_w = 20 m^3/hr,
P_o = 3140 kPa,
T_o = 32°C,
S_g = 0.6,
$(SG)_o$ = 30°API,
$(SG)_w$ = 1.07,
μ_o = 10 cp,
μ_w = 1 cp,

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil,

$(t_r)_o = 10\,\text{min}$,
$(t_r)_w = 10\,\text{min}$.

Problem 10.

Determine the size of a horizontal FWKO given the following data:

$Q_g = 1770\,\text{sm}^3/\text{hr}$,
$Q_o = 320\,\text{m}^3/\text{hr}$,
$\text{Q}_\text{w} = 800\,\text{m}^3/\text{hr}$,
$P_o = 450\,\text{kPa}$,
$T_o = 70°\text{C}$,
$S_g = 0.6$,
$(\text{SG})_o = 30°\text{API}$,
$(\text{SG})_w = 1.07$,
$\mu_o = 10\,\text{cp}$,
$\mu_w = 1\,\text{cp}$,

Droplet removal = 100 microns liquid, 500 microns water, 200 microns oil,

$(t_r)_o = 10\,\text{min}$,
$(t_r)_w = 10\,\text{min}$.

Chapter 6
Mechanical Design of Pressure Vessels

Introduction

Chapters 4 and 5 discuss the concepts for determining the diameter and length of two-phase and three-phase vertical and horizontal separators. This chapter addresses the selection of design pressure rating and wall thickness of pressure vessels. It also presents a procedure for estimating vessel weight and includes some examples of design details.

The purpose of this chapter is to present an overview of simple concepts of mechanical design of pressure vessels that must be understood by a project engineer specifying and purchasing this equipment. Most pressure vessels used in the oil and gas industry are designed and inspected according to the American Society of Mechanical Engineers' Boiler and Pressure Vessel Code (ASME code). Because the ASME code contains much more detail than can be covered in a single chapter of a general textbook such as this one, the project engineer should have access to a copy of the ASME code and should become familiar with its general contents. In particular, Section VIII of the code, "Pressure Vessels," is particularly important. Countries that do not use the ASME code have similar documents and requirements. The procedures used in this chapter that refer specifically to the ASME code are generally applicable in other countries but should be checked against the applicable code.

In U.S. federal waters and the majority of countries with oil and gas operations, all pressure vessels must be designed and inspected in accordance with the ASME code. In some countries, however, there is no such requirement. It is possible to purchase "non-code" vessels in these countries at a small savings in cost. Non-code vessels are normally designed to code requirements (although there is no certainty that this is true), but they are not inspected by a qualified code inspector nor are they

necessarily inspected to the quality standards dictated by the code. For this reason, the use of non-code vessels should be discouraged to assure vessel mechanical integrity.

Design Considerations

Design Temperature

The maximum and minimum design temperatures for a vessel will determine the maximum allowable stress value permitted for the material to be used in the fabrication of the vessel. The maximum temperature used in the design should not be less than the mean metal temperature expected under the design operating conditions. The minimum temperature used in the design should be the lowest expected in service except when lower temperatures are permitted by the rules of the ASME code. In determining the minimum temperature, such factors as the lowest operating temperature, operational upset, auto-refrigeration, ambient temperature, and any other source of cooling should all be considered. If necessary, the metal temperature should be determined by computation using accepted heat transfer procedures or by measurement from equipment in service under equivalent operating conditions.

Design Pressure

The design pressure for a vessel is called its "maximum allowable working pressure" (MAWP). In conversation this is sometimes referred to simply as the vessel's "working pressure." The MAWP determines the setting of the relief valve and must be higher than the normal pressure of the process contained in the vessel, which is called the vessel's "operating pressure." The operating pressure is fixed by process conditions. Table 6-1 recommends a minimum differential between operating pressure and MAWP so that the difference between the operating pressure and the relief valve set pressure provides a sufficient cushion. If the operating pressure is too close to the relief valve setting, small surges in operating pressure could cause the relief valve to activate prematurely.

Some vessels have pressure safety high sensors (PSHs) that shut in the inflow if a higher-than-normal pressure is detected. The use of PSHs is discussed in more detail in the Instrumentation, Process Control and Safety Systems volume of this series. The differential between the maximum operating pressure and the PSH sensor set pressure should be as indicated in Table 6-1, and the relief valve should be set at least 5% or

Table 6-1
Setting Maximum Allowable Working Pressures

Operating Pressure	Minimum Differential Between Operating Pressure and MAWP
Less than 50 psig	10 psi
51–250 psig	25 psi
251–500 psig	10% of maximum operating pressure
501–1000 psig	50 psi
1001 psig and higher	5% of maximum operating pressure

Vessels with high-pressure safety sensors have an additional 5% or 5 psi, whichever is greater to the minimum differential.

5 psi, whichever is greater, higher than the PSH sensor set pressure. Thus, the minimum recommended MAWP for a vessel operating at 75 psig with a PSH sensor would be 105 psig (75 + 25 + 5); the PSH sensor is set at 100 psig and the relief valve is set at 105 psig.

Often, especially for small vessels, it is advantageous to use a higher MAWP than is recommended in Table 6-1. It may be possible to increase the MAWP at little or no cost and thus have greater future flexibility if process changes (e.g., greater throughput) require an increase in operating pressure.

The MAWP of the vessel cannot exceed the MAWP of the nozzles, valves, and pipe connected to the vessel. As discussed in the Plant Piping and Pipeline volume of this series, pipe flanges, fittings, and valves are manufactured in accordance with industry standard pressure rating classes. Table 6-2 is a summary of Material Group 1.1 carbon steel fittings

Table 6-2
Summary ANSI Pressure Ratings Material Group 1.1

	MAWP, psig	
Class	–20°F to 100°F	100°F to 200°F
150	285	250
300	740	675
400	990	900
600	1480	1350
900	2220	2025
1500	3705	3375
2500	6170	5625

manufactured in accordance with American National Standards Institute (ANSI) specification B16.5.

If the minimum MAWP calculated from Table 6-1 is close to one of the ANSI MAWP listed in Table 6-2, it is common to design the pressure vessel to the same MAWP as the ANSI class. For example, the 105-psig pressure vessel previously discussed will have nozzles, valves, and fittings attached to it that are rated for 285 psig (ANSI Class 150). The increase in cost of additional vessel wall thickness to meet a MAWP of 285 psig may be small.

Often, a slightly higher MAWP than that calculated from Table 6-1 is possible at almost no additional cost. Once a preliminary MAWP is selected from Table 6-1, it is necessary to calculate a wall thickness for the shell and heads of the pressure vessel. The procedure for doing this is described in the following section. The actual wall thickness chosen for the shell and heads will be somewhat higher than that calculated, as the shells and heads will be formed from readily available plates. Thus, once the actual wall thickness is determined, a new MAWP can be specified for essentially no additional cost. (There will be a marginal increase in cost to test the vessel to the slightly higher pressure.)

This concept can be especially significant for a low-pressure vessel where a minimum wall thickness is desired. For example, assume the calculations for a 50-psig MAWP vessel indicate a wall thickness of 0.20 in., and it is decided to use 1/4-in. plate. This same plate might be used if a MAWP of 83.3 psig were specified. Thus, by specifying the higher MAWP (83.3 psig), additional operating flexibility is available at essentially no increase in cost. Many operators specify the MAWP based on process conditions in their bids and ask the vessel manufacturers to state the maximum MAWP for which the vessel could be tested and approved.

Maximum Allowable Stress Values

The maximum allowable stress values to be used in the calculation of a vessel's wall thickness are given in the ASME code for many different materials. These stress values are a function of temperature. Section VIII of the ASME code, which governs the design and construction of all pressure vessels with operating pressures greater than 15 psig, is published in two divisions. Each sets its own maximum allowable stress values. Division 1, governing the *design by rules*, is less stringent from the standpoint of certain design details and inspection procedures, and thus incorporates a higher safety factor. The 1998 edition incorporates a safety factor of 4 while the 2001 and later editions incorporate a safety factor of

3.5. The 2001 edition of the code yields higher allowable stresses and thus smaller wall thicknesses. For example, using a material with a 60,000-psi tensile strength, a vessel built under the 1998 edition (safety factor = 4) yields a maximum allowable stress value of 15,000 psi while a vessel built under the 2001 edition (safety factor = 3.5) yields a maximum allowable stress value of 17,142 psi. On the other hand, Division 2 governs the *design by analysis* and incorporates a lower safety factor of 3. Thus, the maximum allowable stress value for a 60,000-psi tensile strength material will become 20,000 psi.

Many companies require that all their pressure vessels be constructed in accordance with Division 2 because of the more exacting standards. Others find that they can purchase less expensive vessels by allowing manufacturers the choice of either Division 1 or Division 2. Normally, manufacturers will choose Division 1 for low-pressure vessels and Division 2 for high-pressure vessels.

The maximum allowable stress values at normal temperature range for the steel plates most commonly used in the fabrication of pressure vessels are given in Table 6-3. For stress values at higher temperatures and for other materials, the latest edition of the ASME code should be referenced.

Determining Wall Thickness

The following formulas are used in the ASME code Section VIII, Division 1 for determining wall thickness:

Wall Thickness—Cylindrical Shells

$$t = \frac{\mathrm{Pr}}{SE - 0.6P}, \tag{6-1}$$

Wall Thickness—2:1 Ellipsoidal Heads

$$t = \frac{Pd}{2SE - 0.2P}, \tag{6-2}$$

Wall Thickness—Hemispherical Heads

$$t = \frac{\mathrm{Pr}}{2SE - 0.2P}, \tag{6-3}$$

Wall Thickness—Cones

$$t = \frac{Pd}{2\cos\alpha(SE - 0.6P)}, \tag{6-4}$$

Table 6-3
Maximum Allowable Stress Value for Common Steels (2007 Edition)

			ASME Section VIII 2007 Edition	
			Div. 1	Div. 2
Metal		**Not Lower Than**	**–20°F**	**–20°F**
Temperature		**Not Exceeding**	**650°F**	**100°F**
Carbon steel plates and sheets	SA-516	Grade 55	15,700	18,300
		Grade 60	17,100	20,000
		Grade 65	18,600	21,700
		Grade 70	20,000	23,300
	SA-285	Grade A	12,900	15,000
		Grade B	14,300	16,700
		Grade C	15,700	18,300
	SA-36		16,600	16,900
Low-alloy steel plates	SA-387	Grade 2, cl.1	15,700	18,300
		Grade 12, cl.1	15,700	18,300
		Grade 11, cl.1	17,100	20,000
		Grade 22, cl.1	17,100	20,000
		Grade 21, cl.1	17,100	20,000
		Grade 5, cl.1	17,100	20,000
		Grade 2, cl.2	20,000	23,300
		Grade 12, cl.2	18,600	21,700
		Grade 11, cl.2	21,400	25,000
		Grade 22, cl.2	21,400	25,000
		Grade 21, cl.2	21,400	25,000
		Grade 5, cl.2	21,400	25,000
	SA-203	Grade A	18,600	21,700
		Grade B	20,000	23,300
		Grade D	18,600	21,700
		Grade E	20,000	23,300
High-alloy steel plates	SA-240	Grade 304	20,000	20,000**
		Grade 304L	16,700	16,700
		Grade 316	20,000	20,000
		Grade 316L	16,700	16,700

Austenitic Stainless set **at 2/3 Yield /** Allowable Stress NOT **3.0 or 3.5 S.F** due to low Yield Strength values relative to ultimate Tensile Strength, 304 UTS 75,000 Yield 30,000
Example: Hydrostatic Testing $1.3 \times 20{,}000 = 26{,}000$ **(Yield is 30,000)** for 304

where

S = maximum allowable stress value, psi (kPa),
t = thickness, excluding corrosion allowance, in. (mm),
P = maximum allowable working pressure, psig (kPa),
r = inside radius before corrosion allowance is added, in. (mm),

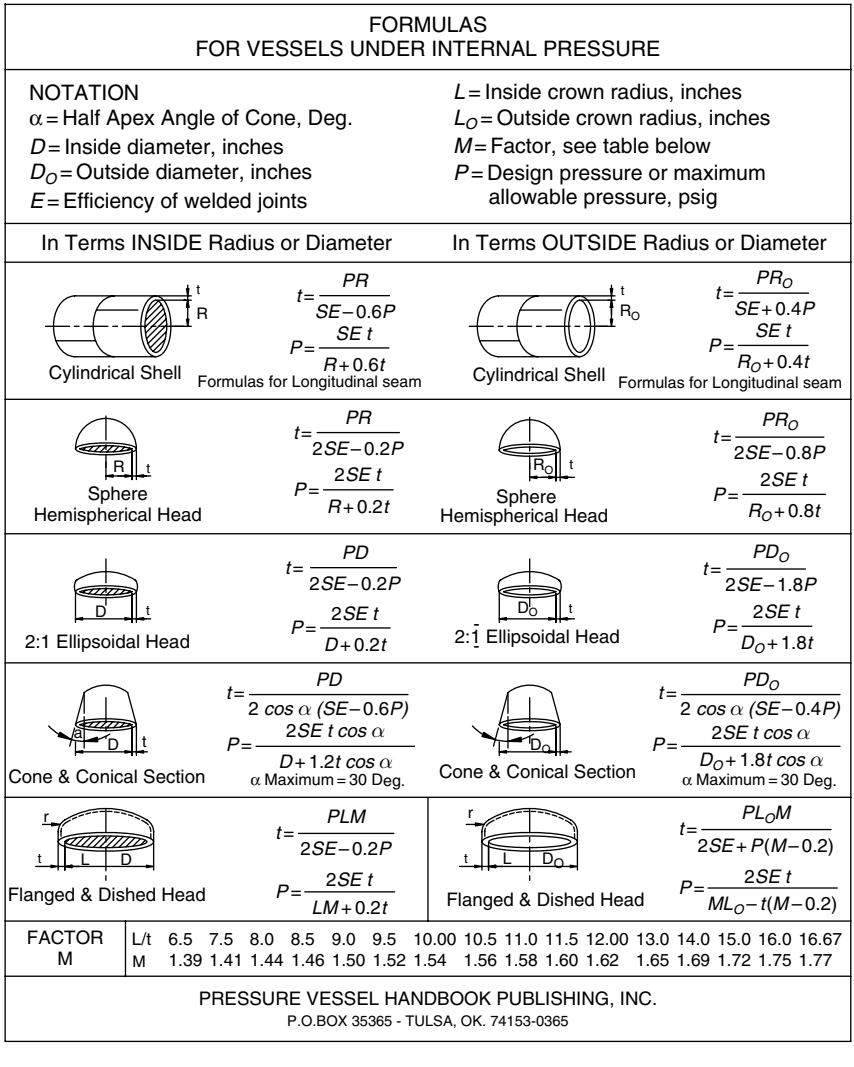


Figure 6-1. Formulas for vessels under internal pressure (ASME Section VIII, Division 1). (Reprinted with permission from *Pressure Vessel Handbook*, Gulf Publishing, Inc., Tulsa, Oklahoma.)

FORMULAS
FOR VESSELS UNDER INTERNAL PRESSURE

1. THE FORMULAS CONFORM TO THE ASME CODE FOR PRESSURE VESSELS, SECTION VIII, DIVISION I.
2. CORROSION ALLOWANCE. If the vessel is subject to thinning by corrosion allowance shall be added to the thickness calculated by the formulas. Code UG-25.
   When allowance for corrosion is provided computing the wall thickness or allowable pressure in terms of the inside dimensions, the allowance shall be added to the length of inside radius or diameter.
3. MAXIMUM ALLOWABLE WORKING PRESSURE for a vessel is the pressure allowable for the weakest element of the vessel. It should be limited not by minor parts but by the shell, heads of flanges.
4. HYDROSTATIC TEST PRESSURE o one and one-half times the maximum allowable working pressure or the design pressure when calculations are not made to determine the maximum allowable working pressure. Code UG-99.
5. DEGREE OF RADIOGRAPHIC EXAMINATION. Condition when full radiography is mandatory are described in the Code UW-11.
6. STRESS VALUES OF MATERIALS, S 1000 psi. Excerpt from Code, table UCS-73

Specification		For Metal Temperature Not Exceeding Deg. F.							
Number	Grade	-20 to 650	700	750	800	850	900	950	1010
SA-283	C	12.8	-	-	-	-		-	-
SA-285	C	13.7	13.2	12.0	10.2	8.3	6.5	-	-
SA-515	55	13.7	13.2	12.0	10.2	8.3	6.5	4.5	2.5
SA-515	60	15.0	14.3	12.9	10.8	8.6	6.5	4.5	2.5
SA-515	65	16.2	15.5	13.8	11.4	8.9	6.5	4.5	2.5
SA-515	70	17.5	16.8	14.7	12.0	9.2	6.5	4.2	2.5
SA-516	55	13.7	13.2	12.0	10.2	8.3	6.5	4.5	2.5
SA-516	60	15.0	14.3	12.9	10.8	8.6	6.5	4.5	2.5
SA-516	65	16.2	15.5	13.8	11.4	8.9	6.5	4.5	2.5
SA-516	70	17.5	16.8	14.7	12.0	9.2	6.5	4.5	2.5

7. CYLINDRICAL SHELL- In the formulas of the form the stress in the longitudinal joint are considered, since usually this governs.

   Stress in the girth seam will govern only when the circumferential joint efficiency is less than half the longitudinal joint efficiency, or when besides the internal pressure additional loadings (wind load, reaction of saddles etc.) arc outing longitudinal bending or tension.

   The formula considering the stress in the girth seam

$$t = \frac{PR}{2SE + 0.4P} \qquad P = \frac{2SEt}{R - 04t}$$

   When the thickness of the shell exceed one half of the inside radius, or P exceeds 0.385 SE. The formulas given in the Code UA-2(c) shall be applied.

8. SPHERE AND HEMISPHERICAL HEAD. When the wall thickness exceeds 0.356 R, or P exceeds 0.665 SE, the formula given in the Code UA-3 shall be applied.
9. ELLIPSOIDAL HEAD. When the ratio of the major and minor axis is other than 2:1. Formulas and factors given in the Code UA-4(c) shall be applied.
10. CONE AND CONICAL SECTION. The formulas given in the form are applicable only when the half apex angle of cone does not exceed 30 degree.
11. ASME FLANGED AND DISHED HEAD. The formulas are applicable when the inside crown radius is not greater than the outside diameter of the head and the inside knuckle radius is not less than 6% of the outside diameter or 3 times the thickness of the head.
12. EFFICIENCY OF WELDED JOINTS (E)
    Excerpt from ASME Code Table UW-12.

No.	TYPE OF JOINT	DEGREE OF EXAMINATION		
		FULL	SPOT	NO
1	Double-welded bolt joint or single welded bolt joint with backing strip which does not remain in place.	1.00	0.85	0.70
2	Single-welded bolt joint with backing strip which remains in place.	0.90	0.80	0.65
3	Single-welded bolt joint without use of backing strip.	—	—	0.60

13. EFFICIENCY (E) TO BE USED IN CALCULATIONS OF SEAMLESS HEADS ASME Code UW-12 (b)

TYPE OF HEAD	TYPE OF JOINT	DEGREE OF EXAMINATION OF HEAD TO SHELL JOINT		
		FULL	SPOT	NO
Hemi Spherical	No 1	1.00	0.85	0.80
	No 2	0.90	0.80	
Others	ANY	1.00	0.85	

14. NOTATION

a	=	Half-apex angle of cone, deg.
D	=	Inside diameter of cone at the large end, inches Inside diameter of ellipsoidal head, inches
D_o	=	Outside diameter of cone at the large end, inches Outside diameter of ellipsoidal head, inches
E	=	Lowest efficiency of any joint in the shell or head; for hemispherical head this includes head to shell joint
L	=	Inside crown radius of flanged and dished head, inches
L_o	=	Outside crown radius of flanged and dished head, inches
P	=	Internal design pressure or maximum allowable working pressure pounds per square inches
r	=	Knuckle radius
R	=	Inside radius of shell or hemispherical head, inches
R_o	=	Outside radius of shell or hemispherical head, inches
S	=	Allowable stress value of material pounds per square, inches
t	=	Thickness of shell or head, inches

Figure 6-1. (Continued)

d = inside diameter before corrosion allowance is added, in. (mm),
E = joint efficiency, see Table 6-4 (most vessels are fabricated in accordance with type of joint no. 1),
α = half the angle of the apex of the cone.

Figure 6-1 summarizes the formulas for pressure vessels under internal pressure (ASME Section VIII, Division 1). Figure 6-2 defines the various types of heads. Most production facility vessels use 2:1 ellipsoidal heads because they are readily available, are normally less expensive, and take up less room than hemispherical heads.

Cone-bottom vertical vessels are sometimes used where solids are anticipated to be a problem. Most cones have either a 90° apex ($\alpha = 45°$) or a 60° apex ($\alpha = 30°$). These are referred to respectively as a "45°" or "60°" cone because of the angle each makes with the horizontal. Equation (6-4) is for the thickness of a conical head that contains pressure.

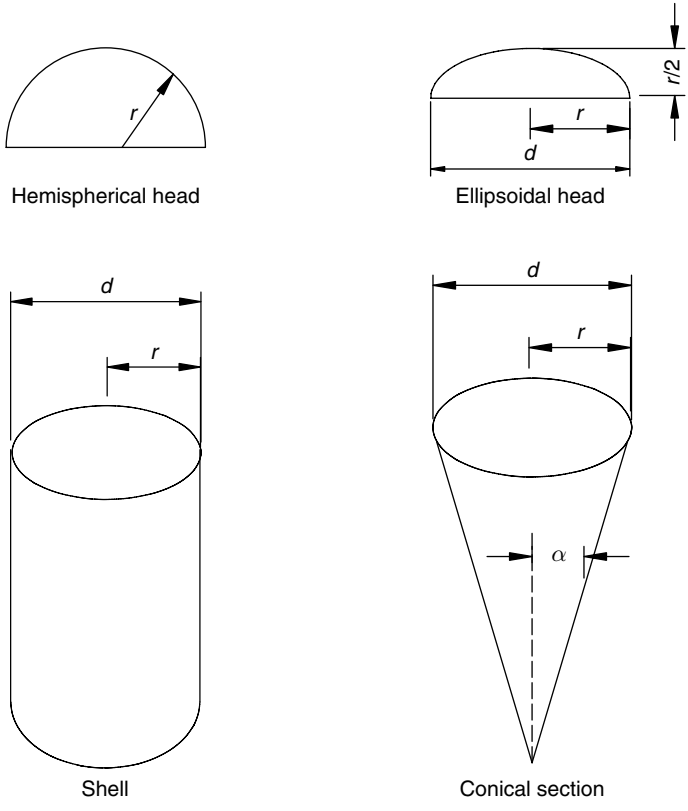


Figure 6-2. Pressure vessel shapes.

Some operators use internal cones within vertical vessels with standard ellipsoidal heads as shown in Figure 6-3. The ellipsoidal heads contain the pressure, and thus the internal cone can be made of very thin steel.

Table 6-4 lists joint efficiencies that should be used in Eqs. (6-1) to (6-4). This is Table UW-12 in the ASME code.

Table 6-5 lists some of the common material types used to construct pressure vessels. Individual operating companies have their own standards, which differ from those listed in this table.

Corrosion Allowance

Typically, a corrosion allowance of 0.125 in. for non-corrosive service and 0.250 in. for corrosive service is added to the wall thickness calculated in Eqs. (6-1) to (6-4).

Table 6-4
Maximum Allowable Joint Efficiencies for Arc and Gas Welded Joints

No.	Type of Joint Description	Limitation	(a) Fully Radiographed[1]	(b) Spot Examined	(c) Not Spot Examined[3]
1	Butt joints as attained by double welding or by other means that will obtain the same quality of deposited weld metal on the inside and outside weld surfaces of UW-35. Welds using metal backing strips that remain in the place are excluded.	None	1.00	0.85	0.70
2	Singled-welded butt joint with backing strip other than those included under (1).	(a) None except as in (b) below (b) Butt weld with one plate offset—for circumferential joints only, see UW-13(c) and Fig. UW-13.1(k)	0.90	0.80	0.65
3	Single-welded butt joint without using backing strip	Circumferential joints only, not over 5/8-in. thick and not over 24-in. outside diameter	—	—	0.60
4	Double full filet lap joint	Longitudinal joints only, not over 3/8-in. thick	—	—	0.55

5	Single full fillet lap joints with plug welds conforming to UW-17.	(a) Circumferential joints[4] for attachment of heads not over 24-in. outside diameter to shells not over 1/2 in. thick (b) Circumferential joints for the attachment to shells of jackets not over 5/8 in. in nominal thickness where the distance from the center of the plug weld to the edge of the plate is not less than 1½ times the diameter of the hole for the plug.	—	—	0.50
6	Single full fillet lap joints without plug welds.	(a) For the attachment of heads convex to pressure to shells not over 5/8-in. required thickness, only with use of fillet weld on inside of shell; or (b) For attachment of heads having pressure on either side to shells not over ¼-in. required thickness with fillet weld on outside of head flange only.	—	—	0.45

[1]See UW-12(a) and UW-51.
[2]See UW-12(b) and UW-52.
[3]The maximum allowable joint efficiencies shown in this column are the weld joint efficiencies multiplied by 0.80 (and rounded off to the nearest 0.05) to effect the basic reduction in allowable stress required by the division for welded vessels that are not spot examined. See UW-12(c).
[4]Joints attaching hemispherical heads to shells are executed.

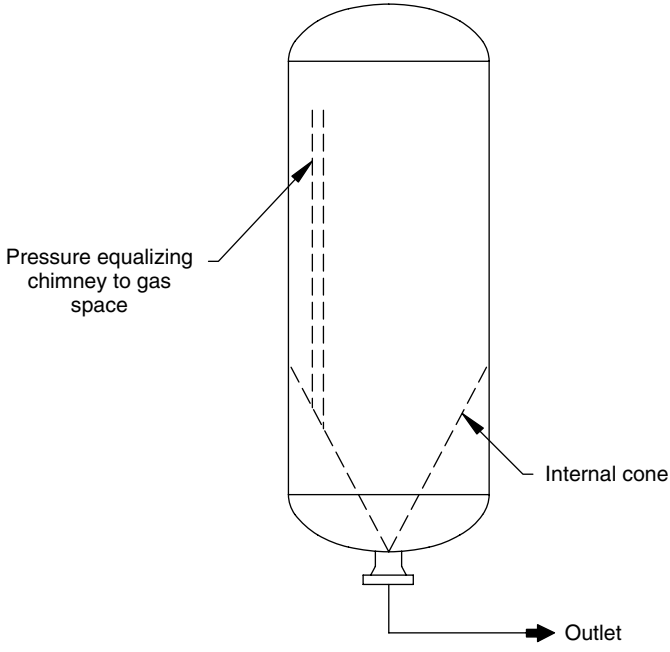


Figure 6-3. Internal cone vessel.

Inspection Procedures

All ASME code vessels are inspected by an approved code inspector. The manufacturer will supply code papers signed by the inspector. The nameplate on the vessel will be stamped to signify it has met the requirements of the code. One of these requirements is that the vessel was pressure tested (1998 edition, 1.5 times the MAWP; 2001 and later editions, 1.3 times the MAWP). However, this is only one of the requirements. The mere fact that a vessel is pressure tested 1.3 or 1.5 times the MAWP does not signify that it has met all the design and quality assurance safety aspects of the code.

It must be pointed out that a code stamp does not necessarily mean that the vessel is fabricated in accordance with critical nozzle dimensions or internal devices as required by the process. The code inspector is only interested in those aspects that relate to the pressure handling integrity of the vessel. The owner must do his own inspection to assure that nozzle locations are within tolerance, vessel internals are installed as designed, coatings are applied properly, etc.

Table 6-5
Materials Typically Specified

	Low Pressure	Common Steel T > −20°F	NACE MR-01-75	Low Temp −50°F < T < 0°	Low Temp FT < −50°F	High CO_2 Service
Plate	SA-36 SA-285-C	SA-516-70	SA-516-70	SA-516-70	SA-240-304	SA-240-16L
Pipe	SA-53-B	SA-106-B	SA-106B	SA-106-B	SA-333-6 TP-304	SA-312 TP-316L
Flanges and fittings	SA-105	SA-105 SA-181-1	SA-105 SA-181-1	SA-350-LF1	SA-182 F-304	SA-182 F-316L
Stud B8M bolts	SA-193-B7	SA-193-B7	SA-193-B7M	SA-320-L7	SA-193-B-8	SA-193-8M
Nuts 8MA	SA-192-2H	SA-194-2H	SA-194-2M	SA-194-4	SA-194-8A	SA-194-MA

Estimating Vessel Weights

It is important to be able to estimate vessel weights, since most cost estimating procedures start with the weight of the vessel. The vessel weight, both empty and full with water, may be necessary to adequately design a foundation or to assure that the vessel can be lifted or erected once it gets to the construction site.

The weight of a vessel is made up of the weight of the shell, the weight of the heads, and the weight of internals, nozzles, pedestals, and skirts. The last two terms are defined in Figure 6-4.

The shell weight can be estimated from

Field Units

$$W = 11dtL, \tag{6-5a}$$

SI Units

$$W = 0.0254dtL, \tag{6-5b}$$

where

W = weight, lb (kg),
d = internal diameter, in. (mm),
t = wall thickness, in. (mm),
L = shell length, ft (m).

The weight of one 2:1 ellipsoidal head is approximately

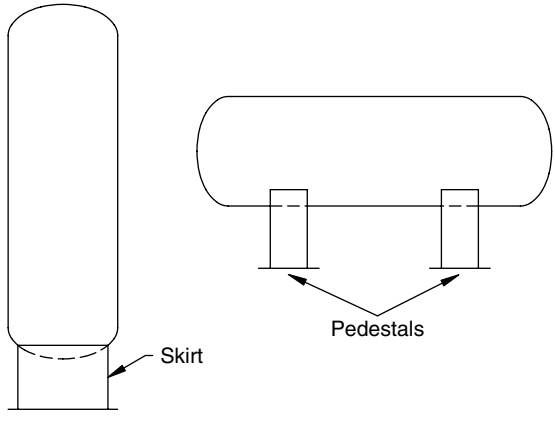


Figure 6-4. Vessel support devices.

Field Units

$$W \approx 0.34td^2 + 1.9td. \tag{6-6a}$$

The weight of a cone is

$$W = \frac{0.23td^2}{\sin\alpha}. \tag{6-7a}$$

SI Units

$$W \approx 9.42 \times 10^{-6} td^2 + 1.34 \times 10^{-3} td \tag{6-6b}$$

The weight of a cone is

$$W \approx 6.37 \times 10^{-6} \frac{td^2}{\sin\alpha}, \tag{6-7b}$$

where α = one-half the cone apex angle.

The weight of nozzles and internals can be estimated at 5 to 10% of the sum of the shell and head weights. As a first approximation, the weight of a skirt can be estimated as the same thickness as the shell (neglecting the corrosion allowance) with a length given by Eq. (6.8) for an ellipsoidal head and Eq. (6.9) for a conical head. For very tall vessels the skirt will have to be checked to assure it is sufficient to support both the weight of the vessel and its appentorances and the overturning moment generated by wind forces.

Field Units

$$L = \frac{0.25d}{12} + 2, \tag{6-8a}$$

$$L = \frac{0.5d}{12\tan\alpha} + 2. \tag{6-9a}$$

SI Units

$$L = 2.5 \times 10^{-4} d + 0.61, \tag{6-8b}$$

$$L = 2.54 \times 10^{-4} \frac{d}{\tan\alpha} + 0.61, \tag{6-9b}$$

where L = skirt length in ft (m).

The weight of pedestals for a horizontal vessel can be estimated as 10% of the total weight of the vessel.

Specification and Design of Pressure Vessels

Pressure Vessel Specifications

Some companies summarize their pressure vessel requirements on a pressure vessel design information sheet such as the one shown in Figure 6-5. Some companies have a detailed general specification for the construction of pressure vessels, which defines the overall quality of fabrication required and addresses specific items such as

- Code compliance
- Design conditions and materials
- Design details
  - Vessel design and tolerances
  - Vessel connections (nozzle schedules)
  - Vessel internals
  - Ladders, cages, platforms, and stairs
  - Vessel supports and lifting lugs
  - Insulation supports
  - Shop drawings
- Fabrication
  - General
  - Welding
  - Painting
  - Inspection and testing
  - Identification stamping
  - Drawings, final reports, and data sheets
  - Preparation for shipment

A copy of this specification is normally attached to a bid request form, which includes a pressure vessel specification sheet such as the one shown in Figure 6-6. This sheet contains schematic vessel drawings and pertinent specifications and thus defines the vessel in enough detail so the manufacturer can quote a price and so the operator can be sure that all quotes represent comparable quality. The vessel connections (nozzle schedules) are developed from mechanical flow diagrams. It is not necessary for the bidder to know the location of the nozzles to submit a quote or even to order material.

Shop Drawings

Before the vessel fabrication can proceed, the fabricator will develop complete drawings and have these drawings approved by the representative of

SEPARATOR DESIGN INFORMATION

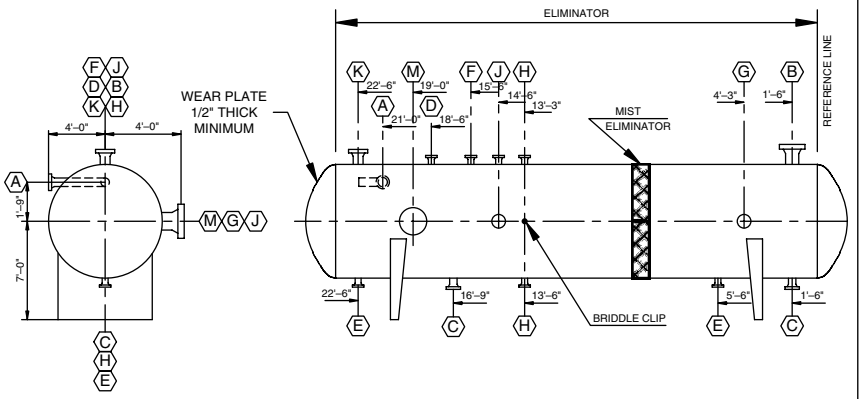
1. Operating Conditions:
   - A. Liquid Volumes
     1. Oil/Condensate: _____ Barrel/Day Gravity: _____ °API
     2. Water: _____ Barrels/Day Sp. Gr.:_____ (Water = 1.0)
   - B. Oil/Condensate Characteristics
     1. Roaming: Nil _____ Moderate _____ Severe _____
     2. Paraffin Problem: No _____Yes_____
     3. Slug Flow: No _____Yes: ____ (If Yes, give details such as maximum liquid rate, slug volume, etc.)_____
   - C. Gas: _____MMscfd Sp. Gr.: _____ (Air - 1.0)
   - D. Operating Temperature (°F):_____Max_____ Min
   - E. Operating Pressure ipsig):_____ Max_____ Min
   - F. H.S Content: _____Mole% CO_2. Content: _____. Mole%
   - G. Geographical Location: _____

II Design Requirements:
   - A. Type:————————Vertical_____ Horizontal_____ Spherical.
     Manufacturer's Recommendation: _____
     _____Two-Phase _____Three-Phase
   - B. Design Pressure: _____ psig at Temperature _____°F
   - C. Type Mist Extractor: (Specify) _____
   - D. Corrosion Allowance: _____(inches)
   - E. Corrosion Allowance for Non-Pressure Internal Parts: _____(inches)
   - F. NACE MR-01-75 Required: No_____ Yes_____
   - G. Special Stress Relieving: No_____ Yes_____
     Specify if Yes: _____
   - H. API RP 14C Safety Systems Required: No_____ Yes_____

III. Coatings:
   - A. External: Mfgr. Std._____ Other_____
     Specify if Other: _____
   - B. Internal: (Specify) _____
   - C. Cathodic Protection: (Specify)_____

IV. Special Instructions:
   - A. Radiographic Inspection: ASME Code _____ Other _____
     Specify if Other: _____
   - B. Hydrostatic Test Pressure: ASME Code _____Other _____
     Specify if Other: _____
   - C. Hardness Testing Requirements: (Specify) _____
   - D. Lifting Lugs: (Specify) _____
   - E. Skid Mounting: (Specify) _____
   - F. Welding Requirement: ASME Code _____ Other _____
     Specify if Other: _____
   - G. Sand Removal System: (Specify) _____
   - H. Other: _____

Figure 6-5. Example of separator design information sheet.



END VIEW

ELEVATION

ITEM. *GAS SCRUBBER* ITEM NO. *MBD-1020*
NO. REQ'D. JOB NO.
PURCHASE ORDER NO. DATE.

DESIGN AND FABRICATION DATA

CONSTRUCTION TO BE IN ACCORDANCE WITH THE LATEST EDITION OF THE
ASME CODE & ADDENDA. *SECTION VIII, DIVISION 2*
CODE SYMBOL. REQUIRED/NOT REQUIRED
DESIGN PRESSURE. 1800 PSIG. AT. –20/100 °F
OPERATING PRESSURE. 1000–1250 PSIG. AT. 60 °F
STRESS RELIEVE. YES/NO/PER CODE RADIOGRAPH. NP/SPOT/100%
JOINT E F F - SHELL. 1.0 HEADS. 1.0
CORROSION ALLOWANCE ALLOWANCE - SHELL 0.125" HEADS. 0.125"
MATERIAL: SHELL. HEADS. *SA - 516 - 70 2:1 ELLPT*
FLANGES. *SA - 516 - 70N* PIPE. *SA - 106 - B*
STUDS. *NOTE 3* NUTS. *SA - 194 - 2H*
GASKETS. *SA - 193 - B7*
SADDLES. *SOFT IRON TYPE R.I.D. MARK "D" CADMIUM PLATED*
HINGES. *YES* LUGS. *YES (2)*
INSULATION THICKNESS. *YES* DAVITS REQUIRED FOR MANHOLES. *NO*
LADDER CLIPS. *NONE* INSULATION RINGS. *NONE*
POINT PER SPEC. *NO* PLATFORM CLIPS. *REQUIRED*

DEMISTER. *NOTE 2*

NOTES:
1. DESIGN, FABRICATIONS, TESTING AND DOCUMENTATION SHALL BE IN ACCORDANCE WITH PARAGON SPECIFICATION
2. THE VANE TYPE MIST ELIMINATOR SHALL BE MANUFACTURED BY ACS INDUSTRIES, INC. (OR APPROVED EQUAL) AND SHALL REMOVE 99% OF ALL DROPLETS 10 MICRONS AND LARGER
3. WELD NECK FLANGES SHALL BE ASTM SA105 INTEGRALLY REINFORCED LONG WELD NECKS ARE ACCEPTABLE

NOZZLE SCHEDULE

MK	NO	SIZE	RATING	TYPE	SERVICE	PROJ.
A	1	12"	900#	RTJ	GAS/CONDENSATE INLET	–
B	1	12"	900#	RTJ	GAS OUTLET	12"
C	2	6"	900#	RTJ	CONDENSATE OUTLET	10"
D	1	2"	900#	RTJ	RELIEF/BLOWDOWN	8'
E	2	3"	900#	RTJ	DRAIN	8'
F	1	2"	900#	RTJ	PRESSURE CONNECTION	8'
G	1	2"	900#	RTJ	TEMPERATURE CONNECTION	8'
H	2	3"	900#	RTJ	LEVEL BRIDDLE	8'
J	2	2"	900#	RTJ	LEVEL BRIDDLE	8'
K	1	8"	900#	RTJ	INSPECTION W/BLIND	10"
M	1	18"	900#	RTJ	MANWAY 18" I.D.	–

PROCESS CONDITIONS

GAS FLOW RATE 200 MMSCFD
GAS SPECIFIC GRAVITY: 0.67 (AIR = 1.0)
HYDROCARBON LIQUID FLOW RATE: 2.5 BBL/MMSCF NORMAL
HYDROCARBON LIQUID SPECIFIC GRAVITY: 0.56 @ OPERATING CONDITIONS (WATER = 1.0)
OPERATING PRESSURE: 1000 PSIG MINIMUM, 1250 MAXIMUM
OPERATING TEMPERATURE: 55°F MINIMUM, 70°F MAXIMUM

NOTES
1. INTERNAL INLET PIPING SHALL BE DESIGNED TO WITHSTAND LIQUID SLUGS ARRIVING AT VELOCITIES AS HIGH AS 45 FT/SEC.
2. VESSEL ORNADINARILY OPERATES EMPTY, BUT LIQUID LEVEL DURING SLUGGING CAN BE AS HIGH AS 42° ABOVE OUTSIDE BOTTOM OF VESSEL

NO.	REVISION	DATE	DRAWN	CHECK	APP'D

ISSUED FOR: CLIENT APPROVAL; BODING; CONSTRUCTION
ENGINEER. DATE.
DRAWN. DATE.
CHECKED. DATE.
APPROVED. DATE.
SCALE. SHEET OF
JOB NO.
CLIENT.
CLIENT JOB NO.

PARAGON
ENGINEERING SERVICES
HOUSTON, TEXAS

PARAGON ENGINEERING SERVICES
MBD - 1020
LP PRODUCTION SEPARATOR
DRAWING NO. MBD - 1020 REV.

Figure 6-6. Example of pressure vessel specification sheet.

the engineering firm and/or the operating company. These drawings are called shop drawings. They will show detailed vessel design and fabrication/welding, nozzle schedules and locations, details of vessel internals, and other accessories. Examples are shown in Figures 6-7 through 6-15. Some typical details are discussed next.

Nozzles

Nozzles should be sized according to pipe sizing criteria, such as those provided in API RP 14E. The outlet nozzle is generally the same size as the inlet nozzle. To prevent baffle destruction due to impingement, the entering fluid velocity is to be limited as

Field Units

$$V_{\text{in}} \leq (3{,}500/\rho_f)^{1/2}, \tag{6-10a}$$

SI Units

$$V_{\text{in}} \leq (5{,}217.7/\rho_f)^{1/2}, \tag{6-10b}$$

where

V_{in} = maximum inlet nozzle fluid velocity, ft/s (m/s),

ρ_f = density of the entering fluid, lb/ft^3 (kg/m^3).

If an interior centrifugal (cyclone) separator is used, the inlet nozzle size should be the same size as the pipe. If the internal design requires the smallest inlet and exit pressure losses possible, the nozzle size should be increased.

Vortex Breaker

As liquid flows out of the exit nozzle, it will swirl and create a vortex. Vortexing would carry the gas out with the liquid. Therefore, all liquid outlet nozzles should be equipped with a vortex breaker. Figure 6-11 shows several vortex breaker designs. Additional designs can be found in the *Pressure Vessel Handbook*. Most designs depend on baffles around or above the outlet to prevent swirling.

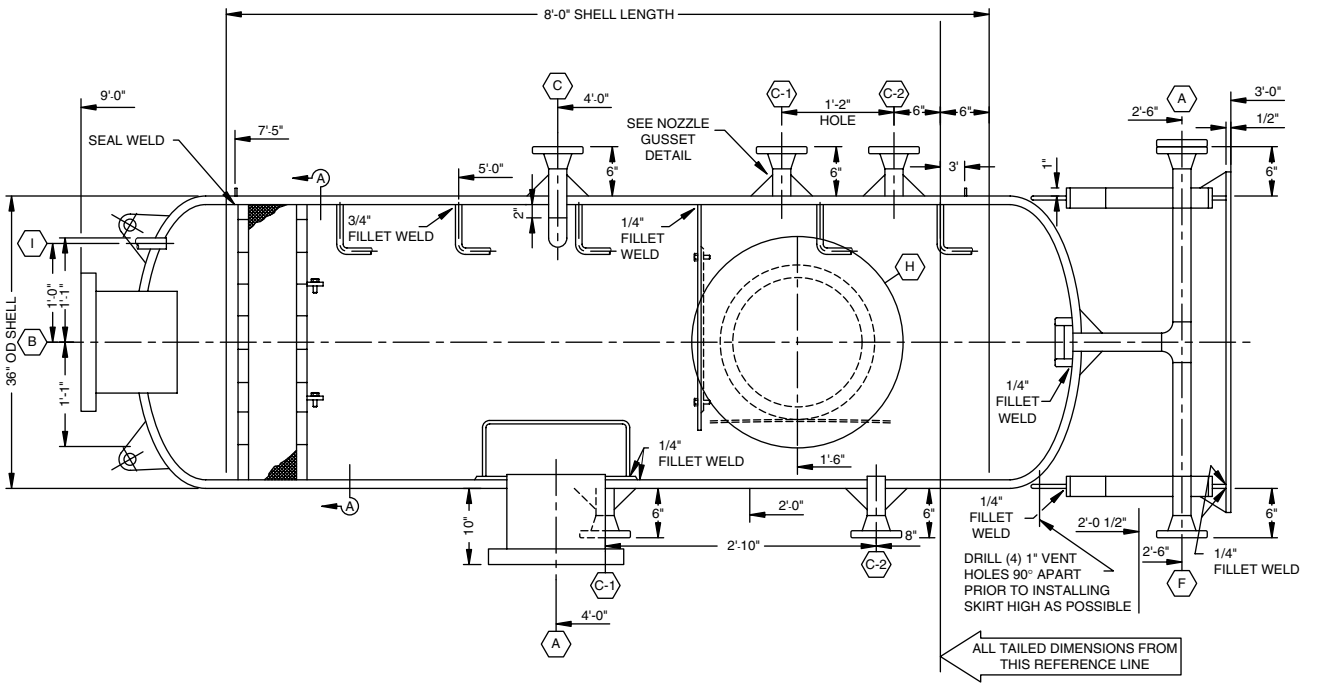
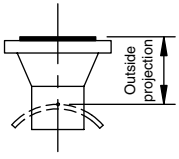


Figure 6-7. Example of pressure vessel shop drawing.

Outside projection, inches using welding neck flange



Nom. pipe size	Pressure rating of flange LB					
	150	300	600	900	1500	2500
2	6	6	6	8	8	8
3	6	6	8	8	8	10
4	6	8	8	8	8	12
6	8	8	8	10	10	14
8	8	8	10	10	12	16
10	8	8	10	12	14	20
12	8	8	10	12	16	22
14	8	10	10	14	16	
16	8	10	10	14	16	
18	10	10	12	14	18	
20	10	10	12	14	18	
24	10	10	12	14	20	

Inside extension

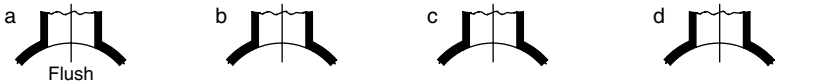


Figure 6-8. Nozzle projections. (Reprinted with permission from *Pressure Vessel Handbook*, Gulf Publishing, Inc., Tulsa, Oklahoma.)

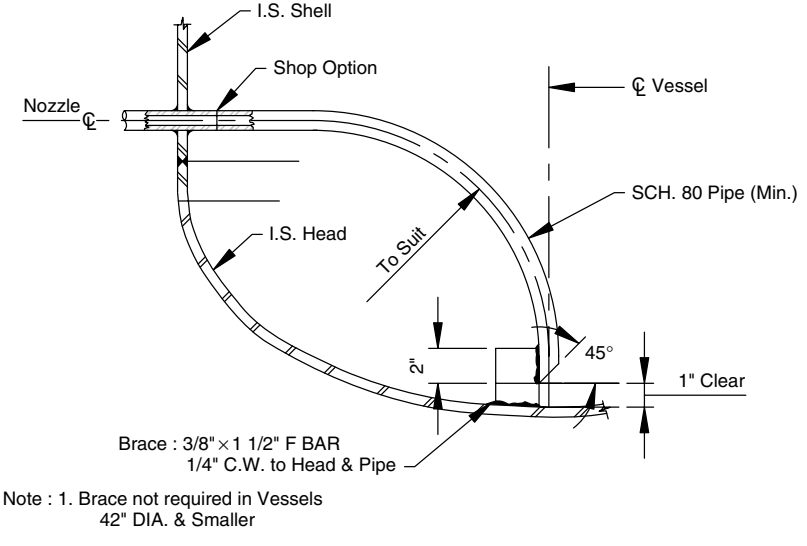


Figure 6-9. Siphon drain.

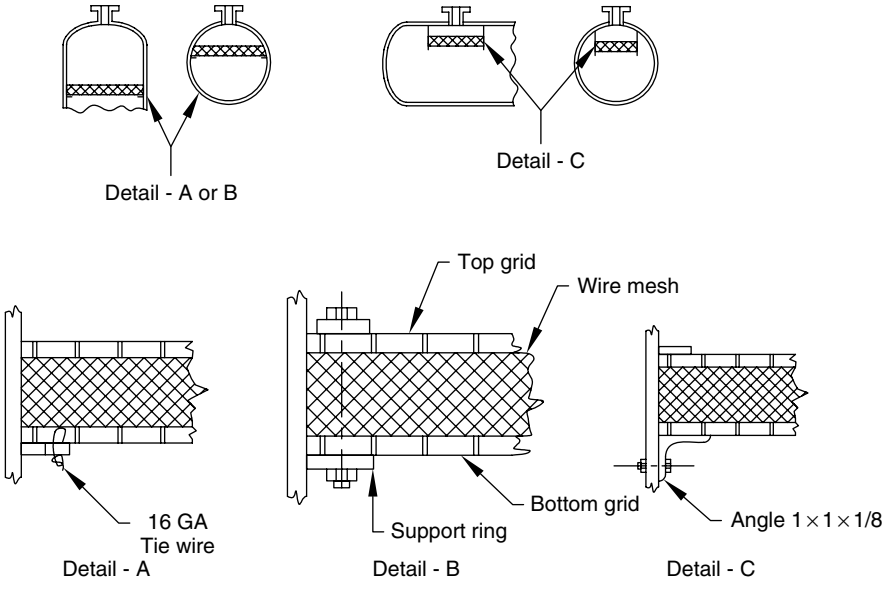


Figure 6-10. Example of supports for mist extractors. (Reprinted with permission from *Pressure Vessel Handbook*, Gulf Publishing, Inc., Tulsa, Oklahoma.)

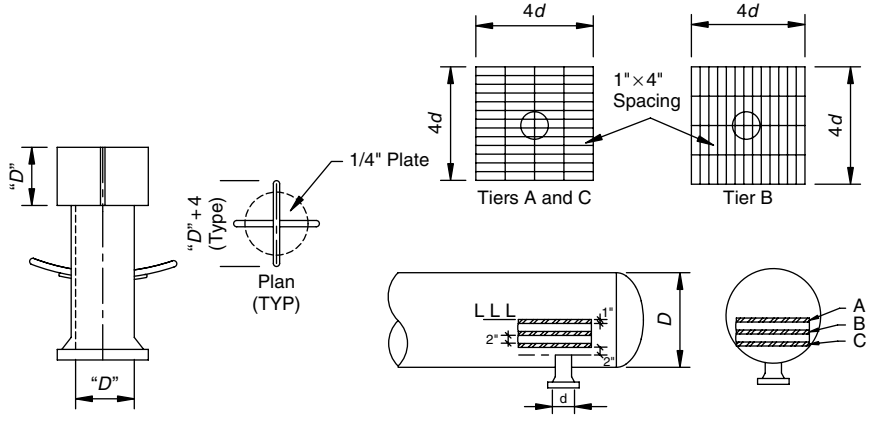
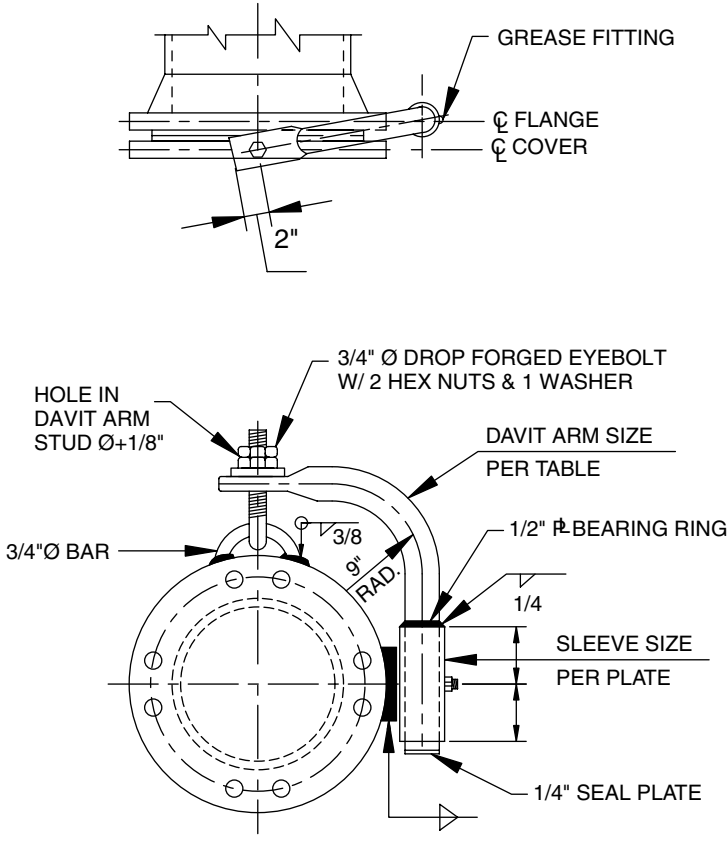


Figure 6-11. Examples of vortex breaker details.



	1 1/2 S/80	2 S/80	2 1/2 S/80	3 S/80
DAVIT SIZE	2 S/80	2 1/2 S/40	3 S/40	3 1/2 S/40
SLEEVE SIZE	16 150#	24 150#	24 300#	20 600#
MANWAY COVER SIZE & RATING	18 150#	18 300#	18 600#	24 600#
	20 150#	20 300#	20 600#	16 900#
	16 300#			18 900#
				20 900#
	ON ANY COVER NOT EXCEEDING			
	325# IN WEIGHT	525# IN WEIGHT	850# IN WEIGHT	1200# IN WEIGHT

Figure 6-12. Examples of horizontal manway cover davit and sleeve detail.

BASE PLATE SCHEDULE		
ANGLE LEG SIZE	"A"	"B"
6"×6"	8"	3 3/8"
5"×5"	7"	2 7/8"
4"×4"	6"	2"
3"×3"	5"	1 3/4"
2 1/2"×2 1/2"	4"	1 1/2"

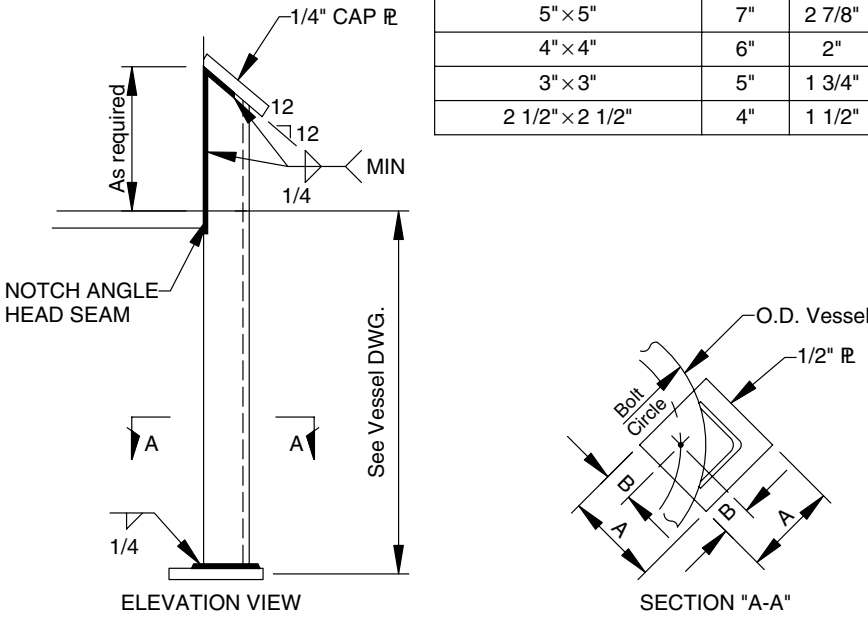


Figure 6-13. Angle support legs.

Manways

Manways are large openings that allow personnel access to the vessel internals for their maintenance and/or replacement. Vessels 36 in. and larger should have a minimum of one 18-in. manway. Vessels 30 in. and smaller should have two 4-in. flanged inspection openings. Manway cover davits should be provided for 12-in. and larger manways for safe and easy opening and closing of the cover. Figure 6-12 shows an example of a manway cover davit and sleeve details.

Vessel Supports

Small vertical vessels may be supported by angle support legs, as shown in Figure 6-13. Larger vertical vessels are generally supported by a

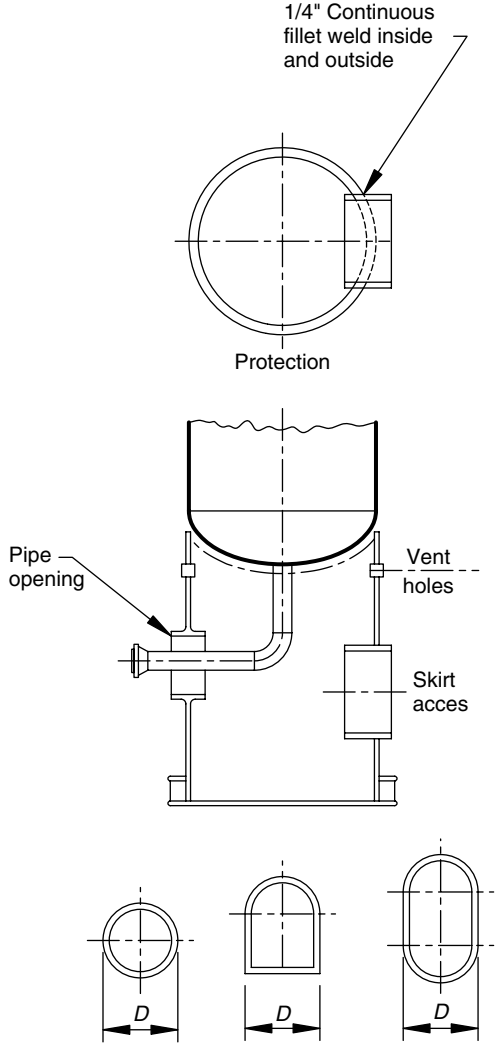


Figure 6-14. Skirt openings. (Reprinted with permission from *Pressure Vessel Handbook*, Gulf Publishing, Inc., Tulsa, Oklahoma.)

skirt support, as shown in Figure 6-14. At least two vent holes, 180° apart, should be provided at the uppermost location in the skirt to prevent the accumulation of gas, which may create explosive conditions. Horizontal vessels are generally supported by a pair of saddle-type supports.

Ladder and Platform

A ladder and platform should be provided if operators are required to climb up to the top of the vessel regularly. An example is shown in Figure 6-15.

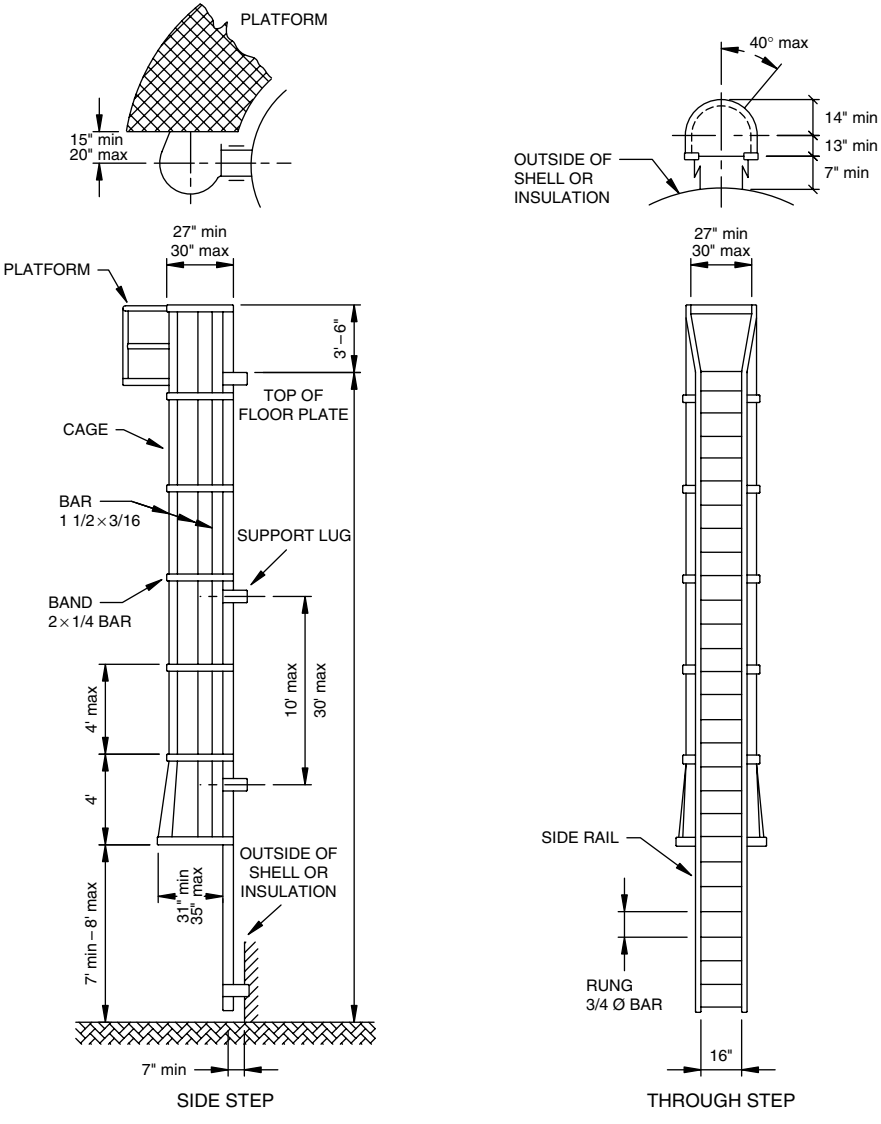


Figure 6-15. Ladders. (Reprinted with permission from *Pressure Vessel Handbook*, Gulf Publishing, Inc., Tulsa, Oklahoma.)

Pressure Relief Devices

All pressure vessels should be equipped with one or more pressure safety valves (PSVs) to prevent overpressure. This is a requirement of both the ASME code and API RP 14C. The PSV should be located upstream of the mist extractor. If the PSV is located downstream of the mist extractor, an overpressure situation could occur when the mist extractor becomes plugged, isolating the PSV from the high pressure, or the mist extractor could be damaged when the relief valve opens. Rupture discs are sometimes used as a backup relief device for the PSV. The disc is designed to break when the internal pressure exceeds the set point. Unlike the PSV, which is self-closing, the rupture disc must be replaced if it has been activated.

Corrosion Protection

Pressure vessels handling salt water and fluids containing significant amounts of H_2S and CO_2 require corrosion protection. Common corrosion protection methods include internal coatings with synthetic polymeric materials and galvanic (sacrificial) anodes. All pressure vessels that handle corrosive fluids should be monitored periodically. Ultrasonic surveys can locate discontinuities in the metal structure, which will indicate corrosion damages.

Example 6-1: Determining the weight of an FWKO vessel (field units)

Determine the weight for the following free-water knockout vessel. It is butt weld fabricated with spot x-ray and to be built to the ASME code Section VIII, Division 1, 1998 edition. A conical head (bottom of the vessel) is desired for ease in sand removal. Compare this weight to that of a vessel without the conical section and that to a vessel with a 1/4-in. plate internal cone.

Design pressure = 125 psig,
Maximum operating temperature = 200°F,
Corrosion allowance = 1/4 in.,
Material = SA516 Grade 70,
Diameter = 10 ft,
Seam-to-seam length above the cone = 12 ft,
Cone apex angle = 60°.

Solution:

Case I—Cone Bottom

(a) Shell:

$$t = \frac{Pr}{SE - 0.6P},$$

$S = 17{,}500\,\text{psi},$ (Table 6-3)

$E = 0.85,$ (Table 6-4)

$$t = \frac{(125)(60)}{(17{,}500)(0.85) - (0.6)(125)}$$

$$= 0.507\,\text{in.},$$

Required thickness $= 0.507 + 0.250 = 0.757\,\text{in.},$

Use $\frac{13}{16}$-in. plate (0.8125),

$$W = 11dtL$$

$$= (11)(120)(0.8125)(12)$$

$$= 12{,}870\,\text{lb}.$$

(b) Head (ellipsoidal 2:1):

$$t = \frac{(125)(120)}{(2)(17{,}500)(0.85) - (0.2)(125)}$$

$$= 0.505\,\text{in.},$$

Required thickness $= 0.505 + 0.250 = 0.755\,\text{in.},$

Use $\frac{13}{16}$-in. plate (0.8125),

$$W \approx 0.34td^2 + 1.9td,$$

$$W = (0.34)(0.8125)(120)^2 + (1.9)(0.8125)(120)$$

$$= 4{,}163\,\text{lb}.$$

(c) Cone:

$$t = \frac{Pd}{2\cos\alpha\,(SE - 0.6P)},$$

$$t = \frac{(125)\,(120)}{(2\cos 30)\,(17{,}500 \times 0.85 - 0.6 \times 125)}$$

$$= 0.585\,\text{in.},$$

Required thickness $= 0.585 + 0.250 = 0.835$ in.,

Use $\frac{7}{8}$-in. plate (0.875),

$$W = \frac{(0.23)\,(0.875)\,(120)^2}{\sin 30}$$

$$= 5{,}796\,\text{lb}.$$

(d) Skirt:

$$\text{Height} = \frac{5}{\tan 30} = 8.66\,\text{ft},$$

Allow 2 ft for access,

Height = 11 ft (The shell wall thickness, neglecting corrosion allowance, is approximately 0.5 inches. Assume 0.5 inches plate),

$W = (11)(120)(0.5)(11) = 7{,}260.$

(e) Summary:

Shell	12,870
Head	4,163
Cone	5,796
Skirt	7,260
Subtotal	30,089
Misc.	5,000
Total	35,089 lb

Case II—2:1 Ellipsoidal Head

(a) Skirt:

$$L = \frac{0.25d}{12} + 2$$
$$= \frac{(0.25)\,(120)}{12} + 2$$
$$= 4.50\,\text{ft},$$
$$W = (11)(120)(0.5)(4.5)$$
$$= 2{,}970\,\text{lb}.$$

(b) Summary:

Shell	12,870
Head-1	4,163
Head-2	4,163
Skirt	2,970
Subtotal	24,166
Misc.	5,000
Total	29,166 lb

Case III—Internal Cone

(a) Internal cone:

$$W = \frac{(0.23)\,(0.25)\,(120)^2}{\sin 30}$$
$$= 1{,}656\,\text{ft}.$$

(b) Shell:

$$\text{Height of cone} = \frac{(10/2)}{\tan 30} = 8.7\,\text{ft},$$
$$\text{Length of shell} = 12 + 8.7 = 20.7\,\text{ft},$$
$$\text{Weight of shell} = (11)(120)(0.8125)(20.7)$$
$$= 22{,}200\,\text{lb}.$$

(c) Summary:

Shell	22,200
Head-1	4,163
Head-2	4,163
Skirt	2,970
Cone	1,656
Subtotal	35,152
Misc.	5,000
Total	40,152 lb

Review Questions

1. The code governing the design and fabrication of pressure vessels is

   a) GPSA Engineering Data Book
   b) Crane Technical Bulletin 410
   c) ASME Boiler and Pressure Vessel Code, Section VIII
   d) API RP 14C
   e) API RP 14E

2. All flowing liquid outlet nozzles should be equipped with a

   a) liquid level controller
   b) double block-and-bleed valve
   c) vortex breaker
   d) level safety low (LSL)
   e) check valve

3. A large opening that allows field maintenance personnel access to the separator internals is called a(n)

   a) integral nozzle
   b) manway
   c) vortex breaker
   d) manhole
   e) hinged closure

4. What is the MAWP for a Material Group 1.1, ANSI Class 600 flange with a temperature of 125°F?

   a) 1480 psig
   b) 1495 psig
   c) 1350 psig
   d) 600 psig
   e) 1440 psig

5. The 2001 edition of the ASME Pressure Vessel code, Section VIII, Division 1 incorporates a safety factor of

   a) 3
   b) 3.5
   c) 4
   d) 4.5
   e) 5

6. A pressure vessel was built using the 1998 edition of the ASME Pressure Vessel code, Section VIII, Division 1. The vessel's maximum temperature is 200°F and the material used was SA-516, Grade 70. Determine the maximum allowable stress value.

   a) 23,30
   b) 17,500
   c) 16,300
   d) 17,400
   e) 18,200

7. Determine the maximum allowable joint efficiency for a double butt weld joint without a backing strip with full radiograph x-ray.

   a) 0.85
   b) 0.70
   c) 1.00
   d) 0.90
   e) 0.65

8. What material would typically be specified for a pressure vessel plate subjected to a temperature of −75°F?

   a) SA-36
   b) SA-240-304
   c) SA-516-70
   d) SA-106-B
   e) SA-312

9. What corrosion protection methods are commonly used in pressure vessels?

   a) sacrificial anodes
   b) internal coatings
   c) liners
   d) cladding
   e) all of the above

Exercises

Problem 1.

Determine the wall thickness for a horizontal separator given the following data:

Design pressure = 1,250 psig,
Design temperature = 150°F,
Vessel's outside diameter (OD) = 36 in.,
ASME code requirements = Section VIII, Division 1, 1998 edition,
Code material = SA-516-70,
Corrosion allowance = 1/8 in.,
Type of joint = double-welded butt joint,
Degree of examination = 100%.

Problem 2.

Using the data in Problem 1, calculate the head thickness for a 2:1 ellipsoidal head.

Problem 3.

Given the data in Problems 1 and 2, calculate the weight of a 36-in. OD by 15-ft horizontal separator.

WEIGHT OF SHELLS & HEADS										
	WALL THICKNESS									
DIAM.	1-3/8"					1-7/16"				
VESSEL	SHELL		HEAD			SHELL		HEAD		
	I.S	O.S	ELIP	F&D	HEMIS	I.S.	O.S.	ELLIP	F&D	HEMIS
12	146	156	142	119	135	206	162	151	126	143
14	225	185	169	148	178	237	193	180	155	188
16	255	215	206	176	228	267	223	220	184	240
18	284	244	239	206	283	289	254	255	220	298
20	313	273	285	239	345	329	285	303	253	363
22	343	303	322	275	412	360	316	342	292	434
24	372	332	364	323	486	390	346	286	337	511
26	402	362	408	364	566	421	377	432	380	594
28	431	391	466	408	651	452	408	493	426	684
30	460	421	527	454	743	482	438	558	481	780
32	490	450	593	502	841	513	469	627	532	882
34	519	479	662	553	945	544	500	699	585	991
36	548	508	734	620	1054	575	531	775	648	1106
38	578	538	812	676	1170	605	561	857	707	1228
40	607	567	892	812	1293	636	592	941	768	1355

Figure 6-16. Weight of shells and heads. (Reprinted with permission from *Pressure Vessel Handbook*, Gulf Publishing, Inc., Tulsa, Oklahoma.)

Problem 4.

Using the "Weight of Shells and Heads" from *Pressure Vessel Handbook* (shown in Figure 6-16 here), determine the weight of the vessel in Problem 3. Compare the results with the results from Problem 3.

Problem 5.

Given the following data, calculate the minimum wall thickness of a 24-in. OD by 10-ft seam-to-seam horizontal separator.

Design pressure = 600 psig,
Design temperature = 110°F,
ASME code requirements = Section VIII, Division 1, 1998 edition,
Code material = SA-516-70,
Corrosion allowance = 1/16 in.,
Type of joint = double-welded butt joint,
Degree of examination = 100%.

Problem 6.

Calculate the minimum head thickness for the vessel in Problem 5.

Problem 7.

Calculate the weight of the vessel in Problem 6.

Problem 8.

Given the following data, calculate the minimum wall thickness of a 406-mm OD by 15-m seam-to-seam horizontal separator.

Design pressure = 10,000 kPa,
Design temperature = 50°C,
ASME code requirements = Section VIII, Division 1, 1998 edition,
Allowable stress = 120,650 kPa,
Corrosion allowance = 3 mm,
Type of joint = double-welded butt joint,
Degree of examination = 100%.

Problem 9.

Calculate the weight of the vessel in Problem 8.

Problem 10.

Calculate the minimum head thickness in Problem 8.

Reference

1. Bednar, H. H., *Pressure Vesel Design Handbook*, Van Nostrand Reinhold Co. 2004.

Chapter 7
Crude Oil Treating and Oil Desalting Systems

CRUDE OIL TREATING SYSTEMS

Introduction

Conditioning of oil-field crude oils for pipeline quality has been complicated by water produced with the oil. Separating water out of produced oil has been performed by various schemes with various degrees of success. The problem of removing emulsified water has grown more widespread and often times more difficult as production schemes lift more water with oil from water-drive formations, water-flooded zones, and wells stimulated by thermal and chemical recovery techniques. The first part of this chapter describes oil-field emulsions and their characteristics, treating oil-field emulsions so as to obtain pipeline quality oil, and equipment used in conditioning oil-field emulsions. The second part of this chapter provides a brief description of oil desalting systems, the importance of mixing, and the equipment used in oil desalting systems.

Equipment Description

Free-Water Knockouts

Most well streams contain water droplets of varying size. If they collect together and settle to the bottom of a sample within 3 to 10 minutes, they are called free water. This is an arbitrary definition, but it is generally used in designing equipment to remove water that will settle out rapidly.

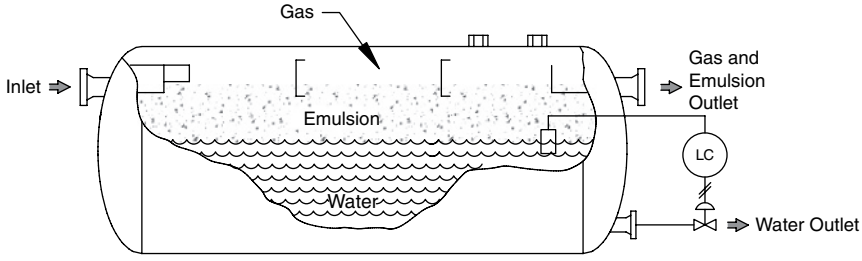


Figure : Freewater knockout Schematic

Figure 7-1. Cutaway of a free-water knockout.

A free-water knockout (FWKO) is a pressure vessel used to remove free water from crude oil streams (refer to Figure 7-1). They are located in the production flow path where turbulence has been minimized. Restrictions such as orifices, chokes, throttling globe valves, and fittings create turbulence in the liquids that aggravate emulsions. Free water, at wellhead conditions, frequently will settle out readily to the bottom of an expansion chamber.

Sizing and pressure ratings for these vessels are discussed in Chapters 4 and 5. Factors affecting design include retention time, flow rate (throughput), temperature, oil gravity (as it influences viscosity), water drop size distribution, and emulsion characteristics. Abnormal volumes of gas in the inlet stream may require proportionately larger vessels as these gas volumes affect the throughput rate. A simple "field check" to determine retention time is to observe a fresh sample of the wellhead crude and the time required for free water to segregate.

In installations where gas volumes vary, a two-phase separator is usually installed upstream of the free-water knockout. The two-phase separator removes most of the gas and reduces turbulence in the free-water knockout vessel. The free-water knockout usually operates at 50 psig (345 kPa) or less due to the vessel's location in the process flow stream. Internals should be coated or protected from corrosion since they will be in constant contact with salt water.

Gunbarrel Tanks with Internal and External Gas Boots

The gunbarrel tank, sometimes called a wash tank, is the oldest equipment used for multiwell onshore oil treating in a conventional gathering station or tank battery. Gunbarrel tanks are very common in heavy crude applications such as in Sumatra and East Kalimantan, Indonesia, Bakersfield,

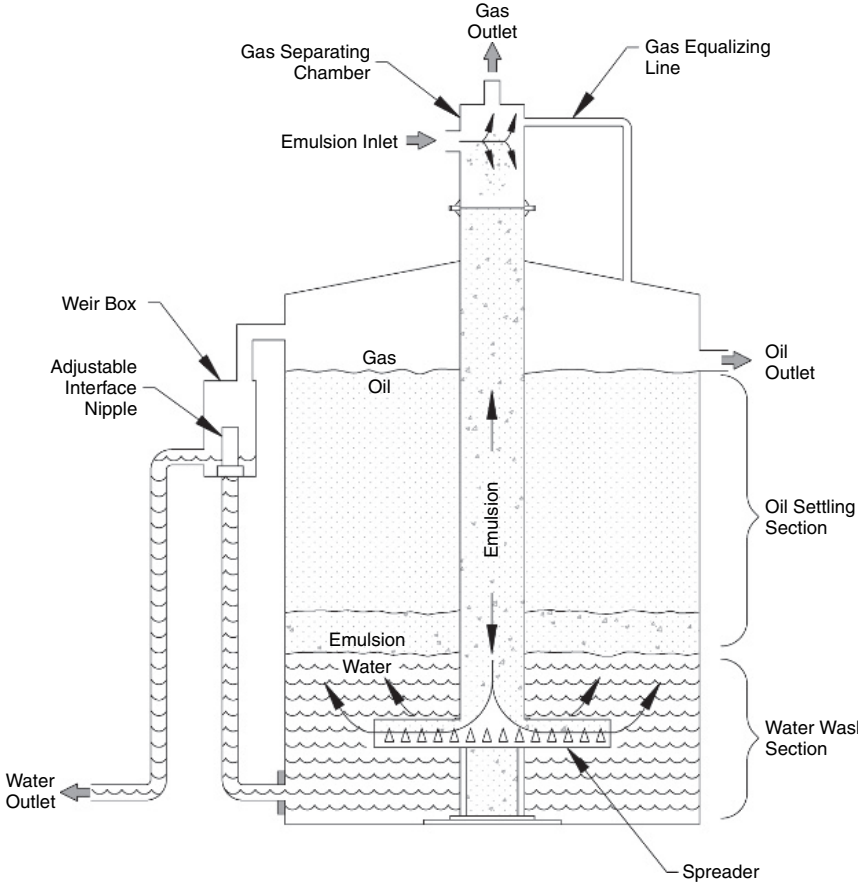


Figure 7-2. Gunbarrel with an internal gas boot.

California, and for low flow rate onshore applications for all crude gravities.

The gunbarrel tank is a vertical flow treater in an atmospheric tank. Figure 7-2 shows a "gunbarrel" tank with an internal gas boot. Typically, gunbarrels have an internal gas separating chamber or "gas boot" extending 6 to 12 ft (2–4 m) above the top of the tank, where gas is separated and vented, and a down-comer extending 2 to 5 ft (0.6–1.5 m) from the bottom of the tank. A variation of the above gunbarrel configuration is a wash tank with an "external" gas boot. This configuration is preferred on larger tanks, generally in the 60,000-barrel range, where attaching an internal gas boot is structurally difficult. In either case, the gunbarrel

tank is nothing more than a large atmospheric settling tank that is higher than downstream oil shipping and water clarifier tanks. The elevation difference allows gravity flow into the downstream vessels.

Because gunbarrels tend to be of larger diameter than vertical heater-treaters, many have elaborate spreader systems that attempt to create uniform (i.e., plug) upward flow of the emulsion to take maximum advantage of the entire cross section. Spreader design is important to minimize the amount of short-circuiting in larger tanks.

The emulsion, flowing from an upstream separator and possibly a heater, enters the top of the gas separation section of the gas boot. The gravity separation section removes flash gas and gas liberated as a result of heating the emulsion. The emulsion flows down the down-comer to a spreader, which is positioned below the oil–water interface. Exiting at the bottom of the down-comer, the emulsion rises to the top of the surrounding layer of water. The water level is controlled by a water leg or automatic level control. The emulsion passage through the water helps collect the entrained water and converts the emulsion into distinct oil and water layers. Oil accumulates at the top and flows out through the spillover line into the oil settling tank. Water flows from the bottom of the tank, up through the water leg, and into a surge or clarifier tank. The settling time in the vessel for the total fluid stream is usually 12 to 24 hr. Most gunbarrels are unheated, though it is possible to provide heat by heating the incoming stream external to the tank, installing heating coils in the tank, or circulating the water to an external or "jug" heater in a closed loop. It is preferable to heat the inlet so that more gas is liberated in the boot, although this means that fuel will be used in heating any free water in the inlet.

The difference in height between the oil spillover line and the external water leg controls the oil-water interface. Example 1 illustrates this design consideration.

Example 7-1: Determination of external water leg height

Given:

Oil gravity @ 60°F	= 36°API,
Water specific gravity	= 1.05,
Height of oil outlet	= 23 ft,
Height of interface level	= 10 ft (for this example),
Height of water outlet	= 1 ft,
Figure 7-3	= Gunbarrel schematic.

Solution:

(1) Determine the oil specific gravity.

$$\text{Oil specific gravity} = \frac{141.5}{131.5 + {}^{\circ}\text{API}}$$
$$= \frac{141.5}{131.5 + 36}$$
$$= 0.845$$

(2) Determine the oil gradient.

Since the charge in the pressure with depth for fresh water is 0.433 psi/ft of depth, the change in pressure with depth of fluid whose specific gravity is SG would be 0.433 (SG); thus, the oil gradient is

$$\text{Oil gradient} = (0.433)(0.845) = 0.366\,\text{psi/ft}.$$

(3) Determine the water gradient.

$$\text{Water gradient} = (0.433)(1.05) = 0.455\,\text{psi/ft}$$

(4) Calculate the height of the oil and the height of the water in the tank.

$$H_o = \text{Height of oil outlet} - \text{height of interface level}$$
$$= 23 - 10$$
$$= 13\,\text{ft},$$
$$H_w = \text{Height of interface level} - \text{height of water outlet}$$
$$= 10 - 1$$
$$= 9\,\text{ft}.$$

(5) Perform a pressure balance.

$$\begin{pmatrix}\text{Hydrostatic Pressure}\\ \text{inside Tank}\end{pmatrix} = \begin{pmatrix}\text{Hydrostatic Pressure}\\ \text{in the Water Leg}\end{pmatrix},$$
$$(13)\,(0.366) + (9)\,(0.455) = (H)\,(0.455)\,,$$
$$H = \frac{(13)\,(0.366) + (9)\,(0.455)}{0.455},$$
$$= 19.5\,\text{ft}.$$

The design details for the spreader, water leg, and gas separation section vary for different manufacturers. These details do not significantly affect the sizing of the tank, provided the spreader minimizes short-circuiting. No matter how careful the design of the spreaders, large wash tanks are very susceptible to short-circuiting. This is due to temperature and density differences between the inlet emulsion and the fluid in the tank, solids deposition, and corrosion of the spreaders.

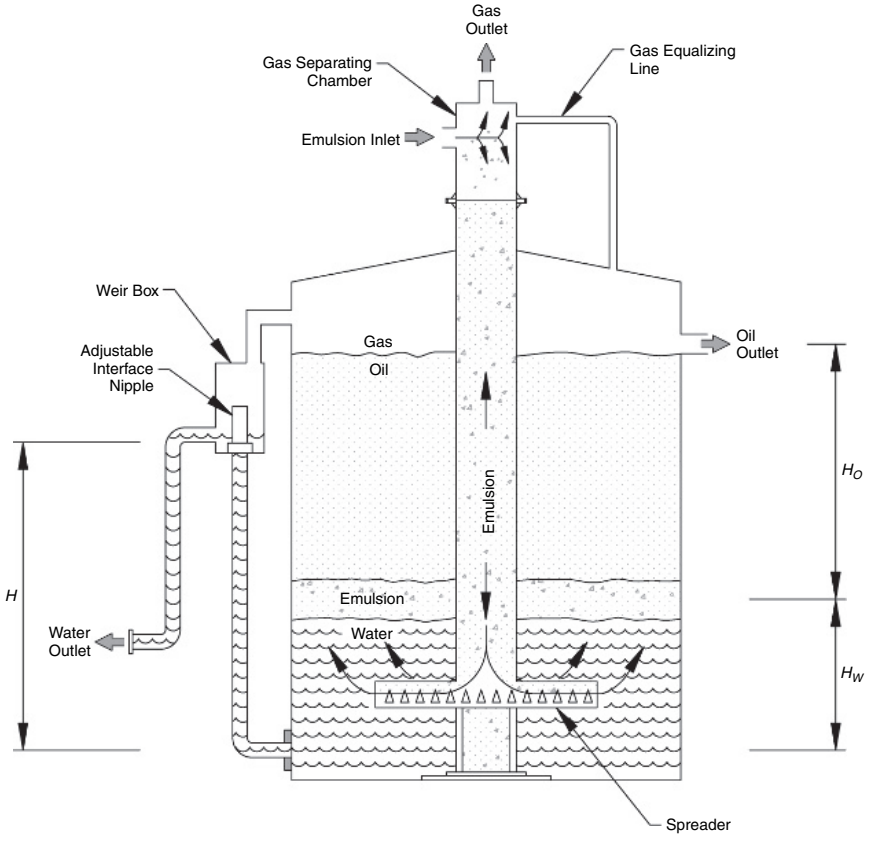


Figure 7-3. Example 1: Determination of external water leg height, *H*.

Standard tank dimensions are listed in API Specification 12F (Shop Welded Tanks), API Specification 12D (Field Welded Tanks), and API Specification 12B (Bolted Tanks). These dimensions are shown in Tables 7-1, 7-2 and 7-3, respectively.

Table 7-1a
Shop Welded Tanks (API Specification 12F)—Field Units

Nominal Capacity (BBL)	Approximate Working Capacity (BBL)	Outside Diameter (ft-in.)	Height (ft-in.)	Height of Overflow Connection (ft-in.)
90	72	7-11	10	9-6
100	79	9-6	8	7-6
150	129	9-6	12	11-6
200	166	12-0	10	9-6
210	200	10-0	15	14-6
250	224	11-0	15	14-6
300	266	12-0	15	14-6
500	479	15-6	16	15-6

Table 7-1b
Shop Welded Tanks (API Specification 12F)—SI Units

Nominal Capacity (BBL)	Approximate Working Capacity (m^3)	Outside Diameter (m)	Height (m)	Height of Overflow Connection (m)
90	11.4	2.41	3.05	2.90
100	12.6	2.90	2.44	2.29
150	20.5	2.90	3.66	3.51
200	26.4	3.66	3.05	2.90
210	31.8	3.05	4.57	4.42
250	35.6	3.35	4.57	4.42
300	42.3	3.66	4.57	4.42
500	76.2	4.72	4.88	4.72

Gunbarrels are simple to operate and, despite their large size, are relatively inexpensive. However, they have a large footprint, which is why they are not used on offshore platforms. Gunbarrels hold a large volume of fluids, which is a disadvantage should a problem develop. When the treating problem is detected in the oil outlet, a large volume of bad oil has already collected in the tank. This oil may have to be treated again, which may require large slop tanks, recycle pumps, etc. It may be beneficial to reprocess this bad oil in a separate treating facility so as to avoid further contamination of the primary treating facility.

Gunbarrels are most often used in older, low-flow-rate, onshore facilities. In recent times, vertical heater-treaters have become so inexpensive

Table 7-2a
Field Welded Tanks (API Specification 12D)—Field Units

Nominal Capacity (BBL)	Design Pressure (oz/in.^{2})		Approximate Working Capacity (BBL)	Nominal Outside Diameter (ft-in.)	Nominal Height (ft-in.)	Height of Overflow Line Connection (ft-in.)
	Pressure	Vacuum				
H-500	8	1/2	479	15-6	16-0	15-6
750	8	1/2	746	15-6	24-0	23-6
L-500	6	1/2	407	21-6	8-0	7-6
H-1000	6	1/2	923	21-6	16-0	15-6
1500	6	1/2	1438	21-6	24-0	23-6
L-1000	4	1/2	784	29-9	8-0	7-6
2000	4	1/2	1774	29-9	16-0	15-6
3000	4	1/2	2764	29-9	24-0	23-6
5000	3	1/2	4916	38-8	24-0	23-6
10,000	3	1/2	9938	55-0	24-0	23-6

Table 7-2b
Field Welded Tanks (API Specification 12D)—SI Units

Nominal Capacity (BBL)	Design Pressure (kPa)		Approximate Working Capacity (m^3)	Nominal Outside Diameter (m)	Nominal Height (m)	Height of Overflow Line Connection (m)
	Pressure	Vacuum				
H-500	3.4	0.2	76.2	4.72	4.88	4.72
750	3.4	0.2	118.6	4.72	7.32	7.16
L-500	2.6	0.2	64.7	6.55	2.44	2.29
H-1000	2.6	0.2	146.8	6.55	4.88	4.72
1500	2.6	0.2	228.6	6.55	7.32	7.16
L-1000	1.7	0.2	124.6	9.07	2.44	2.29
2000	1.7	0.2	282.0	9.07	4.88	4.72
3000	1.7	0.2	439.4	9.07	7.32	7.16
5000	1.3	0.2	781.6	11.79	7.32	7.16
10,000	1.3	0.2	1580.0	16.76	7.32	7.16

that they have replaced gunbarrels in single-well installations. On larger installations onshore in warm weather areas, gunbarrels are still commonly used. In areas that have a winter season it tends to become too expensive to keep the large volume of oil at a high enough temperature to combat potential pour-point problems.

Table 7-3a
Bolted Tanks (API Specification 12B)—Field Units

Nominal Capacity (42-gal BBL)	Number of Rings	Inside Diameter[1]		Height of Shell[2]		Calculated Capacity[3] (42-gal BBL)
		ft	in.	ft	In.	
100	1	9	2 3/4	8	0 1/2	96
200	2	9	2 3/4	16	1	192
300	3	8	2 3/4	24	1 1/2	287
250	1	15	4 5/8	8	0 1/2	266
HIGH 500	2	15	4 5/8	16	1	533
750	3	15	4 5/8	24	1 1/2	799
LOW 500	1	21	6 1/2	8	0 1/2	522
HIGH 1,000	2	21	6 1/2	16	1	1,044
1,500	3	21	6 1/2	24	1 1/2	1,566
LOW 1,000	1	29	8 5/8	8	0 1/2	944
2,000	2	29	8 5/8	16	1	1,987
3,000	3	29	8 5/8	24	1 1/2	2,981
5,000	3	38	7 5/8	24	1 1/2	5,037
10,000	3	54	11 3/4	24	2	10,218

[1] The inside diameter is an approximate dimension. The values shown are 2 in. less than the bottom bolt-circle diameters.
[2] Shell heights shown do not include the thickness of the gasket.
[3] The calculated capacity is based on the inside diameter and height of shell.

Horizontal Flow Treaters

Horizontal flow treaters are not common. Figure 7-4 illustrates one design, which consists of a cylindrical treating tank incorporating internal baffles. The internal baffles establish a horizontal flow pattern in the cylindrical tank, which is more efficient for gravity separation than vertical flow and is less subject to short-circuiting.

The oil, emulsion, and water enter the vessel and must follow the long flow path between the baffles. Separation takes place in the straight flow areas between the baffles. Turbulence coupled with high flow velocities prevents separation at the corners, where the flow reverses direction. Tracer studies indicate that approximately two thirds of the plan area of the tank is effective in oil–water separation.

In addition to gravity separation, the emulsion must be collected and held in the treater for a certain retention time so that the emulsion will break. In horizontal flow treaters, the emulsion collects between the oil and water; however, the horizontal flow pattern tends to sweep the emulsion toward the outlets. The emulsion layer may grow much thicker

Table 7-3b
Bolted Tanks (API Specification 12B)—SI Units

Nominal Capacity (42-gal BBL)	Number of Rings	Inside Diameter[1] (m)	Height of Shell[2] (m)	Calculated Capacity[3] (m^3)
100	1	2.81	2.45	15.3
200	2	2.81	4.90	30.5
300	3	2.81	7.35	45.6
250	1	4.69	2.45	42.3
HIGH 500	2	4.69	4.90	84.7
750	3	4.69	7.35	127.0
LOW 500	1	6.57	2.45	83.0
HIGH 1,000	2	6.57	4.90	166.0
1,500	3	6.57	7.35	249.0
LOW 1,000	1	9.06	2.45	158.0
2,000	2	9.06	4.90	315.9
3,000	3	9.06	7.30	473.9
5,000	3	11.78	7.30	800.8
10,000	3	16.76	7.37	1624.5

[1] The inside diameter is an approximate dimension. The values shown are less than the bottom bolt-circle diameters.
[2] Shell heights shown do not include the thickness of the gasket.
[3] The calculated capacity is based on the inside diameter and height of shell.

at the outlet end of the treater than at the inlet end. Accordingly, it is much easier for the emulsion to be carried out of the vessel with the oil.

Heaters

Heaters are vessels used to raise the temperature of the liquid before it enters a gunbarrel, wash tank, or horizontal flow treater. They are used to treat crude oil emulsions. The two types of heaters commonly used in upstream operations are indirect fired heaters and direct fired heaters.

Both types have a shell and a fire tube. The fire tube contains within it a flame caused by the mixture of air and natural gas ignited by a pilot light and the hot exhaust gases which result from this combustion. The hot external surface of the fire tube heats a bath of liquid in which it is immersed.

Indirect heaters have a third element, which is the process flow coil. Heaters have standard accessories such as burners, regulators, relief valves, thermometers, temperature controllers, etc.

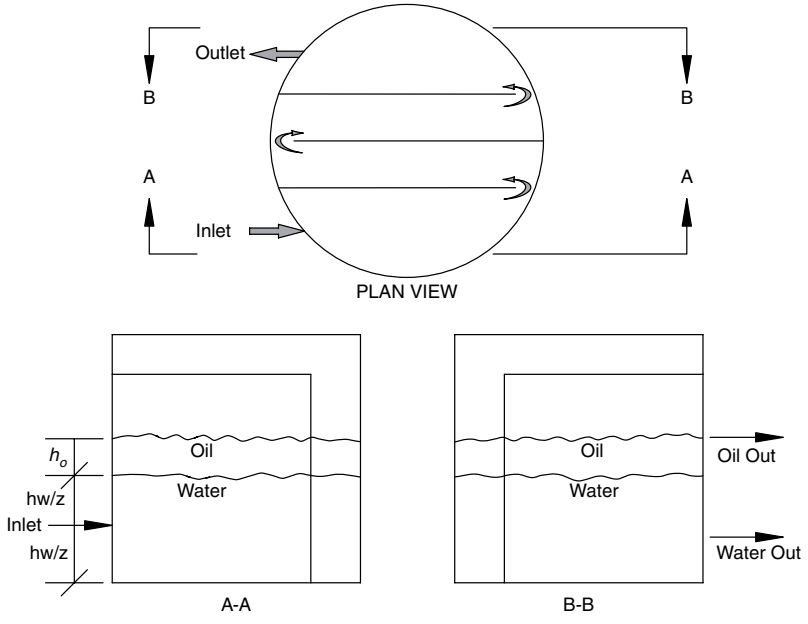


Figure 7-4. Plan view of a cylindrical treating tank incorporating internal baffles that establish horizontal flow.

Indirect Fired Heaters

Figure 7-5 shows a typical indirect fired heater. Oil flows through tubes that are immersed in water, which in turn is heated by a fire tube. Alternatively, heat may be supplied to the water bath by a heating fluid

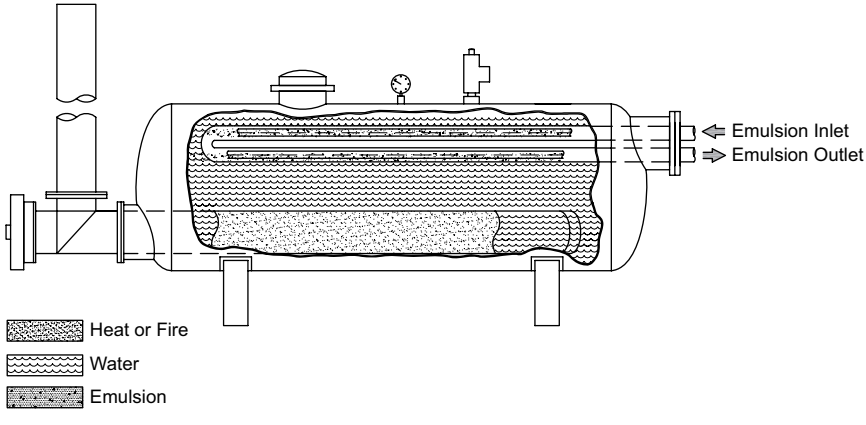


Figure 7-5. Cutaway of a horizontal indirect fired heater.

medium, steam, or electric immersed heaters instead of a fire tube. Indirect heaters maintain a constant temperature over a long period of time and are safer than direct heaters.

Hot spots are not as likely to occur on the fire tube if the calcium content of the heating water is controlled. The primary disadvantage is that these heaters require several hours to reach the desired temperature after they have been out of service.

Direct Fired Heaters

Figure 7-6 shows a typical direct fired heater. Oil flows through an inlet distributor and is heated directly by a fire box. Alternatively, heat may be supplied to the water bath by a heating fluid medium, steam, or an electric immersed heater instead of the fire tube. Direct fired heaters are quick to reach the desired temperature, are efficient (75 to 90%), and offer a reasonable initial cost. Direct fired heaters are typically used where fuel gas is available and high volume oil treating is required. On the other hand, they are hazardous and require special safety equipment. Scale may form on the oil side of the fire tube, which prevents the transfer of heat from the fire box to the oil emulsion. Heat collects in the steel walls under the scale, which causes the metal to soften and buckle. The metal eventually ruptures and allows oil to flow into the fire box, which results in a fire. The resultant blaze, if not extinguished, will be fed by the incoming oil stream.

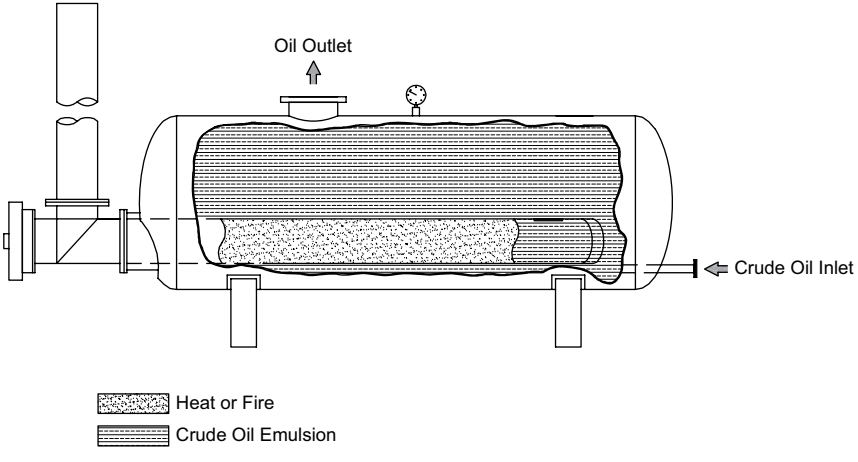


Figure 7-6. Cutaway of a horizontal direct fired heater.

Waste Heat Recovery

A waste heat recovery heater captures waste heat from the exhaust stacks of compressors, turbines, generators, and large engines. These hot exhaust gases can be routed through a tube and immersed in a bath performing the same function as a fire tube. Alternatively, heat exchangers may be used to transfer this heat to a heating fluid medium, which in turn is used to heat the crude oil emulsion.

Heater Sizing

Refer to the Equipment Sizing section of this chapter for details.

Heater-Treaters

Heater-treaters are an improvement over the gunbarrel and heater system. Many designs are offered to handle various conditions such as viscosity, oil gravity, high and low flow rates, corrosion, and cold weather. When compared to gunbarrels, heater-treaters are less expensive initially, offer lower installation costs, provide greater heat efficiency, provide greater flexibility, and experience greater overall efficiency. On the other hand, they are more complicated, provide less storage space for basic sediment, and are more sensitive to chemicals. Since heater-treaters are smaller than other treating vessels, their retention times are minimal (10 to 30 min) when compared to gunbarrels and horizontal flow treaters.

Internal corrosion of the down-comer pipe is a common problem. Build-up of sediment on the walls or bottom of the treater can cause the interface levels to rise and liquid to carry over and/or oil to exit the treater with salt water. Bi-annual inspections should be performed to include internal inspection for corrosion, sediment build-up, and scale build-up.

Vertical Heater-Treaters

The most commonly used single-well treater is the vertical heater-treater, which is shown in Figure 7-7. The vertical heater-treater consists of four major sections: gas separation, free-water knockout, heating and water-wash, and coalescing-settling sections. Incoming fluid enters the top of the treater into a gas separation section, where gas separates from the liquid and leaves through the gas line. Care must be exercised to size this section so that it has adequate dimensions to separate the gas from

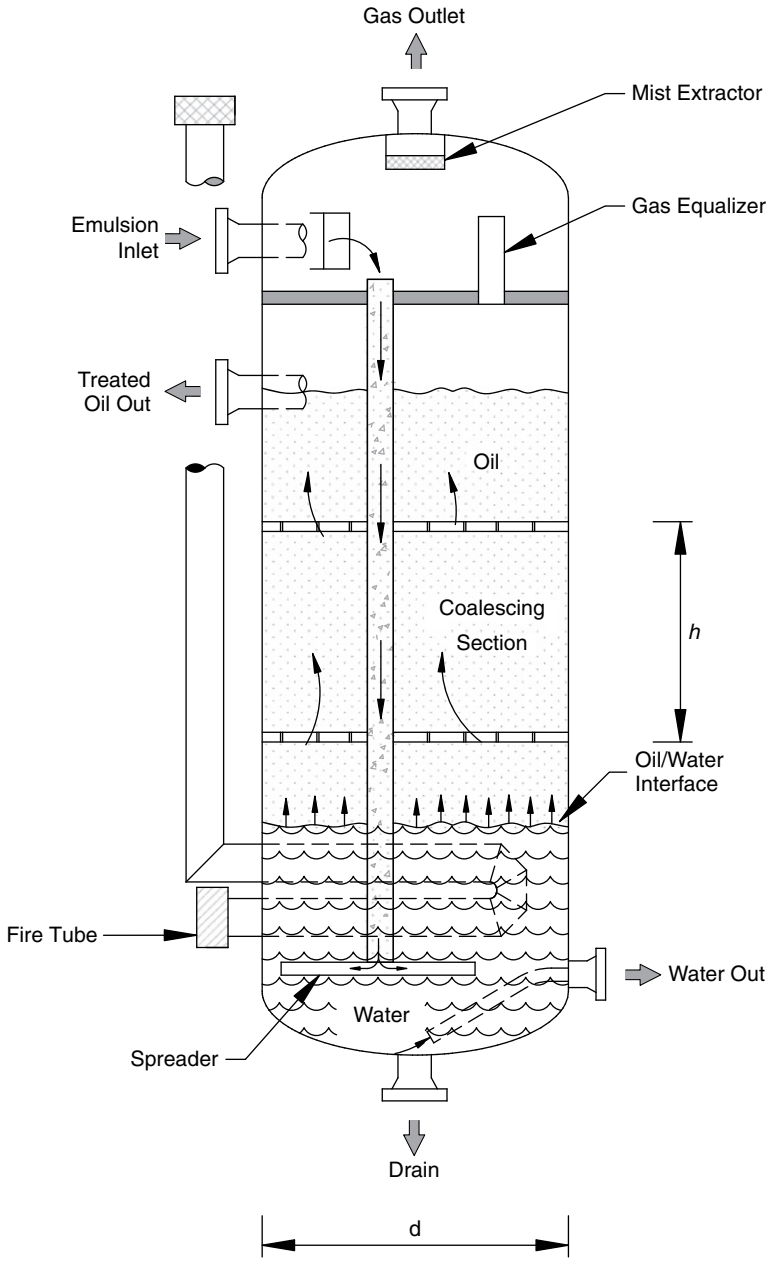


Figure 7-7. Simplified schematic of a vertical heater-treater.

the inlet flow. If the treater is located downstream of a separator, the gas separation section can be very small. The gas separation section should have an inlet diverter and a mist extractor.

The liquids flow through a down-comer to the base of the treater, which serves as a free-water knockout section. If the treater is located downstream of a free-water knockout or a three-phase separator, the bottom section can be very small. If the total well stream is to be treated, this section should be sized for 3 to 5 minutes' retention time to allow the free water to settle out. This will minimize the amount of fuel gas needed to heat the liquid stream rising through the heating section. The end of the down-comer should be slightly below the oil–water interface so as to "water-wash" the oil being treated. This will assist in the coalescence of water droplets in the oil.

The oil and emulsion rise through the heating and water-wash section, where the fluid is heated (refer to Figure 7-8). As shown in Figure 7-9, a fire tube is commonly used to heat the emulsion in the heating and water-wash section. After the oil and emulsion are heated, the heated oil and emulsion enter the coalescing section, where sufficient retention time is provided to allow the small water droplets in the oil continuous phase to coalesce and settle to the bottom. As shown in Figure 7-10, baffles are sometimes installed in the coalescing section to treat difficult emulsions. The baffles cause the oil and emulsion to follow a back-and-forth path up through the treater. Treated oil flows out the oil outlet, at the top of the coalescing section, and through the oil leg heat exchanger, where a valve controls the flow. Heated clean oil preheats incoming cooler emulsion in the oil leg heat exchanger (refer to Figure 7-11). Separated water flows out through the water leg, where a control valve controls the flow to the water treating system (refer to Figure 7-12).

As shown in Figure 7-13, any gas, flashed from the oil due to heating, is captured on the condensing head. Any gas that didn't condense flows through an equalizing line to the gas separation section. As shown in Figure 7-14, a vane-type mist extractor removes the liquid mist before the gas leaves the treater. The gas liberated when crude oil is heated may create a problem in the treater if it is not adequately designed. In vertical heater-treaters the gas rises through the coalescing section. If a great deal of gas is liberated, it can create enough turbulence and disturbance to inhibit coalescence. Equally important is the fact that small gas bubbles have an attraction for surface-active material and hence water droplets. Thus, they tend to keep the water droplets from settling out and may even cause them to carry over to the oil outlet.

The oil level is maintained by pneumatic or lever-operated dump valves. The oil–water interface is controlled by an interface level controller or an adjustable external water leg.

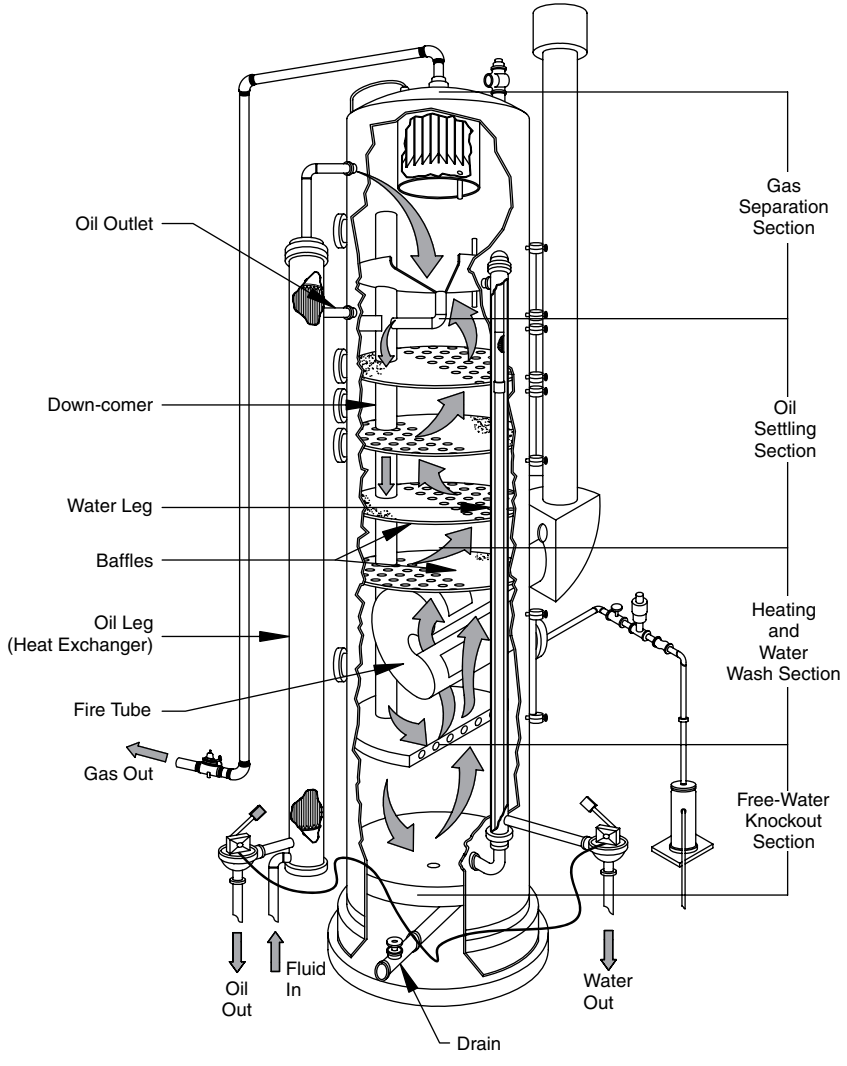


Figure 7-8. Three-dimensional view illustrating oil and emulsion rising through the heating and water-wash.

Standard vertical heater-treaters are available in 20- and 27-ft (6.1- and 8.2-m) heights. These heights provide sufficient static liquid head so as to prevent vaporization of the oil. The detailed design of the treater, including the design of internals (many features of which are patented), should be the responsibility of the equipment supplier.

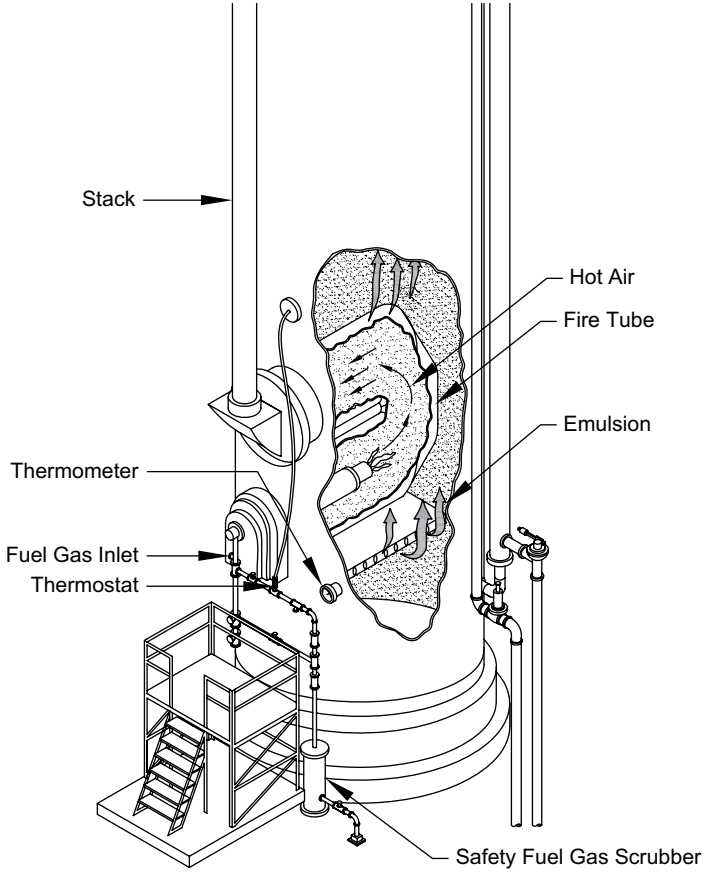


Figure 7-9. Cutaway showing a typical fire-tube that heats the emulsion in the heating and water-wash section.

Coalescing Media

It is possible to use coalescing media to promote coalescence of the water droplets. These media provide large surface areas upon which water droplets can collect. In the past the most commonly used coalescing media was wood shavings or "excelsior," which is also referred to as a "hay section." The wood excelsior was tightly packed to create an obstruction to the flow of the small water droplets and promote random collision of these droplets for coalescence. When the droplets were large enough, they fell out of the flow stream by gravity. Figure 7-15 shows a vertical heater-treater utilizing an excelsior section.

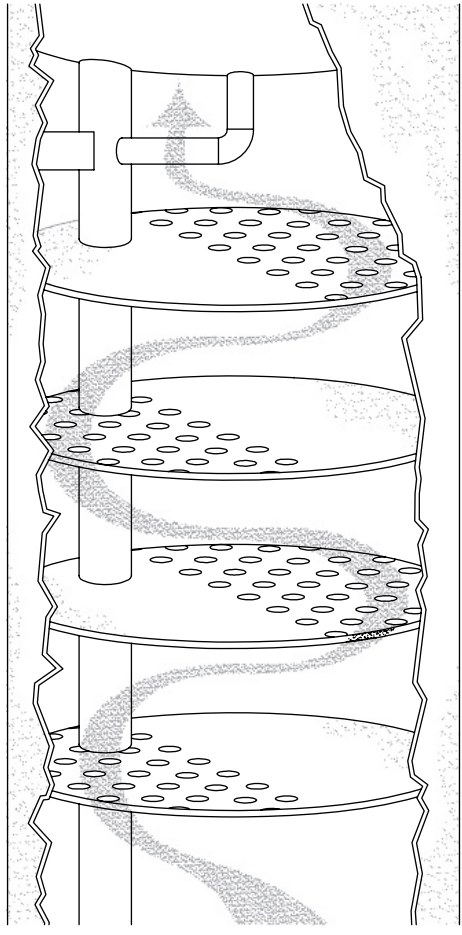


Figure 7-10. Baffles, installed in the coalescing section, cause the emulsion to follow a back-and-forth path up through the oil settling section.

The use of an excelsior section allowed lower treating temperatures. However, these media had a tendency to clog with time and were difficult to remove. Therefore, they are no longer used.

Horizontal Heater-Treaters

For most multiwell flow streams, horizontal heater-treaters are normally required. Figure 7-16 shows a simplified schematic of a typical horizontal heater-treater. Design details vary from manufacturer to manufacturer,

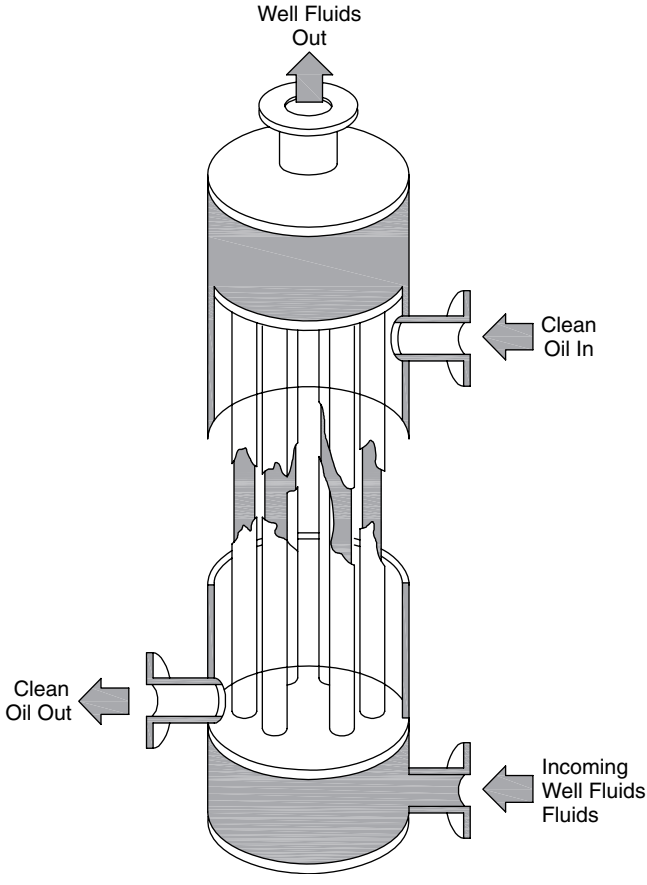


Figure 7-11. Heated clean oil preheats incoming cooler emulsion in the oil leg heat exchanger.

but the principles are the same. The horizontal heater-treater consists of three major sections: front (heating and water-wash), oil surge chamber, and coalescing sections.

Incoming fluids enter the front (heating and water-wash) section through the fluid inlet and down over the deflector hood (refer to Figure 7-17) where gas is flashed and removed. Heavier materials (water and solids) flow to the bottom while lighter materials (gas and oil) flow to the top. Free gas breaks out and passes through the gas equalizer loop to the gas outlet. As shown in Figure 7-18, the oil, emulsion, and free water pass around the deflector hood to the spreader located slightly below the oil–water interface, where the liquid is "water-washed" and the free water

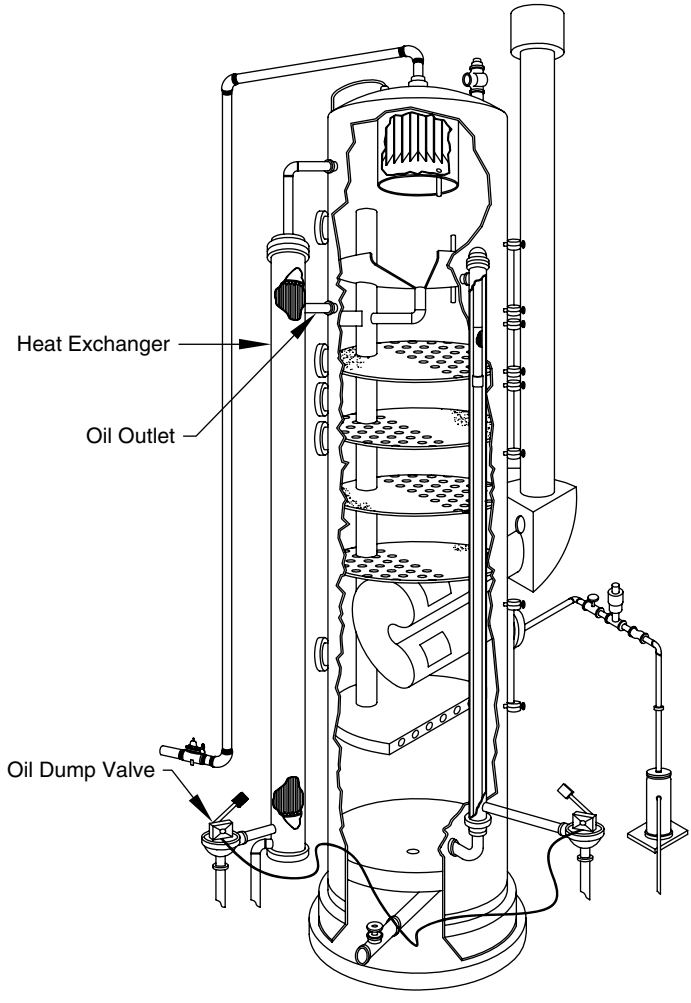


Figure 7-12. Cutaway illustrating oil and water legs.

is separated. For low gas–oil-ratio crudes, blanket gas may be required to maintain gas pressure. The oil and emulsion are heated as they rise past the fire tubes and are skimmed into the oil surge chamber.

As free water separates from the incoming fluids in the front section, the water level rises. If the water is not removed, it will continue to rise until it displaces all emulsion and begins to spill over the weir into the surge section. On the other hand, if the water level becomes too low, the front section will not be able to water-wash the incoming oil and emulsion, which would reduce the efficiency of the treater. Therefore, it is

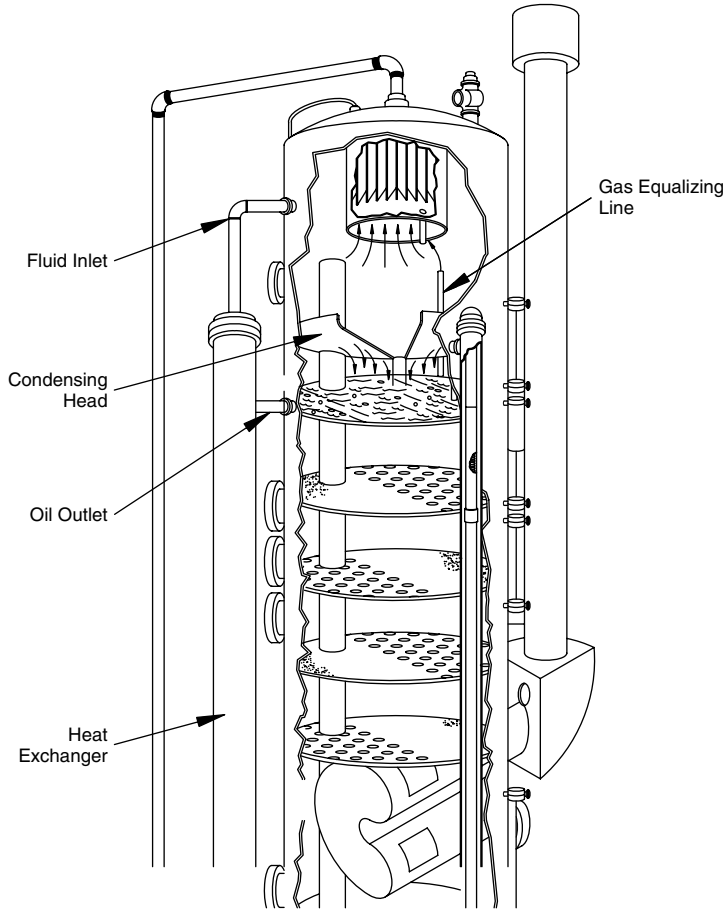


Figure 7-13. Gas, flashed from the oil during heating, is captured on the condensing head.

important to accurately control the oil–water interface in the front section. The oil–water interface is controlled by an interface level controller, which operates a dump valve for the free water (refer to Figure 7-19).

As shown in Figure 7-20, a level safety low shutdown sensor is required in the upper portion of the front (heating and water-wash) section. This sensor assures liquid is always above the fire tube. If the water dump valve malfunctions or fails open, the liquid surrounding the fire tube will drop, thus not absorbing the heat generated from the fire tube and possibly damaging the fire tube by overheating. Thus, if the level above the fire tube drops, the level safety low shutdown sensor sends a signal that closes the fuel valve feeding the fire tube. It is also important to control the

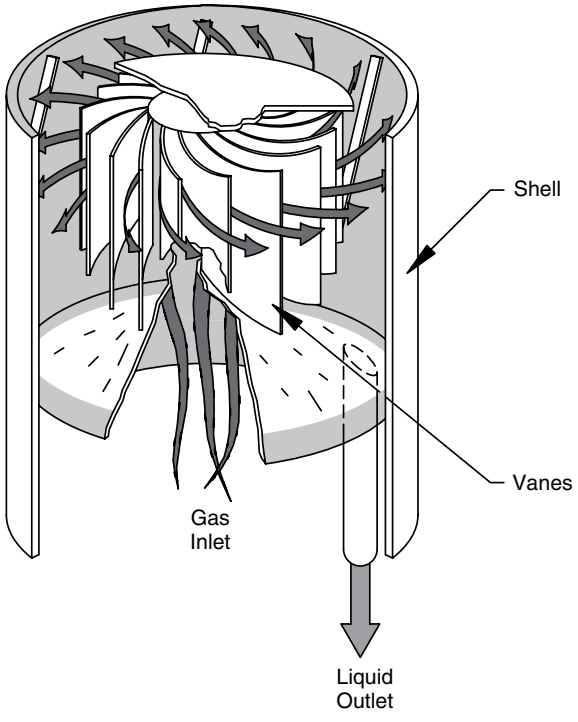


Figure 7-14. Vane-type mist extractor removes the liquid mist before the gas leaves the treater.

temperature of the fluid in the front (heating and water-wash) section. Therefore, a temperature controller, controlling the fuel to the burner or heat source, is required in the upper part of the heating–water-wash section (refer to Figure 7-21).

A level controller, in the oil surge section (refer to Figure 7-22), operates the dump valve on the clean oil outlet line. This dump valve regulates the flow of oil out the top of the vessel, which maintains a liquid packed condition. When the clean oil outlet valve is open, the pressure of the gas in the surge chamber forces the emulsion to flow through the spreader and push the clean oil through the clean oil collector (refer to Figure 7-23). When the clean oil outlet valve closes, the flow of emulsion to the coalescing-settling section stops and gas is prevented from entering the coalescing-settling section (refer to Figure 7-24).

The oil and emulsion flow through a spreader into the back or coalescing section of the vessel, which is fluid packed. The spreader distributes the flow evenly throughout the length of this section. Because it is lighter than the emulsion and water, treated oil rises to the clean oil collector,

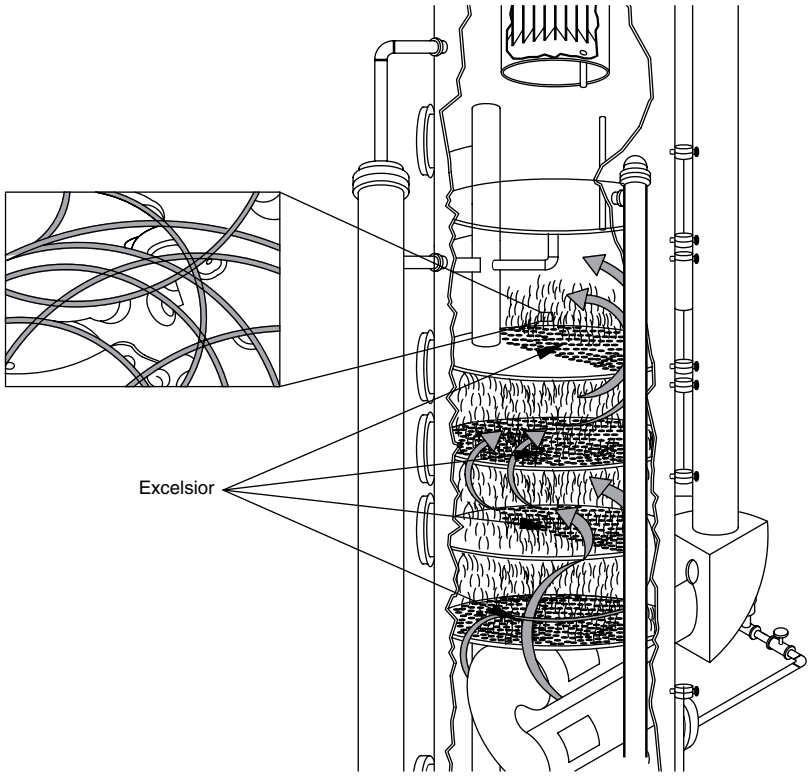


Figure 7-15. Vertical heater-treater fitted with excelsior, between the baffles, which aids in coalescence of water droplets.

where it is collected and flows to the clean oil outlet. The collector is sized to maintain uniform vertical flow of the oil. Coalescing water droplets fall countercurrent to the rising oil continuous phase.

The front (heating and water-wash) section must be sized to handle settling of the free water and heating of the oil. The coalescing section must be sized to provide adequate retention time for coalescing to occur and to allow the coalescing water droplets to settle downward countercurrent to the upward flow of the oil.

Most horizontal heater-treaters built today do not use fire tubes. Heat is added to the emulsion in a heat exchanger before the emulsion enters the treater. In these cases the inlet section of the treater can be fairly short because its main purpose is to degas the emulsion before it flows to the coalescing section.

Some heater-treaters are designed with only the coalescing section. In these cases the inlet is pumped through a heat exchanger to a treater that

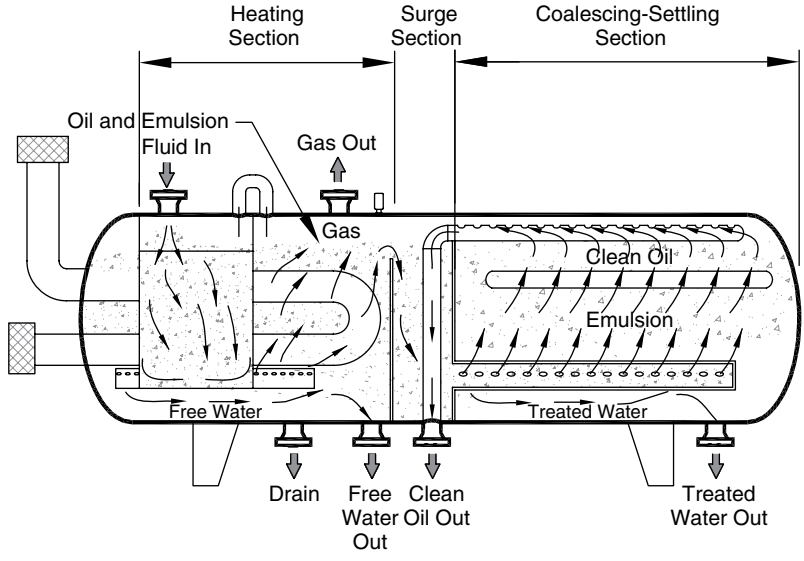


Figure 7-16. Simplified schematic of a horizontal heater-treater.

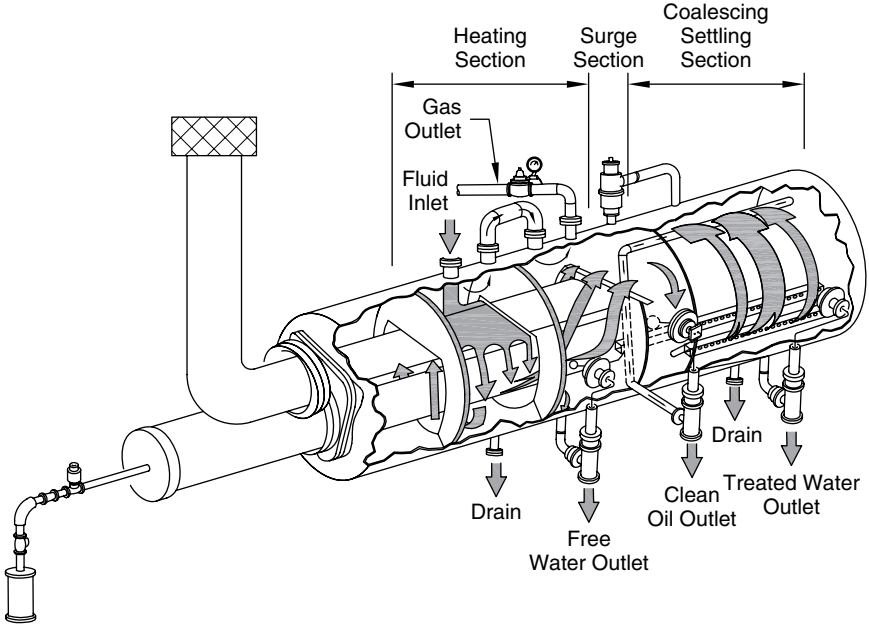


Figure 7-17. Three-dimensional view of a horizontal heater-treater flow pattern.

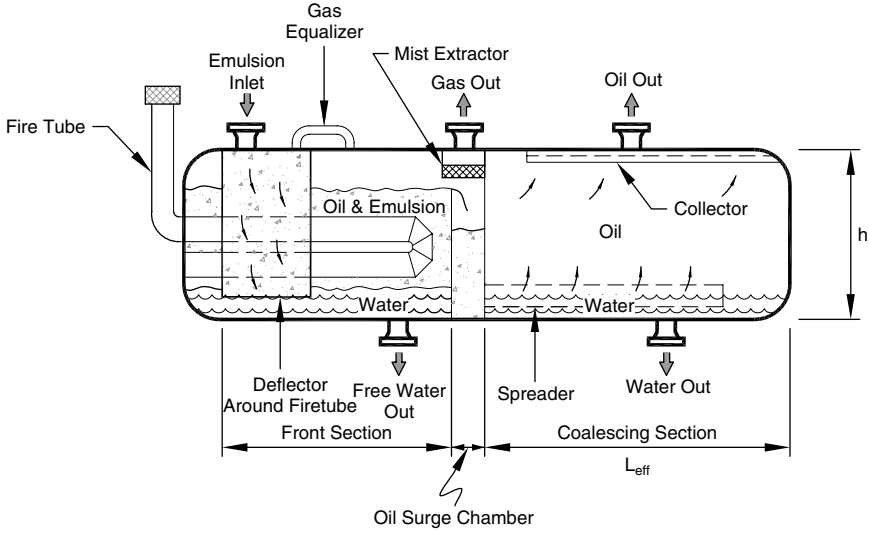


Figure 7-18. Schematic of horizontal heater-treater showing the oil, emulsion, and free water passing around the deflector hood to the spreader located slightly below the oil–water interface where the liquid is "water-washed" and the free water separated.

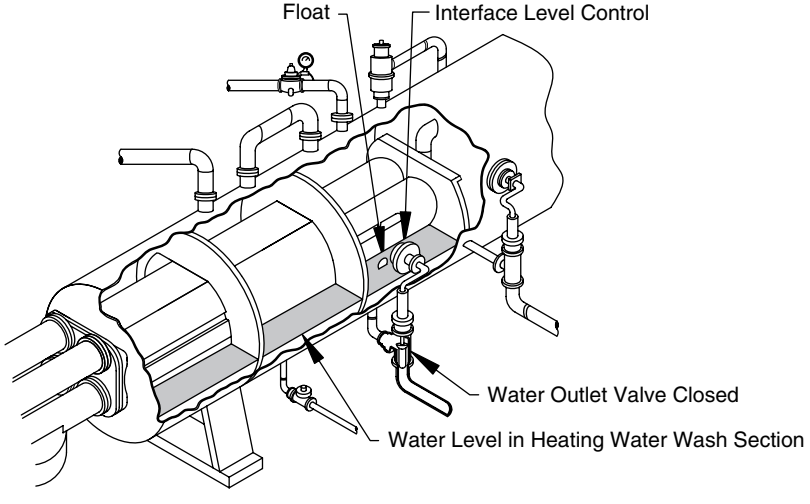


Figure 7-19. Oil–water interface in the heating and water-wash section is controlled by an interface level controller.

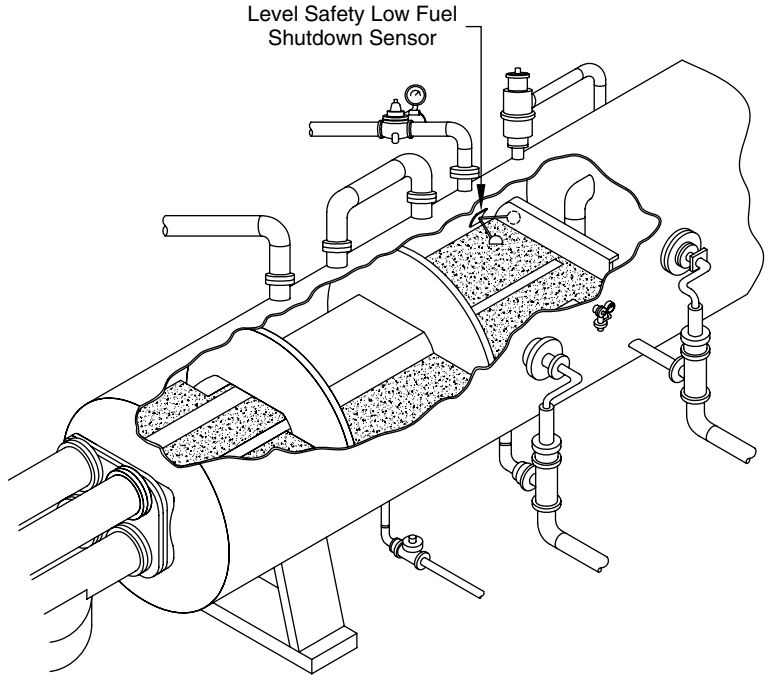


Figure 7-20. Level safety low sensor, located at the top of the heating–water-wash section, shuts off the fuel to the heat source (fire-tube) on low liquid level.

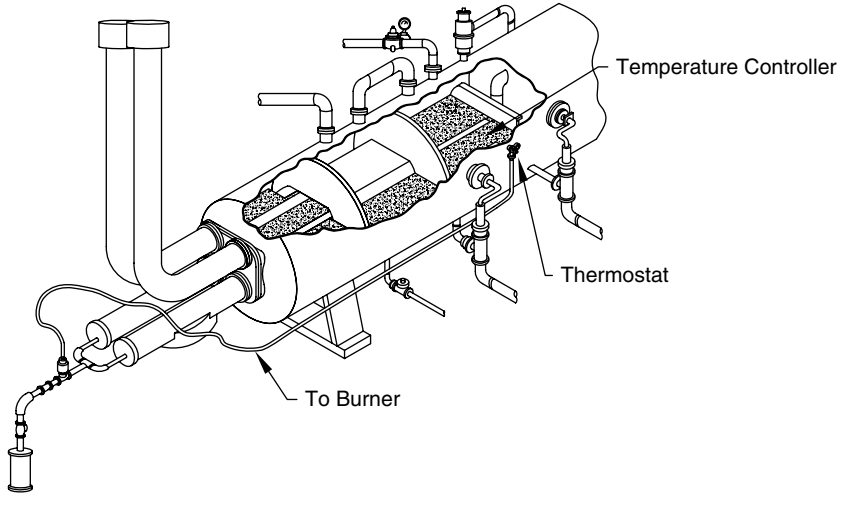


Figure 7-21. Temperature controller, located in the upper part of the heating–water-wash section, controls the fuel to the burner or heat source.

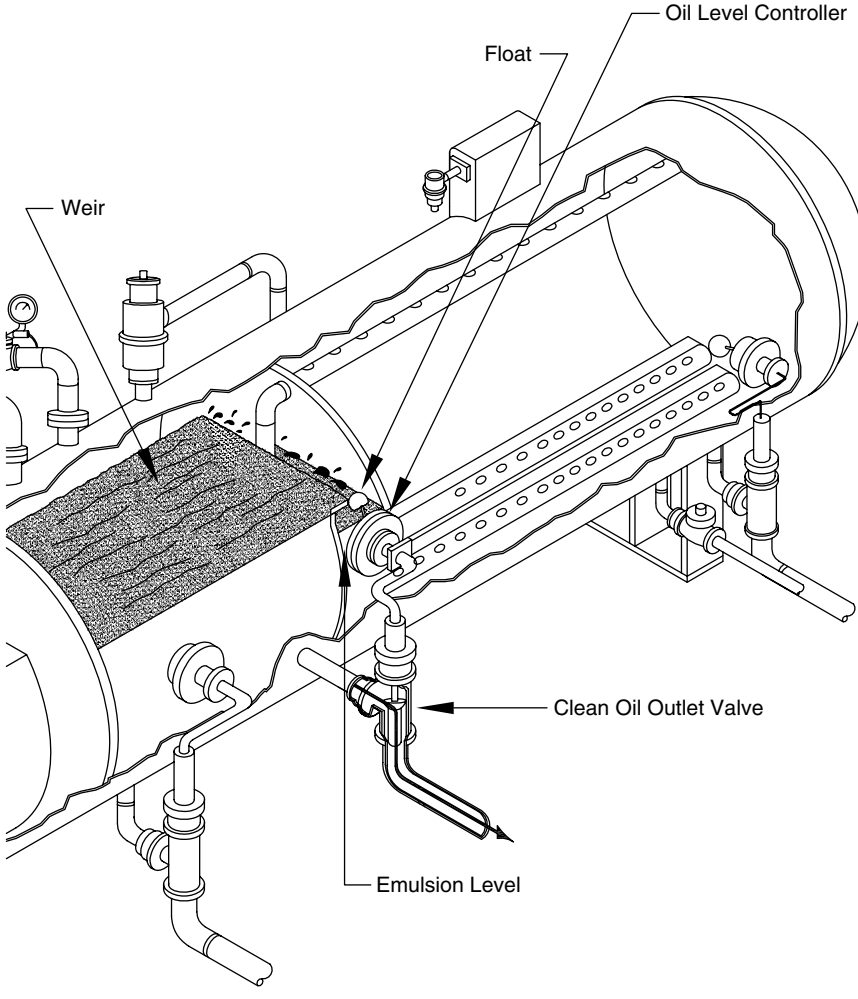


Figure 7-22. Level controller in the oil surge section operates the clean oil dump valve.

operates at a high enough pressure to keep the oil above its bubble point. Thus, the gas will not evolve in the coalescing section of the treater.

Electrostatic Heater-Treaters

Some horizontal heater-treaters add an electrostatic grid in the coalescing section. Figure 7-25 illustrates a simplified schematic of a typical

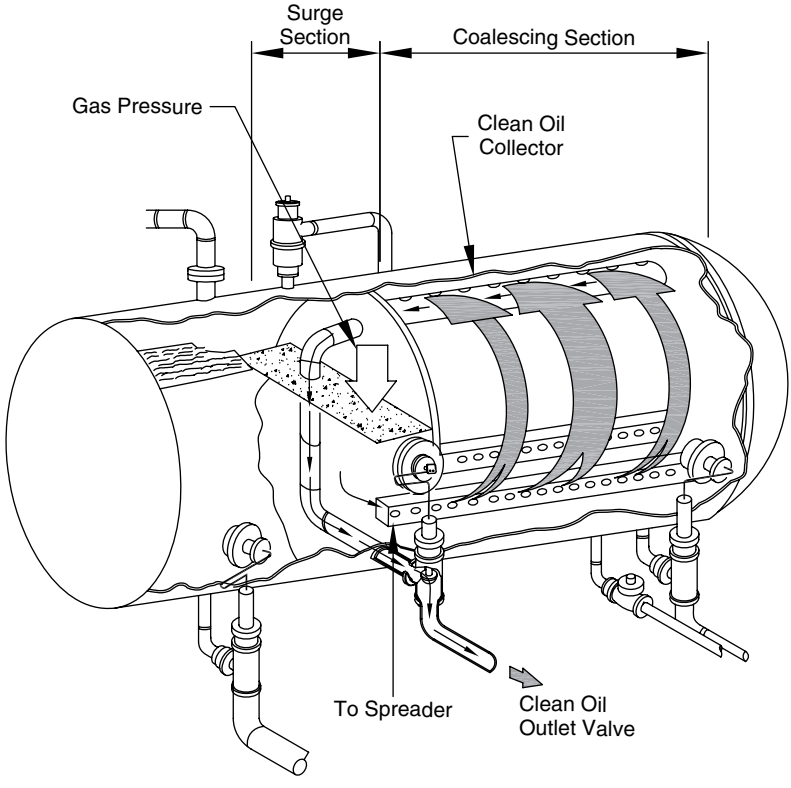


Figure 7-23. Pressure of the gas in the surge section forces the emulsion to flow through the spreader in the coalescing section and push the clean oil out through the clean oil collector.

horizontal electrostatic treater. The flow path in an electrostatic heater-treater is basically the same as in a horizontal heater-treater, except that an electrostatic grid is included in the coalescing-settling section, which helps to promote coalescence of the water droplets.

The electrostatic section contains two or more electrodes, one grounded to the vessel and the other suspended by insulators. An electrical system supplies an electric potential to the suspended electrode. The usual applied voltage ranges from 10,000 to 35,000 VAC, and the power consumption is from 0.05 to 0.10 KVA/ft^2 (0.54 to 1.08 KVA/m^2) of grid. The intensity of the electrostatic field is controlled by the applied voltage and spacing of electrodes. In some installations the location of the ground electrode can be adjusted externally to increase or decrease its spacing to the "hot" electrode. Optimum field intensities vary with applications but generally fall within the range of 1,000 to 4,000 V/in. (39 to 157 V/mm)

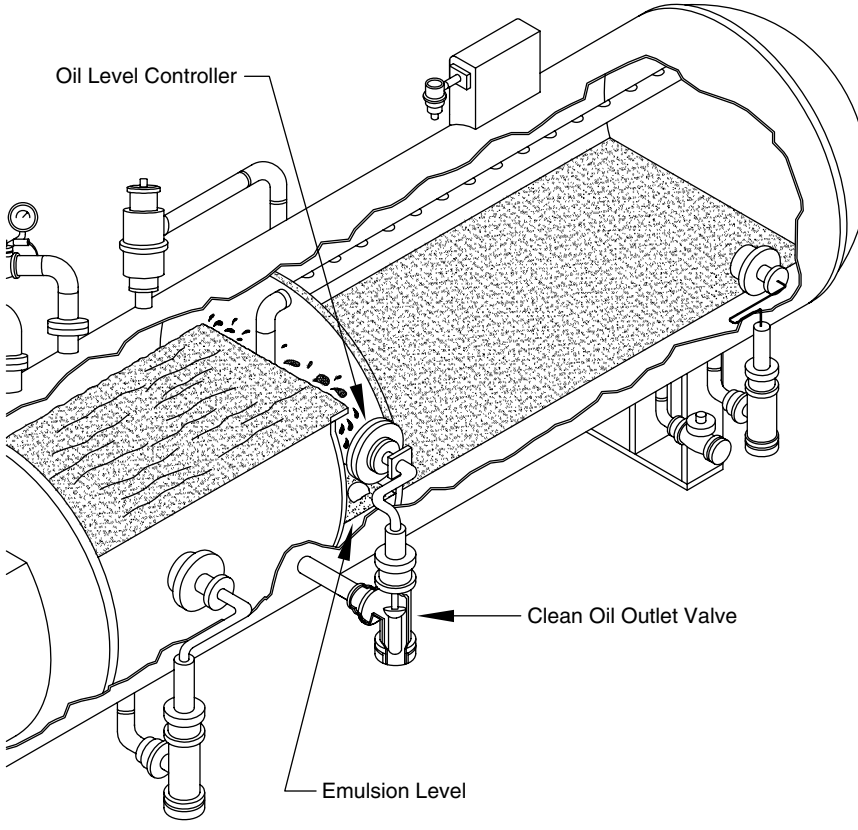


Figure 7-24. When the clean oil dump valve closes, the flow of emulsion to the coalescing settling section stops and the gas is prevented from entering.

of separation. The use of an electric field is most effective whenever the fluid viscosity is less than 50 cp at separating temperature, the specific gravity difference between the oil and water is greater than 0.001, and the electrical conductivity of the oil phase does not exceed 10^{-6} mho/cm.

The electrical control system that supplies energy to the electrodes consists of a system of step-up transformers (either single or three-phase) in which the primary side is connected to a low-voltage power source (208, 220, or 440 V) and secondary windings are designed so that the induced voltage will be of the desired magnitude (refer to Figure 7-26).

As shown in Figure 7-27, oil and small water droplets enter the coalescing section and travel up into the electrostatic grid section, where the water droplets become "electrified" or "ionized" and are forced to collide.

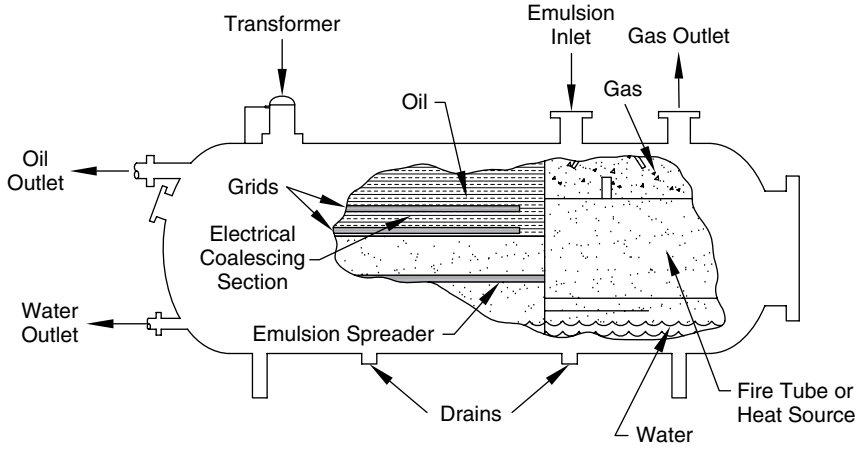


Figure 7-25. Simplified schematic of a horizontal electrostatic heater-treater.

The electrodes have electrical charges that reverse many times a second; thus, the water droplets are placed in a rapid back-and-forth motion. The greater the motion of the droplets, the more likely the water droplets are to collide with each other, rupture the skin of the emulsifying agent, coalesce, and settle out of the emulsion. Because of the forced collisions, electrostatic heater-treaters typically operate at lower temperatures and use less fuel than horizontal heater-treaters. The time in the electronic field is controlled by electrode spacing and the vessel configuration. An electronic field exists throughout the body of the oil within the vessel, even though most coalescing takes place in the more intense fields in the vicinity of the electrodes.

It is imperative that the design of the vessel provide for distribution of the emulsion across the electrical grid. It is also essential to maintain the fluid in the liquid phase in the electrical coalescing section. Gas evolving in the coalescing section will attract the small water droplets in the emulsion, becoming saturated with water and carrying the water up to the oil outlet. In addition, water-saturated vapors, which are highly conductive, will greatly increase the electrical power consumption.

It is also important to prevent the water level from reaching the height of the electrodes. Nearly all produced water contains some salt. These salts make the water a very good conductor of electric currents. Thus, if the water contacts the electrodes, it may short out the electrode grid or the transformer.

Since coalescence of the water droplets in an electric field is dependent on the characteristics of the specific emulsion being treated, sizing of

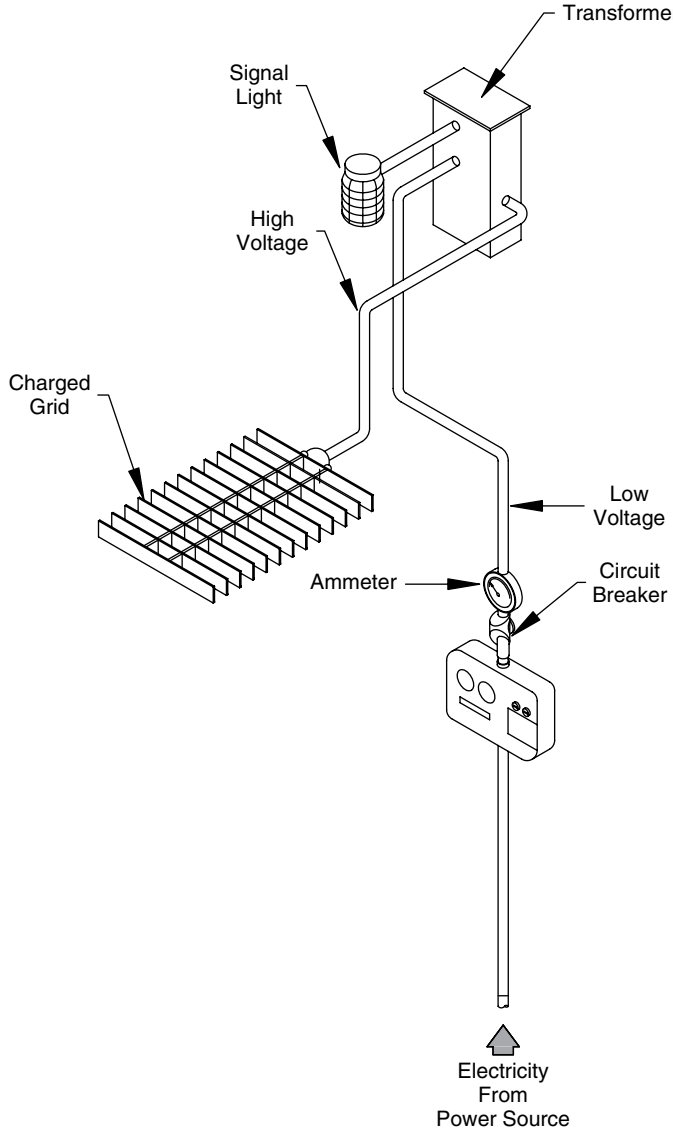


Figure 7-26. Electrical control system of an electrostatic heater-treater.

grid area requires laboratory testing. Field experience tends to indicate that electrostatic treaters are efficient at reducing water content in the crude to the 0.1 to 0.5 percent level. This makes these treaters particularly attractive for desalting operations.

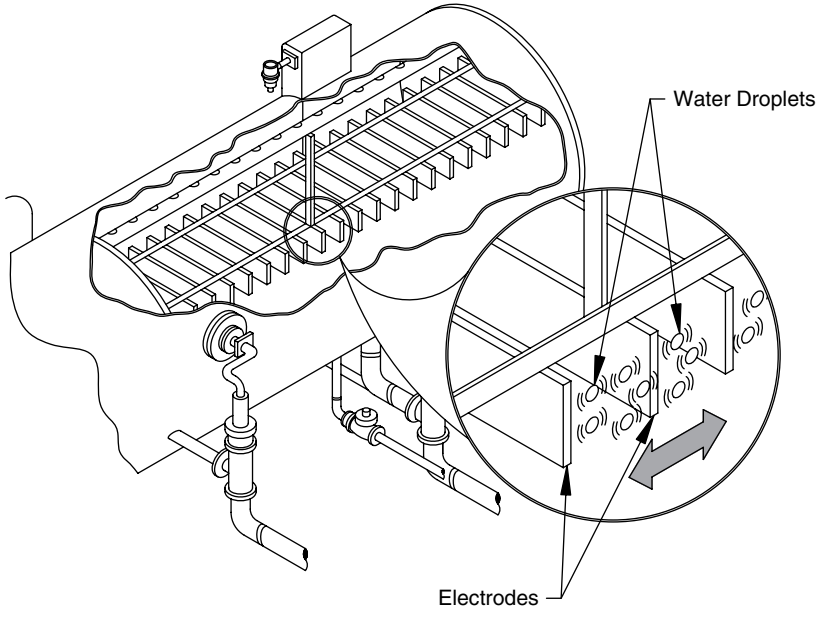


Figure 7-27. Effect of electrical charge on small water droplets in the emulsion.

Oil Dehydrators

The primary factor when designing coalescing units is the loading rate. Vessels are sized for a certain volume flow per unit time per square foot of grid area. Procedures for designing electrostatic grids have not been published. Since coalescence of water droplets in an electric field is so dependent on the characteristics of the particular emulsion to be treated, it is unlikely that a general relationship of water droplet size to use in the settling equations can be developed.

Field experience tends to indicate that electrostatic treaters are effective at reducing water content in the crude to the 0.2 to 0.5% level. This makes them particularly attractive for oil desalting operations. However, for normal crude treating, where 0.5 to 1.0% BS&W is acceptable, it is recommended that the vessel be sized as a horizontal heater-treater, neglecting any contribution from the electrostatic grids. By trial and error after installation, the electric grids may be able to allow treating to occur at lower temperatures or higher flow rates.

Figure 7-28 shows one variation of the electrostatic heater-treater where the vessel only contains the coalescing section with the electrostatic grid. Units configured in this manner are called “oil dehydrators.” These

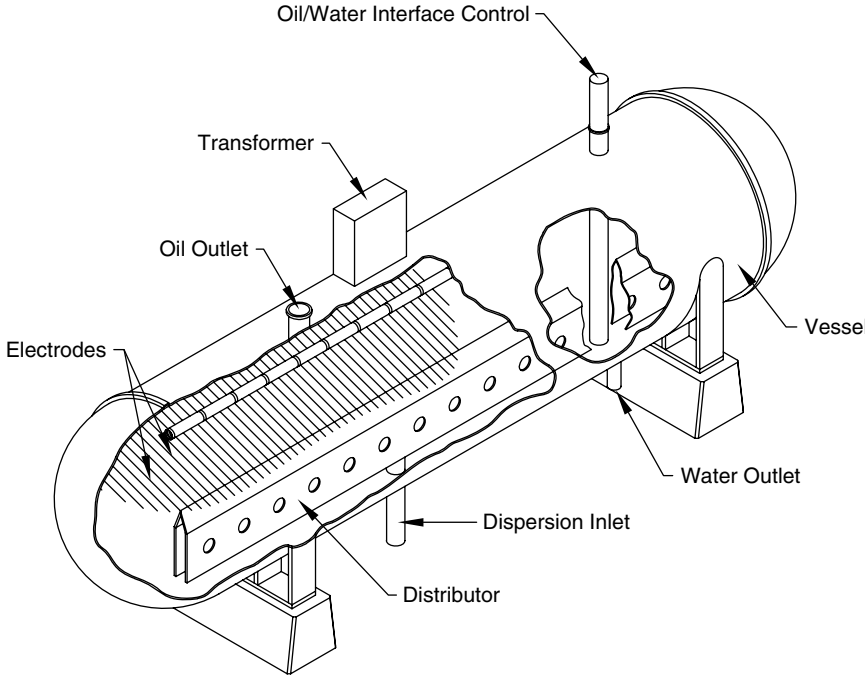


Figure 7-28. Cutaway of a liquid-packed horizontal oil dehydrator.

vessels must have separate upstream vessels for de-gassing, free-water removal and heating. This configuration should be considered when the volume to be treated exceeds 15,000 to 20,000 barrels per day.

Heater-Treater Sizing

Refer to the Equipment Sizing section that follows for details.

Emulsion Treating Theory

Introduction

Removing water from crude oil often requires additional processing beyond the normal oil–water separation process, which relies on gravity separation. Crude oil treating equipment is designed to break emulsions by coalescing the water droplets and then using gravity separation to

separate the oil and water. In addition, the water droplets must have sufficient time to contact each other and coalesce. The negative buoyant forces acting on the coalesced droplets must be sufficient to enable these droplets to settle to the bottom of the treating vessel. Therefore, it's important when designing a crude oil treating system to take into account temperature, time, viscosity of the oil, which may inhibit settling, and the physical dimensions of the treating vessel, which determines the velocity at which settling must occur.

When selecting a treating system, several factors should be considered to determine the most desirable method of treating the crude oil to contract requirements. Some of these factors are

- Stability (tightness) of the emulsion,
- Specific gravity of the oil and produced water,
- Corrosiveness of the crude oil, produced water, and associated gas,
- Scaling tendencies of the produced water,
- Quantity of fluid to be treated and percent water in the fluid,
- Paraffin-forming tendencies of the crude oil,
- Desirable operating pressures for equipment,
- Availability of a sales outlet and value of the associated gas produced.

A common method for separating this "water-in-oil" emulsion is to heat the stream. Increasing the temperature of the two immiscible liquids deactivates the emulsifying agent, allowing the dispersed water droplets to collide. As the droplets collide they grow in size and begin to settle. If designed properly, the water will settle to the bottom of the treating vessel due to differences in specific gravity.

Laboratory analysis, in conjunction with field experience, should be the basis for specifying the configuration of treating vessels. The purpose of this chapter is to present a rational alternative for those instances when laboratory data do not exist or, if it is desirable, to extrapolate field experience.

Emulsions

An emulsion is a stable mixture of oil and water that does not separate by gravity alone. In the case of a crude oil or regular emulsion, it is a dispersion of water droplets in oil. Normal, or regular, oil-field emulsions consist of an oil continuous or external phase and a water dispersed or internal phase. In some cases, where there are high water cuts, such as when a water-drive field has almost "watered out," it is possible to form reverse emulsions with water as the continuous phase and oil droplets as the internal phase. Complex or "mixed" emulsions have been

reported in low-gravity, viscous crude oil. These mixed emulsions contain a water external phase and have an internal water phase mixed in the oil dispersed phase. A stable or "tight" emulsion occurs when the water droplets will not settle out of the oil phase due to their small size and surface tension. Stable emulsions always require some form of treatment. The vast majority of oil treating systems deal with normal emulsions, which is the focus of this chapter.

For an emulsion to exist there must be two mutually immiscible liquids, an emulsifying agent (stabilizer), and sufficient agitation to disperse the discontinuous phase into the continuous phase. In oil production, oil and water are the two mutually immiscible liquids. When oil and water are produced from a well, the fluid stream also contains organic and inorganic materials. These contaminants are preferentially absorbed at the interface between the oil and water phases. Once the contaminants are absorbed at the interface, they form a tough film (skin) that impedes or prevents the coalescence of water droplets. Agitation, sufficient to disperse one liquid as fine droplets through the other, occurs as the well fluids make their way into the well bore, up the tubing, and through surface chokes, down-hole pumps, and gas lift valves. Turbulence caused by the pressure drop across the choke is the primary source of agitation for emulsion formation. However, elimination of the choke, used to control the flow rate of a well, is not a solution to the problem.

The degree of agitation and the nature and amount of emulsifying agent determine the "stability" of the emulsion. Some stable emulsions may take weeks or months to separate if left alone in a tank with no treating. Other unstable emulsions may separate into relatively pure oil and water phases in just a matter of minutes. The stability of an emulsion is dependent on several factors:

- The difference in density between the water and oil phases,
- The size of dispersed water particles,
- Viscosity,
- Interfacial tension,
- The presence and concentration of emulsifying agents,
- Water salinity,
- Age of the emulsion,
- Agitation.

Differential Density

The difference in density between the oil and water phases is one of the factors that determine the rate at which water droplets settle through the continuous oil phase. The greater the difference in gravity, the more

quickly the water droplets will settle through the oil phase. Heavy oils (high specific gravity) tend to keep water droplets in suspension longer. Light oils (low specific gravity) tend to allow water droplets to settle to the bottom of the tank. Thus, the greater the difference in density between the oil and water phases, the easier the water droplets will settle.

Size of Water Droplets

The size of the dispersed water droplets also affects the rate at which water droplets move through the oil phase. The larger the droplet, the faster it will settle out of the oil phase. The water droplet size in an emulsion is dependent upon the degree of agitation that the emulsion is subjected to before treating. Flow through pumps, chokes, valves, and other surface equipment will decrease water droplet sizes.

Viscosity

Viscosity plays two primary roles in the stability of an emulsion. First, as oil viscosity increases, the migration of emulsifying agents to the water droplet's oil–water interface is retarded. This results in larger water droplets being suspended in the oil, which in turn results in less stable emulsions in terms of numbers of small water droplets suspended in the oil. As oil viscosity increases, more agitation is required to shear the larger water droplets down to a smaller size in the oil phase. Thus, the size of the water droplets that must be removed to meet water cut specifications for a given treating system increases as viscosity increases. Second, as viscosity increases, the rate at which water droplets move through the oil phase decreases, resulting in less coalescence and increased difficulty in treating. As oil viscosity increases, the friction encountered by the water droplets moving through the continuous oil phase increases, which in turn impedes separation of the oil and water phases. Thus, higher oil viscosities tend to result in larger water droplets being formed, but impede their separation from the oil continuous phase. This latter effect tends to overpower the former effect making it harder to treat higher viscosity oils.

Interfacial Tension

Interfacial tension is the force that "holds together" the surfaces of the water and oil phases. When an emulsifying agent is not present, the interfacial tension between oil and water is low. When interfacial tension is low, water droplets coalesce easily upon contact. However, when

emulsifying agents are present, they increase the interfacial tension and obstruct the coalescence of water droplets. Anything that lowers the interfacial tension will aid in separation.

Presence and Concentration of Emulsifying Agents

Chemicals (demulsifiers) are normally used to reduce the interfacial tension. Chemical effectiveness is enhanced by mixing, time, and temperature. Adequate mixing and sufficient time are required to obtain intimate contact of the chemical with the dispersed phase. A certain minimum temperature is required to ensure the chemical accomplishes its function. Both viscosity reduction and effectiveness of chemical are dependent on the attainment of a certain minimum temperature. It may well be that the increase in chemical effectiveness is a result of the decrease in viscosity of the oil phase.

Water Salinity

The salinity of the water is a measure of the total dissolved solids in the water phase. As salinity of the water increases, the density of the water increases, which in turn increases the differential density between the water and the oil. The increase in differential density aids in separation of the oil and water phases. Small amounts of salt, or other dissolved solids, in the water phase will appreciably lower the interfacial tension and thus will decrease the difficulty of separating the two phases. To some degree, this phenomenon explains the difficulty of treating water–oil emulsions formed from soft water typically found in many steamflood operations, e.g., Caltex Duri Field Sumatra, Indonesia, and Chevron Texaco Bakersfield, California.

Age of the Emulsion

As emulsions age they become more stable and separation of the water droplets becomes more difficult. The time required to increase stability varies widely and depends on many factors. Before an emulsion is produced, the emulsifying agents are evenly dispersed in the oil. As soon as the water phase is mixed with the oil, the emulsifying agents begin to cluster around the water droplet to form a stable emulsion. While the initial stabilization may occur in a matter of a few seconds, the process of film development may continue for several hours. It will continue until the film

around the droplet of water is so dense that no additional stabilizer can be attracted, or until no stabilizer is left to be extracted from the oil. At such a time the emulsion has reached a state of equilibrium and is said to be aged. The older the emulsion, the more difficult it is to treat. Therefore, emulsion breaking or treating operations are often located as close to the wellhead as possible, so that emulsions formed during flow in the production tubing and wellhead equipment are not allowed to age before treatment.

Agitation

The type and severity of agitation applied to an oil–water mixture determine the water drop size. The more turbulence and shearing action present in a production system, the smaller the water droplets and the more stable the emulsion will be.

Internal and external properties of the stream will change throughout the life of production due to changes in formation characteristics and fluctuations in the ambient conditions encountered on the surface. This partially explains the ever-changing problems associated with emulsion treating. Little or no emulsion exists in oil bearing formations. Emulsions are formed during production on the fluid. The degree of emulsification is dependent on the agitation of the two phases by pumps, chokes, etc.

The above factors determine the "stability" of emulsions. Some stable emulsions may take weeks or months to separate if left alone in a tank with no treating. Other unstable emulsions may separate into relatively clean oil and water phases in just a matter of minutes.

Figure 7-29 shows a normal water-in-oil emulsion. The small water droplets exist within the oil continuous phase. Figure 7-30 shows a close-up of a "skin" (monomolecular film) of emulsifying agent surrounding a water drop, and Figure 7-31 shows two drops touching but unable to coalesce because of the emulsifying-agent "skin" surrounding each drop.

Emulsifying Agents

When studying emulsion stability, it may be helpful to realize that in a pure oil and pure water mixture, without an emulsifying agent, no amount of agitation will create an emulsion. If the pure oil and water are mixed and placed in a container, they quickly separate. The natural state is for the immiscible liquids to establish the least contact or smallest surface area. The water dispersed in the oil forms spherical drops. Smaller drops will coalesce into larger drops, and this will create a smaller interface area for a given volume. If no emulsifier is present, the droplets will

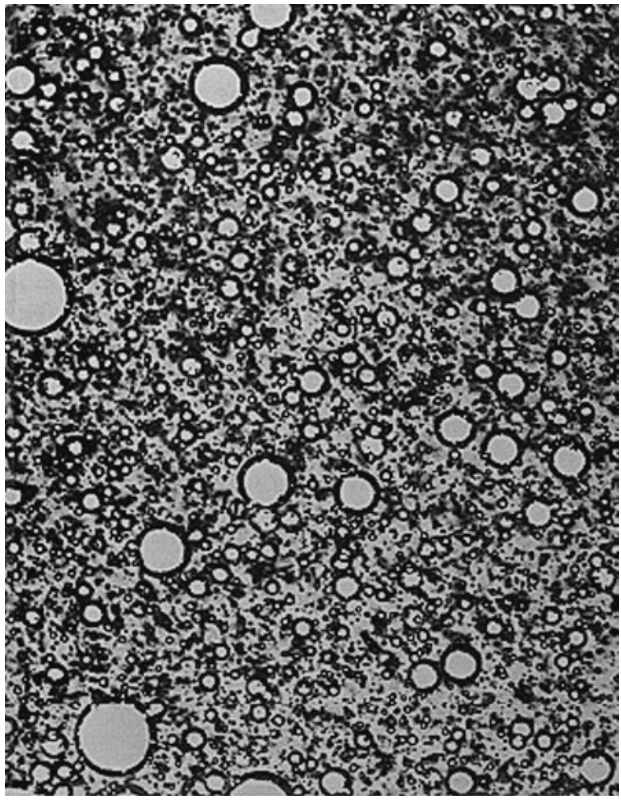


Figure 7-29. Photomicrograph of an oil–in-water emulsion.

eventually settle to the bottom, causing the smallest interface area. This type of mixture is a true "dispersion."

An emulsifying agent in the system is a material, which has a surface-active behavior. Some elements in emulsifiers have a preference for the oil, and other elements are more attracted to the water. An emulsifier tends to be insoluble in one of the liquid phases. It thus concentrates at the interface. There are several ways emulsifiers work to cause a dispersion to become an emulsion. The action of the emulsifier can be visualized as one or more of the following:

- It decreases the interfacial tension of the water droplet, thus causing smaller droplets to form. The smaller droplets take longer to coalesce into larger droplets, which can settle quickly.
- It forms a viscous coating on the droplets, which keeps them from coalescing into larger droplets when they collide. Since coalescence



Figure 7-30. Photomicrograph showing a close-up view of the emulsifying agent skin surrounding a water droplet.

is prevented, it takes longer for the small droplets, which are caused by agitation in the system, to settle out.

- The emulsifiers may be polar molecules, which align themselves in such a manner as to cause an electrical charge on the surface of the droplets. Since like electrical charges repel, two droplets must collide with sufficient force to overcome this repulsion before coalescence can occur.

Naturally occurring surface-active materials normally found in crude oil serve as emulsifiers. Paraffins, resins, organic acids, metallic salts, colloidal silts and clay, and asphaltenes (a general term for material with chemical compositions containing sulfur, nitrogen, and oxygen) are



Figure 7-31. Photomicrograph showing two droplets touching but unable to coalesce because of the emulsifying skin surrounding the droplets.

common emulsifiers in oil fields. Workover fluids and drilling mud are also sources of emulsifying agents.

The type and amount of emulsifying agent have an immediate effect on the emulsion's stability. It has been shown that the temperature history of the emulsion is also important as it affects the formation of paraffins and asphaltenes. The speed of migration of the emulsifying agent to the oil–water interface and the behavior in terms of the strength of the interface bond are important factors. An emulsion treated soon after agitation, or soon after the creation of paraffins and asphaltenes, can be less stable and easier to process if the migration of the emulsifier is incomplete. An aged emulsion may become more difficult to treat because the emulsifying agents have migrated to the oil–water interface.

Normally, the lower the crude viscosity and lighter the crude, the more rapid the aging process will be.

Demulsifiers

Emulsions can be resolved or broken thermally and/or chemically. When we chemically resolve an emulsion, we use a demulsifier or emulsion breaker. These two names are used interchangeably and describe the same chemical. Chemical demulsifiers sold under various trade names, such as Tretolite, Visco, Breaxit, etc., are highly useful in resolving emulsions. Demulsifiers act to neutralize the effect of emulsifying agents. Typically, they are surface-active agents and thus their excessive use can decrease the surface tension of water droplets and actually create more stable emulsions. In addition, demulsifiers for water-in-oil emulsions tend to promote oil-in-water emulsions; therefore, excessive chemical use may cause water treating problems.

Four important actions are required of a demulsifier:

- Strong attraction to the oil–water interface,
- Flocculation,
- Coalescence,
- Solid wetting.

When these actions are present, they promote the separation of oil and water. The demulsifier must have the ability to migrate rapidly through the oil phase to the droplet interface, where it must compete with the more concentrated emulsifying agent. The demulsifier must produce an attraction for similar droplets. In this way large clusters of droplets gather, which, under a microscope, appear like bunches of fish eggs. The oil will take on a bright appearance since small droplets are no longer present to scatter the light rays. At this point the emulsifier film is still continuous. If the emulsifier is weak, the flocculation force may be enough to cause coalescence. This is not true in most cases, and the demulsifier must therefore neutralize the emulsifier and promote a rupture of the droplet interface film. This is the opener that causes coalescence. With the emulsion in a flocculated condition, the film rupture results in rapid growth of water drop size.

The manner in which the demulsifier neutralizes the emulsifier depends upon the type of emulsifiers. Iron sulfides, clays, and drilling muds can be water wet, causing them to leave the interface and be diffused into the water droplet. Paraffins and asphaltenes could be dissolved or altered to make their films less viscous so they will flow out of the way on collision or could be made oil wet so they will be dispersed in the oil.

It would be unusual if one chemical structure could produce all four desirable actions. A blend of compounds is therefore used to achieve the right balance of activity.

The demulsifier selection should be made with the process system in mind. If the treating process is a settling tank, a relatively slow-acting compound can be applied with good results. On the other hand, if the system is a chemical-electric process where some of the flocculation and coalescing action is accomplished by an electric field, there is need for a quick-acting compound, but not one that must complete the droplet-building action.

As field conditions change, the chemical requirements can change. If the process is modified, e.g., very low rates on electrostatic units, the chemical requirements can change. Seasonal changes bring paraffin-induced emulsion problems. Workovers contribute to solids and acid/base contents, which alters the emulsion stability. So no matter how satisfactory a demulsfier is at one point in time, it may not be satisfactory over the life of the field.

The cost to dehydrate crude oil chemically is a function of several factors. First, the ratio of oil to water is important—it is generally easier and, hence, less costly to dehydrate crudes with very high water cuts. Next, the severity of the emulsion is important. A "tight" emulsion consisting of small droplets is much more difficult to break—it has a higher surface area to volume ratio than a "loose" emulsion and, hence, the demulsifier has more work to do to seek out the interface. Next, the residence time available for separation is important. Small residence times inhibit complete separation of water droplets from oil. This may lead to re-entrainment of water as the crude goes from one processing stage to another. The result is ineffective dehydration. Higher temperatures result in lower oil phase viscosities, which enable the demulsifier to migrate to the oil–water interface faster and for coalesced water droplets to drop out easier.

Last, the dehydration cost is directly influenced by chemical selection. Poor chemical selection will result in a non-optimized treatment, which will mean higher costs. Chemical selection is not a simple process—it is best left to suppliers. However, one can assist in the process by providing on-site testing opportunities for chemical suppliers to select the best chemicals for specific applications.

Bottle Test

This is one of the most common, yet least understood, of all the chemical selection tests. Emulsion-breaking chemicals are most commonly tested with a bottle test, which involves mixing various chemicals with samples

of the emulsion and observing the results. Such tests are effective in eliminating some chemicals and selecting those that appear to be more efficient. Bottle tests also provide an estimate of the amount of chemical required and an estimate of the settling time required for a treating vessel. Bottle tests should be performed on a representative sample as soon as the sample is obtained because of the possible detrimental effects of aging. These tests should be performed at conditions that are as close to field treating conditions as possible. Synthetic water should not be used in place of produced water in bottle tests because the produced water may have very different properties, and it may contain impurities that are not present in the synthetic water.

While candidate chemicals and approximate dosages can be determined in bottle tests, the dynamic nature of the actual flowing system requires that several candidates be field-tested. In actual conditions, the emulsion undergoes shearing through control valves, coalescence in flow-through pipes, and changes to the emulsion that occur inside the treating vessel as a result of inlet diverters, water-wash sections, etc. Static bottle tests cannot model these dynamic conditions.

As well as determining the potential dehydration performance of a demulsifier, the bottle test can also be used to investigate chemical incompatibilities. Here, the performance of a demulsifier is evaluated on a chemical-free sample and then on a sample of crude, which includes the other production chemicals at their respective dose rates. The change in performance, if any, is recorded and the chemical discarded if incompatibilities exist. Another aspect of incompatibility may also be determined, namely, in which order the chemicals should be injected. If the bottle tester is experienced, this order of injection, which will produce subtle changes in the bottle test results, can be investigated and an optimum injection order determined.

Field Trial

Having selected a promising demulsifier candidate, a field trial should be carried out to test the chemical's ability to operate in a dynamic system. In the field test, the flexibility of the demulsifier to process changes can be established. This data will be useful when the chemical is used in full-scale operation. In most field trial situations, the demulsifier being tested is first used in conjunction with a test separator system. This enables the supplier to look at the response of the chemical to one or more wells and to provide the tester an idea of the true field dosage. If this preliminary scenario is successful, the chemical can then be dosed into the full system and optimized for different well configuration and flow rates. In the field

trial, the chemical's response to system upsets can be determined and, hence, an operating response can be set.

Field Optimization

After a successful field trial, a full-scale field optimization is carried out. Here, the chemical performance is monitored routinely as are the possible side effects of under-or overdosing, such as separator interface build-up. It may be that if the field produces through two or more platforms, injection locations and dose rates may need to be optimized for each location.

Changing the Demulsifier

As crude characteristics change over the life of a field, the performance of the demulsifier chemical will change also. Typically, when fields first produce water, the emulsions formed are difficult to break. As the field ages and the water cut increases, the stability of the emulsion and even the emulsifying agents themselves may change. Hence, it is usual to investigate demulsifier performance every 2 to 3 years. In some cases where a step change in water cut is experienced, it may be prudent to investigate demulsifier performance more frequently. In most cases a quick bottle test is all that is required to determine if the current chemical is still optimum. If not, a full bottle test to find a more effective chemical can be undertaken.

Demulsifier Troubleshooting

The most common problem with demulsifiers is overdosing. Poor treatment, dirty water, and interface pad build-up are all symptoms of overdosing an optimum chemical. Overdosing can occur by a step increase in dose rate, e.g., going from 5 to 20 ppm, or by a gradual accumulation of chemical in the system. The latter is most often seen in high water cut systems where a small change from optimum can result in dirty water. The gradual accumulation of chemical usually occurs at the separator interface and is often difficult to detect. However, highly variable water quality caused by intermittent interface sloughing is often a clue to this scenario.

Other problems with demulsifiers can be that their viscosity changes with temperature. Most demulsifiers are viscous chemicals whose ability to be pumped can drop dramatically with reduced temperature. If this

is the case, it may be prudent to ask the chemical supplier to produce a "winterized" version of the chemical. This is often done by reducing the percentage of active ingredient and adding more solvent carrier. If this is the solution, the dose rates will need to be reoptimized for best performance.

Another common problem with demulsifiers is their apparent lack of treatment "range." It is not uncommon for a field demulsifier to have a different performance standard for different wells within a field. In some cases "rogue wells" may exist, which are basically untreatable by the optimum demulsifier for the rest of the system. In these cases two demulsifiers may be used or the original demulsifier may be injected at a higher dose rate or even downhole in the rogue well. The bottle test will often indicate rogue wells and their best treatment solution.

Demulsifiers and corrosion inhibitors are often the cause of poor dehydration performance. Corrosion inhibitors are surfactant chemicals that often act as emulsifying agents, thus making the demulsifier work harder. In cases of conflict, it is usually easier to blend a new demulsifier or change the injection points of the chemicals than it is to replace the corrosion inhibitor. However, in some North Sea fields the opposite was true. Corrosion inhibitor replacement was the best way to deal with the incompatibility problem.

As there are no online analyzers for demulsifier performance, one must monitor the facilities for changes in water or crude quality that may be attributed to poor demulsifier performance. Chemical suppliers can help here by giving us the anticipated system response to incompatibilities and over- or underdosing. They should get this information from the bottle test and the demulsifier field trial.

Emulsion Treating Methods

General Considerations

Treating processes and equipment should not be selected until the physical characteristics of the oil and water have been determined and a study of the effect of available chemicals on the emulsion has been made.

The water remaining in the crude after the free water has settled out is considered to be in an emulsified state. Emulsified oil is removed by one or more treating processes. Treating refers to any process designed to separate crude oil from water and foreign contaminates carried along with it from the reservoir. Emulsion treating processes require some combination of the following: chemical addition, settling time, heat, and electrostatic coalescing.

Chemical Addition

The purpose of treating chemicals is to induce coalescence so that the oil and water will separate rapidly. Surface-active agents are absorbed at the oil–water interface, rupture the tough film (skin) surrounding the water droplets, and/or displace the emulsifying agent and force the emulsifying agent back into the oil phase.

There is no universal chemical that will break all emulsions equally well. Determining the correct chemical to use is commonly done by a chemical sales representative using a bottle test (discussed earlier in this chapter).

Amount of Chemical

The amount of chemical required cannot be predicted accurately from bottle tests. The only reliable method of determining the amount of chemical to use is to run tests in the field. When changing to a new chemical or starting up a new treating system, one must first use an excess (1 quart per 100 barrels) of chemical and then gradually reduce the amount to the minimum amount that will produce the desired results. When determining the amount of chemical to add, one must make certain no other changes are being made in the facility. Temperature should remain constant during the test; otherwise, it is impossible to determine which change, chemical or temperature, has caused a certain effect.

The amount of chemical added can vary from 1 gallon per 150 barrels to 1 gallon per 1000 barrels. Concentrations higher than 1 gallon per 250 barrels should be investigated for possible errors such as incorrect chemical being used or the method of chemical addition being wrong. Too much chemical can be the cause of a very tight emulsion that will not break down.

Chemicals should be added continuously as possible during the entire production period and at a rate related to the production rate. Even though some residual chemical is held in the treater or gunbarrel, chemicals cannot be batched and be expected to do an adequate treatment. Chemicals cannot act properly unless they are thoroughly mixed with the emulsion. The farther upstream, a minimum of 200 feet, from a treater or gunbarrel the chemicals are added the better the mixing and thus the better the treatment. The ideal location for injection is at the manifold before the fluid enters a separator. In some cases an emulsion that is difficult to treat may break quite easily if a chemical pump is set at the well. It is not uncommon for one well in a field to cause most of the trouble. Setting

a pump at this well can increase efficiency and reduce the amount of chemicals required to break the emulsion.

Bottle Test Considerations

The best demulsifier is the compound that results in the most rapid and complete separation of the phases at a minimum concentration. The important characteristics in the bottle test will be dictated by the production needs and the behavior of the system.

Water Drop-Out Rate

In high water volume systems a chemical that creates a fast water drop-out rate is necessary to make the system function as designed. When free-water knockouts are used, the speed of water drop-out may become the most important factor. Chemicals with fast water drop-out characteristics are sometimes incomplete in treatment requiring other chemicals for final separation. In low water volume systems (fields with facilities having longer than normal residence times), the rate of water drop-out may be of minor significance in selecting the best demulsifier. In all cases, the rate of water drop-out should be noted and recorded.

Sludge

When oil, water and sediment collect together without breaking to separate water, oil and solid phases, the result is called sludge. Sludge is stabilized by finely divided solids and other contaminates to form pads that cause a secondary emulsion located between the oil and water. Depending upon the system and sludge stability, interface sludge may or may not cause a problem. Loose interface sludge can be detected by swirling the test bottle about its axis, and if the material is loose, it will break.

Interface

The desired interface is one that has shiny oil in contact with the water (mirror interface). The interface, when using a new chemical, should be as good as, if not better than, that formed by the chemical being replaced.

Water Turbidity

The turbidity (clarity) of the water is very difficult to interpret in the bottle test and correlate to facility behavior. When the chemical effects

in the bottle are pronounced and reproducible, some correlation can be expected. Clear water is definitely the desired result.

Oil Color

Emulsions have a hazy appearance when compared to the bright color of treated oil. As a crude oil emulsion separates, the color tends to brighten. Brightening of oil can be encouraging, but it can also be deceptive if taken as the sole qualification for chemical selection. While bright color is no guarantee of a successful chemical, lack of it assures that the compound is not worthy of further consideration.

Centrifuge Results

An important quality in the final evaluation is the centrifuge results. It is always good practice to make a centrifuge grind out to accurately determine the final amount of BS&W entrained in the oil.

Chemical Selection

A thorough understanding of the treating equipment and its contribution to the treatment are necessary before chemical selection can be made. If little agitation is available, a fast-acting chemical is necessary. If a free-water knockout vessel is used, the water drop-out rate will be very important. If heat is unavailable, the chemical must work at ambient temperatures. Different types of vessels require different chemical actions.

Settling Tank or "Gunbarrel"

Speed is not too important since both tanks usually have a high volume-to-throughput ratio. The chemical may continue acting over a relatively long period. An interface layer often develops but usually stabilizes at some acceptable thickness. An interface layer in a gunbarrel sometimes aids the treating process in that it acts as a filter for solids and unresolved emulsions. Fresh oil containing a demulsfier passing up through the interface layer helps treat the interface and prevents an excessive build-up.

Vertical Heater-Treater

The speed of chemical action is important since the volume-to-throughput ratio is usually lower than a gunbarrel or settling tank. With the higher throughput, it is harder to stabilize an interface layer, so more complete

treatment is necessary in a shorter time period. Solids control may be important in controlling the interface.

Horizontal Heater-Treater

The speed of chemical action is important due to its high throughput. The large interface area and shallow depth require that the interface be fairly clean. Since this treater can tolerate only very little interface accumulation, the chemical treatment must be complete. Since solids tend to collect at the interface, the chemical must also effectively de-oil any solids so that they may settle out by gravity.

Settling Time

Following the addition of treating chemicals, settling time is required to promote gravity settling of the coalescing water droplets. Figure 7-32 illustrates the effects of time on coalescence. Emulsion treating equipment designed to provide sufficient time for free water to settle include three-phase separators, free-water knockouts, heater-treaters, and gunbarrels

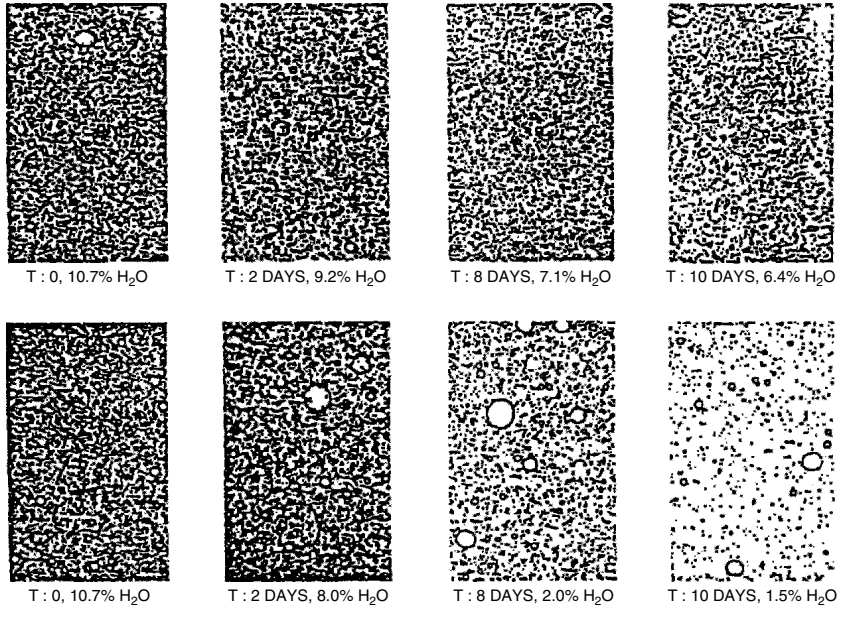


Figure 7-32. Effect of time on coalescence. Top: emulsion without chemicals. Bottom: emulsion with demulsifier added.

with an internal or external gas boot. The time necessary for free water to settle is affected by differential density of the oil and water, viscosity of the oil, size of the water droplets, and relative stability of the emulsion.

Coalescence

The process of coalescence in oil treating systems is time-dependent. In dispersions of two immiscible liquids, immediate coalescence seldom occurs when two droplets collide. If the droplet pair is exposed to turbulent pressure fluctuations, and the kinetic energy of the oscillations induced in the coalescing droplet pair is larger than the energy of adhesion between them, the contact will be broken before coalescence is completed.

Experiments with deep-bed gravity settlers indicate that the time to "grow" a droplet size due to coalescence can be estimated by the following equation:

$$t = \frac{\pi}{6}\left(\frac{d^j - (d_o)^j}{\phi K_s}\right), \tag{7-1}$$

where

d_o = initial droplet size, microns,
d = final droplet size, microns,
ϕ = volume fraction of the dispersed phase,
K_S = empirical parameter for the particular system,
j = an empirical parameter that is always larger than 3 and dependent on the probability that the droplets will "bounce" apart before coalescence occurs,
t = time required to grow a droplet of size d, min.

When the energy of oscillations is very low so that "bouncing" of droplets approaches 0, j approaches 3. Assuming a value of 4, the minimum time required to obtain a desired particle diameter can be expressed as

$$t = \frac{\pi}{6}\left(\frac{d^4 - (d_o)^4}{\phi K_s}\right). \tag{7-2}$$

Assuming d_o is small relative to the droplet size we wish to "grow" by coalescence in our gravity settler, Eq. (7-2) can be approximated:

$$t = \frac{d^4}{2\phi K_s}. \tag{7-3}$$

The following qualitative conclusions for coalescence in a gravity settler can be drawn from this relationship:

- A doubling of residence time increases the maximum size drop grown in a gravity settler less than 19%. If $j > 4$, the growth in droplet diameter will be even slower. For this reason, after an initial short coalescence period, adding additional retention time is not very effective for making the oil easier to treat. Very often engineers will attribute improved performance in large gunbarrel tanks to retention time when it is really due to slowing the oil velocity. This allows smaller droplets of water to separate in accordance with Stokes' law.
- The more dilute the dispersed phase, the greater the residence time needed to "grow" a given particle size will be. That is, coalescence occurs more rapidly in concentrated dispersions. This is the reason that oil is "water-washed" by entering the treating vessel below the oil–water interface in most gunbarrels and treaters. Flocculation and coalescence therefore occur most effectively at the interface zone between oil and water.

Viscosity

The viscosity of the oil continuous phase is extremely important in sizing a treater. Stokes' law, used to determine the settling velocity of a water droplet settling through the continuous oil phase, includes the oil viscosity. As the oil viscosity increases, the settling velocity of a given droplet decreases. This requires that the treater size be increased.

The oil viscosity also affects coalescence of the water droplets. As the oil viscosity increases, there is more resistance to random motion of the water droplets. Therefore, the droplets do not move as fast or as far. This decreases the energy and the frequency of droplet collisions. Thus, it is more difficult to grow large water droplets in the vessel. As the oil viscosity increases, it is also more difficult to shear the oil droplets that coalesce in the piping leading to the vessel and in the water-wash section of the vessel. The net effect is that increasing the oil viscosity increases the size of the minimum water droplet that must be removed.

By far the best situation is to have oil viscosity versus temperature data for a particular oil to be treated. Alternately, data from other wells in the same field can usually be used without significant error. This viscosity versus temperature data may be plotted on special ASTM graph paper. Such plots are usually straight lines, unless the oil has a high cloud point. The viscosity may then be predicted at any temperature.

Laboratory testing of a particular oil at various temperatures is the most reliable method of determining how an oil behaves. ASTM D 341

outlines a procedure where the viscosity is measured at two different temperatures and then, either through a computation or on special graph paper, the viscosity at any other temperature can be obtained.

As a rule, with crude of 30°API and higher, the viscosity is so low that normally it may be difficult to find any information on file regarding a specific crude viscosity. Between 30°API and 11°API, the viscosity becomes more important, until in some cases it is impossible to process very low gravity crudes without a diluent to reduce the viscosity. The use of a diluent is not unusual for crude oil below 14°API.

With virtually any crude oil the viscosity change with temperatures can be an excellent guide to minimum crude processing temperatures. An ASTM chart of the viscosity versus temperature is useful to detect the paraffin formation or cloud point of the crude as shown in Figure 7-33. This normally establishes a minimum temperature for the treating process. There are examples of 30°API crude and higher that have pour points of 80 to 90°F (27 to 32°C). Crude oils of this type are common in the Uinta and Green River Basins of the United States as well as in Southeast Asia.

If no data are available, the oil viscosity may be estimated by a variety of methods from the temperature and oil gravity. These methods, however, are not very accurate, as the viscosity is a function of the oil composition and not strictly the oil gravity. In fact, two oils with the same gravity at the same temperature may have viscosities that are orders of magnitude apart.

In the absence of any laboratory data, Figures 7-34 and 7-35 may be used to estimate oil viscosities. Additional correlations that can be used to estimate crude viscosity given its gravity and temperature are discussed in Chapter 3.

Heat Effects

Adding heat to the incoming oil–water stream is the traditional method of separating the phases. The addition of heat reduces the viscosity of the oil phase, allowing more rapid settling velocities in accordance with Stokes' law of settling. For some emulsifying agents, such as paraffins and asphaltenes, the addition of heat deactivates, or dissolves, the emulsifier and thus increases its solubility in the oil phase. Treating temperatures normally range from 100–160°F(38–70°C). In treating of heavy crudes the temperature may be as high as 300°F(150°C).

Adding heat can cause a significant loss of the lower-boiling-point hydrocarbons (light ends). This results in "shrinkage" of the oil, or loss of volume. The molecules leaving the oil phase may be used as fuel, vented, or compressed and sold with the gas. Even if they are sold with the gas,

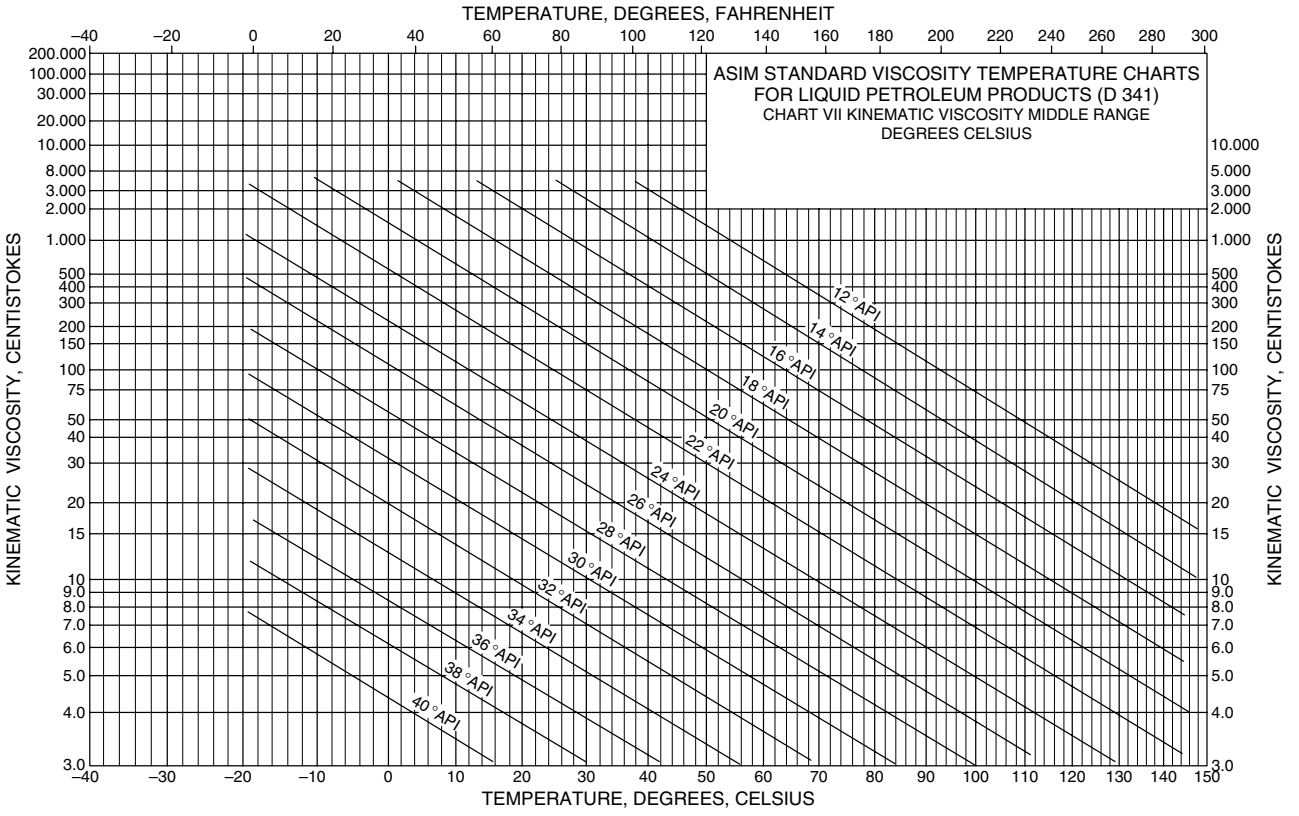


Figure 7-33. Viscosity versus temperature for several crude oils. (Courtesy of ASTM D-341.)

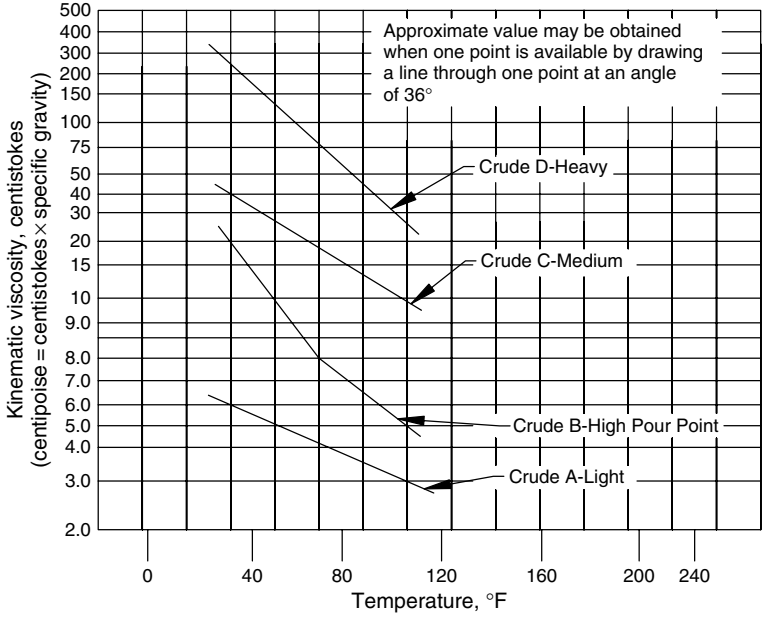


Figure 7-34. Typical oil viscosity versus temperature and gravity for estimating purposes, field units.

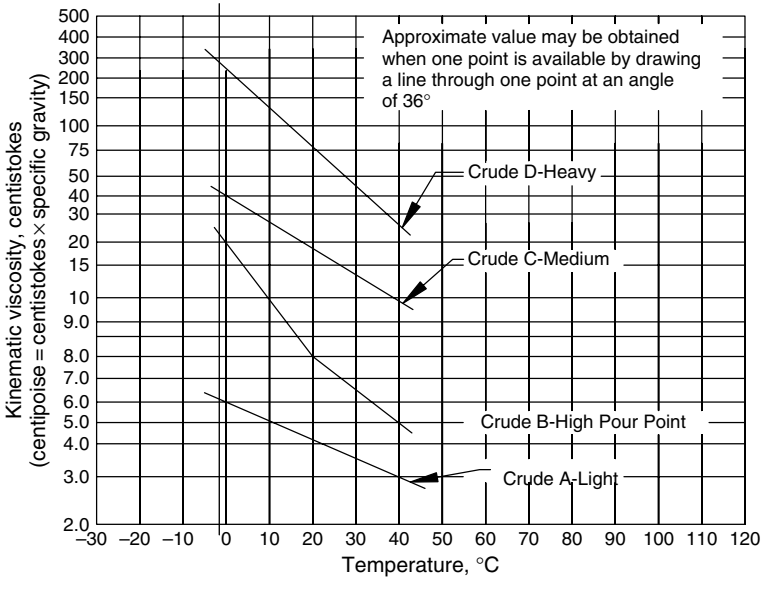


Figure 7-35. Typical oil viscosity versus temperature and gravity for estimating purposes, SI units.

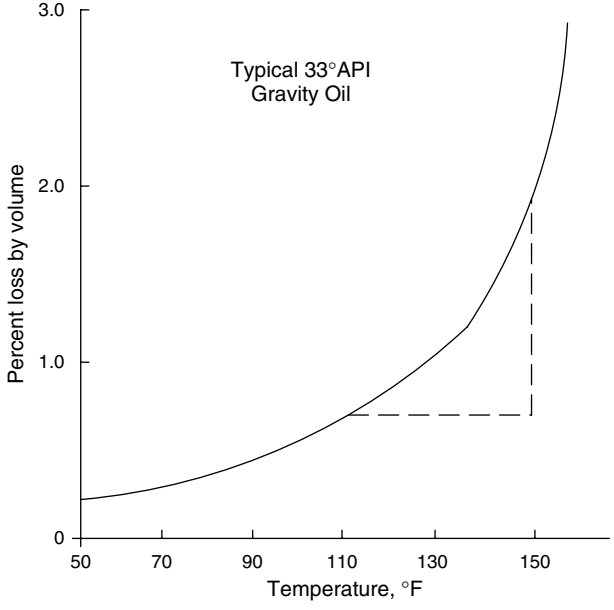


Figure 7-36. Percent loss by volume as a function of temperature for a 33°API gravity crude oil.

there will probably be a net loss in income realized by converting liquid volume into gas volume. Figure 7-36 shows the amount of shrinkage that may be expected from a typical 33°API gravity crude oil.

Increasing the temperature at which treating occurs has the disadvantage of making the crude oil that is recovered in the storage tank heavier and thus decreasing its value. Because the light ends are boiled off, the remaining liquid has a lower API gravity. Figure 7-37 shows the API gravity loss for a typical crude oil.

Increasing the temperature may lower the specific gravity, at the treater operating pressure, of both the oil to be treated and the water that must be separated from it. However, depending on the properties of the crude, it may either increase or decrease the difference in specific gravity as shown in Figure 7-38.

In most cases, if the treating temperature is less than 200°F (93°C), the difference between the oil and water specific gravities (ΔSG) is constant and thus can be neglected. The variation of oil specific gravity with temperature is approximately as shown in Figure 7-39. The specific gravity for water is given in Figure 7-40.

Finally, it takes fuel to provide heat, and the cost of fuel must be considered. Thus, while heat may be needed to treat the crude adequately,

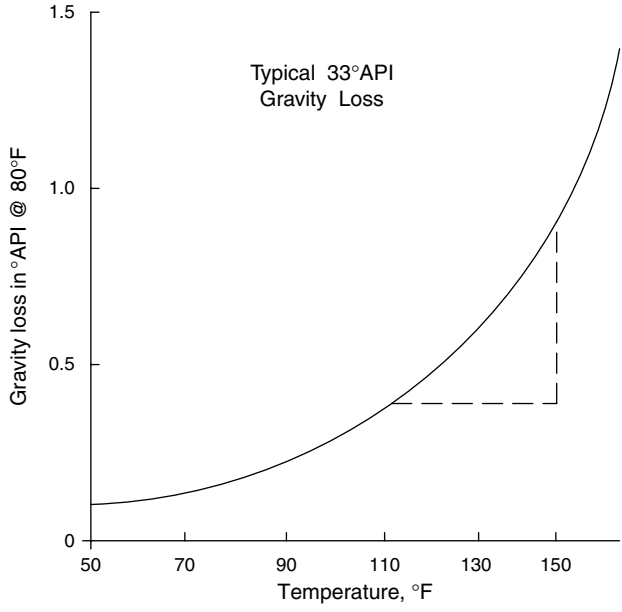


Figure 7-37. API gravity loss as a function of temperature for a 33°API gravity crude oil.

the less heat that is used, the better. Using data from a 1983 study, Table 7-4 illustrates the overall economic effect of treating temperature for a lease that produces 21,000 BOPD (139 m^3/hr) of a 29°API crude.

The gas liberated when crude oil is heated may create a problem in the treating equipment if the equipment is not properly designed. In vertical heater-treaters and gunbarrels the gas rises through the coalescing section. If much gas is liberated, it can create enough turbulence and disturbance to inhibit coalescence. Perhaps more important is the fact that the small gas bubbles have an attraction for surface-active material and hence for the water droplets. The bubbles thus have a tendency to keep the water droplets from settling and may even cause them to carry over to the oil outlet.

The usual oil-field horizontal heater-treater tends to overcome the gas liberation problem by coming to equilibrium in the heating section before introducing the emulsion to the settling-coalescing section. Some large crude processing systems use a fluid-packed, pump-through system that keeps the crude well above the bubble point. Top-mount degassing separators above electrostatic coalescers have been used in some installations.

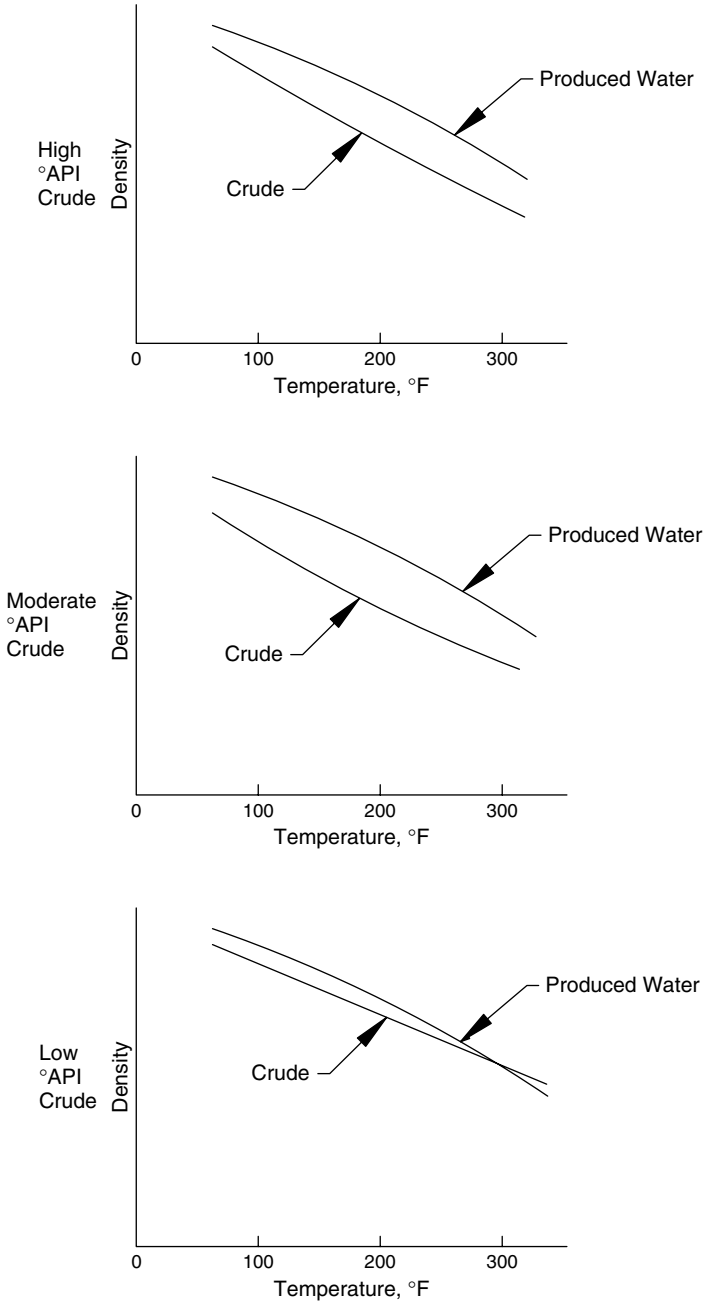


Figure 7-38. Relationship of specific gravity and temperature for a high, moderate, and low °API gravity crude.

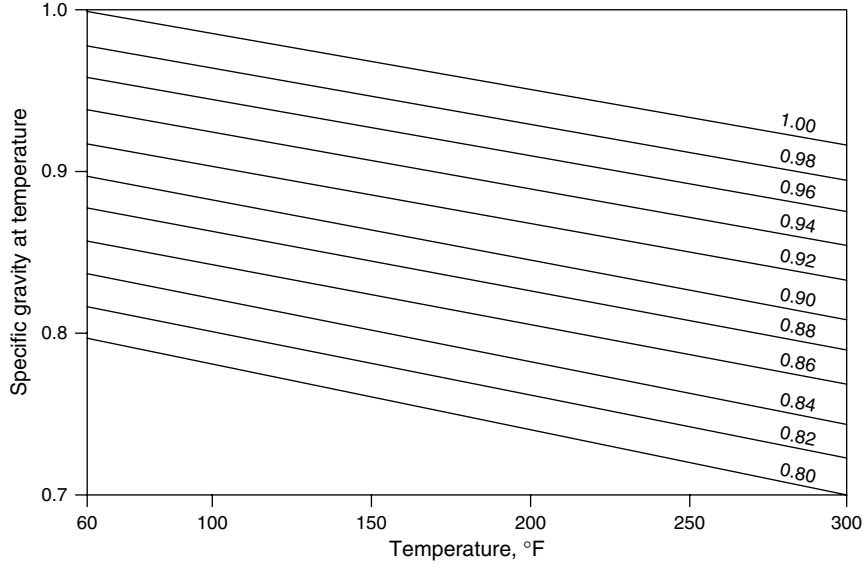


Figure 7-39. Variation of specific gravity of petroleum fractions with temperature. (Adapted from the GPSA *Engineering Data Book*.)

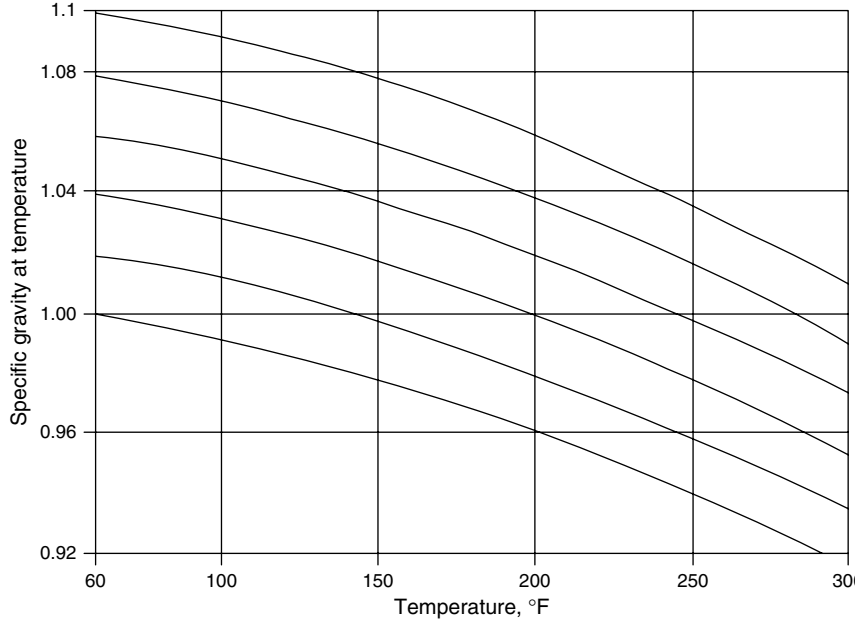


Figure 7-40. Variation of specific gravity of water with temperature.

Table 7-4
Economic Effect of Treating at a Higher Temperature for a Specific Field

Increase in NGL Value: Component	Volume for 120°F Treater Temperature	Volume for 100°F Treater Temperature	Difference	Price* per Unit	Change NGL Rev
Methane	163 mcfh	162 mcfh	1 mcfh	$2.44	2.44/hr
Ethane	1,835 gal/hr	1,802 gal/hr	33 gal/hr	0.232	7.66/hr
Propane	1,653 gal/hr	1,527 gal/hr	126 gal/hr	0.439	55.31/hr
Butane	1,086 gal/hr	930 gal/hr	156 gal/hr	0.672	104.83/hr
Pentane+	1,251 gal/hr	968 gal/hr	283 gal/hr	0.749	211.97/hr
					$382.21/hr
				$/Day	
		Total NGL Revenue Gain		$9,173.04	
		Net Value to Producer		$3,057.68	
Volume Shrinkage 317 BOPD × $19.52/bbl				(6,187.84)	
API Gravity Loss: 20,931 BPD × $0.15/bbl × 7/10 (API change) =				(2,197.75)	
Fuel Cost 125 Mcfd × $2.44/Mcf =				(306.00)	
TOTAL LEASE REVENUE LOSS:				$(5,633.91)	

Source: V. L. Heiman et al.: "Maximize Revenue by Analyzing Crude Oil Treating" Society of Petroleum Engineers of AIME, SPE 12206 (October 1983).
* February 1983 prices.

If properly and prudently done, heating an emulsion can greatly benefit water separation. However, if a satisfactory rate of water removal can be achieved at the minimum temperature delivered into a process, there may be no reason to suffer the economic penalties associated with adding heat.

Electrostatic Coalescers

Coalescing of the small water drops dispersed in the crude can be accomplished by subjecting the water-in-oil emulsion to a high-voltage electrical

field. When a non-conductive liquid (oil) containing a dispersed conductive liquid (water) is subjected to an electrostatic field, the conductive particles or droplets are caused to combine by one of three physical phenomena:

- The droplets become polarized and tend to align themselves with the lines of electric force. In so doing, the positive and negative poles of the droplets are brought adjacent to each other. Electrical attraction brings the droplets together and causes them to coalesce.
- Droplets are attracted to an electrode due to an induced charge. In an A-C field, due to inertia, small droplets vibrate over a larger distance than larger droplets promoting coalescence. In a D-C field the droplets tend to collect on the electrodes, forming larger and larger drops until eventually they fall by gravity.
- The electric field tends to distort and thus weaken the film of the emulsifier surrounding the water droplets. Water droplets dispersed in oil and subjected to a sinusoidal alternating-current field will be elongated along the lines of force during the first half cycle. As they are relaxed during the low-voltage portion, the surface tension will pull the droplets back toward the spherical shape. The same effect is obtained in the next half of the alternating cycle. The weakened film is thus more easily broken when droplets collide, making coalescence more likely.

Whatever the actual mechanism, the electric field causes the droplets to move about rapidly in random directions, which greatly increases the chances of collision with another droplet. When droplets collide with the proper velocity, coalescence occurs.

The attraction between water droplets in an electric field is given by

$$F = \frac{K_s \varepsilon^2 (d_m)^6}{S^4} \text{ (with } S \geq d_m\text{)}, \tag{7-4}$$

where

F = attractive force between droplets,
K_S = constant for system,
ε = voltage gradient,
d_m = diameter of droplets,
S = distance between droplets.

This equation indicates that the greater the voltage gradient is, the greater the forces causing coalescence will be. However, experimental data show that at some gradient the water droplet can be pulled apart and a strong emulsion can be developed. For this reason, electrostatic treaters

are normally equipped with a mechanism for adjusting the gradient in the field.

Water Droplet Size and Retention Time

The droplet diameter is the most important single parameter to control to aid in water settling since this term is squared in Stokes' law's settling equation. A small increase in diameter will create a much larger increase in settling velocity. Thus, in sizing treating equipment, it is necessary to predict a droplet diameter, which must be separated from the oil to meet a desired BS&W specification.

It would be extremely rare to have laboratory data of droplet coalescence for a given system. Qualitatively, we would expect droplet size to increase with retention time in the coalescing section and with heat input, which excites the system, leading to more collisions of small droplets. Droplet size could be expected to decrease with oil viscosity, which inhibits the movement of the particles and decreases the force of the collision. While it may be possible to predict the droplet size at the inlet to the treater, the shearing that occurs at the inlet nozzle and inlet diverter coupled with the coalescence that occurs at the oil–water interface cannot be determined. The treater represents a dynamic process, which cannot be adequately simulated by static laboratory tests.

The coalescence equation indicates that the oil–water interface zone is where nearly all of the coalescence occurs. Except for providing some minimal time for initial coalescence to occur, increasing retention time in a crude oil treating system may not be very cost-effective. Consequently, in most systems one would not expect retention time to have a significant impact on increasing the water droplet diameter.

The effect of temperature on water droplet size distribution is small. The temperature does, however, have a large effect on the oil viscosity. Since temperature and retention time have relatively small effects, an empirical relationship can be proposed relating droplet size distribution to oil viscosity alone. This relationship assumes sufficient retention time has been provided so initial coalescence can occur. Typically, retention times vary from 10 to 30 minutes, but values outside this range are also common.

If the water droplet size distribution in the oil to be treated were known, it would be possible to predict the size of droplets that must be removed to assure that a specific amount of water remains in the treated oil. Therefore, a relationship exists between the design BS&W content of the treated oil and the droplet size that must be removed for a set droplet size distribution. Since the droplet size distribution is a function

of viscosity as stated above, the droplet size to be removed is related to both the required BS&W and the oil viscosity.

Treater Equipment Sizing

General Considerations

The major factors controlling the sizing of emulsion treating equipment are

- Heat input required,
- Gravity separation considerations,
- Settling equations,
- Retention time equations,
- Water droplet size.

Heat Input Required

The heat input and thus the fuel required for treating depend on the temperature rise, amount of water in the oil, and flow rate. Heating water requires about twice as much energy as it does to heat oil. For this reason, it is beneficial to separate any free water from the emulsion to be treated with either a free-water knockout located upstream of the treater or an inlet free-water knockout system in the treater itself.

Assuming that the free water has been separated from the emulsion, the water remaining is less than 10% of the oil, and the treater is insulated to minimize heat losses, the required heat input can be determined from

Field Units

$$q = 16 Q_o \Delta T \; [0.5\,(\mathrm{SG})_o + 0.1], \tag{7-5a}$$

SI Units

$$q = 1100 Q_o \Delta T \; [0.5\,(\mathrm{SG})_o + 0.1], \tag{7-5b}$$

where

q = heat input, BTU/hr (kW),
Q_o = oil flow rate, BOPD (m^3/hr),
ΔT = increase in temperature, °F(°C),
SG_o = specific gravity of oil relative to water.

Derivation of Equations (7-5a) and (7-5b)

The general heat transfer equation is expressed by

$$q = WC\Delta T, \tag{7-6}$$

where

q = heat (BTU/hr) (kW),
W = flow rate, lb/hr (kg/hr),
C = specific heat (BTU/lb-°F) (approximately 0.5 for oil and 1.0 for water) (J/kg°C),
ΔT = temperature increase, °F(°C).

Field Units

Since water weighs 350 lb/bbl (1000 kg/m^3),

$$W = \frac{350}{24}(\mathrm{SG})_l Q_l,$$

where

SG_l = specific gravity of the liquid,
Q_l = liquid flow rate (BPD).

The total energy required is determined from

$$q = q_o + q_w + q_{\text{lost}},$$

where

q = total energy required to heat the stream,
q_o = energy required to heat the oil
$= [(350/24)(\mathrm{SG})_o Q_o](0.5)\Delta T$,
q_w = energy required to heat the water
$= [(350/24)(\mathrm{SG})_w Q_w](1.0)\Delta T$,
q_{lost} = energy lost to surroundings, assume 10% of total heat input (q).

Substituting gives us

$$q = (350/24)[(\mathrm{SG})_o Q_o(0.5) + (\mathrm{SG})_w Q_w]\Delta T + (0.1)q.$$

Assume 10% water and specific gravity water = 1:

$$q = 16 Q_o \Delta T[0.5(\mathrm{SG})_o + (0.1)].$$

SI Units

Since water weighs 1000 kg/m^3, we have

$$W = 1000(\mathrm{SG})_l Q_l,$$

where

SG_l = specific gravity of the liquid,
Q_l = liquid flow rate, m^3/hr.

The total energy required is determined from

$$q = q_o + q_w + q_{\text{lost}},$$

where

q = total energy required to heat the stream,
q_o = energy required to heat the oil
$= [(1000)(\mathrm{SG})_o Q_o](0.5)\Delta T$,
q_w = energy required to heat the water
$= [(1000)(\mathrm{SG})_w Q_w](1.0)\Delta T$,
q_{lost} = energy lost to surroundings, assume 10% of total heat input (q).

Substituting gives us

$$q = (1000)[(\mathrm{SG})_o Q_o(0.5) + (\mathrm{SG})_w Q_w]\Delta T + (0.1)q.$$

Assume 10% water and specific gravity water = 1:

$$q = 1100 Q_o \Delta T[0.5(\mathrm{SG})_o + (0.1)].$$

Gravity Separation Considerations

Most oil-treating equipment relies on gravity to separate water droplets from the oil continuous phase, because water droplets are heavier than the volume of oil they displace. However, gravity is resisted by a drag force caused by the droplets' downward movement through the oil. When the two forces are equal, a constant velocity is reached, which can be computed from Stokes' law as (Stokes' law was derived in Chapter 4).

Field Units

$$V_t = 1.78 \times 10^{-6} \frac{(\Delta SG)\, d_m^2}{\mu}, \tag{7-7a}$$

SI Units

$$V_t = 5.56 \times 10^{-7} \frac{(\Delta SG)\, d_m^2}{\mu}, \tag{7-7b}$$

where

V_t = downward velocity of the water droplet relative to the oil continuous phase, ft/s (m/s),
d_m = diameter of the water droplet, microns,
ΔSG = difference in specific gravity between the oil and water,
μ = dynamic viscosity of the oil continuous phase, centipoise (cp).

Several conclusions can be drawn from Stokes' law:

- The larger the size of a water droplet, the larger the square of its diameter and, thus, the greater its downward velocity will be. That is, the bigger the droplet size, the less time it takes for the droplet to settle to the bottom of the vessel and thus the easier it is to treat the oil.
- The greater the difference in density between the water droplet and the oil phase, the greater the downward velocity will be. That is, the lighter the crude, the easier it is to treat the oil. If the crude gravity is 10°API and the water is fresh, the settling velocity is zero, as there is no gravity difference.
- The higher the temperature, the lower the viscosity of the oil and, thus, the greater the downward velocity will be. That is, it is easier to treat the oil at high temperatures than at low temperatures (assuming a small effect on gravity difference due to increased temperature).

Settling Equations

The specific gravity difference between the dispersed water droplets and the oil should result in the water "sinking" to the bottom of the treatment vessel.

Since the oil continuous phase is flowing vertically upward in both vertical and horizontal treaters previously described, the downward velocity of the water droplet must be sufficient to overcome the velocity of the oil traveling upward through the treater. By setting the oil velocity equal

to the water settling velocity, the following general sizing equations may be derived:

Horizontal Vessels:

Field Units

$$dL_{\text{eff}} = 438 \frac{FQ_o\mu_o}{(\Delta\text{SG})\, d_m^2}, \tag{7-8a}$$

SI Units

$$dL_{\text{eff}} = 5.0 \times 10^5 \frac{FQ_o\mu_o}{(\Delta\text{SG})\, d_m^2}. \tag{7-8b}$$

If the treater has a spreader and a collector, then the spreader/collector short-circuiting factor is 1. If the treater lacks the spreader, collector, or both, then "F" should be some value greater than 1.

Derivation of Equations (7-8a) and (7-8b)

Field Units

V_t and V_o are in ft/s, d_m in microns, μ in cp,

$$V_t = V_o,$$

$$V_t = \frac{1.78 \times 10^{-6}\,(\Delta\text{SG})\, d_m^2}{\mu},$$

Q is in ft^3/s, A in ft^2, Q_o in BPD, d in in.,

$$V_o = \frac{Q}{A},$$

$$Q = 6.49 \times 10^{-5} Q_o,$$

$$A = \left(\frac{d}{12}\right)(L_{\text{eff}}),$$

$$V_o = 7.79 \times 10^{-4} \left(\frac{Q_o}{dL_{\text{eff}}}\right),$$

$$dL_{\text{eff}} = 438 \left(\frac{Q_o\mu}{(\Delta\text{SG})\, d_m^2}\right).$$

SI Units

V_t and V_o are in m/s, d_m in microns, μ in cp,

$$V_t = V_o,$$

$$V_t = \frac{5.56 \times 10^{-7} (\Delta SG)\, d_m^2}{\mu},$$

Q is in m^3/s, A in m^2, Q_o in m^3/hr, d in mm,

$$V_o = \frac{Q}{A},$$

$$Q = \frac{Q_o}{3600},$$

$$A = \left(\frac{d}{1000}\right)(L_{\text{eff}}),$$

$$V_o = (0.2778)\left(\frac{Q_o}{dL_{\text{eff}}}\right),$$

$$dL_{\text{eff}} = 5.0 \times 10^5 \left(\frac{Q_o \mu}{(\Delta SG)\, d_m^2}\right).$$

Vertical Vessels:

Field Units

$$d = 81.8\left[\frac{FQ_o\mu_o}{(\Delta SG)\, d_m^2}\right]^{1/2} \tag{7-9a}$$

SI Units

$$d = 25,230\left[\frac{FQ_o\mu_o}{(\Delta SG)\, d_m^2}\right]^{1/2} \tag{7-9b}$$

Note that the height of the coalescing section for a vertical treater does not enter into the settling equation. The cross-sectional area of flow for the upward velocity of the oil is a function of the diameter of the vessel alone. This is a limiting factor in the capacity of vertical treaters. In a horizontal vessel, the cross-sectional area for flow for the upward velocity of the oil is a function of the diameter times the length of the coalescing section.

Gunbarrels

The equations for gunbarrels are similar to those for vertical treaters since the flow pattern and geometry are the same. However, gunbarrel tanks experience a great deal of short-circuiting due to uneven flow distribution. This is a result of the large tank diameter. The sizing equation for gunbarrels includes a short-circuiting factor "F." This factor accounts for imperfect liquid distribution across the entire cross section of the treating vessel or tank and is a function of the flow conditions in the vessel. The larger the retention time, the larger the short-circuiting factor will be.

Field Units

$$d = 81.8\left[\frac{FQ_o\mu_o}{(\Delta\text{SG})\,d_m^2}\right]^{1/2}, \tag{7-10a}$$

SI Units

$$d = 25,230\left[\frac{FQ_o\mu_o}{(\Delta\text{SG})\,d_m^2}\right]^{1/2}, \tag{7-10b}$$

where

d = minimum vessel internal diameter, in. (mm),
Q_o = oil flow rate, BOPD (m^3/hr),
μ_o = oil viscosity, cp,
L_{eff} = length of coalescing section, ft (m),
ΔSG = difference in specific gravity between oil and water (relative to water),
d_m = diameter of water droplet, microns,
F = short-circuiting factor

Horizontal Flow Treaters

In horizontal flow settling, the water droplets settle perpendicular to the oil flow. By setting the oil retention time equal to the water settling time, the following equation may be used:

Field Units

$$wL_{\text{eff}} = 800\left(\frac{Q_o\mu_o}{(\Delta\text{SG})\,d_m^2}\right), \tag{7-11a}$$

SI Units

$$wL_{\text{eff}} = 9.0 \times 10^5 \left(\frac{Q_o \mu_o}{(\Delta \text{SG})\, d_m^2} \right), \tag{7-11b}$$

where

L_{eff} = effective length for separation, ft (m),
w = effective width of flow channel, in. (mm).

The effective length is normally 75% of the separation length available. For example, in Figure 7-41 the effective length is 75% of the sum of L1 through L4. The effective width is approximately 80% of the actual channel width.

Note that the height of the flow channel drops out of Eqs. (7-11a) and (7-11b). This is because the oil retention time and the water settling time are both proportional to the height. Also note that these equations assume an F of approximately 1.8.

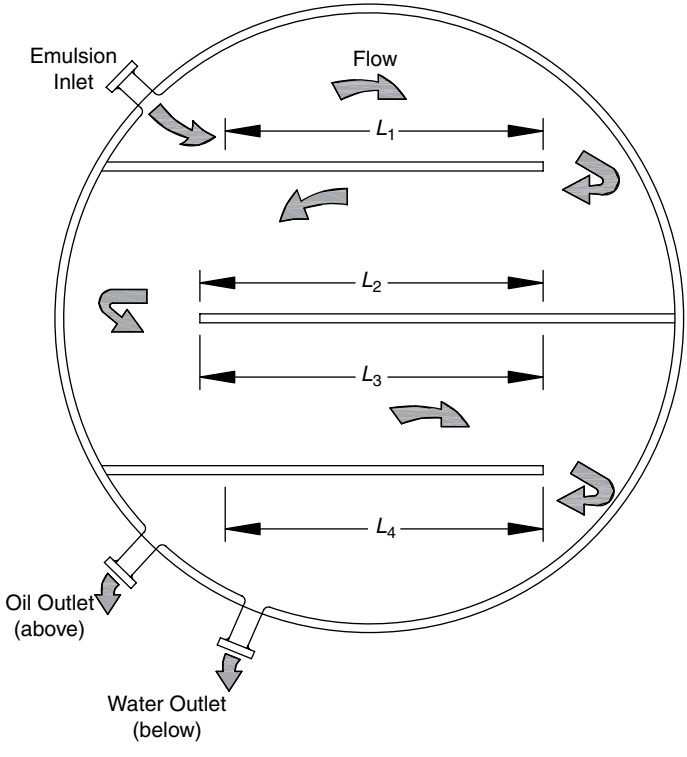


Figure 7-41. Plan view of a cylindrical treating tank using horizontal flow.

Derivation of Equations (7-10a) and (7-10b) and (7-11a) and (7-11b)

Field Units

V_t and V_o are in ft/s, d_m in microns, μ in cp,

$$V_t = V_o,$$

$$V_t = \frac{1.78 \times 10^{-6} (\Delta SG)\, d_m^2}{\mu},$$

Q is in ft^3/s, A in ft^2, Q_o in BPD, d in in.,

$$V_o = \frac{Q}{A},$$

$$V_o = 0.0119 \left(\frac{Q_o}{d^2} \right),$$

$$d^2 = 6{,}690 \left(\frac{Q_o \mu}{(\Delta SG)\, d_m^2} \right),$$

$$d = 81.8 \left(\frac{Q_o \mu}{(\Delta SG)\, d_m^2} \right)^{1/2}.$$

SI Units

V_t and V_o are in ft/s (m/s), d_m in microns, μ in cp,

$$V_t = V_o,$$

$$V_t = \frac{5.56 \times 10^{-7} (\Delta SG)\, d_m^2}{\mu},$$

Q is in m^3/s, A in m^2, Q_o in m^3/hr, d in mm,

$$V_o = \frac{Q}{A},$$

$$V_o = 353.6 \left(\frac{\mu}{5.556 \times 10^{-7} (\Delta SG)\, d_m^2} \right),$$

$$d^2 = 636{,}429{,}086 \left(\frac{Q_o \mu}{(\Delta SG)\, d_m^2} \right),$$

$$d = 25{,}230 \left(\frac{Q_o \mu}{(\Delta SG)\, d_m^2} \right)^{1/2}.$$

Retention Time Equations

The oil must be held at temperature for a specific period of time to enable de-emulsifying the water-in-oil emulsion. This information is best determined in the laboratory but, in the absence of such data, 20 to 30 minutes is a good starting point.

The retention time in the coalescing-settling section of a treater is the volume of the coalescing-settling section divided by the oil flow rate. The volume of the coalescing-settling section is a function of the square of the vessel diameter and the length of the flow path of the coalescing section.

Depending on the specific properties of the stream to be treated, the geometry required to provide a certain retention time may be larger or smaller than the geometry required to satisfy the settling equation. The geometry of the vessel is determined by the larger of the two criteria. The equations for retention time are as follows.

Horizontal Vessels:

Field Units

$$d^2 L_{\text{eff}} = \frac{Q_o\,(t_r)_o}{1.05} \tag{7-12a}$$

SI Units

$$d^2 L_{\text{eff}} = \frac{Q_o\,(t_r)_o}{3.535 \times 10^{-5}} \tag{7-12b}$$

Vertical Vessels:

Field Units

$$d^2 h = \frac{(t_r)_o\, Q_o}{0.12} \tag{7-13a}$$

SI Units

$$d^2 h = \frac{(t_r)_o\, Q_o}{4.713 \times 10^{-8}} \tag{7-13b}$$

Part of the overall vessel height is required to provide for water retention. The removal of oil from the water is not a primary concern. Equations can be derived for water retention similar to the equations for oil

retention. Assuming that a short-circuiting factor is not critical, the height required for water retention can be derived.

The height of water required to provide a given retention time defines the distance between the down-comer exit and the water outlet. The height to the oil–water interface may be much greater due to the need to provide space for fire tubes. The height of the coalescing section, and thus the overall height of the vessel, is most often determined by the need to maintain the oil at the oil–water interface above its bubble-point pressure. Thus, most vertical heater-treaters have much higher oil retention times than necessary for coalescence alone.

Gunbarrels:

Field Units

$$d^2 h = \frac{F\,(t_r)_o\,Q_o}{0.12}, \tag{7-14a}$$

SI Units

$$d^2 h = \frac{F\,(t_r)_o\,Q_o}{4.713 \times 10^{-8}}, \tag{7-14b}$$

where

t_r = retention time, min,
Q_o = oil flow, BOPD (m^3/hr),
h = height of the coalescing section, in. (mm),
F = short-circuiting factor

Horizontal Flow Treaters

The potential for short-circuiting in high tanks is great. Therefore, it is normally assumed that the height limit to consider in calculating retention time is 50% of the actual flow channel width. Providing higher flow channels neither increases the effective retention time nor increases the ability to separate water droplets from the oil.

To provide a specified oil retention time requires a certain volume based on flow rate as follows:

Field Units

$$hwL_{\text{eff}} = 0.56\,(t_r)_o\,Q_o, \tag{7-15a}$$

SI Units

$$hwL_{\text{eff}} = 1.67 \times 10^4 \, (t_r)_o \, Q_o, \tag{7-15b}$$

where h = effective height of the flow channel, in. (mm).

Derivation of Equations (7-12a) and (7-12b)

Field Units

t is in s, V in ft^3, Q in ft^3/s, D in ft, d in in., L_{eff} in ft,

$$t = \frac{V}{Q}.$$

Assuming only 75% of the cross-sectional area is effective, we find that

$$V = 0.75\left(\frac{\pi D^2 L_{\text{eff}}}{4}\right)$$

$$= \frac{(0.75)\,\pi d^2 L_{\text{eff}}}{(4)\,(144)},$$

$$Q = 6.49 \times 10^{-5} Q_o,$$

$$d^2 L_{\text{eff}} = 0.0159 Q_o t,$$

$(t_r)_o$ is in min,

$$t = 60(t_r)_o,$$

$$d^2 L_{\text{eff}} = \frac{Q_o \, (t_r)_o}{1.05}.$$

SI Units

t is in s, V in m^3, Q in m^3/s, D in m, d in mm, L_{eff} in m,

$$t = \frac{V}{Q}.$$

Assuming only 75% of the cross-sectional area is effective, we find that

$$V = 0.75\left(\frac{\pi D^2 L_{\text{eff}}}{4}\right)$$

$$= \frac{(0.75)\,\pi d^2 L_{\text{eff}}}{(4)\,(1000)},$$

$$Q = \frac{Q_o}{3600},$$

$$d^2 L_{\text{eff}} = (471.5)\,Q_o t,$$

$(t_r)_o$ is in min,

$$t = 60(t_r)_o,$$

$$d^2 L_{\text{eff}} = \frac{Q_o\,(t_r)_o}{3.535 \times 10^{-5}}.$$

Equations (7-13a and b), (7-14a and b), and (7-15a and b) are derived in the same manner as the retention time equation for horizontal separators.

Water Droplet Size

In order to develop a treater design procedure, the water droplet size to be used in the settling equation to achieve a given outlet water cut must be determined. It would be extremely rare to have laboratory data of the droplet size distribution for a given emulsion as it enters the coalescing section of the treater. Qualitatively, we would expect the minimum droplet size that must be removed for a given water cut to (1) increase with retention time in the coalescing section, (2) increase with temperature, which tends to excite the system, leading to more collisions of small droplets, and (3) increase with oil viscosity, which tends to inhibit the formation of small droplets from shearing that occurs in the system.

We have seen that, after an initial period, increasing the retention time has a small impact on the rate of growth of particles. Thus, for practically sized treaters with retention times of 10 to 30 minutes, retention time would not be expected to be a determinant variable. Intuitively, one would expect viscosity to have a much greater effect on coalescence than temperature.

Assuming that the minimum required size of droplets that must be settled is a function only of oil viscosity, equations have been developed correlating this droplet size and oil viscosity [1]. The authors used data

from three conventional treaters operating with 1% water cuts. Water droplet sizes were back-calculated using Eqs. (7-8a) and (7-8b). The calculated droplet sizes were correlated with oil viscosity, and the following equations resulted:

$$d_{\text{mi}}\% = 200\mu^{0.25}, \qquad \mu_o < 80\,\text{cp}, \tag{7-16}$$

where

$d_{\text{mi}}\%$ = diameter of water droplet to be settled from the oil to achieve 1% water cut, microns,
μ = viscosity of the oil phase, cp.

Using the same procedure, the following correlation for droplet size was developed for electrostatic treaters:

$$d_{\text{mi}}\% = 170\mu^{0.4}, \qquad 3\,\text{cp} < \mu_o < 80\,\text{cp}. \tag{7-17}$$

For viscosities below 3 cp, Eq. (7-16) should be used. The two equations intersect at 3 cp, and electrostatic treaters would not be expected to operate less efficiently in this range. Additionally, the data from which the electrostatic treater droplet size correlation was developed did not include oil viscosities less than 7 cp.

The same authors also investigated the effect of water cut on minimum droplet size. Data from both conventional and electrostatic treaters over a range of water cuts were used to back-calculate an imputed droplet size as a function of water cut, resulting in the following equation:

$$\frac{d_m}{d_{\text{mi}}\%} = W_c^{0.33}, \tag{7-18}$$

where

d_m = diameter of water droplet to be settled from the oil to achieve a given water cut (W_c), microns,
W_c = water cut, percent.

As the volume of a sphere is proportional to the diameter cubed, Eq. (7-18) indicates that the water cut is proportional to the droplet diameter cubed.

It must be stressed that the above equations should be used only in the absence of other data and experience. These proposed relationships are based only on limited experimental data. An approximate sizing relationship, derived from Eqs. (7-16) and (7-17), are given in Figures 7-42 and 7-43 in terms of the flow rate of emulsion (given in BPD) flowing

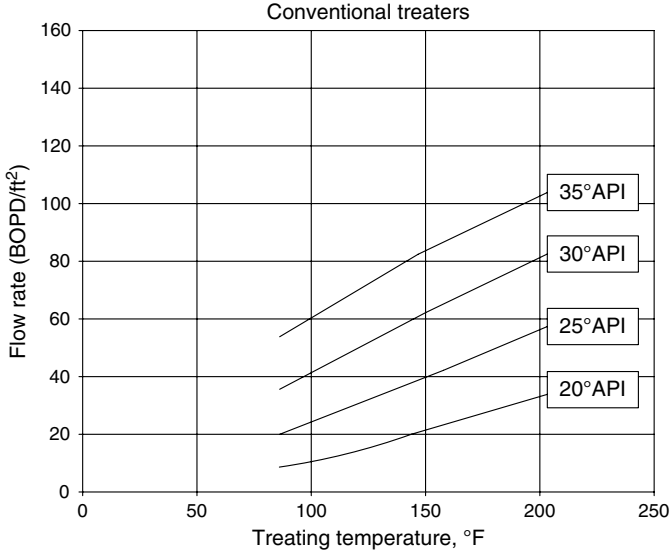


Figure 7-42. Flow rate vs. treating temperature for conventional treaters.

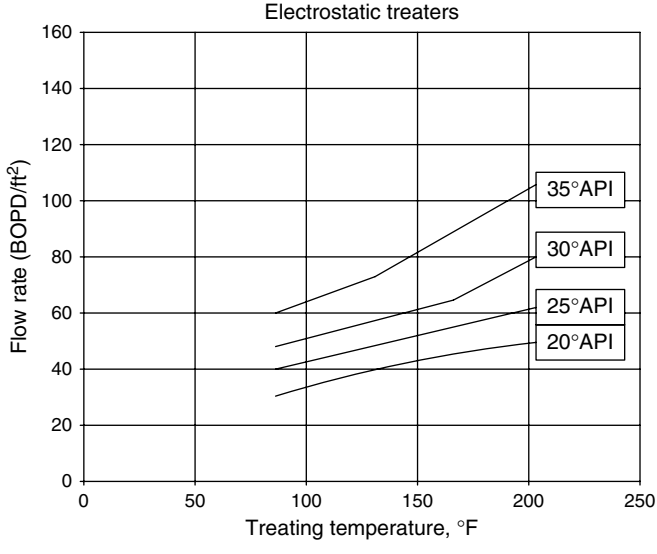


Figure 7-43. Flow rate vs. treating temperature for electrostatic treaters.

vertically through a horizontal cross-sectional area of one square foot. For a horizontal treater with vertical flow through the coalescing section, the flow area can be approximated as the diameter of the vessel times the length of the coalescing section.

Design Procedure

In specifying the size of a treater, it is necessary to determine the diameter (d), length or height of the coalescing section (L_{eff} or h), and treating temperature or fire-tube rating. As we have seen, these variables are interdependent, and it is not possible to arrive at a unique solution for each. The design engineer must trade the cost of increased geometry against the savings from reducing the treating temperature.

The equations previously presented provide tools for arriving at this trade-off. However, because of the empirical nature of some of the underlying assumptions, engineering judgment must be utilized in selecting the size of treater to use.

General Design Procedure

1. Choose a treating temperature.
2. Determine the heat input required from Eqs. (7-5a) and (7-5b).
3. Determine oil viscosity at treating temperature. In the absence of laboratory data, Chapter 3 provides correlations that can be used to estimate crude viscosity given its gravity and temperature.
4. Select a type of treater, and size the treater using the appropriate design procedure below.
5. Choose the design minimum droplet size that must be separated from experimental data, analogy to other treaters in service or Eqs. 7-16, 7-17 and 7-18.
6. Repeat the above procedure for different treating temperatures.

Design Procedure for Vertical Heater-Treaters and Gunbarrels (Wash Tanks with Internal/External Gas Boot)

1. Calculate the minimum treater diameter using Eqs. (7-9a and b) or (7-10a and 7-10b).
2. For various diameters greater than the minimum, calculate the height required in the coalescing-settling section using Eqs. (7-13a and b) or (7-14a and b).

3. Calculate the height required to keep the oil above its bubble-point pressure if the emulsion is heated after the gas has been separated from it. For most standard applications, a 20- or 27-ft (6.1- or 8.2-m) seam-to-seam length will provide ample height for all sections of the treater.
4. Select a standard size treater from vendor literature that meets the above requirements. Note: Standard size guidelines are presented in this chapter and in the example presented in the next section.

Design Procedure for Horizontal Heater-Treaters

1. For various standard diameters, develop a table of effective lengths versus standard diameters, using Eqs. (7-8a) and (7-8b) for settling.
2. For the same diameters used in step 1, calculate the effective lengths required using Eqs. (7-12a) and (7-12b) for retention time.
3. Select a treater, which satisfies the larger effective length requirements for the selected diameter.

Design Procedure for Horizontal-Flow Treaters

1. Calculate several combinations of w and L_{eff} using Eqs. (7-11a) and (7-11b) for settling.
2. For each combination of w and L_{eff} used in step 1, calculate the h required for the specified retention time using Eqs. (7-12a) and (7-12b).
3. Select a combination of w and L_{eff} for which the calculated h is less than one-half the flow width.

This above procedure allows the production facility engineer to choose the major sizing parameters of heater-treaters when little or no laboratory data are available. This procedure does not give the overall dimensions of the treater, which must include inlet gas separation and free-water knockout sections. However, it does provide a method for specifying a fire-tube capacity and a minimum size for the coalescing section (where the treating actually occurs) and provides the engineer with the tools necessary to evaluate specific vendor proposals.

Figure 7-44 provides standard dimensions, pressure ratings, and fire-box ratings for vertical and horizontal heater-treaters. Figure 7-45 is a

Typical Vertical Treater Dimensions and Pressures

Outside Diameter ft	Shell Length Head Seam to Head Seam to ft = 6 in.	Minimum Design Pressure psig
3	10.12 or 15	50
4	10.12.20 or 27 1/2	50
6	12.20. or 27 1/2	50
8	20 or 27 1/2	40
10	20 or 27 1/2	40

Typical Horizontal Treater Dimensions and Pressures

Outside Diameter ft	Shell Length Head Seam to Head Seam to ft = 6 in.	Minimum Design Pressure psig
3	10.12 or 15	50
4	10.12 or 15	50
6	12.15 or 20	50
8	15.20.25 or 30	50
10	20.30.40.50 or 60	50
12	30.40.50 or 60	50

TYPICAL FIREBOX RATINGS

Outside Diameter ft	Vertical Minimum Area sq ft	Vertical Heat Duty BTU/hr	Horizontal Minimum area sq ft	Horizontal Heat Duty BTU/hr
3	10	100,000	15	150,000
4	25	250,000	25	250,000
6	50	500,000	50	500,000
8	100	1,000,000	75	750,000
10	125	1,250,000	200	2,000,000
12	-	-	320	3,200,000

These are suggested minimum requirements. Heat requirements of the process and fluid characteristics should be considered in sizing the firebox.

Figure 7-44. Standard dimensions, pressure ratings, and fire-box ratings for vertical and horizontal heater-treaters.

VFH OIL PROCESSING UNIT CAPACITIES

Shell size (Length × Diameter)	Fire-tube capacity (BTU/Hr)	Fire-tubes (Number and O.D)	Oil (Barrel per hour)	Oil (Barrel per day)	Free Water (Barrels per day)	Gas (MM sct/d)
STANDARD MODEL - 8° to 42°API GRAVITY OIL						
6'×15'	550,000	1–18"	15–75	360–1800	1500	0.5–1
6'×20'	1,000,000	1–18"	35–120	800–2800	2,200	0.5–1
6'×30'	1,500,000	1–18"	70–180	1500–4000	3,500	0.5–1
8'×20'	1,375,000	1–24"	40–130	1000–3200	3,500	1.5–2
8'×25'	1,750,000	1–24"	60–160	1400–3800	4,500	1.5–2
8'×25'	2,500,000	2–18"	60–160	1400–3800	4,500	1.5–2
9'×30'	2,000,000	1–24"	75–200	1800–4800	5,500	1.5–2
9'×30'	3,000,000	2–18"	75–200	180–4800	5,500	1.5–2
8'×35'	2,500,000	1–24"	85–220	2000–5300	6,500	1.5–2
8'×35'	3,750,000	2–18"	85–220	2000–5300	6,500	1.5–2
10'×25'	2,250,000	2–18"	100–250	2400–6000	5,000	2–3
10'×25'	3,000,000	2–24"	100–250	2400–6000	5,000	2–3
10'×30'	2,500,000	2–18"	120–320	2900–7700	6,000	2–3
10'×30'	3,500,000	2–24"	120–320	2900–7700	6,000	2–3
10'×35'	2,750,000	2–18"	150–400	3600–9600	6,500	2–3
10'×35'	3,750,000	2–24"	150–400	3600–9600	6,500	2–3
10'×40'	2,750,000	2–18"	190–500	4500–12000	7,000	2–3
10'×40'	3,750,000	2–24"	190–500	4500–12000	7,000	2–3
12'×45'	5,000,000	2–24"	190–500	4500–12000	8,500	2–3
12'×75'	6,000,000	2–24"	200–520	4800–12500	10,500	2–3
12'×45'	7,500,000	3–24"				
12'×50'	9,400,000	3–24"				
12'×55'	9,400,000	3–24"				
12'×60'	10,500,000	3–24"	PLEASE CONSULT C-E NATCO ENGINEERING			
12'×65'	10,500,000	3–24"	FOR CAPACITIES OF 12' DIAMETER UNITS			
12'×70'	13,000,000	3–24"				
12'×75'	13,000,000	3–24"				
ROCKY MOUNTAIN MODEL - 28°C to 42°API GRAVITY OIL 50 PSI WORKING PRESSURES						
6'×15'	650,000	1–18"	12–60	300–1400	1000	0.5–1
6'×20'	850,000	1–18"	30–90	700–2200	1500	0.5–1
6'×25'	1,250,000	1–18"	36–100	900–2500	1700	0.5–1
25 AND HIGHER PSI WORKING PRESSURES AVAILABLE						
8'×15'	750,000	1–24"	20–75	500–1800	1600	1.5–2
8'×15'	1,000,000	2–18"	30–90	700–2100	1600	1.5–2
8'×20'	1,375,000	1–24"	40–100	1000–2400	2300	1.5–2
8'×20'	1,500,000	2–18"	60–130	1500–3200	2000	1.5–2
8'×25'	1,250,000	1–24"	75–150	1800–3600	3000	1.5–2
8'×25'	1,750,000	2–18"	75–150	1800–3600	3000	1.5–2
8'×30'	1,500,000	1–24"	80–170	2000–4200	3600	1.5–2
8'×30'	1,750,000	2–18"	80–170	2000–4200	3600	1.5–2
8'×35'	1,650,000	1–24"	90–200	2200–4800	4200	1.5–2

Figure 7-45. Typical vendor supplied vertical heater-treater capacity table.

typical horizontal heater-treater table supplied by an equipment manufacturer. Figure 7-46 is a typical electrostatic heater-treater capacity table supplied by an equipment manufacturer.

ELEKTROSTATIC TREATER CAPACITIES 15 ° to 42 °API GRAVITY OIL								
Shell size (Length × Diameter)	Fire-tube capacity (Bt/Hr)	Fire-tubes (Number and O.D)	Oil				Free Water (barrels per Day)	Gas (MM sctd)
			Bbls/Hr		Bbls/Day			
			AC	AC/DC	AC	AC/DC		
6'×15'	550,000	1.18"	20–100	24–120	480–2400	576–2880	500–1500	0.5–1
6'×20'	1,000,000	1.18"	20–100	24–120	480–2400	576–2880	500–1500	0.5–1
8'×15'	750,000	1.24"	50–180	60–261	1200–4320	1400–5184	600–1800	1.5–2
8'×15'	1,100,000	2.18"	50–180	60–261	1200–4320	1400–5184	600–1800	1.5–2
8'×20'	1,300,000	1.21"	100–230	120–276	2400–5520	2880–6624	800–2400	1.5–2
8'×20'	2,000,000	2.18"	100–230	120–276	2400–5520	2880–6624	800–2400	1.5–2
8'×25'	1,500,000	1.24"	125–250	150–300	3000–600	3600–7200	800–2400	1.5–2
8'×25'	2,250,000	2.18"	125–250	150–300	3000–600	3000–6000	800–2400	1.5–2
10'×20'	2,000,000	2.18"	140–280	168–336	3360–6720	4032–8064	1000–3000	2–3
10'×20'	2,500,000	2.24"	140–280	168–336	3360–6720	4032–8064	1000–3000	2–3
10'×20'	3,000,000	3.18"	140–280	168–336	3360–6720	4032–8064	1000–3000	2–3
10'×25'	2,000,000	2.18"	175–430	210–516	4200–10320	5040–12384	1000–3000	2–3
10'×25'	2,500,000	2.24"	175–430	210–516	4200–10320	5040–12384	1000–3000	2–3
10'×25'	3,000,000	3.18"	175–430	210–516	4200–10320	5040–12384	1000–3000	2–3
10'×30'	2,000,000	2.18"	200–580	240–696	4800–13920	5760–16704	1000–3000	2–3
10'×30'	2,500,000	2.24"	200–580	240–696	4800–13920	5760–16704	1000–3000	2 3
10'×30'	3,000,000	3.18"	200–580	240–696	4800–13920	5760–16704	1000–3000	2–3
10'×35'	3,000,000	2.24"	200–580	240–696	4800–13920	5760–16704	1500–4500	2–3
10'×35'	3,750,000	3.18"	200–580	240–696	4800–13920	5760–16704	1500–4500	2–3
10'×40'	3,750,000	2.24"	350–730	420–876	8400–17520	10080–21024	2000–6000	3–5
10'×45'	5,000,000	2.24"	350–730	420–876	8400–17520	10080–21024	2500–7500	3–5
10'×50'	6,000,000	2.24"	350–730	420–876	8400–17520	10080–21024	3000–9000	3–5

Figure 7-46. Typical vendor-supplied horizontal electrostatic heater-treater capacity table.

Examples

Example 7-2: Sizing a horizontal-treater (field units)

Given:

Oil flow rate = 5,000 BOPD
Inlet oil temperature = 80°F
Water SG = 1.04
Inlet BS&W = 10%
Outlet BS&W = 1%

Solution:

1. **Settling Equation**. Investigate treating at 80°F, 100°F, 120°F.

Treating Temperature	80°F	100°F	120°F
ΔSG	0.165	0.165	0.165
μ_o	40	15	9
d_m	503	394	346
dL_{eff}	2,098	1,283	998

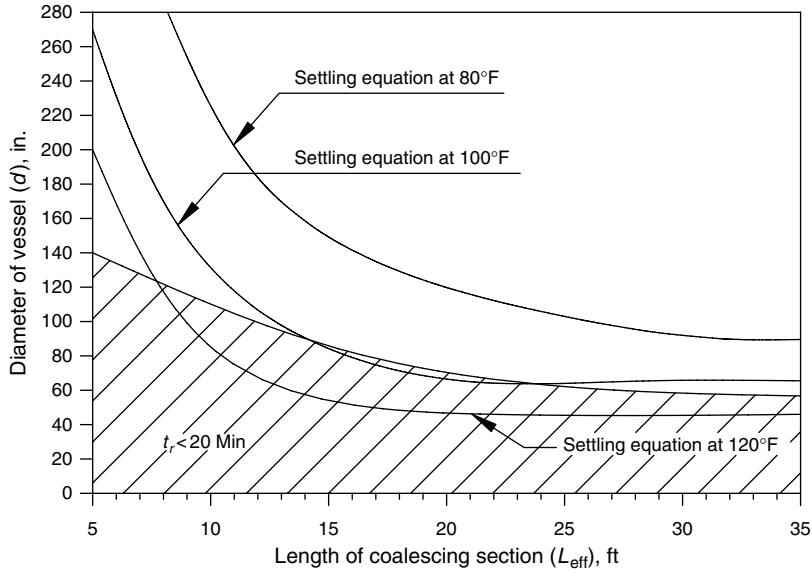


Figure 7-47. Example 7-2: Horizontal heater field units.

2. **Retention Time Equation**. Plot computations of d and L_{eff} with retention times less than 20 minutes.

$$d^2 L_{\text{eff}} = 20(5{,}000)/1.05 = 95{,}238$$

The shaded area of Figure 7-47 represents combinations of d and L_{eff} with $t_r < 20$ min.

3. **Heat Required**

$$q = (16)(5{,}000)(\Delta T)[(0.5)(0.876) + 0.1]$$
$$= 43{,}040(\Delta T)$$

Substituting treating temperature values of 80°F, 100°F, and 120°F and substituting initial oil temperature value of 80°F will yield values of heat required of 0, 0.86, and 1.72 MMBtu/h.

4. **Selection**. Choose any combination of d and L_{eff} that is not in the shaded area. Read corresponding treating temperature.

Example solutions are

Treating Temperature (°F)	d (in.)	L_{eff} (ft)	Heat Required (MMBtu/h)
80°F	144	15	0.00
	120	18	
	96	22	
100°F	96	14	0.86
	72	20	
120°F	96	10	1.72
	72	20	

An economical solution would be a 72-in.-diameter treater with a 20-ft coalescing section and a 0.86-MMBtu/h firetube capacity. Given the nature of empirical design procedures, crude could possibly be treated at 80°F. The additional fire-tube capacity will allow a temperature of 100°F if required by field conditions.

Example 7.3: Sizing a horizontal treater (SI units)

Given:

Oil flow rate $= 33\,m^3/hr$
Inlet oil temperature $= 27°C$
Water SG $= 1.04$
Inlet BS&W $= 10\%$
Outlet BS&W $= 1\%$
Retention time $= 20\,min$

Solution:

1. **Settling Equation**. Investigate treating at 27°C, 38°C, and 49°C.

Treating Temperature	27°C	38°C	49°C
ΔSG	0.165	0.165	0.165
μ_o	40	15	9
d_m	503	394	346
dL_{eff}	15,860	9,700	7,540

2. **Retention Time Equation**. Plot computations of d and L_{eff} with retention times less than 20 minutes.

$$d^2 L_{eff} = 20(5{,}000)/1.05 = 95{,}238$$

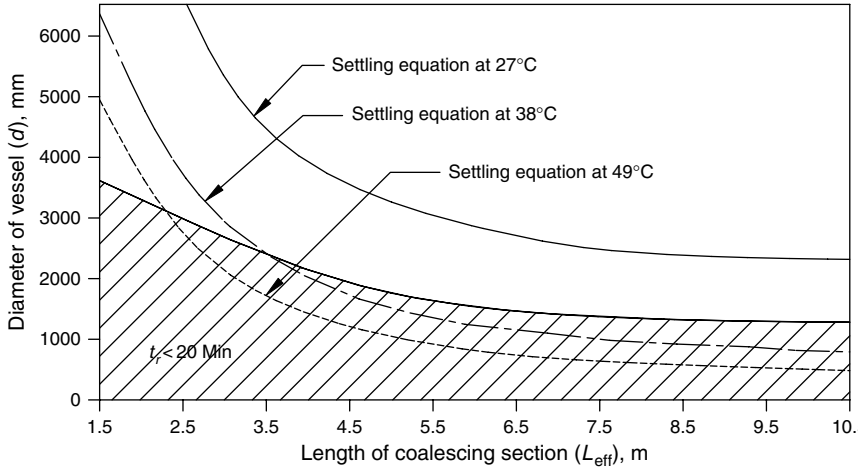


Figure 7-48. Example 7-3: Horizontal heater SI units.

The shaded area of Figure 7-48 represents combinations of d and L_{eff} with $t_r < 20$ min.

3. **Heat Required**

$$q = (1100)(33)(\Delta T)[(0.5)(0.876) + 0.1]$$
$$= 19{,}529(\Delta T)$$

Substituting treating temperature values of 27°C, 38°C, and 49°C and substituting initial oil temperature value of 27°C will yield values of heat required of 0, 0.90, and 1.82 MKJ/h.

4. **Selection**. Choose any combination of d and L_{eff} that is not in the shaded area. Read corresponding treating temperature.

Example solutions are

Treating Temperature (°C)	d (mm)	L_{eff} (m)	Heat Required (MKJ/h)
27°C	3658	4.6	0.00
	3048	5.5	
	2438	6.7	
38°C	2438	4.3	0.90
	1829	6.1	
49°C	2438	3.0	1.82
	1829	6.1	

An economical solution would be a 1829-mm-diameter treater with a 6.1-m coalescing section and a 0.9-MKJ/h fire-tube capacity. Given the nature of empirical design procedures, crude could possibly be treated at 27°C. The additional fire-tube capacity will allow a temperature of 38°C if required by field conditions.

Example 7.4: Sizing a vertical treater (field units)

Given:

Oil gravity = 40°API, 0.875 SG
Oil flow rate = 2,000 BOPD
Inlet oil temperature = 90°F
Water SG = 1.04
Inlet BS&W = 10%
Outlet BS&W = 1%

Solution:

1. **Settling Equation**. Investigate treating at 90°F, 100°F, 120°F.

Treating Temperature	90°F	100°F	120°F
ΔSG	0.215	0.215	0.215
μ_o	7.0	5.1	3.3
d_m	325	301	270
d	64	59	53

2. **Retention Time**. Plot computations of d and h with retention times less than 20 minutes.

$$d^2h = (20)(2{,}000)/0.12 = 333{,}333$$

The shaded area of Figure 7-49 represents combinations of d and h with $t_r < 20$ min.

3. **Heat Required**

$$q = (16)(2{,}000)(\Delta T)[(0.5)(0.825) + 0.1]$$
$$= 16{,}400(\Delta T)$$

4. **Selection**. Choose any combination of d and h that is not in the shaded area. Read the corresponding treating temperature.

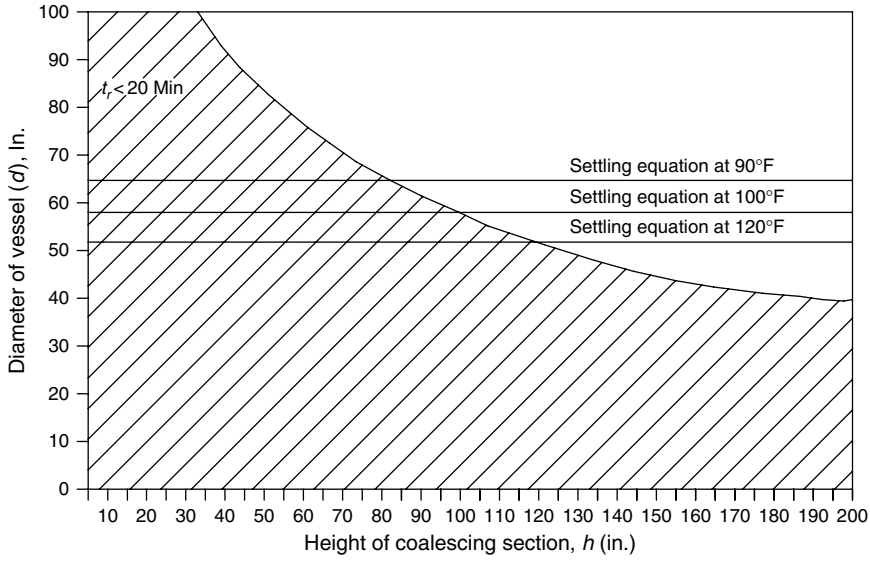


Figure 7-49. Example 7-4: Vertical heater field units.

Example solutions are

Treating Temperature (°F)	*d* (in.)	*h* (in.)	Heat Required (MMBtu/h)
120°F	53	120	0.49
100°F	59	100	0.16
90°F	64	90	0.0

An economical solution would be a 60-in.-diameter treater with a 100-in.-high coalescing section and a 0.16-MMBtu/h fire-tube capacity. In actual service, crude may not require heating at all. The fire-tube capacity will allow a treating temperature of 100°F if required by field conditions.

Example 7.5: Sizing a vertical treater (SI units)

Given:

Oil gravity = 40°API, 0.875 SG
Oil flow rate = 13 m^3/hr
Inlet oil temperature = 32°C
Water SG = 1.04

Inlet BS&W = 10%
Outlet BS&W = 1%

Solution:

1. **Settling Equation**. Investigate treating at 32°C, 38°C, 49°C.

Treating Temperature	32°C	38°C	49°C
ΔSG	0.215	0.215	0.215
μ_o	7.0	5.1	3.3
d_m	325	301	270
d	1626	1499	1346

2. **Retention Time**. Plot computations of d and h with retention times less than 20 minutes.

$$d^2h = (20)(2{,}000)/0.12 = 333{,}333$$

The shaded area of Figure 7-50 represents combinations of d and h with $t_r < 20$ min.

3. **Heat Required**

$$q = (1100)(13)(\Delta T)[(0.5)(0.825) + 0.1]$$
$$= 7{,}329(\Delta T)$$

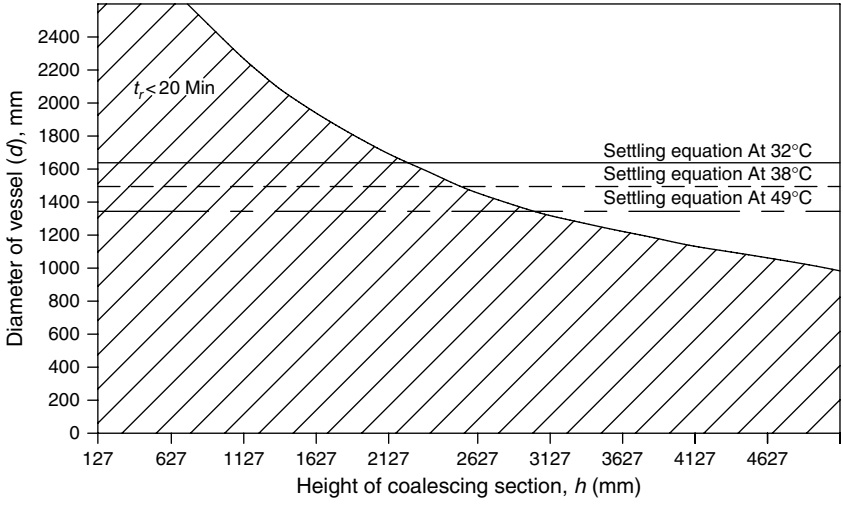


Figure 7-50. Example 7-5: Vertical heater SI units.

4. **Selection.** Choose any combination of d and h that is not in the shaded area. Read the corresponding treating temperature.
   Example solutions are

Treating Temperature (°C)	d (mm)	h (mm)	Heat Required (MKJ/h)
49°C	1346	3048	0.56
38°C	1499	2540	0.20
32°C	1626	2286	0.0

   An economical solution would be a 1524-mm-diameter treater with a 2540-mm-high coalescing section and a 0.2-MKJ/h fire-tube capacity. In actual service, crude may not require heating at all. The fire-tube capacity will allow a treating temperature of 38°C if required by field conditions.

Practical Considerations

Successful treatment of emulsions, depending on specific emulsion characteristics, can be treated by low temperature with or without adding chemicals, or chemicals with or without heat. Some fields having high water cut (e.g., 95%) can be treated successfully without heat or chemicals, but require extremely long retention times. It is better to use chemicals instead of heat from the standpoints of installation, maintenance, and operating costs. The following discussion provides some general guidelines to help one select the right oil treating equipment configuration for a specific application.

Gunbarrels with Internal/External Gas Boot

Gunbarrels (wash tank with internal/external gas boot) should be considered when isolated, high-salt-water percentage production is indicated, provided retention time requirements do not make gunbarrel sizing impractical. When used without heat, the vessel should provide ample settling time, e.g., 12 to 24 hr. Sufficient retention time allows some storage of basic sediment during cold weather when chemical efficiency declines. The basic settlement is cleaned from the tank during warm weather and by periodically rolling (circulating) the gunbarrel.

Heater-Treaters

A heater-treater should be considered in fields requiring heat to break the emulsion. Good practice is to install a slightly larger (+10%) heater-treater than is necessary. This allows extra capacity for unforeseeable production increases (normally water), reduction in the amounts of treating chemical used, and startup of a cold unit. A reduction in chemical cost can easily pay for the additional cost of a larger treater in a few years. Depending on the characteristics of the oil and the efficiency of the chemical, retention times range between 10 to 60 minutes.

Electrostatic Heater-Treaters

An electrostatic heater-treater should be considered in fields with maximum salt content specifications imposed [10 to 30 lb per thousand barrels (PTB)], any time the BS&W must be reduced below 0.5%, and offshore facilities where space and/or heat is limited.

Other configuration considerations that the designer may be required to evaluate are free-water knockout instead of a gunbarrel and using an electrostatic heater-treater instead of a heater-treater. Applying the basic principles presented in the section coupled with sound engineering judgment will allow the designer to select the most appropriate selection.

OIL DESALTING SYSTEMS

Introduction

The process of removing water-soluble salts from an oil stream is called oil desalting. Nearly all crude oil contains entrained water, which almost always contains dissolved salts, specifically sodium chloride, magnesium, and calcium. The majority of the produced salt water is removed by separation and the oil treating process. However, a small amount of entrained water remains in the crude oil. The crude oil is sent to the refinery where it is heated as part of the various refinery processes. The entrained water is driven off as steam. However, the salt in the water does not leave with the steam but crystallizes and remains suspended in the oil, or may deposit as scale within heat exchange equipment. In addition, the entrained salt crystals will usually deactivate catalyst

beds and plug downstream processing equipment. Due to these problems, refineries usually require the crude oil salt content be reduced to very low levels prior to processing. Refineries usually achieve the reduced salt content by specifying in purchase contracts a maximum salt content, as well as maximum water content. A common salt specification would be 10 to 20 pounds per thousand barrels ($0.003\,kg/m^3$ to $0.006\,kg/m^3$). To satisfy the refinery specification, upstream production facilities may be required to perform oil desalting.

This part of the chapter describes the methods and equipment commonly used to desalt crude oil.

Equipment Description

Desalters

Since the salt content is directly related to the amount of residual water, the best desalters remove as much water as possible. Any device that removes water from oil can be used as a desalter. However, the majority of desalters employed are horizontal electrostatic treaters. These treaters will produce the lowest residual water level of all treaters. Figure 7-25 illustrates a conventional horizontal electrostatic treater of the type typically used in desalting operations. Because very low water contents are required, the crude is usually pumped through the desalter at pressures above its bubble point. In addition, the temperature of the crude to be desalted is determined by upstream heat exchangers. Thus there is need for an inlet degassing and heating section as shown in the typical oil field horizontal electrostatic treater discussed earlier.

Mixing Equipment

Globe Valves

A manual globe throttling valve is one of the simplest methods to promote the mixing of dilution water and salt water. The pressure drop resulting from forcing the oil and water through this manual valve is used to shear the water droplets and mix the droplets in the oil. The major disadvantage of any manual valve is its inability to automatically adjust for changes in oil flow rate. As the flow rate varies, the pressure drop, and thus the mixing efficiency, varies. Therefore, if the oil flow rate increases significantly, the pressure drop may increase to the point where the resulting mixed emulsion is impossible to treat.

It is possible to automate the globe valve to avoid "over mixing". A differential pressure controller is used to control the pressure drop through the globe valve. This system automatically adjusts for changing flow rates and maintains a set pressure drop. Since this system's set point can be adjusted in the field, it allows an operator to optimize its performance.

The conventional single-ported and balanced double-ported globe valves are commonly used and yield good results. The valve body should be line size. If a single-port valve is used, some form of premixing should be provided prior to the valve to ensure an even distribution of the water droplets to control the droplet size distribution. Double-port globe valves yield lower pressure drops than single-port valves, and may eliminate the need for premixing.

The pressure drop through the mixing valve varies from 10 to 50 psi (70 to 340 kPa). The required pressure drop can be decreased if a premixing device is installed upstream of the mixing valve. The reason for this is that the premixing device distributes the water in the oil, and the valve will be required only to shear the droplets. If the mixing must both shear and distribute the water, higher pressure drops are necessary.

Spray Nozzles

Upstream premixing is commonly performed with either spray nozzles or static mixers.

As shown in Figure 7-51, one common method of premixing the water and oil involves using a system of spray nozzles. Water is pumped through

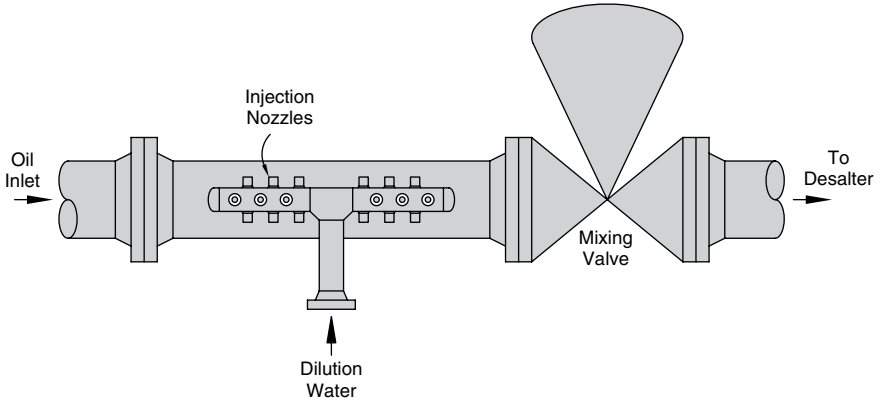


Figure 7-51. Schematic of a spray nozzle system for premixing water and oil.

the nozzles and then distributed throughout the oil stream. These systems are effective and are usually less expensive than static mixers.

Static Mixers

Static mixers use pieces of corrugated plate, as shown in Figure 7-52. These mixers typically divide into many parallel paths which divide and recombine as the flow passes through the mixer. The alternate layers of corrugations are perpendicular to each other so that the fluid must pass through a series of relatively small openings. This mixer shears the water droplets to a much smaller size than the old mixers. These mixers produce a narrow range of droplet sizes. This is a result of two opposing phenomena. Large droplets are sheared by the mixing action in the small openings, while at the same time these mixers provide large surface areas where small droplets may collect and coalesce. Theoretically, the coalescing ability improves the performance of the dehydration equipment due to the reduction in the number of very small droplets which makes dehydration easier and decreases the chances of creating a stable, untreatable emulsion during the mixing process.

Static mixers are sized to provide an average droplet size using empirical equations based on test data. The average droplet size for desalting is roughly between 250 and 500 microns. The average droplet size is a function of the oil flow rate. The primary disadvantage of static mixers is that they may not be adjusted as the flow varies. Therefore, if the oil

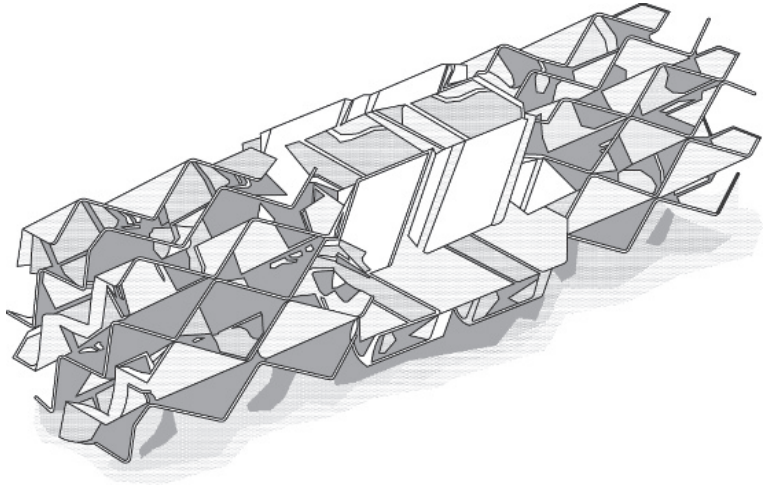


Figure 7-52. New static mixer.

flow will vary over a range of 3 to 1, or more, static mixers should not be used as the only mixing device.

Process Description

Most of the salt contained in crude oil is dissolved in the small water droplets. Since water is the salt carrier, removing the water will remove the salt from the crude. The salt content of the water is expressed as parts per million (ppm) equivalent sodium chloride. Salinity may range from 0 to over 150,000 ppm. Desalting is required when the amount of salt contained in the entrained water after treating is higher than some specified amount.

For example, assume a heater-treater is used for dehydration and it yields oil that is 0.5% water, each thousand barrels of dehydrated oil includes 5 bbl of water. If we next assume the water has a low salt content, say 10,000 ppm NaCl, then each barrel of water would contain approximately 3.5 pounds of salt. With 5 bbls of water per thousand barrels of oil, the oil would then contain approximately 17.5 PTB (pounds per thousand barrels). If the purchase agreement specified 10 PTB or less, some desalting, or a more efficient dehydrator, would be required.

In this example, an electrostatic treater might be all that is required to achieve an oil outlet that contains less than 0.3% water. This example assumed a low salt content. If the water had a high salt content, say 200,000 ppm NaCl, there would be approximately 70 pounds of salt per barrel of water (lb/bbl). In this case, even dehydrating to 0.1% leaves 70 PTB. To reach the required 10 PTB, desalting would be required.

The desalting process involves two steps. The first step is to mix fresh water with entrained produced water. This will lower the produced water salinity by diluting the salt. The second step is dehydration which is the removal of the water from the crude. This dilution and dehydration produces a lower salinity in the residual water in the crude oil. The dilution water in desalting does not have to be fresh. Any water with a lower salt content than the produced water can be used.

Single-Stage Desalting

Figure 7-53 is a schematic of a single-stage desalting system. In this system, the dilution water is injected into the oil stream and then mixed. The oil then enters the desalter where the water is removed. To reduce dilution water requirements, the crude oil may be dehydrated prior to the desalting process. This removes the bulk of the produced water prior to desalting.

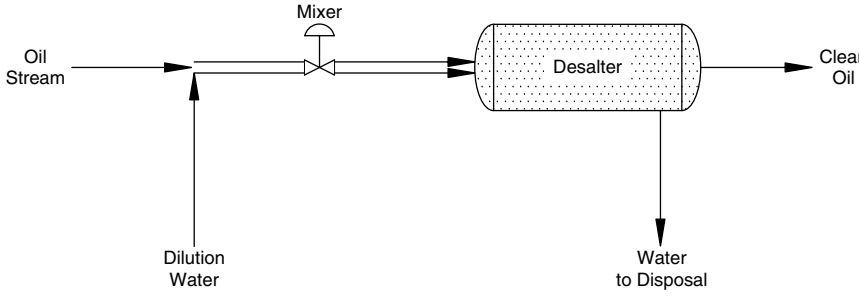


Figure 7-53. Schematic of a single-stage desalting system.

Two-Stage Desalting

Figure 7-54 is a schematic of a two-stage desalting system with dilution water recycling capability. This system is similar to the dehydrator and desalter system described in the previous section. The only difference is that the water removed in the second stage is pumped back to the first stage. The addition of this recycle provides for some dilution of the salt water prior to the first stage. This further reduces the dilution water requirement compared to a single-stage dehydrator and desalter system.

If further desalting is needed, it is possible to add more stages in a similar manner.

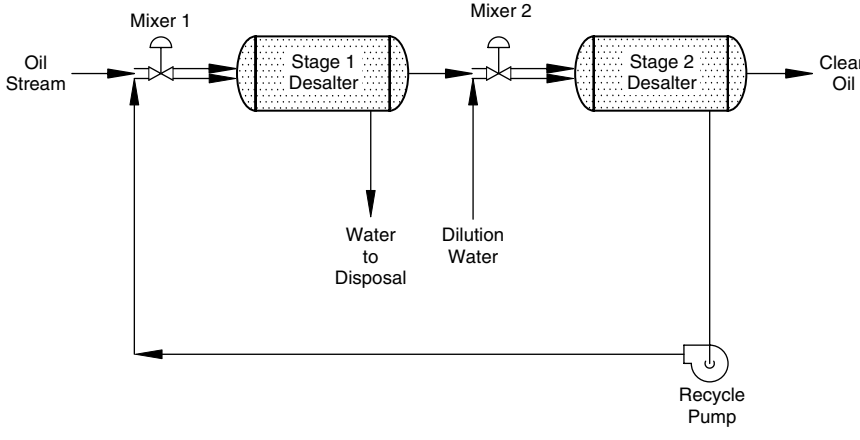


Figure 7-54. Schematic of a two-stage desalting system with a recycle stream.

Nomenclature

A = cross-sectional area, ft^2 (m^2)
C = specific heat, BTU/lb-°F (J/Kg°C)
API = API gravity of oil, °API
D = vessel's internal diameter, ft (m)
d = vessel's internal diameter, in. (mm)
d_m = drop diameter, microns (μ)
F = short circuit factor
H = height of liquid volume, ft (m)
h = height of liquid volume, in. (mm)
H_o = height of oil pad, ft (m)
h_o = height of oil pad, in. (mm)
H_w = height from water outlet to interface, ft (m)
h_w = height from water outlet to interface, in. (mm)
L_{eff} = effective length of the vessel, ft (m)
L_{ss} = vessel length seam-to-seam, ft (m)
P = pressure, psia (kPa)
q = heat input, BTU/hr (kW)
Q = flow rate, ft^3/s (m^3/s)
Q_g = gas flow rate, MMscfd (std m^3/hr)
Q_l = liquid flow rate, BPD (m^3/hr)
Q_o = oil flow rate, BPD (m^3/hr)
Q_w = water flow rate, BPD (m^3/hr)
SG = oil specific gravity
SG_o = specific gravity of water
SG_w = specific gravity of oil
T = temperature, °R (K)
T = temperature, °F (°C)
t = time, s
t_o = oil retention time or settling time, s
t_r = liquid retention time, min
$(t_r)_o$ = oil retention time, min
$(t_r)_w$ = water retention time, min
t_w = water retention time or settling time, s
V = volume, ft^3(m^3)
V_g = gas velocity, ft/s (m/s)
V_l = average liquid velocity, ft/s (m/s)
V_o = oil volume, ft^3(m^3) or oil velocity, ft/s (m/s)

V_t = terminal settling velocity of the droplet, ft/s (m/s)
V_w = water volume, ft^3(m^3)
V_l = average liquid velocity, ft/s (m/s)
W_c = water cut, percent
W = flow rate, lb/hr
w = width of flow, mm
Z = gas compressibility factor, dimensionless
ΔP = pressure loss, psi (kPa)
ΔSG = difference in specific gravity
μ = viscosity of continuous phase, cp (Pa s)
μ_o = viscosity of oil phase, cp (Pa s)
μ_w = viscosity of water phase, cp (Pa s)
ρ = density of the continuous phase, lb/ft^3 (kg/m^3)
ρ_g = density of the gas at the temperature and pressure in the separator, lb/ft^3 (kg/m^3)
ρ_l = density of liquid, lb/ft^3(kg/m^3)
ρ_o = oil density, lb/ft^3(kg/m^3)
ρ_w = water density, lb/ft^3(kg/m^3)

Review Questions

1. List the three things necessary for an emulsion to exist.

2. Match the following terms:

_____ Reverse emulsion	a. Water-in-oil
_____ Stable emulsion	b. Easy to separate
_____ Normal emulsion	c. Oil-in-water
_____ Unstable emulsion	d. Difficult to separate and requires treating

3. List the four primary methods of separating water from crude.

a) _____
b) _____
c) _____
d) _____

4. Agents absorbed at the water–oil interface to lower the interfacial tension are called

   a) emulsions
   b) surfactants
   c) coalescers
   d) destabilizers
   e) B and D

5. The most efficient method to determine the chemicals to best treat an emulsion is accomplished by

   a) a bottle test
   b) electrostatic coalescence
   c) heating
   d) agitation
   e) injection

6. The quantity of demulsifier necessary to produce the desired degree of treatment is influenced by

   a) _____
   b) _____
   c) _____
   d) _____

7. When selecting a chemical to treat an emulsion,

   a) speed is generally not a consideration for a gunbarrel or wash tank
   b) speed becomes a more important consideration for a vertical treater
   c) speed is not an important consideration for a horizontal heater-treater
   d) all of the above
   e) A and B only

8. Settling time

   a) is usually required after chemical addition
   b) is dependent upon the differential density of the oil and water
   c) is the time required for free water to separate from the emulsion
   d) all of the above
   e) A and C

9. The application of heat in the treating process

   a) reduces the size of the treating vessel
   b) may vaporize the light hydrocarbons in the oil
   c) is very expensive
   d) all of the above
   e) A and B

10. One of the most effective pieces of crude oil treating equipment used to remove only free water from the flow stream is a(n)

    a) FWKO
    b) heater-treater
    c) electrostatic treater
    d) gunbarrel
    e) wash tank with an internal gas boot

11. The indirect-fired heater differs from the direct-fired heater in that the indirect-fired heater

    a) is less expensive
    b) is more expensive
    c) has a process flow coil
    d) all of the above
    e) B and C

12. The prime consideration for using direct-fired heaters is

    a) efficiency and cost
    b) less storage space for BS&W
    c) more complicated
    d) all of the above
    e) A and B

13. Figure 7-55 is a schematic of a vertical heater-treater. Label each line with the appropriate identification from the group of devices locates at the bottom of the figure.

14. Figure 7-56 is a schematic of a horizontal electrostatic treater. Label each line with the appropriate identification from the group of devices located at the bottom of the figure.

Fire Tube	Baffles	Emulsion
Heat Exchanger	Drip Trap	Gas
Pressure Equalizing Line	Adjustable Siphon Nipple	Water
Safety Relief Valve	Thermostat	Solids
Mist Extractor	Back Pressure Valve	Well Fluids
Fuel Gas Scrubber	Down-comer	Clean Oil

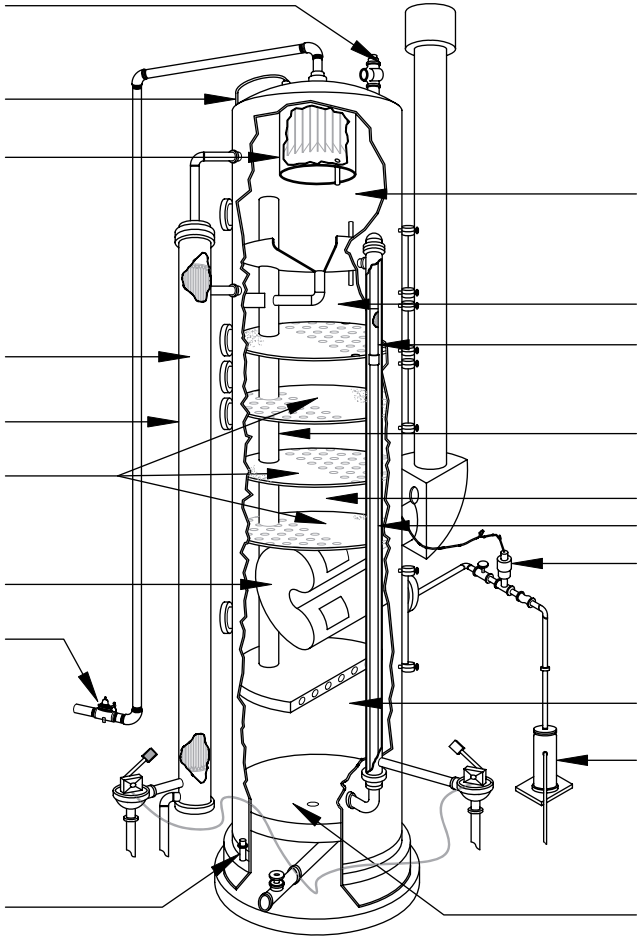


Figure 7-55. Review Question 13, schematic of a vertical heater-treater.

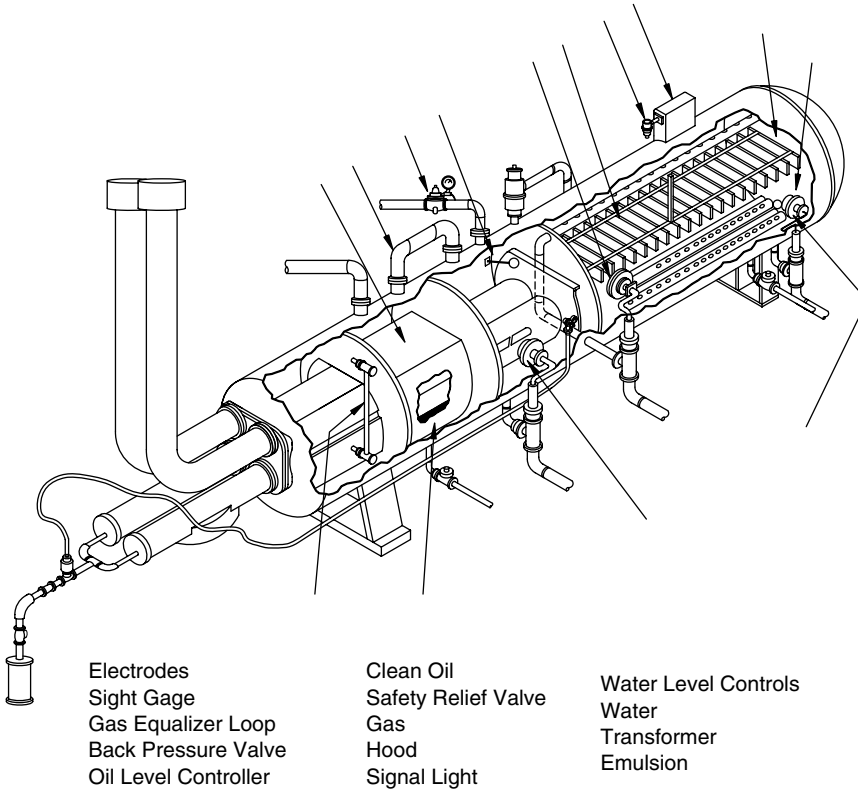


Figure 7-56. Review Question 14, schematic of a horizontal heater-treater.

Exercises

Problem 1.

Given the following data, determine the height of the external leg for a gunbarrel.

Oil gravity	= 45°API,
Water gravity	= 1.05,
Oil outlet height	= 30 ft,
Interface height	= 18 ft,
Water outlet height	= 1.5 ft.

Problem 2.

Given the following data, determine the height of the external leg for a gunbarrel.

Oil gravity = 45°API,
Water gravity = 1.05,
Oil outlet height = 9.15 m,
Interface height = 5.5 m,
Water outlet height = 45 cm.

Problem 3.

Given the following data, select a standard firebox rating for a heater-treater.

Oil flow rate (Q_o) = 1000 BOPD,
Oil gravity = 0.875,
Inlet temperature = 80°F,
Treating temperature = 160°F.

a. 250,000 BTU/hr
b. 575,000 BTU/hr
c. 650,000 BTU/hr
d. 1,225,000 BTU/hr
e. 3,200,000 BTU/hr
f. none of the above

Problem 4.

Determine the size of a horizontal heater-treater given the following data:

Q_o = 5,000 BOPD,
Q_w = 500 BWPD,
P_o = 35 psig,
T_o = 80°F,
$(SG)_o$ = 30°API,
$(SG)_w$ = 1.07,
μ = 10 cp,
$(t_r)_o$ = 25 min.

Investigate at the following temperatures: 110°, 130°, and 150°F.

Problem 5.

Determine the size of a vertical heater-treater given the following data:

Q_o = 2,500 BOPD,
Q_w = 200 BWPD,
P_o = 35 psig,
T_o = 80°F,
$(SG)_o$ = 35°API,
$(SG)_w$ = 1.05,
μ = 10 cp.

Investigate at the following temperatures: 110°, 120°, and 140°F.

Problem 6.

Determine the size of a horizontal heater-treater given the following data:

Q_o = 795 m^3/hr,
Q_w = 80 m^3/hr,
P_o = 6 kPa,
T_o = 35°C,
$(SG)_o$ = 30°API,
$(SG)_w$ = 1.07,
μ = 10 cp,
$(t_r)_o$ = 20 min.

Investigate at the following temperatures: 40°, 55°, and 65°C.

Problem 7.

Determine the size of a vertical heater-treater given the following data:

Q_o = 395 m^3/hr,
Q_w = 40 m^3/hr,
P_o = 5 kPa,
T_o = 35°C,
$(SG)_o$ = 30°API,
$(SG)_w$ = 1.07,
μ = 10 cp,
$(t_r)_o$ = 25 min.

Investigate at the following temperatures: 40°, 50°, and 60°C.

Problem 8.

Determine the size of an "off-the-shelf" FWKO vertical vessel given the following data:

Q_o = 2,000 BOPD,
Q_w = 5,000 BWPD,
P_o = 50 psig,
T_o = 160°F,
$(SG)_o$ = 30°API,
$(SG)_w$ = 1.07,
μ = 10 cp,
$(t_r)_o$ = 20 min.

Problem 9.

Design an oil treating gathering station for the following field:

Given:	
Field size	320 acres,
Well spacing	40 acres,
Number of wells	8,
Gas–oil ratio (maximum)	200 ft^3/bbl,
Q_o (max. per well for 12 years)	100 BOPD,
Q_l (max. per well)	1000 BOPD,
$(SG)_o$	30°API,
$(SG)_w$	1.07,
μ	10 cp,
P_o	30 psig,
T_o	55°F,
Pipeline requirements	less than 1% BS&W.

Reservoir data:

Strong water-drive reservoir, initially wells will flow water-free, but within 1 to 4 years they will begin to produce water, eventually increasing to over 99% at abandonment after a 20-year producing life.

Assumptions:

1) Gas production is negligible.
2) Some pipeline oil storage tanks and emergency storage tank will be required with any treating system selected.
3) Operator coverage will be 7 days per week, twice a day, once in the morning and once in the evening.

4) If a gunbarrel system is selected, a free-water knockout must be used to remove free water and a standard 350,000-BTU/hr fire-box heater must be used to raise the temperature of the fluid from 55°F to 120°F.

Bottle test results indicate that one quart of chemical product will have the following results:

1) Per 100 bbl of oil will remove salt water down to 6% emulsion after 9 min.
2) Per 50 bbl, down to 3% in 9 min.
3) Per 150 bbl, down to 8% in 9 min.

Available equipment, including installed costs:

Vertical two-phase separator (can handle 10,000 BLPD and 1.6 MMscfd) USD 60,000,

Free-water knockouts,

$10' \times 30'$	USD 90,000
$8' \times 20'$	USD 70,000
$8' \times 15'$	USD 60,000
$6' \times 20'$	USD 45,000
$6' \times 15'$	USD 40,000

Three-phase separators:

$10' \times 30'$	USD 110,000
$8' \times 20'$	USD 90,000
$8' \times 15'$	USD 80,000
$6' \times 20'$	USD 70,000
$6' \times 15'$	USD 60,000

Vertical heater-treater (can handle up to 875 BOPD, treating below 0.5% BS&W, the following emulsions):

1) 3% emulsion	USD 140,000 + USD 40,000/yr fuel
2) 6% emulsion	USD 160,000 + USD 40,000/yr fuel
3) 8% emulsion	USD 200,000 + USD 40,000/yr fuel

Electrostatic heater-treater (can handle up to 875 BOPD, treating below 0.5% BS&W, the following emulsions)

1) 3% emulsion	USD 140,000 + USD 40,000/yr fuel
2) 6% emulsion	USD 160,000 + USD 40,000/yr fuel
3) 8% emulsion	USD 200,000 + USD 40,000/yr fuel

Gunbarrels (12 hours' retention time)

1) 750 BBL USD 120,000
2) 500 BBL USD 100,000

Heaters

350,000 BTU/hr USD 40,000 + USD 50,000/yr fuel

Determine:

1) Select your recommended configuration of equipment that will produce pipeline-quality oil.
   At a minimum, the treating system design should include the following considerations:

   a) Chemical injection may be necessary.
   b) Some method of primary separation to remove the gas should be provided.
   c) A vessel capable of providing pipeline-quality oil is required.

2) Size and select the vessels for the equipment configuration.
3) Assign installation and operating cost values and total the associated costs.

Comments on configuration selection:

The configuration selected can be influenced by many considerations if several configurations are economically near-equal. The fewer number of vessels for maintenance should be a strong consideration. Very often, long delivery lead times on one type of equipment may require selection of another configuration, if that lead time will significantly delay the startup of oil sales from a new field. All of these and other factors should be considered prior to configuration selection and equipment requisitioning.

Reference

1. M. E. Thro and K. E. Arnold, "Water Droplet Size Determination for Improved Oil Treater Sizing," Society of Petroleum Engineers 69th Annual Technical Conference and Exhibition, New Orleans, LA, 1994.

Chapter 8
Crude Stabilization

Introduction

The liquids that are separated from the gas stream during initial separation may be flowed directly to a tank or may be "stabilized" in some fashion. As was discussed in Chapter 2, these liquids contain a large percentage of methane and ethane, which will flash to gas in the tank. This lowers the partial pressure of all other components in the tank and increases their tendency to flash to vapors. Stabilization is the process of increasing the amount of intermediate (C_3 to C_5) and heavy (C_{6+}) components in the liquid phase. In an oil field this process is called crude stabilization and in a gas field it is called condensate stabilization.

In almost all cases the molecules have a higher value as liquid than as a gas. Crude oil streams typically contain a low percentage of intermediate components. Thus, it is not normally economically attractive to consider other alternatives to multistage separation to stabilize the crude. In addition, the requirement to treat the oil at high temperature is more important than stabilizing the liquid and may require the flashing of both intermediate and heavy components to the gas stream.

Gas condensate, on the other hand, may contain a relatively high percentage of intermediate components. Thus, some sort of condensate stabilization should be considered for each gas well production facility.

The most common method used to remove the light components from hydrocarbon liquids before the liquid enters a stock tank or a pipeline is stage separation. Separation followed by weathering in a stock tank is the simplest method of stabilization, but it is often the most efficient method. A stabilizer can achieve a stable specification product with a higher liquid recovery, but usually results in higher capital expenditures' (CAPEX) and operating expenses (OPEX). The addition of a stabilizer requires additional space which is normally not a factor for onshore applications, but may be a major consideration for an offshore installation.

Produced hydrocarbons from wells normally flow to a separator for removal of the hydrocarbon gas. The hydrocarbon crude or condensate oil outflow from the separator usually goes through additional stages of separation or treatment before reaching the sales point. In each of these stages the liquid reaches near equilibrium at a different condition of pressure and temperature thus to some extent "stabilizing" the crude or condensate.

The following methods of crude stabilization are normally used:

- Multi-stage separation
- Weathering in a stock tank
- Heater-treater after separation
- Stabilizer.

The method one selects for stabilization depends primarily on contract specifications and economics. Factors that favor the installation of a stabilization unit include:

- An oil contract specification that requires a low crude vapor pressure that cannot easily be obtained by stage separation.
- A sour crude with a contract specification that limits the H_2S content to less than 60 ppm.
- Condensate production with 50°API or higher and flow rates in excess of 5,000 bpd (m^3/hr).

This chapter reviews basic principles involved in stabilization, the various process schemes used to stabilize a liquid hydrocarbon stream, and the equipment used in the stabilization process.

Basic Principles

Phase-Equilibrium Considerations

Before one can effectively analyze any stabilization process, they should have an understanding of the nature of the equilibrium relationships in multi-component hydrocarbon systems and the regions in which each calculation is applicable. Nearly all hydrocarbon processing operations involve some form of equilibrium between vapor and liquid phases. As was discussed in detail in Chapter 3, the distribution of individual components between phases was correlated in terms of equilibrium ratios, or K values, which are functions of the temperature, pressure, and composition of the system.

A typical pressure-temperature phase diagram is shown in Figure 8-1. As was shown in Chapter 3, a diagram of this type can be drawn from any

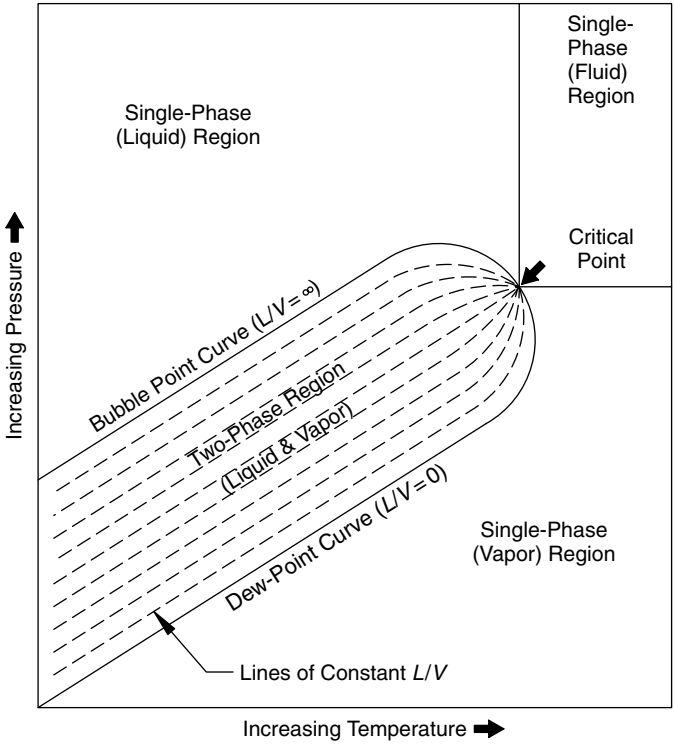


Figure 8-1. Phase diagram for a multi-component hydrocarbon system.

system of fixed composition. Actual pressure and temperature coordinates will obviously be different for various compositions. The equilibrium relationships apply only at pressure and temperature combinations in the two-phase region, or in the area between the bubble point and dew point curves (refer to Figure 8-1). The bubble point curve represents the point at which the first bubble of vapor forms from a liquid-phase system. The dew point curve represents the point at which the first drop of liquid forms from a vapor-phase system of fixed composition. In the high pressure-low temperature region, the hydrocarbon mixture will be in a single liquid phase; in the high temperature-low pressure region, the hydrocarbon mixture will be in a single vapor phase. Equilibrium relationships are not applicable in the single phase regions.

Referring to Figure 8-1, the point where the bubble point and dew point converge is defined as the critical point. In the region above the critical pressure and temperature, the hydrocarbon mixture exists as a single phase where the vapor and liquid phases are indistinguishable.

Equilibrium separation involves the two-phase region between the bubble point and dew point curves. Equilibrium calculations are often referred to as "flash" calculations, and are based upon a combination of the vapor-liquid equilibrium relationship and material balance equations. Flash calculations allow one to determine the amount of vapor and liquid at any point in the system. Flash calculations, including the calculation of the bubble point and dew point, are discussed in detail in Chapter 3.

Flash Calculations

Flash calculations allow one to determine the amount of hydrocarbon vapor and liquid at any point in the process. At a specific pressure and temperature, each component of a hydrocarbon mixture will be in equilibrium. The mole fraction of any component in the vapor phase depends not only on the pressure and temperature but also on the partial pressure of that component.

As discussed in Chapter 2, the fraction of any one component that flashes to gas at any stage in a process is a function of the temperature, pressure, and composition of the fluid at that stage. Thus, the amount of vapor depends upon the total composition of the fluid, because the mole fraction of any one component in the gas phase is a function of every other component in that phase.

Process Schemes

Multi-Stage Separation

Figure 8-2 shows a multi-stage separation system. This is the most common method of separating oil and gas. This system typically requires from two to four separation stages, each occurring in a separator vessel and is described in Chapter 2.

Oil Heater-Treaters

Three-phase separators, which utilize gravity separation, often are not adequate to separate the water from the oil. Heating the emulsion is commonly used to break the emulsion. As discussed in Chapter 7, heater treaters not only improve the oil-water separation process, but also stabilize the crude by vaporizing the light hydrocarbons prior to the crude flowing to an atmosphere pressure storage tank. Utilizing heater-treaters

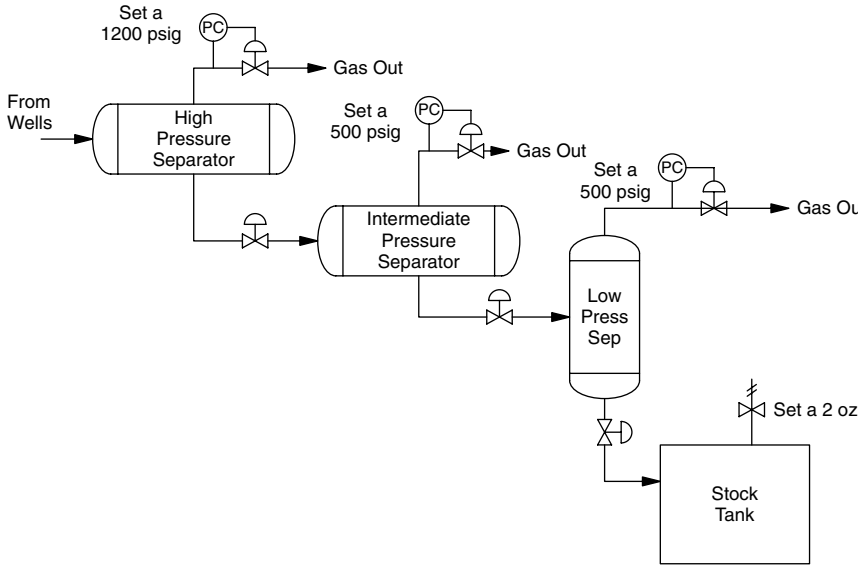


Figure 8-2. Schematic of a three-stage separation system.

alone often results in higher than desired losses of intermediate components to the vapor phase when the hot crude is flashed entering the storage tank.

The crude departing the treater can be cooled before going to the storage tank by exchanging heat with the colder emulsion upstream of the treater. This will lead to fewer vapor losses and will help stabilize the intermediate components when the crude is flashed at storage tank conditions. For small flow rates, the oil-treating temperature is kept as low as possible to prevent stock tank losses, since the treated oil will normally go directly to the stock tank without cooling.

Liquid Hydrocarbon Stabilizer

It is possible to stabilize a hydrocarbon liquid at constant pressure by successively flashing the hydrocarbon liquid at increasing temperatures as shown in Figure 8-3. At each successive stage the partial pressure of the intermediate components is higher than it could have been at that temperature if some of the lighter components had not been removed by the previous stage. It would be very costly to arrange a process as shown in Figure 8-3 and thus never done. Instead, the same effect can

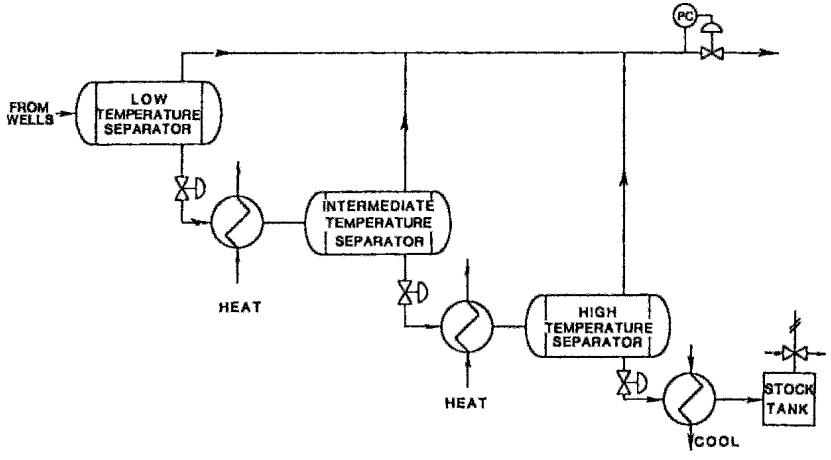


Figure 8-3. Multiple flashes at constant pressure and increasing temperature.

be obtained in a tall, vertical pressure vessel with a cold temperature at the top and a hot temperature at the bottom. This unit is called a "stabilizer."

A stabilizer applies the same principles as multi-stage separation except that the flashes take place in a stabilizer tower operating at a constant pressure, but with varying temperatures. The stabilizer tower is normally a trayed vertical pressure vessel; however, structured packing may also be used. As heat is added to the bottom of the stabilizer tower, vapors are generated on the bottom tray. The hot vapors rise to the tray above, where they bubble through the liquid. The liquid is heated by the hot vapors, which vaporize some of the hydrocarbon liquid. The vapors, in turn, are cooled by the liquid, and a portion of the vapor is condensed. This process of vaporization and condensation is repeated on each tray in the stabilizer tower. As the liquids fall down the stabilizer tower, the heavier hydrocarbons are condensed so that the hydrocarbon liquids leaving the stabilizer tower contain almost none of the light hydrocarbon components, and the vapor leaving the top of the stabilizer tower contains almost none of the heavier components.

The vapor pressure of the liquid hydrocarbon leaving the bottom of the tower is controlled by controlling the stabilizer tower pressure and bottom temperature. At a constant pressure, the liquid hydrocarbon product's vapor pressure can be increased by lowering the bottom temperature, or decreased by increasing the bottom temperature.

Figure 8-4 illustrates a liquid hydrocarbon stabilizer system. The well stream flows to a high pressure, three-phase separator. Liquids containing

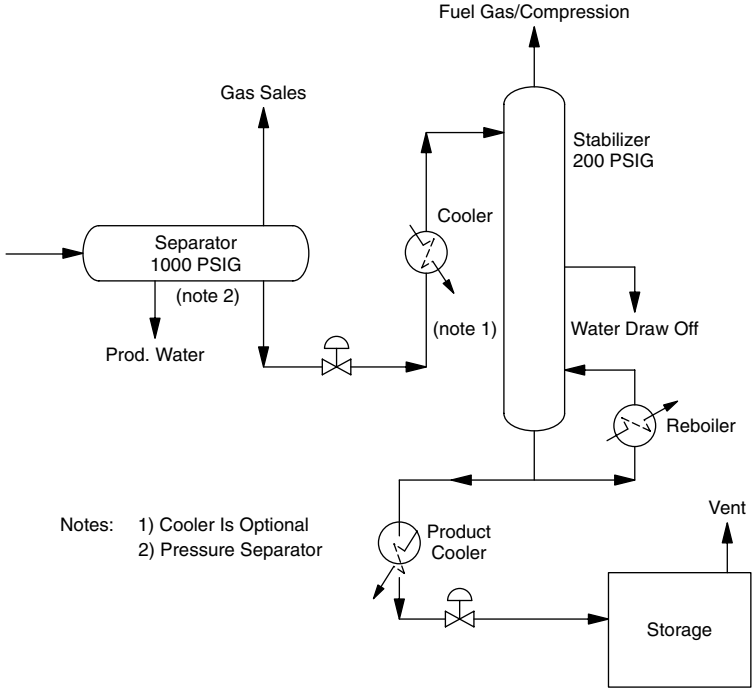


Figure 8-4. Schematic of a typical cold-feed stabilization system.

a high fraction of light ends are cooled and enter the stabilizer tower at a pressure between 100 to 200 psi (700 to 1,400 kPa).

As the hydrocarbon liquid falls from tray to tray in the stabilizer tower, it is heated by the hot gases bubbling through the liquid. On each tray some of the liquids are vaporized and some of the hot gases are condensed. The liquids falling down the stabilizer tower become richer and richer in heavy hydrocarbon components and leaner and leaner in light hydrocarbons. At the bottom of the stabilizer tower, some of the liquids are cycled to a reboiler where they receive heat to provide the necessary bottom temperature which is normally in the range of 200° to 400°F (90° to 200°C). The reboiler could be a direct-fired bath, an indirect-fired bath, or a heating media exchanger. For a specific bottom product's vapor pressure, a lower stabilizer tower operating pressure requires a lower bottom temperature, but more compression is required for the overhead vapors.

The hydrocarbon liquid leaving the stabilizer tower at the bottom tray temperature is in equilibrium with the vapors and is at its bubble point. The liquid leaving the stabilizer tower is cooled before going to storage or pipeline. The hydrocarbon vapors leaving the top of the stabilizer tower

are in equilibrium with the liquids on the top tray and are at their dew point.

One design consideration that needs to be addressed in the design of a stabilizer system is whether to use a cold feed or reflux. A cold-feed stabilizer without reflux such as that shown in Figure 8-4 does not achieve as good a split between the light and heavy components as a column with reflux (see Figure 8-5 and the following discussion); thus, recoveries are not as high. However, a stabilizer with reflux requires additional equipment, higher CAPEX, and higher OPEX, but achieves a higher recovery. Descriptions of both a cold-feed stabilizer and a stabilizer with reflux follow.

Cold-Feed Stabilizer

A conventional stabilizer tower is a distillation column with a reboiler, but no overhead condenser (refer to Figure 8-4). The lack of an overhead condenser means that there is no liquid reflux from the overhead stream. Thus, the feed is introduced on the top tray and must provide all the cold liquid for the stabilization tower. Since the feed is introduced on the top tray, it is important to minimize the flashing of the feed so that intermediate components are not lost overhead. To lower the feed stream temperature and reduce flashing, a cooler is sometimes added on the inlet feed stream.

Adding a cooler on the inlet feed stream lowers the temperature of the inlet hydrocarbon liquid, lowers the fraction of intermediate components that flash to vapor on the top tray and increases the recovery of these components in the liquid bottoms. However, the colder the feed, the more heat is required from the reboiler to remove light components from the liquid bottoms. If too many light components remain in the liquid, the vapor pressure limitations for the liquid may be exceeded. Light components may also encourage flashing of intermediate components (by lowering their partial pressure) in the storage tank. There is a balance between the amount of inlet cooling and the amount of reboiling required.

The hydrocarbon liquid out the bottom of the stabilizer tower must meet a specified vapor pressure. The stabilizer tower is designed to maximize the molecules of intermediate components in the liquid without exceeding the vapor pressure specification. This is accomplished by driving the maximum number of molecules of methane and ethane out of the liquid and keeping as much of the heavier ends as possible from going out with the gas. The hot liquid from the stabilizer is at its bubble point at the pressure and temperature in the stabilizer. It must be cooled sufficiently to avoid flashing when it enters the atmospheric storage tank.

Given inlet composition, pressure, and temperature, a stabilization tower temperature and the number of trays that produce a liquid with a specified vapor pressure can be chosen as follows:

1. Assume an initial split of components in the inlet that yields the desired vapor pressure. That is, assume a split of each component between the tower overhead (gas) and bottoms (liquid). There are various rules of thumb that can be used to estimate this split in order to give a desired vapor pressure. Once the split is made, both the assumed composition of the liquid and the assumed composition of the gas are known.
2. Calculate the temperature required at the base of the tower to develop this liquid. This is the temperature at the bubble point for the stabilizer tower pressure and for the assumed outlet composition. Since the composition and pressure are known, the temperature at its bubble point can be calculated.
3. Calculate the composition of the gas in equilibrium with the liquid. The composition, pressure, and temperature of the liquid are known, and the composition of the gas that is in equilibrium with this liquid can be calculated.
4. Calculate the composition of the inlet liquid falling from Tray 1. Since the composition of the bottom liquid and gas in equilibrium with the liquid is known, the composition of the feed to this tray is also known. This is the composition of the liquid falling from Tray 1.
5. Calculate the temperature of Tray 1. From an enthalpy balance, the temperature of the liquid falling from Tray 1, and thus the temperature of the flash on Tray 1, can be calculated. The composition is known, the enthalpy can be calculated. Enthalpy must be maintained, so the enthalpy of the liquid of known composition falling from Tray 1 must equal the sum of the enthalpies of the liquid and gas flashing from it at known temperature.
6. This procedure can then be carried on up the tower to Tray N, which establishes the temperature of the inlet and the gas outlet composition.
7. From the composition of the inlet and gas outlet the liquid outlet composition can be calculated and compared to that assumed in step 1.
8. The temperature or number of trays can then be varied until the calculated outlet liquid composition equals the assumed composition, and the vapor pressure of the liquid is equal to or less than that assumed. If the vapor pressure of the liquid is too high, the bottom temperature must be increased.

The overhead gas can be used as fuel, or compressed and included with the sales gas. Any water that enters the column in the feed stream will collect in the middle of the column due to the range of temperatures involved. This water cannot leave with the bottom product or with the overhead stream; therefore, provisions should be made to remove this water from a tray near the middle of the column. The heating of the liquid hydrocarbon in the stabilizer tower acts as a demulsifier to remove water from hydrocarbon liquid. The excellent water-separating ability of the stabilizer usually eliminates the need for a hydrocarbon liquid dehydration system.

Stabilizer with Reflux

Figure 8-5 shows a typical stabilizer system with reflux and a feed/bottom heat exchanger. In this configuration, the well fluid is heated by the bottom product and injected into the stabilizer tower, below the top, where the temperature in the stabilizer tower is equal to the temperature of the feed. The stabilizer tower's top temperature is controlled by cooling and condensing part of the hydrocarbon vapors leaving the stabilizer and pumping the resulting hydrocarbon liquids back to the tower. This replaces the cold feed configuration and allows better control of the overhead product and, consequently, slightly higher recovery of the heavier components. This configuration minimizes the amount of flashing.

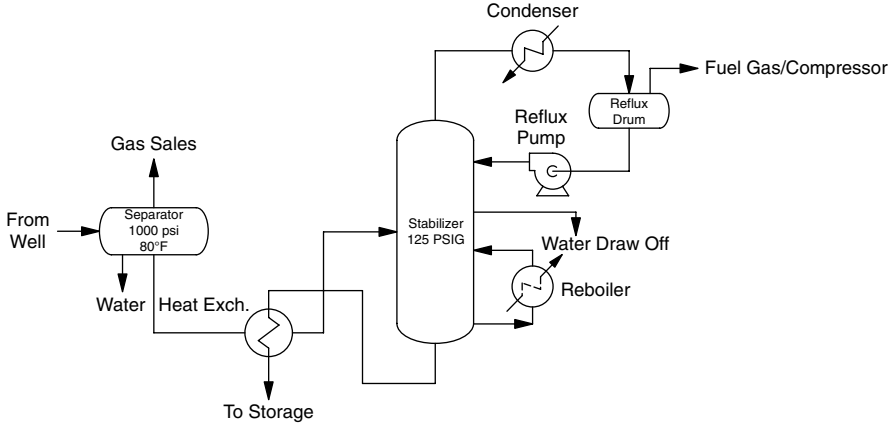


Figure 8-5. Schematic of a typical crude stabilization with reflux and feed/bottom heat exchanger.

The principles of this configuration are the same as in a cold-feed stabilizer or any other stabilizer tower. As the liquid falls through the tower, it goes from tray to tray, and gets increasingly richer in the heavier components and increasingly leaner in the lighter components. The stabilized hydrocarbon liquid is cooled in the heat exchanger by the feed stream before flowing to the stock tank or pipeline.

At the top of the stabilizer tower intermediate components going out with the gas are condensed, separated, pumped back to the stabilizer tower, and sprayed down on the top tray. This liquid is called "reflux," and the two-phase separator that separates it from the hydrocarbon liquid from the gas is called a "reflux tank" or "reflux drum." The reflux performs the same function as the cold feed in a cold feed stabilizer. Cold liquid hydrocarbons strip out the intermediate components from the gas as the gas rises.

The heat required at the reboiler depends upon the amount of cooling done in the condenser. The colder the condenser, the purer the product, and the larger the percentage of the intermediate components that will be recovered in the separator and kept from going out with the gas. The hotter the bottom temperature, the greater the percentage of light components boiled out of the bottoms. The greater the percentage of light components boiled out of the bottoms liquid, the lower the vapor pressure of the bottoms liquid.

A heat balance around the stabilizer tower is part of the design. The heat leaves the stabilizer tower in the form of vapors out the top, and the liquid bottom product has to be balanced by the heat entering in the feed and the reboiler. If the stabilizer tower has a reflux, this amount of heat has to be added to the column balance.

A stabilizer tower with reflux will recover more intermediate components from the gas than a cold-feed stabilizer tower. However, it requires more equipment to purchase, install, and operate. This additional cost must be justified by the net benefit of the incremental hydrocarbon liquid recovery, less the cost of natural gas shrinkage and loss of heating value, over that obtained from a cold-feed stabilizer.

Equipment Description

Stabilizer Tower

The stabilizer tower is a fractionation tower using trays or packing. Figure 8-6 shows a stabilizer tower with bubble cap trays. Trays, structured packing, or random packing are used in the tower to promote intimate contact between the vapor and liquid phases, thereby permitting

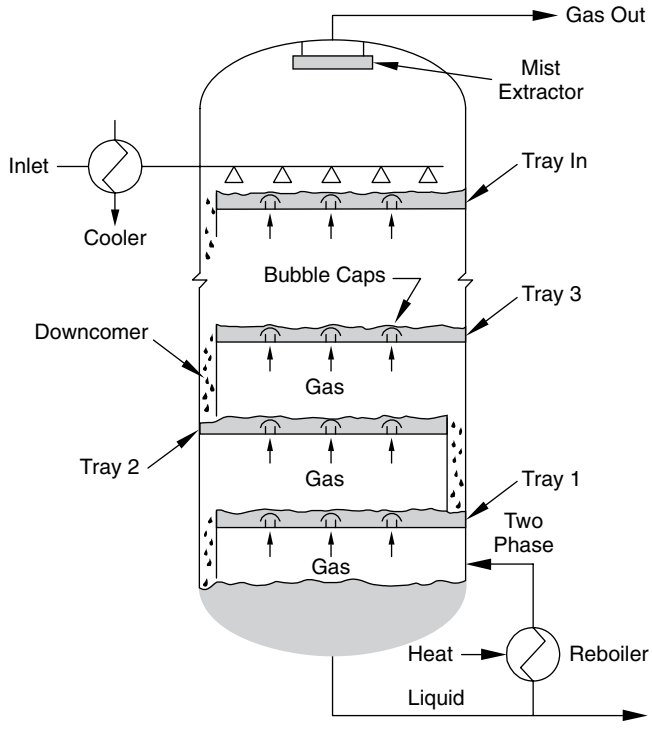


Figure 8-6. Schematic of a stabilizer tower.

the transfer of mass and heat from one phase to the other. The feed to the stabilizer tower normally enters near the top of a cold-feed stabilizer, and at or near the tray where the stabilizer tower conditions and feed composition most nearly match the inlet feed conditions, in stabilizer towers with reflux. The liquids in the stabilizer tower fall down through the down-comer, across the tray, over the weir and into the down-comer to the next tray. The temperature on each tray increases as the liquids drop from tray to tray. Hot gases come up the stabilizer tower and bubble through the liquid on the tray above, where some of the heavier components in the gas are condensed and some of the lighter components in the liquid are vaporized. The gas gets leaner and leaner in heavy hydrocarbons as it travels up the stabilizer tower; the falling liquids get richer and richer in the heavier hydrocarbon components. The vapors leaving the top of the stabilizer tower contain a minimum amount of heavy hydrocarbons, and the liquid leaving the bottom of the tower contains a minimum of light hydrocarbons. Stabilizer columns commonly operate at pressures between 100 to 200 psig (700 to 1,400 kPa).

Trays and Packing

The number of actual equilibrium stages determines the number of flashes that will occur. The more stages, the more complete the split, but the taller and more costly the tower. Most stabilizers will normally contain approximately five theoretical stages. In a refluxed tower, the section above the feed is known as the rectification section, while the section below the feed is known as the stripping section. The rectification section normally contains about two equilibrium stages above the feed, and the stripping section normally contains three equilibrium stages.

Trays

For most trays, liquid flows across an "active area" of the tray and then into a "down-comer" to the next tray below, etc. Inlet and/or outlet weirs control the liquid distribution across the tray. Vapor flows up the stabilizer tower and passes through the tray active area, bubbling up through (and thus contacting) the liquid flowing across the tray. The vapor distribution is controlled by:

- Perforations in the tray deck (sieve trays),
- Bubble caps (bubble cap trays), or
- Valves (valve trays).

Trays operate within a hydraulic envelope. At excessively high vapor rates, liquid is carried upward from one tray to the next (essentially backmixing the liquid phase in the stabilizer tower). For valve trays and sieve trays, a capacity limit can be reached at low vapor rates when liquid falls through the tray floor rather than being forced across the active area into the down-comers. Because the liquid does not flow across the trays, it misses contact with the vapor, and the separation efficiency drops dramatically.

Trays are generally divided into four categories:

- Sieve trays,
- Valve trays,
- Bubble cap trays, and
- High capacity/high efficiency trays.

Sieve Trays Sieve trays are the least expensive tray option. In sieve trays, vapor flowing up through the tower contacts the liquid by passing through small perforations in the tray floor (Figure 8-7b). Sieve trays rely on vapor velocity to exclude liquid from falling through the perforations in the tray floor. If the vapor velocity is much lower than design, liquid

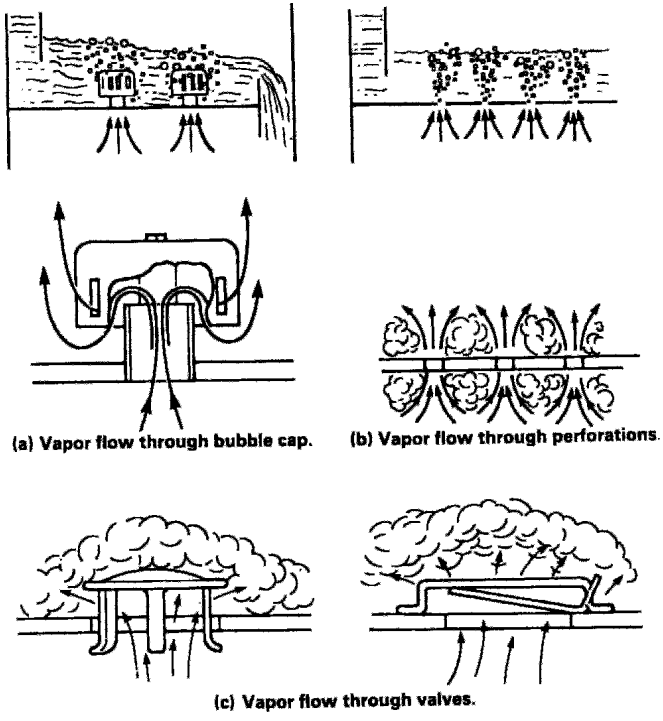


Figure 8-7. Vapor flow through trays.

will begin to flow through the perforations rather than into the downcomer. This condition is known as weeping. Where weeping is severe, the equilibrium efficiency will be very low. For this reason, sieve trays have a very small turndown ratio.

Valve Trays Valve trays are essentially modified sieve trays. Like sieve trays, holes are punched in the tray floor. However, these holes are much larger than those in sieve trays. Each of these holes is fitted with a device called a "valve." Vapor flowing up through the tower contacts the liquid by passing through valves in the tray floor (Figure 8-7c). Valves can be fixed or moving. Fixed valves are permanently open and operate as deflector plates for the vapor coming up through the tray floor. For moving valves, vapor passing through the tray floor lifts the valves and contacts the liquid. Moving valves come in a variety of designs, depending on the manufacturer and the application. At low vapor rates, valves will close, helping to keep liquid from falling through the holes in the deck. At sufficiently low vapor rates, a valve tray will begin to weep. That is, some liquid will leak through the valves rather than flowing to the tray

down-comers. At very low vapor rates, it is possible that all the liquid will fall through the valves and no liquid will reach the down-comers. This severe weeping is known as "dumping." At this point, the efficiency of the tray is nearly zero.

Bubble Cap Trays In bubble cap trays, vapor flowing up through the tower contacts the liquid by passing through bubble caps (Figure 8-7a). Each bubble cap assembly consists of a riser and a cap. The vapor rising through the tower passes up through the riser in the tray floor and then is turned downward to bubble into the liquid surrounding the cap. Because of their design, bubble cap trays cannot weep. However, bubble cap trays are also more expensive and have a lower vapor capacity/higher pressure drop than valve trays or sieve trays.

High Capacity/High Efficiency Trays High capacity/high efficiency trays have valves or sieve holes or both. They typically achieve higher efficiencies and capacities by taking advantage of the active area under the down-comer. At this time, each of the major vendors have their own version of these trays, and the designs are proprietary.

Bubble Cap Trays vs. Valve Trays At low vapor rates, valve trays will weep. Bubble cap trays cannot weep (unless they are damaged). For this reason, it is generally assumed that bubble cap trays have nearly an infinite turndown ratio. This is true in absorption processes (e.g., glycol dehydration), in which it is more important to contact the vapor with liquid than the liquid with vapor. However, this is not true of distillation processes (e.g., stabilization), in which it is more important to contact the liquid with the vapor.

As vapor rates decrease, the tray activity also decreases. There eventually comes a point at which some of the active devices (valves or bubble caps) become inactive. Liquid passing these inactive devices gets very little contact with vapor. At this point, it is possible that liquid may flow across the entire active area without ever contacting a significant amount of vapor. This will result in very low efficiencies for a distillation process. Nothing can be done with a bubble cap tray to compensate for this.

However, a valve tray can be designed with heavy valves and light valves. At high vapor rates, all the valves will be open. As the vapor rate decreases, the valves will begin to close. With light and heavy valves on the tray, the heavy valves will close first, and some or all of the light valves will remain open. If the light valves are properly distributed over the active area, even through the tray activity is diminished at low vapor rates, what activity remains will be distributed across the tray. All liquid flowing across the tray will contact some vapor, and mass transfer

will continue. Of course, even with weighted valves, if the vapor rate is reduced enough, the tray will weep and eventually become inoperable. However, with a properly designed valve tray this point may be reached after the loss in efficiency of a comparable bubble cap tray. So, in distillation applications, valve trays can have a greater vapor turndown ratio than bubble cap trays.

Tray Efficiency and Stabilizer Height In general, stabilizer trays generally have a 70% equilibrium stage efficiency. That is, 1.4 actual trays are required to provide one theoretical stage. The spacing between trays is a function of the spray height and the down-comer backup (the height of clear liquid established in the down-comer). The tray spacing will typically range from 20 to 30 inches (with 24 inches being the most common), depending on the specific design and the internal vapor and liquid traffic. The tray spacing may increase at higher operating pressures (greater than 165 psia) because of the difficulty in disengaging vapor from liquid in the active areas of the tray.

Packing

Packing typically comes in two types: random and structured. Liquid distribution in a packed bed is a function of the internal vapor/liquid traffic, the type of packing employed, and the quality of the liquid distributors mounted above the packed bed. Vapor distribution is controlled by the internal vapor/liquid traffic, by the type of packing employed, and by the quality of the vapor distributors located below the packed beds.

Packing material can be plastic, metal, or ceramic. Packing efficiencies can be expressed as height equivalent to a theoretical plate (HETP).

Random Packing A bed of random packing typically consists of a bed support (typically a gas injection support plate) upon which pieces of packing material are randomly arranged (they are usually poured or dumped onto this support plate). Bed limiters, or hold-downs, are sometimes set above random beds to prevent the pieces of packing from migrating or entraining upward. Random packing comes in a variety of shapes and sizes. For a given shape (design) of packing, small sizes have higher efficiencies and lower capacities than large sizes.

Figure 8-8 shows a variety of random packing designs. An early design is known as a Rasching ring. Rasching rings are short sections of tubing and are low-capacity, low-efficiency, high-pressure drop devices. Today's industry standard is the slotted metal (Pall) ring. A packed bed made of 1-inch slotted metal rings will have a higher mass transfer efficiency and a higher capacity than will a bed of 1-inch Rasching rings. The HETP

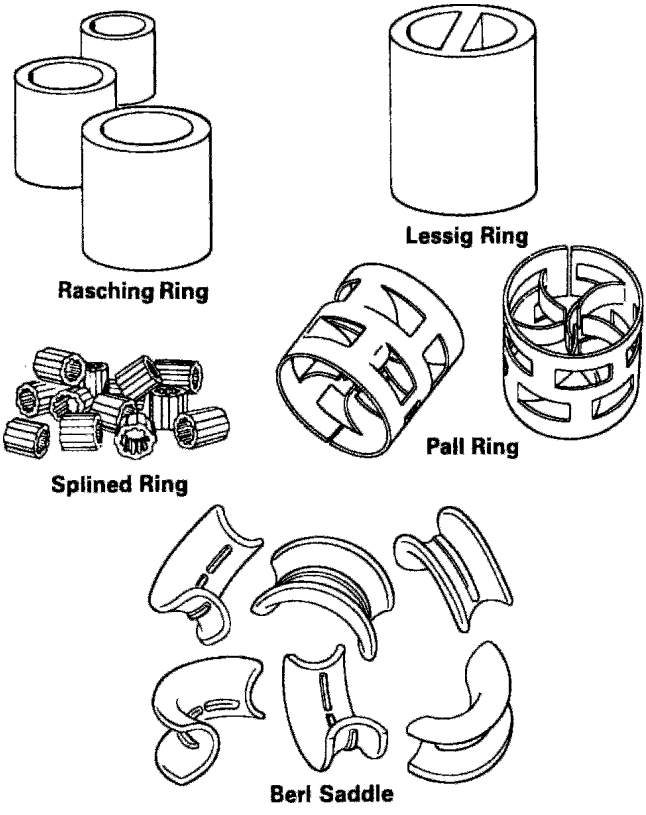


Figure 8-8. Various types of random packing.

for a 2-inch slotted metal ring in a stabilizer is about 36 inches. This is slightly more than a typical tray design, which would require 34 inches (1.4 trays × 24-inch tray spacing) for one theoretical plate or stage.

Structured Packing A bed of structured packing consists of a bed support upon which elements of structured packing are placed. Beds of structured packing typically have lower pressure drops than beds of random packing of comparable mass transfer efficiency. Structured packing elements are composed of grids (metal or plastic) or woven (metal or plastic) or of thin vertical crimped sheets (metal, plastic, or ceramic) stacked parallel to each other. Figure 8-9 shows examples of the vertical crimped sheet style of structured packing.

The grid types of structured packing have very high capacities and very low efficiencies, and are typically used for heat transfer or for vapor

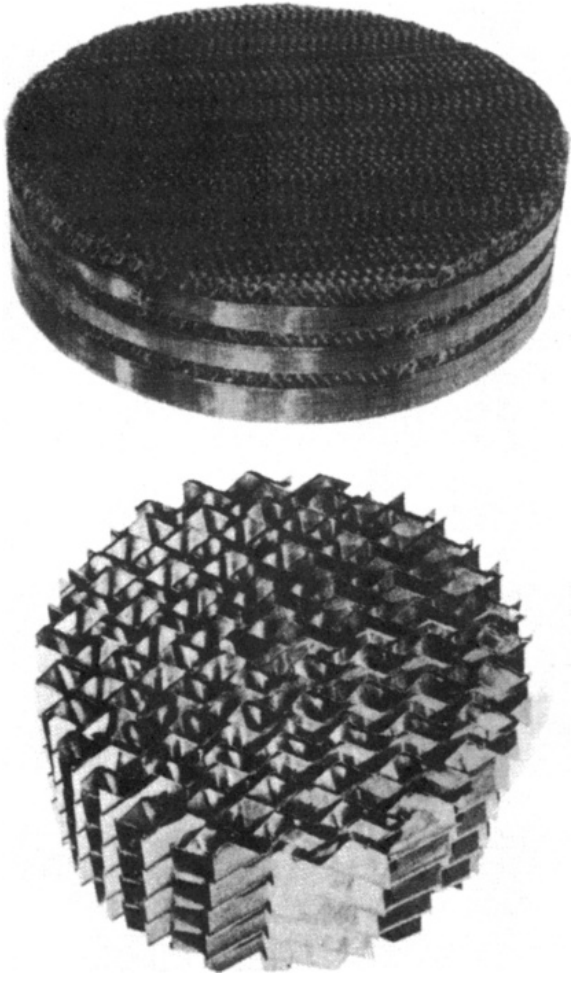


Figure 8-9. Structured packing can offer better mass transfer than trays. (Courtesy of Koch Engineering Co., Inc.)

scrubbing. The wire mesh and the crimped sheet types of structured packing typically have lower capacities and higher efficiencies than the grid type.

Trays or Packing

There is no umbrella answer. The choice is dictated by project scope (new tower or retrofit), current economics, operating pressures, anticipated operating flexibility, and physical properties.

Distillation Service For distillation services, as in hydrocarbon stabilization, tray design is well understood, and many engineers are more comfortable with trays than with packing. In the past, bubble cap trays were the standard. However, they are not commonly used in this service anymore. Sieve trays are inexpensive but offer a very narrow operating range when compared with valve trays. Although valve trays offer wider operating range than sieve trays, they have moving parts and so may require more maintenance. High capacity/high efficiency trays can be more expensive than standard valve trays. However, high capacity/high efficiency trays require smaller diameter stabilization towers, so they can offer significant savings in the overall cost of the distillation tower. The high capacity/high efficiency tray can also be an ideal candidate for tower retrofits in which increased throughputs are required for existing towers.

Random packing has traditionally been used in small diameter (<20 inches) towers. This is because it is easier and less expensive to pack these small diameter towers. However, random packed beds are prone to channeling and have poor turndown characteristics when compared with trays. For these reasons, trays were preferred for tower diameters greater than 20 inches. In recent years an improved understanding of the impact of the high pressure on packing performance has been gained. Improved vapor and liquid distributor designs and modified bed heights have made the application of packing to large-diameter, high-pressure distillation towers more common. A properly designed packed bed system (packing, liquid distributors, vapor distributors) can be an excellent choice for debottlenecking existing distillation towers.

Stripping Service For stripping service, as in a glycol or amine contactor, bubble cap trays are the most common. In recent years, there has been a growing movement toward crimped sheet structured packing. Improved vapor and liquid distributor design in conjunction with structured packing can lead to smaller-diameter and shorter stripping towers than can be obtained with trays.

Stabilizer Reboiler

The stabilizer reboiler boils the bottom product from the stabilizer tower. The source of all heat used to generate vapor in a stabilizer is the reboiler. The boiling point of the bottom product is controlled by controlling the heat input of the reboiler. Together with the stabilizer operating pressure, this action controls the vapor pressure of the bottom product.

The reboiler may be either a kettle-type or a thermo-siphon type reboiler. Reboiler temperatures typically range from 200° to 400°F

(90° to 200°C) depending on operating pressure, bottom product composition, and vapor pressure requirements. Its important to note that reboiler temperatures should be kept to a minimum to decrease the heat requirements, limit salt buildup, and prevent corrosion problems.

Maintaining stabilizer operating pressures below 200 psig (1,400 kPa) will result in reboiler temperatures below 300°F (150°C). A water-glycol heating medium can then be used to provide heat. Higher stabilizer pressures require the use of steam or hydrocarbon-based heating mediums. However, operating at high pressures decreases the flashing of the feed when entering the stabilizer tower and decreases the amount of feed cooling required. In general, a liquid hydrocarbon stabilizer should be designed to operate between 100 to 200 psig (700 and 1,400 kPa).

Selection of a stabilizer heat source depends on the medium and tower operating pressure. The source of reboiler heat should be considered when a crude stabilizer is being evaluated. If turbine generators or compressors are installed nearby, then waste heat recovery should be considered.

Stabilizer Cooler

The stabilizer cooler is used to cool the bottom product leaving the tower before it goes to a tank or pipeline. The temperature of the bottom product may be dictated by contract specification or by efforts to prevent loss of vapors from an atmospheric storage tank.

For a stabilizer with a reflux system, the bottom product may be cooled by exchanging heat with the feed to the stabilizer.

Stabilizer Reflux System

The stabilizer reflux system consists of a reflux condenser, reflux accumulator, and reflux pumps. The system is designed to operate at a temperature required to condense a portion of the vapors leaving the top of the stabilizer. The temperature range is determined by calculating the overhead vapor's dew point temperature. The heat duty required is determined by the amount of reflux required.

The type of exchanger selected for the reflux depends on the design temperature required to condense the reflux. The lower the operating pressure of the stabilizer, the lower the temperature required for condensing the reflux. In most installations, air-cooled exchangers may be used. Some installations may require refrigeration and a shell-and-tube exchanger configuration.

The reflux accumulator consists of a two-phase separator with several minutes of retention time to allow separation of the vapors and liquids. The reflux accumulator is normally located below the reflux condenser, with the line sloped from the condenser to the accumulator. The reflux accumulator must be located above the reflux pumps to provide the necessary net positive suction head (NPSH) required by the pumps. The size of the reflux accumulator depends on the amount of reflux required and the total amount of vapors leaving the stabilization tower.

Reflux pumps are sized to pump the required reflux from the reflux accumulator back to the top of the stabilizer tower. These pumps are normally designed for a pressure drop of 50 psi (340 kPa). Depending upon the reflux circulation rate, two 100 percent pumps or three 50 percent pumps may be installed. This allows either a 100 percent spare or a 50 percent spare pump.

Stabilizer Feed Cooler

An inlet feed cooler may be required if a cold feed stabilizer tower is used. Calculations are required to determine the design feed temperature and the heat duty exchanger. This exchanger is usually a shell-and-tube type with some type of refrigerant required to cool the feed sufficiently.

Stabilizer-Heater

A feed heater may be required for stabilizers with a reflux system. If a feed heater is used, it is normally a shell-and-tube type exchanger that exchanges heat between the cold feed and the hot bottom product, which is then cooled before going to storage or pipeline.

The selection of equipment and the decision whether to use cold-feed or a reflux system depends on a number of factors. The availability of heat sources for reboiler and streams for cooling the system influence the final decision. Economics of product recovery, CAPEX, and OPEX are major considerations.

Stabilizer Design

It can be seen from the previous description that the design of both a cold-feed stabilizer and a stabilizer with a reflux is a rather complex and involved procedure. Distillation computer simulations are available that can be used to optimize the design of any stabilizer if the properties

of the feed stream and desired vapor pressure of the bottom product are known. Cases should be run of both a cold-feed stabilizer and one with reflux before a selection is made. Because of the large number of calculations required, it is not advisable to use hand calculation techniques to design a distillation process. There is too much opportunity for computational error.

Normally, the contract specification will specify a maximum Reid Vapor Pressure (RVP). This pressure is measured according to a specific American Society of Testing Materials (ASTM) testing procedure. A sample is placed in an evacuated container such that the ratio of the vapor volume to the liquid volume is 4 to 1. The sample is then immersed in a 100°F liquid bath. The absolute pressure then measured is the RVP of the mixture.

Since a portion of the liquid was vaporized to the vapor space, the liquid will have lost some of its lighter components. This effectively changes the composition of the liquid and yields a slightly lower vapor pressure than the true vapor pressure of the liquid at 100°F. Figure 8-10 can be used to estimate true vapor pressure at any temperature from a known RVP.

The inherent error between true vapor pressure and RVP means that a stabilizer designed to produce a bottom liquid with a true vapor pressure equal to the specified RVP will be conservatively designed. The vapor pressures of various hydrocarbon components at 100°F are given in Table 8-1.

The bottom temperature of the stabilization tower can be approximated if the desired pressure of the liquid is known. The vapor pressure of a mixture is given by:

$$VP = \Sigma[VP_n \times MF_n] \tag{8-2}$$

Where VP = vapor pressure of mixture, psia
VP_n = vapor pressure of component n, psia
MF_n = mole fraction of component n in liquid

To estimate the desired composition of the bottom liquid, the vapor pressure of the different components at 100°F can be assumed to be a measure of the volatility of the component. Thus, if a split on n-C_4 is assumed, the mole fraction of each component in the liquid can be estimated from:

$$L_n = [F_n(\text{n-}C_4\text{split})]/(RV_n) \tag{8-3}$$

$$MF_n = L_n/[\Sigma(L_n)] \tag{8-4}$$

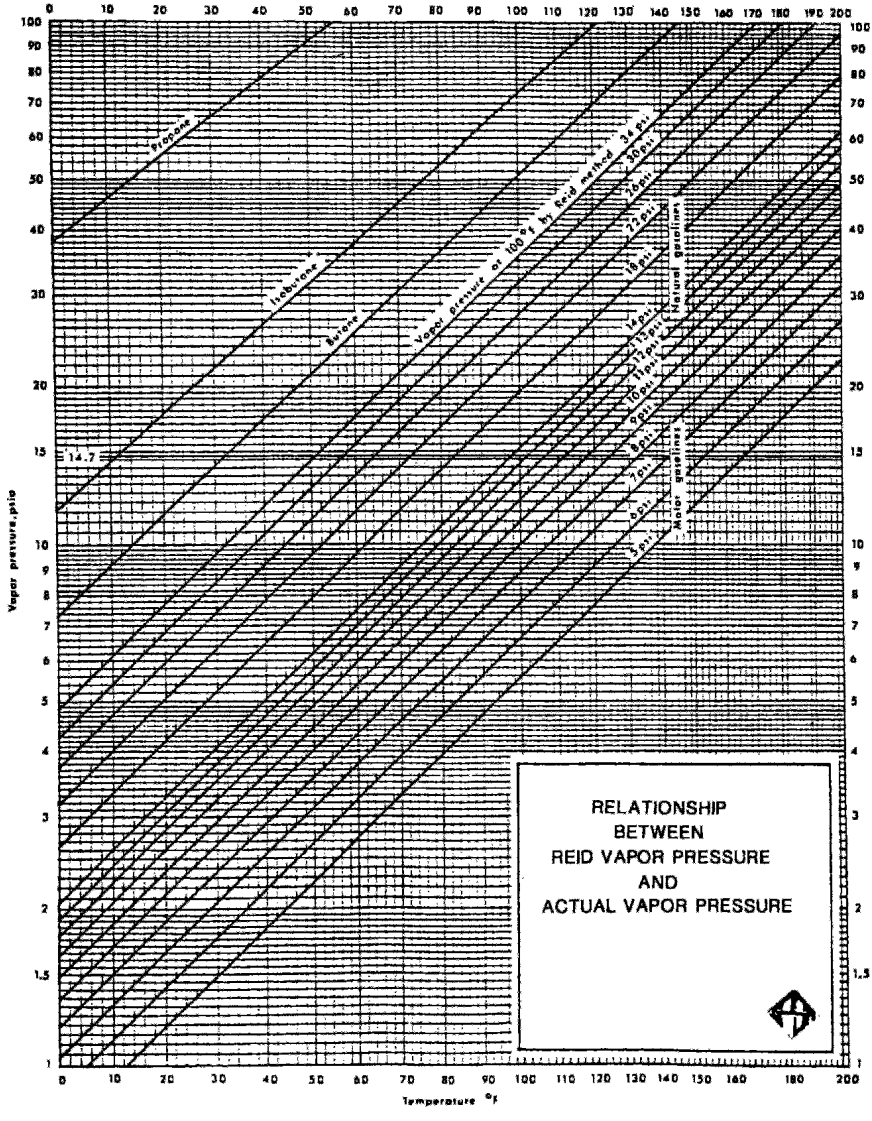


Figure 8-10. Relationship between Reid vapor pressure and actual vapor pressure. (From Gas Processors Suppliers Association, Engineering Data Book, 9th Edition.)

Where F_n = total number of moles of component n in the feed
L_n = total number of moles of component n in the bottom liquid divided by moles of n-C_4 in feed
(n-C_4 split) = relative volatility of component n from Table 8-1

Table 8-1
Vapor Pressure and Relative Volatility of Various Components

Component	Vapor Pressure at 100°F (psia)	Relative Volatility
C_1	5000	96.9
C_2	800	15.5
C_3	190	3.68
i-C_4	72.2	1.40
n-C_4	51.6	1.00
i-C_5	20.4	0.40
n-C_5	15.6	0.30
C_6	5.0	0.10
C_{7+}	0.1	0.0
CO_2	—	infinite
N_2	—	infinite
H_2S	394	7.64

To determine the composition of the bottom liquid, assume a split of n-C_4 and compute MF_n from Equations 8-3 and 8-4. The vapor pressure can then be computed from Equation 8-2. If the vapor pressure is higher than the desired RVP choose a lower number for the n-C_4 split. If the calculated vapor pressure is lower than the desired RVP, choose a higher number for the n-C_n split. Iterate until the calculated vapor pressure equals the desired RVP.

The bottom temperature can then be determined by calculating the bubble point of the liquid described by the previous iteration at the chosen operating pressure in the tower. This is done by choosing a temperature, determining equilibrium constants from Chapter 3 and computing:

$$C = \Sigma(L_n \times K_n) \tag{8-5}$$

If C is greater than 1.0, the assumed temperature is too high. If C is lower than 1.0, the assumed temperature is too low. By iteration a temperature can be determined where $C = 1.0$.

Typically, bottom temperatures will range from 200–400°F depending on operating pressure, bottom composition, and vapor pressure requirements. Temperatures should be kept to a minimum to decrease the heat requirements, limit salt buildup, and prevent corrosion problems.

Stabilizer As a Gas-Processing Plant

A gas-processing plant is designed to recover ethane, butane, and other natural gas liquids from the gas stream. A stabilizer also recovers some portion of these liquids. The colder the temperature of the gas leaving the overhead condenser in a reflux stabilizer, or the colder the feed stream in a cold-feed stabilizer, and the higher the pressure in the tower, the greater the recovery of these components as liquids. Indeed, any stabilization process that leads to recovery of more molecules in the final liquid product is removing those molecules from the gas stream. In this sense, a stabilizer may be considered as a simple form of a gas-processing plant.

It is difficult to determine the point at which a condensate stabilizer becomes a gas plant. Typically, if the liquid product is sold as a condensate, the device would be considered a condensate stabilizer. If the product is sold as a mixed natural gas liquid stream (NGL) or is fractionated into its various components, the same process would be considered a gas plant.

Chapter 9
Produced Water Treating Systems

Introduction

When hydrocarbons (crude oil, condensate, and natural gas) are produced, the well stream typically contains water produced in association with these hydrocarbons. The produced water is usually brine, brackish, or salty in quality but in rare situations may be nearly "fresh" in quality. The water must be separated from the hydrocarbons and disposed of in a manner that does not violate established environmental regulations. Typically, the produced water is separated from the hydrocarbons by passing the well stream through process equipment such as three-phase separators, heater-treaters, and/or a free-water knockout vessel. These gravity separation devices do not achieve a full 100% separation of the hydrocarbons from the produced water. The produced water separated from the hydrocarbons in these gravity separation devices will contain 0.1 to 10 volume percent of dispersed and dissolved hydrocarbons. Produced water treating facilities are used to further reduce the hydrocarbon content in the produced water prior to final disposal.

Regulatory standards for overboard disposal of produced water into offshore surface waters vary from country to country. Failure to comply with such regulations can often result in civil penalties, large fines, and lost or deferred production. Intentional violation of these regulations can result in criminal prosecution of officers and other individuals acting on behalf of the company who intentionally neglected compliance. Currently, regulations require the "total oil and grease" content of the effluent water to be reduced to levels ranging between 15 mg/l to 50 mg/l depending upon the host country. For U.S. offshore operations, the current standard is 29 mg/l.

Disposal of produced water into onshore surface waters is generally prohibited by environmental regulations. Onshore disposal typically

requires the produced water effluent to be injected into a saltwater disposal well. Onshore produced water treating for hydrocarbon removal prior to injection into a saltwater disposal well is not commonly regulated; however, a regulatory permit is typically required before initiating any substrata water disposal injection project. The permitting procedure helps to safeguard subsurface drinking water supplies by assuring that disposal wells are drilled and completed in a manner so the fresh drinking water supply zones are isolated from communication with any brackish or salty water zones. Dispersed hydrocarbon removal to very low levels, comparable to those required for offshore discharge may still be required to prepare the water for downstream solids filtering equipment.

The purpose of this chapter is to present the engineer with a procedure for selecting the appropriate type of equipment for treating oil from produced water and to provide the theoretical equations and empirical rules necessary to size the equipment. When this design procedure is followed, the engineer will be able to develop a process flow sheet, determine equipment sizes, and evaluate vendor proposals for any wastewater treating system once the discharge quality, the produced water flow rate, the oil specific gravity, the water specific gravity, and drainage requirements are determined.

Disposal Standards

Offshore Operations

Standards for the disposal or produced water to surface waters both onshore and offshore are developed by governmental regulatory authorities. Table 9-1 summarizes offshore disposal standards for several countries. The standards are current as of this writing.

Table 9-1
Worldwide Produced Water Effluent Oil Concentration Limitations

Ecuador, Colombia, Brazil	30 mg/l All facilities
Argentina and Venezuela	15 mg/l New facilities
Indonesia	25 mg/l Grandfathered facilities
Malaysia, Middle East	30 mg/l All facilities
Nigeria, Angola, Cameroon, Ivory Coast	50 mg/l All facilities
North Sea, Australia	30 mg/l All facilities
Thailand	50 mg/l All facilities
USA	29 mg/l OCS water
	Zero discharge inland water

In addition to placing limits on the oil content, regulatory agencies generally specify an analytical method for determining the oil content. A number of analytical methods are available, and they produce different amounts of oil measured and reported for the same sample. Analytical methods are discussed in Appendices A–C.

Produced water toxicity is regulated only in the United States, where a government permit is necessary to limit the toxicity of produced water discharged into the waters.

Onshore Operations

Disposal of produced water into freshwater streams and rivers is generally prohibited except for the very limited cases where the effluent is low in salinity. Some oil-field brines might kill freshwater fish and vegetation due to high salt content.

Regulatory agencies generally require that produced water from onshore operations be disposed of by subsurface injection, although there are limited exceptions. In addition to requiring subsurface disposal, regulatory agencies regulate the completion and operation of the disposal wells.

Characteristics of Produced Water

Produced water contains a number of substances, in addition to hydrocarbons, that affect the manner in which the water is handled. The composition and concentration of substances may vary between fields and even between different production zones within a single field. The terminology used for concentration is milligrams per liter (mg/l), which is mass per volume ratio and is approximately equal to parts per million (ppm). Some of the important produced water constituents are discussed in this section.

Dissolved Solids

Produced waters contain dissolved solids, but the amount varies from less than 100 mg/l to over 300,000 mg/l, depending on the geographical location as well as the age and type of reservoir. In general, water produced with gas is condensed water vapor with few dissolved solids and will be fresh with a very low salinity. Aquifer water produced with gas or oil will be much higher in dissolved solids. Produced water from hot reservoirs tends to have higher TDS concentrations while cooler reservoirs tend to have lower levels of TDS.

Dissolved solids are inorganic constituents that are predominantly sodium (Na^+) cations and chloride (Cl^-) anions. Other common cations are calcium (Ca^{2+}), magnesium (Mg^{2+}), and iron (Fe^{2+}), while barium (Ba^{2+}), potassium (K^+), strontium (Sr^+), aluminum (Al^{3+}), and lithium (Li^+) are encountered less frequently. Other anions present are bicarbonate (HCO_3^-), carbonate (CO_3^{2-}), and sulfate (SO_4^-).

All produced water treating facilities should have water analysis data for each major reservoir and for the combined produced water stream. Especially important are constituents that could precipitate to form scales.

Precipitated Solids (Scales)

The more troublesome ions are those that react to form precipitates when pressure, temperature, or composition changes occur. These are the well-known deposits that form in tubing, flowlines, vessels, and produced water treating equipment.

Mixing of oxygenated deck drain water with produced water should be avoided because this may result in the formation of calcium carbonate ($CaCO_3$), calcium sulfate ($CaSO_4$), and iron sulfide (FeS_2) scale, along with oil-coated solids.

Calcium Carbonate ($CaCO_3$)

Calcium carbonate ($CaCO_3$) precipitate can be formed by mixing two dissimilar waters, but the usual cause is the reduction in pressure and release of dissolved carbon dioxide from produced water. This increases the produced water's pH, which reduces the solubility of $CaCO_3$ and leads to scale precipitate. Temperature effects are equally important since $CaCO_3$ is less soluble at higher temperatures and will form a deposit in heat exchangers, heaters, and treaters. Its solubility in fresh water is approximately 1,000 mg/l at 60°F (15°C) and diminishes to 230 mg/l as temperature is increased to 200°F (93°C). Fortunately, higher salinity increases $CaCO_3$ solubility in produced water to a value greater than that given above for pure water.

Calcium Sulfate ($CaSO_4$)

Calcium sulfate ($CaSO_4$) is one of several sulfate scales and is also called gypsum. Like $CaCO_3$, it can form either as a result of mixing dissimilar waters or naturally as a result of changes in temperature and pressure as the water travels from the subsurface to the surface treating facility. $CaSO_4$ solubility is at its maximum level of 2,150 mg/l at approximately

100°F (38°C) and diminishes to 2,000 mg/l as it cools to 60°F (15°C). The solubility of $CaSO_4$ also declines with increasing temperature above 100°F with its solubility reducing to 1,600 mg/l at 200°F (93°C). $CaSO_4$ also increases in solubility as the salinity of the produced water increases.

Iron Sulfide (FeS_2)

Iron sulfide (FeS_2) is a product of corrosion caused by waters containing dissolved hydrogen sulfide coming into contact with equipment fabricated from carbon steel or iron materials. Mixing water containing iron cations (Fe^{2+}) with another water containing hydrogen sulfide will also result in an FeS_2 precipitate.

Barium and Strontium Sulfate ($BaSO_4$ and $SrSO_4$)

Barium and strontium sulfate ($BaSO_4$ and $SrSO_4$) are much less soluble than calcium sulfate, but they are not as common in produced waters. $BaSO_4$ solubility is quite low, having a value of approximately 3 mg/l over the range from 100°F (15°C) to 200°F (93°C). $SrSO_4$ solubility is 129 mg/l at 77°F (25°C) and diminishes to 68 mg/l as the solution temperature increases to 257°F (125°C). If a produced water stream containing appreciable quantities of barium or strontium ions is mixed with sulfate-rich water, barium and/or strontium scaling can be expected. These waters are incompatible due to this scaling characteristic and should not be mixed.

Scale Removal

Hydrochloric acid can be used to dissolve calcium carbonate and iron sulfide scales. However, iron sulfide chemically reacts with hydrochloric acid and produces hydrogen sulfide, a highly toxic gas having the odor of rotten eggs. Due to the high toxicity of hydrogen sulfide, safety provisions need to be implemented.

Calcium sulfate is not soluble in hydrochloric acid, but chemicals are available that will convert it to an acid-soluble form that can then be removed by the acid. This process is slow; however, because a two-step process must be repeated to strip the scale layer by layer. Thus, the removal of calcium sulfate is more difficult than the removal of calcium carbonate.

Practical means of dissolving barium or strontium sulfate are not available. These hard scales can be removed by mechanical means, which is

a time-consuming process. Mechanical removal of scale can create a disposal problem for the resulting waste material and possibly could result in contamination by naturally occurring radioactive materials (NORM).

Controlling Scale Using Chemical Inhibitors

Scale-inhibiting chemicals are available to retard or prevent all types of scale. They mostly function by enveloping a newly precipitated crystal, thereby retarding growth. Common scale inhibitors include

- Inorganic phosphates (inexpensive but only applicable at low temperature),
- Organic phosphate esters (easy to monitor but limited to temperatures below 100°F),
- Phosphates (easy to monitor and have a higher thermal stability to 150°F),
- Polymers (best thermal stability and effectiveness, but difficult to monitor).

Sand and Other Suspended Solids

In addition to scale particles, produced water often contains other suspended solids. These include formation sand and clays, stimulation (fracturing) proppant, or miscellaneous corrosion products. The amount of suspended solids is generally small unless the well is producing from an unconsolidated formation, in which case large volumes of sand can be produced. Produced sand is often oil wet and its disposal is a problem. Sand removal is discussed in a later subsection entitled "Equipment Description."

Small amounts of solids in produced water may or may not create problems in water treating depending on the particle size and its relative attraction to the dispersed oil. If the physical characteristics and electronic charge of such solids result in an attraction to the dispersed oil droplets, the solid particles can attach to the dispersed oil droplets to stabilize emulsions, thereby preventing coalescence and separation of the oil phase. The combined specific gravity of the resulting oil/solid droplet can be approximately equal to that of the produced water, and gravity separation becomes difficult if not impossible.

The concentration of suspended solids can be monitored with 0.45-micron Millipore filter test, and residue can be analyzed for mineral content in an attempt to identify the source of the solid.

When solids are present, the following practices should be applied:

- Chemical treatment must be used to "break" the electronic attraction between the solid particle and the oil droplet.
- Equipment design must incorporate solids removal ports, jets, and/or plates.
- Oil measurement techniques not affected by the solids should be used.
- Solids are likely to be oil coated, and offshore disposal may be prohibited, as is the case in the United States. This applies to solids removed from desanders or vessels, not to the solids suspended in the water.
- Water injection for disposal should be made into a disposal zone that has high enough pore space openings to prevent the suspended solids from plugging. Consideration should be given to using filtration equipment to remove the larger particles prior to injection into the disposal well. Periodic back flowing and acidizing are generally needed to maintain disposal wells if filtration is not applied.
- Water-flood injection for pressure maintenance and additional recovery often requires filtration (to remove suspended solids). Water injection pressures typically must be maintained at pressure levels below the fracture gradient pressure of the formation.

Dissolved Gases

The most important gases found in produced water include natural gas (methane, ethane, propane, and butane), hydrogen sulfide, and carbon dioxide. In the reservoir the water can be saturated with these gases at relatively high pressures. As the produced water flows up the wells, most of these gases flash to the vapor phase and are removed in primary separators and stock tanks. The pressures and temperatures at which the produced water is separated from the main oil, condensate, and/or natural gas streams will impact the quality of dissolved gas that will be contained in the produced water stream feeding the water treating facilities. The higher the separation pressure, the higher the quantity of dissolved gases will be. An inverse relationship holds true for the effects of temperature: the higher the separation temperature, the lower the quantity of dissolved gases will be.

Natural gas components are slightly soluble in water at moderate to high pressures and will be present in the produced water stream. The solubility of natural gas (primarily methane) is illustrated graphically as a function of pressure, temperature, and specific gravity of the water in Chapter 3. It is interesting to note that natural gas components have an affinity for the dispersed oil droplets, and this principle is applied to

the design of gas flotation equipment commonly used in produced water treating systems.

If hydrogen sulfide is present in the produced reservoir fluid, or if sulfate reducing bacteria are a problem in the reservoir or production equipment, hydrogen sulfide will likewise be present in the produced water stream. Hydrogen sulfide is corrosive, can cause iron sulfide scaling, and is extremely toxic if inhaled. The toxicity of hydrogen sulfide hinders operation and maintenance of equipment, particularly when the vessels must be opened for adjustments, as in the case when weir adjustments are required in gas flotation cells. Special training and life support breathing equipment are recommended for use by personnel when such activities result in exposure to hydrogen sulfide. Additionally, iron sulfide (the corrosion product of hydrogen sulfide) presents a potential fire hazard since it is prone to auto-ignition when exposed to air or other sources of oxygen.

If carbon dioxide is present in the produced reservoir fluid, it too will be present in the produced water. Carbon dioxide is corrosive and can cause $CaCO_3$ scaling. On the other hand, removal of CO_2 and H_2S will result in increased pH, which could lead to scaling.

Oxygen is not found naturally in produced water. However, when the produced water is brought to the surface and exposed to the atmosphere, oxygen will be absorbed into the water. Water containing dissolved oxygen can cause severe and rapid corrosion, solids generation from oxidation reactions, and oil weathering that inhibits cleanup. To prevent this, a natural gas blanket should be maintained on all of the production and water treating tanks and vessels used within the process.

Seawater is often used as the source of water for water floods and water injection pressure maintenance projects offshore. Seawater contains considerable amounts of dissolved oxygen and some carbon dioxide. Bacteria in untreated seawater may also be a problem. The oxygen and carbon dioxide must be removed from the source water by either vacuum de-aeration or gas stripping prior to injection. Facilities used in treating water for water flood and pressure maintenance injection are not covered in this text.

Oil in Water Emulsions

Most emulsions encountered in the oil field are water droplets in an oil continuous phase and are called “normal emulsions.” The water is dispersed in the form of very small droplets ranging between 100 to 400 microns in diameter. Oil droplets in a water continuous phase are

known as "reverse emulsions" and can occur in produced water treating operations.

If the emulsion is unstable, the oil droplets will coalesce when they come in contact with each other and form larger droplets, thus breaking the emulsion. An unstable emulsion of this type will break within minutes.

A stable emulsion is a suspension of two immiscible liquids in the presence of a stabilizer or emulsifying agent that acts to maintain an interfacial film between the phases. Chemicals, heat, settling time, and electrostatics are used to alter and remove the film and cause emulsion breakdown. Untreated stable emulsions can remain for days or even weeks.

Emulsion breakers for water-in-oil emulsions, also known as destabilizers or demulsifiers, are oil-soluble and are added to the total well stream ahead of the process equipment. Being oil-soluble, the emulsion breaker is carried with the crude. Thus, if the emulsion is not broken in the first-stage separator, the chemical has additional time to act in the subsequent separators and the stock tank.

Oil-in-water emulsions can be broken by "reverse emulsion breakers," which are special destabilizers or demulsifiers. These are similar to the conventional emulsion breakers except that they are water-soluble. Reverse emulsion breakers are generally injected into the water stream after the first oil–water separation vessel. Typical concentrations are in the 5- to 15-ppm range, and overtreating should be avoided because these chemicals can stabilize an emulsion.

The emulsions in produced water will become oil in the form of dispersed droplets after the emulsion film is broken. The droplets will coalesce to yield an oily film that can be separated from the produced water using gravity settling devices such as skim vessels, coalescers, and plate separators. However, small droplets require excessive gravity settling time, so flotation cells or acceleration enhanced methods such as hydrocyclones and centrifuges are used. Equipment selection is based on the inlet oil's droplet diameter and concentration.

Dissolved Oil Concentrations

Dissolved oil is also called "soluble oil," representing all hydrocarbons and other organic compounds that have some solubility in produced water. The source of the produced water affects the quantity of the dissolved oil present. Produced water derived from gas/condensate production typically exhibits higher levels of dissolved oil. In addition, process water condensed from glycol regeneration vapor recovery systems contains

aromatics including benzene, toluene, ethyl benzene, and xylenes (BTEX) that are partially soluble in produced water.

Gravitational-type separation equipment will not remove dissolved oil. Thus, a high level of total oil and grease could be discharged if the produced water source contains significant quantities of dissolved oil. Produced water streams containing high concentrations of dissolved oil can be recycled to a fuel separator to help reduce the quantity of dissolved oil in the water effluent. Other technologies, such as bio-treatment, adsorption filtration, solvent extraction, and membranes, are currently being evaluated by industry for removing dissolved oil, but such processes are not yet readily available for commercial applications.

It is essential that actual water test analysis data for dissolved and dispersed oil concentrations are needed in the planning stage prior to designing a water treating facility for a specific application. If the design engineer assumes a value for the dissolved oil content without first having obtained actual water test analysis for the specific produced water stream to be treated, the facility design may not be capable of treating the effluent water to meet regulatory compliance specifications. Therefore, lab testing is required first.

The solubility of crude oil in produced water has not been extensively documented, but the solubility of several hydrocarbons can illustrate the potential range. Field experience indicates that solubility does not change appreciably with the temperatures used during water treating, specifically from 77 to 167°F (25 to 75°C). Solubility does increase significantly, however, as temperatures rise above 167°F (75°C).

The effect of high salinity on reducing the solubility of dissolved hydrocarbons implies that produced water from gas well and gas processing sources should be mixed with the saltiest brine available to reduce the dissolved oil concentration. The dissolved hydrocarbons would be forced out of solution from the water into the vapor phase or into a dispersed oil droplet removed by gravity separation equipment.

Water chemistry and hydrocarbon solubility are also related to toxicity. Dissolved saturated paraffinic (aliphatic) petroleum hydrocarbons have low solubilities in water and have not demonstrated toxicity. Aromatics, such as benzene, toluene, ethyl benzene, and xylene, are more soluble and more toxic.

Dispersed Oil

Dispersed oil can consist of oil droplets ranging in size from about 0.5 microns in diameter to greater than 200 microns in diameter. The oil droplet size distribution is one of the key parameters influencing the

produced water treating performance. According to Stokes' law, the rising velocity of an oil droplet is proportional to the square of the droplet diameter. For equipment that operates on the principle of Stokes' law, the diameter of the oil droplet has a major effect on the separation and removal of the oil droplet from the water.

The capability of a given de-oiling device or system to remove and recover dispersed oil decreases as the droplet size decreases. Oil droplet size distribution is a fundamental characteristic of produced water and must be considered in designing and sizing treating systems to meet regulatory standards for effluent water compliance.

Figure 9-1 is an example of a typical histogram of an oil droplet distribution. The histogram divides the particle counts into discrete size ranges along the horizontal axis. The number and size of the ranges are determined by the equipment used to obtain the data. The height of the vertical bars corresponds to the volume percentage of oil droplets in each range. A particle distribution curve is constructed by connecting the tops of the bars at the midpoint of each size range.

Figure 9-2 illustrates a typical volume distribution curve. The volume percentage of the particles is equal to or smaller than each specified size that is plotted. The vertical axis scale is from 0 to 100% since the data are plotted cumulatively.

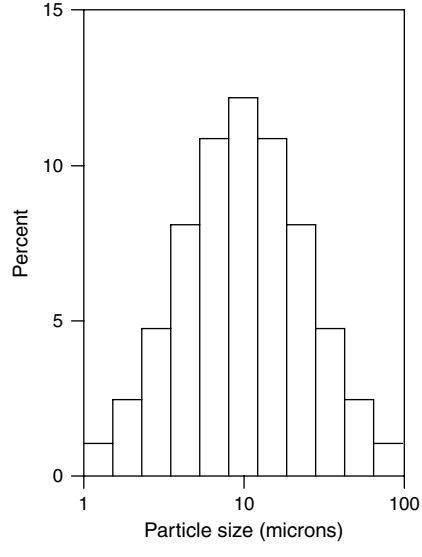


Figure 9-1. Histogram of oil droplet distribution.

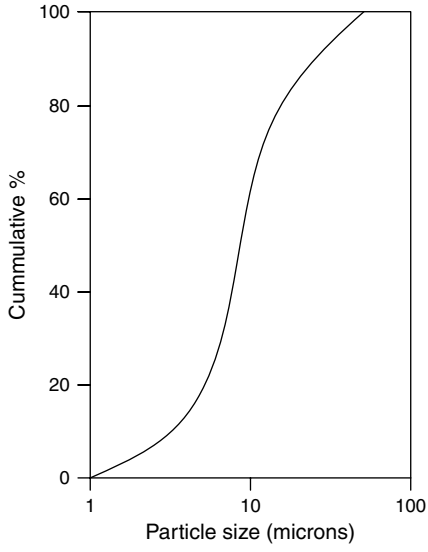


Figure 9-2. Typical oil volume distribution curve.

The dispersed oil droplet size distribution may vary from point to point in a produced water system, and from one system to another. The size distribution is affected by interfacial tension, turbulence, temperature, system shearing (pumping, pressure drop across pipe fitting, etc.), and other factors. The droplet size distributions should be measured in the field when troubleshooting and/or upgrading systems, whenever possible.

In the absence of data, the generalized relationship in Figure 9-3 can be used for oil droplet size distributions. Since the distribution is linear, it places more volume in smaller-diameter droplets. However, because this straight-line relationship is a very rough estimate, field data should be used whenever possible. For produced water effluent from a three-phase separator, a maximum oil droplet diameter of 250 to 500 microns and an oil content of 1000 to 2000 mg/l can be used in the absence of field data. For first phase de-oiling equipment, an oil droplet diameter of 30 microns with inlet total oil levels less than 100 mg/l can be assumed for produced water feed to final treating equipment. Operational experience in the area may also provide reliable data from similar existing facilities that can be used to estimate inlet oil concentrations and droplet size distributions.

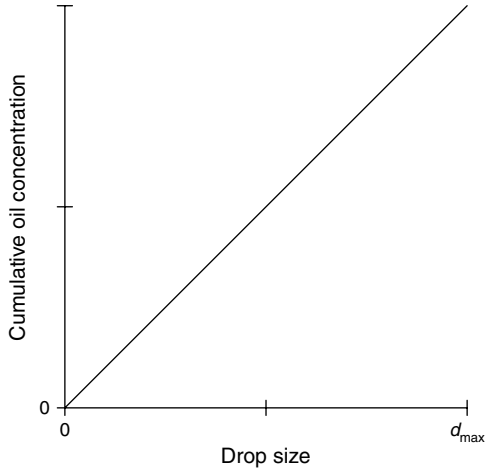


Figure 9-3. Droplet size distribution for design.

Toxicants

Produced water can exhibit toxicity to marine organisms in laboratory tests. The potential for toxic effects in the natural environment is one reason for concern about the environmental effects of produced water discharges.

The toxicity of produced water is determined by exposing groups of test organisms to a series of produced water concentrations in seawater for a fixed period of time. The cumulative effect is measured as a function of concentration. The object of the test is to observe effects of the test organisms such as mortality, reduction in rate of growth, and reduction in ability to reproduce.

Test results are expressed either as the maximum concentration of produced water that will produce no effect on the test organisms [the "No Observable Effect Concentration" (NOEC)] or as the concentration that produces a 50% effect on the test organisms.

The effect can be mortality, in which case the test result is referred to as an "LC-50," which stands for the concentration that is lethal to half of the test organisms. A numerically lower LC-50 indicates a more toxic effluent. For effects on growth, or other indicators, the test result is expressed as an "EC-50," which stands for the concentration that produces an effect on 50% of the test organisms.

Produced water toxicity varies widely. For a given effluent and test organism, higher concentrations are needed to produce observable effects

during shorter exposure times. Some produced waters are essentially nontoxic, requiring concentrations in excess of 10% to produce effects in four-day exposure tests. As a general rule, the most toxic produced waters can cause acute effects during 24-hour exposures at concentrations as low as 0.5%, although such effects are generally seen only at concentrations in excess of 1%. Acute effects in 96-hour tests can be seen at concentrations as low as 0.1%. A study of the acute toxicity to mysid shrimp in the Gulf of Mexico produced waters found a range of LC-50 values of 0.1% to 86% produced water, with an average of 19%. Seven-day exposure tests are used to measure "chronic" toxicity of produced waters for regulatory purposes in the United States. The average chronic "NOEC" for four Gulf of Mexico produced waters was found to be 1.6%.

Field studies show produced water discharged into the open ocean is diluted to concentrations of 1% or less within a few meters of the discharge pipe. Dilution to concentrations below a few tenths of a percent typically occurs within 300 feet (100 meters) of the discharge pipe. Furthermore, the produced water plume occupies only a small fraction of the water column and is constantly moving due to local currents. As a result, it is highly unlikely that organisms in the marine environment will be exposed to elevated produced water concentrations for the long exposure times used in laboratory toxicity tests. The rapid initial dilution of produced water discharges and long exposure times needed to cause observable toxicity greatly reduce the potential for toxic effects on marine life from produced water discharges into the open ocean.

Proper outfall design can significantly reduce the potential for toxic effects from produced water discharges. Outfalls should be positioned such that the effluent plume does not contact the sea bottom. Bottom contact greatly reduces the rate of dilution and makes it possible for the produced water to have a direct impact on the organisms on the ocean floor. Diffusers (multiport outfalls) may be used to increase dispersion and reduce the potential for plume contact with the sea floor.

Produced water may contain dispersed oil, dissolved oil, metals, ammonia, treating chemicals, and salts. Each of these constituents could act as a source of toxicity. Published research results indicate that organic compounds in produced water are significant factors in toxicity but not the source of toxicity in all cases. Common industry practice for water treating is to reduce the dispersed oil content of produced water effluent and as a result may not fully treat all sources of toxicity.

Techniques for produced water treatment for toxicity reduction are still under development. Novel treatment technologies may yet be applied, but good water management of existing facilities will certainly contribute to the overall control of toxicity. Water treatment chemicals, which reduce the dispersed oil content in the effluent water, can also contribute to

the toxicity of such effluent. Other chemicals used to control scale, biofouling, and corrosion can also contribute to effluent water toxicity, but are commonly required to reduce equipment maintenance costs. As a result, optimization of the chemicals application program can help control effluent toxicity as well reduce the overall chemical usage costs.

Produced water toxicity is regulated only in the United States, where government permit limits the toxicity of produced water that can be discharged in the Gulf of Mexico.

Naturally Occurring Radioactive Materials

Naturally occurring radioactive materials (NORM) can be transported to the surface in produced water and can be found in production wastes, equipment, and solids at production facilities. At offshore locations, dissolved NORM are discharged along with produced water. Because of concern over human exposure to environmental radiation, oil-field NORM have received regulatory attention and managing waste has become a significant cost factor for the industry.

Oil-field NORM result from the presence of uranium and thorium in hydrocarbon bearing formations. Many oil and gas bearing formations contain shales that have higher than average concentrations of uranium and thorium. These elements occur in chemical forms that are not water-soluble under reservoir conditions. U^{238} and Th^{232} decay into different isotopes of radium (Ra^{236} and Ra^{228}). These radium isotopes further decay into the radioactive gas called radon (Rn^{232}). Both radium and radon are soluble at very low and harmless levels in formation water under reservoir conditions and can be transported to the surface along with oil, gas, and produced water.

Once produced water leaves the reservoir, decreases in temperature and pressure can lead to the precipitation of scale which can trap and concentrate the radium and its decay products and particulates in production equipment. It can accumulate as hard scales, sludge, or tank bottoms which can be harmful to humans if during the process of cleaning the scale is invested. Radium is often associated with barium scales since radium and barium are in the same chemical family. These accumulated NORM containing solids when they are cleaned out of vessels and other equipment during maintenance must be disposed of in a controlled fashion. Some equipment items cannot be readily decontaminated and are subject to special handling procedures. Oil-field NORM scales are an environmental concern because of the potential for human exposure to ionizing radiation. The radium and radium decay products in oil-field NORM present a hazard only if taken into the body by ingestion or inhalation.

The external radiation from equipment or waste containing NORM is almost never a significant concern. The discharge of radium in produced water is of concern because it may accumulate in seafood consumed by humans. Since no established safe level exists for the intake of radium, any consumption of radium in food is of potential concern. However, for the case of radium discharged in produced water, risk assessment studies show that consumption of fish caught near produced water outfalls will not pose an unacceptable human health risk, even in the worst cases.

Radon in produced fluids partitions into the gas phase during primary separation and enters the gas processing stream. Radon's boiling point is between that of ethane and propane, and thus radon is concentrated in the natural gas liquids fraction (this is generally a problem only in a gas plant fractionation section).

Radon and its decay products may be found in any equipment that comes into contact with natural gas or natural gas liquids.

Regulations governing NORM focus on equipment and wastes containing NORM rather than produced water. Regulations generally specify a limit on the external radiation level from wastes or equipment above which the material must be treated as NORM and cannot be released for unrestricted use without prior decontamination. Regulations also specify the maximum acceptable radium concentration in wastes and soils for unrestricted release or disposal. Existing regulations do not limit the radium concentration in offshore produced water discharges. Operators in the U.S. Gulf of Mexico are required to measure and report the radium concentrations of their effluents to the EPA.

NORM accumulations in production equipment can be controlled in some situations but cannot be eliminated entirely. Since NORM are incorporated in scale and other precipitates, reduced NORM accumulation is a benefit of a properly managed scale control program. NORM cannot be made nonradioactive. Consequently, the emphasis in NORM waste management is on identification, control, and volume reduction. NORM site remediation activities are directed at reducing the potential for human exposure to hazardous amounts of radioactive material.

Bacteria

Most produced waters contain bacteria but generally in small amounts. Measurement is done according to API RP 38, "Recommended Practice for Biological Analysis of Subsurface Injection Waters." The type and number of bacteria are important when selecting a biocide program. All bacteria have many strains, and some will be immune to a specific

bactericide. Thus, continued testing and periodic change of chemical may be needed. The types of bacteria are

- **Aerobic** bacteria which require oxygen and are present in large quantities when seawater or surface water is used for water-flood injection. Chlorine, usually from a hypochlorite generator, is used for control.
- **Anaerobic** bacteria which grow in the absence of oxygen. One strain is the sulfate reducing bacteria (SRB) that excrete sulfide ions that form hydrogen sulfide. The associated corrosion of equipment, safety hazard from H_2S, particle plugging, potential H_2S souring of a water-flood zone, and unsightly aesthetics of iron sulfide as well as sulfide smell cause these bacteria to be a major problem. A rigorous, permanent biocide program using commercial bactericides, or a chemical (glutaraldehyde, formaldehyde, or acrolein), is needed.
- **Facultative** bacteria which can grow in an aerobic or anaerobic environment. Their presence can create conditions aiding the growth of SRB. Specialized chemical selection is needed for control.

The API test uses a standard culture media for specific bacteria. Other media have been used or different techniques have been applied to estimate the quantity of bacteria. Field tests show the following:

- If the total bacteria count is less than 10,000 per ml (and no SRB are present), bacteria shouldn't be a problem.
- If the total bacteria count is greater than 100,000 per ml, plugging of filter media or formation rock is possible and biocide control should be used.
- If the SRB count is greater than 100 per ml, treatment should be initiated for a critically important injection system; counts of 100 to 1000 per ml will require some treatment to prevent injection well plugging; counts greater than 10,000 per ml will require a rigorous program of biocide control.

Protected locations are preferred sites for bacteria growth. In pipelines, pigging may be useful to remove sediments that would otherwise shield bacteria from biocides. Any place where water lies stagnant offers an ideal site for the establishment of bacterial colonies. These places include the bottoms of vessels, ahead of blind flanges, beneath corrosion products in lines, and in "rat-holes" of well bores. Growth is affected by oil or water treating chemical selection because the SRB not only require sulfate but also need a nutrient, which can be supplied by the carbon, nitrogen, or phosphorous in chemicals. SRB counts of a test sample with the intended chemical concentration in the produced water should be done before implementing changes.

System Description

Table 9-2 lists the various methods employed in produced water treating systems and the types of equipment that employ each method. Figure 9-4 shows a typical produced water treating system configuration. Produced water will always have some form of primary treating prior to disposal. This system could take the form of a skim tank, skim vessel, CPI, cross-flow separator, or gas flotation unit. Other than the gas flotation unit, all of these devices employ gravity separation techniques. Depending upon the severity of the treating problem, secondary treatment may be required. Secondary treatment could utilize a CPI, a cross-flow separator, or a gas flotation unit. Liquid-liquid hydrocyclones or centrifuges are often used either in a single stage or with an upstream or downstream skim vessel or flotation unit.

Offshore, produced water can be piped directly overboard after treating, or it can be routed through a disposal pile or a skim pile. Water from the deck drains must be treated for removal of "free" oil. This is normally done in a skim vessel called a sump tank. Water from the sump tank is either combined with the produced water or routed separately for disposal overboard.

Table 9-2
Produced Water Treating Equipment

Method	Equipment Type	Approximate Minimum Drop Size Removal Capacities (Microns)
Gravity separation	Skimmer tanks and vessels API separators Disposal piles Skim piles	100–150
Plate coalescence	Parallel plate interceptors Corrugated plate interceptors Cross-flow separators Mixed-flow separators	30–50
Enhanced coalescence	Precipitators Filters/coalesces Free-flow turbulent coalesces	10–15
Gas flotation	Dissolved gas Hydraulic dispersed gas Mechanical dispersed gas	10–20
Enhanced gravity separation	Hydrocyclones Centrifuges	15–30
Filtration	Multimedia membrane	1+

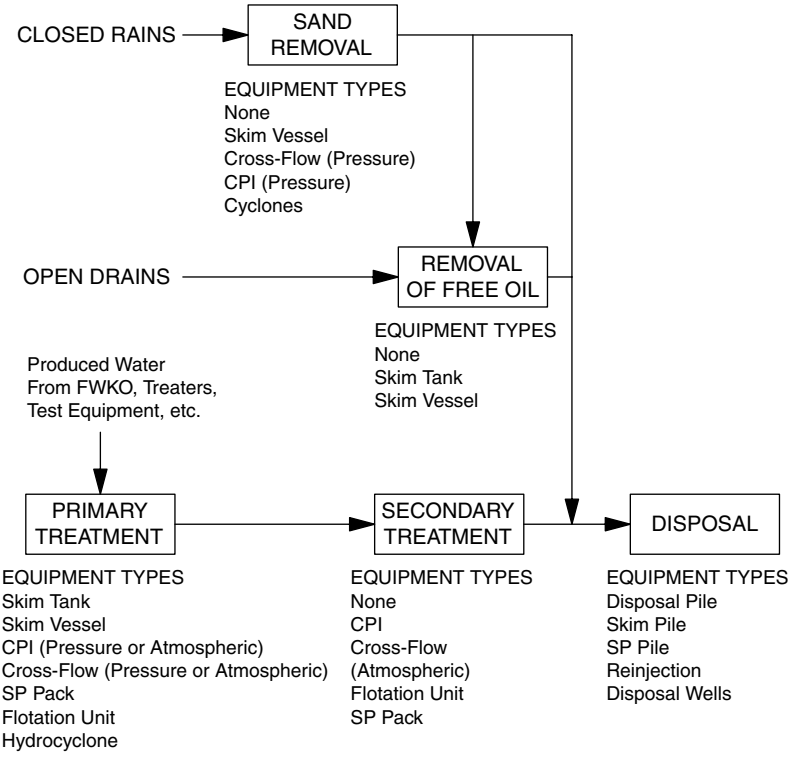


Figure 9-4. Typical produced water treating system.

Onshore, the water is normally re-injected in the formation or pumped into a disposal well. In the past, particularly in dry climates in countries with emerging environmental regulations, small amounts of produced water were disposed of in an evaporation pit. This practice has virtually been ceased and thus will not be discussed any further in this text.

For safety considerations, closed drains, if they exist in the process, should never be tied into atmospheric drains and should be routed to a pressure vessel prior to entering an atmospheric skim tank or pile. This should be done in a skim vessel, with or without a CPI or cross-flow separator, in a pressure vessel.

Theory

The function of all water treating equipment is to cause the oil droplets, which are dispersed in the water continuous phase, to separate and float to the surface of the water so they can then be removed. In gravity

separation units, the difference in specific gravity causes the oil to float to the surface of the water. The oil droplets are subjected to continuous dispersion and coalescence during the trip up the well bore through the surface chokes, flow lines, control valves, and process equipment. When energy is put into the system at a high rate, the drops are dispersed to smaller sizes. When the energy input rate is low, small droplets collide and join together in the coalescence process.

The three basic phenomena that are used in the design of common produced water treating equipment are gravity separation, coalescence, and flotation. Dispersion also affects the design but to an unpredictable degree. In the past filtration has been tried, but, due to high maintenance costs, has been found to be unsatisfactory.

Gravity Separation

Most commonly used water treating equipment depends on the forces of gravity to separate the oil droplets from the water continuous phase. The oil droplets, being lighter than the volume of water they displace, have a buoyant force exerted upon them. This is resisted by a drag force caused by their vertical movement through the water. When the two forces are equal, a constant velocity is reached, which can be computed from Stokes' law as

Field Units

$$V_o = \frac{1.78 \times 10^{-6} \, (\Delta \mathrm{SG}) \, (d_m)^2}{\mu_w}, \tag{9-1a}$$

SI Units

$$V_o = \frac{5.556 \times 10^{-7} \, (\Delta \mathrm{SG}) \, (d_m)^2}{\mu_w}, \tag{9-1b}$$

where

V_o = rising vertical velocity of the oil droplet relative to the water continuous phase, ft/s (m/s),
d_m = diameter of the oil droplet, microns (μ),
ΔSG = difference in specific gravity of oil and water relative to water,
μ_w = viscosity of the water continuous phase, cp.

Several conclusions can be drawn from this simple equation:

1. The larger the size of an oil droplet, the larger the square of its diameter and, thus, the greater its vertical velocity will be. That is, the bigger the droplet size, the less time it will take for the droplet to rise to a collection surface and thus the easier it will be to treat the water.
2. The greater the difference in density between the oil droplet and the water phase, the greater the vertical velocity will be. That is, the lighter the crude, the easier it will be to treat the water.
3. The higher the temperature, the lower the viscosity of the water and, thus, the greater the vertical velocity will be. That is, it is easier to treat the water at high temperatures than at low temperatures.

Theoretically, Stokes' law should apply to oil droplets as small as 10 microns. However, field experience indicates that 30 microns sets a reasonable lower limit on the droplet sizes that can be removed. Below this size, small pressure fluctuations, platform vibrations, etc. tend to impede the rise of the oil droplets to the coalescing surface.

Coalescence

The process of coalescence in water treating systems is more time-dependent than the process of dispersion. In a dispersion of two immiscible liquids, immediate coalescence seldom occurs when two droplets collide. If the droplet pair is exposed to turbulent pressure fluctuations, and the kinetic energy of the oscillations induced in the droplet pair is larger than the energy of adhesion between them, the contact will be broken before coalescence is completed. If the energy input into the system is too great, dispersion will occur, as discussed below. If there is no energy input, then the frequency of droplet collision, which is necessary to initiate coalescence, will be low, and coalescence will occur at a very low pace.

Most water treating equipment, with the exception of flotation units and hydrocyclones, consists of vessels in which the oil droplets rise to a surface due to gravity forces. From a process standpoint, these are considered "deep bed gravity settlers." Experiments with deep bed gravity settlers (refer to Chapter 7 for further discussion) yield the following two qualitative conclusions:

- Doubling the residence time causes only a 10% increase in the maximum size droplet that will be grown in a gravity settler.

- The more dilute the dispersed phase (oil), the greater the residence time required to grow a given particle size will be. That is, coalescence occurs more rapidly in concentrated dispersions.

From these conclusions it shows that after an initial period of coalescence in a settler, additional retention time has a rapidly diminishing ability to cause coalescence and to capture oil droplets.

Dispersion

The term "dispersion" refers to the process of a discontinuous phase (oil) being split into small droplets and distributed throughout a continuous phase (water). This dispersion process occurs when a large amount of energy is input to the system in a short period of time. This energy input overcomes the natural tendency of two immiscible fluids to minimize the contacting surface area between the two fluids.

The dispersion process is diametrically opposed by coalescence, which is the process in which small droplets collide and combine into larger droplets. As the oil and water mixture flows through the piping, these two processes are simultaneously occurring. In the piping a droplet of oil splits into smaller droplets when the kinetic energy of its motion is larger than the difference in surface energy between the single droplet and the two smaller droplets formed from it. While this process is occurring, the motion of the smaller oil droplets causes coalescence to occur. Therefore, it should be possible to define statistically a maximum droplet size for a given energy input per unit mass and time at which the rate of coalescence equals the rate of dispersion.

One relationship for the maximum particle size that can exist at equilibrium was proposed by Hinze as follows:

$$d_{\max} = 432 \left(\frac{t_r}{\Delta P} \right)^{2/5} \left(\frac{\sigma}{\rho_w} \right)^{3/5}, \tag{9-2}$$

where

$d_{\max}$ = diameter of droplet above whose size only 5% of the oil volume is contained, microns,
σ = surface tension, dynes/cm,
ρ_w = density, g/cm^3,
ΔP = pressure drop, psi,
t_r = retention time, min.

From Eq. (9-2), it can be seen that the greater the pressure drop and, thus, the shear forces that the fluid experiences in a given period of time while flowing through the treating system, the smaller the maximum oil droplet diameter will be. That is, large pressure drops that occur in small distances through chokes, orifices, throttling globe control valves, descanters, etc. result in smaller droplets.

Equation (9-2) is presented to illustrate the factors that affect drop size distribution in the system. The equation can be applied to determine a maximum droplet size that can exist downstream of a control valve or any other device that causes a large pressure drop.

The dispersion process is theoretically not instantaneous. However, it appears from field experience to occur very rapidly. For design purposes it could be assumed that whenever large pressure drops occur, all droplets larger than d_{max} will instantaneously disperse. This is, of course, a conservative approximation.

Unfortunately, Eq. (9-2) cannot be used directly to predict the coalescence of droplets that occur in piping with high-pressure drops downstream of a process component in which dispersion takes place. This is because the coalescence to a new d_{max} determined in Eq. (9-2) is time-dependent, and there is currently no basis to estimate the time required to grow d_{max}.

Flotation

Flotation is a process that involves the injection of fine gas bubbles into the water phase. The gas bubbles in the water adhere to the oil droplets. The buoyant force on the oil droplet is greatly increased by the presence of the gas bubble. Oil droplets are then removed when they rise to the water surface, where they are trapped in the resulting foam and skimmed off the surface. Experimental results show that very small oil droplets (greater than 10 microns) in a very dilute suspension can be removed by flotation. High percentages (90% +) of oil removal are achieved in very short times.

Figure 9-5 shows a cross section of a one-cell and a three-cell hydraulic inductor dispersed gas flotation unit. Clean water from the effluent is pumped to a recirculation header (E) that feeds a series of venturi eductors (B). Water flowing through the eductor sucks gas from the vapor space (A) that is released at the nozzle (G) as a jet of small bubbles. The bubbles rise, causing flotation in the chamber (C), forming a froth (D) that is skimmed with a mechanical device at (F).

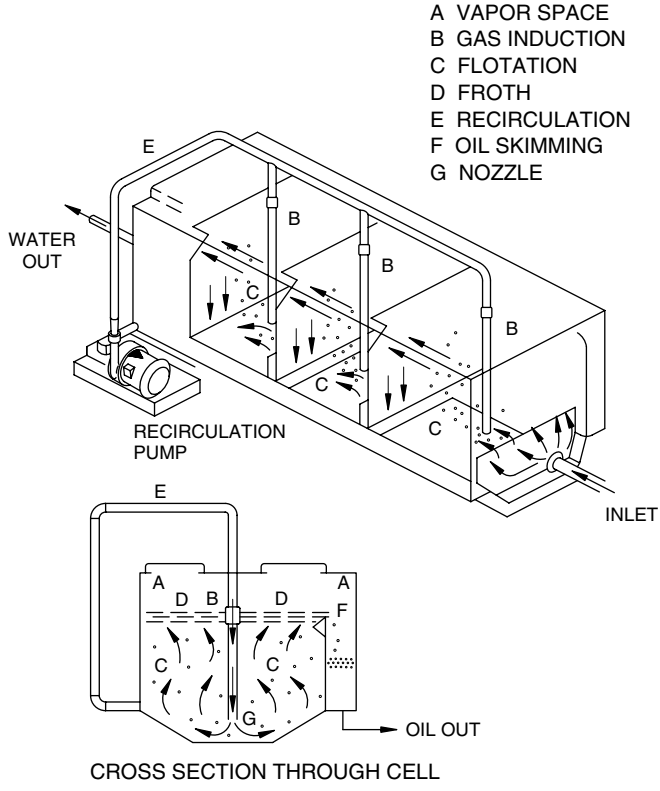


Figure 9-5. Dispersed gas flotation unit with inductor.

It would be extremely difficult to develop a precise mathematical model of the process occurring in the zones identified in this cross section. However, with the aid of some liberal assumptions, it is possible to develop a qualitative model of the efficiency of such a cell and gain an understanding of the importance of various parameters. The efficiency of a specific cell with constant geometry can be approximated through the use of Eqs. (9-3) through (9-5). Since these equations are presented to provide a qualitative "feel" for the effects of various parameters on flotation cell efficiency, units are not listed. In using these equations, however, one must use parameters with consistent units.

$$E = \frac{C_i - C_o}{C_i}, \tag{9-3}$$

$$E = \frac{K}{Q_w + K}, \tag{9-4}$$

$$K = \frac{6\pi K_p r^2 h q_g}{q_w d_b}, \tag{9-5}$$

where

E = efficiency per cell,
C_i = inlet oil concentration,
C_o = outlet oil concentration,
Q_w = liquid flow rate, BPD,
K_p = mass transfer coefficient,
r = radius of mixing zone,
h = height of mixing zone,
q_g = gas flow rate,
q_w = liquid flow through the mixing zone,
d_b = diameter of gas bubble.

The following conclusions can be drawn from Eq. (9-5):

1. Removal efficiency is independent of the influent oil concentration or the oil droplet size distribution.
2. Decreasing the diameter of the gas bubbles without changing the gas flow rate increases the efficiency.
3. Increasing the gas flow rate increases the efficiency.
4. Increasing the bulk flow rate decreases the efficiency.

Equation (9-5) cannot be used directly. It depends on the design details of the particular unit, which is under the control of the manufacturer, and depends on the mass flow transfer coefficient, which is a function of the composition and chemical treatment of the liquid. Most manufacturers attempt to design each cell for a typical efficiency in excess of 50%. The overall efficiency of a multiple cell flotation unit can be calculated from Eq. (9-6):

$$E_t = 1 - [1 - E]^n, \tag{9-6}$$

where

E_t = overall efficiency,
n = number of stages or cells.

For an average design efficiency of 50% per stage, the following overall efficiencies may be calculated:

# of Cells (N)	Overall Efficiency (E_t)
1	0.50
2	0.75
3	0.87
4	0.94
5	0.97

Most flotation units consist of three or four cells. Using more cells may not be cost-effective for the small performance increases shown above.

As an additional consideration, each cell must have some retention time so that the gas bubbles may have time to rise to the liquid surface. It is recommended that a minimum water retention time of one minute be provided in each cell.

Flotation units function best if the water flow through the unit is smooth. Therefore, it is recommended that throttling level controls be used to control the level in the upstream components of the system and in the flotation unit.

Filtration

Flow of produced water through a properly selected filter media will cause the small droplets of oil to contact and attach to the filter fibers. Depending on the media design and thickness these droplets will either stay trapped in the media or eventually "grow" as other droplets contact them. At some point the droplets will become large enough so that the drag forces on the droplet created by the bulk water flow through the media cause the now larger droplets to be stripped from the media. These larger droplets are then more easily separated by gravity settling downstream of the media. This action is called "filter/coalescing."

It is also possible to design the filter media to drop the oil. The media is cleaned periodically by stopping the flow and backwashing that is, flowing at very high velocities in the reverse direction for a short period of time. Thus a standard filter such as those described in Chapter 10 can be used.

Equipment Description and Sizing

Skim Tanks and Skim Vessels

The simplest form of primary treating equipment is a skim (clarifier) tank or vessel; refer to Figure 9-6. These items are normally designed to provide long residence times during which coalescence and gravity separation can occur. Skim tanks can be used as atmospheric tanks, pressure vessels, and surge tanks ahead of other produced water treating equipment.

The terminology used to describe the different equipment often is a source of great confusion. A "skim (clarifier) tank" is the terminology used to describe a tank that is used to remove dispersed oil. "Settling tanks," however, is the terminology used to describe tanks whose primary purpose is to remove entrained solids. On the other hand, "wash tanks" function as a free-water knockout or gunbarrel and are used when the incoming stream contains 10 to 90% oil. They are designed to make only a rough separation of the oil and water. The water from wash tanks is generally sent to a skim (clarifier) tank or another unit to remove the remaining oil.

If the desired outlet oil concentration is known, the theoretical dimensions of the vessel can be determined. Unlike the case of separation, with skim vessels one cannot ignore the effects of vibration, turbulence, short-circuiting, etc. American Petroleum Institute (API) Publication 421, *Management of Water Discharges: Design and Operation of Oil-Water Separators*, uses short-circuit factors as high as 1.75 and is the basis upon which many of the sizing formulas in this chapter were derived.

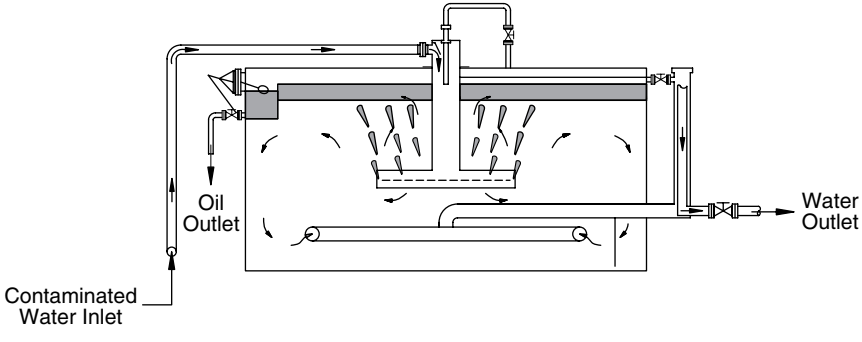


Figure 9-6. Schematic of a skimmer tank.

Configurations

Skim vessels can be either vertical or horizontal in configuration.

Vertical

In vertical skimmers the oil droplets must rise upward countercurrent to the downward flow of the water. Some vertical skimmers have inlet spreaders and outlet collectors to help even the distribution of the flow, as shown in Figure 9-7. The oil, water, and any flash gases are introduced below the oil–water interface. Small amounts of gas liberated from the water help to "float" the oil droplets. In the quiet zone between the spreader and the water collector, some coalescence can occur, and the

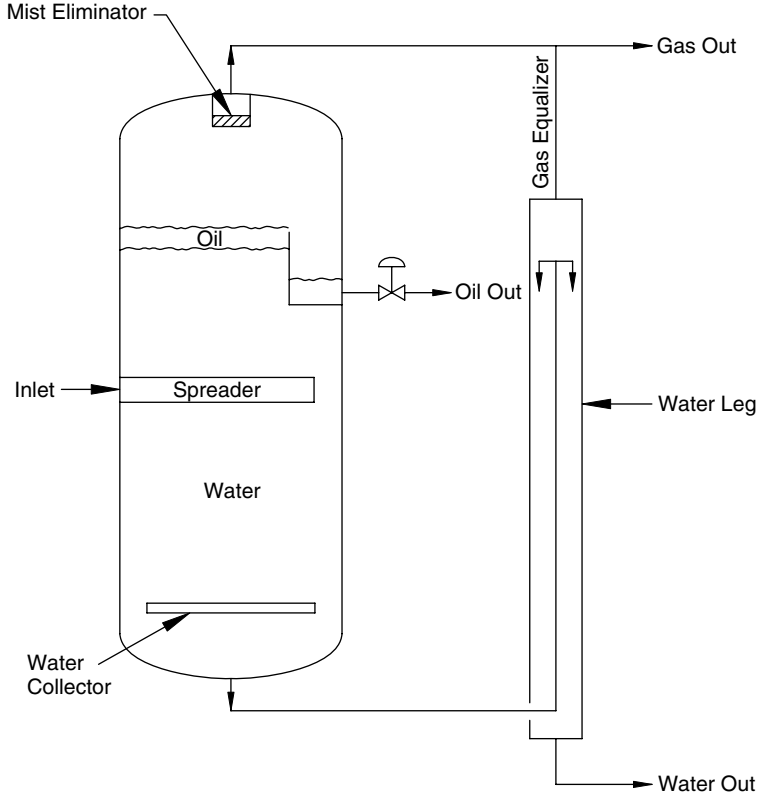


Figure 9-7. Schematic of a vertical skimmer vessel.

buoyancy of the oil droplets causes them to rise counter to the water flow. Oil will be collected and skimmed off the surface.

The thickness of the oil pad depends on the relative heights of the oil weir and the water leg and on the difference in specific gravity of the two liquids. Often, an interface level controller is used in place of the water leg.

Horizontal

In horizontal skimmers the oil droplets rise perpendicular to the flow of the water, as shown in Figure 9-8. The inlet enters in the water section so that the flashed gases may act as a dissolved gas flotation cell. The water flows horizontally for most of the length of the vessel. Baffles could be installed to straighten the flow. Oil droplets coalesce in this section of the vessel and rise to the oil–water surface, where they are captured and eventually skimmed over the oil weir. The height of the oil can be controlled by interface control, by a water leg similar to that shown in Figure 9-7, or by a bucket and weir arrangement.

Horizontal vessels are more efficient at water treating because the oil droplets do not have to flow countercurrent to the water flow. However, vertical skimmers are used in instances where

1. Sand and other solid particles must be handled. This can be done in vertical vessels with either the water outlet or a sand drain off the bottom. Experience with elaborately designed sand drains in large

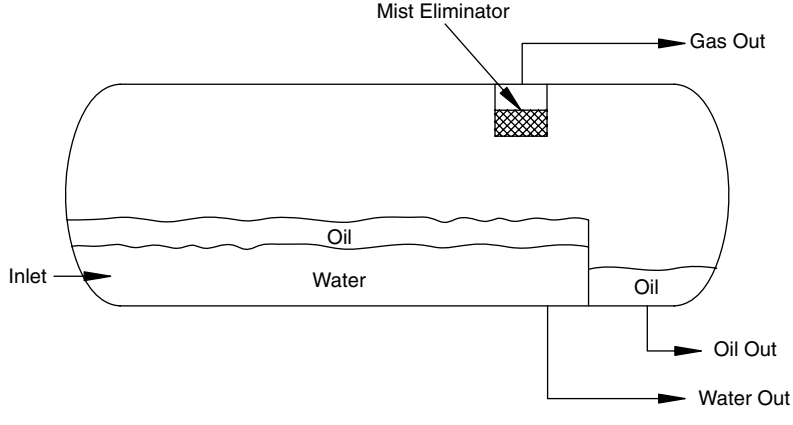


Figure 9-8. Schematic of a horizontal skimmer vessel.

horizontal vessels is expensive, and they have been only marginally successful in field operations.
2. Liquid surges are expected. Vertical vessels are less susceptible to high-level shutdowns due to liquid surges. Internal waves due to surging in horizontal vessels can trigger a level float even though the volume of liquid between the normal operating level and the high-level shutdown is equal to or larger than that in a vertical vessel. This possibility can be minimized through the installation of stilling baffles in the vessel.

It should be noted that vertical vessels have some drawbacks that are not process-related and that must be considered in making a selection. For example, the relief valve and some of the controls may be difficult to service without special access platforms and ladders. The vessel may have to be removed from a skid for trucking due to height restrictions.

Pressure Versus Atmospheric Vessels

The choice of pressure versus atmospheric vessel for the skimmer tank is not determined solely by the water treating requirements. The overall needs of the system must be considered in this decision. Pressure vessels are more expensive than tanks. However, they are recommended where

1. Potential gas blow-by through the upstream vessel dump system could create too much back-pressure in an atmospheric vent system.
2. The water must be dumped to a higher level for further treating and a pump would be needed if an atmospheric vessel were installed.

Due to the potential danger from overpressure and potential gas venting problems associated with atmospheric vessels, pressure vessels are preferred downstream of pressurized three-phase separators. However, an individual cost/benefit decision must be made for each application, taking into account all the requirements of the system.

Retention Time

Skim tanks are often used as the primary produced water treating equipment. The oil concentration of the inlet water entering the skim tank ranges from 500 to 10,000 mg/l. A minimum residence time of 10 to 30 minutes should be provided to assure that surges do not upset the system

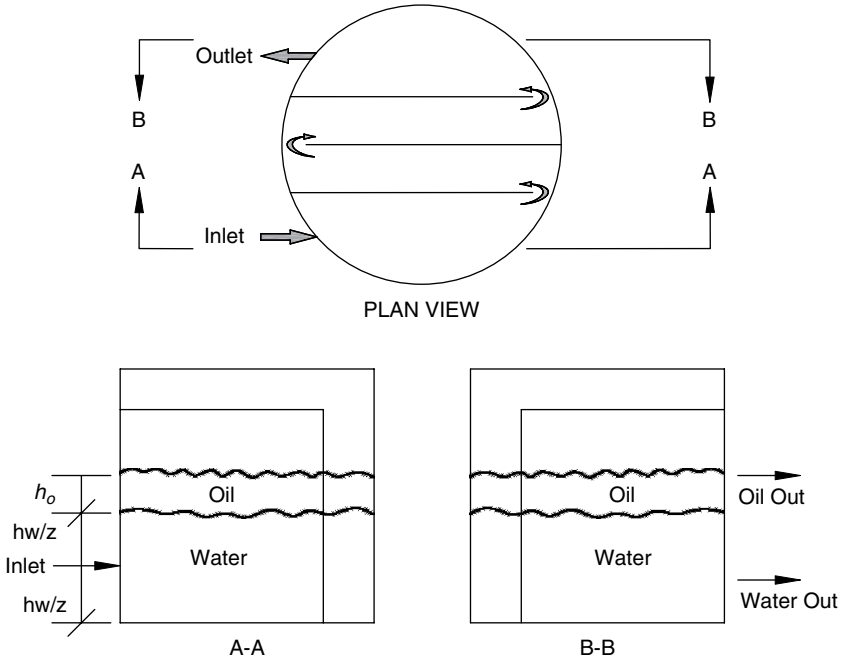


Figure 9-9. Schematic of a vertical skim tank with baffles.

and to provide for some coalescence. The minimum droplet size removal is in the 100- to 300-micron range. As previously discussed, the potential benefits of providing much more residence time will probably not be cost-efficient beyond this point. Skimmers with long residence times require baffles to attempt to distribute the flow and eliminate short-circuiting. Tracer studies have shown that skimmer tanks, with carefully designed spreaders and baffles, exhibit improved flow behavior. Figure 9-9 is a schematic of a vertical skim tank with baffles.

Performance Considerations

Several factors can affect the performance of a skim tank. Some of the more important factors include

- Carefully designed inlet and outlet distributors significantly improve the performance of a skim tank.
- Higher inlet water temperatures improve the oil removal due to a reduction in the bulk water phase viscosity.

- A short, wide, "stocky" tank is preferred over a tall, slender tank because it offers a lower downward water velocity, which aids in gravity separation.
- Unbaffled tanks are inefficient due to short-circuiting.
- Short-circuiting is reduced by installing a single vertical baffle.
- Horizontal baffles improve skim tank performance; however, to achieve maximum benefit, they should be installed as close to the horizontal as possible and caution should be used during operating maintenance not to alter the baffle arrangement.
- Often water treating chemicals, such as flocculants, are added upstream of the skim vessel. These chemicals work effectively to remove the smaller oil droplets by attaching to the oil droplets and causing them to rise to the oil–water interface in the skim vessel. However, if the chemical dosage is not carefully monitored, especially when the water rate decreases, an excess of chemical flocculants will result in a froth layer at the oil–water interface. This froth can cause the level controller to malfunction, leading to oil potentially spilling out of the vessel. Therefore, if chemical injection is used, its dosage should be carefully controlled.

Skim vessels are recommended when

- Pressure reduction from a separator is required to protect downstream produced water treating equipment.
- Degassing water, catching oil slugs or controlling surges is desired and the skim vessel is between the upstream separator and downstream produced water treating equipment.
- An existing vessel can be converted or space is available for a new vessel.
- The inlet oil concentration is high and the effluent must be reduced to 250 mg/l for the downstream equipment.
- Solid contaminants are in the inlet stream.

Skim vessels are not recommended when

- Influent oil droplet sizes are mostly below 100 microns.
- Size and weight are the primary considerations.
- Offshore structure (platform, tension leg, etc.) movement could generate waves in the vessel.
- Water temperature is very cold due to long subsea pipelines connected to other platforms.

Skimmer Sizing Equations

Horizontal Cylindrical Vessel: Half-Full

The required diameter and length of a horizontal cylinder operating 50% full of water can be determined from Stokes' law as follows:

Field Units

$$dL_{\text{eff}} = \frac{1{,}000\, Q_w \mu_e}{(\Delta\text{SG})\,(d_m)^2}, \tag{9-7a}$$

SI Units

$$dL_{\text{eff}} = 1{,}145{,}734 \frac{Qw\mu}{(\Delta\text{SG})\, dm^2}, \tag{9-7b}$$

where

d = vessel internal diameter, in. (mm),
Q_w = water flow rate, bwpd (m^3/hr),
μ_w = water viscosity, cp,
d_m = oil droplet diameter, microns,
L_{eff} = effective length in which separation occurs, ft (m)
ΔSG = difference in specific gravity between the oil and water relative to water.

Derivation of Equation (9-7)

Oil droplets must settle vertically upward through horizontally flowing water. t_o and t_w are in s, d_m in microns, μ_w in cp, and d in in. (mm).

Field Units

$$t_o = t_w,$$

$$t_o = \frac{d}{(24)\, V_o},$$

$$V_o = \frac{1.78 \times 10^{-6}\, (\Delta\text{SG})\, d_m^2}{\mu_w}.$$

L_{eff} is in ft, Q in ft^3/s, and A in ft^2.

$$t_w = \frac{L_{\text{eff}}}{V_w},$$

$$V_w = \frac{Q}{A},$$

Q_w is in BPD.

$$Q = \frac{Q_w\,(5.61)}{(24)\,(3{,}600)},$$

$$A = \frac{1}{2}\frac{\pi d^2}{(4)\,(144)}.$$

Use an efficiency factor of 1.8 for turbulence and short-circuiting.

$$dL_{\text{eff}} = 70\frac{Q_w \mu_w}{(\Delta\text{SG})\,d_m^2}$$

SI Units

$$t_o = t_w,$$

$$t_o = \frac{d}{2{,}000\,V_o},$$

$$V_o = \frac{5.556 \times 10^{-7}\,(\Delta\text{SG})\,d_m^2}{\mu_w}.$$

L_{eff} is in m, Q in m^3/s, and A in m^2.

$$t_w = \frac{L_{\text{eff}}}{V_w},$$

$$V_w = \frac{Q}{A}.$$

Q_w is in m^3/hr.

$$Q = \frac{Q_w}{3{,}600},$$

$$A = \frac{1}{2}\frac{\pi d^2}{(4)\,(1{,}000)^2}.$$

Use an efficiency factor of 1.8 for turbulence and short-circuiting.

$$dL_{\text{eff}} = 1{,}145{,}734\frac{Q_w\mu}{(\Delta\text{SG})\,d_m^2}$$

Any combination of L_{eff} and d that satisfies this equation will be sufficient to allow all oil particles of diameter d_m or larger to settle out of the water.

In addition to the settling criteria, a minimum retention time should be provided to allow coalescence. As stated earlier, increasing the retention time beyond that required for initial coalescence is not cost-effective for increasing oil droplet diameter. However, some initial retention time can be cost-effective in increasing the oil droplet size distribution. Typically, retention times vary from 10 to 30 minutes. It is recommended that a retention time of not less than 10 minutes be provided in skimmers that have no means of promoting coalescence. To ensure that the appropriate retention time has been provided, the following equation must also be satisfied when one selects d and L_{eff}. However, only those combinations that also satisfy the following retention time criteria should be chosen:

Field Units

$$d^2L_{\text{eff}} = 1.4(t_r)_w Q_w, \tag{9-8a}$$

where $(t_r)_w$ is retention time, in min.

SI Units

$$d^2L_{\text{eff}} = 4.2 \times 10^4 (t_r) Q_w, \tag{9-8b}$$

where $(t_r)_w$ is retention time, in min. This equation was derived in Chapter 4.

The choice of correct diameter and length can be obtained by selecting various values for d and L_{eff} for both Eqs. (9-7) and (9-8). For each d, the larger L_{eff} must be used to satisfy both equations.

The relationship between the L_{eff} and the seam-to-seam length of a skimmer depends on the physical design of the skimmer internals. Some approximations of the seam-to-seam length may be made based on experience as follows:

$$L_{\text{ss}} = \frac{4}{3}L_{\text{eff}}, \tag{9-9}$$

where L_{ss} is seam-to-seam length, in m (ft).

This approximation must be limited in some cases, such as vessels with large diameters. Therefore, the L_{eff} should be calculated using Eq. (9-9) but must be equal to or greater than the values calculated using the following equations:

Field Units

$$L_{\text{ss}} = L_{\text{eff}} + 2.5. \tag{9-10a}$$

SI Units

$$L_{\text{ss}} = L_{\text{eff}} + 0.76. \tag{9-10b}$$

Field Units

$$L_{\text{ss}} = L_{\text{eff}} + \frac{d}{24}. \tag{9-11a}$$

SI Units

$$L_{\text{ss}} = L_{\text{eff}} + \frac{d}{2{,}000}. \tag{9-11b}$$

Equation (9-10) will govern only when the calculated L_{eff} is less than 7.5 ft (2.3 m). The justification for this limit is that some minimum vessel length is always required for oil and water collection before dumping. Equation (9-11) governs when one-half the diameter in feet exceeds one-third of the calculated L_{eff}. This constraint ensures that even flow distribution can be achieved in short vessels with large diameters.

Horizontal Rectangular Cross-Section Skimmer

Similarly, the required width and length of a horizontal tank of rectangular cross section can be determined from Stokes' law using an efficiency factor of 1.9 for turbulence and short-circuiting:

Field Units

$$WL_{\text{eff}} = 70\frac{Q_w \mu_w}{(\Delta \text{SG})\,(d_m)^2}, \tag{9-12a}$$

SI Units

$$WL_{\text{eff}} = 950\frac{Q_w\mu_w}{(\Delta\text{SG})\,(d_m)^2}, \tag{9-12b}$$

where

W = width, ft (m),
L_{eff} = effective length in which separation occurs, ft (m).

Derivation of Equation (9-12)

Oil droplets must settle virtually upward through horizontally flowing water. t_o and t_w are in s; d_m in microns; μ in cp, L_{eff}, H, and W in ft (m); Q in ft^3/s (m^3/s); A in ft^2(m^2); and Q_w in BPD (m^3/hr).

Field Units

$$t_o = t_w,$$

$$t_o = \frac{H}{V_o},$$

$$V_o = \frac{1.78\times10^{-6}\,(\Delta\text{SG})\,d_m^2}{\mu_w},$$

$$t_w = \frac{L_{\text{eff}}}{V_w},$$

$$V_w = \frac{Q}{A},$$

$$Q = \frac{Q_w\,(5.61)}{(24)\,(3{,}600)},$$

$$A = HW,$$

$$WL_{\text{eff}} = 36.5\frac{Q_w\mu_w}{(\Delta\text{SG})\,d_m^2}.$$

Using an efficiency factor of 1.9 for turbulence and short-circuiting,

$$WL_{\text{eff}} = 70\frac{Q_w\mu_w}{(\Delta\text{SG})\,d_m^2}.$$

SI Units

$$t_o = t_w,$$

$$t_o = \frac{H}{V_o},$$

$$V_o = \frac{5.556 \times 10^{-7}\,(\Delta \mathrm{SG})\,d_m^2}{\mu_w},$$

$$t_w = \frac{L_{\mathrm{eff}}}{V_w},$$

$$V_w = \frac{Q}{A},$$

$$Q = \frac{Q_w}{3{,}600},$$

$$A = HW,$$

$$WL_{\mathrm{eff}} = 500\frac{Q_w \mu_w}{(\Delta \mathrm{SG})\,d_m^2}.$$

Using an efficiency factor of 1.9 for turbulence and short-circuiting,

$$WL_{\mathrm{eff}} = 70\frac{Q_w \mu_w}{(\Delta \mathrm{SG})\,d_m^2}.$$

Note that Eq. (9-12) is independent of height. This is because the oil settling time and the water settling retention time are both proportional to the height. Typically, the height of the water flow is limited to less than one-half the width to assure good flow distribution. With this assumption, the following equation can be derived to ensure that sufficient retention time is provided. If the height-to-width ratio is set to 50%, then the following retention time equation applies:

Field Units

$$W^2 L_{\mathrm{eff}} = 0.008(t_r)_w Q_w. \tag{9-13a}$$

SI Units

$$W^2 L_{\mathrm{eff}} = 0.0333(t_r)_w Q_w. \tag{9-13b}$$

Derivation of Equation (9-13)

t_w is in s, L_{eff} in ft (m), V_w in ft/s (m/s), Q in ft^3/s (m^3/s), Q_w in BPD (m^3/hr); and H, W, and L_{eff} in ft (m).

Field Units

$$t_w = \frac{L_{\text{eff}}}{V_w},$$

$$V_w = \frac{Q}{A},$$

$$Q = \frac{Q_w\,(5.61)}{(24)\,(3{,}600)},$$

$$A = HW,$$

$$t_w = 15{,}401\frac{HWL_{\text{eff}}}{Q_w}.$$

H is limited to $0.5W$; $(t_r)_w$ is in min.

$$W^2 L_{\text{eff}} = 0.008(t_r)_w Q_w.$$

SI Units

$$t_w = \frac{L_{\text{eff}}}{V_w},$$

$$V_w = \frac{Q}{A},$$

$$Q = \frac{Q_w}{3600},$$

$$A = HW,$$

$$t_w = 3{,}600\frac{HWL_{\text{eff}}}{Q_w}$$

H is limited to $0.5W$; $(t_r)_w$ is in min.

$$W^2 L_{\text{eff}} = 0.0333(t_r)_w Q_w$$

The choice of W and L that satisfies both requirements can be obtained graphically. The height of water flow, H, is set equal to 0.5 W.

As with horizontal cylindrical skimmers, the relationship between L_{eff} and L_{ss} is dependent on the internal design. Approximations of the L_{ss} of rectangular skimmers may be made using Eq. (9-9) and (9-10). However, the L_{ss} must also be limited by the following:

$$L_{\text{ss}} = L_{\text{eff}} + \frac{W}{20}. \tag{9-14}$$

As before, the L_{ss} should be the largest of Eqs. (9-9), (9-10), and (9-14).

Vertical Cylindrical Skimmer

One can determine the required diameter of a vertical cylindrical tank by setting the oil rising velocity equal to the average water velocity as follows:

Field Units

$$d^2 = 6{,}691F\frac{Q_w\mu}{(\Delta\text{SG})\,d_m^2}, \tag{9-15a}$$

SI Units

$$d^2 = 6.365\times 10^8 F\frac{Q_w\mu}{(\Delta\text{SG})\,d_m^2}, \tag{9-15b}$$

where F is a factor that accounts for turbulence and short-circuiting.

For small-diameter skimmers [48 in. or less (1.2 m or less)], the short-circuiting factor should be equal to 1.0. Skimmers with diameters greater than 48 in. (1.2 m) require a value for F. Inlet and outlet spreaders and baffles affect the flow distribution in large skimmers; therefore, they affect the value of F. It is recommended that for large-diameter skimmers, F should be set equal to $d/48$. Substituting this into Eq. (9-15) gives the following:

Field Units

$$d = 140\frac{Q_w\mu_w}{(\Delta\text{SG})\,d_m^2}. \tag{9-16a}$$

SI Units

$$d = 5.3\times 10^9\frac{Q_w\mu_w}{(\Delta\text{SG})\,d_m^2}. \tag{9-16b}$$

Derivation of Equation (9-15)

Oil droplets must settle vertically upward through vertically downward flowing water. V_o and V_w are in ft/s (m/s), d_m in microns, μ in cp, Q in ft^3/s (m^3/s), A in ft^2(m^2), Q_w in BPD (m^3/hr), and d in in. (mm).

Field Units

$$V_o = V_w,$$

$$V_o = \frac{1.78 \times 10^{-6}\,(\Delta SG)\,d_m^2}{\mu_w},$$

$$V_w = \frac{Q}{A},$$

$$Q = \frac{Q_w\,(5.61)}{(24)\,(3{,}600)},$$

$$A = \frac{\pi d^2}{(4)\,(144)},$$

$$d^2 = 6{,}691\,F\frac{Q_w\mu}{(\Delta SG)\,d_m^2}.$$

SI Units

$$V_o = V_w,$$

$$V_o = \frac{5.556 \times 10^{-7}\,(\Delta SG)\,d_m^2}{\mu_w},$$

$$V_w = \frac{Q}{A},$$

$$Q = \frac{Q_w}{3{,}600},$$

$$A = \frac{\pi d^2}{4\,(1{,}000)},$$

$$d^2 = 6.365 \times 10^8\,F\frac{Q_w\mu}{(\Delta SG)\,d_m^2}.$$

The height of the water column in a vertical skimmer can be determined for a selected d from retention time requirements:

Field Units

$$H = 0.7\frac{(t_r)_w Q_w}{d^2}, \tag{9-17a}$$

SI Units

$$H = 21{,}218\frac{(t_r)_w Q_w}{d^2}, \tag{9-17b}$$

where H is the height of the water, in ft (m).

Derivation of Equation (9-17)

t_w is in s, $(t_r)_w$ in min, H in ft (m), and V_w in ft/s (m/s).

Field Units

$$t_w = \frac{H}{V_w},$$
$$V_w = \frac{Q}{A},$$
$$A = \frac{\pi d^2}{4\,(144)},$$
$$t_w = 60\,(t_r)_w,$$
$$H = 0.7\frac{(t_r)_w Q_w}{d^2}.$$

SI Units

$$t_w = \frac{H}{V_w},$$
$$V_w = \frac{Q}{A},$$
$$A = \frac{\pi d^2}{4\,(1{,}000)},$$
$$t_w = 60\,(t_r)_w,$$
$$H = 21{,}218\frac{(t_r)_w Q_w}{d^2}.$$

The height of the oil pad in both vertical and horizontal skimmers typically ranges from 2 to 6 in. (50 to 150 mm). It is important to remember that the purpose of a water skimmer is to remove oil from water and produce as clean a water stream as possible. The quality of the skimmed oil from a skimmer is a secondary consideration. In fact, skimmed oil streams typically contain 20 to 50% water. The objective is to maximize the water treating ability of the skimmer. Maintaining a minimum oil pad thickness accomplishes this objective.

Coalescers

Several different types of devices have been developed to promote the coalescence of small dispersed oil droplets. These devices use gravity separation similar to skimmers but also induce coalescence to improve the separation. Thus, these devices can either match the performance of a skimmer in less space or offer improved performance in the same space.

Plate Coalescers

The use of flow through parallel plates to help gravity separation in skim tanks was pioneered in the late 1950s as a method of modifying existing refinery horizontal rectangular cross-section separators to treat oil droplets less than 150 microns in diameter. Various configurations of plate coalescers have been devised. These are commonly called parallel plate interceptors (PPI), corrugated plate interceptors (CPI), or cross-flow separators. All of these depend on gravity separation to allow the oil droplets to rise to a plate surface where coalescence and capture occur. Plate coalescers overcome the size and weight disadvantage of skim tanks by enhancing coalescence of the oil droplets, thereby substantially increasing their rise velocities. Consequently, plate coalescers require smaller cross-sectional areas, thus providing space and weight gains over skim tanks.

As shown in Figure 9-10, flow is split between a number of parallel plates spaced 1/2 to 2 in. (1.2 to 5 cm) apart. To facilitate capture of the oil droplets, the plates are inclined to the horizontal, which promotes oil droplet coalescence into films, and to guide the oil to the top for entrapment into channels, thereby preventing remixing with the water. The plates provide a surface for the oil droplets to collect and for solids particles to settle.

Figure 9-11 shows that an oil droplet entering the space between the plates will rise in accordance with Stokes' law. At the same time, the oil droplet will have a forward velocity equal to the bulk water velocity.

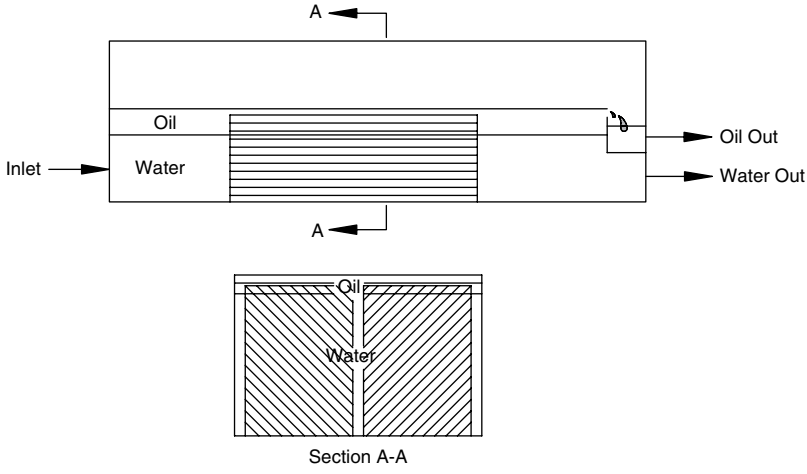


Figure 9-10. Schematic of a parallel plate interceptor.

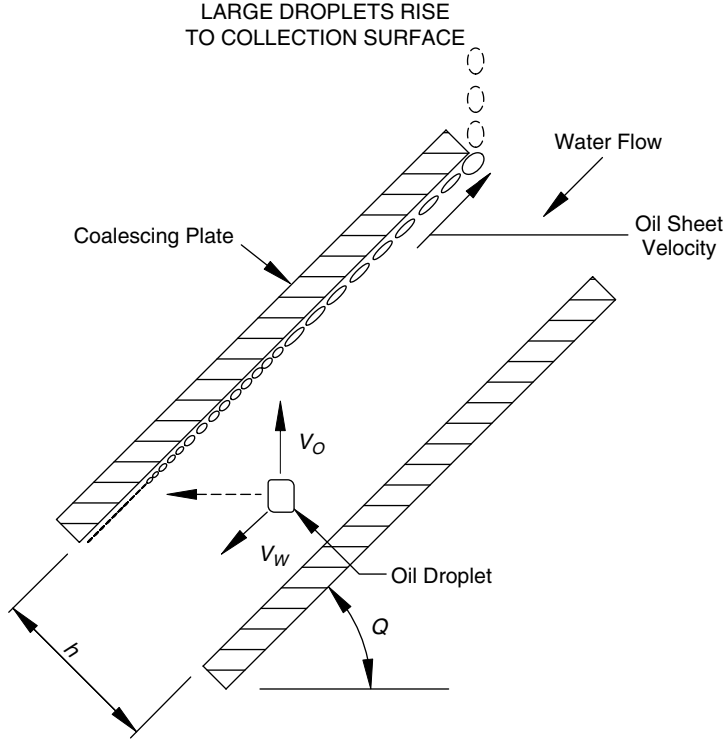


Figure 9-11. Cross section showing plate coalescer operation.

By solving for the vertical velocity needed by a particle entering at the base of the flow to reach the coalescing plate at the top of the flow, the resulting droplet diameter can be determined.

It is important to note that Stokes' law should apply to oil droplets as small in diameter as 1 to 10 microns. However, field experience indicates that 30 microns sets a reasonable lower limit on the droplet sizes that can be removed. Below this size small pressure fluctuations, platform vibration, etc. tend to impede the rise of the droplets to the coalescing surface.

Parallel Plate Interceptor (PPI)

The first form of a plate coalescer was the parallel plate interceptor (PPI), as shown in Figure 9-9. This involved installing a series of plates parallel to the longitudinal axis of an API separator (a horizontal rectangular cross-section skimmer). The plates form a "V" when viewed along the axis of flow so that the oil sheet migrates up the underside of the coalescing plate and to the sides. Sediments migrate toward the middle and down to the bottom of the separator, where they are removed. The interplate spacing can be small, which would allow packing more plates inside a vessel, which would in turn maximize the area for oil droplets to coalesce. However, this spacing would increase the probability of plugging the interspaces with solids. As a compromise, a distance of $^3/_4$ in. is typically used. The angle of inclination for the plates is generally established at 45°.

Corrugated Plate Interceptor (CPI)

The most common form of parallel plate interceptor used in production operations is the corrugated plate interceptor (CPI). This is a refinement of the PPI in that it takes up less plan area for the same particle size removal, it makes sediment handling easier, and it has the added benefit of being cheaper than a PPI.

Figure 9-12 shows the flow pattern of a typical downflow CPI design. Water enters the inlet nozzle (1), where solids flow downward and settle in the primary collection box (2). Water and oil flow up and through a perforated distribution baffle plate (3). The CPI pack (4) receives oily water. Oil rises out of the flow path to the underside of the ridge and coalesces into a film moving upward opposite the bulk water flow. A thick layer of oil is allowed to collect until it flows over an adjustable weir (5) into an oil collection box for removal. Light solids and sludge separation is simultaneously accomplished and falls to the lower plate surface along the gutters and collects at the bottom (6), where it is removed. After exiting

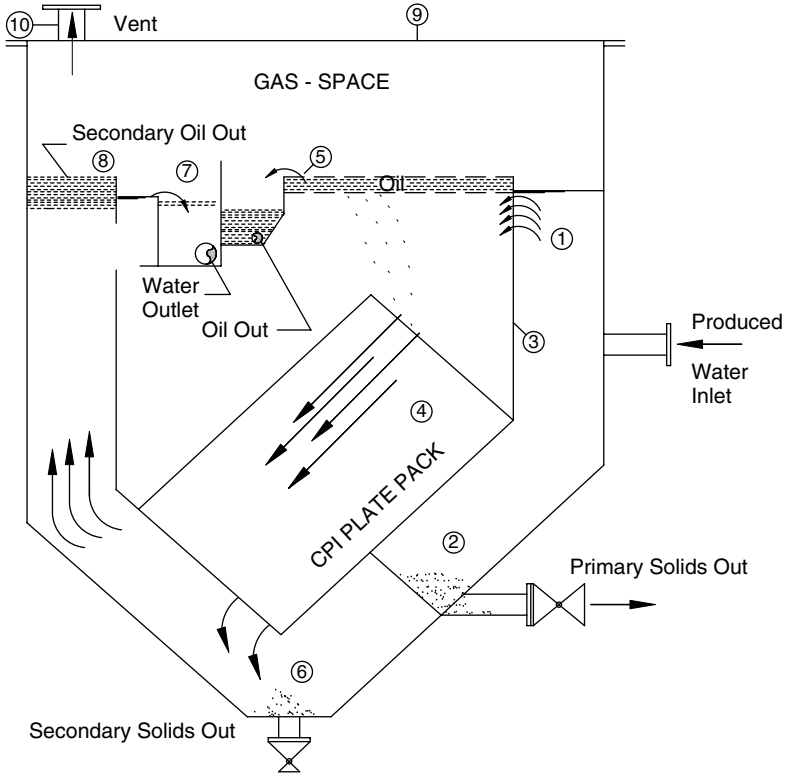


Figure 9-12. Schematic showing flow pattern of a typical down-flow CPI design.

the CPI pack, the water moves upward and flows over an adjustable weir (7) into the water removal collection box. A secondary oil removal outlet (8) is located above the water outlet. A gasketed cover (9) allows for gas blanket operation. It is also supplied with an adequately sized vent nozzle (10).

In CPIs the parallel plates are corrugated (like roofing material), and the axes of the corrugations are parallel to the direction of flow. Figure 9-13 shows a typical CPI pack. The plate pack is inclined at an angle of 45° and the bulk water flow is forced downward. The oil sheet rises upward counter to the water flow and is concentrated in the top of each corrugation. When the oil reaches the end of the plate pack, it is collected in a channel and brought to the oil–water interface.

In areas where sand or sediment production is anticipated, the sand should be removed prior to flowing through a standard CPI. Because of the required laminar flow regime, all plate coalesces are efficient sand settling devices.

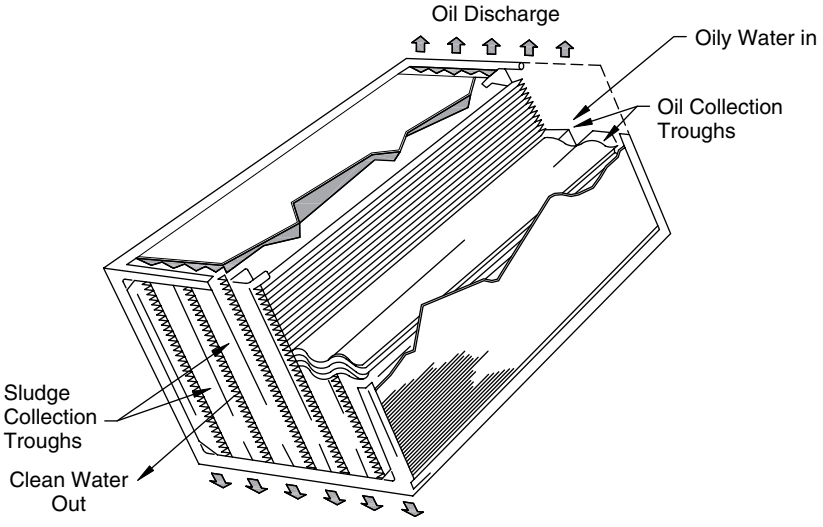


Figure 9-13. CPI plate pack.

Experience has shown that oil wet sand may adhere to a 45° slope. Therefore, the sand may adhere to and clog the plates. In addition, the sand collection channels installed at the end of the plate pack cause turbulence that affects the treating process and are themselves subject to sand plugging. To eliminate the above problems, an "upflow" CPI unit employing corrugated plates, spaced a minimum of 1 in. (2.5 cm) apart with a 60° angle of inclination, may be used. Water jets for sand removal should also be installed. Figure 9-14 is a schematic showing the flow pattern of a typical upflow CPI design. Figure 9-15 compares the flow pattern of an upflow and downflow CPI pack.

The main components of a CPI plate separator are

- Separator basin,
- CPI plate pack,
- Oil and effluent weir,
- Basin cover,
- Solids hopper,
- Inlet and outlet nozzles.

The ***separator basin*** and its internals are generally made of carbon steel plate with at least a 3/16-in. thickness. The basin edges are welded. All carbon steel external and internal surfaces are blast cleaned and painted with epoxy paint.

The ***CPI plate packs*** are constructed of chlorinated polyvinylchloride (CPVC), polyvinylchloride (PVC), polypropylene (PP), fiberglass

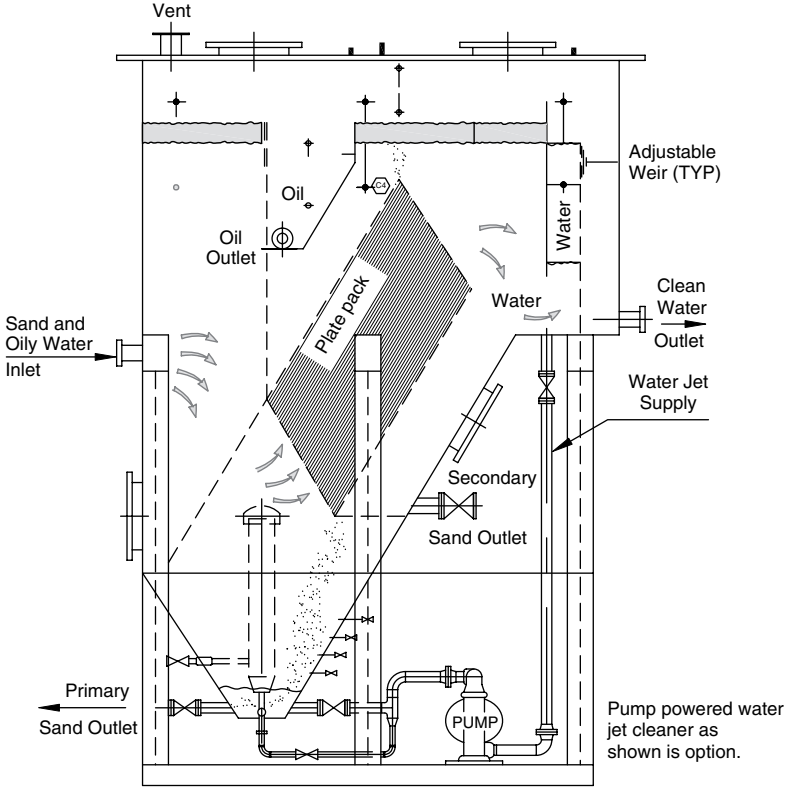


Figure 9-14. Schematic showing flow pattern of a typical upflow CPI design.

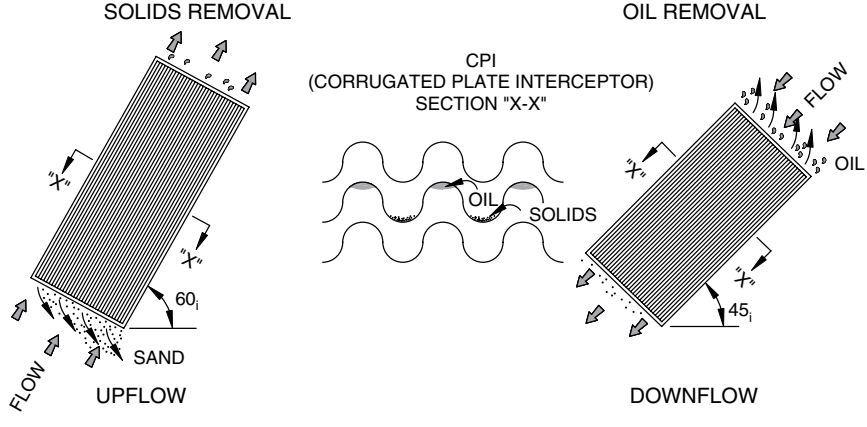


Figure 9-15. Upflow versus downflow flow pattern.

reinforced polyester, carbon steel, galvanized steel, or various grades of stainless steel. Stainless steel plate packs can be used up to temperatures as high as 350°F (125°C), whereas the polymer plates are limited to about 140°F(55°C). The plate pack usually has a 316 SS frame for robustness and easy removal during maintenance. Polypropylene plates have an inherently oleophilic property that attracts oil, thus promoting coalescence. Polypropylene also repels water, which adds the downward flow of sludge, thus reducing chances of sludge fouling.

The ***oil weir*** is a bucket type and made of carbon or stainless steel. The ***effluent weir*** is a plate type and its height is adjustable.

The ***basin cover*** is normally made of carbon steel, heavy-duty galvanized steel, or lightweight fiberglass reinforced plastic (FRP) with 3/16-in. thickness.

The ***solids hopper*** may be conical or dish-shaped for cylindrical separators, or shaped like an inverted pyramid for rectangular separators.

The vessel should be leak tested prior to coating. The assembled package should be dry function tested to ensure proper operation. Any plastic piping should also be hydrotested.

Cross-Flow Devices

Equipment manufacturers have modified the CPI configuration for horizontal water flow perpendicular to the axis of the corrugations in the plates, as shown in Figure 9-16. This modification allows the plates to be put on a steeper angle to facilitate sediment removal and to enable the plate pack to be more conveniently packaged in a pressure vessel. The latter benefit may be required if gas blow-by through an upstream dump valve could cause relief problems with an atmospheric tank.

Cross-flow devices can be constructed in either horizontal or vertical pressure vessels. The horizontal vessels require less internal baffling, as the ends of almost every plate conduct the oil directly to the oil–water interface and the sediments to the sediment area below the water flow area. However, as shown in Figure 9-17, the pack is long and narrow and, therefore, it requires an elaborate spreader and collection device to force the water to travel across the plate pack in plug flow. The inlet oil droplets may shear in the spreader, which would make separation more difficult. This configuration would be preferred when a pressure vessel in a high-pressure system is needed.

Vertical units, although requiring collection channels on one end to enable the oil to rise to the oil–water interface and on the other end to allow the sand to settle to the bottom, can be designed for more efficient sand removal.

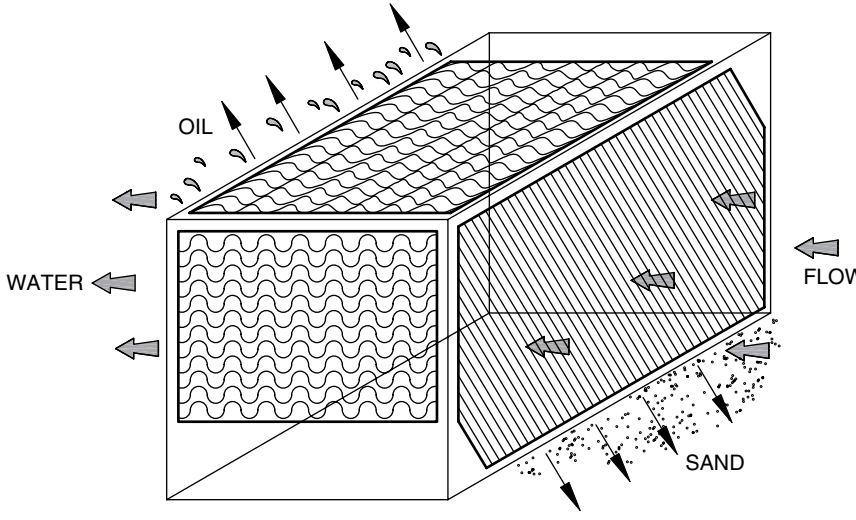


Figure 9-16. Schematic showing flow pattern of cross-flow plate pack.

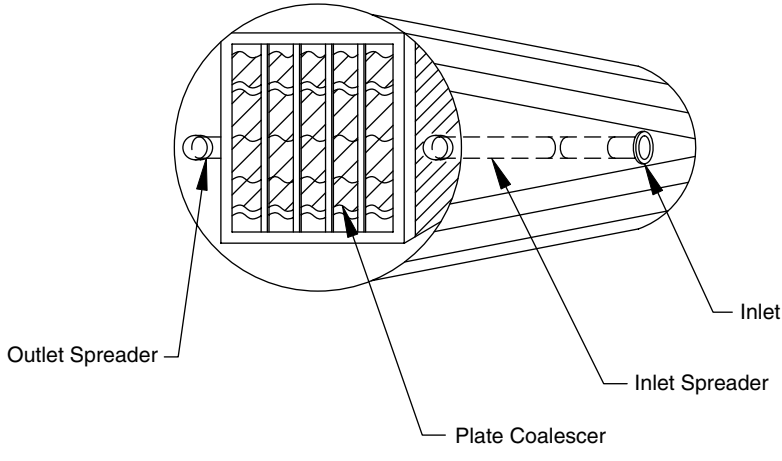


Figure 9-17. Schematic showing cross-flow device installed in a horizontal pressure vessel.

The cross-flow device may be installed in an atmospheric vessel, as shown in Figure 9-18, or in a vertical pressure vessel.

CPI separators are generally cheaper and more efficient at oil removal than cross-flow separators. However, cross-flow separators should be considered where a pressure vessel is preferred or where high sand production is expected and the sand is not removed upstream of the water treating equipment.

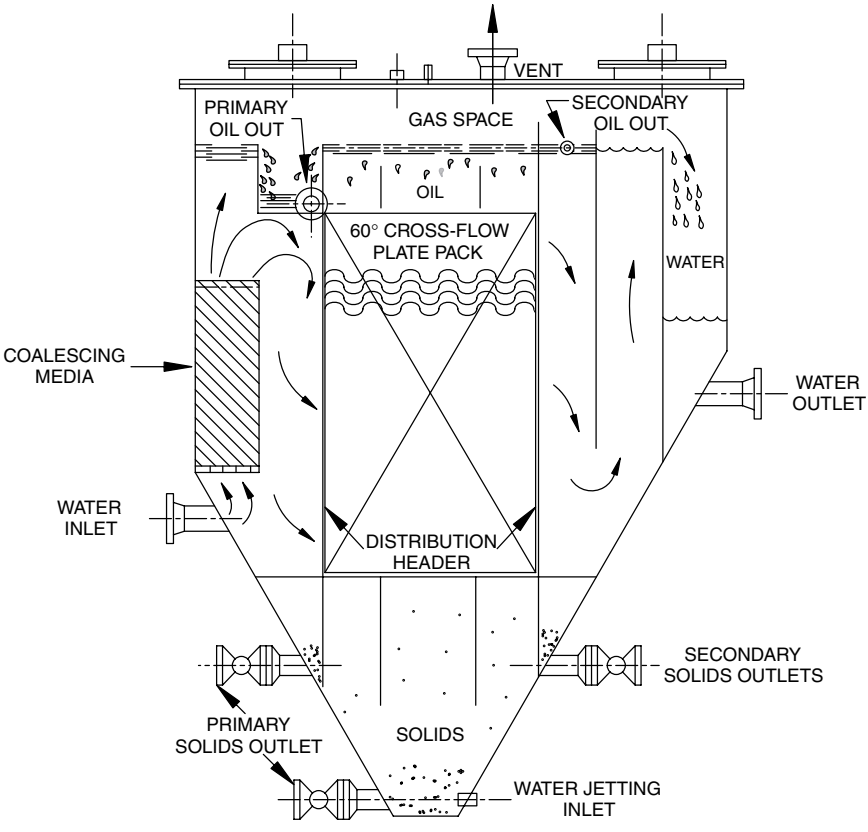


Figure 9-18. Schematic showing cross-flow device installed in atmospheric vessel.

Performance Considerations

Flow direction considerations include

- ***Downflow.*** For efficient oil removal the downflow configuration is preferred. In this case the plate pack is inclined at a 45° angle, provided the solids content is not significant.
- ***Upflow.*** If the production stream contains a significant amount of solid particles, upflow CPIs with the pack inclined at 60° to the horizontal are preferred. The higher plate slope provides about 25% greater runoff force and a 30% lower erosion rate than the industry-standard 45° plate slope.
- ***Cross flow.*** Cross flow should be considered where the use of a pressure vessel is preferred and solids and oil removal is desired.

Plate separators generally exhibit the following advantages:

- They require very little maintenance. Coalescing packs can be easily removed as complete modules for inspection and cleaning, if necessary.
- They have smaller size and weight requirements than skim vessels because of the effect of the closely spaced inclined plates.
- They can accept fairly high concentrations of oil or solids in the inlet feed. The inlet oil influent can be as high as 3,000 mg/l.
- They can separate oil droplets down to about 30 microns.
- They have a sand removal ratio of 10:1, that is, if a CPI unit captures 50-micron oil droplets, it will also capture solid particles as low as 5 microns.
- They are totally enclosed, thereby eliminating vapor losses and reducing fire hazards.
- CPIs are more efficient at oil removal than cross-flow separators are.
- They are simple and inexpensive in comparison to some of the other types of produced water treating devices, for example, flotation units.
- They have no moving parts and do not require power.
- They are easy to cover, due to their small size, and retain hydrocarbon vapors.
- They are easy to install in a pressure vessel, which helps to retain hydrocarbon vapors and protect against overpressure due to failure of an upstream level control valve.

The disadvantages of plate separators include

- They are not effective for streams with slugs of oil.
- They cannot effectively handle large amounts of solids and emulsified streams.

Plate separators are recommended when

- Water flow rate is steady or feed is from a pump.
- Size and weight are not constraints.
- Utilities and equipment are available to periodically clean the plate packs.
- Influent oil content is high and oil concentration must be reduced to 150 mg/l for effective second-stage treating in a downstream unit.
- Solid contaminants are not significant in the waste stream, and sand content is less than 110 ppm.

Plate separators are not recommended when

- Influent droplet sizes are mostly below 30 microns.
- Size and weight are the primary considerations.

- Sand particle diameters are less than 25 microns, and solids removal is a primary objective.

Selection Criteria

Plate separators are effective to approximately 30 microns. Vendor-supplied nomographs can be used to estimate the performance of CPIs. Figure 9-19 presents a relationship among the liquid inflow temperature,

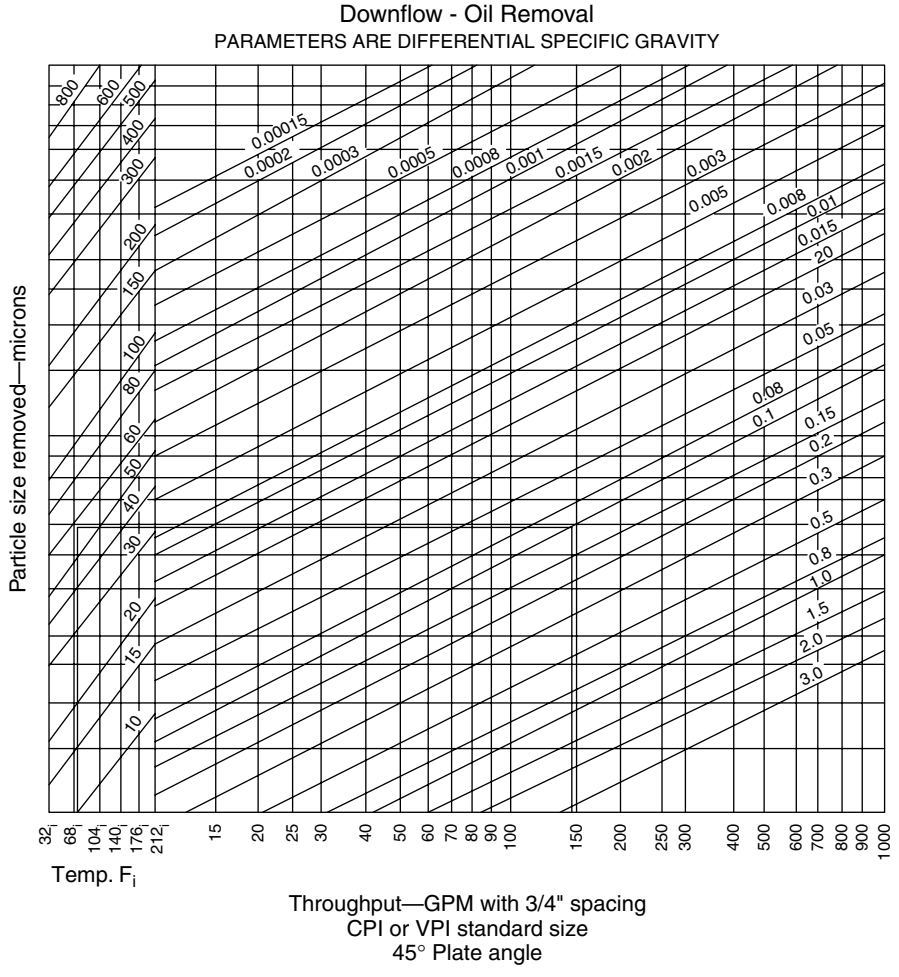


Figure 9-19. Nomograph for downflow CPI.

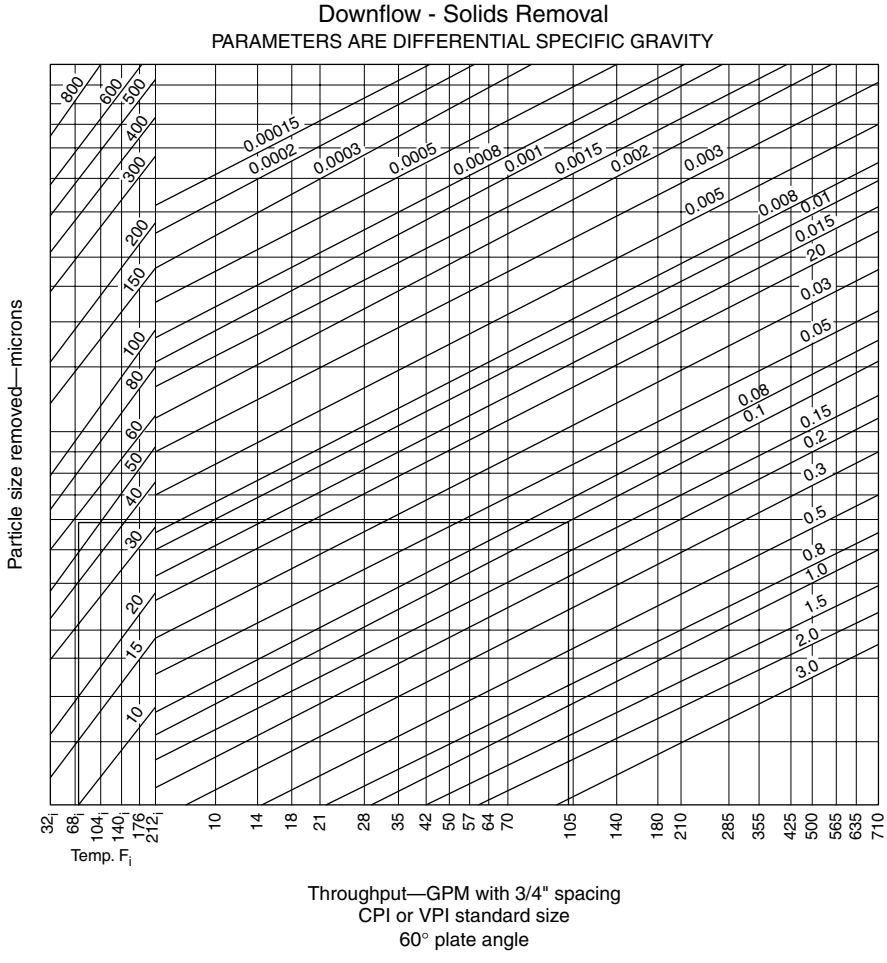


Figure 9-20. Nomograph for upflow CPI.

particle size removed, differential specific gravity of the oil and water, and capacity for downflow oil removal.

For example, produced water flowing at a rate of 150 gpm (5,143 bbl/day) per CPI pack with $^3/_4$-in. spacing, a differential specific gravity of 0.1, and a flowing temperature of 68°F will remove a particle of about 60 microns. Similarly, Figure 9-20 is a nomograph for upflow solids and oil removal, and Figure 9-21 has the performance relationship for cross-flow oil removal.

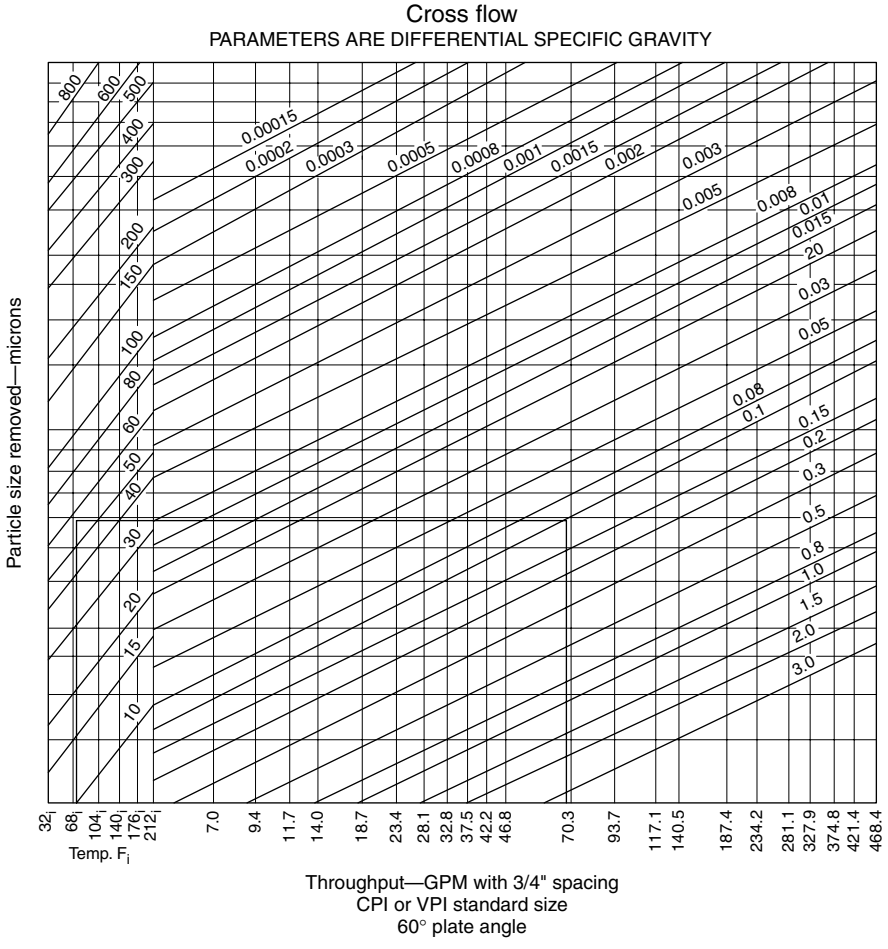


Figure 9-21. Nomograph for cross-flow CPI.

Coalescer Sizing Equations

The general sizing equation for a plate coalescer with flow either parallel to or perpendicular to the scope of the plates for droplet size removal is

Field Units

$$HWL = \frac{4.8 Q_w h \mu_w}{\cos\theta \, (d_m)^2 \, (\Delta \mathrm{SG})}, \tag{9-18a}$$

SI Units

$$HWL = \frac{0.794 Q_w h \mu_w}{\cos\theta \, (d_m)^2 \, (\Delta SG)}, \tag{9-18b}$$

where

d_m = design oil droplet diameter, microns,
Q_w = bulk water flow rate, bwpd (m^3/hr),
h = perpendicular distance between plates, in. (mm),
μ_w = viscosity of the water, cp,
θ = angle of the plate with the horizontal,
H, W = height and width of the plate section perpendicular to the axis of water flow, ft (m),
L = length of plate section parallel to the axis of water flow, ft (m),
ΔSG = difference in specific gravity between the oil and water relative to water.

Derivation of Equation (9-18)

The oil droplet must rise to the underside of the coalescing plate. T_o and t_w are in s, V_w in ft/s (m/s), Q in ft^3/s (m^3/s), A in ft^2 (m^2), H, W, and L in ft (m), and Q_w in BPD (m^3/hr).

Field Units

$$t_o = t_w,$$

$$t_w = \frac{L_{\text{eff}}}{V_w},$$

$L_{\text{eff}} = 0.7L$ (that is, only 70% of the actual length of the pack is effective in the settling process),

$$V_w = \frac{Q}{A},$$

$$Q = \frac{Q_w \, (5.61)}{(24) \, (3{,}600)},$$

$A = 0.9HW$ (the plate material itself takes up 10% of the flow area),

$$t_w = \frac{(0.7)\,L\,(0.9)\,HW\,(24)\,(3{,}600)}{5.61Q_w},$$

h is in in.

$$t_o = \frac{h}{12\cos\theta} \times \frac{1}{V_o},$$

$$V_o = \frac{1.78 \times 10^{-6}\,(\Delta\text{SG})\,d_m^2}{\mu},$$

$$HWL = \frac{4.8Q_w h\mu}{\cos\theta\,(\Delta\text{SG})\,d_m^2}.$$

SI Units

$$t_o = t_w,$$

$$t_w = \frac{L_{\text{eff}}}{V_w},$$

$L_{\text{eff}} = 0.7L$ (that is, only 70% of the actual length of the pack is effective in the settling process),

$$V_w = \frac{Q}{A},$$

$$Q = \frac{Q_w}{3{,}600},$$

$A = 0.9HW$ (the plate material itself takes up 10% of the flow area),

$$t_w = \frac{(0.7)\,L\,(0.9)\,HW\,(3{,}600)}{Q_w},$$

h is in mm,

$$t_o = \frac{h}{1{,}000\cos\theta} \times \frac{1}{V_o},$$

$$V_o = \frac{5.556 \times 10^{-7}\,(\Delta\text{SG})\,d_m^2}{\mu},$$

$$HWL = \frac{0.794Q_w h\mu}{\cos\theta\,(\Delta\text{SG})\,d_m^2}.$$

Experiments have indicated that the Reynolds number for the flow regime cannot exceed 1,600 with four times the hydraulic radius as the characteristic dimension. Based on this correlation, the minimum H times W for a given Q_w can be determined from

Field Units

$$HW = 14 \times 10^4 \frac{Q_w h\,(\mathrm{SG})_w}{\mu_w}. \tag{9-19a}$$

SI Units

$$HW = 8.0 \times 10^{-4} \frac{Q_w h\,(\mathrm{SG})_w}{\mu_w}. \tag{9-19b}$$

Derivation of Equation (9-19)

μ'_w is in lb-s/ft^2 (kgs/m^2), ρ_w in lb/ft^3 (kg/m^3), R in ft (m), and D in ft (m). R is the hydraulic radius.

Field Units

$$R = \frac{\text{area between plates}}{\text{wetted perimeter}}$$

$$R = \frac{hW}{12} \times \frac{1}{2\left(W + \frac{h}{12}\right)} \cong \frac{h}{24},$$

$$\mathrm{Re} = \text{Reynolds number},$$

$$\mathrm{Re} = \frac{V_w D \rho_w}{\mu' g},$$

$$D = 4R,$$

$$V_w = \frac{5.61}{(24)\,(3{,}600)\,(0.9)} \frac{Q_w}{HW},$$

μ is in cp,
$\mu'_w = 2.088 \times 10^{-5}\mu$,
$\mathrm{Re} = 1{,}600$,
$\rho_w = 62.4(\mathrm{SG})$,

$$HW = 7.0 \times 10^{-4} \frac{Q_w h\,(\mathrm{SG})_w}{\mu}.$$

To account for surges from control valves, use a safety factor of 2. Therefore,

$$HW = 14 \times 10^{-4} \frac{Q_w h\,(\mathrm{SG})_w}{\mu}.$$

SI Units

$$R = \frac{\text{area between plates}}{\text{wetted perimeter}}$$

$$R = \frac{hW}{1000} \times \frac{1}{2\left(W + \frac{h}{1000}\right)} \cong \frac{h}{24},$$

$\mathrm{Re} =$ Reynolds number,

$$\mathrm{Re} = \frac{V_w D \rho_w}{\mu' g},$$

$$D = 4R,$$

$$V_w = \frac{Q_w}{(3{,}600)\,(0.9)\,(HW)},$$

μ is in cp,
$\mu'_w = 0.0001\mu$,
$\mathrm{Re} = 1{,}600$,
$\rho_w = 1{,}000(\mathrm{SG})$,

$$HW = 4.0 \times 10^{-4} \frac{Q_w h\,(\mathrm{SG})_w}{\mu}.$$

To account for surges from control valves, use a safety factor of 2. Therefore,

$$HW = 8.0 \times 10^{-4} \frac{Q_w h\,(\mathrm{SG})_w}{\mu}.$$

CPI Sizing

For CPIs, plate packs come in standard sizes with $H = 3.25$ ft (1 m), $W = 3.25$ ft (1 m), $L = 5.75$ ft (1.75 m), $h = 0.69$ in., and $\theta = 45°$. The size of the CPIs is determined by the number of standard plate packs installed. To arrive at the number of packs needed, the following equation is used:

Field Units

$$\text{number of packs} = 0.077\frac{Q_w\mu}{(\Delta\text{SG})\,(d_m)^2}. \tag{9-20a}$$

SI Units

$$\text{number of packs} = 11.67\frac{Q_w\mu}{(\Delta\text{SG})\,(d_m)^2}. \tag{9-20b}$$

To ensure that the Reynolds number limitation is met, the flow through each pack should be limited to approximately 20,000 bwpd.

It is possible to specify a 60° angle of inclination to help alleviate the solids plugging problem inherent in CPIs. This requires a 40% increase in the number of packs according to the following equation:

Field Units

$$\text{number of packs} = 0.11\frac{Q_w\mu}{(\Delta\text{SG})\,(d_m)^2}. \tag{9-21a}$$

SI Units

$$\text{number of packs} = 16.68\frac{Q_w\mu}{(\Delta\text{SG})\,(d_m)^2}. \tag{9-21b}$$

Cross-Flow Device Sizing

Cross-flow devices obey the same general sizing equations as plate coalesces. Although some manufacturers claim greater efficiency than CPIs, the reason for this is not apparent from theory, laboratory, or field tests; as a result, verification is unavailable. If the height and width of these cross-flow packs are known, Eq. (9-18) can be used directly. It may be necessary to include an efficiency term, normally 0.75, in the denominator on the right side of Eq. (9-18) if the dimensions of H or W are large and a spreader is needed.

Both horizontal and vertical cross-flow separators require spreaders and collectors to uniformly distribute the water flow among the plates. For this reason, the following equation has been developed assuming a 75% spreader efficiency term:

Field Units

$$HWL = \frac{6.4 Q_w h \mu_w}{\cos\theta\,(\Delta SG)\,(d_m)^2}. \tag{9-22a}$$

SI Units

$$HWL = \frac{1.06 Q_w h \mu_w}{\cos\theta\,(\Delta SG)\,(d_m)^2}. \tag{9-22b}$$

Example 9-1: Determining the dispersed oil content in the effluent water from a CPI plate separator

Given:

Feed water flow rate = 25,000 bbl/day @ 125°F,
Feed water specific gravity = 1.06 @ 125°F,
Feed water viscosity = 0.65 cp @ 125°F,
Dispersed oil concentration = 650 mg/l,
Dissolved oil concentration = 10 mg/l,
Total oil & grease = 660 mg/l.

The dispersed oil droplet size distribution in feed water is as follows:

Microns	<40	40–60	60–80	80–100	100–120	>120
Vol. %	9 14	30	35	10	2	100%

A vendor has quoted that one of its standard plate packs would be capable of reducing the total oil and grease content of the effluent water to less than 200 mg/l. The vendor's standard plate pack has the following geometric specification:

$H = 3.25\,\text{ft}, \quad W = 3.25\,\text{ft}, \quad L = 5.75\,\text{ft}, \quad h = 0.69\,\text{in.}, \quad \theta = 45°.$

Calculate the total oil and grease content in effluent water from the plate pack to check the vendor's quoted performance.

Solution:

In order to calculate the total oil and grease in the effluent water, we must first determine the smallest oil droplet size that can be removed in the vendor's standard plate pack at the design conditions given. Equation (9-20) was derived for the given plate pack geometric configuration

assumed in this example calculation. Rearranging Eq. (9-20) to solve for the minimum oil droplet size (d_m), we have the following results:

$$d_m = \sqrt{0.077 \frac{Q_w \mu}{\Delta SG\,(\text{no. of packs @ 45 deg})}}$$

$$= \sqrt{0.077 \frac{(25{,}000)\,(0.65)}{(1.06 - 0.75)\,(1)}}$$

$$= 63.5 \textit{ microns.}$$

The volume percent of the dispersed oil removed by the plate pack is determined by summing the volume percents of dispersed oil droplets contained in the feed water that are greater than or equal to 63.5 microns (see dispersed oil droplets size distribution data given). Therefore,

$$\text{vol. \% removed} = \left(\frac{80 - 63.5}{80 - 60}\right)(30) + (35) + (10) + (2)$$

$$= 71.75\%.$$

Calculating the dispersed oil content in the effluent water from the plate pack (C_{out}):

$$C_{\text{out}} = (650)\,(100\% - 71.75\%)$$

$$= 183.6\,\text{mg/l}.$$

Since the plate pack does not remove any of the dissolved oil, the total oil and grease content in the effluent water from the plate pack is equal to 183.6 mg/l plus 10 mg/l, or 193.6 mg/l. Therefore, the vendor's quoted performance looks to be correct.

Oil/Water/Sediment Coalescing Separators

The oil/water/sediment coalescing separator is an enhancement of the cross-flow configuration in that it utilizes a two-step process to separate small oil droplets and solids from the well stream. The coalescing packs used are cross flow in design rather than downflow or upflow. The units can be configured in either an atmospheric pressure tank (see Figure 9-22) or a vertical pressure vessel (as shown in Figure 9-23).

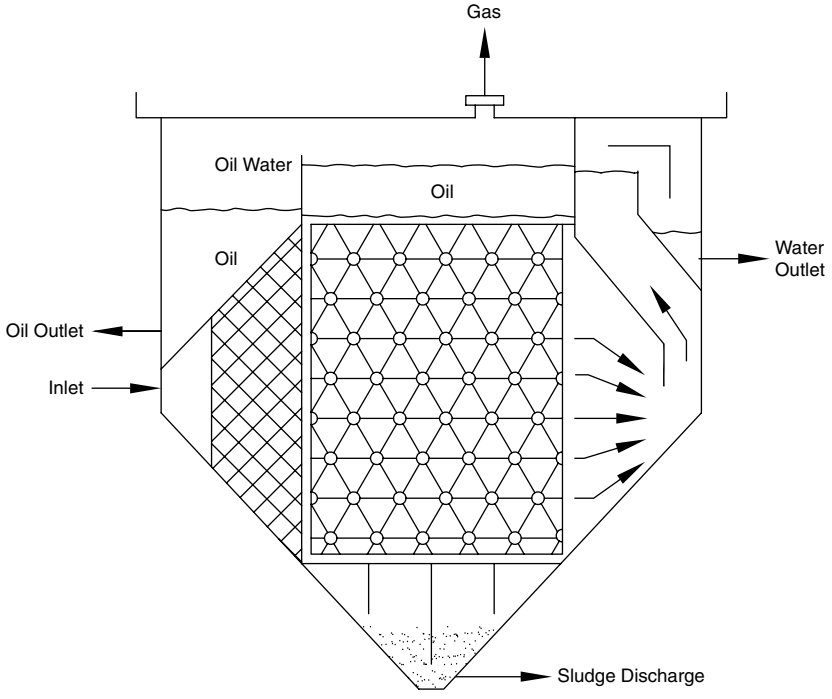


Figure 9-22. Schematic of an oil/water/sediment coalescing tank.

Both configurations use an inlet flow distributer/coalescer pack and a cross-flow plate pack.

The inlet flow distributer/coalescer pack evenly spreads the inlet flow over the full height and width of the separator pack. Flow through this pack is mildly turbulent, thus creating opportunities for the oil droplets to coalesce into larger ones.

The cross-flow plate pack receives flow from the distributer/coalescer pack. It consists of mutually supportive, inclined plates oriented in a hexagonal configuration. Laminar flow is established and maintained as water flows in a sinusoidal path across the pack from the inlet to the outlet. Oil rises into the top of hexagons and then along the plate's surface to the oil layer that is established at the top of the pack. The sludge slides down the plates and drops into a discreet sludge hopper in the bottom of the separator.

Standard spacing of cross-flow plate packs is 0.80 in., with optional available spacing of either 0.46 or 1.33 in. The pack is inclined 60° to lessen plugging. More coalescing sites are offered to the dispersed oil droplets due to the hexagonal pattern of the pack.

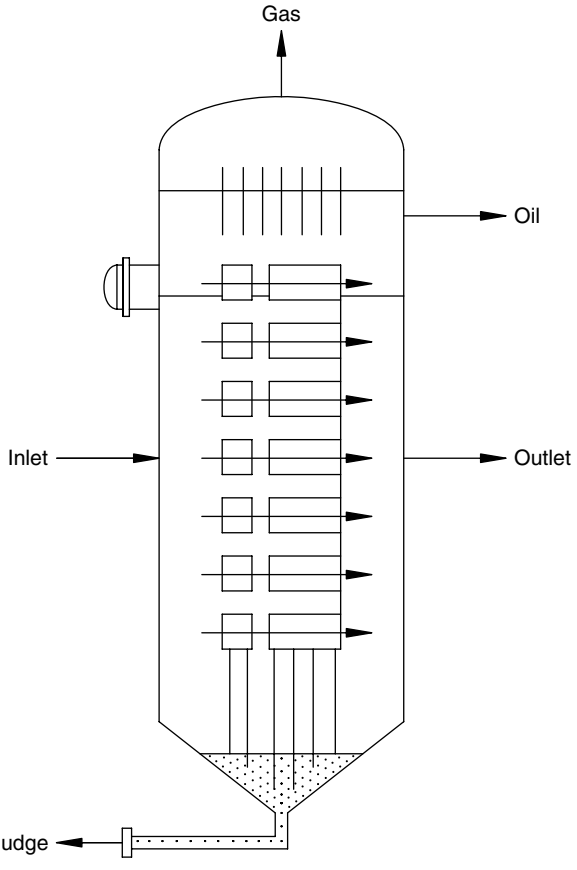


Figure 9-23. Schematic of an oil/water/sediment coalescing pressure vessel.

Coalescing pack materials include polypropylene, polyvinyl chloride, stainless steel, and carbon steel. Due to its oleophilic nature (enhances oil removal capabilities and resists plugging and fouling of the pack), polypropylene packs are commonly used up to 150°F (66°C). Above this temperature, the polypropylene loses pack integrity and chemical degradation begins. Stainless steel and carbon steels are used in temperatures above 150°F (66°C) and environments that contain large amounts of aromatic hydrocarbons.

Oil/Water/Sediment Sizing

The geometry of plate spacing and length can be analyzed for this configuration using Eq. (9-18) and the techniques previously discussed.

Performance Considerations

The oil/water/sediment coalescing separator exhibits the same advantages and disadvantages as plate separators. The one additional improvement is that the minimum oil droplet size that can be removed is 20 microns.

Skimmer/Coalescers

Several designs that are marketed for improving oil–water separation rely on installing coalescing plates or packs within horizontal skimmers or free-water knockouts to encourage coalescence and capture of small oil droplets within the water continuous phase. Coalescers act to accumulate oil on a preferentially oil wet surface where small droplets can accumulate. These larger oil droplets can be either collected directly from the oil wet surface or stripped from the oil wet surface and separated from the water phase using some type of gravity-based equipment.

Coalescing equipment may either be housed in a separate vessel or, more commonly, be installed in a coalescing pack contained in a gravity vessel. Figure 9-24 shows a schematic of a horizontal FWKO with coalescing pack. The plates in a CPI or cross-flow vessel may be fabricated

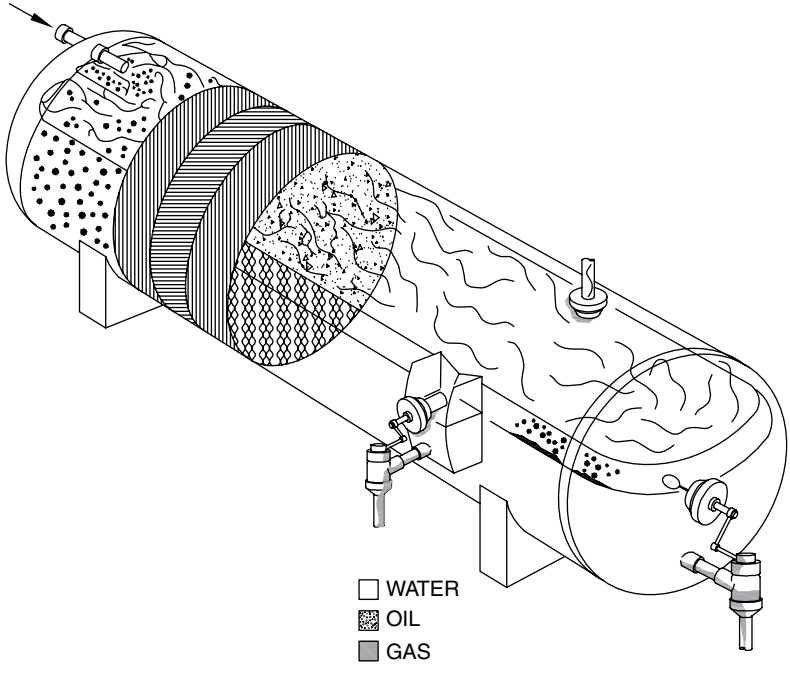


Figure 9-24. Schematic of an FWKO with a coalescing pack.

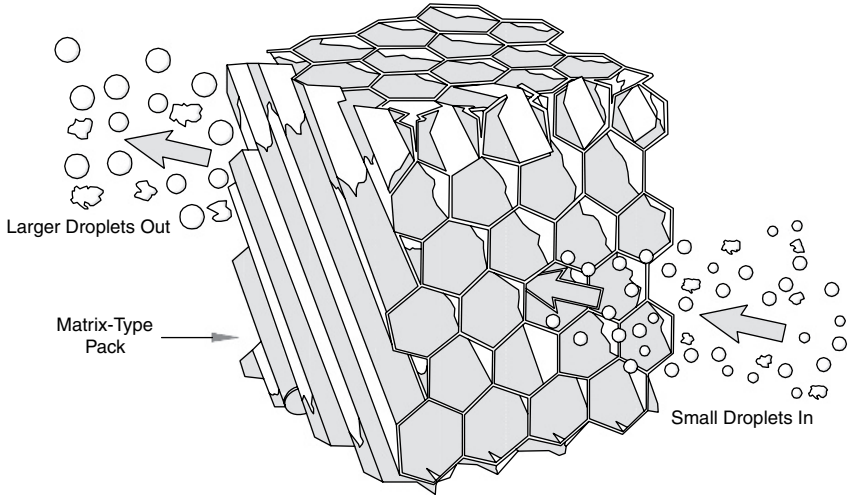


Figure 9-25. Structured packing serving as a coalescer.

of an oleophilic (oil wetting) material and thereby serve as both a gravity separation device and a coalescing device. Figure 9-25 is a cross section of structured packing serving as a coalescer.

The oil wetting surface may also occur in the form of a fibrous pack or as a collection of granules. Coalescers of this type resemble filters but serve to "grow" rather than to capture oil droplets.

Matrix Type

Mats of fibers have the advantage of large surface areas and easy fabrication. Oleophilic materials are spun into thin fibers, and the fibers are collected into a pack, across which the oily water flows. Oil droplets stick to the fibers and coalesce. Figure 9-26 illustrates the coalescence process on a fibrous mat. The coalesced droplets can easily be collected after they emerge from the mat using a gravity-based separator. Figure 9-27 is an example of such an arrangement.

Loose Media

Oleophilic material can also be fabricated into loose media, and the media collected in a vessel. If the material is fabricated in a granular form and assembled into a deep bed gravity settler, the deep bed filter can also perform a coalescing function.

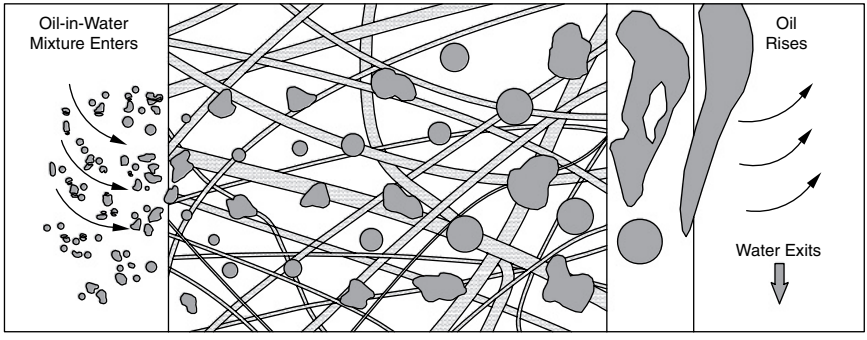


Figure 9-26. Oil coalescence on a fibrous mat.

The geometry of plate spacing and length can be analyzed for each of these designs using Eq. (9-18) and the techniques previously discussed. The packs cover the entire inside diameter of the vessel unless sand removal internals are required. Pack lengths range from 2 to 9 ft, depending on the service.

Performance Considerations

Coalescers are used to improve the performance of other gravity-based separation equipment. They are specified by the equipment vendor as an integral part of the water treating system, or they may be added as a retrofit to improve the performance of an existing system. Coalescers are particularly useful when the oil droplet size in the incoming water is small as a result of excess shearing in upstream piping or valves.

Coalescers can be used when

- An existing low-pressure separator, skimmer, or plate separator can be retrofitted with a coalescing section.
- The coalescing section is accessible for cleaning or replacement.
- The inlet oil droplet size is less than 50 microns and larger droplets are desired.
- Coalescers can also serve as a skimmer (the limitations listed for skimmers are applicable).

Coalescers should not be used when

- Inlet droplet sizes are less than 10 microns.
- Inlet droplet sizes are greater than 100 microns.
- Size and weight are primary considerations.

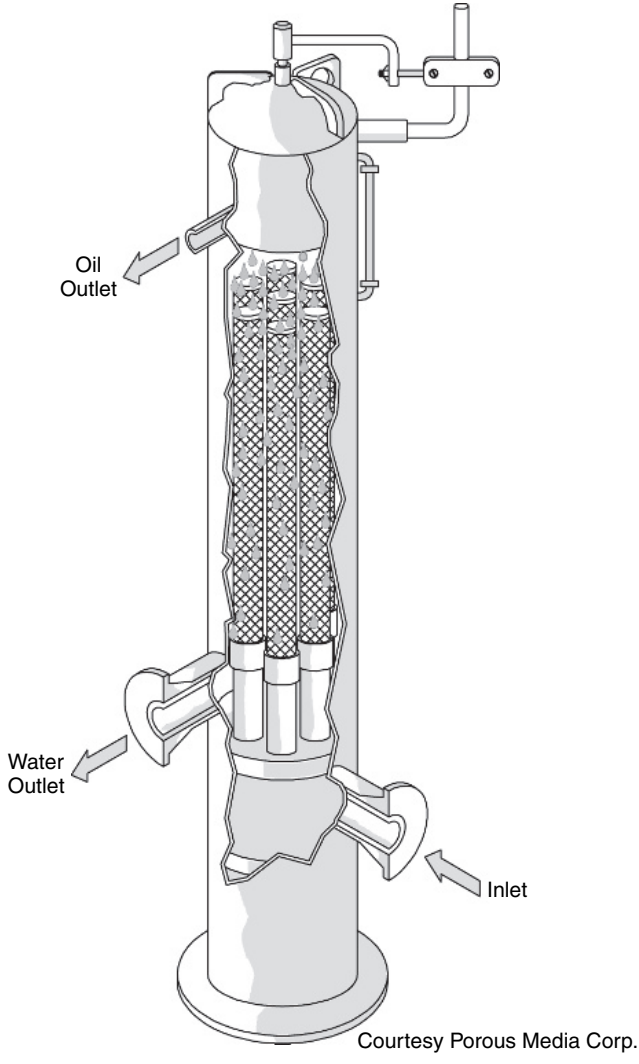


Figure 9-27. Collection of oil from a matrix separator.

Precipitators/Coalescing Filters

Precipitators are obsolete and would not be used in a new installation. In the past, it was common to direct the water to be treated through a bed of excelsior (straw) or another similar medium, as shown in Figure 9-28, to aid in the coalescing of oil droplets. However, the coalescing medium

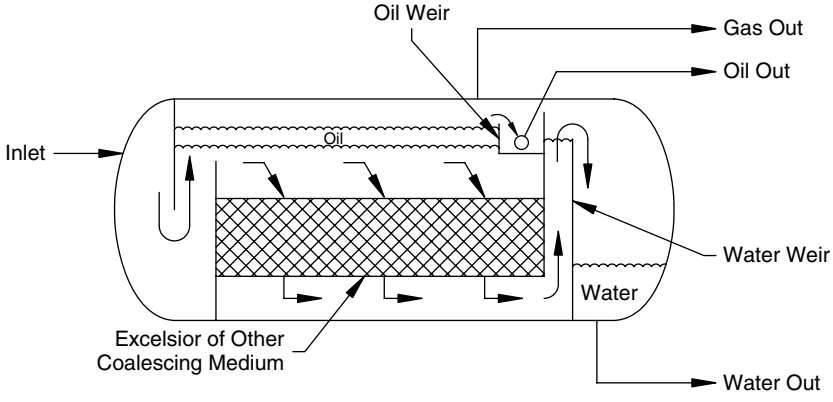


Figure 9-28. Schematic of a precipitator.

has a tendency to clog. Many of these devices in oil-field service have the medium removed. In such a case they actually act like a vertical skimmer since the oil droplets must flow countercurrent to the downward flow of the water through the area where the medium was originally located.

Coalescers, as shown in Figure 9-29, are similar in design to a precipitator except that they usually employ a larger gravity separation section

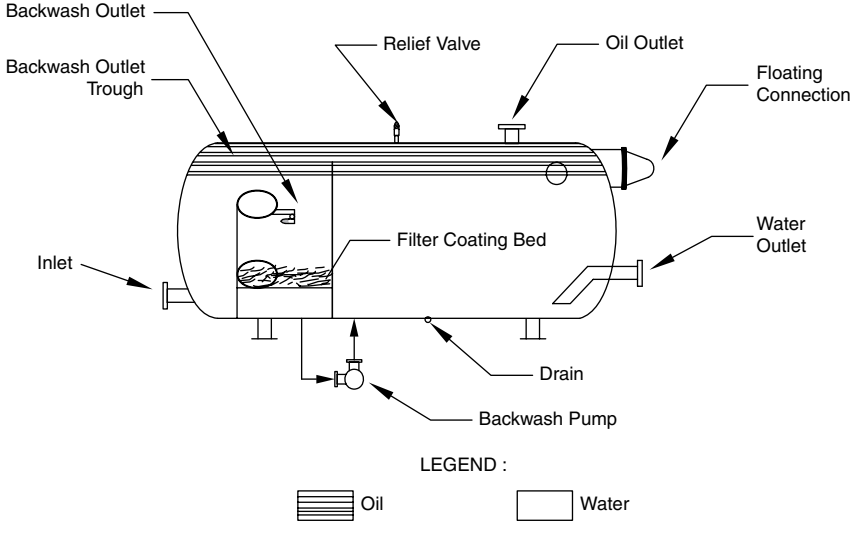


Figure 9-29. Schematic of a coalescer.

than a precipitator and utilize a back-washable filter bed for coalescing and some sediment removal. The filter media are designed for automatic backwash cycles. They are extremely efficient at water cleaning, but clog easily with oil and are difficult to backwash. The backwash fluid must be disposed of, which leads to further complications.

Some operators have had success with filters employing sand and other filter media in onshore operations where the backwash fluid can be routed to large settling tanks, and where the water has already been treated to 25–75-mg/l oil. Applications of this type are typical when the produced water will be re-injected as for a water flood. See Chapter 10 for a more complete description of filter construction.

Free-Flow Turbulent Coalescers

The plate coalescing devices discussed above use gravity separation followed by coalescence to treat water. Plate coalescers have the disadvantage of requiring laminar flow and closely spaced plates in order to capture the small oil droplets and keep them from stripping the coalesced sheet. They are thus susceptible to plugging with solids.

Free-flow turbulent coalescers are a type of device that is installed inside or just upstream of any skim tank or coalescer to promote coalescence. These devices had been marketed and sold under the trade name SP Packs. They are no longer available for sale but the concept can still be employed in water treating system design. As shown in Figure 9-30, SP Packs force the water flow to follow a serpentine pipe-like path sized to create turbulence of sufficient magnitude to promote coalescence, but not so great as to shear the oil droplets below a specified size. SP Packs are less susceptible to plugging since they require turbulent flow (high Reynolds numbers), have no closely spaced passages, and have a pipe path similar in size to the inlet piping.

SP Packs are designed to coalesce oil droplets to a defined drop size distribution, with a d_{max} of 1,000 Microns. They can be created by sizing a series of short runs of pipe with a diameter sized to create a Reynold's number of 50,000 containing 6–10 short radius 180 degree bends. Each run of straight pipe should be 30 to 50 pipe diameters long. Increasing the d_{max} from a typical value of 250 microns in a normal inlet to a skim tank or coalescer to 1,000 microns significantly reduces the size of the skimmer required. In addition, the need for retention time in the skimmer is not as important, since coalescence has occurred prior to the skimmer. As a result, retention times in the skimmer may be reduced to the 3- to 10-minute range.

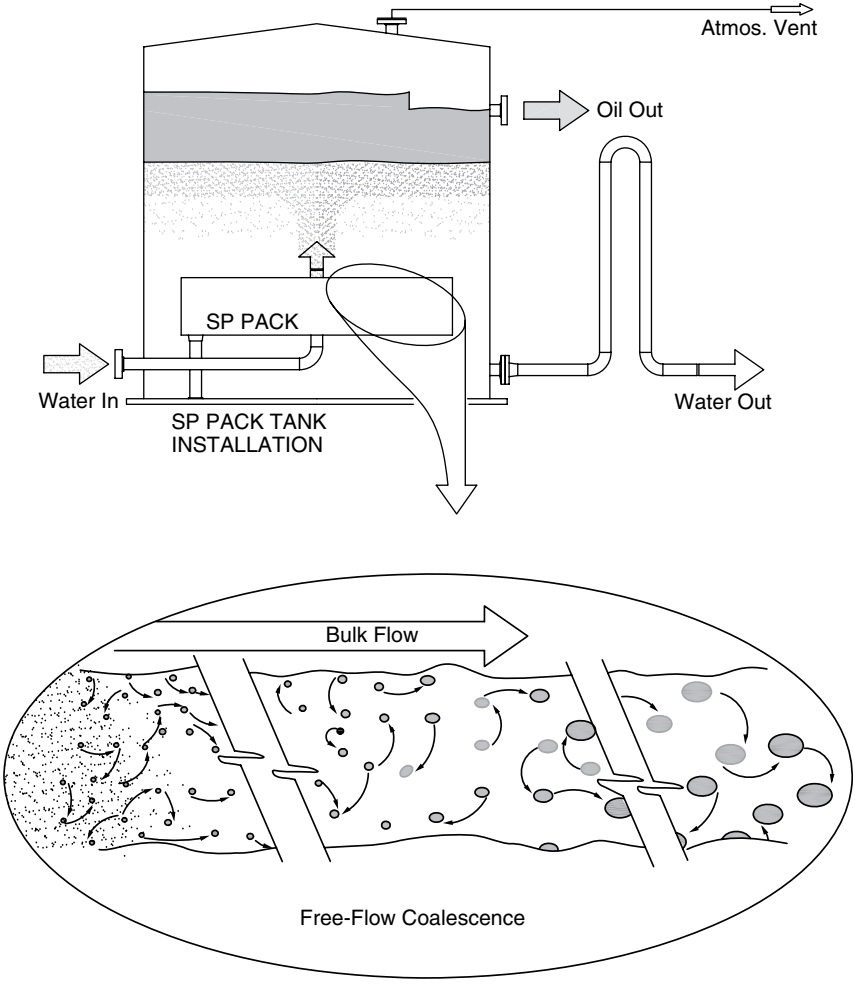


Figure 9-30. Principles of operation of an SP Pack.

SP Packs may be effectively multistaged, as shown in Figure 9-31. As shown in Figure 9-32, a two-stage system may consist of an SP Pack, a skim vessel, a second SP Pack, and a second identical skim vessel. One SP Pack and skimmer combination constitutes one stage of coalescence and separation. The second SP Pack coalesces the small oil droplets in the first skimmer's outlet; then the second skimmer may remove the larger oil droplets.

The addition of the SP Pack greatly improves the oil removal in the second skimmer because of coalescence. If the second SP Pack were not

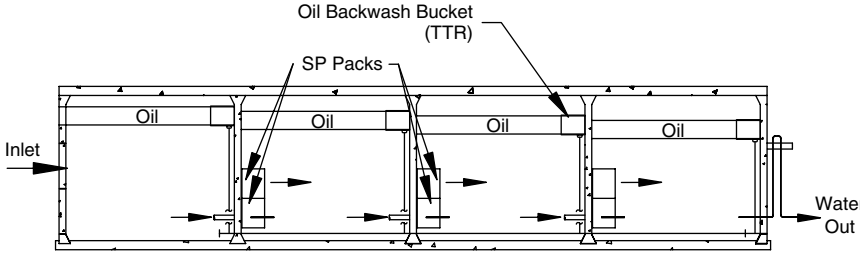


Figure 9-31. SP Pack installed in a horizontal flume.

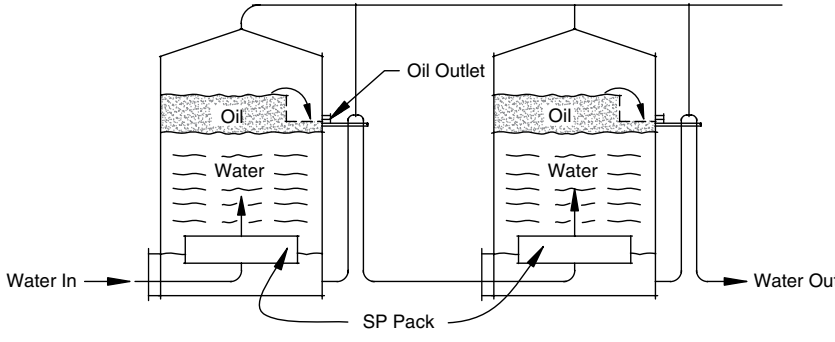


Figure 9-32. SP Packs installed in a series of staged tanks.

in the system, all the large oil droplets would be removed in the first skimmer and the second skimmer would remove little oil. Any number of stages in series may be used in the system.

SP Packs can also be used as retrofit components to improve the performance of existing water treating systems. Deck drainage may also be routed through an SP Pack prior to the drain sump or disposal pile.

SP Pack systems may be economical onshore where space is available for large skim tanks. Offshore SP Packs may be used for small water rates, roughly 5,000 bwpd ($33\,m^3$/hr). If space is available offshore, larger flow-rate applications may prove economical (for example, if the facility is mounted on a barge, and, as with any new device, its selection for an application must be made carefully, with an understanding of the potential risks and potential benefits).

As shown in Figure 9-33, the SP Pack is placed inside any gravity settling device (skimmers, clarifiers, plate coalescers, etc.), and by growing a larger drop size distribution, the gravity settler is more efficient at removing oil, as shown in Figure 9-34.

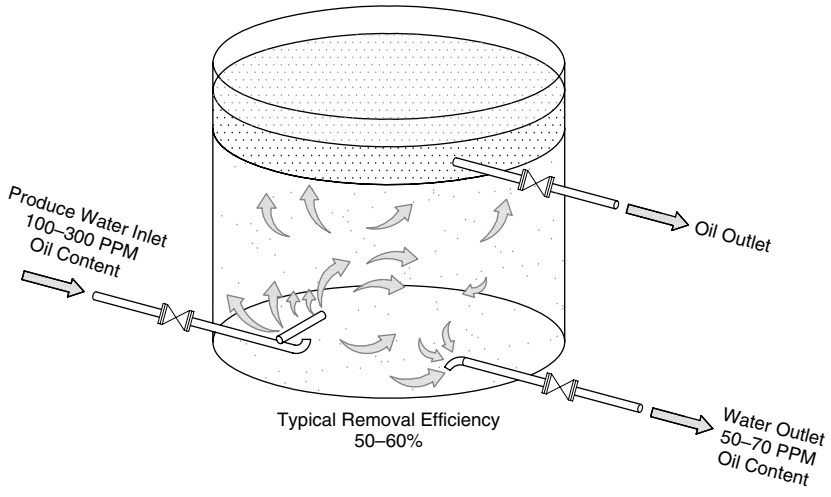


Figure 9-33. SP Pack installed in a clarifier skim tank.

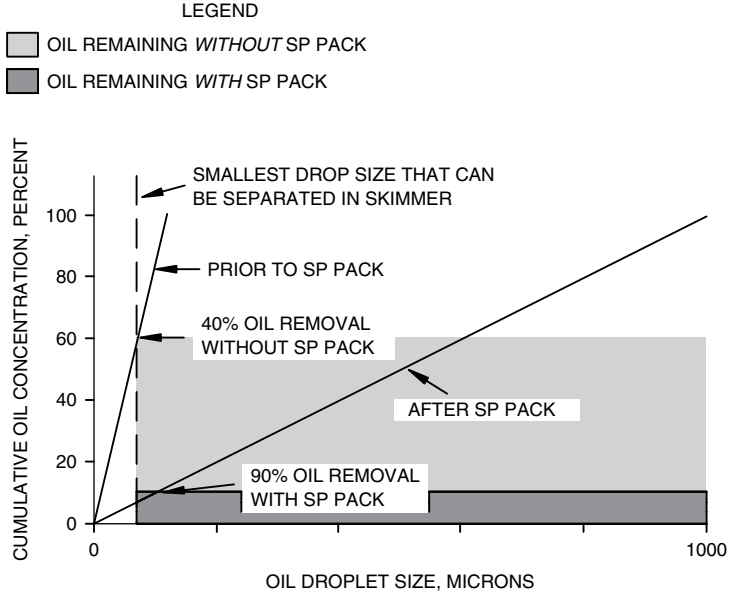


Figure 9-34. The SP Pack grows a larger droplet size distribution, thus allowing the skimmer to recover more oil.

Performance Considerations

The efficiency in each stage is given by

$$E = \frac{C_i - C_o}{C_i}, \tag{9-23}$$

where

C_i = inlet concentration,
C_o = outlet concentration.

Since the drop size distribution developed by the SP Pack can be conservatively estimated as a straight line,

$$E = 1 - \frac{d_m}{d_{\max}}, \tag{9-24}$$

where

d_m = drop size that can be treated in the stage,
$d_{\max}$ = maximum size drop created by the SP Pack
= 1,000 microns for standard SP Packs.

The overall efficiency of a series staged installation is then given by

$$E_t = 1 - (1 - E)^n, \tag{9-25}$$

where n is the number of stages.

Figures 9-35 and 9-36 illustrate the increased oil removal efficiency of an SP Pack installed in various sized tanks.

Flotation Units

Flotation units are the only commonly used water treating equipment that does not rely on gravity separation of the oil droplets from the water phase. Flotation units employ a process in which fine gas bubbles are generated and dispersed in water, where they attach themselves to oil droplets and/or solid particulates. The gas bubbles then rise to the vapor–liquid interface as oily foam, which is then skimmed from the water interface, recovered, and then recycled for further processing. The effective specific gravity of the oil–gas bubble combination is significantly lower than that of a standalone oil droplet. Accordingly to Stokes' law, the resulting rising velocity of the oil–gas bubble combination is greater than that of

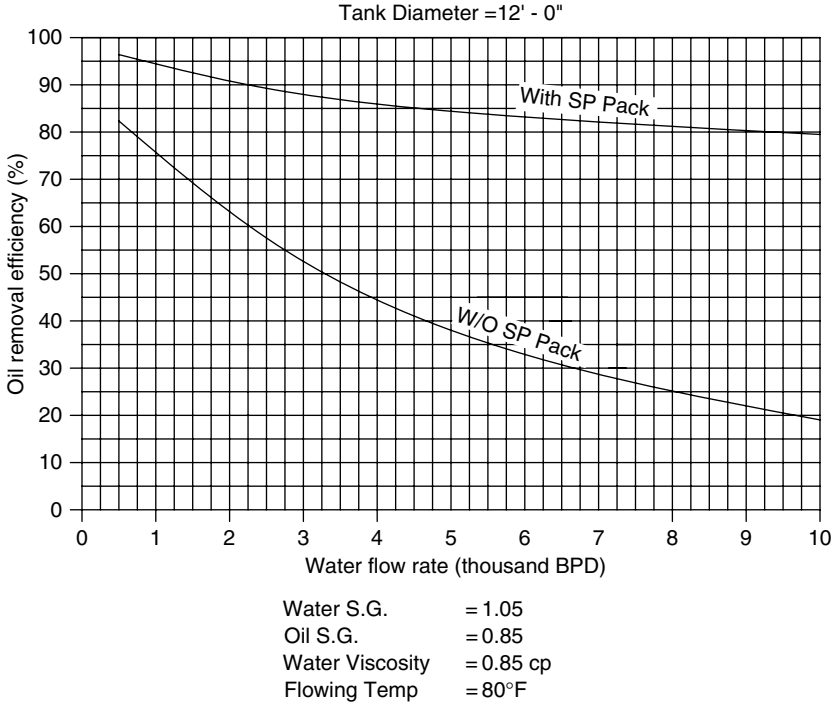


Figure 9-35. Improved oil removal efficiency of an SP Pack installed in a 12'-0" tank.

a standalone oil droplet acting to accelerate the oil–water separation process. Flotation aids such as coagulants, polyelectrolytes, or demulsifiers are added to improve performance.

Two distinct types of flotation units have been used; they are distinguished by the method employed in producing the small gas bubbles needed to contact the water. These are dissolved gas units and dispersed gas units.

Dissolved Gas Units

Dissolved gas designs take a portion of the treated water effluent and saturate the water with natural gas in a high-pressure "contactor" vessel. The higher the pressure, the more gas that can be dissolved in the water. Gas bubbles are formed by flashing dissolved gas into the produced water. As a result, the bubbles are much smaller (10 to100 microns) than for induced gas flotation (100 to 1,000 microns). On the other hand, the

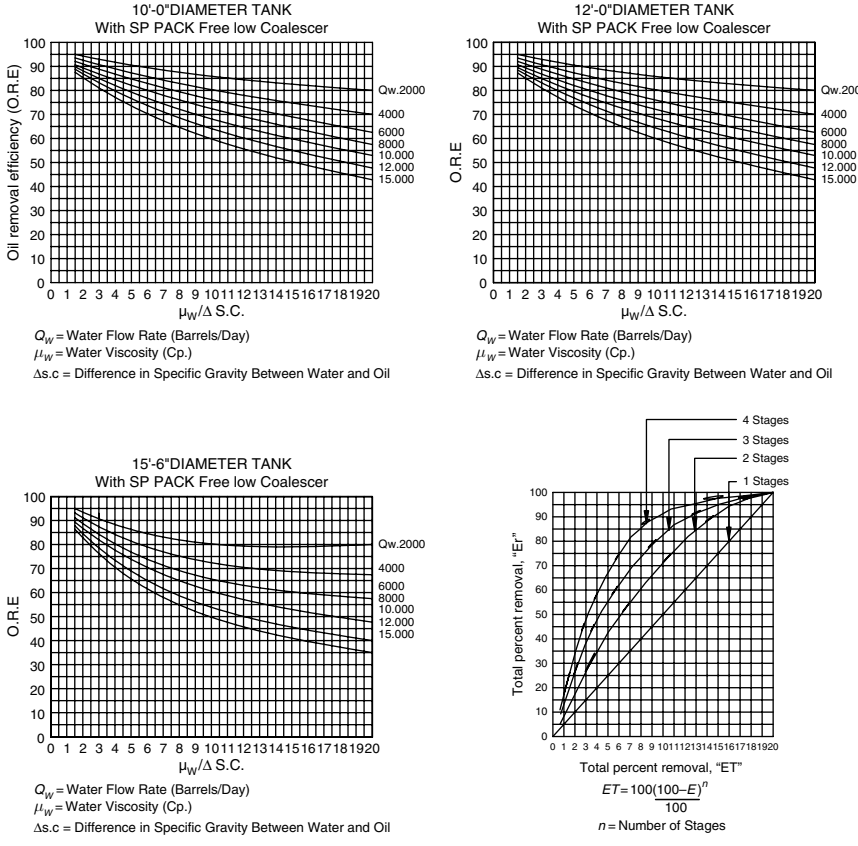


Figure 9-36. Oil removal efficiencies of various size tanks.

gas volumes are limited by the solubility of the gas in water and are much lower than for dispersed gas flotation.

Most units are designed for a 20- to 40-psig (140- to 280-kPa) contact pressure. Normally, 20 to 50% of the treated water is recirculated for contact with the gas. The gas saturated water is then injected into the flotation tank as shown in Figure 9-37. The dissolved gas breaks out of the oily water solution when the water pressure is flashed (reduced) to the low operating pressure of the gas flotation unit, in small-diameter bubbles that contact the oil droplets in the water and bring them to the surface in froth. This type of flotation unit typically has not worked well in the oil field.

Dissolved gas units have been used successfully in refinery operations where air can be used as the gas, where large areas are available for the equipment, and where the water to be treated is, for the most part,

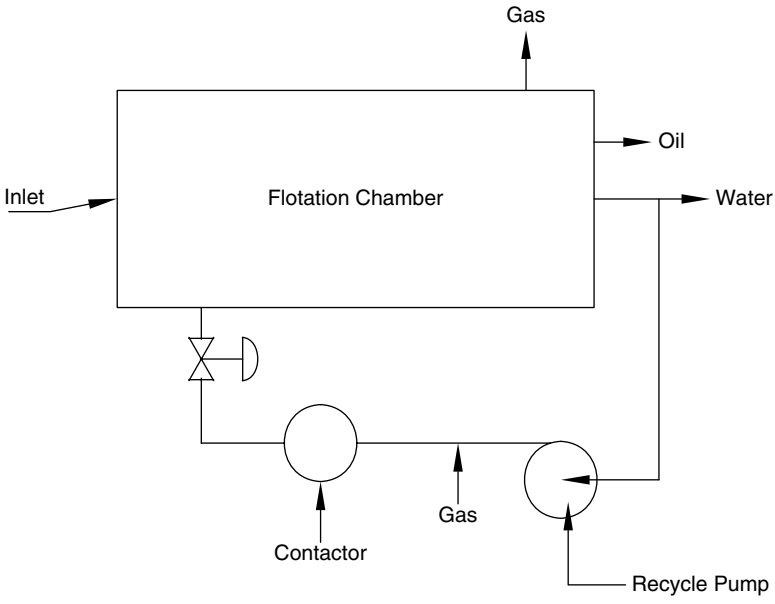


Figure 9-37. Schematic of a dissolved gas flotation process system.

oxygenated fresh water. In treating produced water, it is desirable to use natural gas to exclude oxygen, to avoid creating an explosive mixture, and to minimize corrosion and bacteria growth. This requires the venting of the gas or installation of a vapor recovery unit. In addition, the high dissolved solids content of produced water has created scale problems in these units. Field experiences with dissolved natural gas units in production operations have not been as successful as experience with dispersed gas units.

Design parameters are recommended by the individual manufacturers but normally range from 0.2 to 0.5 scf/barrel (0.036 to 0.89 Sm^3/m^3) of water to be treated and flow rates of treated plus recycled water of between 2 and 4 gpm/ft^2 (4.8 and 9.8 m^3/m^3). Retention times of 10 to 40 min and depths of between 6 and 9 ft (1.8 to 2.7 m) are specified.

Dissolved gas units are common in chemical plant operations, but, for the following reasons, they are seldom used in producing operations:

- They are larger than dispersed gas units and they weigh more, so they have limited application offshore.
- Many production facilities do not have vapor recovery units and, thus, the gas is not recycled.
- Produced water has a greater tendency to cause scale in the bubble-forming device than the freshwater that is normally found in plants.

Dispersed Gas Units

In dispersed gas units, gas bubbles are dispersed in the total stream either by the use of a hydraulic inductor device or by a vortex set up by mechanical rotors. There are many different proprietary designs of dispersed gas units. All require a means to generate gas bubbles of favorable size and distribution into the flow stream, a two-phase mixing region that causes a collision to occur between the gas bubbles and the oil droplets, a flotation or separation region that allows the gas bubbles to rise to the surface, and a means to skim the oily froth from the surface. Figure 9-38 shows the regions in which the above four processes occur. These processes and the regions in which they occur are as follows:

- Gas Circulation Path (A) and Fluid Circulation Path (B) (Bubble generation),
- Two-Phase Mixing Region (1) (Attachment of oil droplets to the bubbles),
- Flotation (Separation) Region (2) (Rise of the bubbles to the surface applying Stokes' law),
- Skimming Region (3) (Bubble collapse and oil skimming).

Gas bubble/oil droplet attachment can be enhanced with the use of polyelectrolyte chemicals. These flotation aid chemicals can also be used to cause bubble/solid attachments, and thus flotation units can be used

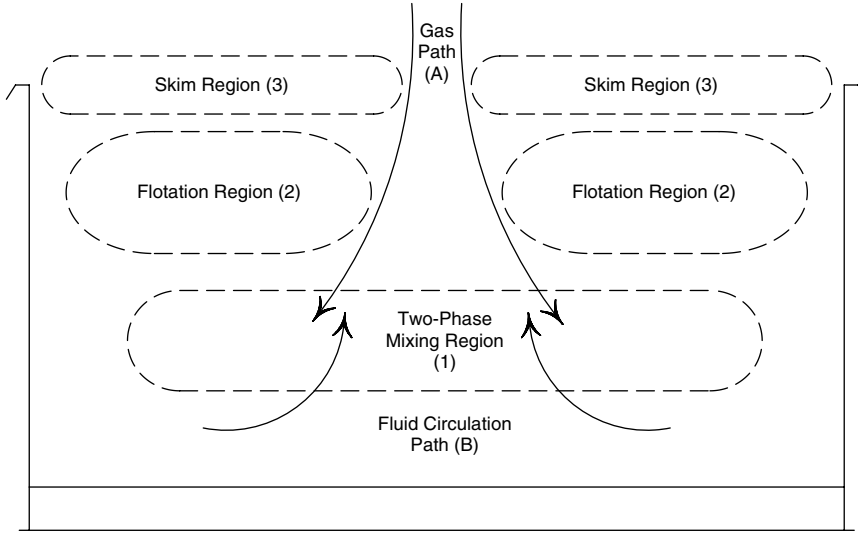


Figure 9-38. Dispersed gas flotation cell mechanics.

to remove solids as well. These chemicals are typically added to the water to yield a chemical concentration level between 1 to 10 ppm in the feed water. These chemicals are classified as surfactants, which tend to migrate to the bubble surface where they enhance the intermolecular forces between the bubbles and the oil droplets. In addition, if the oil has emulsifying tendencies, de-emulsifiers may also have to be added, in the 20- to 50-ppm range. Chemical treatment programs are highly location specific, and an effective treatment for one oil–water system may be ineffective for another. On-site bench-scale flotation chemical screening test using a pilot flotation device should be carried out for each application. Equipment manufacturers and chemical suppliers are generally equipped to perform this screening.

For the oil droplets to become attached to the bubbles, the bubbles and droplets must come into intimate contact. This contact is promoted by a highly turbulent region, generally located near the bubble generators. Studies have shown that attachment is enhanced by small gas bubbles, large oil droplets, and high bubble concentration.

To operate efficiently, the unit must generate a large number of small gas bubbles. Tests indicate that bubble size decreases with increasing salinity. At salinities above 3%, bubble size appears to remain constant, but oil recovery often continues to improve. Most oil-field waters contain sufficient dissolved solids to create favorable flotation bubble sizes. The low water salinity associated with gas condensates may make the application of gas flotation to gas condensate fields more difficult than for oil fields. Some steam-flood produced waters contain 2,000 to 5,000 ppm of dissolved salts and would tend to generate large, less effective bubbles.

Figure 9-39 shows the effect of gas bubble size on the oil droplet capture rate. The smaller the bubble size, the greater the chance of capture will be. Typical mean bubble sizes range between 50 to 60 microns.

Oil removal is dependent to some extent on oil droplet size. Flotation has very little effect on oil droplets that are smaller in diameter than 2 to 5 microns. Thus, it is important to avoid subjecting the influent to large shear forces (e.g., level control valves) immediately upstream of the unit. It is best to separate control devices from the unit by long lengths of piping (at least 300 diameters) to allow pipe coalescence to increase droplet diameter before flotation is attempted. Above 10 to 20 microns, the size of the oil droplet does not appear to affect oil recovery efficiency, and thus elaborate inlet coalescing devices are not needed.

High gas bubble concentration (fraction of the gas–water mixture that is vapor) increases the oil recovery. Field tests demonstrate that oil removal improves as the cumulative gas–water ratio increases. Table 9-3 shows the effects of installing multiple cells in series.

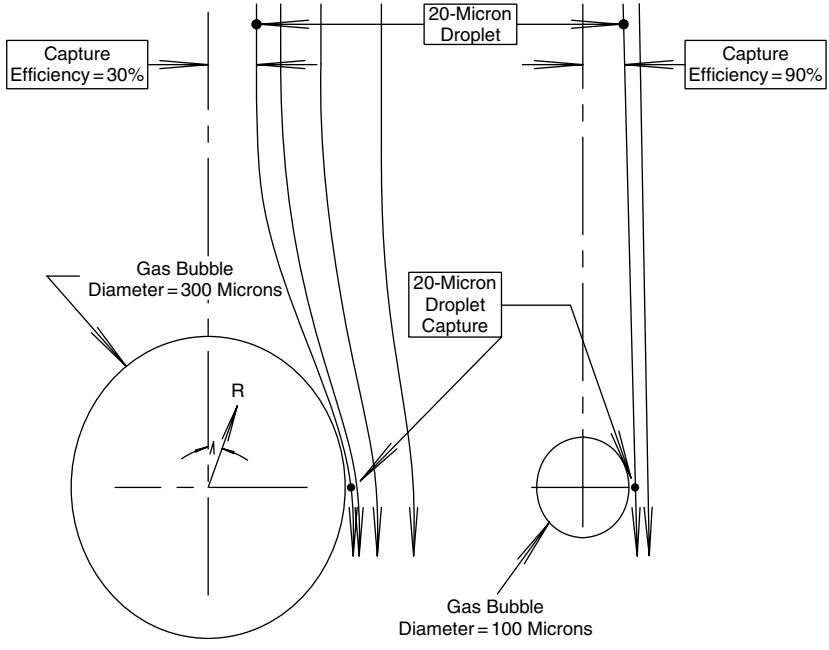


Figure 9-39. Effect of gas bubble size on oil droplet rate.

Table 9-3
Effects of Increased Gas Concentration on Oil Recovery

Water Locations in Machine	Cumulative Gas–Water Ratio, ft^3/bbl	PPM Oil in Treated Water
Inlet water	0	225
Cell no. 1 effluent	9	96
Cell no. 2 effluent	18	50
Cell no. 3 effluent	27	20
Cell no. 4 effluent	34	14
Discharge cell effluent	35.0	14

The rise and separation of oily bubbles from water require a relatively quiescent zone so that bubbles are not remixed into the bulk fluid. The rising velocity of the bubbles must exceed both turbulent velocities and any net downward bulk velocity.

The oily bubbles rise to the surface as an oily foam, which is then skimmed from the surface of the bulk water phase. This skimming process acts to

collapse the foam, which further concentrates the oily phase. Skimming is usually achieved by a combination of weirs and skim paddles that move the oily foam to the edge of the cell and over the weir. The weir height relative to the position and speed of the skim paddles must be adjusted to prevent both excessive foam build-up on the bulk water surface and excessive water carryover into the oil skim bucket located below the weir.

Hydraulic Induced Units

Hydraulic induced flotation units induce gas bubbles by gas aspiration into the low-pressure zone of a venturi tube. Figure 9-40 shows a schematic cross section of a hydraulic induced flotation unit. Clean water from the effluent is pumped to a recirculation header (E) that feeds a series of Venturi educators (B). Water flowing through the eductors sucks gas from the vapor space (A) that is released at the nozzle (G) as a jet of small bubbles. The bubbles rise, causing flotation in the chamber (C), forming a froth (D) that is skimmed with a mechanical device at (F).

Hydraulic induced units are available with one, three, or four cells. Figure 9-41 shows the flow path through a three-cell unit. These devices use less power and less gas than mechanical rotor units. Gas–water ratios less than $10 ft^2/bbl$ at the design flow rate are used. The volume of gas dispersed in the water is not adjustable, so throughputs less than design result in higher gas–water ratios.

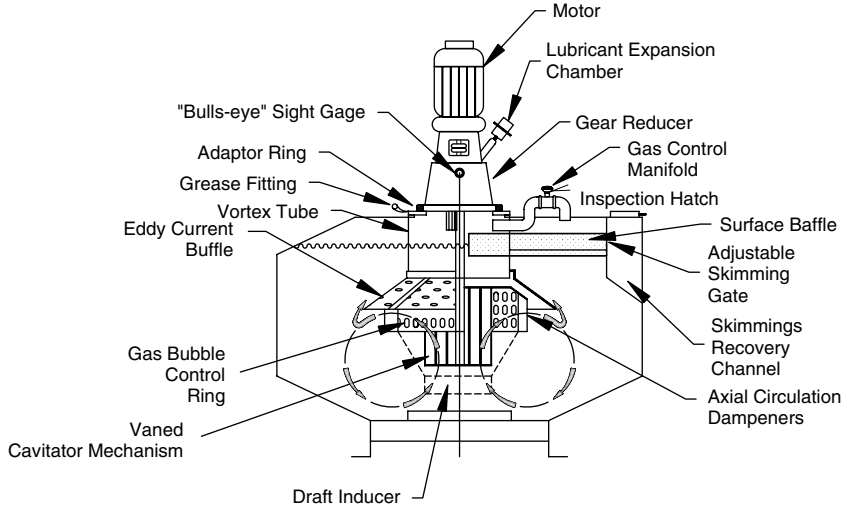


Figure 9-40. Schematic of a hydraulic induced gas flotation unit.

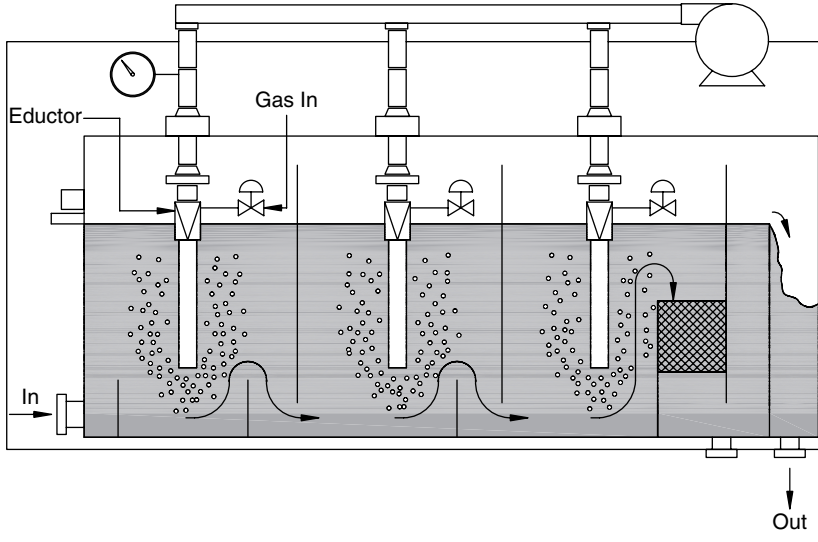


Figure 9-41. Schematic showing the flow path through a hydraulic induced flotation unit.

Hydraulic induced units are less complex than the mechanical induced units. The required water recycle rate to drive the eductor varies with both the design capacity of the unit and between different manufacturers, but is generally around 50%. Eductor design is proprietary and varies considerably, in both hydraulic design and mechanical placement between manufacturers. Figure 9-42 is a sketch of an eductor. Control of bubble size and distribution is much more difficult than for mechanical units. Stage efficiencies for hydraulic induced units have a tendency to be lower than those of mechanical units.

Mechanical Induced Units

Mechanical induced flotation units induce gas bubbles into the system by entrainment of gas in a vortex generated by a stirred paddle. Figure 9-43 shows a cross section of a dispersed gas flotation cell that utilizes a mechanical rotor. The rotor creates a vortex and vacuum within the vortex tube. Shrouds assure that the gas in the vortex mixes with and is entrained in the water. The rotor and draft inducer causes the water to flow as indicated by the arrows in this plane while also creating a swirling motion. A baffle at the top directs the froth to a skimming tray as a result of this swirling motion.

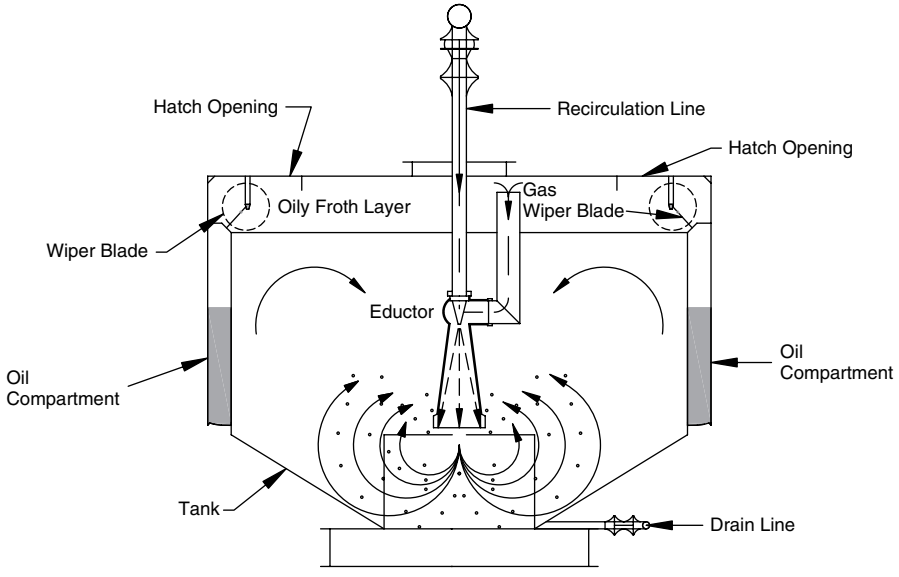


Figure 9-42. Cross section of a hydraulic inductor.

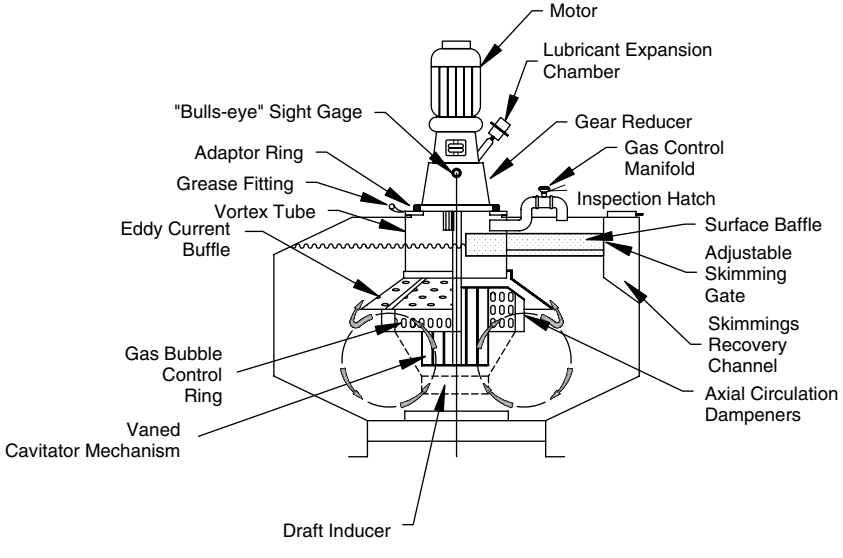


Figure 9-43. Cross section of a mechanical induced dispersed gas flotation unit.

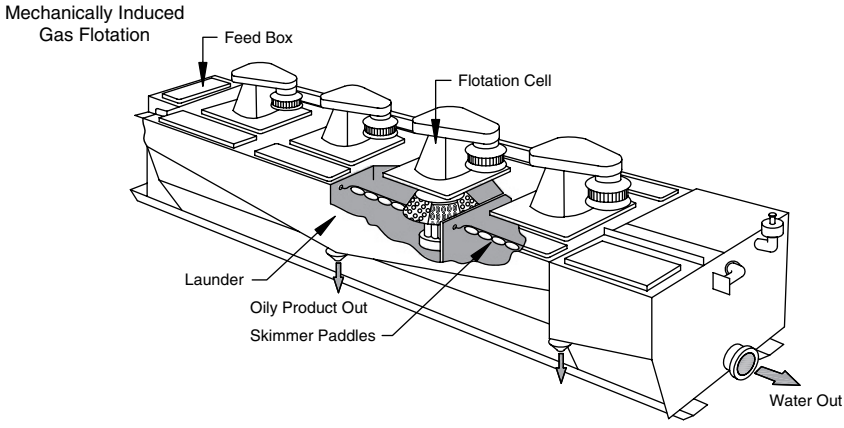


Figure 9-44. Cross section of a four-cell mechanical induced flotation unit.

Most mechanical induced units contain three or four cells. Figure 9-44 illustrates a four-cell unit. Bulk water moves in series from one cell to the other by underflow baffles. Each cell contains a motor-driven paddle and associated bubble generation and distribution hardware. Field tests have indicated that the high intensity of mixing in each cell creates the effect of plug flow of the bulk water from one cell to the next. That is, there is virtually no short-circuiting or breakthrough of a part of the inlet flow to the outlet weir box.

The mechanical complexity makes mechanical induced flotation units the most maintenance-intensive of all gas flotation configurations. As a result of the need for motor shaft seals on penetrations to the cell, mechanical induced flotation units have traditionally operated very near atmospheric pressure.

Each of the above processes assumes the inlet water to the flotation unit is already at atmospheric pressure. When the upstream primary separator operates at elevated pressures, substantial gas saturation of the produced water may already exist. In these cases, flashing to atmospheric pressure may be sufficient to generate bubbles without added gas saturation.

Other Configurations

The combination of dissolved gas flotation and CPIs has been attempted recently, with injection of a recycled portion of the effluent from the CPI into the influent stream. Little field data are available on this design; therefore, it is not recommended at this time because, when the dissolved

air breaks out of solution, turbulence that can adversely affect the action of the CPI is created.

There are many types of configurations with complex flow patterns and number of cells ranging from one to five. Some designs have multiple eductors per cell. Some have recirculation rates through the eductors that may be several multiples of the bulk water throughput rate. The concept described above should give the engineer some guidance to add in understanding the pros and cons of each manufacture's proprietary designs.

One new design when shows some promise uses a "Sparger." A Sparger introduces gas from an external high pressure source similar to that of an aerator in an aquarium. Porous media nozzles are used to form very small bubbles, the size of which is controlled somewhat by the pore size in the media. For the most effective attachment of oil droplets to these sparged bubbles, the bubble size should be approximately the same as the smallest oil droplets to be removed. Due to the small bubbles that are used in sparging, long fluid residence times on the order of 10 minutes are required. Sparging generates some mechanical complexity due to the needs for a separate pressurized gas supply and for numerous porous media nozzles that may be prone to plugging. On the other hand, using multiple spargers could generate smaller bubbles, greater flow rates and better gas mixing with the produced water than other designs. The detriment is that the porous media could plug with time leading to high maintenance costs and poor availability.

Sizing Dispersed Gas Units

It can be shown mathematically that an efficient design must have a high gas induction rate, a small-diameter induced gas bubble, and relatively large mixing zone. The design of the nozzle or rotor, and of the internal baffles, is thus critical to the unit's efficiency. The nozzles, rotors, and baffles for these units are patented designs.

As measured in actual field tests, these units operate on a constant percent removal basis. Within normal ranges their oil removal efficiency is independent of inlet concentration or oil droplet diameter.

Field tests indicate that a properly designed unit with a suitable chemical treatment program should have oil removal efficiency between 40–55% per active cell and an overall efficiency of about 90%. An excellently designed system might exhibit an efficiency as high as 95%, while a poorly designed, poorly operated unit, or difficult oil–water chemistry could easily degrade performance to as low as 80%. Equation (9-25) verifies the above efficiencies. For example, Eq. (9-25) shows that a three-cell unit can be expected to have an overall efficiency of 87% while

a four-cell unit can be expected to have an overall efficiency of 94%. The unit's actual efficiency will depend on many factors that cannot be controlled or predicted in laboratory or field tests.

Each cell is designed for approximately 1 minute's retention time to allow the gas bubbles to break free of the liquid and form the froth at the surface. Each manufacturer gives the dimensions of its standard units and the maximum flow rate based on this criteria.

Graphs of the dispersed oil concentrations in the effluent water versus dispersed oil concentrations in the inlet feed stream are shown in Figure 9-45 for representative efficiencies achievable in a typical four-cell dispersed gas flotation unit. For inlet concentrations less than about 200 mg/l, the oil removal efficiency declines slightly. At low oil inlet concentrations, it becomes more difficult for the flotation unit to achieve intimate contact and interaction between the gas bubbles and dispersed oil droplets. As a result, Figure 9-45 may understate the effluent concentrations for influent oil concentrations less than 200 mg/l.

Depending on the oil concentration in the influent and the quality requirements in the effluent, flotation may or may not serve as a stand-alone process in produced water treating. Water qualities coming from a primary production separator tend to be in the 500- to 2,000-ppm range. As can be seen from Figure 9-45, a well-designed gas flotation unit would be limited to an effluent quality in the 30- to 80-ppm range when used as the sole water treating unit downstream of the primary

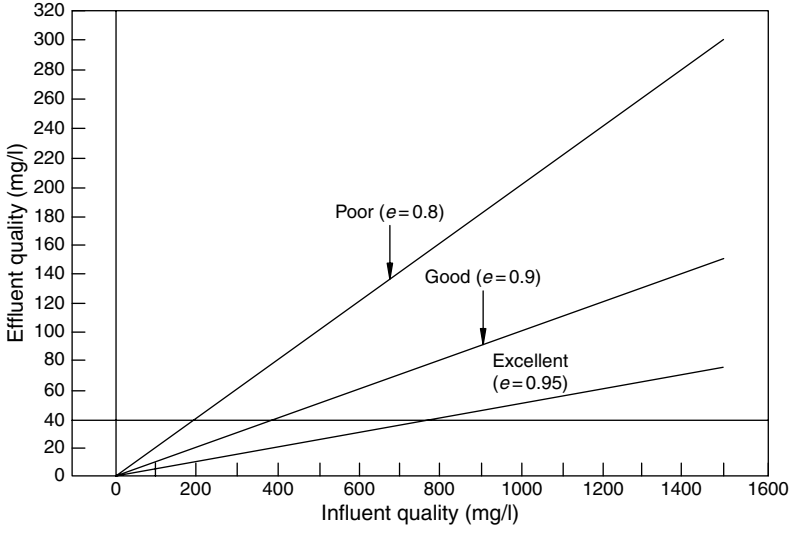


Figure 9-45. Effluent quality versus influent quality.

separator. Since the separation efficiency is reasonably independent of the influent oil concentration, upsets in the primary separator operation could make a significant difference in the gas flotation effluent quality. In order to meet the effluent quality established by the authorities having jurisdiction, normally in the 30- to 50-ppm range, it is usually necessary to combine a gas flotation unit with some unit between the flotation unit and the primary separator, such as a corrugated plate interceptor (CPI).

Gas flotation units require a customized chemical treatment program to achieve adequate results. If produced water originates from several sources in variable quantities, the development of a chemical treatment program may be difficult.

Skimmed oily water volumes are typically 2 to 5% of the machine's rated capacity and can be as high as 10% when there is a surge of water flow into the unit. Since skimmed fluid volume is a function of weir length exposure over time, operation of the unit at less than design capacity increases the water residence time but does not decrease the skimmed fluid volumes.

Gas flotation units normally include multiple cells in series. If the mechanical or hydraulic aeration unit in any cell fails, the water merely flows through that cell with little or no oil separation. As a result, mechanical failure in a single cell causes a degradation in performance. For example, a four-cell unit with a mechanical failure in one cell becomes a three-cell unit capable of separating 87.5% of the dispersed oil in the feed stream [i.e., $1-(1-0.5)^3=0.875$].

Gas flotation equipment is typically purchased as a prefabricated unit selected from a vendor's list of standard size units, rather than being custom specified and designed for each specific application. As a result the bulk of the design is performed by the vendor, and relatively little design opportunity exists for the user. A tabulation of representative sizes, weights, horsepower, and residence times for both hydraulically and mechanically induced flotation units is illustrated in Table 9-4.

Performance Considerations

Several factors that should be taken into account to maintain performance include

- The cells must be properly leveled on initial installation, and this level condition must be maintained. Since the skimming depends on proper operation of a weir, small out-of-level conditions will prevent proper skimming of oil. Movement of the flotation cells can also set up liquid surges that can prevent proper skimming.

- Liquid levels must be carefully controlled to permit proper weir operation. Level control system parameters must be carefully set to prevent liquid level oscillations. Throttling valves are preferred over snap acting valves on both the water inlet and outlet. The flow disturbances caused by the rapid opening and closing of the snap acting valves may generate level disturbances. Gravity flow of the inlet feed stream to the gas flotation unit is preferred over pumping. The high shearing action created in a pump will break up the larger oil droplets into smaller droplets, making separation more difficult.
- Many induced flotation units, particularly mechanical flotation units, operate at pressures within a few ounces of atmospheric pressure. The walls are thin and have numerous penetrations for motor shafts and observation hatches. As a result of the simplicity of design, air can easily enter the units around the paddle or if observation hatches are left open. Oxygen in the water treating system increases the corrosion rate in the unit as well as all downstream carbon steel equipment and can cause the formation of a reddish precipitate resulting from oxidation of dissolved iron in the treated water. To avoid corrosion

Table 9-4
Characteristics of Representative Gas Flotation Units

IGF Type	Company	Brand	Model	Power HP	Length ft (ss)	Width ft (OD)
Hydraulic IGF	Wemco	ISF	30X	15	20.5	4.5
	ESI	Tridair	75X	30	29	5.5
	Monosep	Verisep	160X	50	33	7.5
			DL-100	7.5	15.5	5
			DL-200	20	20.5	9
			DL-500	50	23	14
			3MV	3	15	3.5
			10MV	8	15	6
			10MV	25	31.3	9.5
Mechanical IGF	Serck	Depurator	SB-020	6	14	2.5
	Baker		SB-100	8	21	4.5
	Petrolite		SB-500	50	37	7
	Wemco		GFS-5	12	22	3.5
			GFS-10	20	27	5
			GFS-45	40	37	6
			36	12	14.4	3.5
			56	20.5	26.5	5.7
			84X	60.5	34.1	8.9
			144X	120.5	64.2	12

Table 9-4
Characteristics of Representative Gas Flotation Units—cont'd

IGF Type	Brand	Model	Fluid Volume		Nominal Flow		Residence min
			ft^3	gal	bwpd	ft^3min	
	ISF	30X	326	2439	10,200	39.8	8.2
		75X	689	5154	25,700	100.2	6.9
		160X	1457	10,900	54,800	213.7	6.8
	Tridair	DL-100	304	2274	10,000	39.0	7.8
Hydraulic IGF		DL-200	433	3239	20,000	78.0	5.6
		DL-500	1602	11,985	50,000	195.0	8.2
	Varisep	3MV	144	1077	3,000	11.7	12.3
		10MV	424	3172	10, 000	39.0	10.9
		50MV	1798	13,451	50,000	195.0	9.2
		SB-020	32	240	2,000	7.8	4.1
		SB-100	209	1563	10,000	39.0	5.4
		SB-500	898	6714	50,000	195.0	4.6
Mechanical IGF		GFS-5	308	2304	5,000	19.5	15.8
		GFS-10	540	4040	10,000	39.0	13.8
		GFS-45	1332	9965	45,000	175.5	7.6
	Depurator	36	81	606	1,720	6.7	12.1
		56	346	2588	10,300	40.2	8.6
		84X	1390	10,399	50,000	195.0	7.1
		144X	5832	43,629	1,714,000	668.3	8.7

and the precipitate, care should be taken to avoid oxygen ingress. Hatches should be left closed as much as possible, and the integrity of shaft seals should be maintained.

- Proper chemical treatment is essential to the operation of gas flotation. Care must be taken to ensure that the chemical injection facilities are operating as expected and that proper dosage is administered and mixed, both to promote sufficient separation and to prevent excessive chemical use. The customized chemical treatments involving polyelectrolytes, de-emulsifiers, scale inhibitors, and corrosion inhibitors may result in chemical incompatibilities, either between chemicals or between chemical treatments and flotation cell materials. These incompatibilities may be compounded by propagation through the treatment facility of any chemicals added upstream of the water treating system. Units should be monitored

for any unexpected sludge or precipitates or for unexpectedly high corrosion or elastomer deterioration rates.
- Stripping of acid gases (H_2S, CO_2) in an induced gas flotation unit can cause a pH increase, which may result in scaling. If this is the case, minimizing the gas flow can help reduce this problem.

Field tests indicate the following performance findings:

- Induced gas flotation units remove almost 100% of oil droplets 10–20 microns and above and have some effect on oil droplets in the 2- to 5-micron range.
- Oil removal efficiency depends of chosing the correct chemical and chemical dosage (refer to Figure 9-46).
- Mechanical units tend to be more efficient than hydraulic units with influent concentrations, from 50 to 150 mg/l, while hydraulic units are more efficient above 500 mg/l. Both units are equally efficient between 150 to 500 mg/l.
- The performances of all induced gas flotation units are relatively insensitive to flow-rate variations between 70 to 125% of the design flow rate.
- Mechanically induced units appear to tolerate greater throughput rate fluctuations than hydraulic induced units.
- The separation efficiency of all units depends on the influent concentration.
- Changing the water temperature from ambient to 140°F (60°C) results in a slight improvement in oil recovery at normal pH values.

Gas flotation units should be used when

- The inlet oil concentrations are not too high (250–500 mg/l).
- The effluent discharge requirements are not too severe (25–50 mg/l).
- Chemical companies are available to formulate an appropriate chemical treatment program.
- Power costs are low or moderate.

Gas flotation units should not be used when

- Equipment size and weight are prime considerations.
- The unit is subject to accelerations and tilting such as floating production facilities.
- The water stream to be treated is comprised of multiple water sources having significantly varying water chemistry and dispersed oil characteristics.
- Service support from water treating chemical vendors is limited.
- Very low effluent oil concentrations are required.
- Power costs are high.

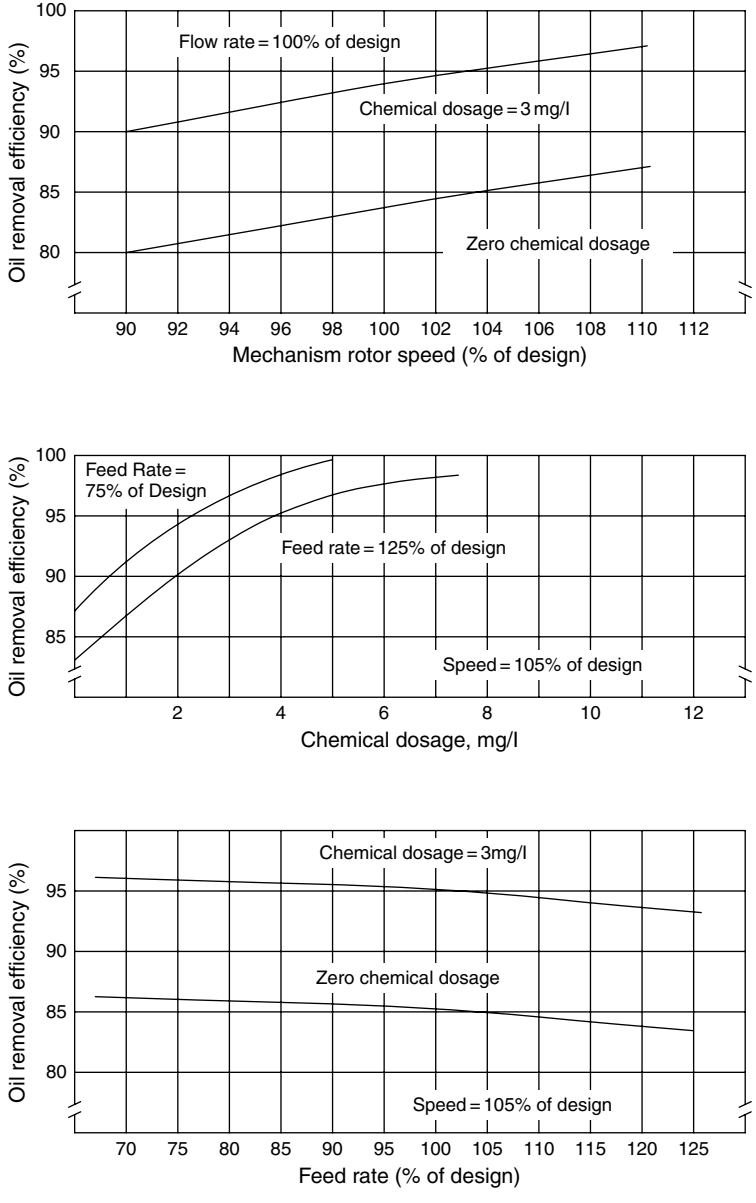


Figure 9-46. Oil removal efficiencies of mechanical induced flotation units.

Hydrocyclones

General Considerations

Since the early 1980s, hydrocyclones have been used in produced water treatment to de-oil the water prior to discharge. Hydrocyclones used to de-oil the water are referred to as "liquid-liquid de-oiling" hydrocyclones. Liquid-liquid hydrocyclones are further classified as static or dynamic hydrocyclones.

Operating Principles

Hydrocyclones, sometimes called "enhanced gravity separators," use centrifugal force to remove oil droplets from oily water. As shown in Figure 9-47, a de-oiling static hydrocyclone typically consists of liner(s) contained within a pressure retaining outer vessel or shell. The liner consists of the following four sections: a cylindrical swirl chamber, a concentric reducing section, a fine tapered section, and a cylindrical tail section. Figure 9-48 shows a typical multiliner vessel.

Oily water enters the cylindrical swirl chamber through a tangential inlet nozzle (Figure 9-49), creating a high-velocity vortex with a reverse-flowing central core. The fluid accelerates as it flows through the

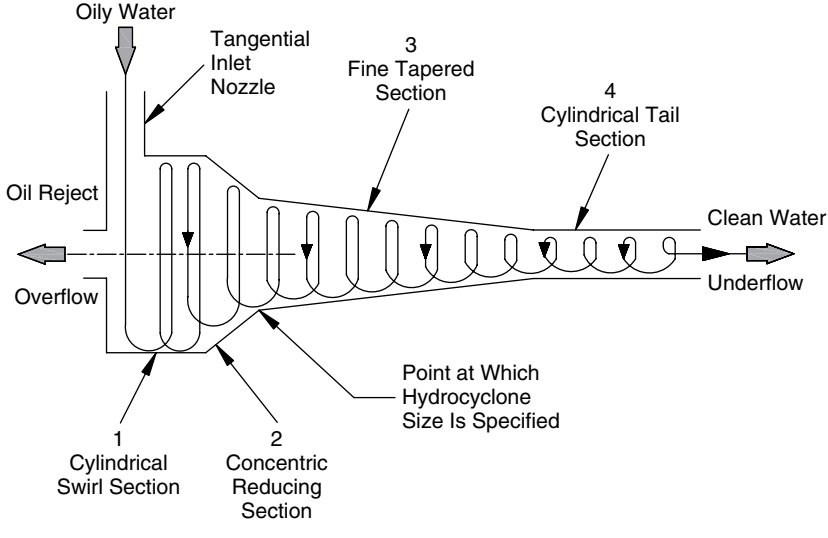


Figure 9-47. Liquid-liquid static hydrocyclone separation liner.

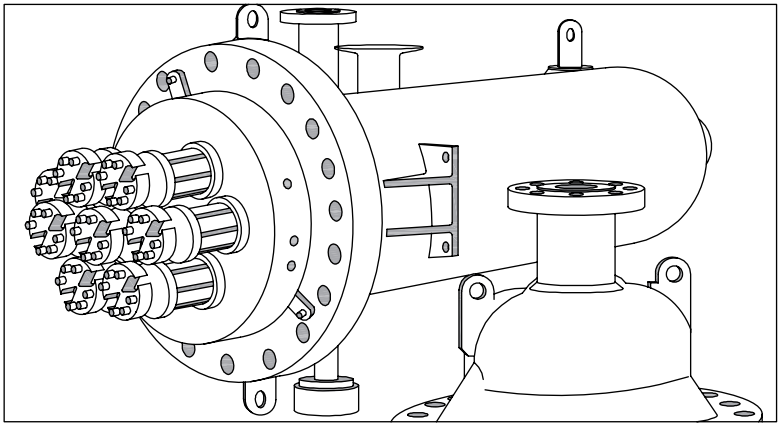


Figure 9-48. Multiliner hydrocyclone vessel.

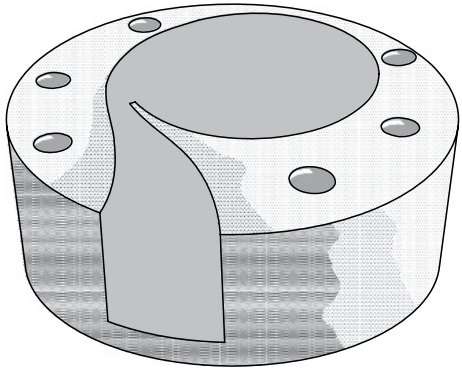


Figure 9-49. Tangental inlet nozzle.

concentric reducing section and the fine tapered section. The fluid then continues at a constant rate through the cylindrical tail section. Larger oil droplets are separated from the fluid in the fine tapered section, while smaller droplets are removed in the tail section. Centripetal forces cause the lighter-density droplets to move toward the low-pressure central core, where axial reverse flow occurs. The oil is removed through a small-diameter port located in the head of the hydrocyclone and is known as the "reject stream" or "overflow." Clean water is removed through the downstream outlet and is know as the "underflow."

The separation mechanism inside a hydrocyclone is governed by Stokes' law. However, in a hydrocyclone, the gravitational force is orders

of magnitude (between 1,000–2,000 gs) higher than that available in conventional gravity-based separation equipment. A high-velocity vortex with a reverse-flowing central core is set up by entry through a specially designed tangential inlet(s) (Figure 9-49). The fluid is accelerated (thereby offsetting the frictional losses) through the concentric reducing and fine tapered sections of the cyclone—where the bulk of separation occurs—into the cylindrical tail section where smaller, slower-moving droplets are recovered. The size of a hydrocyclone (for example, 35 mm or 60 mm) refers to the diameter at transition between the concentric reducing and fine tapered section of the cyclone (see Figure 9-47).

A hydrocyclone can be oriented either horizontally or vertically although horizontal orientation is more common. The horizontal orientation requires more plan area (deck space) but is more convenient for maintenance (about a 42-in. clearance is required to remove the liners from the vessel). The energy required to achieve separation is provided by the differential pressure across the cyclone. A minimum of 100 psi is generally needed. Higher pressures are preferable when available. The reject stream is on the order of 1 to 3 volume percent of the inlet. Only about 10% (by volume) of the reject stream is oil, the rest being water. The reject stream may be directed back to the separator through a low-shear progressive cavity pump. It should be noted, however, that in certain field applications oil-field chemicals have caused swelling of the rubber stator of these pumps, leading to poor performance. In such situations, a low-speed single-stage centrifugal pump with an open impeller may be suitable.

Many hydrocyclone installations typically include a de-gassing vessel downstream of the clean product water outlet. The vessel provides a short residence time serving essentially as a single gas flotation unit. The vessel also provides oil slug-catching volume in case of major upsets and additional residence time for emulsion-breaking chemicals.

Static Hydrocyclones

Static hydrocyclones require a minimum pressure of 100 psi to produce the required velocities. Manufacturers make designs that operate at lower pressures, but these models have not always been as efficient as those that operate at higher inlet pressures. If a minimum separator pressure of 100 psi is not available, a low-shear pump should be used (e.g., a progressive cavity pump) and sufficient pipe should be used between the pump and the hydrocyclone to allow pipe coalescence of the oil droplets. As is the case with flotation units, hydrocyclones do not appear to work well with oil droplets less than 10 to 20 microns in diameter.

The performance of a static hydrocyclone is chiefly influenced by the reject ratio and the pressure drop ratio. The reject ratio is defined as the ratio of the reject fluid rate (volume of oil and water discharged from the reject outlet) to the total inlet volume flow rate, expressed as a percentage. The reject orifice size is fixed (typically 2 mm), and the reject ratio is controlled by back pressure, directly proportional to the pressure drop ratio, on the reject outlet stream. The recommended reject ratio for hydrocyclones is 1 to 3% of feed flow, generally about 2%. Although a lower reject ratio (1%) is sufficient to give the optimum efficiency, 2% provides a safety margin to ensure that the efficiency is not affected by upset conditions, such as fluctuations in pressure drop across the unit or a slight surge in oil concentration, which could affect efficiency. To maintain efficiency at a lower reject ratio, the reject port diameter must be made smaller, which increases the probability of it becoming blocked. Operating above the optimum reject ratio does not impair oil removal efficiencies. The PDR refers to the ratio of the pressure difference between the inlet and reject outlets and the difference between the inlet and the water outlet. A PDR of between 1.4 and 2.0 is usually desired. Performance is also affected by inlet oil droplet size, concentration of inlet oil, differential specific gravity, and inlet temperature. Temperatures greater than 80°F result in better operation.

Although the performance of hydrocyclones varies from facility to facility (as with flotation units), an assumption of 90% oil removal is a reasonable number for design. Often the unit will perform better than this, but for design it would be unwise to assume this will happen. Performance cannot be predicted more accurately from laboratory or field testing because it is dependent on the actual shearing and coalescing that occur under field flow conditions and on impurities in the water, such as residual treating and corrosion chemicals and sand, scale, and corrosion products, which vary with time.

Hydrocyclones are excellent coalescing devices, and they actually function best as a primary treating device followed by a downstream skim vessel that can separate the 500- to 1,000-micron droplets that leave with the water effluent. A simplified P&ID for a hydrocyclone is shown in Figure 9-50.

Advantages of static hydrocyclones include that (1) they have no moving parts (thus, minimum maintenance and operator attention are required), (2) their compact design reduces weight and space requirements when compared to those of a flotation unit, (3) they are insensitive to motion (thus, they are suitable for floating facilities), (4) their modular design allows easy addition of capacity, and (5) they offer lower operating costs when compared to flotation units, if inlet pressure is available.

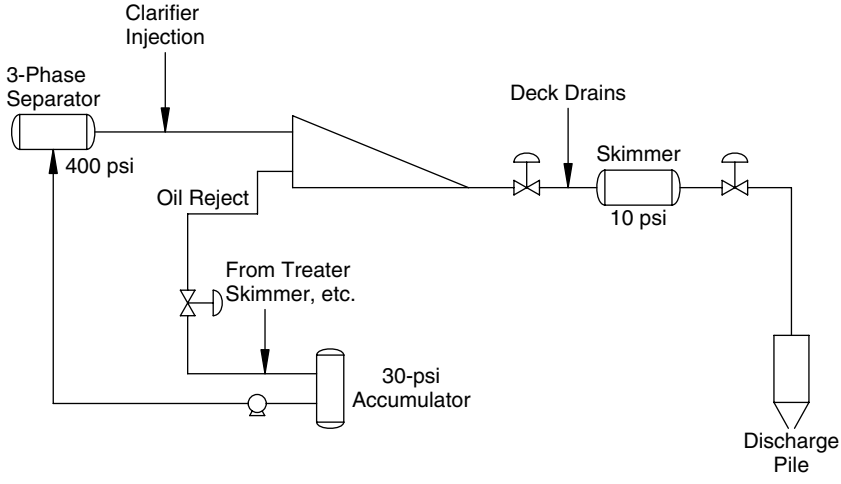


Figure 9-50. Simplified P&ID showing a hydrocyclone used as preliminary treating device.

Disadvantages include the need to install a pump if oil is available only at low pressure and the tendency of the reject port to plug with sand or scale. Sand in the produced water will cause erosion of the cones and increase operating costs.

Performance of hydrocyclones is also affected by the following parameters:

- **Oil drop size (at fixed concentrations).** Efficiency generally decreases as the oil droplet size is reduced. This is consistent with Stokes' law, where a smaller droplet will move less rapidly toward the hydrocyclone core. Droplets below a certain size (about 30 microns) are not captured by the hydrocyclone and, therefore, as the median feed oil droplet size decreases, more of the smaller droplets escape and the efficiency drops. Restrictions (valves, fittings, etc.) and pumps causing droplet shearing in the incoming flow should be avoided.
- **Differential specific gravity.** At a constant temperature, the hydrocyclone oil removal efficiency increases as the salinity increases and/or the crude specific gravity decreases. As the specific gravity difference between water and oil increases, a greater driving force for oil removal in the hydrocyclone occurs.
- **Inlet temperature.** The temperature of the produced water inlet stream determines the viscosity of the oil and water phases and the density difference between the two phases. As temperature increases, the water viscosity of water decreases slightly, while the density difference increases more substantially. This is because oil density

decreases at a faster rate than the water density. Temperatures greater than 80°F result in better operation.

- **Inlet flow rate.** The centrifugal force induced in the hydrocyclone is a function of the flow rate. At low flow rates, insufficient inlet velocity exists to establish a vortex and separation efficiency is low. Once the vortex is established, the efficiency increases slowly as a function of the flow rate to a point where the pressure at the core approaches atmospheric. Any further increase in the flow rate hinders oil flow from the reject outlet and causes efficiency to decline. In addition, a high flow rate can cause shearing of the droplets. This maximum flow rate is the "capacity" of the chamber. Flow rate is controlled by back pressure on the underflow outlet. The ratio of maximum to minimum flow rate, as determined by the lowest separation efficiency acceptable and the available pressure drop, is the "turndown ratio" for a given application.

Dynamic Hydrocyclones

The major difference between static and dynamic hydrocyclones is that in the dynamic hydrocyclone an external motor is used to rotate the outer shell of the hydrocyclone, whereas in a static hydrocyclone the outer shell is stationary and feed pressure supplies the energy to accomplish separation of oil from water (no external motor is required).

As shown in Figure 9-51, a dynamic hydrocyclone consists of a rotating cylinder, axial inlet and outlet, reject nozzle, and external motor. The rotation of the cylinder creates a "free vortex." The tangential speed is inversely proportional to the distance to the centerline of the cyclone. Since there is no complex geometry that requires a high pressure drop, dynamic units can operate at lower inlet pressures (approximately 50 psig) than static units. In addition, the effect of the reject ratio is not as important in dynamic units as it is in static units.

Dynamic hydrocyclones have found few applications relative to hydrocyclones because of poor cost–benefit ratio.

Selection Criteria and Application Guidelines

Hydrocyclones can be applied when

1. Median oil particle size is in excess of 30 microns.
2. Produced water feed pressure is at least 100 psig.

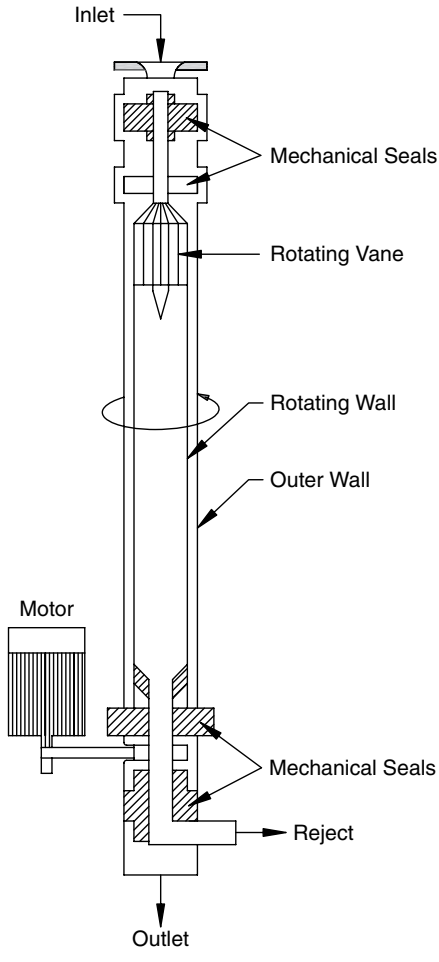


Figure 9-51. Liquid-liquid dynamic hydrocyclone separation.

3. Platform deck space is a critical consideration. Hydrocyclones have low size and weight requirements compared to other water treating equipment for the same capacity.
4. Platform motion is significant, such as tension leg platforms or floating production facilities, since the hydrocyclone is insensitive to motion. Other devices, such as flotation cells, are adversely impacted since platform motion makes accurate level control (using weirs or other control devices) difficult.
5. An appreciable quantity of solids is not present.
6. An appreciable amount of free gas is not present.
7. The flow rate and feed water oil concentration are fairly constant.

8. Low equipment maintenance is desired. Since a hydrocyclone has no moving parts, its maintenance requirements are fairly low.
9. Power constraints exist. Hydrocyclones do not require any outside energy supply, except for a low HP (about 5 HP) reject-recycle pump.

They are not applicable when

1. A tight emulsion exists, with a median oil droplet size less than 30 microns (manufacturer's claim that newer high-efficiency liners are capable of removing 20 microns).
2. The feed water pressure is less than 100 psig. A pump would be required to develop adequate pressure to use a hydrocyclone. However, pumps can cause the oil droplets to shear, making it more difficult for separation by a hydrocyclone.
3. The difference in specific gravity between the oil and water is relatively low, that is, heavy crude is being produced.
4. Considerable sand is entrained in the produced water. The sand could potentially plug the reject orifice and also cause erosion of the liner.

Sizing and Design

The performance of hydrocyclones is measured in terms of oil removal efficiency (E).

$$\text{product oil removal efficiency:} \quad \frac{E = [(C_i - C_o)(100)]}{C_i},$$

where

C_i = dispersed oil concentration in feed water,
C_o = dispersed oil concentration in effluent water.

Figure 9-52 shows generalized removal efficiency curves of a hydrocyclone. For a typical case (30°API oil and 1.05 SG water), the differential specific gravity is 0.17 and the removal efficiency would be 92% of 40-micron, 85% of 30-micron, and 68% of 20-micron oil droplets.

Figure 9-53 shows a typical control scheme for a hydrocyclone.

Disposal Piles

Disposal piles are large-diameter (24- to 48-in.) open-ended pipes attached to the platform and extending below the surface of the water.

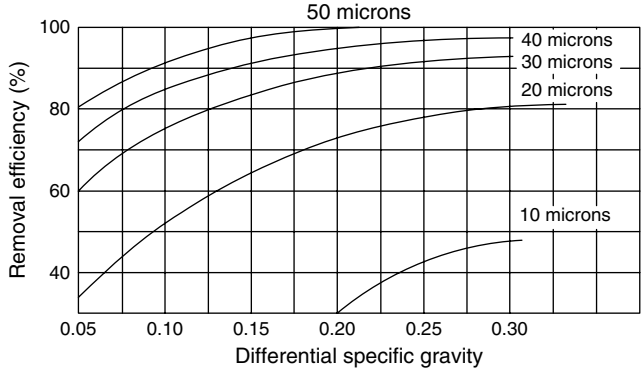


Figure 9-52. Generalized performance curves for a hydrocyclone.

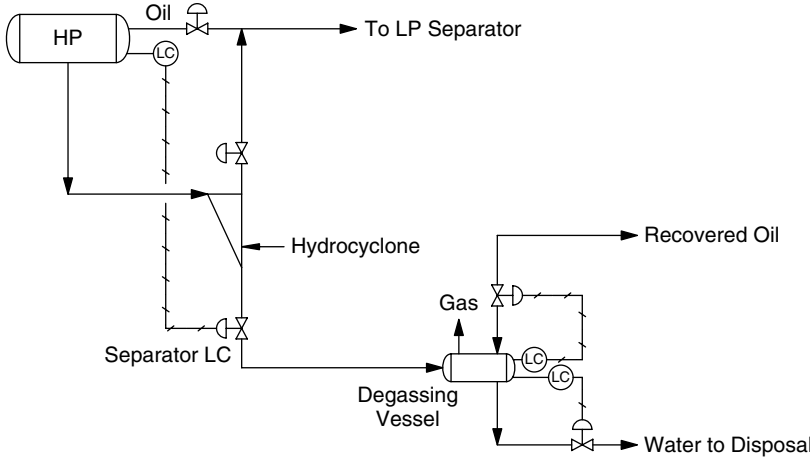


Figure 9-53. Typical control scheme for a hydrocyclone.

Their main uses are to (1) concentrate all platform discharges into one location, (2) provide a conduit protected from wave action so that discharges can be placed deep enough to prevent sheens from occurring during upset conditions, and (3) provide an alarm or shutdown point in the event of a failure that causes oil to flow overboard.

Most authorities having jurisdiction require all produced water to be treated (skimmer tank, coalesced, or flotation) prior to disposal in a disposal pile. In some locations, disposal piles are permitted to collect treated produced water, treated sand, liquids from drip pans and deck drains, and as a final trap for hydrocarbon liquids in the event of equipment upset.

Disposal piles are particularly useful for deck drainage disposal. This flow, which originates either from rainwater or wash-down water, typically contains oil droplets dispersed in an oxygen-laden freshwater or saltwater phase. The oxygen in the water makes it highly corrosive, and commingling with produced water may lead to scale deposition and plugging in skimmer tanks, plate coalescers, or flotation units. The flow is highly irregular and would thus cause upsets throughout these devices. Finally, this flow must gravitate to a low point for collection and either is pumped up to a higher level for treatment or treated at that low point. Disposal piles are excellent for this purpose. They can be protected from corrosion, they are by design located low enough on the platform to eliminate the need for pumping the water, they are not severely affected by large instantaneous flow-rate changes (effluent quality may be affected to some extent, but the operation of the pile can continue), they contain no small passages subject to plugging by scale build-up, and they minimize commingling with the process since they are the last piece of treating equipment before disposal.

Disposal Pile Sizing

The produced water being disposed of has been treated in vessels having the capability of treating smaller droplets than those that can be predicted to settle out in the relatively slender disposal pile. Small amounts of separation will occur in the disposal pile due to coalescence in the inlet piping and in the pipe itself. However, no significant treating of produced water can be expected.

Most authorities having jurisdiction require that deck drainage be disposed of with no free oil. If the deck drainage is merely contaminated rainwater, the disposal pile diameter can be estimated from the following equation, assuming the need to separate 150-micron droplets:

Field Units

$$d^2 = \frac{0.3\,(Q_w + 0.356 A_D R_W + Q_{WD})}{(\Delta \mathrm{SG})}, \tag{9-26a}$$

SI Units

$$d^2 = \frac{28{,}289\,(Q_w + 0{,}001 A_D R_W + Q_{WD})}{(\Delta \mathrm{SG})}, \tag{9-26b}$$

d = pile internal diameter, in. (mm),
Q_w = produced water rate (if in disposal pile), bwpd (m^3/hr),
A_D = plan area of the deck, ft^2 (m^2),
R_W = rainfall rate, in./hr (mm/hr)
= 2 in./hr for Gulf of Mexico (50 mm/hr for GOM),
ΔSG = difference in specific gravity between oil droplets and rain water,
Q_{WD} = wash-down rate, BPD (m^3/hr)
Q_{WD} = 1,500 N (9.92 N),
N = number of 50-gpm (189.25-l/min) wash-down hoses.

Derivation of Equation (9-26)

From the equation for a vertical skim tank with $F = 1.0$:

Field Units

$$d_2 = 6{,}691 \frac{Q_g \mu_w}{(\Delta SG)\, d_m^2},$$

d_m = 150 microns,
Q_w = produced water rate if it is disposed of in pile + rainfall rate or wash-down rate, BPD,

$$Q_R = \frac{R_W}{12} A_D \frac{|24|}{(5.61)},$$

Q_R = rainfall rate, BPD,
R_w = rainfall rate, in./hr,
A_D = deck area, ft^2,

$$d^2 = \frac{0.3\,(Q_w + 0{,}356 A_D R_W + Q_{WD})}{(\Delta SG)}.$$

SI Units

$$d_2 = 6.365 \times 10^8 \frac{Q_g \mu_w}{(\Delta SG)\, d_m^2},$$

d_m = 150 microns,
Q_w = produced water rate if it is disposed of in pile + rainfall rate or wash-down rate, m^3/hr,

$$Q_R = \frac{R_W}{1000} \times A_D,$$

Q_R = rainfall rate, mm^3/hr,
R_w = rainfall rate, mm/hr,
A_D = deck area, m^2,

$$d^2 = \frac{28{,}289\,(Q_w + 0{,}001 A_D R_W + Q_{WD})}{(\Delta SG)}.$$

In Eq. (9-26) either the wash-down rate or the rainfall rate should be used as it is highly unlikely that both would occur at the same time. The produced water rate is only used if produced water is routed to the pile for disposal.

The disposal pile length should be as long as the water depth permits in shallow water to provide for maximum oil containment in the event of a malfunction and to minimize the potential appearance of any sheen. In deep water the length is set to assure that an alarm and then a shutdown signal can be measured before the pile fills with oil.

These signals must be high enough so as not to register tide changes. The length of pile submergence below the normal water level required to assure that a high level will be sensed before the oil comes within 10 feet of the bottom is given by

Field Units

$$L = \frac{(H_T + H_S + H_A + H_{SD})\,SG_0}{(\Delta SG)} + 10, \tag{9-27a}$$

SI Units

$$L = \frac{(H_T + H_S + H_A + H_{SD})\,SG_0}{(\Delta SG)} + 0.6, \tag{9-27b}$$

where

L = depth of pile below mean water level MWL (submerged length), ft (m),
H_T = normal tide range, ft (m),
H_S = design annual storm surge, ft (m),
H_A = alarm level (usually 2 ft), ft (m),
H_{SD} = shutdown level (usually 2 ft or 0.6 m), ft (m),
$(\mathrm{SG})_o$ = specific gravity of the oil relative to water.

Derivation of Equation (9-27)

Fields Units

$$(H_W + H_T + H_S + H_A + H_{\mathrm{SD}})\,\rho_o = H_W \rho_W,$$
$$(H_W)\,(\mathrm{SG})_o + (H_T + H_S + H_A + H_{\mathrm{SD}})\,(\mathrm{SG})_o = (H_W)\,(\mathrm{SG})_w,$$
$$H_W = \frac{(H_T + H_S + H_A + H_{\mathrm{SD}})\,(\mathrm{SG})_o}{(\Delta \mathrm{SG})}.$$

Pile length should be 10 feet (3 m) longer than oil column height.

Field Units

$$L = H_W + 10 = \frac{(H_T + H_S + H_A + H_{\mathrm{SD}})\,(\mathrm{SG})_o}{(\Delta \mathrm{SG})} + 10$$

SI Units

$$L = H_W + 0.6 = \frac{(H_T + H_S + H_A + H_{\mathrm{SD}})\,\mathrm{SG}_o}{(\Delta \mathrm{SG})} + 0.6$$

It is possible in shallow water to measure the oil–water interface for alarm or shutdown with a bubble arrangement and a shorter pile. However, this is not recommended where water depth permits a longer pile. To minimize wave action effects a minimum pile length of about 50 feet is required. Figure 9-54 is a schematic showing disposal pile length.

Skim Piles

The skim pile is a type of disposal pile. As shown in Figure 9-55, flow through the multiple series of baffle plates creates zones of no flow that

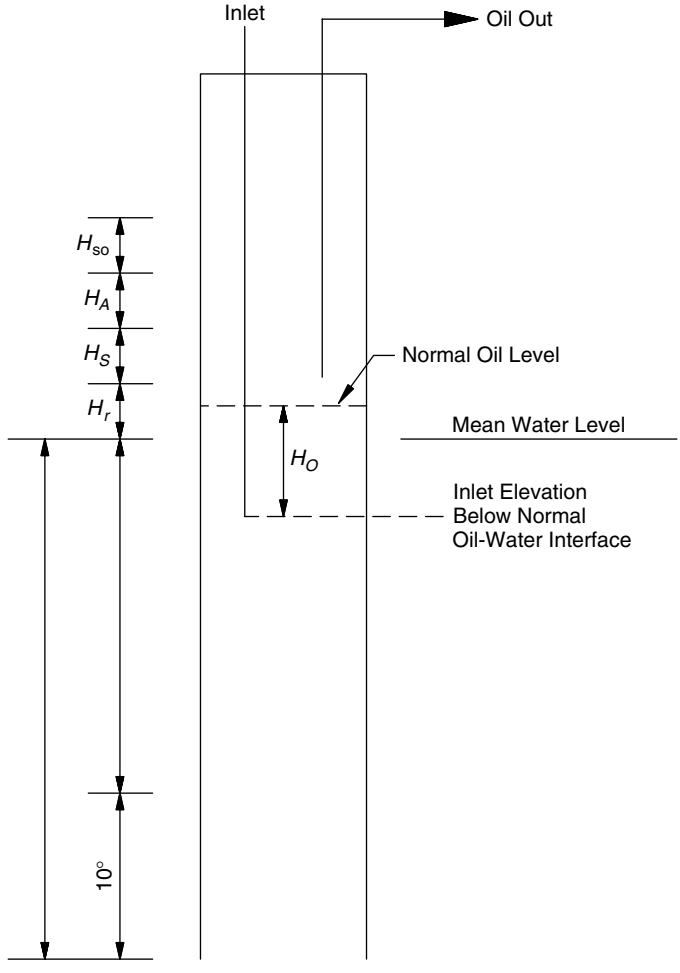


Figure 9-54. Disposal pile length.

reduce the distance a given oil droplet must rise to be separated from the main flow. Once in this zone, there is plenty of time for coalescence and gravity separation. The larger droplets then migrate up the underside of the baffle to an oil collection system.

Besides being more efficient than standard disposal piles, from an oil separation standpoint, skim piles have the added benefit of providing for some degree of sand cleaning. Most authorities having jurisdiction state that produced sand must be disposed of without "free oil." It is doubtful that sand from a vessel drain meets this criterion when disposed of in a standard disposal pile.

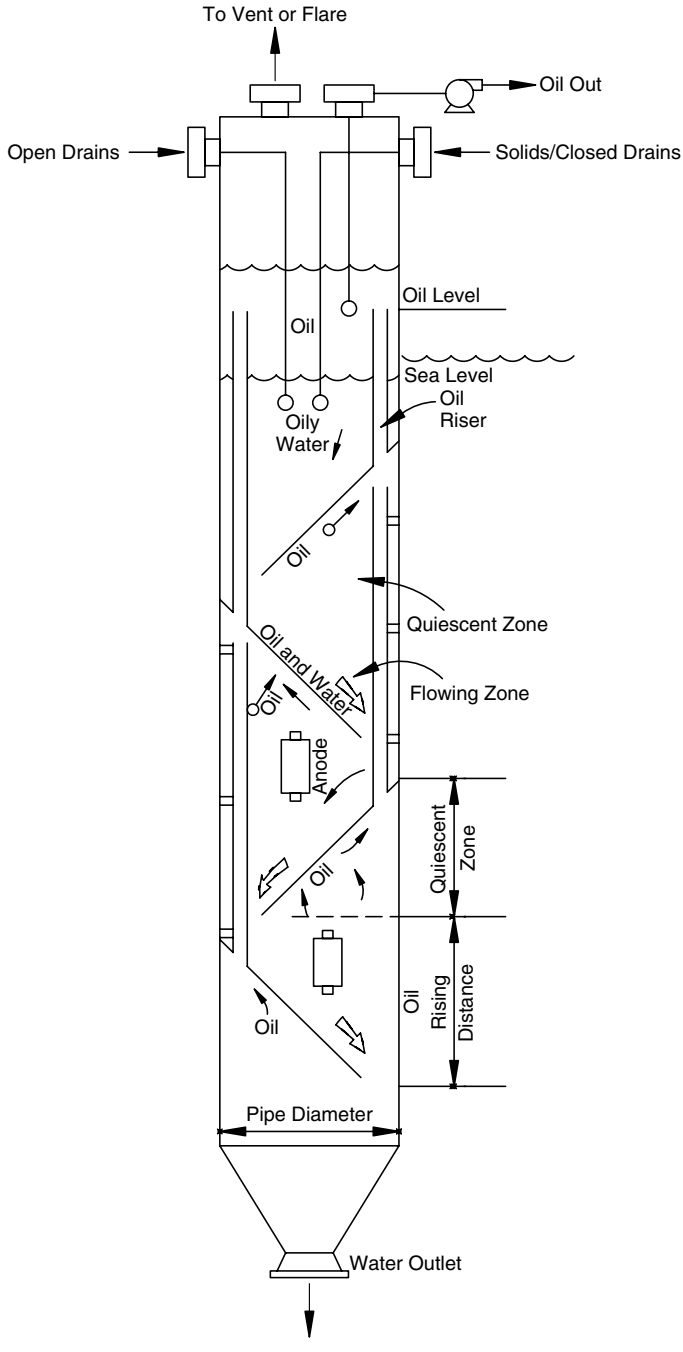


Figure 9-55. Cross section showing flow pattern of a skim pile.

Sand traversing the length of a skim pile will abrade on the baffles and be water-washed. This can be said to remove the free oil that is then captured in a quiescent zone.

Skim Pile Sizing

The determination of skim pile length is the same as that for any other disposal pile. Because of the complex flow regime, a suitable equation has yet to be developed to size skim piles for deck drainage. However, field experience has indicated that acceptable effluent is obtained with 20 minutes' retention time in the baffled section of the pile. Using this and assuming that 25% of the volume is taken up by the coalescing zones, we have the following:

Field Units

$$d^2 L' = 19.1\,(Q_w + 0.356 A_D R_W + Q_{\mathrm{WD}})\,, \tag{9-28a}$$

SI Units

$$d^2 L' = 565{,}811\,(Q_w + 0.001 A_D R_W + Q_{\mathrm{WD}})\,, \tag{9-28b}$$

where L' is the length of baffle section, in ft [the submerged length is $L' + 15\,\mathrm{ft}\,(L' + 4.6\,\mathrm{m})$ to allow for an inlet and exit from the baffle section].

Derivation of Equation (9-28)

t_w is in s, Q in $\mathrm{ft}^3/\mathrm{s}\,(\mathrm{m}^3/\mathrm{s})$, d in in. (mm), V in $\mathrm{ft}^3\,(\mathrm{m}^3)$, Q_w in BPD $(\mathrm{m}^3/\mathrm{hr})$, and L' in ft (m).

Field Units

$$t_W = \frac{Vol}{Q},$$

$$Vol = \frac{\pi d^2 L'}{(4)\,(144)} \times \frac{3}{4},$$

$$Q = \frac{5.61}{(24)\,(3{,}600)}\,(Q_w + 0.356 A_D R_W + Q_{\mathrm{WD}})\,,$$

$$t_w = 60(t_r)_w = 1{,}200,$$
$$d^2L' = 19.1(Q_w + 0.356A_D R_w + Q_{WD}).$$

SI Units

$$t_W = \frac{Vol}{Q},$$
$$Vol = \frac{\pi d^2 L'}{(1{,}000)^2\,(4)} \times \frac{3}{4},$$
$$Q = (1/3{,}600)(Q_w + 0.001A_D R_w + Q_{WD}),$$
$$t_w = 60(t_r)_w = 1{,}200,$$
$$d^2L' = 565{,}811(Q_w + 0.001A_D R_w + Q_{WD}).$$

Drain Systems

A drain system that is connected directly to pressure vessels is called a "pressure" or "closed" drain system. A drain system that collects liquids that spill on the ground is an "atmospheric," "gravity," or "open" drain. The liquid in a closed drain system must be assumed to contain dissolved gases that flash in the drain system and can become a hazard if not handled properly. In addition, it must be assumed that a closed drain valve could be left open by accident. Once the liquid has drained out of the vessel, a large amount of gas will flow out of the vessel into the closed drain system (gas blow-by) and will have to be handled safely. Thus, closed drain systems should always be routed to a pressure vessel and should never be connected to an open drain system.

Liquid gathered in an open drain system is typically rainwater or wash-down water contaminated with oil. With the oil usually circulated back into the process, every attempt should be made to minimize the amount of aerated water that is recycled with the oil. This goal is best achieved by routing open drains to a sump tank that has a gas blanket and operates as a skimmer. To keep gas from the skimmer from flowing out the drain, a water seal should be built into the inlet to the sump tank. Water seals should also be installed on laterals from separate buildings or enclosures to keep the open drain system from being a conduit of gas from one location in the facility to another.

Information Required for Design

Effluent Quality

The U.S. Environmental Protection Agency (EPA) establishes the maximum amount of oil and grease content in water that can be discharged into navigable waters of the United States. In other locations local governments or governing bodies will establish this criterion.

Examples of worldwide produced water effluent oil concentration limitations include

Ecuador, Colombia, Brazil, Argentina,

Venezuela:	30 mg/l all facilities
Indonesia:	15 mg/l new facilities
	25 mg/l "grandfathered" facilities
Malaysia, Middle East:	30 mg/l all facilities

Nigeria, Angola, Cameroon,

Ivory Coast:	50 mg/l all facilities
North Sea, Australia:	30 mg/l all facilities
Thailand:	50 mg/l all facilities
USA:	29 mg/l OCS waters
	zero discharge inland waters

- Produced water flow rate (Q_w, bwpd)
- Specific water gravity of produced water $(SG)_w$. Assume 1.07 if data are not available.
- Wastewater viscosity at flowing temperatures (μ, cp). Assume 1.0 cp if data are not available.
- Concentration of oil in water to be treated (mg/l or ppm). This is best determined from field samples or laboratory data.
- Specific gravity of oil at flowing temperature $(SG)_o$
- Particle size distribution curve for oil droplets in the produced water. This is best determined from field samples or laboratory data and an analysis of drop size management due to dispersion and coalescence in the piping system.
- Design rainfall rate (R_w, in./hr). Assume 2 in./hr in the Gulf of Mexico.
- Flow rate for wash-down (Q_{WD}, BPD). Assume 1,500 bwpd per 50-gpm wash-down hose.
- Particle size distribution curve for "free oil" droplets in deck drainage.
- Concentration of soluble oil at discharge conditions (mg/l or ppm).

Influent Water Quality

Produced Water

The first step in choosing a water treating system is to characterize the influent water streams. It is necessary to know both the *oil concentration* in this stream and the *particle size distribution* associated with this concentration. This is best determined from field samples and laboratory data.

Various attempts have been made to develop design procedures to determine oil concentration in water outlets from properly designed free-water knockouts and theaters. A conservative assumption would be that the water contains less than 1,000 to 2,000 mg/l of dispersed oil.

It is possible to theoretically trace the particle size distribution up the tubing, through the choke, flow lines, manifolds, and production equipment into the free-water knockout using equations presented in previous sections. However, many of the parameters needed to solve these equations, especially those involving coalescence, are unknown.

Because of the dispersion through the water dump valve, the oil size distribution at the outlet of a free-water knockout or heater-treater is not a significant design parameter. From the dispersion theory it can be shown that after passing through the dump valve a maximum droplet diameter on the order of 10 to 50 microns will exist no matter what the droplet size distribution was upstream of this valve.

If there were sufficient time for coalescence to occur in the piping downstream of the dump valve, then the maximum droplet diameter would be defined by Eq. (9-2) prior to the water entering the first vessel in the water treating system.

The solution of this equation requires the determination of surface tension. The surface tension of an oil droplet in a water continuous phase is normally between 1 and 50 dynes/cm. It is not possible to predict the value without actual laboratory measurements in the produced water. Small amounts of impurities in the produced water can lower the surface tension significantly from what might be measured in synthetic water. In addition, as these impurities change with time, so will the surface tension. In the absence of data it is recommended that a maximum diameter of between 250 and 500 microns be used for design.

It is clear that there will be distribution of droplet sizes from zero to the maximum size, and this distribution will depend upon parameters unknown at the time of initial design. Experimental data indicate that a conservative assumption for design would be to characterize the distribution by a straight line, as shown in Figure 9-52.

Soluble Oil

In every system substances that show up as "oil" in the laboratory test procedure will be dissolved in the water. This is especially true where samples are acidized for "stabilization" prior to extraction with a solvent. This soluble oil cannot be removed by the systems discussed in this chapter. The soluble oil concentration should be subtracted from the discharge criteria to obtain a concentration of dispersed oil for design. Soluble oil concentrations as high as 1,000 mg/l have been measured on rare occasions.

Deck Drainage

Federal regulations and most authorities having jurisdiction require that "free oil" be removed from deck drainage prior to disposal. It is extremely difficult to predict an oil drop size distribution for rainwater or washdown water that is collected in an open drain system, and regulations do not define what size droplet is meant by "free oil."

Long-standing refinery practice is to size the drain water treating equipment to remove all oil droplets 150 microns in diameter or larger. If no other data are available, it is recommended that this be used in sizing sumps and disposal piles.

Equipment Selection Procedure

It is desirable to bring information included earlier into a format that can be used by the design engineer in selecting and sizing the individual pieces of equipment needed for a total water treating system. Federal regulations and most authorities having jurisdiction require that produced water from the free-water knockout receive at least some form of primary treatment before being sent to a disposal pile or skim pile. Deck drainage may be routed to a properly sized disposal pile that will remove "free oil."

Every water treating system design must begin with the sizing, for liquid separation of a free-water knockout, heater treated, or three-phase separator. These vessels should be sized in accordance with the procedures discussed in previous chapters.

With the exception of these restraints the design engineer is free to arrange the system as he or she sees fit. There are many potential combinations of the equipment previously described. Under a certain set of circumstances, it may be appropriate to dump the water from a free-water

knockout directly to a skim tank for final treatment before discharge. Under other circumstances a full system of plate coalescers, flotation units, and skim piles may be needed. In the final analysis the choice of a particular combination of equipment and its sizing must rely rather heavily on the judgment and experience of the design engineer. The following procedure is meant only as a guideline and not as a substitute for this judgment and experience. Many of the correlations presented herein should be refined as new data and operating experience become available. In no instance is this procedure meant to be used without proper weight given to operational experience in the specific area.

1. Determine the oil content of the produced water influent. In the absence of other information, 1,000 to 2,000 mg/l could be assumed.
2. Determine the dispersed oil effluent quality. In the absence of other information, use 23 mg/l for design in the Gulf of Mexico and other similar areas (29 mg/l allowed less 6 mg/l dissolved oil).
3. Determine oil drop size distribution in the influent produced water stream. Use a straight-line distribution with a maximum diameter of 250 to 500 microns in the absence of better data.
4. Determine the oil particle diameter that must be treated to meet effluent quality required. This can be calculated as effluent quality divided by influent quality times the maximum oil particle diameter calculated in step 3.
5. If there is a large amount of space available (as in an onshore location), consider an SP Pack system. Proceed to step 10. If the answer to step 4 is less than 30 to 50 microns, a flotation unit or cyclone is needed. Proceed to step 6. If the answer to step 4 is greater than 30 microns, a skim tank or plate coalescer could be used as a single stage of treatment, but this is not really recommended. Proceed to step 9.
6. Determine flotation cell influent quality from the required effluent quality assuming 90% removal. Influent quality is effluent quality desired times 10.
7. If required flotation cell influent quality is less than quality determined in step 1, determine the particle diameter that must be treated in skim tank or plate coalescers to meet this quality. This can be calculated as the flotation cell influent quality divided by the influent quality determined in step 1 times the maximum particle diameter calculated in step 3.
8. Determine effluent from hydrocyclone, assuming that it is 90% efficient, and determine the particle diameter that must be treated in the downstream skim vessel, assuming that $d_{\max} = 500$. This

value can be calculated as 500 times the dispersed oil effluent quality (step 1) divided by the effluent concentration from the hydrocyclone.

9. Determine skimmer dimensions.
   a. Choose horizontal or vertical configuration.
   b. Choose pressure vessel or atmospheric vessel.
   c. Determine size. Refer to appropriate equations.
10. Determine overall efficiency required, efficiency per stage, and number of stages for an SP Pack system.
11. Determine plate coalescers' dimensions.
    a. Choose CPI or cross-flow configuration.
    b. Determine size. Refer to appropriate equations.
12. Choose skim tank, SP Pack, or plate coalescers for application, considering cost and space available.
13. Choose method of handling deck drainage.
    a. Determine whether rainwater rate or wash-down rate governs design.
    b. Size disposal pile for drainage assuming 150-micron drop removal. Refer to appropriate equations.
    c. If disposal pile diameter is too large,
       i. Size sump tank to use with disposal pile (refer to appropriate equations), or
       ii. Size skim pile (refer to appropriate equations).

Equipment Specification

Once the equipment types are selected using the previous procedure, the design equations presented in Chapter 6 can be used to specify the main size parameters for each of the equipment types.

Skim Tank

1. Horizontal vessel designs. The internal diameter and seam-to-seam length of the vessel can be determined. The effective length of the vessel can be assumed to be 75% of the seam-to-seam length.
2. Vertical vessel designs. The internal diameter and height of the water column can be determined. The vessel height can be determined by adding approximately 3 feet to the water column height.

SP Pack System

The number and size of tanks can be determined. Alternatively, the dimensions and number of compartments in a horizontal flume can be specified.

CPI Separator

The number of plate packs can be determined.

Cross-Flow Devices

The acceptable dimensions of the plate pack area can be determined. The actual dimensions depend on the manufacturers' standard sizes.

Flotation Cell

Information is given to select a size from the manufacturers' data.

Disposal Pile

The internal diameter and length can be determined. For a skim pile the length of the baffle section can be chosen.

Example 9-2: Design the produced water treating system for the data given

Given:

40°API
5,000 bwpd (33 m^3/hr)
Deck size is 2,500 ft^2 (232.3 m^2)
48 mg/l discharge criteria (48 mg/l)
Water gravity-feeds to system

Step 1. Assume 6 mg/l soluble oil, and oil concentration in produced water is 1,000 mg/l.
Step 2. Effluent quality required is 48 mg/l. Assume 6 mg/l dissolved oil. Therefore, effluent quality required is 42 mg/l.

Step 3. Assume maximum diameter of oil particle (d_{max}) = 500 microns.

Step 4. Using Figure 9-52, the size of oil droplet that must be removed to reduce the oil concentration from 1,000 mg/l to 42 mg/l is

$$\frac{d_m}{500} = \frac{42}{1{,}000},$$
$$d_m = 21 \text{ microns}.$$

Step 5. Consider an SP series tank treating system. See step 10. If SP Packs are not used, since $d_m < 30$ microns, a flotation unit or hydrocyclone must be used. Proceed to step 6. (Note: since d_m is close to 30 microns, it may be possible to treat this water without a flotation unit. We will take the more conservative case for this example.)

Step 6. Since the flotation cell is 90% efficient, in order to meet the design requirements of 42 mg/l it will be necessary to have an influent quality of 420 mg/l. This is lower than the 1,000-mg/l concentration in the produced water assumed to in step 1. Therefore, it is necessary to install a primary treating device upstream of the flotation unit.

Step 7. Using Figure 9-52, the size of oil droplet that must be removed to reduce the oil concentration from 1,000 mg/l to 420 mg/l is

$$\frac{d_m}{500} = \frac{420}{1{,}000},$$
$$d_m = 210 \text{ microns}.$$

Step 8. Inlet to water treating system is at too low a pressure for a hydrocyclone. Size a skim vessel upstream of the flotation unit.

Step 9. Skim vessel design. Pressure vessel is needed for process considerations (e.g., fluid flow, gas blow-by).

a. Assume horizontal pressure vessel.

Settling equation

Field Units

$$dL_{\text{eff}} = \frac{1{,}000 Q_w \mu_w}{(\Delta \text{SG})\,(d_m)^2},$$
$$\mu_w = 1.0 \text{ (assumed)},$$
$$(\text{SG})_w = 1.07 \text{ (assumed)},$$

$$(\mathrm{SG})_o = 0.83 \text{ (calculated)},$$

$$dL_{\text{eff}} \frac{(1{,}000)\,(5{,}000)\,(1.0)}{(0.24)\,(210)^2} = 472.$$

Assume various diameters (d) and solve for L_{eff}.

d (in.)	L_{eff} [ft (m)]	Actual Length [ft (m)]
24	19.7	26.3
48	9.8	13.1
60	7.9	10.5

Retention time equation

Assume retention time of 10 minutes.

$$d^2 L_{\text{eff}} = 1.4(t_r)_w Q_w,$$

$$d^2 L_{\text{eff}} = (1.4)(10)(5{,}000) = 7{,}000.$$

d (in.)	L_{eff} [ft (m)]	Actual Length [ft (m)]
48	30.4	40.4
72	13.5	17.9
84	9.9	13.1
96	7.6	10.1

SI Units

$$dL_{\text{eff}} = \frac{1{,}145{,}734 Q_W \mu_W}{(\Delta \mathrm{SG})\,(d_m)^2},$$

$$\mu_w = 1.0 \text{ (assumed)},$$

$$(\mathrm{SG})_w = 1.07 \text{ (assumed)},$$

$$(\mathrm{SG})_o = 0.83 \text{ (calculated)},$$

$$dL_{\text{eff}} = \frac{(1{,}145{,}734)\,(33)\,(1.0)}{(0.24)\,(210)^2} = 3{,}572.$$

Assume various diameters (d) and solve for L_{eff}.

(*d*) (mm)	L_{eff} (m)	Actual Length (m)
609.6	5.9	8.0
1219.2	3.0	4.0
1542	2.3	3.2

Retention time equation

Assume retention time of 10 minutes.

$$d^2 L_{\text{eff}} = 42{,}441 (t_r)_w Q_w,$$
$$d^2 L_{\text{eff}} = (42{,}441)(10)(33).$$

d (mm)	L_{eff} (m)	Actual Length (m)
1219	9.4	12.3
1829	4.2	5.5
2134	3.1	4.0
2438	2.4	3.1

b. Assume vertical pressure vessel.

Field Units

Settling equation

$$d^2 = 6{,}691 F \frac{Q_w \mu_w}{(\Delta \text{SG})\,(d_m)^2},$$
$$F = 1.0 \text{ assumed},$$
$$d^2 = \frac{(6{,}691)\,(1.0)\,(5{,}000)\,(1.0)}{(0.24)\,(210)^2},$$
$$d = 56.22 \text{ in.}$$

Retention time equation

$$H = 0.7 \frac{(t_r)_w\, Q_w}{d^2} \approx 0.7 \frac{(10)\,(5{,}000)}{d^2}$$

d (in.)	L_{eff} (ft.)	Seam-to-Seam Height (ft.)
60	9.72	12.7
66	8.03	11.0
72	6.75	9.8

SI Units

Settling equation

$$d^2 = 6{,}365 \times 10^8 \frac{Q_w \mu_w}{(\Delta SG)\,(d_m)^2},$$

$$F = 1.0 \text{ assumed},$$

$$d^2 = \frac{\left(6{,}365 \times 10^8\right)(1.0)\,(33)\,(1.0)}{(0.24)\,(210)^2},$$

$$d = 1{,}409 \text{ mm}.$$

Retention time equation

$$H = \frac{21{,}218\,(t_r)_w\,Q_w}{d^2} \approx 21{,}218\frac{(10)\,(33)}{d^2}$$

d (mm)	L_{eff} (m)	Seam-to-Seam Height (m)
1,524	3.0	3.9
1,676.4	2.5	3.4
1,829	2.1	3.0

A vertical vessel 60 in. (1,524 mm) × 12.5 ft (3.8 m) or 72 in. (1,829 mm) × 10 ft (3 m) would satisfy all the parameters. Depending on cost and space considerations, we recommend a 72-in. (1,829-mm) × 10-ft (3-m) vertical skimmer vessel for this application.

Step 10. Investigate SP Packs in tanks as an option. Calculate overall efficiency required:

Field Units

$$E_t = \frac{1{,}000 - 42}{1{,}000} = 0.958.$$

Assume 10-ft- (3-m-) diameter vertical tanks:

$$d_m^2 = 6{,}691F\frac{Q_w\mu_w}{(\Delta SG)\,(d^2)},$$
$$d_m^2 = \frac{6{,}691\,(2)\,(5{,}000)\,(1.0)}{(0.24)\,(120)^2},$$
$$d_m = 139.$$

Assume SP Pack grows 1,000-micron drops:

$$E = 1 - \frac{d_m}{1{,}000}$$
$$= 0.861.$$

One acceptable choice is two 10-ft- (3-m-) diameter SP tanks in series.

$$E_t = 1 - (1 - 0.861)^2 = 0.981$$

SI Units

$$E_t = \frac{1{,}000 - 42}{1{,}000} = 0.958$$

Assume 10-ft- (3-m-) diameter vertical tanks:

$$d_m^2 = 6{,}365 \times 10^8 F\frac{Q_w\mu_w}{(\Delta SG)\,(d^2)},$$
$$d_m^2 = \frac{6{,}365 \times 10^8\,(2)\,(33)\,(1.0)}{(0.24)\,(3{,}000)^2},$$
$$d_m = 139\,\text{microns}.$$

Assume SP Pack grows 1,000-micron drops:

$$E = 1 - \frac{d_m}{1{,}000}$$
$$= 0.861.$$

One acceptable choice is two 10-ft- (3-m-) diameter SP tanks in series.

$$E_t = 1 - (1 - 0.861)^2 = 0.981$$

Step 11. Check for alternate selection of CPI.

Field Units

$$\text{number of packs} = 0.07\frac{Q_w \mu_w}{(\Delta\text{SG})\, d_m^2}$$
$$= \frac{(0.077)\,(5{,}000)\,(1.0)}{(0.24)\,(210)^2}$$
$$= 0.04 \text{ packs},$$
$$Q_w < 20{,}000 : \text{use 1 pack CPI.}$$

SI Units

$$\text{number of packs} = 11.6\frac{Q_w \mu_w}{(\Delta\text{SG})\, d_m^2}$$
$$= \frac{(11.67)\,(33)\,(1.0)}{(0.24)\,(210)^2}$$
$$= 0.04 \text{ packs},$$
$$Q_w < 132 : \text{use 1 pack CPI.}$$

Step 12. Recommended skimmer vessel over CPI as skimmer will take up about same space, will cost less, and will not be susceptible to plugging. Note that it would also be possible to investigate other configurations such as skim vessel, SP Pack, CPI, etc. as alternatives to the use of a flotation unit.

Step 13. Sump design. Sump is to be designed to handle the maximum of either rainwater or wash-down hose rate.

a. Rainwater rate:

Field Units

Assume

R_w = rainfall rate; 2 in./hr,
A_D = deck area; 2500 ft^2,
$Q_w = 0.356 A_D R_W$
$= (0.356)(2{,}500)(2)$
$= 1{,}780$ bwpd.

SI Units

Assume

$$\begin{aligned} R_W &= 50.8 \text{ mm/hr}, \\ A_D &= 232.3\,\text{m}^2, \\ Q_w &= 0.001 A_D R_W \\ &= (0.001)(232.3)(50.8) \\ &= 11.8\,\text{m}^3/\text{hr}. \end{aligned}$$

b. Wash-down rate:

Field Units

Assume

$$\begin{aligned} N &= 2, \\ Q_{WD} &= 1{,}500 \text{ N} \\ &= 1{,}500\ (2) \\ &= 3{,}000 \text{ bwpd}. \end{aligned}$$

Assume

$$\begin{aligned} N &= 2, \\ Q_{WD} &= 9.92 \text{ N} \\ &= 9.92\ (2) \\ &= 19.84\,\text{m}^3/\text{hr}. \end{aligned}$$

The minimum design usually calls for two hoses.
Because freshwater enters the sump via the drains, the sump tank must be sized using a specific gravity of 1.0 and a viscosity of 1.0 for freshwater.

c. Assume horizontal rectangular cross-section sump.

Settling equation:

Field Units

$$WL_{\text{eff}} = 70\frac{Q_w \mu_w}{(\Delta\text{SG})\, d_m^2},$$

$$WL_{\text{eff}} = 70\frac{(3,000)\,(1.0)}{(0.150)\,(150)^2},$$

$$WL_{\text{eff}} = 62.2,$$

W = width, ft (m),
L_{eff} = effective length in which separation occurs, ft (m),
H = height of tank, which is 1.5 times higher than water level within tank, or $0.75W$.

Tank Width (ft)	Tank L_{eff} (ft)	Seam-to-Seam Length	Height (ft)
4	15.6	20.0	3.0
5	12.4	16.2	3.8
6	10.4	13.5	4.5

SI Units

$$WL_{\text{eff}} = 950\frac{Q_w \mu_w}{(\Delta\text{SG})\, d_m^2},$$

$$WL_{\text{eff}} = 950\frac{(19.84)\,(1.0)}{(0.150)\,(150)^2},$$

$$WL_{\text{eff}} = 5.6,$$

W = width, ft (m),

L_{eff} = effective length in which separation occurs, ft (m),

H = height of tank, which is 1.5 times higher than water level within tank, or 0.75 W.

Tank Width (m)	Tank L_{eff} (m)	Seam-to-Seam (1.2 L_{eff})	Height (m)
1.2	4.7	6.2	0.9
1.5	3.7	4.9	1.1
1.8	3.1	4.1	1.4

A horizontal tank 6 ft (1.83 m) by 14 ft (4.3 m) by 5 ft (1.52 m) would satisfy all design parameters.

d. If it is determined that the dimensions of the sump tank are inappropriately large for the platform, an SP Pack can be added upstream of the sump tank to increase oil droplet size by approximately two times the inlet droplet size.

Therefore, the sump tank size with an SP Pack can be determined by

Field Units

$$WL_{\text{eff}} = \frac{70\,(3{,}000)\,(1.0)}{(0.15)\,(300)^2}$$

$$= 15.6,$$

W = width, ft (m),

L_{eff} = effective length in which separation occurs, ft (m),
H = height of tank, which is 1.5 times higher than water level within tank, or 0.75 W.

Tank Width (ft)	Tank L_{eff} (ft)	Seam-to-Seam Length	Height (ft)
3	5.2	6.7	2.3
4	3.9	5.1	3.0
5	3.1	4.0	3.8

SI Units

$$WL_{eff} = \frac{950\,(19.84)\,(1.0)}{(0.15)\,(300)^2}$$

$$= 1.4,$$

W = width, ft (m),

L_{eff} = effective length in which separation occurs, ft (m),
H = height of tank, which is 1.5 times higher than water level within tank, or 0.75 W.

Tank Width (m)	Tank L_{eff} (m)	Seam-to-Seam ($1.2\,L_{eff}$)	Height (m)
0.9	1.56	2.0	0.68
1.2	1.2	1.6	0.9
1.5	0.93	1.2	1.13

A horizontal tank (with an SP Pack) 4 ft (1.2 m) by 4 ft (1.2 m) by 5 ft (1.5 m) would satisfy all design parameters. It can be seen that by adding an SP Pack, sump tank sizes can be substantially reduced.

Step 14. Recovered Oil Tank. Assume a cylindrical tank with a retention time of 15 minutes and a process flow of 10% of the design water flow for flotation cells and a process flow of 5% of the design meter flow for skim vessels.

Field Units

$$
\begin{aligned}
Q_w &= (0.10)(5{,}000) + (0.05)(5{,}000) \\
&= 750\,\text{BPD} \\
&= \frac{0.7\,(t_r)\,Q_w}{d_2}, \\
H &= \frac{7{,}875}{d^2}, \\
H &= \frac{0.7\,(15)\,(750)}{d^2}.
\end{aligned}
$$

Vessel Diameter (in.)	Effective Length (ft)	Seam-to-Seam Length (ft)
30	8.8	11.8
36	6.1	9.1
42	4.5	7.5

SI Units

$$
\begin{aligned}
Q_w &= (0.10)(33) + (0.05)(33) \\
&= 4.95\,\text{m}^3/\text{hr}, \\
H &= \frac{21{,}218\,(t_r)\,Q_w}{d_2}, \\
H &= \frac{21{,}218\,(15)\,(4.95)}{d^2} \\
&= \frac{1{,}575{,}437}{d^2}.
\end{aligned}
$$

Assume various diameters (d) and solve for liquid heights (H). $L_{ss} = L_{eff} + 3\,\text{ft}$ $L_{eff} + 0.9\,\text{m}$.

Vessel Diameter (in.)	Effective Length (ft)	Seam-to-Seam Length (ft)
762	2.7	3.6
914	1.9	2.8
1067	1.4	2.3

A vertical vessel 36 in. (914 mm) × 6 ft (1.8 m) would satisfy all design parameters.

Nomenclature

A_D = plan area of the deck, $ft^2(m^2)$
C_i = inlet oil concentration
d = vessel's internal diameter, in. (mm)
d = final droplet size, microns (μ)
d_b = diameter of gas bubble
d_m = oil droplet's diameter, microns (μ)
d_{max} = diameter of droplet above whose size only 5% of the oil volume is contained, microns (μ)
d_o = initial droplet size, microns (μ)
d_r = oil droplet diameter to be removed, microns (μ)
E = efficiency per cell
E_t = overall efficiency
F = factor that accounts for turbulence and short-circuiting
H = height of water, ft (m)
h = height of mixing zone, ft (m)
H_A = alarm level, ft (m)
H_o = height of oil pad, ft (m)
H_S = design annual storm surge, ft (m)
H_{SD} = shutdown level, ft (m)
H_T = normal tide range, ft (m)
H_w = maximum height of oil below MWL, ft (m)
j = an empirical parameter that is always larger than 3 and depends on the probability that the droplets will bounce apart before coalescence takes place
K_p = mass transfer coefficient
K_s = empirical parameter for the particular system
L = length of plate section parallel to the axis of water flow, ft (m)
L = depth of pile below mean water level, ft (m)
L' = length of baffle section, ft (m)
L_{eff} = effective length in which separation occurs, ft (m)
L_{ss} = seam-to-seam length, ft (m)
N = number of plate packs
N = number of 50-gpm wash-down hoses
n = number of stages or cells
q_g = gas flow rate
Q_w = water flow rate, bwpd (m^3/hr)
q_w = liquid flow rate through mixing zone
Q_{WD} = wash-down rate, bwpd (m^3/hr)
r = radius of mixing zone
R_W = rainfall rate, in./hr (mm/hr)

SG_o = specific gravity of the oil relative to water
SG_w = specific gravity of the produced water
$SG_{w'}$ = specific gravity of the sea water
$(t_r)_w$ = retention time, min
t_r = retention time, min
V_o = vertical velocity of the oil droplet relative to the water continuous phase, ft/s (m/s)
W = width, ft (m)
α_w = fractional cross-sectional area of water
β_w = fractional water height within vessel
γ = height-to-width ratio, H/W
ΔP = pressure drip, psi (kPa)
ε = mixing parameter equivalent to the work done on a fluid per unit mass per unit time, cm^2/s^3
θ = angle of the plate with the horizontal
μ_w = water viscosity, cp (Pas)
ρ_w = water density, g/cm^3
σ = surface tension, dynes/cm
ϕ = volume fraction of the oil phase

Review Questions

1. List five methods of produced water treating equipment:

   a) _____
   b) _____
   c) _____
   d) _____
   e) _____

2. List the approximate minimum drop size removal capabilities for the following equipment types:

   a) Gravity separation _____
   b) Plate coalescence _____
   c) Enhanced coalescence _____
   d) Gas flotation _____
   e) Enhanced gravity separation _____

3. Produced water will always have some form of primary treating prior to disposal. This system could take the form of a skim tank, skim vessel, CPI, cross-flow separator, SP Pack, or gas flotation unit. Depending upon the severity of the treating problem,

secondary treatment may be required. List three examples of secondary treatment that are commonly used:

a) _____
b) _____
c) _____

4. List the three basic phenomena that are used in the design of common produced water treating equipment:

a) _____
b) _____
c) _____

5. The dispersion process is diametrically opposed by coalescence, which is the process in which small droplets collide and combine into larger droplets. True or false?
6. Flotation is a process

a) Involving the injection of fine gas bubbles into the water phase
b) Where the gas bubbles in the water adhere to the oil droplets
c) Where the oil droplets are removed when they rise to the water surface, where they are trapped in the resulting foam and skimmed off the surface
d) That is capable of removing small oil droplets (greater than 10 microns)
e) All of the above apply
f) Only A, B, and C apply

7. Skim vessels are not recommended when

a) Influent oil droplet sizes are mostly below 100 microns
b) Size and weight are the primary considerations
c) Offshore structure movement could generate waves in the vessel
d) Water temperature is very cold due to long subsea pipelines connected to other platforms

8. Plate separators generally exhibit the following advantages:

a) Require very little maintenance
b) Can be easily removed as complete modules for inspection and cleaning, if necessary
c) Have smaller size and weight requirements than skim vessels
d) Can accept fairly high concentrations of oil or solids in the inlet feed
e) All of the above

9. Plate separators are recommended when

   a) Water flow rate is steady or feed is from a pump
   b) Size and weight are not constant
   c) Utilities and equipment are available to periodically clean the plate packs
   d) Influent oil content is high and oil concentration must be reduced to 150 mg/l for effective second-stage treating in a downstream unit
   e) Solid contaminants are not significant in the waste stream and sand content is less than 110 ppm

10. Gas flotation units should be used when

   a) Only proven technology is to be used
   b) The inlet oil concentrations are not too high (250–500 mg/l)
   c) The effluent discharge requirements are not too severe (25–50 mg/l)
   d) Chemical companies are available to formulate an appropriate chemical treatment program
   e) Power costs are low or moderate
   f) Oil/water density differences are low, such as heavy oils

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Chapter 10
Water Injection Systems

Introduction

Oil-field waters usually contain impurities. These impurities are classified as dissolved minerals, dissolved gases, or suspended solids. Suspended solids can be naturally occurring, generated by precipitation of dissolved solids, generated as products of corrosion, or created by microbiological activity. Changes in temperature, pressure, pH, or the mixing of waters from different sources may cause *scaling*, which is precipitation of dissolved solids. Suspended solids may settle out of the water stream or may be carried as a suspension in flowing water.

The two primary sources of freshwater are surface water and groundwater. A portion of the rain or melting snow and ice at the earth's surface soaks into the ground, while part of it collects in ponds and lakes or runs off into creeks and rivers. This latter portion is termed "surface water."

Water encountered in production operations usually comes from separated produced water or from wells specifically drilling into a subsurface water aquifer. The latter is often called "source water," is often brackish, and may contain a large quantity of dissolved solids.

This chapter provides information about equipment selection and sizing for removing suspended solids and dissolved gases from water. The water's source affects the types and amounts of contaminants in the water. For example, produced water will be contaminated by some hydrocarbons.

The treatment of water to remove calcium and magnesium dissolved solids ("water softening") is important, especially if the water is to be used as boiler feed water for the generation of steam, as in a steam flood. Nevertheless, a discussion of processes and equipment for water softening and removing other dissolved solids is beyond the scope of this chapter.

The removal of suspended solids and dissolved gases from water may be desirable for a variety of reasons, the most common of which are to prepare the water for injection into a producing formation and to

minimize the corrosion and solids build-up in surface equipment. Prior to injecting water, it may be important to remove solids above a certain size to minimize damage to the formation caused by solids plugging. This plugging can limit injection volumes, increase pump horsepower requirements, or lead to fracturing of the reservoir rock. Dissolved gases such as oxygen in the water may promote bacteria growth within the formation, or they may speed the process of corrosion.

The presence of oxygen or hydrogen sulfide (H_2S) in water can lead to the formation of FeS, Fe_2O_3, elemental sulfur particles, and scale. These solid particles may form after the water is already downstream of solid removal equipment. Without proper consideration of dissolved gases, the benefits of installing solids removal equipment can thus be partially negated.

In any solids removal system, there is a need for equipment to handle the bulk solids or sludge removed from the water and a procedure for removal of these solids. For many common water injection systems, the amount of bulk solids to be removed can be rather large. If the solids are free of oil, they may be disposed of in slurry piped to pits onshore or overboard offshore. If coated with oil, they may require treating prior to disposal. Treating oil from solids is beyond the scope of this chapter. Oil is normally separated from produced solids by abrasion in hydrocyclones or by washing with detergents or solvents.

Selection of a specific design of water treating system for removing suspended solids and dissolved gases from a water source requires establishing the year-round quality of the water source. This determination normally requires that tests be performed to identify the amount of dissolved gases [primarily oxygen and hydrogen sulfide (H_2S)] present in the water, the total mass of suspended solids and their particle size distribution, and the amount of oil present in the source. In addition, if a source of water is to be injected into a reservoir, it must be checked to ensure that it is compatible with reservoir water; that is, that under reservoir conditions, dissolved solids will not precipitate in sufficient quantity to plug the well or reservoir. Similarly, if two sources of water are to be mixed on the surface, they must be checked for compatibility under surface conditions of pressure, temperature, and pH. The tests are normally performed by laboratories that specialize in offering these services. (Determination of allowable concentration and particle size of solids and the acceptable level of dissolved gases in injection water is beyond the scope of this text.)

First, the theory involved in the various processes for removal of solids and dissolved gases from water is discussed. Next, the equipment used in both processes is discussed, and, finally, a design procedure to follow in selecting the equipment for a specific application is presented.

Treating water for solids removal and for removal of dissolved gases are really two separate concepts, using two separate sets of theories and equipment. They are combined in this chapter only because both are usually considered together when designers plan a water treating system for water injection. With this presentation the designer can select the design required to prepare any water stream for several common uses. Water softening, potable water making, and boiler feed water preparation, however, are several important water treating topics that are not within the scope of this chapter.

Solids Removal Theory

Removal of Suspended Solids from Water

For a variety of reasons, it may be desirable to remove suspended solids from a water stream. This removal is most commonly done as part of a water injection system for water-flood or enhanced oil recovery. It may also be necessary to remove suspended solids prior to injecting produced water in disposal wells.

Two different principles have been used to develop equipment for removing suspended solids from water. Gravity settling uses the density difference between the solid particle and the water to remove the solids; filtration traps the solids within a filter medium that allows water to pass.

The quantity of suspended solids in a water stream is normally expressed in milligrams per liter (mg/l) or parts per million (ppm) by weight (mg/l divided by water specific gravity equals ppm). The size of the suspended particles is usually expressed as a diameter stated in units of micrometers (10^{-6} meters), also called microns. The capability of the equipment or filters to remove suspended solids is expressed in terms of removal of a percentage of all suspended solids having a diameter greater than a specified micron size. These values will usually range from 150 microns (μ) for gravity separators to less than 0.5 microns (μ) for filters. Suspended solids less than 40 microns (μ) in diameter cannot be seen with the naked eye. Figure 10-1 shows relative sizes for a variety of common materials.

Gravity Settling

The force of gravity may be used to remove solid particles from water if the density of these particles is not the same as the density of the water. Typically, solid particles have a density greater than water; therefore,

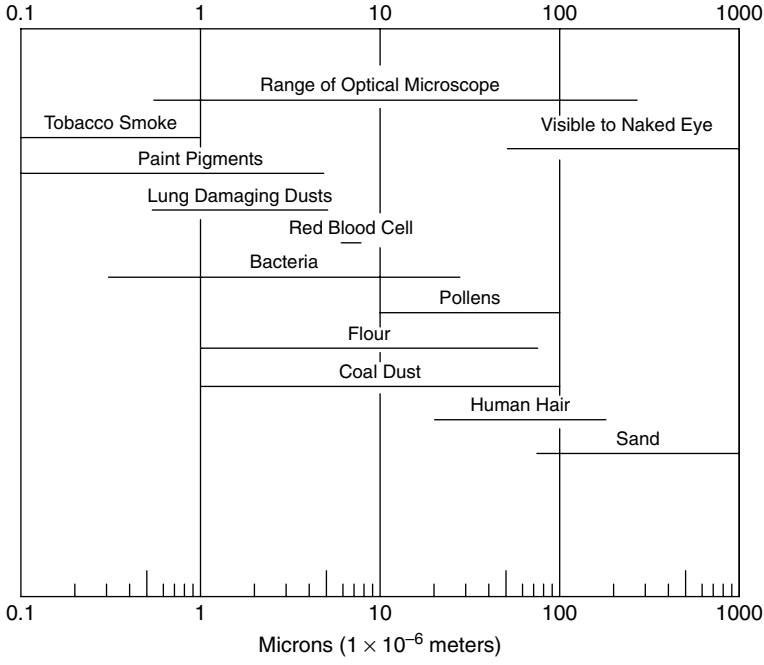


Figure 10-1. Relative sizes of common materials.

they fall relative to the water due to the force of gravity. The terminal settling velocity is such that the gravitational force on the particle equals the drag force resisting its motion due to friction. Assuming the particle is roughly spherical, the drag force may be determined as follows:

$$F_D = C_D A_\rho \left[\frac{V_t^2}{2_g}\right], \tag{10-1}$$

where

F_D = drag force, lb (kg),
C_D = drag coefficient,
A = cross-sectional area of particle, $ft^2 (m^2)$,
ρ = density of the continuous phase, $lb/ft^3 (kg/m^3)$,
V_t = terminal setting velocity of the particle, ft/s (m/s),
G = gravitational constant, 32.2 $ft/s^2 (9.81 m/s^2)$.

The terminal settling velocity of small particles through water is low, and flow around the particle is laminar. Therefore, Stokes' law may be applied to determine the drag coefficient as follows:

$$C_D = \frac{24}{\mathrm{Re}} \tag{10-2}$$

where Re is the Reynolds number.

By equating the gravitationally induced negative buoyant force with the drag force, one may derive the following equation for calculating the terminal settling velocity of the particle:

Fields Units

$$V_t = \frac{1.78 \times 10^{-6}(\Delta \mathrm{SG}) d_m^2}{\mu}, \tag{10-3a}$$

SI Units

$$V_t = 5.44 \times 10^{-10} \frac{(\Delta \mathrm{SG}) d_m^2}{\mu}, \tag{10-3b}$$

where

ΔSG = difference in specific gravity of the particle and the water,
d_m = particle diameter, microns (μ),
μ = viscosity of the water, cp (Pas).

Equation (10-3) may be used to size any of several types of equipment designed to use gravity settling. Such devices as settling ponds, pits, flumes, and tanks are commonly used onshore where space is available.

Parallel plate interceptors, such as CPIs (corrugated plate interceptors) and cross-flow separators, can be effective at removing suspended solids from water. However, the solids tend to cling to the plates and plug the plate pack. For this reason, parallel plate interceptors are not normally used to remove large quantities of solids from water.

Other devices such as hydrocyclones and centrifuges also take advantage of the density differences between the water and the suspended solid particles. These devices induce centrifugal forces in the water, causing the heavy solid particles to move away from the axis of rotation. Gravity settling in large tanks and vessels relies on low fluid velocities and large particle sizes (greater than 10 microns) to be most effective.

Flotation Units

Fines, oil, and oil wetted solids, which cannot be removed by gravity settling, may be removed by gas flotation units. In these units, a bubble attaches to the contaminant particle, lowering the effective weight of the particle and allowing it to rise to the surface, where it is removed by skimming.

Flotation units can be classified as dissolved gas or induced gas, depending on the gas supply mechanism. Dissolved gas flotation (DGF) introduces a gas/water solution in saturation at high pressure into the wastewater stream. Induced gas flotation (IGF) forms gas bubbles and provides turbulence for mixing by rotating mechanical diffusers or by recirculating a portion of the water through gas eductors. See Chapter 9 for a more complete description of flotation units.

Although flotation is a very common process used in the mining industry to separate metals from crushed rock slurries, it is not commonly used for solids removal in production operations.

Filtration

Filtration can be used to remove suspended solid particles from water by passing the water through a porous filter medium. As the water passes through the small pores in the filter medium, particles larger than the pores become trapped. The size of the pores in a filter medium determines the smallest particles that may be trapped.

Suspended solids are separated from fluids via three mechanisms: inertial impaction, differential interception, and direct interception.

Inertial Impaction

Particles (1 to 10 microns) in a fluid stream have mass and velocity and, hence, have a momentum associated with them. As the liquid and entrained particles pass through a filter media, the liquid stream will take the path of least resistance to flow and will be diverted around the fiber. The particles, because of their momentum, tend to travel in a straight line and, as a result, those particles located at or near the center of the flow line will strike or impact upon the fiber and be removed. Figure 10-2 illustrates this process. The fluid stream, shown as solid lines, flows around the filter fibers while the particles continue along their path, shown as dashed lines, and strike the fibers. Generally, larger particles will more readily deviate from the flow lines than small ones. In practice, however, because of the differential densities of the particles and fluids

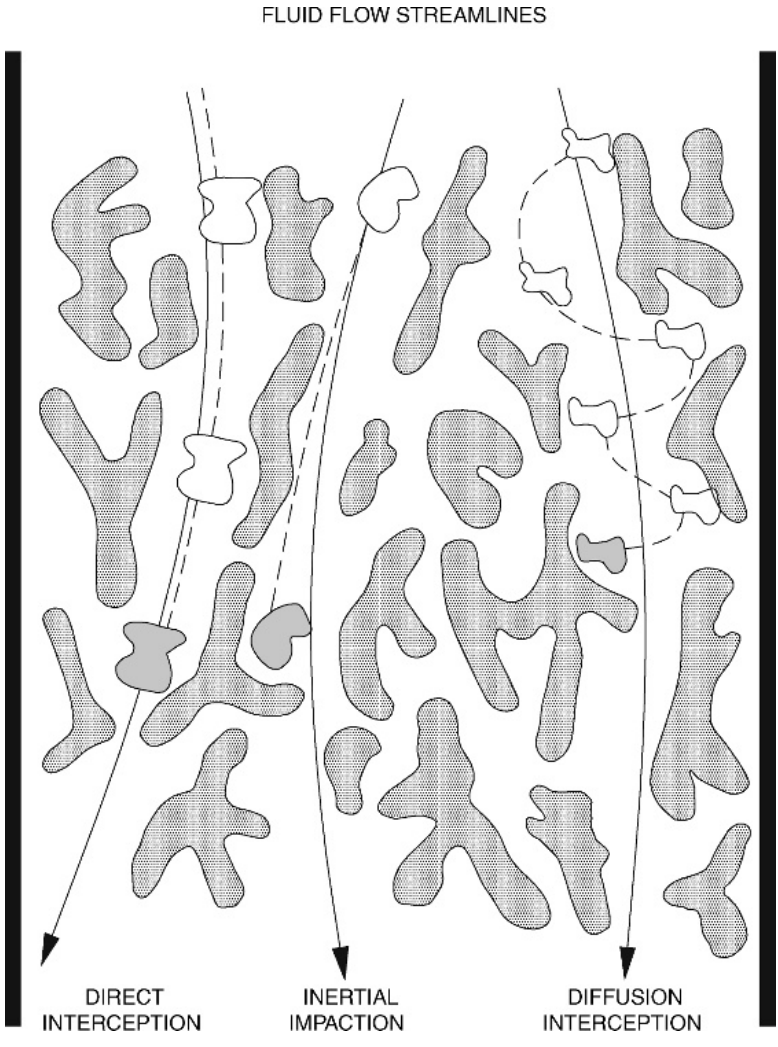


Figure 10-2. Filtration mechanisms.

are very small, deviation from the liquid flow line is much less and hence inertial impaction in liquid filtration plays a relatively small role.

Diffusional Interception

For particles that are extremely small (i.e., those with very little mass and less than 0.3 microns in diameter), separation can result from diffusional interception. In this process, particles are in collision with the

liquid molecules. These frequent collisions cause the suspended particles to move in a random fashion around the fluid flow lines. Such movement, which can be observed microscopically, is called "Brownian motion." Brownian motion causes these smaller particles to deviate from the fluid flow lines and hence increase the likelihood of their striking the fiber surface and being removed. Figure 10-2 shows the particle flow characterized by Brownian motion and impacting the filter fibers. As with inertial impaction, diffusional interception has a minor role in liquid filtration because of the inherent nature of liquid flow, which tends to reduce the lateral movement or excursions of the particle away from the fluid flow lines.

Direct Interception

While inertial impaction and diffusional interception are not as effective in liquid service as in gas service, direct interception is equally as effective in both and is the desired mechanism for separating particles from liquids. In a filter medium, one observes not a single fiber, but rather an assembly of a large number of such fibers. These fibers define openings through which the fluid passes. If the particles in the fluid are larger than the pores or openings in the filter medium, they will be removed as a result of direct interception. Figure 10-2 portrays this removal mechanism. Direct interception is easily understood in the case of a woven mesh filter with uniform pores and no thickness or depth; once a particle passes through an opening, it proceeds unhindered downstream. Yet such a filter will collect a very significant proportion of particles whose diameter is smaller than the openings or pores of the medium. Several factors that help account for this collection are

- Most suspended particles, even if quite small when viewed from some directions, are irregular in shape and hence can "bridge" an opening.
- A bridging effect can also occur if two or more particles strike an opening simultaneously.
- Once a particle has been stopped by a pore, that pore is at least partially occluded and subsequently will be able to separate even smaller particles from the liquid stream.
- Specific surface interactions can cause a small particle to adhere to the surface of the internal pores of the medium. For example, a particle considerably smaller than a pore is likely to adhere to that pore provided the two surfaces are oppositely charged. A very strong, negatively charged filter can cause a positive charge to be induced on a less strongly charged negative particle.

Direct interception can also obviously occur in filters in which the pore openings are not uniform, but instead vary in size (but within carefully controlled limits) throughout the thickness of the filter medium, resulting in a tortuous flow path.

Filter Types

In recent years it has become increasingly common to classify filters and filter media as either "depth type" or "surface type." Unfortunately, filter manufacturers have been unable to agree upon an "official" definition of the terms. As a result, much misunderstanding is encountered in the field on this subject. Hopefully, this discussion will separate fact from fiction.

A number of available filter media provide different pore sizes for solids removal. Depending on their construction, filter media may be divided into either nonfixed-pore or fixed-pore types.

Nonfixed-Pore Structure Media

Nonfixed-pore structure filters depend principally on the filtration mechanisms of inertial impaction and/or diffusional interception to trap particles within the spaces of their internal structure.

The nonfixed filters are constructed of nonrigid media. Variations in pressure drop through nonfixed filters may cause minor deformation or movement of the filter medium, potentially changing the size of some of the pores in the medium (hence the name "nonfixed pore"). Nonfixed-pore filters are by far the most common type of filters and include the following:

- Unbonded fiberglass cartridges,
- Cotton-wound or sock filters,
- Molded cellulose cartridges,
- Spun-wound polypropylene cartridges,
- Sand and other granular media beds,
- Diatomaceous earth filters.

Nonfixed-pore structure-type filters depend not only on trapping but also on adsorption to retain particles. As long as the dislodging force exerted by the fluid is less than the force retaining the particle, the particle will remain attached to the medium. However, when such a filter has been onstream for a length of time and has collected a certain amount of particulate matter, a sudden increase in flow and/or pressure can overcome these retentive forces and cause the release downstream of some of the

particles. This unloading will frequently occur after the filter has been in use for some time and can give a false impression of long service life for the filter.

Most nonfixed-pore structure filters are subject to media migration. This means that parts of the filter medium become detached and continue to pass downstream, contaminating the effluent (fluid that has passed through the filter).

Fixed-Pore Structure Media

Fixed-pore media filters consist of either layers of medium or a single layer of medium having depth, depend heavily on the mechanism of direct interception to do their job, and are so constructed that the structural portions of the medium cannot distort and that the flow path through the medium is tortuous. It is true that such filters retain some particles by adsorption as a result of inertial impaction and diffusional interception. It is also true that they contain pores larger than their removal rating. However, pore size is controlled in manufacture so that quantitative removal of particles larger than a given size can be assured.

Fixed-pore filters are constructed such that the pore size does not change. Such filters represent relatively new technology in filter medium construction for oil-field use and include the following:

- Resin-impregnated cellulose cartridges,
- Resin-bonded glass fiber cartridges,
- Continuous polypropylene cartridges.

As solids are trapped in a filter, some of the available flow paths are blocked. This blockage causes an increased pressure drop through the filter and may cause minor movement within the filter medium, which can result in unloading and/or media migration. (Unloading refers to previously trapped solids being released downstream; media migration refers to portions of the media being released downstream.) Media migration almost always has some unloading associated with the release of media material. These two phenomena usually cause a sudden decrease in the pressure drop through the filter. It should be noted that, by definition, a fixed-pore filter does not exhibit unloading or media migration.

Eventually, a filter collects solids until the pressure drop is too large for continued operation. At this point, the filter medium must be replaced or cleaned. The amount of solids a filter may remove per unit volume is referred to as the filter's "solids loading." Different types of filters may have vastly different solids loading capabilities.

A filter's solids loading capacity is affected by the filter design and the particular medium used. Figure 10-3, for example, shows portions

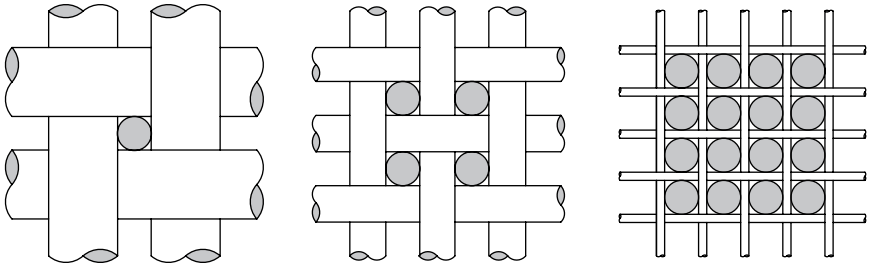


Figure 10-3. Fiber diameter affects filter's solids capacity.

of three filters using different fiber media, which can be glass fiber, cellulose, cotton, or polypropylene. All three sections represent the same filter area with the same pore sizes. The only difference among the three is the fiber diameter used to form the medium. The right side represents a filter with 16 times as many pores per unit volume as the top filter; its solids loading should thus be much larger. This is true even though the filters may be made of the same materials and may appear the same to the naked eye.

Surface Media

A surface or screen filter is one in which all pores rest on a single plane, which therefore depends largely upon direct interception to separate particles from a fluid. Only a few filters on the market today, for example, woven wire mesh, woven cloth, and certain membrane filters, qualify as surface filters.

A surface or screen filter will stop all particles larger than the largest pore opening. While particles smaller than the largest pore may be stopped because of factors previously discussed (bridging, etc.), there is no guarantee that such particles will not pass downstream. Woven wire mesh filters are currently available with openings down to 5 microns.

Summary of Filter Types

It should be apparent that classifying filters as depth or surface is meaningless. Almost all filters exhibit "depth" when viewed under a microscope. A more meaningful classification of filters is as follows:

1. Nonfixed-pore structures have pores whose dimensions increase at high pressure ("wound," low-density filters).

2. Fixed-pore structures have pores that do not increase in size at high pressures (most membrane filters).
3. Screen media (woven cloth or screens).

Fixed-pore structure filters are superior for most purposes when compared with the screen type. They combine high dirt capacity per unit area with both absolute removal of particles larger than a given size and minimum release of collected particles smaller than this rated size under impulse conditions.

Nonfixed-pore structure filters do not have absolute ratings, are subject to media migration, and can unload particles on impulse. Thus, often a fixed-pore guard filter is installed downstream of these filters. On the other hand, nonfixed pore filter such as multi-media filter can be designed for much higher solids loadings by varying the size of the filter media through its depth. The initial layers with larger pore sizer remove the greatest weight of solids (those having the largest diameters) while successive layers remove smaller and smaller diameter solids from the flow stream.

Removal Ratings

Particular attention should be paid to filter rating. Various rating systems have evolved to describe the filtration capabilities of filter elements. Unfortunately, there is no generally accepted rating system, and this tends to confuse the filter user. Several of the rating systems now in use are described below.

Nominal Rating

Many filter manufacturers rely on a nominal filter rating, which has been defined by the National Fluid Power Association (NFPA). The NFPA states, that the nominal filter rating is "An arbitrary micron value assigned by the filter manufacturer, based upon removal of some percentage of all particles of a given size or larger. It is rarely well defined and not reproducible." In practice, a "contanimant" is introduced upstream of the filter element, and subsequently the effluent flow (flow downstream of the filter) is analyzed microscopically. A given nominal rating of a filter means that 98% by weight of the contaminant above the specified size has been removed; 2% by weight of the contaminant has passed downstream.

Note that this is a gravimetric test rather than a particle count test. Counting particles upstream and downstream is a more meaningful way to measure filter effectiveness. The various tests used to give

nonfixed-pore structure filters a nominal rating yield results that are misleading. Typical problems are as follows:

1. The 98% contaminant removal by weight is determined by using a specific containment at a given concentration and flow. If any one of the test conditions is changed, the test results could be altered significantly.
2. The 2% of the contaminant passing through the filter is not defined by the test. It is not uncommon for a filter with a nominal rating of 10 μ to pass particles downstream ranging in size from 30 to over 100 μ.
3. Test data are often not reproducible, particularly among different laboratories.
4. Some manufacturers do not base their nominal rating on 98% contaminant removal by weight, but instead a contamination removal efficiency of 95%, 90%, or even lower. Thus, it often happens that a filter with an absolute rating of 10 μ is actually finer than another filter with a nominal rating of 5 μ. Therefore, it is always advisable to check the criteria upon which a nominal rated filter is based.
5. The very high upstream contaminant concentrations used for such tests are not typical of normal system conditions and produce misleading high-efficiency values. It is common for a wire-mesh filter medium with a mean (average) pore size of 15 μ to pass a 10-μ nominal specification. However, at normal system contaminations, this same filter medium will pass almost all 10-μ particles.

Therefore, one cannot assume that a filter with a nominal rating of 10 μ will retain all or most particles 10 μ or larger. Yet some filter manufacturers continue to use only a nominal rating both because it makes their filters seem finer than they actually are and because it is impossible to place an absolute rating on a nonfixed-pore structure.

Absolute Rating

The NFPA defines an absolute rating as follows: "The diameter of the largest hard spherical particle that will pass through a filter under specified test conditions. It is an indication of the largest opening in the filter element." Such a rating can be assigned only to an integrally bonded medium.

There are several recognized tests for establishing the absolute rating of a filter. What test is used will depend on the manufacturer, on the type of medium to be tested, or sometimes on the processing industry. In all cases the filters have been rated by a "challenge" system. A filter

is challenged by pumping through a suspension of a readily recognized contaminant (e.g., glass beads or a bacterial suspension) and both the influent and effluent examined for the presence of the test contaminant.

The challenge tests are destructive tests—i.e., the challenged filter cannot be used after the test. Consequently, integrity tests for filters have been established, which are nondestructive and correlate with the destructive challenge test. In other words, if the test filter was successfully integrity tested by the nondestructive test, that would mean it would pass the destructive challenge test. However, after passing the integrity test, the filter element can be placed in service and will provide the user with the results claimed by the filter manufacturer.

Beta (β) Rating System

While absolute ratings are clearly more useful than nominal ratings, a more recent system for expressing filtration rating is the assignment of Beta ratio values. Beta ratios are determined using the Oklahoma State University "OSU F-2 Filter Performance test." The test, originally developed for use on hydraulic and lubricating oil filters, has been adapted by many filter manufacturers for rapid semi-automated testing of filters for service with aqueous liquids, oils, or other fluids.

The Beta rating system is simple in concept and can be used to measure and predict the performance of a wide variety of filter cartridges under specified conditions. The rating system is based on measuring the total particle counts at several different particle sizes, in both the influent and effluent streams. A profile of removal efficiency is then given for the filter.

The Beta value is defined as follows:

$$\beta_X = \frac{\text{number of particles of a given size and larger in influent}}{\text{number of particles of a given size and larger in effluent}},$$

where X is the particle size, in microns.

The percent removal efficiency at a given particle size can be obtained directly from the Beta value and can be calculated as follows:

$$\%\ \text{removal efficiency} = \frac{\beta - 1}{\beta}(100).$$

The relationship between Beta values and percent removal efficiency is illustrated in Table 10-1.

Usually a $\beta = 5{,}000$ can be used as an operational definition of an absolute rating.

Table 10-1
Beta Ratio and Removal Efficiency Comparison

No. of Particles per ml ≥ 10 m				
Filter	**Influent**	**Effluent**	**Beta Ratio B_{10}**	**Removal Efficiency %**
A	10,000	5,000	2	50
B	10,000	100	100	99
C	10,000	10	1,000	99.9
D	10,000	2	5,000	99.98
E	10,000	1	10,000	99.99

The Beta values allow comparison of removal efficiencies at different particle sizes for different cartridges in a meaningful manner.

The type of filter medium, its rating (nominal, absolute, or Beta), and its solids loading are thus all important in selecting a filter. The designer should pay particular attention to the precise meaning of the manufacturer's rating.

Choosing the Proper Filter

Among the more important factors that must be taken into consideration when choosing a filter for a particular application are the size, shape, and hardness of the particles to be removed, the quantity of those particles, the nature and volume of the fluid to be filtered, the rate at which the fluid flows, whether the flow is steady, variable, and/or intermittent, the system pressure and whether that pressure is steady or variable, the available differential pressure, the compatibility of the medium with the fluid, the fluid temperature, the properties of the fluid, the space available for particle collection, and the degree of filtration required. Let's examine how some of these factors affect filter selection.

Nature of Fluid

The materials from which the medium, the cartridge hardware, and the housing are constructed must be compatible with the fluid being filtered. Fluids can corrode the metal core of a filter cartridge or a pressure vessel, and the corrosion will in turn contaminate the fluid being filtered. Thus, it is essential to determine whether a fluid is acid, alkali, aqueous, oil- or solvent-based, etc.

Flow Rate

Flow rate through a filter is dependent on two general parameters, pressure drop available (ΔP) and resistance to flow at the filter media (R). Flow rate depends directly on pressure drop and inversely on resistance. Thus, for a constant R, the greater the pressure, the greater the flow. For a constant ΔP, lowering the resistance increases the flow.

Pressure drop can come from any number of sources and is usually expressed as pounds per square inch (psi) or bar. All other factors being equal, if the pressure drop available for a fluid is increased, then the flow rate of that fluid through the media will increase.

Viscosity is the resistance of a fluid to the motion of its molecules among themselves; in other words, a measure of the thickness or "flowability" of a fluid. Water, ether, and alcohol have low viscosities; heavy oils and syrup have high viscosities. Viscosity affects resistance directly. If all other conditions remain constant, since flow through the media is laminar doubling the viscosity in a filter system gives twice the original resistance to flow. Consequently, as viscosity increases, the pressure required to maintain the same flow rate increases. Centipoise is the unit of measurement comparing the viscosity of a fluid with that of water, which has a viscosity of 1 centipoise at 70°F.

Temperature

The temperature at which filtration will occur can affect the viscosity of the fluid, the corrosion rate of the housing, and the filter medium compatibility. Viscous fluids generally become less viscous as temperature increases. If a fluid is too viscous, it may be advisable to preheat the fluid and to install heater bands in the filter. Thus, it is important to determine the viscosity of a fluid at the temperature at which filtration will occur.

High temperature also tends to accelerate corrosion and to weaken the gaskets and seals of filter housings. Very often disposable filter media cannot withstand high temperatures, particularly for prolonged periods of time. It is for this reason that one must often choose porous metal, cleanable filters.

Pressure Drop

Everything a fluid flows through or by contributes resistance to the flow of that fluid in an additive fashion. The pressure losses due to flow of

the fluid through the tubing, piping, etc., couple with the pressure loss through the filter to produce resistance.

Resistance to flow through a clean filter will be caused by the filter housing, cartridge hardware, and filter medium. For a fluid of given viscosity, the smaller the diameter of the pores or passages in the medium, the greater the resistance to flow will be. When a fluid meets resistance in the form of a filter, the result is a drop in pressure downstream of that filter, and the measurement of the pressure drop across the filter is called the differential pressure, or ΔP. Thus, for all practical purposes the terms "pressure drop," "differential pressure," and "ΔP" are synonymous.

The more resistance a filter medium offers to fluid flow, the greater the differential pressure at constant flow will be. Since flow is always in the direction of the lower pressure, the differential pressure will cause fluid to flow. Thus, it is differential pressure that moves the fluid through the filter assembly and overcomes resistance to flow and ΔP.

In designing a filtration system, one must provide a sufficient pressure source not only to overcome the resistance of the filter, but also to permit flow to continue at an acceptable rate as the medium plugs so as to use fully the effective solids holding capacity of the filter. If the ratio of initial clean pressure drop through the filter to total available pressure is disproportionately high, unacceptable flow will quickly result as the filter plugs even though the medium's capacity for collecting solids has not been exhausted. When this occurs, the proper solution is usually to increase the inlet pressure by increasing pump head capacity or, as an alternative, to reduce clean pressure drop by increasing filter size.

The maximum allowable pressure drop is the limit beyond which the filter might fail structurally should additional system pressure be applied to maintain adequate flow. This limit is always specified by the filter manufacturer.

In choosing a pressure source, one must take into consideration the resistance to flow of the filter—both constant resistance components (filter housing and element hardware) and the variable resistance components of the filter cake and medium. As filtration proceeds at constant flow, there will be an increase in pressure drop made up of a constant component and an increasing variable component. Eventually, the increasing pressure drop component becomes so large that either the flow is reduced below design levels, or the filter is structurally damaged. In a well designed system enough pressure drop should be available to maintain design flow to near the filter's maximum allowable pressure drop.

If a pressure head exists downstream, as for example in an elevated receiver, this must be overcome without limiting the available pressure drop for the filter. In such cases, a check valve should be installed downstream of the filter to prevent reverse pressure from damaging the filter media.

The pressure drop across the filter assembly can be reduced by increasing the size of the assembly. This allows an increased number of filter elements to be installed which in turn increases the total throughput. This is usually an economical approach for continuous processes where the increase in the larger filter assembly, and thus total throughput, is offset by the cost of using multiple smaller assemblies with the same throughput.

Surface Area

The life of most screen and fixed-pore structure filters is greatly increased as their surface areas are increased. To understand why this is so, let us look at two filters of identical medium (thus subject to the same pressure drop limit) that pass the same fluid at the same flow rate.

The first filter has a surface area of $5\,ft^2$ and collects a 0.005-in.-thick (128-μ) filter cake in a 24-hour period. After 24 hours most of the pores are plugged, the pressure drop is 75 psi, and the useful life of the filter has been exhausted.

Let's increase the surface area of the filter to $30\,ft^2$ and calculate the useful life. If a filter with a surface of $5\,ft^2$ collects a filter cake of 0.005 in. in 24 hours, then at the same flow rate, a filter of $30\,ft^2$ will collect that same filter cake thickness in x hours. Thus,

$$\frac{5}{24} = \frac{30}{x},$$

$$5x = (30)(24),$$

$$x = 144\,\text{hr}.$$

While the 30-ft^2 filter has collected a filter cake of 0.005 inches in 144 hr, its useful life will not be exhausted because the pressure drop will not have reached 75 psi [there are 6 times as many pores to plug $(30/5 = 6)$]. Since the flow rate per ft^2 of filter area is in the ratio of 5/30, the pressure drop across the 30-ft^2 filter will be $(5/30)(75\,\text{psi}) = 12.5\,\text{psi}$.

If the 30-ft^2 filter has a pressure drop of 12.5 psi in 144 hr, then it will have a pressure drop of 75 psi in x hr. Thus,

$$\frac{12.5}{144} = \frac{75}{x},$$

$$12.5x = (75)(144),$$

$$x = 864 \text{ hr}.$$

The life of the 30-ft^2 filter is therefore 36 times that of the 5-ft^2 filter (864/24). If one calculates the square of the area ratio $(30/5)^2$, the answer is 36.

The benefit of opting for a filter assembly with a large surface area can be expressed as follows:

Let T = Volume of liquid which can be treated for a filter with area A, (ft^2). Then

$$T_1 = T_2 \left(\frac{A_1}{A_2}\right)^n,$$

where n is a life extension factor between 1 and 2. The life extension factor, n, will approach 2 provided that

1. The filter cake is not compressible. If the filter cake is compressible, n will tend to be nearer to 1.
2. The collected cake does not become a finer filter than the medium itself (i.e., collect finer solids as it builds up). To the extent that the filter cake acts as a finer filter than the medium itself, n will tend to approach 1.
3. The solids collected are relatively uniform in particle diameter.

From the above it is apparent that an increase in surface area will yield at least a proportional increase in service life. Under favorable circumstances, the ratio of service life may approach the square of the area ratio. In most cases, a filter user will save money in the long run by paying the higher initial cost of a larger filter assembly.

Void Volume

Void volume is always of great importance. All other factors being equal, the medium with the greatest void volume is most desirable because it

will yield the longest life and lowest initial clean pressure drop per unit thickness. As the fiber diameter decreases, the void volume increases, assuming constant pore size. Other factors, however, such as strength, compressibility as pressure is applied (which reduces void volume), compatibility of the medium with the fluid being filtered, cost of medium, cost of constructing that medium into a usable filter, etc., must all be considered when designing a filter for a particular application.

Degree of Filtration

The filter chosen for a given application must be able to remove contamination from the fluid stream to the degree required by the process involved. Once the size of the contaminants to be removed has been determined, it is possible to choose a filter with the particle removal characteristics needed to do the job. Choosing a filter with a pore size finer than required can be a costly mistake. Remember, the finer the filtration, the more rapid the clogging and the higher the cost will be.

The filter selected must be able to retain particles removed from the subject fluid. Depth-type filters of the type whose pores can increase in size as pressure is increased are subject to unloading. With surface filters or fixed structure filters, one selects a medium that will not change its structure under system-produced stress.

Prefiltration

The purpose of a prefilter or other device to remove bulk quantities of significant higher size is to reduce overall operating cost by extending the life of the final filter. Extending final filter life may not in itself be sufficient to justify prefiltration; overall cost reduction is usually the principal consideration.

Field experience indicates that for most applications where the solids are of near uniform size it is better to increase the final filter area rather than provide a prefilter. This is because increasing the final filter area always yields a longer cycle and lower operating costs. Doubling the area of the final filter will result in two to four times the life. On the other hand, installing settling devices, hydrocyclone desanders, or large pore space sand filters upstream of filters designed to remove very small particles is often a more economical solution than just increasing the size of the final “polishing” filter.

Coagulants and Flocculation

Suspended matter in water may contain very small particles that will not settle out by gravity or that may pass through filters. These particles may be removed by a coagulation-and-flocculation process. Coagulation is the process of destabilization by charge neutralization, and flocculation is the process of bringing together the destabilized or coagulated particles to form a larger agglomerate, or floc.

Coagulation and flocculation results are difficult to predict based on a water analysis; therefore, laboratory jar tests are performed to simulate the coagulation and flocculation condition. The laboratory data are then used to determine the basis for design and efficient operation. The tests are run to establish

- Optimum pH for coagulation,
- Most effective coagulation and coagulation aid,
- Most effective coagulation dosage and order of chemical addition,
- Coagulation and flocculation time,
- Settling time or flocculation time,

Chemicals used include

- Chlorine
- Bentonite (for low-turbidity water)
- Primary inorganic coagulants
- pH adjustment chemicals
- Polyelectrolytes.

Chlorine addition may assist coagulation by oxidizing organic contaminants that have dispersing properties. Waters with high organic content require high coagulant demand. Chlorination prior to addition of coagulant feed may reduce the required coagulant dosage.

The term "polyelectrolytes" refers to all water-soluble organic polymers used for clarification of water by coagulation. The available water-soluble polymers may be classified as anionic, cationic, or approaching neutral charge. They are typically long-chain, high-molecular-weight polymers with many charge sites to aid in coagulation and flocculation. Violent mixing of polyelectrolytes may break the chains and cause them to be less effective. However, some mixing is required to ensure that the chemical and solids come into contact. Turbulent flow in piping provides sufficient mixing if the chemical is injected far enough upstream of the equipment.

Chemicals can be added to the water in clarifiers, which are tanks that contain mixers that cause sufficient turbulence to create contact between

the chemical and the solids. Coagulation/flocculation chemicals can also be added in flotation units to aid in attaching gas bubbles to the solid particles. Polyelectrolytes added to the feed stream to filtration units have proven effective at increasing filtration efficiency.

Measuring Water Compatibility

Scale deposits are usually salts or oxides of calcium, magnesium, iron, copper, and aluminum. Common scale deposits may consist of calcium carbonate ($CaCO_3$), calcium phosphate ($CaPO_4$), calcium silicate ($Ca_2Si_2O_4$), calcium sulfate ($CaSO_4$), or magnesium hydroxide [$Mg(OH)_2$], magnesium phosphate ($MgPO_4$), and magnesium silicate ($Mg_2Si_2O_4$).

The tendency of water to form scale or cause corrosion is measured by either the Langelier Scaling Index (LSI), which is also called the Saturation Index, or the Ryznar Stability Index (RSI), which is also called the Stability Index (Table 10-2).

The LSI deals with the conditions at which given water is in equilibrium with calcium carbonate and provides a qualitative indication of the

Table 10-2
Saturation Index

To determine:	
pCa:	Locate ppm value for Ca as $CaCO_3$ on the ppm scale. Proceed horizontally to the left diagonal line down to the pCa scale.
pAlk:	Locate ppm value for "M" Alk as $CaCO_3$ on the ppm scale. Proceed horizontally to the right diagonal line down to the pAlk scale.
Total solids:	Locate ppm value for total solids on the ppm scale. Proceed horizontally to the proper temperature line and up to the "C" scale.
Example:	
Temp. = 140 °F, pH = 7.80	pCa = 2.70
Ca hardness = 200 ppmw	pAlk = 2.50
M alkalinity = 160 ppmw	C at 140 °F = 1.56
Total solids = 400 ppmw	Sum = pH3 = 6.76
	Actual pH = 7.50
	Difference = 1.04
	= Saturation index

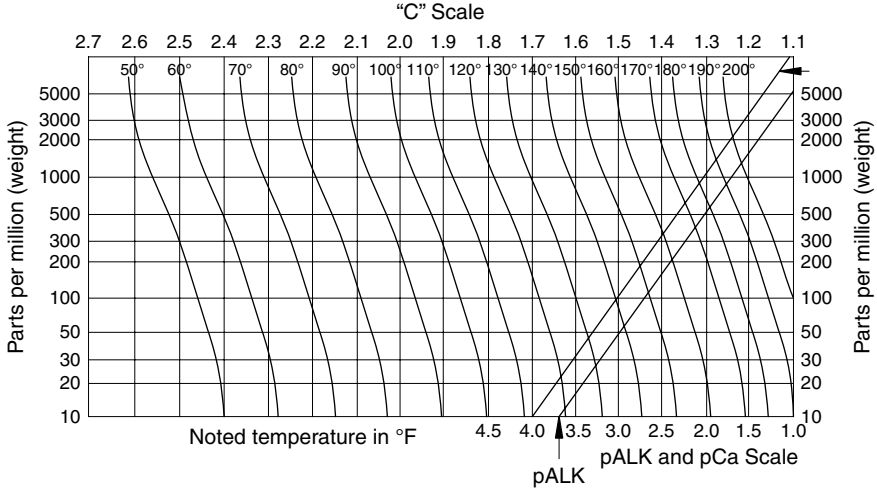


Figure 10-4. Langelier Saturation Index Chart (reprinted from GPSA Engineering Data Book, courtesy of Betz Laboratories, Inc.).

tendency of calcium carbonate to deposit or dissolve. The index is determined by subtracting from the actual pH of the water sample (pHA), a computed value (pHS) based on the ppm of calcium hardness as $CaCO_3$, alkalinity hardness as $CaCO_3$, and total solids, as shown in Figure 10-4. If the index is positive, calcium carbonate will tend to deposit. If it is negative, calcium carbonate will tend to dissolve.

The Stability Index (RSI) is given by

$$\text{RSI} = 2\ \text{Ph}_\text{S} - \text{pH}_\text{A}.$$

When the index is less than 6, scaling can be expected; an index between 6 and 7 indicates a stable water. A pH greater than 7 indicates potential corrosion problems.

Scaling may be controlled by the following: blow-down to limit build-up of solids concentrations; acid treatment to reduce the water alkalinity; and use of commercial scale inhibitors such as polyphosphates, phosphonates, and polymers.

Solids Removal Equipment Description

This section describes several different processes designed to remove suspended solids or dissolved gases from water. As stated, the most common reason for such water treatment is associated with water injection;

therefore, the subject of water injection will also be discussed throughout this chapter. The principles and equipment involved can also be applied when the water must be treated for other reasons.

Figure 10-5 shows a schematic of steps that may be required to prepare water for injection. As shown in the figure, the choice of process is affected by the water source.

Almost every oil or gas production facility must deal with some produced water at some time during its production life. In many facilities the water may be disposed of once its hydrocarbon content is reduced to acceptable levels. When water is to be injected, it may be necessary to treat the water to attain low levels of oil (on the order of 25 to 50 mg/l) to prevent impairment of the injection formation.

Typically, produced water may require filtration to remove dispersed oil. Filtering will remove all but a very small amount of the dispersed oil. It will be necessary, however, to clean the water to less than approximately 50 mg/l suspended oil so that oil does not create plugging problems in the filters. The small amount of suspended oil in the water will be removed by the filter and will contaminate the filter backwash.

Surface water is a common water source for water-flood or other injection projects. If surface water is available, it may be cheaper to obtain than subsurface water. However, surface water may require more treatment than other water sources; and, since it is fresh, it may cause swelling of clays in some formations.

The surface water should be free of large contaminants such as plant or marine life. The strainer in Figure 10-5 is intended to prevent such material from entering treatment facilities. Surface water is exposed to the atmosphere; therefore, it contains dissolved oxygen. Typically, the oxygen requires removal to minimize corrosion and bacterial growth. The oxygen concentration within the water will vary, depending on the water temperature; therefore, de-aeration equipment must be designed to remove the maximum anticipated oxygen content. Oxygen concentrations of approximately 8 ppm are typical for most surface waters.

Chemical injection of biocides is of particular importance for treating surface water. The water contains microscopic marine life, bacteria, plankton, and algae, which cannot be allowed to continue to grow within treating equipment.

Injection of other chemicals may also be required for other reasons. Corrosion inhibitor may be required to minimize deterioration of the surface facilities. Bactericide may be used on either a periodic or a continuous basis to minimize bacteria growth. Chemicals such as oxygen scavengers may be injected to react with contaminants to remove or change them. Similarly, chemicals may be injected to prevent the formation of scale

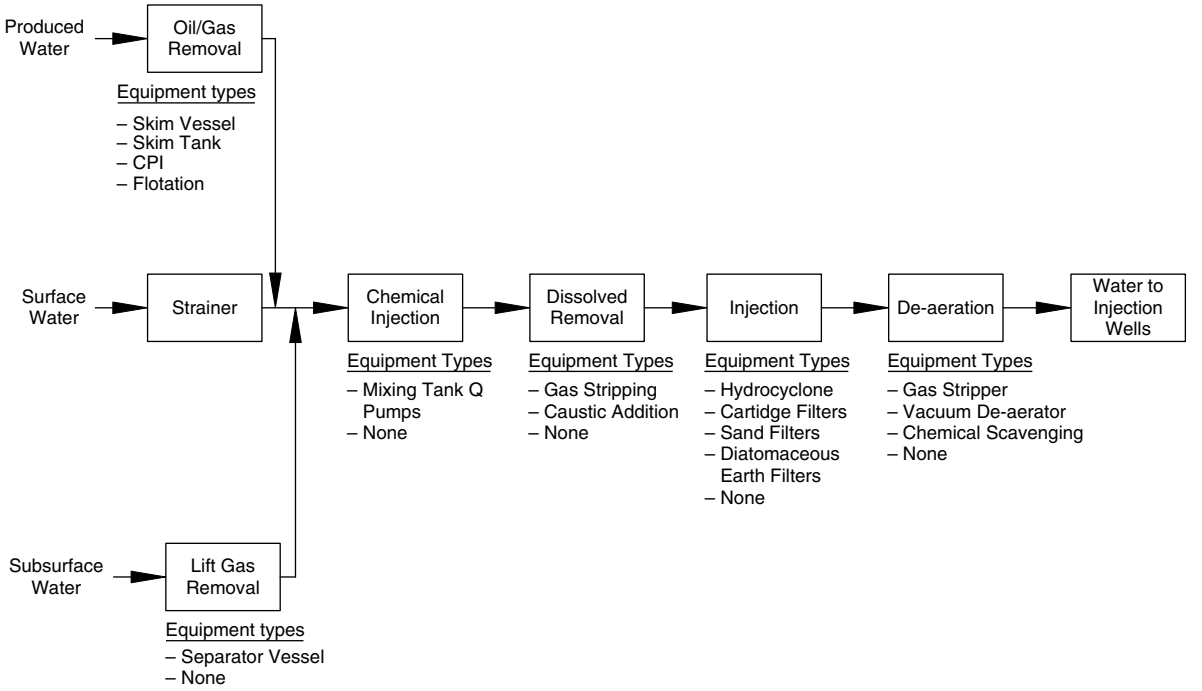


Figure 10-5. Water injection system treatment steps and equipment types.

within the surface equipment or the precipitation of solids under reservoir conditions.

Subsurface water from source water wells may also be used. Typically, if water is available at all, the water zone will be close to the surface, compared with hydrocarbon zones. Therefore, the cost of drilling and maintaining source water wells is typically much lower than the same cost for producing wells. However, the water rarely will flow to the surface under pressure; therefore, it must be pumped or gas lifted.

If the water is gas lifted to the surface, separation facilities will be required. Typically, a simple two-phase separator is sufficient, since small levels of dissolved natural gas in the water are not harmful to the equipment or the injection formation. If the gas used to lift the water contains acid gases or oxygen, then some treating may be necessary to remove or neutralize these harmful gas components.

Subsurface water is typically the least expensive source water to treat. However, the cost of drilling source water wells and the pumping or gas lifting expenses may make this the most expensive water to obtain.

Water from any source contains dissolved minerals and salts in addition to the suspended solids. Generally, these dissolved salts will remain in solution and are thus not a problem. However, if produced water is to be mixed with other source water as part of a waterflood, the water compatibility should be verified. Mixing other waters with produced water or changing the produced water's pH or temperature may cause dissolved solids to precipitate.

The compatibility of the injected water and the water in the injection formation should also be checked to ensure that the conditions within the reservoir will not cause scale to form. Chemicals may be needed to inhibit scale formation under down-hole conditions.

Filtration to remove suspended solids, no matter what the source, is intended to minimize plugging of the formation. Solids that are in the injected water and are larger than a certain size may plug the formation at the well bore, causing the surface injection pressure to rise or the flow rate injected to fall.

The degree of filtration required depends on the permeability and pore size of the injection formation. The final selected filtration design should be one that will minimize formation plugging and reduce the frequency of remedial well work. This selection is an economic decision balancing the cost of remedial well work against the cost of the filter system. In cases where the water is being injected into disposal wells, filtration requirements might be relaxed. Disposal wells are typically less expensive to drill or work over and more readily fractured than wells injecting the water into a producing formation. Therefore, the economic risks of

plugging a disposal well may be less than those associated with a well injecting the water into the producing formation.

The test to check for compatibility of injection water with the receiving formation may consist of

- A chemical analysis of the proposal injection water to indicate basic cations and anions present,
- A core plug test to determine the maximum particle size that can be injected into the formation without undue plugging,
- A core plug test to determine and establish permissible injection rates and pressures.

Gravity Settling Tanks

The simplest treating equipment for removing solids from water is a gravity settling tank or vessel, which may be designed in either a vertical or horizontal configuration. In vertical settling tanks, the solid particles must fall countercurrent to the upward flow of the water. A typical vertical gravity settling vessel is shown in Figure 10-6. The water enters the vessel and flows upward to the water outlet. Solids fall countercurrent to the water and collect in the bottom. As shown, large-diameter vessels or tanks should have spreaders and collectors to distribute the water flow and minimize short-circuiting.

For small-diameter gravity settling vessels, the collected solids may be removed by periodically opening the sand drain shown in Figure 10-6. The use of a cone bottom rather than an elliptical head allows more complete removal of solids through the drain. Typically, an angle of 45 to 60° from the horizontal is used for the cone to overcome the angle of internal resistance of the sand and allow natural flow of solids when the drain is opened.

Any flash gases that evolve from the water leave the settling vessel through the gas outlet at the top of the vessel. The volume of flash gas must be kept to a minimum so the gas does not adversely affect the removal of small solids particles. If large amounts of gas are flashed, the small gas bubbles can adhere to solids particles and carry them to the water surface. The solids then may be carried out the water outlet.

In horizontal settlers, the solids fall perpendicular to the flow of the water, as shown in Figure 10-7. The inlet is often introduced above the water section so that flash gases may be separated from the water prior to separating the solids from the water.

The collected solids must be periodically removed from the vessel; thus, several drains may be placed along the length of a horizontal vessel.

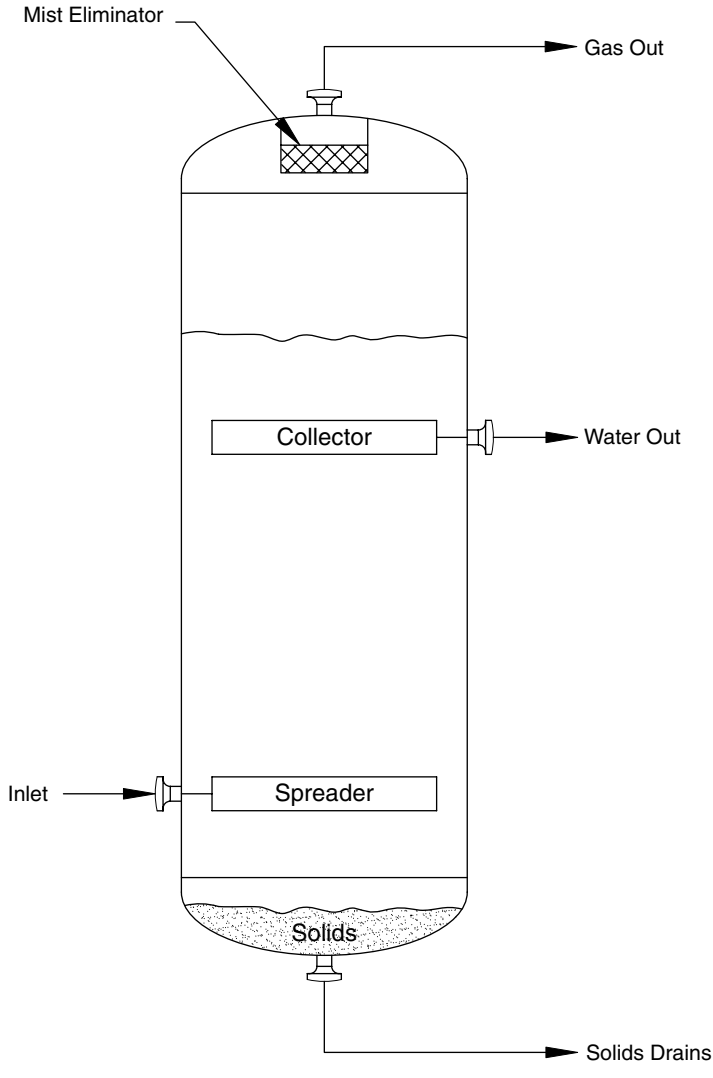


Figure 10-6. Schematic of vertical gravity settling vessel.

Since the solids will have an angle of repose of 45 to 60°, the drains must be spaced at very close intervals and operated frequently to prevent plugging. The addition of sand jets in the vicinity of each drain to fluidize the solids while the drains are in operation is expensive, but sand jets proved successful in keeping drains open. Alternatively, the vessel may have to be shut down so that solids may be manually removed through

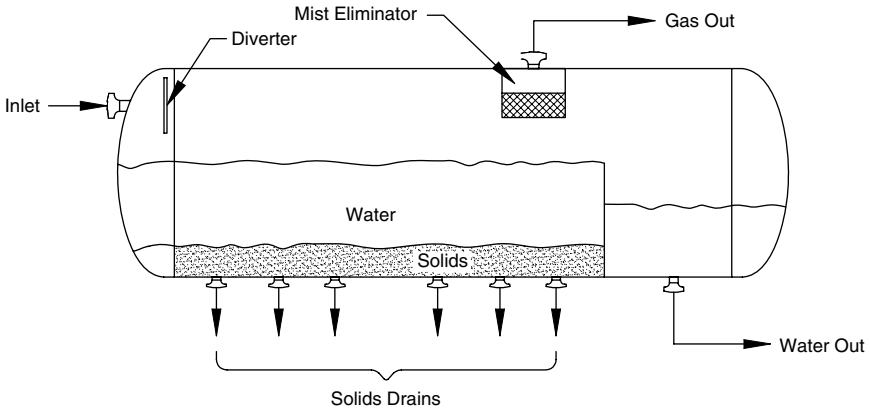


Figure 10-7. Schematic of horizontal gravity settling vessel.

a manway. Although effective, this method can be extremely expensive and time-consuming.

Horizontal vessels are more efficient at solids separation because the solid particles do not have to fall countercurrent to the water flow. However, other considerations, such as the difficulty of removing solids, must be kept in mind when such a configuration is selected. Horizontal vessels require more plan area to perform the same separation as vertical vessels. Most small horizontal vessels have less liquid surge capacity. For a given change in liquid surface elevation, there is typically a larger increase in liquid volume for a horizontal vessel than for a vertical vessel sized for the same flow rate. However, the geometry of most small horizontal vessels causes any high-level shutdown device to be located close to the normal operating level. In large-diameter [greater than 6 ft (1.8 m)] horizontal vessels and in vertical vessels, the shutdown could be placed much higher, allowing the level controller and dump valve more time to react to a surge.

It should be pointed out that vertical vessels have some drawbacks that are not process-related; these must be considered in making a selection. For example, the relief valve and some of the controls may be difficult to service without special ladders and platforms. The vessel may have to be removed from a skid for trucking due to height restrictions.

The choice of a pressure vessel versus an atmospheric tank for a solids-settler depends on the overall needs of the system. Although pressure vessels are more expensive than tanks, they should be considered when potential gas blow-by through an upstream vessel dump system could create too much backpressure in an atmospheric tank's vent system;

or when the water must be dumped to a higher elevation for further treating and a pump would be needed if an atmospheric tank were installed.

For gravity settling of solids, water retention time does not directly affect the solids removal, and only settling theory must be considered. Some small retention time is required for evolved gases to flash out of solution and reach equilibrium. This process usually requires less than 30 seconds; therefore, retention time criteria rarely govern vessel size.

Gravity settling theory is useful for settling large-diameter solids. It is normally used where there is a high solids (greater than 50-m) flow rate of large-diameter solids that might otherwise quickly overload equipment designed to separate smaller-diameter solids from the liquid stream.

Horizontal Cylindrical Gravity Settlers

The required diameter and length of a horizontal cylindrical settler can be determined from Stokes' law as follows:

Field Units

$$dL_{\text{eff}} = 1000\frac{\beta_w Q_w \mu_w}{\alpha_w(\Delta\text{SG})d_m^2}, \tag{10-4a}$$

SI Units

$$dL_{\text{eff}} = 1.2\times 10^9\frac{\beta_w Q_w \mu_w}{\alpha_w(\Delta\text{SG})d_m^2}, \tag{10-4b}$$

where

d = vessel's internal diameter, ft (m),
L_{eff} = effective length in which separation occurs, ft (m),
Q_w = water flow rate, BWPD (m^3/hr),
μ_w = water viscosity, cp,
d_m = particle diameter, microns μ,
ΔSG = difference in specific gravity between the particle and water relative to water,
β_w = fractional water height within the vessel (h_w/d),
α_w = fractional cross-sectional area of water,
h_w = water height, in. (m).

Equation (10-4) assumes a turbulence and short-circuiting factor of 1.8. Any combination of d and L_{eff} that satisfies this equation will be sufficient to allow all particles of diameter d_m or larger to settle out of the water.

The fractional water height and fractional water cross-sectional areas are related by the following equation:

$$\alpha_w = (1/180)\cos^{-1}[1-2\beta_w] - (1/\pi)[1-2\beta_w]\sin\left[\cos^{-1}(1-2\beta_w)\right]. \tag{10-5}$$

By selecting a fractional water height within the vessel, one may calculate the associated fractional cross-sectional area using Eq. (10-5); the resulting values may then be used in Eq. (10-4).

In addition to the settling criteria, a minimum retention time should be provided to allow the water and flash gases to reach equilibrium. Typically, retention times to reach equilibrium are less than 30 seconds. Although retention time requirements rarely govern the size of a settling vessel, these requirements should be checked. To ensure that the appropriate retention time has been provided, the following equation must also be satisfied when selecting d and L_{eff}:

Field Units

$$d^2 L_{\text{eff}} = \frac{(t_r)_w Q_w}{1.4\alpha_w} \tag{10-6a}$$

SI Units

$$d^2 L_{\text{eff}} = 21{,}000\frac{(t_r)_w Q_w}{1.4\alpha_w}. \tag{10-6b}$$

The choice of the correct diameter and length can be obtained by selecting various values for d and L_{eff} for both Eqs. (10-5) and (10-6). For each d the larger L_{eff} must be used to satisfy both equations.

The relationship between the L_{eff} and the seam-to-seam length of a settler is dependent on the settling vessel's internal physical design. Some approximations of the seam-to-seam length may be made based on experience as follows:

$$L_{\text{ss}} = (4/3)L_{\text{eff}}, \tag{10-7}$$

where L_{ss} is the seam-to-seam length. This approximation must be limited in some cases, such as vessels with large diameters. Therefore, the

L_{ss} should be calculated using Eq. (10-7), but it must exceed the value calculated using the following equations:

Field Units

$$L_{ss} = L_{eff} + 2.5, \tag{10-8a}$$

where $L_{eff} < 7.5$ ft.

SI Units

$$L_{ss} = L_{eff} + 0.76. \tag{10-8b}$$

Field Units

$$L_{ss} = L_{eff} + \frac{d}{24}. \tag{10-9a}$$

SI Units

$$L_{ss} = L_{eff} + \frac{d}{2000}. \tag{10-9b}$$

Equation (10-8) only governs when the calculated L_{eff} is less than 2.3 m (7.5 ft). The justification of this limit is that some minimum vessel length is always required for smoothing the water inlet and outlet flow. Equation (10-9) governs when one-half the diameter in feet exceeds one-third of the calculated L_{eff}. This constraint ensures that even flow distribution can be achieved in short vessels with large diameters.

Horizontal Rectangular Cross-Sectional Gravity Settlers

Similarly, the required width and length of a horizontal tank of rectangular cross section can be determined from Stokes' law as

Field Units

$$WL_{eff} = 70\frac{Q_w \mu_w}{(\Delta SG) d_2^m}, \tag{10-10a}$$

SI Units

$$WL_{\text{eff}} = 9.7 \times 10^5 \frac{Q_w \mu_w}{(\Delta \text{SG}) d_2^m}, \tag{10-10b}$$

where

W = width, ft (m),
L_{eff} = effective light in which separation occurs, ft (m).

Equation (10-10) assumes a turbulence and short-circuiting factor of 1.9. Note that Eq. (10-10) is independent of height because the particle settling time and the water retention time are both proportional to the height. Typically, the height is limited to one-half of the width to promote even flow distribution.

As before, an equation may be developed to ensure that sufficient retention time is provided. If the height-to-width ratio is set, then the following retention time equation applies:

Field Units

$$W^2 L_{\text{eff}} = \frac{0.004 (t_r)_w Q_w}{\gamma}, \tag{10-11a}$$

SI Units

$$W^2 L_{\text{eff}} = \frac{(t_r)_w Q_w}{60\gamma}, \tag{10-11b}$$

where

γ = height-to-width ratio (H_w/W),
H_w = height of the water, ft (m).

As with horizontal cylindrical settlers, the relationship between L_{eff} and L_{ss} depends on the internal design. Three approximations of the L_{ss} of rectangular settling vessels may be made using Eq. (10-4). However, the L_{ss} must be limited by Eq. (10-5) and the following:

$$L_{\text{ss}} = L_{\text{eff}} + W/2. \tag{10-12}$$

As before, the L_{ss} should be the largest of Eqs. (10-7), (10-8), and (10-12).

Vertical Cylindrical Gravity Settlers

The required diameter of a vertical cylindrical tank can be determined by setting the settling velocity equal to the average water velocity as follows:

Field Units

$$d^2 = 6700F\frac{Q_w\mu_w}{(\Delta\text{SG})d_2^m}, \tag{10-13a}$$

SI Units

$$d^2 = 6.5\times 10^{11}F\frac{Q_w\mu_w}{(\Delta\text{SG})d^m}, \tag{10-13b}$$

where F is the factor that accounts for turbulence and short-circuiting. For small-diameter settlers, the short-circuiting factor should be equal to 1.0. Settlers with diameters greater than 48 in. (1.22 m) require a larger value for F. Inlet and outlet spreaders and baffles affect the flow distribution in large settlers and therefore affect the value of F. It is recommended that, for large-diameter settlers, F should be set equal to $d/48$. Substituting this into Eq. (10-13) gives the following:

Field Units

$$d = 140\frac{Q_w\mu_w}{(\Delta\text{SG})d_2^m}, \tag{10-14a}$$

SI Units

$$d = 5.3\times 10^{9}\frac{Q_w\mu_w}{(\Delta\text{SG})d_2^m}, \tag{10-14b}$$

where $d > 48$ in. (1.22 m). Equation (10-14) applies only if the settler diameter is greater than 48 in (1.22 m). For smaller settlers, Eq. (10-13) should be used and F should equal 1.0.

The height of water column in feet can be determined for a selected d from retention time requirements:

Fields Units

$$H = 0.7\frac{(t_r)_w Q_w}{d^2}, \tag{10-15a}$$

SI Units

$$H_w = 21{,}000\frac{(t_r)_w Q_w}{d^2}, \tag{10-15b}$$

where H is the height of water, in ft (m).

Plate Coalescers

The equations for sizing the various configurations are identical to those presented in Chapter 8 and can be used directly, where d_m is the diameter of the solid particle (and not the oil droplet diameter) and ΔSG is the difference in specific gravity between the solid and water (and not between oil and water).

Plate coalescers are not addressed in this section because they have a tendency to plug and are thus not recommended for solids separation.

Hydrocyclones

Hydrocyclones, also called desanders or desilters, operate by directing the water into a cone through a tangential inlet that imparts rotational movement to the water. Figures 10-8a and 10-8b show a hydrocyclone cone and an assembly of eight cones.

The rotary motion generates centrifugal forces toward the outside of the cone, driving the heavy solids to the outer perimeter of the cone. Once the particles are near the wall, gravity draws them downward to be rejected at the apex of the cone. The resulting heavy slurry is then removed as "underflow." The clear water near the center of the vortical motion is removed through an insert at the centerline of the hydrocyclone, called a "vortex finder," and passes out as "overflow" through the top of the cone.

The advantage of hydrocyclones is that the centrifugal forces separate particles without the need for large settling tanks. Operationally, hydrocyclones are good at removing solids with diameters of approximately 35 microns and larger.

A major drawback of hydrocyclones is that during upsets in flow or pressure drop, the rotary motion in the cone may be interrupted, possibly causing solids to carry over into the overflow liquid. Other drawbacks are wear problems, large pressure drops, and limited ability to handle surges in flow. Some manufacturers offer replaceable liners to handle wear problems.

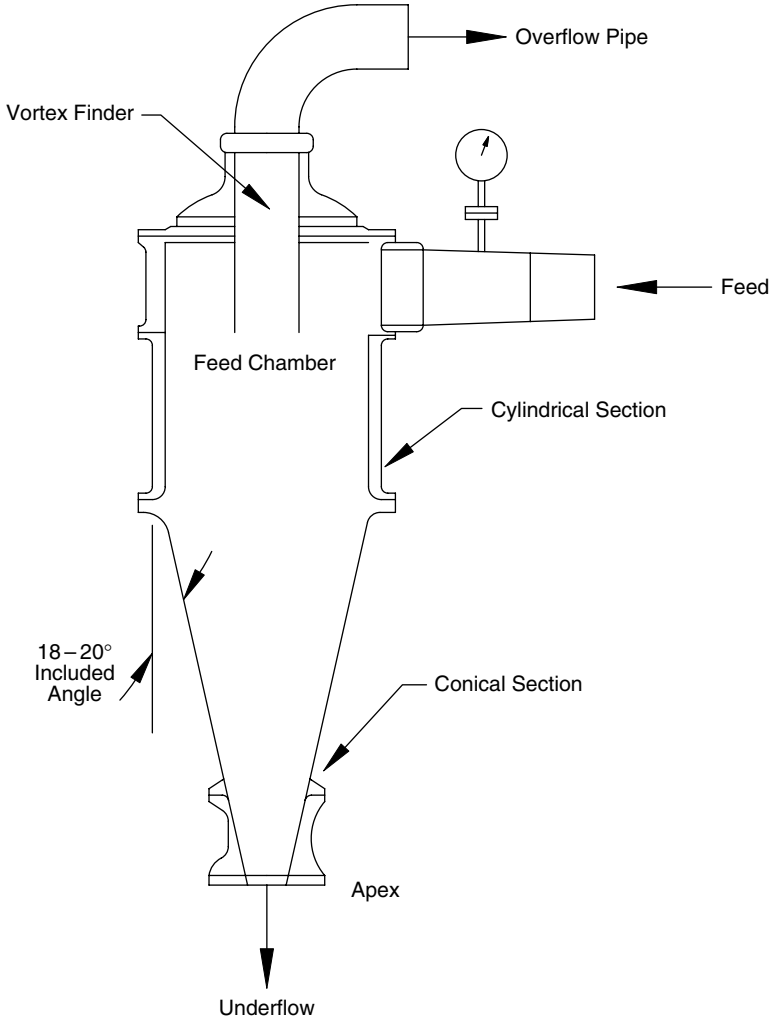


Figure 10-8a. Schematic of a hydrocyclone.

Hydrocyclones are rarely used as the only solids removal device, although they can remove very high loadings of solids, making them very useful as a first step in solids removal. If filters are used as a second step, the hydrocyclone can greatly lengthen the filters' cycle time. At the same time, the filters can provide removal of the smaller-diameter solids and protect against carryover from the hydrocyclones during upsets.

The ability of a hydrocyclone to separate a certain diameter solid particle (fineness of separation) is affected by the differential pressure

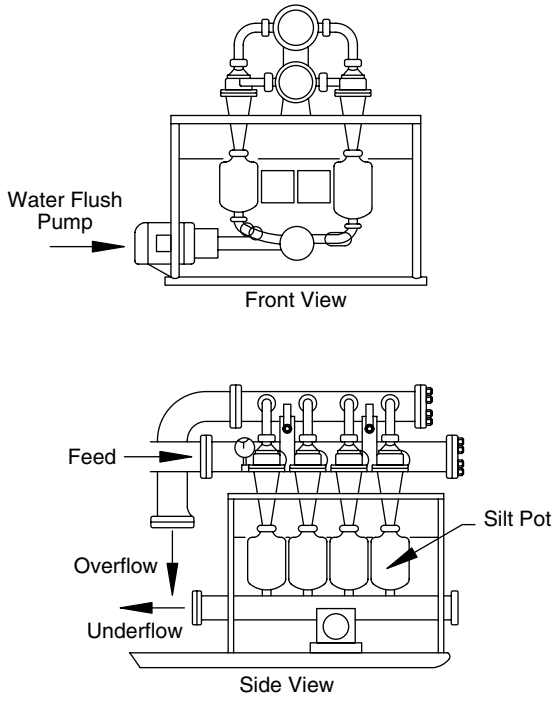


Figure 10-8b. Schematic of a hydrocyclone.

between the inlet and overflow, the density difference between the solid particles and the liquid, and the geometry and size of the cone and inlet nozzle. The pressure drop through the cone is the critical variable in terms of affecting fineness of separation and is itself a function of flow rate. Thus, the lower the flow rate, the lower the pressure drop and the coarser the separation. A minimum flow must be maintained through each cone to create the required pressure drop and rotary motion to ensure separation. Typically, hydrocyclones are operated with a 25- to 50-psi (140- to 275-kPa) pressure drop.

Many theoretical and empirical equations have been proposed for calculating fineness of separation. All reduce to the following form for a hydrocyclone of fixed proportions:

$$d_{50} = K\left[\frac{D^3\mu}{Q_w \Delta \mathrm{SG}}\right]^{\frac{1}{2}}, \tag{10-16}$$

where

D = major diameter of hydrocylone cone,
D_{50} = solid particle diameter that is recovered 50% to the overflow and 50% to the underflow, microns (μ),
μ = slurry viscosity, cp (Pas),
Q_w = slurry flow rate, BPD (m^3/hr),
ΔSG = difference in specific gravity between the solid and the liquid,
K = proportionally and shape constant.

The diameter of solid particle that is recovered 1 to 3% to the overflow and 97 to 99% to the underflow is

$$d_{99} = 2.2\, d_{50}. \tag{10-17}$$

The flow rate through a hydrocyclone of fixed proportions handling a specified slurry is given by

$$Q_w = K^I(\Delta P)\frac{1}{2}, \tag{10-18}$$

where

Q_w = flow rate, BWPD (m^3/hr),
K^I = proportionally and shape constant,
ΔP = pressure drop, psi (kPa).

Equations (10-16), (10-17), and (10-18) can be used to approximate the performance of a hydrocyclone for different flow conditions, if its performance is known for a specific set of flow conditions.

Solids discharge in the underflow slurry is performed in either an open or a closed system. With the open system, the slurry is rejected through an adjustable orifice at the apex of the cone to an open trough. The orifice can be adjusted to regulate the flow rate of the water leaving with the solids. The open system can allow oxygen entry into the system.

In the closed system, a small vessel called a "silt pot" is connected to the apex, which remains open. A valve is located at the bottom of the silt pot and is normally closed. Solids pass through the apex and collect in the bottom of the silt pot. The valve at the bottom of the silt pot is opened periodically to reject the solids. The opening and closing of this valve can be manual or automatic.

Hydrocyclone units may be put on line individually, thus providing some ability to account for changes in flow rate. When specifying a hydrocyclone unit, the design engineer must provide the following information:

- Total water flow rate,
- Particle size to be removed and the percentage of removal required,
- Concentration, particle size distribution, and specific gravity of particles in the feed,
- Design working pressure of the hydrocyclone,
- Minimum pressure drop available for the hydrocyclone.

With this information the designer can select equipment from various manufacturers' catalog descriptions.

Centrifuges

Centrifuges can be used to separate low-gravity solids or very high percentages of high-gravity solids. The principle involved is the same as in a hydrocyclone in that centrifugal force rapidly separates solids from the liquid. Centrifuges typically require extensive maintenance and can handle only small liquid flow rates. For these reasons centrifuges are not commonly used in water treating applications.

Flotation Units

It is possible to remove small particles using dispersed or dissolved gas flotation devices. These units are primarily used for removing suspended hydrocarbons from water. Gas is normally dispersed into the water or released from solution in the water, forming bubbles approximately 30 to 120 microns (μ) in diameter. The bubbles form on the surfaces of the suspended particles, creating particles whose average density is less than that of water. These rise to the surface and are mechanically skimmed. In the feed stream, chemicals called "float aids" are normally added to the flotation unit to aid in coagulation of solids and attachment of gas bubbles to the solids. The optimum concentration and chemical formulation of float aids are normally determined from batch tests in small-scale plastic flotation models on site. Because of the difficulty of predicting particle removal efficiency with this method, it is not normally used to remove solids from water in production facilities.

Disposable Cartridge Filters

Cartridge filters are simple and relatively lightweight; and they can be used to meet a variety of filtration requirements. A typical cartridge filter vessel is shown in Figure 10-9. The water enters the top section and must flow through one of the filter elements to exit through the lower section of the vessel. The top head of the vessel is bolted so that the cartridges can be changed when the pressure drop across them reaches an upper limit. A relief valve can be included in the vessel to prevent excessive differential pressure between the upper and lower sections of the vessel.

Filter cartridges are available in a wide variety of materials, and they provide a range of performance options. Cartridges are available with manufacturers' particle size ratings from 0.25 microns (μ) to any larger particle size. When selecting a filter cartridge, the designer must determine what the manufacturer's rating actually means in terms of removal percentage.

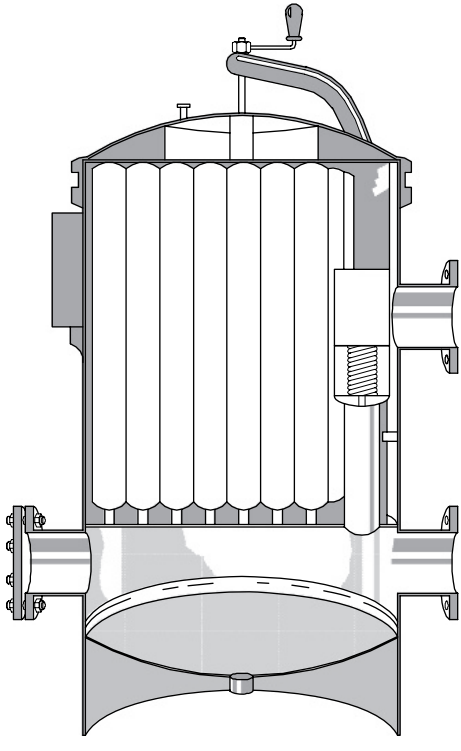


Figure 10-9. Cartridge filter vessel. (Courtesy of Perry Equipment Corp.)

Filter cartridge solids removal performances and allowable flow rates vary greatly from manufacturer to manufacturer, even if the cartridges are made of the same material. Therefore, it is difficult to develop generalized relationships between the water flow rate and filter area. As a result, it is necessary to rely on manufacturers' information when selecting and sizing a cartridge filter system.

In designing a water treatment system that includes cartridge filters, it may be desirable to select a fixed-pore filter medium and absolute rated filters. The fixed-pore cartridges provide more consistent particle removal efficiencies from one cartridge to the next than do nonfixed-pore cartridges. The fixed-pore type also prevents solids unloading and media migration during periods of high differential pressure. Fixed-pore filters are usually given absolute ratings by their manufacturers.

Nonfixed-pore cartridges may be used, but the differential pressure across the filters must be monitored closely. High differential pressures may cause solids unloading and media migration. If either occurs, the pressure drop through the filter will decrease and may be below the limit when the cartridge is scheduled to be changed. Therefore, the operator checking the pressure drop will believe that the cartridges are functioning correctly, even though large amounts of solids may have been released to the downstream water.

Solids unloading may be avoided by using a high differential pressure switch to continuously monitor the pressure drop or by changing the cartridges when the pressure drop is still small compared to the maximum pressure drop recommended by the manufacturer. The resulting frequent changing of the cartridges may result in excessive operating costs if the early change-out method is used.

Typically, cartridge filters have low solids-loading limits, so the cartridges can absorb only a relatively small amount of solids before they must be changed. Manufacturers have developed special cartridges to improve solids loading. Pleated construction of a thin filter medium such as paper or cotton fabric greatly increases the effective filter surface area of the cartridge. The increased surface area provides for higher flow rates and solids-loading capacities than a cylindrical cartridge of the same medium.

Some cartridges use a multilayered design of media such as fiberglass, which provides in-depth filtration. The layers of media have progressively smaller pores as the water moves from the outside to the inside of the cartridge. As the pore size changes, particles are trapped at different depths within the filter, allowing higher solids loadings but typically decreasing flow rates slightly.

Since cartridge filters have low solids-loadings capacities, it is common to install primary solids removal equipment upstream of the cartridge filters. Typical systems include either a hydrocyclone or a sand filter

followed by the cartridge filter. The upstream equipment removes the larger solids and reduces the amount of solids that the cartridges must remove, therefore extending the time between cartridge changes.

A spare filter vessel may be provided so that cartridges may be changed without reducing water flow rates. Any number of vessels can be used to provide the required number of cartridges, but the most common system arrangements include three 50% vessels or four 33% vessels. The number of filter vessels selected depends on a cost analysis and on operating preference.

Other factors to consider in the selection of cartridge filters are the type of filter medium and its characteristics. As an example, polypropylene cartridges are a better selection than cotton for water service, since cotton swells. The compatibility of filter membranes and binders with chemical additives or impurities in the water should be checked. The designer should contact specific manufacturers for detailed information.

When specifying a cartridge filter unit the following information should be included:

- Maximum water flow rate,
- Particle size to be removed by filtration and the percentage of removal required,
- Solids concentration in the inlet water,
- Design working pressure of the filter vessel,
- Maximum pressure drop available for filtration.

Backwashable Cartridge Filters

Backwashable cartridge filters are available in a variety of designs using metal screens, permeable ceramic, or consolidated sand as a filter medium. Filters of this type are simple and lightweight like the disposable cartridge filters, but they have the additional advantage of being backwashable. The media used in backwashable filters typically provide filtration of particles between 10 and 75 microns (μ).

Backwashable cartridge filters have low solids-loading limits; therefore, they have potentially short intervals between backwash cycles. It is important not to expose backwashable filters to differential pressures over approximately (170 kPa) because the particles may become too deeply imbedded in the pores to be removed by backwashing. With proper maintenance and repeated backwashing, this type of filter may last up to two years.

Regeneration or backwashing involves flowing clean water through the filter in the opposite direction of the normal filtration. Backwashable filters often require an acid backwash as well. The solids trapped in

the filter media are then forced out of the filter and carried away with the backwash fluid. This process is quicker and may be less costly than changing cartridges. The flow rate of fluid required for backwash is specified by the manufacturer.

The disadvantage of this system is that filtered water must be stored and then pumped through the filter. The resulting backwash fluid must then be directed to another storage medium. A method and equipment for disposing of the backwash fluid, which can be contaminated with oil or acid used in the backwash cycle, must also be provided.

Filters of this type are available in a variety of designs, including the cartridge filter vessel in Figure 10-9. Alternatively, each cartridge may be in a separate housing and the housings may be manifolded on a skid. With the manifold design, it is possible to backwash individual filters while the other filters continue to operate normally.

The designer should contact manufacturers for detailed information on selecting filters of this type. When specifying a backwashable cartridge filter, the designer should include the following:

- Maximum water flow rate,
- Particle size to be removed by filtration and the percentage of removal required,
- Solids concentration in the inlet water,
- Design working pressure of the filter vessel,
- Maximum pressure drop available for filtration.

Granular Media Filters

The terms "granular media filter" and "sand filter" refer to a number of filter designs in which fluid passes through a bed of granular medium. Typically, these filters consist of a pressure vessel filled with the filter media, as shown in Figure 10-10. Media support screens prevent the media solids from leaving the filter vessel.

The water to be filtered may flow either downward (down-flow) or upward (up-flow) through the media. As the water passes through the media, the small solids are trapped in the small pores between the media particles. Down-flow filters may be designed as either "conventional" (see Figure 10-11) or "high flow rate" (see Figure 10-12). Conventional down-flow filters are normally designed for an approximate flow rate range of 1 to 8 gpm/ft^2 (2.5 to 20 $m^3/hr\,m^2$), while high-flow-rate types may have flow rates as high as 20 gpm/ft (249 $m^3/hr\,m^2$). Up-flow filters (see Figure 10-13), on the other hand, are limited to less than 8 gpm/ft^2 (20 $m^3/hr\,m^2$) because higher flow rates may fluidize the media bed and, in effect, backwash the media.

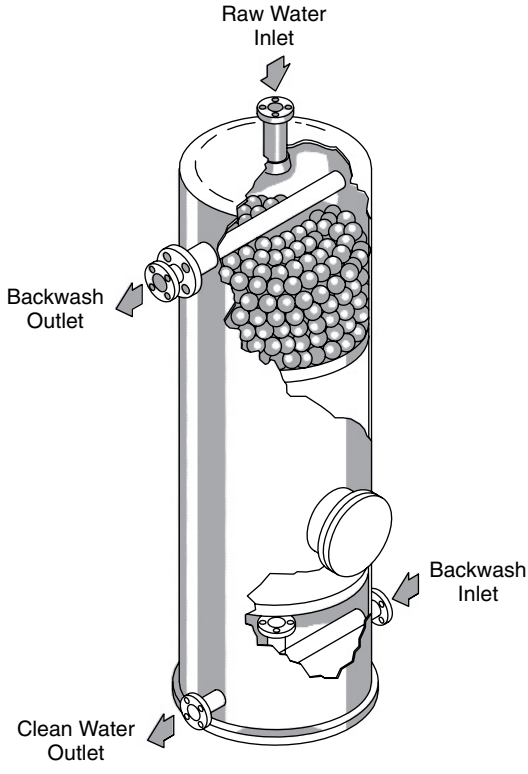


Figure 10-10. Down-flow granular media filter. (Courtesy of CE Natco.)

The advantage of the high-flow-rate filter over the conventional down-flow filter is that, at higher velocities, a deeper penetration of the bed is achieved, allowing a higher solids loading (weight of solids trapped per cubic foot of bed). This factor results in both a longer interval between backwashing and a smaller-diameter vessel. The disadvantage is that, with deeper penetration, inadequate backwashing may allow formation of permanent clumps of solids that gradually decrease the filter capacity. If fouling is severe, the filter media must be chemically cleaned or replaced.

Granular media filters must be cleaned periodically by backwashing to remove filter solids. The process involves fluidizing the bed to eliminate the small pore spaces in which solids were trapped during filtration. The small solids are then removed with the backwash fluid through a media screen that prevents loss of media solids. The filter media may be fluidized by flowing water upward through the filter at a high rate or by introducing the water through a nozzle that produces high velocities and turbulence within the filter vessel. Recycle pumps may be used to pump

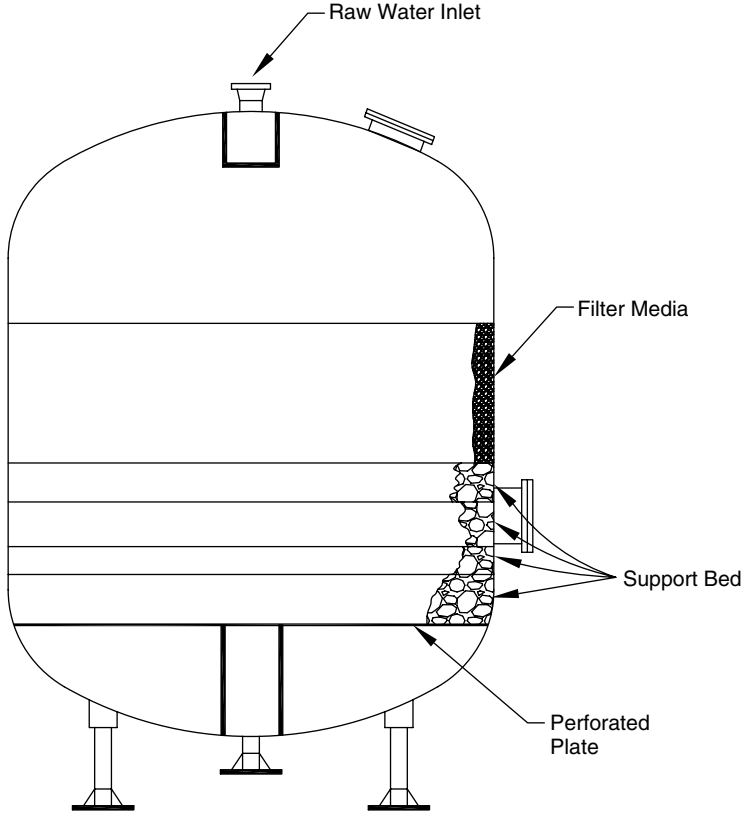


Figure 10-11. Conventional graded bed filter.

water through the fluidization nozzle to decrease the total water volume required to fluidize the filter media. As with backwashable cartridge filters, the backwash fluid must be collected for disposal.

The backwashing process is usually initiated because of a high pressure drop through the filter. Alternatively, the filter may be backwashed on a regular schedule, provided the pressure drop limit is not exceeded between backwash cycles. The cycle time for a sand filter depends on the water's solids content and the allowable solids loading of the individual filter. Conventional down-flow filters with flow rates of less than $8\,\text{gpm/ft}^2 (20\,\text{m}^3/\text{hr}\,\text{m}^2)$ typically can remove 1/2 to 1 1/2 lb/ft^2 (2.4 to $7.3\,\text{kg/m}^2$) of solids of filter media before backwashing. High-flow-rate filters may remove up to $4\,\text{lb/ft}^2 (19.5\,\text{kg/m}^2)$ prior to backwashing because the high water velocity forces small solids farther into the media bed, increasing the effective depth of the filter and thus the number

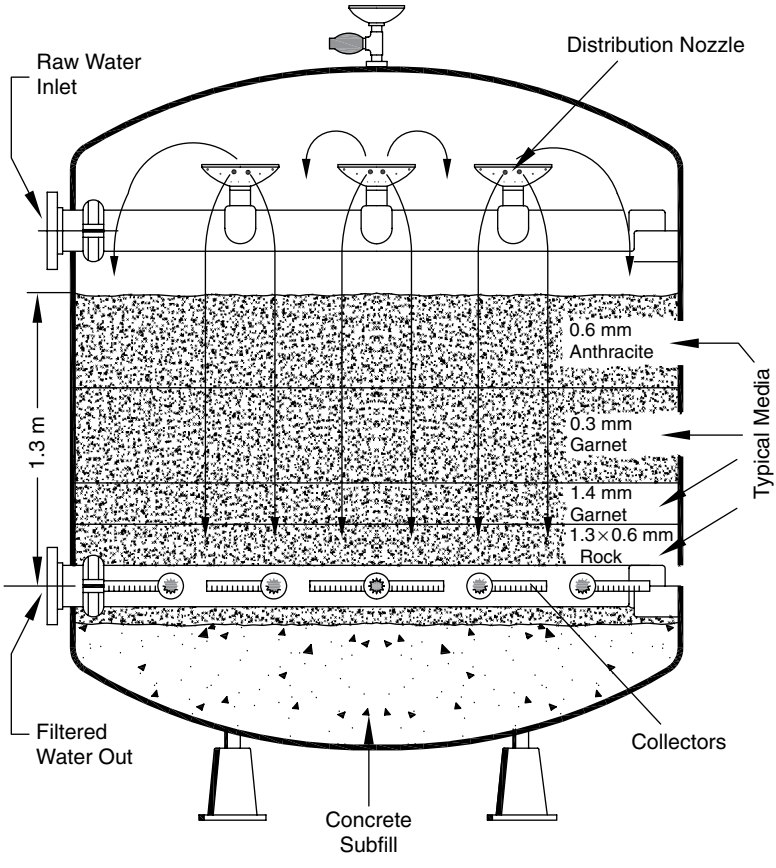


Figure 10-12. Deep bed down-flow (multimedia) filter.

of pores available to trap solids. Up-flow filters may remove up to $6\,\text{lb/ft}^2 (29.3\,\text{kg/m}^2)$ because the upward flow loosens and partially fluidizes the bed, allowing greater penetration by the small solids.

The decision to use down-flow or up-flow filters is normally governed by the influent suspended solids content and the preferred time between backwash cycles. Down-flow filters are normally used when the suspended solids content of the influent is below 50 mg/l, and up-flow filters are used for a suspended solids content range of 50 to 500 mg/l. Table 10-3 provides a comparison of typical influent flow rates and solids loadings.

Granular media filters fall in the category of nonfixed-pore filters because the filter media are not held rigidly in place. Thus, if not backwashed promptly, granular media filters can unload previously filtered

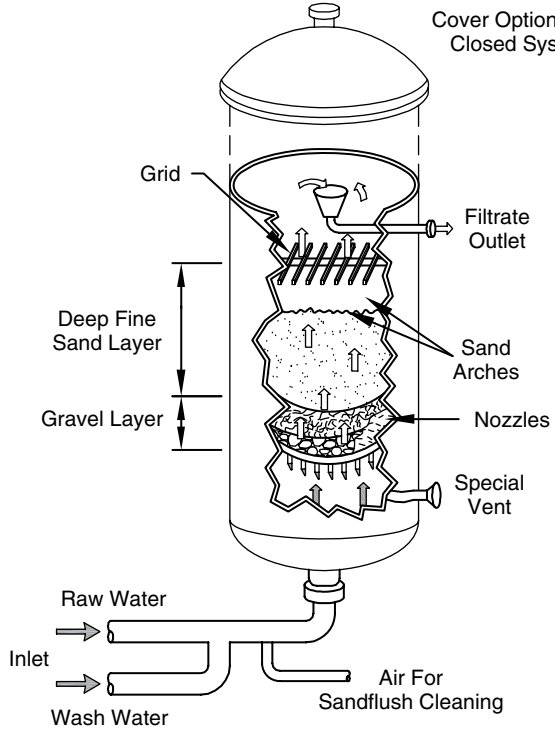


Figure 10-13. Deep bed up-flow filter.

Table 10-3
Typical Parameters for Granular Bed Filters

	Flow Rate		Solids Loading*	
Type	(m^3/hr m^2)	(gpm/ft^2)	(kg/m^2)	(lb/ft^2)
Conventional down-flow	2.4–19.6	1–8	2.4–7.3	0.5–1.5
High-rate down-flow	19.6–48.9	8–20	7.3–19.5	1.5–4
Up-flow	14.7–29.3	6–12	19.5–48.8	4–10

* Weight of solids trapped per unit area of media prior to backwashing.

solids. Media migration, however, is usually not a problem because media screens are usually built into the filter vessel, preventing the media from leaving the filter vessel.

Granular media filters use sand, gravel, anthracite, graphite, or pecan or walnut shells. The filter bed may be made of a single material or

of several layers of different materials to increase the solids loading by forcing the water through progressively smaller pores.

The pore size distribution within a granular media filter is variable, depending on the random distribution of the media solids after backwashing. Because of their variable pore size, granular media filters cannot be given an absolute rating. Typically, granular media filters can consistently remove 95% of all 10-μ and larger solids.

Backwash flow rates vary with specific filter designs and are specified by the manufacturer. Some designs require an initial air or gas scour [10- to 15-psig (69- to 103-kPa) supply] to fluidize the bed. This is especially true for filters handling produced waters that contain suspended hydrocarbons that can coat the filter media. Several cycles of scour followed by flushing may be required during the backwash operation. Detergents may also be needed to aid in cleaning the filter media.

Raw water is usually used for backwash. When the backwash cycle is complete, water is allowed to flow through the filter for a period of time until the effluent quality stabilizes. Only then is the filter put back on stream.

Filters work by trapping the solid particles within their pore structure. A filter's ability to trap particles smaller than the pore space may be greatly aided by the addition of polyelectrolytes and filter aids. These chemicals promote coagulation in the line leading to the filter and aid the formation of a chemical or ionic bond between these small particles and the filter medium. For example, a specific filter may be capable of removing 90% of the 10-μ and larger particles without chemicals and 98% of the 2-μ and larger particles with chemicals.

Granular media filters are commonly used as a first filtration step (normally called "primary filtration") prior to cartridge filters (known as "secondary filtration"). This type of system works well because the granular media filter removes the bulk of the large solids, thus increasing the cycle time for replacing cartridges. The cartridge filters then remove the small solids to the required size. In addition, the cartridges catch any solids released by the sand filter due to unloading. Tables 10-4 and 10-5 provide typical operating and design parameters for two types of granular media filters.

Specific manufacturers should be contacted to select a standard granular media filter and obtain detailed sizing and operating information. To select a granular media filter, the designer should specify the following:

- Maximum water flow rate,
- Particle size to be removed by filtration and the percentage of removal required,
- Solids concentration in the inlet water,
- Design working pressure of the filter vessel,
- Maximum pressure drop available for filtration.

Table 10-4
Typical Operating and Design Parameters for a Specific Up-Flow Filter

A. Operating Parameters	
Service rate	14.6 to 29.3 $m^3/hr\,m^2$ (6 to 12 gpm/ft^2)
Chemical treatment	Polyelectrolytes at 0.5 to 5 ppm Determine if needed by bench tests
Flush rate	Temperature-dependent (34.2 to 48.9 $m^3/hr\,m^2$, or 14 to 20 gpm/ft^2)
Regeneration time sequence	
Cycle 1:	
Drain	2 to 5 minutes (drain water to top of sand bed)
Fluidize bed	5 minutes with air or natural gas
Flush	10 to 20 minutes (until water is clear)
Cycle 2:	
Drain	3 to 5 minutes (drain water to top of sand bed)
Fluidize bed	5 minutes with air or natural gas
Flush	10 to 20 minutes (until water is clear)
Settle	5 minutes
Prefilter	15 to 20 minutes depending on water quality
B. Design Parameters	
Service rate	14.6 to 29.3 $m^3/hr\,m^2$ (6 to 12 gpm/ft^2) of filter area
Inlet solids	Will hold up to 49 kg of solids m^2 (10 lb/ft^2) of filter area (400 ppm maximum)
Inlet oil	Up to 50 ppm
Total outlet solids	2 to 5 ppm without chemical treatment 1 to 2 ppm with chemical treatment
Outlet oil	Less than 1 ppm
Cycle length	2-day minimum
Fluidize gas flow	55 to 90 m^3/hr per m^2 (3 to 5 cfm/ft^2) surface area (supply pressure of 83 to 109 kPa (12 to 15 psig))
Freeboard area	50 to 70% of total media depth
Bed expansion	Approximately 30% during flush cycle
Particle size removal	By theory, can be calculated from smallest sand (Barkman and Davidson)
C. Miscellaneous Data	

1. If inlet water contains above 15 ppm oil, a solvent or surfactant wash may be required during regeneration cycle number 1.
2. Sizing of media
   1st layer: 32 to 38 mm gravel, 101 mm thick (1 1/4 to 1 1/2 in. gravel, 4 in. thick)
   2nd layer: 10 to 16 mm gravel, 254 mm thick (3/8 to 5/8 in. gravel, 10 in. thick)
   3rd layer: 2 to 3 mm sand, 305 mm thick (2 to 3 mm sand, 12 in. thick)
   4th layer: 1 to 2 mm sand, 1524 mm thick (1 to 2 mm sand, 60 in. thick)

Table 10-5
Typical Operating and Design Parameters for a Specific Down-Flow Filter

A. Operating Parameters

Service rate	11.0 $m^3/hr\,m^2$ (4.5 gpm/ft^2)
Chemical treatment	20 ppm blend of cationic polyelectrolyte and sodium laminate
Regeneration	
Backwash	4 minutes at 41.6 $m^3/hr\,m^2$ (17 gpm/ft^2)
Rinse	4 minutes at 11.0 $m^3/hr\,m^2$ (4.5 gpm/ft^2)

B. Design Parameters

Service rate	4.9 $m^3/hr\,m^2$ (2 gpm/ft^2)
Inlet solids	< 20 ppm
Inlet oil	< 10 ppm

C. Miscellaneous Data

1. Sizing of media

	Thickness		Size		
Kind	(mm)	(in.)	(mm)	(in.)	Specific Gravity
Anthracite (top)	457	18	1.0 to 1.1	—	1.5
Sand	229	9	0.45 to 0.55	—	2.6
Garnet	76	3	0.2 to 0.3	—	4.2
Garnet	76	3	1.0 to 2.0	—	4.2
Gravel	76	3	4.8 × No. 10 Mesh	3/16 × No. 10 Mesh	2.6
Gravel	76	3	9.5 × 4.8	3/8 × 3/16	2.6
Gravel	76	3	19.0 × 9.5	3/4 × 3/8	2.6
Gravel	76	3	38.1 × 19.0	1 1/2 × 3/4	2.6
Rock	76	3	50.8 × 38.1	2 × 1 1/2	2.6

2. Filter may need detergent in backwash.
3. High amounts of oil during upset conditions may necessitate solvent washing the filter media.
4. Media can stick together and form balls with excessive chemicals or oil in the inlet and may require the bed to be replaced or cleaned.
5. Backwash rates in excess of 41.6 $m^3/hr\,m^2$ (17 gpm/ft^2) may cause carryover of anthracite, especially when backwash water is cold.

Diatomaceous Earth Filters

For filtration of 0.5- to 1.0-μ particles, diatomaceous earth (DE) filters may be used. In the past DE filters were commonly used for removing very fine solids because they were the least costly filters available in this range. Recently, manufacturers have developed cartridge filters that can effectively remove 0.25-μ solids; this type of filter is becoming more popular than DE filters.

DE filters remove solids by forcing the water through a filter cake of diatomaceous earth. The filter cake is built up on thin wire screens of corrosion-resistant materials such as stainless steel, Monel, or inconel. A large number of wire screens, called "leaves," may be arranged within the vessel to provide a large surface area for filtration. Typically, the flow rate through DE filter screens ranges from 0.5 to 1 gpm/ft^2 (1.2 to 2.4 $m^3/hr\,m^2$). A DE leaf filter is shown in Figures 10-14a and 10-14b. This process involves precoating the leaves with a thin layer of DE, which is introduced as slurry (see Figures 10-15a and 10-15b).

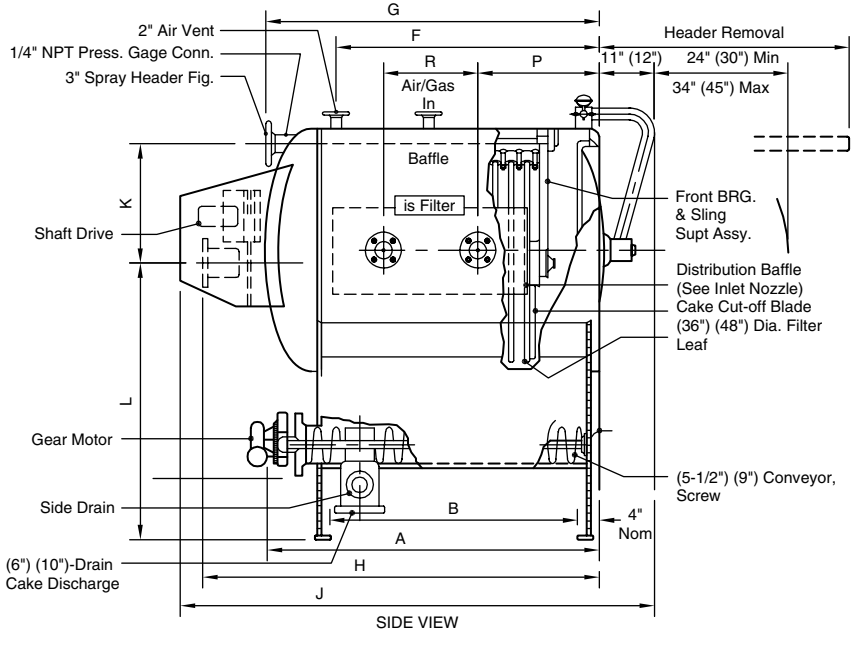


Figure 10-14a. DE filter. (Courtsey of US Filter Corp.)

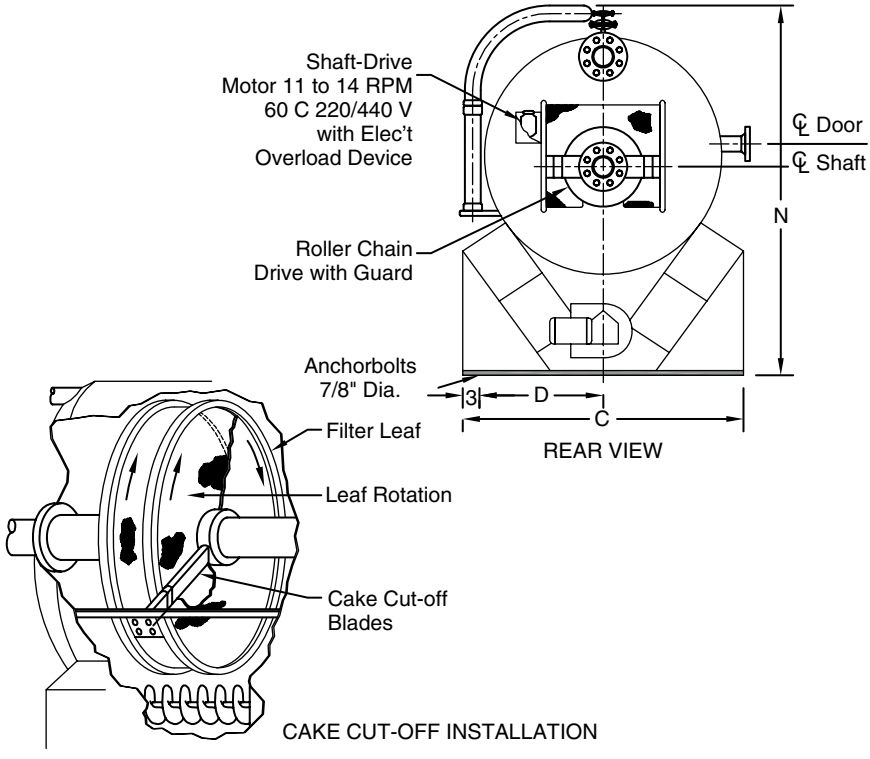


Figure 10-14b. DE filter. (Courtesy of US Filter Corp.)

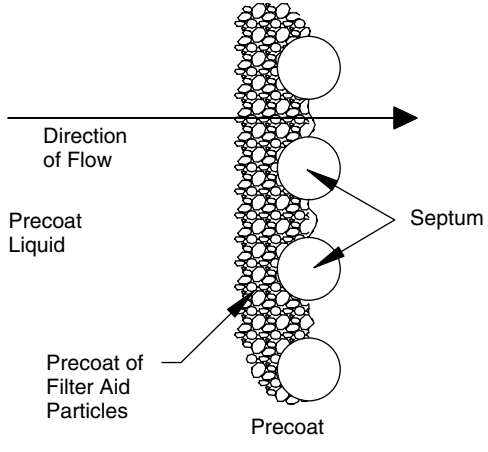


Figure 10-15a. Principles of DE filtration. (Courtesy of Johns Manville Corp.)

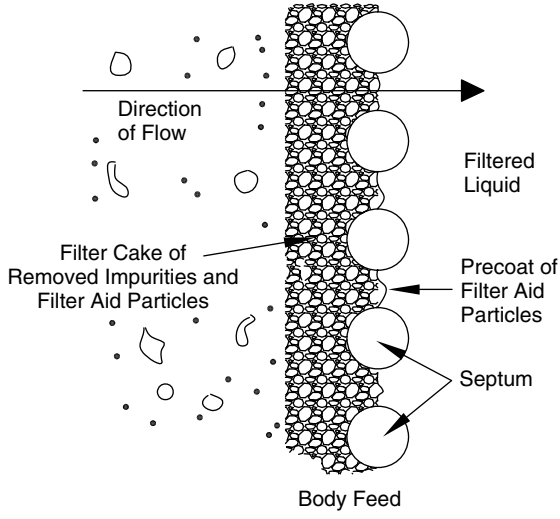


Figure 10-15b. Principles of DE filtration. (Courtesy of Johns Manville Corp.)

After the precoat, the water is introduced and filtration begins. A filter aid such as DE and cellulose fiber must be mixed with the water to promote an even build-up of filter cake and to maintain the filter cake's permeability. This combination is called "body feed." The weight of body feed should be roughly equal to the weight of the solids to be filtered. When the pressure drop reaches the high limit, usually between 25 and 35 psig (170 and 240 kPa), the filter cake must be backwashed from the leaves and the process started over with the precoat.

DE filters require slurry mixing tanks, injection pumps, and large quantities of body feed in addition to the filter vessel itself. Therefore, these systems are expensive to install and to operate, and they require much more space than do other filters.

If the precoat layer is not applied evenly to all leaves, significant amounts of solids may be released downstream. DE filters, like sand filters, are the nonfixed-pore type and suffer from unloading and media migration. In fact, unloading is typically more common with DE filters than with other filters. In addition to unloading and media migration, pressure fluctuations may cause portions of the filter cake to be lost from the leaves. The loss of filter cake then allows solids to pass downstream until the cake is again built up. Normally, guard filters are provided downstream of DE filters to protect against such leakage. Table 10-6 provides operating and design parameters for a typical DE filter.

Table 10-6
Operating and Design Parameters for a Typical DE Filter

A. Operating Parameters	
Service rate	1.2 to 2.4 $m^3/hr\,m^2$ (0.5 to 1 gpm/ft^2)
DE bodyfeed	2 to 5 ppm DE/ppm suspended solids
Regeneration time sequence	
Drain	1 to 5 minutes
Sluice	5 minutes
Fill and add precoat	3 minutes
Circulate	5 to 15 minutes
DE precoat:	
Amount	0.5 to 1.0 kg/m^2 (10 to 20 $lb/100\,ft^2$)
Filter slurry	30 to 60% water
Circulate rate	2.4 to 4.9 $m^3/hr\,m^2$ (1 to 2 gpm/ft^2) (4.5 fps)
B. Design Parameters	
Service rate	1.2 $m^3/hr\,m^2$ (0.5 gpm/ft^2)
Inlet solids	<20 ppm
Inlet oil	<10 ppm
Total outlet solids	<1 ppm
Regenerate at 20-psig pressure drop across filter	
C. Miscellaneous Data	
Wet bulk density DE	240 to 320 kg DE/m^3 (15 to 20 lb DE/ft^3)
Dry bulk density DE	112 to 240 kg DE/m^3 (7 to 15 lb DE/ft^3)
Specific gravity DE	2.3
Cycle length	2 to 3 days
Screen material	Polypropylene, plain weave, 33×42 count 630 deniar warp, twist direction 3.5 Z, weight 201 g/m^2 (5.92 oz/yd^2), heat set for permeability of 730 $m^3/hr\,m^2$ (40 cfm/ft^2) and scoured. Stainless steel can be used.

Note: Bulk density of Perlite filter aid is one-half that of DE. When Perlite is used, the above guidelines should be adjusted on an equivalent volume.

Chemical Scavenging Equipment

Chemical scavenging systems typically require chemical storage facilities, mix tanks, and injection pumps. Depending on the injection rate, the storage facilities may simply be drum racks or a small atmospheric tank. Premixed chemical scavengers may also be purchased in drums or bulk tanks if the quantities used are relatively small.

To select the best method of storage and mixing and to size injection pumps, it is necessary to calculate the chemical injection rate. The following method provides an estimate of chemical usage based on the reaction stoichiometry. Specific chemical suppliers and equipment manufacturers should be contacted to assist in making final equipment selections.

The required injection rate of chemical scavenger may be calculated as follows:

Field Units

$$W_{\text{cs}} = 1.09 \times 10^{-5}\ Q_w\ \text{SG}_w\ Co_2\ R\ \text{MW}_{\text{cs}}, \tag{10-19a}$$

SI Units

$$W_{\text{cs}} = 7.5 \times 10^{-4}\ Q_w\ \text{SG}_w\ Co_2\ R\ \text{MW}_{\text{cs}}, \tag{10-19b}$$

where

W_{cs} = mass flow rate of chemical scavenger, lb/day (kg/day),
Q_w = water flow rate, BWPD (m^3/hr),
SG_w = water specific gravity,
Co_2 = inlet oxygen concentration in water, ppm,
R = stoichiometric reaction ratio between the scavenger and oxygen lb mol/hr O_2 (kg mol/hr O_2),
MW_{cs} = chemical scavenger molecular weight, lb/mol.

Equation (10-19) indicates the mass flow rate of the chemical scavenger's active ingredient. The pump injection rate depends on the concentration of active ingredient in the mixed chemical solution. Chemical manufacturers can assist in determining the best solution concentration and the resulting volumetric injection rate.

The required injection rate of catalyst may be calculated as follows:

Field Units

$$W_c = 7.7 \times 10^{-7}\ Q_w\ \text{SG}_w\ C_{ca} \tag{10-20a}$$

SI Units

$$W_c = 5.3 \times 10^{-2}\ Q_w\ \text{SG}_w\ C_{ca}, \tag{10-20b}$$

where

W_c = mass flow rate of catalyst (CoC1_2), lb/day (kg/day),
C_{ca} = catalyst concentration, ppm (normally $C_{\text{ca}} = 0.001$ ppm).

Again, the volumetric injection rate depends on the mixed solution concentration of the catalyst. Manufacturers may be able to provide a premixed solution of scavenger and catalyst. This solution should be considered because it will decrease the amount of storage, mixing, and injection equipment required.

Nomenclature

A = cross-sectional area of the particle, $ft^2 (m^2)$
C_{ca} = catalyst concentration, ppm
C_D = drag coefficient
Co_2 = inlet oxygen concentration in water, ppm
d = vessel's internal diameter, in. (m)
d_m = particle diameter, μ
d_{50} = particle diameter that is recovered 50% to the overflow and 50% to underflow
d_{99} = particle diameter that is recovered 1% to the overflow and 99% to the underflow
F = factor that accounts for turbulence and short-circuiting
F_D = drag force, lb (kg)
g = gravitational constant, $32.2 ft/s^2$ $(9.81 m/s^2)$
H = height of water, ft (m)
H_w = height of the vessel water, in. (m)
h_w = water height, in. (m)
K = proportionality and shape constant for solids particle removal
K = proportionality and shape constant for flow rate vs. pressure drop
L_{eff} = effective length in which separation occurs, ft (m)
L_{ss} = seam-to-seam length, ft (m)
MW_{cs} = chemical scavenger molecular weight, lb mol/hr (kg mol/hr)
P = operating pressure, psia (kPa)
Q = volumetric vapor flow rate at p_{op} and T, actual ft^3/min (m^3/hr)
Q_g = gas flow rate, MMscfd (std m^3/hr)
Q_w = water flow rate, BWPD (m^3/hr)
R = stoichiometric reaction ratio between the scavenger and oxygen, lb mol/hr O_2 (kg mol/hr O_2)
Re = Reynolds number
SG_w = water specific gravity

T = operating temperature, °R(K)
$(t_r)_w$ = water retention time, min
V_t = terminal settling velocity of the particle, ft/s (m/s)
W = width, ft (m)
W_c = mass flow rate of catalyst ($CoC1_2$), lb/day (kg/day)
W_{cs} = mass flow rate of chemical scavenger, lb/day (kg/day)
α_w = fractional cross-sectional area of water
β_w = fractional water height within the vessel (h_w/d)
γ = height to width ratio, (H_w/W)
ΔP = pressure drop, psi (kPa)
ΔSG = difference in specific gravity of the particle and the water
μ_w = water viscosity, cp (Pas)
ρ = density of the continuous phase, lb/ft^3(kg/m^3)

Appendix A
Definition of Key Water Treating Terms

Introduction

This section discusses many of the key terms typically used in produced water treating systems. Most of these terms are defined within Chapters 9 and 10 when initially introduced. This appendix is not intended to be a comprehensive listing of all terms.

Produced Water

The well stream from the reservoir typically contains varying quantities of water that is commonly referred to as "produced water." The produced water source can be from (1) an aquifer layer underlying the oil and or natural gas zones, (2) connate water found within the reservoir formation sand matrix, (3) water vapor condensing from the gas phase as the result of Joule-Thompson expansion/cooling effects occurring from pressure reduction up the well bore and across wellhead chokes, (4) water-bearing formations not directly in communication with the hydrocarbon reservoir, or (5) a combination of the same. Produced water is typically salty and contains varying quantities of dissolved inorganic compounds and salts, suspended scales and other particles, dissolved gases, dissolved and dispersed liquid hydrocarbons, various organic compounds, bacteria, toxicants, and trace quantities of naturally occurring radioactive materials. Other miscellaneous sources of water from within the processing facilities (e.g., from drains, glycol regeneration units, etc.) are sometimes mixed with produced water for treatment and disposal.

Regulatory Definitions

The terminology for "total oil and grease," "dispersed oil," and "dissolved oil" may vary with location and specific test standard used by the authorities having jurisdiction. These terms should be applied with caution and should conform to the regulations and test standards applicable to the specific location.

Oil Removal Efficiency

Produced water treating equipment performance is commonly described in terms of its "oil removal efficiency." This efficiency considers only the removal of dispersed oil and neglects the dissolved oil content. For example, if the equipment removes half of the dispersed oil contained in the influent produced water, it is said to have a 50% oil removal efficiency. For a specific piece of equipment or an overall system, the oil removal efficiency can be calculated using the following equation:

$$E = [1 - (C_o/C_i)] \times 100,$$

where

E = oil removal efficiency, %,
C_o = dispersed oil concentration in the water outlet (effluent) stream, ppm (mg/l),
C_i = dispersed oil concentration in the water inlet (influent) stream, ppm (mg/l).

The performance can be described by determining the inlet and outlet oil concentrations and the associated oil droplet size distributions at the equipment inlet and outlet. This information can then be used to define the oil removal efficiency for any given oil droplet size or range of droplet sizes. This concept is further discussed in Chapter 9.

Total Oil and Grease

"Total oil and grease" is defined as the combination of both the dispersed and dissolved liquid hydrocarbons and other organic compounds (i.e., "dissolved oil" plus "dispersed oil") contained in produced water. This term is referenced in certain regulatory standards and is commonly used

to evaluate water treating system design. Total oil and grease consists of normal paraffinic, asphaltic, and aromatic hydrocarbon compounds plus specialty compounds from treating chemicals. The measurement of total oil and grease is dependent on the analysis method used.

Dispersed Oil

Produced water contains hydrocarbons in the form of dispersed oil droplets, which, under proper conditions, can be coalesced into a continuous hydrocarbon liquid phase and then separated from the aqueous phase using various separation devices. The diameters of these oil droplets can range from over 200 microns to less than 0.5 microns and may be surrounded by a film (emulsifier) that impedes coalescence. The relative distribution of droplet sizes is an important design parameter and is influenced by the hydrocarbon properties, temperature, down-hole operating conditions, presence of trace chemical contaminants, upstream processing and pipe fittings, control valves, pumps, and other equipment that act to create turbulence and shearing action. These oil droplets are collectively defined as "dispersed oil."

Conventional water treating systems commonly used by the oil and gas industry remove only the "dispersed oil." This text focuses on the design of water treating systems that remove and recover "dispersed oil."

Dissolved Oil

Produced water contains hydrocarbons and other organic compounds that have dissolved within the aqueous phase and cannot be recovered by conventional water treating systems. Fatty acids are likely to be present within paraffinic oils and naphthenic acids within asphaltic oils. These organic acids, aromatic components, polar compounds (also called non-hydrocarbon organics), and certain treating chemicals are slightly soluble in water and collectively make up the organic compounds found in solution of the aqueous phase. The portion of these components that are dissolved into the produced water is defined as "dissolved oil." Dissolved oil is microscopically indistinguishable within the aqueous phase since it is solution at the molecular level and cannot be separated from the produced water by means of coalescence and/or gravity separation devices.

Treatment methods for removal of dissolved oil are not covered in this text. However, the oil and gas industry is currently evaluating treatment

methods such as bio-treatment, air stripping, adsorption filtration, and membranes. These designs are typically prototypical in nature and require a larger capital investment, a greater maintenance work effort, and more space and may result in by-products having disposal problems more onerous than those associated with the disposal of dissolved oil. The confined space typically available on an offshore platform presents a real challenge in developing a suitable water treating process for removing dissolved oil from produced water.

Dissolved Solids

Several inorganic compounds are soluble in water. The total measure of these compounds found in solution with produced water is referred to as "total dissolved solids" (TDS). When these compounds are found in solution with the produced water, they are referred to as "dissolved solids." The most common water-soluble compound in produced water is sodium chloride. A number of other compounds collectively comprise the dissolved solids contained in produced water. These are discussed in Chapter 9.

Suspended Solids

Produced water and oil contain very small particulate solid matter held in suspension in the liquid phase by surface tension and electrostatic forces. This solid matter is referred to as a "suspended solid" and may consist of small particles of sand, clay, precipitated salts and flakes of scale, and products of corrosion such as iron oxide and iron carbonate. When suspended solids are measured by weight or volume, the composite measurement is referred to as the "total suspended solids" (TSS) content. Chapter 9 provides a detailed discussion on suspended solids.

Scale

Under certain conditions, the dissolved solids precipitate or crystallize from the produced water to form solid deposits in pipe and equipment. These solid deposits are referred to as "scale." The most common scales include calcium carbonate, calcium sulphate, barium sulphate, strontium sulphate, and iron sulfide. Scale is further discussed in Chapter 9.

Emulsion

An "emulsion" is an oil and water mixture that has been subjected to shearing resulting in the division of oil and water phases into small droplets. Most emulsions encountered in the oil field are water droplets in an oil continuous phase and are referred to as "normal emulsions." Oil droplets in a water continuous phase are referred to as "reverse emulsions." Emulsions are further discussed in Chapter 8.

Appendix B
Water Sampling Techniques

Sampling Considerations

Any water analysis method is only as good as the "sample" used to represent the effluent stream. Sampling of a continuously flowing stream containing two or more phases (e.g., oil and water) is difficult unless the mixture is completely emulsified or is a very fine stable dispersion. Since the sampling techniques for oil concentration measurement and particle size distribution differ in some aspects, they are described separately here.

Sample Gathering for Oil Concentration Measurement

Generally, the larger the sample the more likely it is to be representative. However, for practical reasons, the sample size varies from 15 ml to about 1l. Typically, the smaller samples are used for daily analysis, whereas the larger samples are used for monthly regulatory compliance purposes. The smaller the residual oil droplets, the more evenly dispersed they are likely to be. Care should be exercised to avoid sampling the surface of a liquid (since this is not truly representative). "Isokinetic" (which means equal linear velocity) sampling in midstream is the best, but is rarely possible. The sample probe must be inserted so that the velocity profile remains undisturbed, thereby getting a realistic particle distribution and, thus, a higher accuracy. The general guidelines are

- Flush the sample line thoroughly and take the sample quickly.
- Sample after a pump or a similar turbulent area where the stream is well mixed.
- Obtain the sample from a liquid-full vertical pipe, if possible.

Sample bottles should be scrupulously clean and preferably of glass. Oil or other organic material can adhere to the walls of a plastic container and give erroneous readings. Never use a metal container or a metal cap. The water can corrode it and become contaminated with corrosion products. Bottles used at oily sites or handled by an operator with oily hands can have thin surface films, and washing can leave detergent residue, both of which can give rise to erroneous and high oil readings.

General guidelines one should follow to improve measurement accuracy are as follows:

- Use only glass or inert plastic (e.g., Teflon) stoppers. Cork or other absorbent materials must not be used unless covered with aluminum foil.
- Do not rinse or overflow the bottle with the sample because an oil film will appear on the bottle and give a false reading.
- Cap the sample and prepare a label immediately with an indelible, smear-proof marking pen. Attach it to the bottle immediately.
- Analyze the entire sample and wash the bottle with solvent.

The person taking the sample must be well trained and experienced and be able to recognize a spoiled or unrepresentative sample. Samples must be correctly labeled immediately after being taken and any abnormal circumstances must be noted on the sample. If any doubt exists, the sample should be discarded and a new one taken in a fresh container.

The sampling frequency depends on the practicality of sampling at each site or may also be specified by the authorities having jurisdiction. A manned installation would require a higher-analysis frequency than an unmanned site, which may be less accessible. For example, in the United States, the EPA states, "The sample type shall be a 24-hour composite consisting of the arithmetic average of results of 4 grab samples taken over a 24-hour period. If only one sample is taken for any one month, it must meet both the daily and monthly limits. Samples shall be collected prior to the addition of any seawater to the produced water waste stream."

Sample Storage for Oil Concentration Measurement

If possible, perform the sample analysis as soon as the sample is obtained. If immediate analysis is not possible as with certain samples (normally for regulatory reporting) that are sent onshore for analysis, then acidify the sample to pH 2 using hydrochloric acid (HCl) to preserve the sample against bacterial action and/or dissolve the precipitated calcium carbonate, which could cause difficulties separating the solvent phase from the water. Acidification causes a higher total oil and grease concentration. This is

because acid reacts with organic salts to liberate organic acids, which are then extracted into the solvent. This gives a higher reading in the analysis. Therefore, when a sample has been acidified, the solvent extract should pass through a silica-gel column or similar material to remove these polar substances (organic acids).

In the case of analysis done immediately (for daily measurement), the sample should be acidified only if the approved analysis procedure requires it.

Sample Gathering for Particle Size Analysis

The following points are applicable when obtaining a sample for particle size analysis:

- Flush the sample line thoroughly and take the sample quickly.
- Obtain the sample from a liquid-full vertical pipe, if possible.
- Use only glass or inert plastic (e.g., Teflon) stoppers. Cork or other absorbent materials must not be used unless covered with aluminum foil.
- Do not rinse or overflow the bottle with the sample as this can put an oil film on the bottle and give a false reading.
- Cap the sample and prepare a label immediately with an indelible, smear-proof marking pen. Attach it to the bottle immediately.

In addition to the above factors, other important factors that need to be considered when measuring particle size distribution are

1. **Avoid shearing of oil droplets across a sample valve.** Typically, when sampling produced water the sample valve is never fully open, since full flow is so intense that sampling may be almost impossible. Consequently, the flow is restricted by controlling the valve partially open. In this situation, the produced water is subjected to choking from high pressure down to atmospheric conditions. The shearing within the sampling valve causes the oil droplets in the sample to break up into smaller droplets.
2. **Avoid a "dissolved gas flotation effect."** The produced water sample is depressurized as it passes through the choke valve, and gas is liberated as minute gas bubbles. These bubbles may coalesce with the oil droplets so the oil droplets adhere to the gas bubbles and rise to the sample surface. This phenomenon is called "dissolved gas flotation." It will not alter the total concentration of

oil in the sample, but could split the oil present into two separate fractions:

- Dispersed oil, remaining in the water and not affected by the gas bubbles,
- Free oil, formed as a thin film on the water surface caused by the flotation.

The free oil formed may easily adhere to the walls of the sample container as a thin and almost invisible film. This film is easily lost as the initial sample is split into subsamples. The fraction of oil in the sample ending up as free oil may be as high as 50 to 80%.

One way of avoiding droplet shear and gas flotation effects is to conduct an on-line sample measurement. This technique, however, requires specialized equipment, such as the Melvern Mastersizer, and is constrained in terms of the maximum pressure it can tolerate. Alternatively, a sample pressure cylinder (bomb) can be used to avoid droplet shear and the gas flotation effect. A sample in a stainless steel cylinder with a needle valve and a ball valve minimize the shearing of droplets during sampling.

3. **Avoid coalescence of oil droplets by stabilizing the sample.** A sample used for droplet size measurement may need to be stabilized to avoid coalescence of the small oil droplets to larger ones. The propensity of a droplet to coalesce increases as the oil concentration of the sample increases. This stabilization becomes more important when measuring samples with a high oil concentration. Stabilization may be achieved by diluting the sample with a known amount of water. This reduces the chance of droplet coalescence and also stabilizes the sample by reducing the salt concentration of the sample. Further stabilization of the sample may be achieved by the addition of a viscous polymer solution and/or surfactant (e.g., 2% sodium dodecyl sulphate). However, one should be very cautious when adding such chemicals since the wrong surfactant could actually promote coalescence.
4. **Sample may contain solids and other non-oil particles in addition to the oil droplets.** Produced water samples often contain solids and other non-oil particles in addition to oil droplets. To determine only the oil droplet size distribution, one must first determine the size distribution for all particles within the sample and then determine the size distribution of particles left behind in the sample after solvent extraction. Solvent extraction removes all of the oil droplets from the sample, leaving behind only the solids and non-extractable non-oil particles. One can then block out those size ranges (corresponding to the solids and non-oil particles) from the initial size distribution to obtain a more representative size distribution.

Appendix C
Oil Concentration Analysis Techniques

Introduction

Several analytical techniques measure the amount of oil and grease in water. These techniques may be broadly classified as either gravimetric or infrared (IR) absorbance methods and are described in detail here. These methods are based on the extraction of oil and grease into a solvent. A sample may contain suspended solids, which have to be filtered. In this case, the sample must first undergo solvent extraction followed by filtration of the extract.

Several different solvents have been used. These include petroleum ether, diethyl ether, chloroform, and carbon tetrachloride. Of these, petroleum ether and diethyl ether are highly flammable, whereas chloroform (although a very good solvent) and carbon tetrachloride are toxic. Thus, these solvents are not recommended for use. Currently, 1,1,2-trichloro, trifluoroethane (Freon 113) is used when infrared (IR) absorbance is used for analysis. However, these solvents are being phased out because of potential interference with the ozone layer in the atmosphere. Studies are currently under way to find a replacement solvent. Potential candidates include hexane, cyclohexane, methylene chloride, perchloroethylene, and a commercial hydrochlorofluorocarbon (DuPont 123). When the gravimetric technique is used for analysis, 1,1,1-trichloroethane or dichloroethylene may also be used.

"Total oil and grease" is defined by the measurement procedure stipulated by the authorities having jurisdiction. Important variations that could give different results for the same sample are

- **Number of extractions performed on the water sample.** Multiple extractions with Freon on the same water sample will generally give a higher oil concentration than a single extraction.

- **The solvent-to-sample ratio.** A higher solvent-to-sample ratio will also give a higher oil concentration.
- **Determination of IR absorbance at multiple wavelengths.** This variation will give a higher oil concentration as opposed to absorbance measurement at a single wavelength.
- **Use of silica gel.** This variation is discussed ahead.

Determination of Dissolved Oil and Grease

The dissolved oil and grease content is first determined by measuring the total oil and grease content and then subtracting the measured dispersed oil and grease content. The measured dispersed oil is obtained by removal of the dissolved oil and grease from the solvent with silica gel. This can be expressed by the following equation:

$$\text{dissolved oil and grease} = \text{total oil and grease} - \text{dispersed oil and grease}. \quad \text{(C-1)}$$

Dissolved organic matter is generally either polar or of low molecular weight. To remove the dissolved organic matter, the solvent extract is contacted (either in an adsorption column or by intimate mixing) with activated silica gel (e.g., Florisil) or alumina. The materials not adsorbed by the silica gel are described as "dispersed" oil and grease.

In the absence of silica gel, filtration can be used to remove the dispersed oil and grease. In this technique the filtrate, obtained after passing through a 0.45-μ filter paper, is left behind on the filter paper. The U.S. EPA uses the term "petroleum hydrocarbons" for dispersed oil and grease.

The choice of measuring total oil and grease versus dispersed oil and grease is important from an operational standpoint because conventional water treating technology can reduce the concentration of dispersed oil and grease, but not the concentration of dissolved oil and grease. However, measurement techniques are strictly specified in some countries (e.g., the United States) and left open to negotiation or operator discretion in others.

Gravimetric Method

This method is the U.S. EPA-required method to measure oil and grease in produced water for regulatory compliance purposes in the United States. A detailed procedure can be found in EPA 413.1.

In this method the water sample is extracted with Freon 113 and the extract evaporated to remove the solvent. The weight of the residue is related to the concentration of oil in the water sample. Generally, for the same water sample, this method gives a lower value for oil and grease concentration than the IR absorbance method because of the loss of volatile organics during the evaporation process.

Advantages

- Stipulated by the EPA as the technique to be used to measure oil in produced water discharge for regulatory purposes.
- Method is simple and well understood.

Disadvantages

- Requires samples be collected and preserved according to EPA protocol for shipment to an onshore laboratory.
- Is time-consuming.
- Uses Freon (a CFC) as a solvent.
- Lower limit of measurement is 5 mg/l.
- Not applicable to light hydrocarbons that volatilize below 70°C.

Infrared (IR) Absorbance Method

For infrared absorbance methods (e.g., EPA 413.2 and EPA 418.1), the water sample is extracted with Freon 113. The IR absorbance of the extract is measured at single, or multiple, wavelength(s) to give the oil concentration. In this method, the water sample is often acidified to prevent any salts from precipitating out (e.g., iron sulfide). IR absorbance at multiple wavelengths results in a higher oil concentration measurement than if a single wavelength was used.

Advantages

- Fast and convenient for offshore surveillance.
- The lower limit of measurement is 0.2 mg/l.

Disadvantages

- Uses Freon (a CFC).
- The lower limit is 0.2 mg/l.

Analysis of Variance of Analytical Results

In 1975, the U.S. EPA had a set of six different samples analyzed by a number of laboratories using both the gravimetric and IR methods. Scatter in reported analysis included sampling errors as well as analytical errors. The true values were taken to be the average of the reported values (excluding those of extreme scatter) and are as follows:

Sample	**1**	**2**	**3**	**4**	**5**	**6**
Gravimetric	201.7	209.3	95.5	77.6	33.7	19.1
Infrared	261.3	265.4	131.0	110.0	50.7	23.0

(Note: Error is defined as the difference of an observation from the best obtainable estimate of the true value, which in this case is the arithmetic mean.)

It is interesting to note that the gravimetric value is lower than the infrared value of each of the samples. This is expected since the solvent evaporation step in the gravimetric process causes some loss of volatile organics, leading to lower results than the IR method. Errors ranged from 0 to 241%. The worst laboratory had errors between 49% and 98%, whereas the best laboratory had errors up to 8%.

Particle Size Analysis

The oil droplet size distribution is one of the key parameters influencing water treating equipment selection. Therefore, accurate measurement of the oil droplet size distribution is an important task. Another important parameter is quantifying the size distribution upstream and downstream of production equipment, such as control valves.

Oil and other particles in produced water range in size from less than 1 μ up to several hundred microns. Although many particles found in produced water are not spherical, for practical purposes, the particles are represented by equivalent spheres.

Droplet Size Measurement Equipment

Three different types of equipment are commonly used for droplet size measurement. Each has its advantages and disadvantages. First, establish the information desired before selecting the equipment type:

1. **Coulter Counter.** The Coulter Counter consists of two electrodes immersed in a beaker of sample water, which contains a sufficient number of dissolved ions to easily conduct an electrical current. The negative electrode is located inside a glass tube, which is sealed except for a tiny hole or orifice on the side of the tube. The positive electrode is located in the water sample beaker. A constant electrical current is passed from the positive electrode to the negative electrode through the orifice. When a non-conductive particle passes through the orifice, a change in electrical resistance occurs between the two electrodes which is proportional to the particle volume.

   A fixed volume of water containing suspended particles is forced through the orifice. As each particle passes through the orifice, the increased resistance results in a voltage that is proportional to the particle volume. The series of pulses produced by a series of particles passing through the orifice are electronically scaled and counted, yielding a particle size distribution. One must realize that the particle "diameter" given by the counter is the diameter of a fictitious equivalent sphere with the same volume as the real particle.

   This equipment has some limitations because the size range is limited. In addition, the samples have to be suspended in an electrolyte solution, which can prove difficult if the sample is totally soluble in the solution. However, the Coulter Counter does provide both frequency and volume distributions against volumetric particle size.
2. **Light (laser) scattering counters.** These include instruments that are based on the principle of light absorption/total scatter, or light blockage, to detect particles in a fluid. Water flows through a sensor cell, and as each particle passes through the intense beam of light in the sensor, light is scattered. The instrument measures the magnitude of each scattered light pulse, which is proportional to the surface area of the particle. The particle diameter determined by the instrument in this case is the diameter of a sphere with the same surface area as the particle.

   Laser diffraction systems are widely accepted due to their ease of use, wide size range, and simple sample preparation. However, as an optical technique, it is still subject to variations in response from particle shape and refractive index and is unable to give frequency

information (that is, the number of particles within a given size range). However, these techniques can provide relative frequency information (that is, percent of total particle volume within a given size range).

3. **Microscopy.** In this technique, the droplet size distribution is determined by observing the water sample under a microscope and visually measuring the size of the droplets. Often a magnified photograph is also used for visual determination. This technique has the advantage of being able to distinguish between oil droplets and non-oil particles. A microscope also helps to see first hand if there are any extreme shape factors. However, the technique generally uses a very small sample volume and therefore may not be representative.

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Glossary of Terms

Acid gas	H_2S and/or CO_2 contained in or extracted from a natural gas.
Accumulator	A vessel used to collect and store liquids.
°API gravity	An arbitrary scale expressing the relative density of liquid petroleum products. The measuring scale is calibrated in degrees API (°API) and is calculated by the following formula: $$\text{Deg API} = \frac{141.5}{\text{SG}(@\ 60°\text{F})} - 131.5.$$
Artificial lift	Mechanical means of raising a crude oil in a well to the surface, including sucker-rod pump, hydraulic pump, gas lift, and electrical submersible pump.
Atmospheric pressure	The pressure exerted on the earth by the earth's atmosphere. A pressure of 760 mm of mercury, 29.92 in. of mercury, or 14.696 psia is used as a standard for some measurements. The various state regulatory bodies have set other standards for use in measuring the legal volume of natural gas that is sold or processed. Atmospheric pressure may also refer to the absolute ambient pressure at any given location.

Bad oil	Crude with a BS&W content in excess of pipeline spec.
Boiling range	Range of boiling point temperatures used to characterize a cut.
Bubble point	The temperature at a given pressure or the pressure at a given temperature at the instant the first bubble of gas is formed in a given liquid.
Cannula	A large-bore hypodermic needle attached to a syringe; used to remove samples from liquid layers.
Chromatography	A technique for sample analysis where individual components of a batch sample, carried by an inert gas stream, are selectively sorbed and disrobed on a sorbent column at different rates in relation to equilibrium coefficients. Separated components are quantitatively detected as they leave the sorbent column.
Clean crude	Crude oil containing no BS&W.
Collector pipe	Perforated or slotted pipe used to remove treated oil as uniformly as possible at top of coalescing section.
Compressibility factor	A factor usually expressed as Z, which gives the ratio of the actual volume of gas at a given temperature and pressure to the volume of gas when calculated by the ideal gas law without any consideration of the compressibility factor.
Conditioning	See "processing."
Connate water	Formation water held in the pores by capillary action; water originally contained in sedimentary rocks at the time of deposition.
Continuous phase	See "emulsion."
Control valve	Valve used to control flow rate of a fluid entering or leaving a process component.
Convergence pressure	The pressure at given temperature for a hydrocarbon system of fixed composition at which the vapor–liquid equilibria values of the various components in the system become unity.

	The convergence pressure is used to adjust the vapor-liquid system under consideration.
Cricondenbar	The highest pressure at which vapor and liquid phases can be identified in a multicomponent system.
Cricondentherm	The highest temperature at which vapor and liquid phases can be identified in a multicomponent system.
Critical pressure	The pressure necessary to condense a vapor at its critical temperature.
Critical temperature	The highest temperature at which a pure element or compound can exist as a liquid. Above this temperature, the fluid is a gas and cannot be liquefied regardless of the pressure applied.
Crude oil	Unrefined liquid petroleum.
Cubic equation	Equation of state with three constants.
Custody transfer	Transfer of ownership of oil or gas streams, usually at some arbitrary location in the field.
Cut	A petroleum fraction containing numerous individual compounds that is characterized by average properties such as boiling point range, °API, SG, etc.
Cyclone	A cone-shaped separator that uses centrifugal force to separate two immiscible phases.
Dehydration	The act or process of removing water from gases or liquids.
Demulsifier	Demulsifiers or demulsifying chemicals are a mixture of chemicals used to break the emulsion by destroying or weakening the stabilizing film around the dispersed drops.
Dense phase	Fluid existing above both the cricondenbar pressure and the critical temperature.
Desalting	The act or process of removing salts from crude oils.

Desulfurization	The process by which sulfur and sulfur compounds are removed from gases or liquid hydrocarbon mixtures.
Dew point	The temperature at any given pressure or pressure at a given temperature at which liquid initially condenses from a gas or vapor. It is specifically applied to the temperature at which the water vapor starts to condense from a gas mixture (water dew point) or at which hydrocarbon starts to condense (hydrocarbon dew point).
Direct heater	A heater in which fire-tube contacts the process fluid directly.
Dispersed phase	See "emulsion."
Drive	Pressure tending to cause an oil in reservoir to flow through the rock pores to the well bore and upwards through the tubing to the surface; common types of drive are free gas cap, dissolved gas, water, and gravity.
Dry gas	(1) Gas containing little or no hydrocarbons commercially recoverable as liquid product. Gas in this definition preferably should be called "lean gas." (2) Gas whose water content has been reduce by a dehydration process (rare usage).
Dual emulsion	An emulsion in which the continuous phase is oil and the dispersed phase is an oil-in-water emulsion.
Electrodes or grid	Plates or rods used to establish the electric field in electrostatic treaters.
Electrostatic	Treater using electrostatic fields in the oil treater coalescing area.
Emulsified water	Water that will not separate readily from a water-in-crude emulsion.
Emulsifier	In addition to oil and water, a third substance—called an emulsifier or emulsifying agent—must be present for a stable emulsion to be produced. These emulsifiers usually exist as a film on the surface of the dispersed drops.
Emulsion	A combination of two immiscible liquids. One liquid is broken up into droplets and is known as

	the discontinuous, dispersed, or internal phase. The other liquid that surrounds the drops is the continuous or external phase.
Equation of state	An equation relating the pressure, temperature, and specific volume of a fluid.
Error	Set-point value—process output.
Excelsior	Fibrous material used to separate water from oil in a heater-treater.
External phase	See "emulsion."
Flash point	The lowest temperature at which vapor from a hydrocarbon liquid will ignite.
Free water	Water that separates readily (in <5 min) from a produced crude oil.
Gain	Ratio of controller output to error.
Gas anchor	A short section of tubing that extends down from an insert sucker-rod pump and is used to separate gas from oil before it enters the pump to prevent gas locking.
Gas-condensate field	A petroleum field or reservoir in which the hydrocarbons in the formation exist in a vapor state under high temperature. A lowering of the temperature causes a condensation of the heavier hydrocarbons, which will then not be produced with the gas.
Gas constant	A constant number, which mathematically is the product of the total volume and the total pressure, divided by the absolute temperature for one mole of any ideal gas or mixture of ideal gases at any temperature. $PV/T = R$.
Gathering lines	The network of pipelines that carry gas/oil from the wells to the processing plant or other separation equipment.
Gauging	Measurement of oil in a storage tank.
Grasshopper	Vertical pipe arrangement on the outside of an atmospheric crude oil tank that controls internal water–oil interfacial level by manipulation of its height.

Gun barrel	Settling tank or wash tank, with built-in gas boot.
Handling	See "processing."
Hay	See "excelsior."
Head	Pressure due to a height of fluid.
Heater-treater	A vessel used to dehydrate crude oil that uses chemicals, settling, and heat.
Heating baffle hood or shroud	A baffle that surrounds the fire-tubes and is designed to minimize heating of free water in a heater-treater.
Heating value	The amount of heat developed by the complete combustion of a unit quantity of a material.
Heave	Vertical motion of a ship or floating platform.
Hexane (or Heptanes) plus	The portion of a hydrocarbon fluid mixture or the last component of a hydrocarbon analysis that contains the hexanes (or heptanes) and all hydrocarbon heavier than the hexanes (or heptanes).
Hydrate	A solid material resulting from the combination of hydrocarbon with water under pressure.
Indirect heater	A heater in which the fire-tube heats a liquid that, in turn, heats the process fluid.
Injection of gas	Putting gas into the formation by force (pressure).
Innage	Crude oil contained in a tank between the tank bottom and the oil surface; as contrasted to outage (see "outage").
Interface	Two uses: (1) the surface area of the drops in an emulsion; (2) the area between two separated phases in a vessel.
Interface pad	A layer of solid accumulated at the interface between relatively pure water and oil layers.
Internal phase	See "emulsion."
Interphase drain	A perforated pipe or other device used to removed the solid phase accumulated at the oil–water interface in a treater.
Inverse emulsion	See "reverse emulsion."

Joule-Thomson	The change in gas temperature that occurs when the gas is expanded at constant enthalpy from a higher pressure to a lower pressure. The effect for most gases at normal pressure, except hydrogen and helium, is a cooling of the gas.
K value	Ratio of mole fraction of a component in vapor to that in liquid.
Knockout	Separator that removes (1) free water from crude oil or (2) total liquids from a gas stream.
Knockout drops	A demulsifier used to separate BS&W from a crude oil emulsion sample; allows determination of BS&W.
Lean gas	(1) The residue gas remaining after recovery of natural gas liquids in a gas processing plant. (2) Unprocessed gas containing little or no recoverable natural gas liquids.
Light ends	The low-boiling, easily evaporated components of a hydrocarbon liquid.
Loose emulsion	An unstable or easily broken emulsion.
Manifold	A pipe with one or more inlets and two or more outlets, or vice versa.
Mercaptan	A compound sometimes found in gas and gas liquids which must be reduced by removal or conversion to conform to specification. Any of a series of compounds of the alcohol and phenols, but containing sulfur in place of oxygen. (R represents an alkyl group or radical.)
Molecular sieve	A synthetic zealot (essentially silica-alumina) used in adsorption processes.
Natural gas	Gaseous petroleum.
Offset	Set-point—process output after control action.
Oil-field	Surface area overlying an oil reservoir.
Oil-in-water (o/w) emulsion	An emulsion consisting of oil drops dispersed in a continuous water phase.

Outage	Space in a tank between the oil surface and the top of the tank; also called "ullage."
Overdosing	Adding excess or too much demulsifier.
Plate-fin exchangers	Heat exchangers, which use thin sheets of metal to separate the hot and cold fluids instead of tubes.
Pentane-plus	A hydrocarbon mixture consisting mostly of normal pentane (C_5H_{12}) and heavier components extracted from natural gas.
Petroleum	Hydrocarbons (gas and oil) obtained from underground reservoirs.
Pigging	A procedure of forcing a solid object through a pipeline for cleaning purposes.
Pipeline oil	A crude oil that meets all pipeline specs such as °API, S content, pour point, S&W content, RVP, etc.
Pitch	Angular motion of a ship or floating platform.
Pressure maintenance	Injection of gas into a formation to keep up the pressure.
Processing	All unit operations performed on wellhead fluids in the field.
Produced water	Water produced with crude oil or gas. It is usually classified as entrained or free. Entrained or emulsified water does not settle out readily. Free water settles within 5 min.
Proportional band	$\dfrac{100}{\text{Controller Gain}}$
Prover	Device used to calibrate a flow meter.
Raw gas	Unprocessed gas or the inlet gas to a plant.
Raw mix liquids	A mixture of natural gas liquid prior to fractionation. Also called "raw make."
Recompressor	A compressor used from some particular service, such as compressing residue gas; implies restoring of pressure level of a stream that has been subjected to pressure reduction.
Regular emulsion	A water-in-oil (w/o) emulsion.

Relief system	The system for temporarily releasing excess fluid, usually gas, to avoid a pressure in excess of the design pressure for the particular equipment.
Reservoir	Subsurface, permeable rocks body containing crude oil and/or natural gas.
Retrograde condensate (vaporization)	Condensate or vaporization that is reverse of usual behavior. Condensation caused by a decrease in pressure or increase in temperature. Vaporization caused by an increase in pressure or decrease in temperature. Can only occur in mixture.
Reverse emulsion	An oil-in-water (o/w) emulsion.
Roll	Angular motion of a ship or a floating platform.
RVP (Reid vapor pressure)	A vapor pressure for liquid products as determined by ASTM test procedure D-323. The Reid vapor pressure is reported as pound per sq in. at 100°F. The RVP is always less than the true vapor pressure at 100°F.
Sales gas	A gas that meets all specifications for sales.
Sand pans	Inverted troughs or angle's baffles used to aid sand and sediment removal from treaters.
Scrubber	A separator that removes small amounts of liquid from a gas stream.
Sensor	Measuring instrument.
Separator	Vessel used to split a multiphase well stream into a gas stream and one or more liquid streams.
Separator gas	Same as associated gas.
Shrinkage	Reduction in volume of oil as gas is evolved from it.
Solution gas	Gas that is dissolved in crude oil, either in a reservoir or in the producing equipment.
Sour gas or oil	A gas or oil containing H_2S or mercaptans above a specified concentration level.

Specific gravity	The ratio of the mass of given volume of a substance to that of an equal volume of another substance used as standard. Unless otherwise stated, air is used as the standard for gases and water for liquids and the volumes measured at 60°F and atmospheric pressure (15.56°C and 101.325 kPa).
Spreaders	Perforated pipes or channels used to inject emulsions as uniformly as possible throughout the treater's cross section.
Stabilization	Removing volatile compound from a crude oil to reduce its bubble-point pressure (and its RVP).
Stabilizer	A name for a fractionation system that stabilizes any liquid (i.e., reduces the vapor pressure so that the resulting liquid is less volatile).
Stable emulsions	Require an active treatment for breaking or phase separation to occur.
Steam flooding	EOR method for shallow, heavy oil deposits in which high-temperature steam is injected into the formation to make the oil more easily produced.
Stock-tank oil	Oil remaining after stage-separation train or stabilization (i.e., after dissolved gas has been released).
Strapping	Measuring and recording the dimension of a storage tank.
Sulfur	A yellow, nonmetallic chemical element. In its elemental state, called "free sulfur," it has a crystalline or amorphous form. In many gases and oil streams, sulfur may be found in volatile sulfur compounds (i.e., hydrogen sulfide, sulfur oxides, mercaptans, carbonyl sulfide).
Surge	Motion of a ship or floating platform; pressure pulse in a pipeline.
Surge factor	Equipment is usually sized using the maximum flow rate expected during predicted life of facility. Generally, accepted practice is to add a surge factor (20–50%) to handle short-term fluctuations.
Sway	Motion of a ship or floating platform.

Sweet	This refers to the near or absolute absence of objectionable sulfur compounds in either gas or liquid as defined by given specification standard.
Sweetening	Act or process of removing H_2S and other sulfur compounds.
Tight emulsion	A very stable or hard-to-break emulsion.
Trap	Gas–oil separator, usually horizontal.
Treating	Removing undesirable components or properties from a fluid.
Vapor pressure	The pressure exerted by a liquid when confined in a specified tank or test apparatus.
V/L ratio	Vapor–liquid equilibrium ratio.
Water cut	Volume % water in crude oil–water mixture.
Water-in-oil (w/o) emulsion	In vast majority of cases, crude oil emulsions consist of water drops dispersed in a continuous oil phase. Also called "regular" or "normal emulsion."
Water leg or water siphon	Piping system for removing water from a treater at a controlled rate. Also called "grasshopper."
Wet gas	Natural gas that yields hydrocarbon condensate (does not usually refer to water content). Also called "rich" gas.
Wetting	Refers to adhesion or sticking of a liquid to a solid surface. If the solid surface (grain of reservoir rock, fines, etc.) is covered preferentially by oil, the surface is called "oil wetted." If water is preferentially attracted, the surface is "water wetted."
Yaw	Angular motion of a ship or floating platform.

Common Abbreviations

ACT	Automatic custody transfer; see LACT
AG	Acid gas
AGA	American Gas Association

AIME	American Institute of Mining, Metallurgical, and Petroleum Engineers
AISI	American Iron & Steel Institute
ANSI	American National Standards Institute
API	American Petroleum Institute—national trade association of United States petroleum industry, a private standardizing and lobbying organization
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATG	Automatic tank gauging system
atm	Atmosphere
bbl	Barrel (42 U.S. gallons). The oil industry standard for volumes of oil and its products; always reduced to 60°F and vapor pressure of the liquid
BEP	Best efficiency point (for a centrifugal pump)
Bhp	Brake horsepower
BLM	Bureau of Land Management—U.S. government agency that regulates petroleum production onshore
blpd	Barrels of liquid per day
B_o	Formation volume factor
BOPD	Barrels of oil per day
BPD	Barrels per day
Brf	Barrels of reservoir fluid
BS&W	Basic sediment and water; water and other contaminants present in crude oil
BTEX	Benzene, toluene, ethyl benzene, and xylene
Bscf	Billions of standard cubic feet
bsto	Barrels of stock-tank oil
BTU	British thermal unit
BWPD	Barrels of water per day

C_1	Methane
C_2	Ethane
C_3	Propane
C_4's	Butanes
C_5's	Pentanes
C_6	Hexanes
C_{6+}	Hexanes and heavier
C_7	Heptanes
C_{7+}	Heptanes and heavier
C_8	Octanes
CAAA	Clean Air Act Amendments
CF	Characterization factor
cfm	Cubic feet per minute
CI	Controller input
CMA	Chemical Manufacturers Association
CMV	Corrected meter volume
CO	Controller output
cp	Centipoise
CV	Control valve
CW	Continuous-welded
°API	Degrees API gravity
°F	Degrees Fahrenheit
°C	Degrees Celsius
DOE	Department of Energy
DOT	Department of Transportation
EBHAZOP	Experienced-based HAZOP
ECT	Environmental control technology
EOR	Enhanced oil recovery
EPA	Environmental Protection Agency

EODR	Electro optical distance ranging
EOS	Equation of state
ERW	Electric resistance welded
ft/sec	Feet per second
FERC	Federal Energy Regulatory Commission
FIA	Fire Insurance Association
FMA	Factory Mutual Association
FRP	Fiber-reinforced plastic
FVF	Formation volume factor
FWKO	Free-water knockout
gal	U.S. gallon
GHV	Gross heating value
GLC	Gas–liquid chromatography
GLR	Gas–liquid ratio, expressed as scf/bbl
GOM	Gulf of Mexico
GOR	Gas–oil ratio, combined gas released from stage separation of oil, expressed as scf/Bsto
GOSP	Gas–oil separation plant
gph	Gallons per hour
GPM	Gallons liquefiable hydrocarbons per 1000 scf of natural gas
gpm	Gallons per minute; describes liquid flow rate
GPSA	Gas Processors Supplier Association
gr	Grain (7000 gr = 1 lb)
GSC	Gas–solid chromatography
HAZIN	Hazards identification
HAZOPS	Hazards Operability Study
HC	Hydrocarbon
HCL	Higher combustion limit

HHV	Higher heating value
HP	High pressure
hp	Horsepower
hp-h, hp-hr	Horsepower-hour
HTG	Hydrostatic tank gauging
H_2O	Water
H_2S	Hydrogen sulfide
i-C_4	Isobutane
i-C_5	Isopentane
ID	Inside diameter
ISA	Instrument Society of America
ISO	International Standards Organization
J-T	Joule-Thomson (constant enthalpy) expansion
kW	Kilowatts
kWh	Kilowatts-hour
LACT	Lease automatic custody transfer
LC	Level control
LCL	Lower combustion limit
LCV	Level control valve
Lb	Pounds
Lbmol	Pound mole
LED	Light emitting diode
LET	Lowest expected temperature
LHV	Lower heating value
LMTD	Log mean temperature difference
LNG	Liquefied natural gas; primarily C_1 with lesser amounts of C_2 and C_3

LP	Low pressure
LPG	Liquefied petroleum gas, C_3-C_4 mix
mA	Milliampere
MAWP	Maximum allowable working pressure
Mcf	Sloppy equivalent for Mscf
Mcfd	Thousand cubic feet per calendar day
MF	Meter factor
MIGAS	Ministry of Oil and Gas (Indonesia)
MMcf	Same as MMscf
MMcfd	Millions of standard cubic feet
MMscfd	MMscf per day
MMS	Minerals Management Service
MPT	Minimum pipeline temperature
Mscf	Thousand standard cubic feet
Mscfd	Mscf per day
MW	Molecular weight
N, N_2	Nitrogen
NACE	National Association of Corrosion Engineers
NBS	National Bureau of Standards, now NIST
n-C_4	Normal butane
n-C_5	Normal pentane
NFPA	National Fire Protection Association
NGL	Natural gas liquids; includes ethane, propane, butanes, pentanes, or mixture of these
NHV	Net heating value
NIST	National Institute for Standard and Technology, formerly NBS
NORM	Naturally occurring radioactive materials
NPDES	National Pollution Discharge Elimination System

NPS	National pipe standard
NPSH	Net positive suction head
NPSHA	Net positive suction head available
NPSHR	Net positive suction head required
OCS	Outer continental shelf
OD	Outside diameter
ORLM	Optical reference line method
OSHA	Occupational Safety and Health Administration
OTM	Optical triangulation method
PCV	Pressure control valve
PD	Positive displacement (e.g., a PD pump)
PE	Polyethylene
PI	Proportional-integral
PID	Proportional-integral-derivative
PP	Polypropylene
PR	Peng-Robinson equation of state
ppm	Parts per million
ppmv	Parts per million by volume
ppmw	Parts per million by weight
psi	Pounds per square inch
psia	Pounds per square inch absolute
psig	Pounds per square inch gauge
PTB	Pounds of salt per thousand barrels of clean crude oil
PTV	Prover true volume
PTT	Petroleum Authority of Thailand
PVC	Polyvinyl chloride
RK	Redlich-Kwong equation of state
RP	Recommended practice (e.g., API RP 14 E)

Rpm	Revolutions per minute
RVP	Reid vapor pressure
S	Sulfur
SAW	Submerged arc welded
S&W	Sediment and water
SCADA	Supervisory control and data acquisition
Scf	Standard cubic foot; means of expressing volume of natural and other gases. The volume at 60°F and 14.696 psia (ideal gas) for process calculations. For sales purposes, it may be defined differently by law in some states in the United States
Scfm	Standard cubic feet per minute
SDV	Shut-down valve
SDWA	Safe Drinking Water Act
SF	Shrinkage factor
SG	Specific gravity
SI	Abbreviation for (1) shut in, (2) Système International (French for "International System of Units")
SP	Set point
SPE	Society of Petroleum Engineers
SRB	Sulfate-reducing bacteria
stbo	Stock-tank barrels of crude oil
TAPS	Trans-Alaska Pipeline System
TBP	True boiling point
TEG	Triethylene glycol
TVP	True vapor or bubble-point pressure
TTEG	Tetra Ethylene Glycol
UMSRK	Usdin-McAuliffe form of the SRK equation of state
UIC	Underground injection control
UOP K	Universal Oil Products K factor
USGS	United States Geological Survey

VLE	Vapor–liquid equilibrium
VRU	Vapor recovery unit
WC	Water column (e.g., $h_w = 80$ in. WC)
WMT	Waste-management technology
WOR	Water–oil ratio

Greek

Δ	Increment or difference

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